DATA SCIENCE AND VISUALIZATION

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MOTIVATION

The purpose of computing is insight, not numbers.

…. of **visualization** is insight, not **pictures**

DEFINITION

Visualization

1. Formation of mental visual images

2. The act or process of interpreting in visual terms or of putting into visible form

DEFINITION

Visualization is the process that

Transforms (abstract) **data** into

interactive graphical representation for the purpose of

exploration, confirmation, or presentation.

GOOD DATA VISUALIZATION

Makes data **accessible**

Combines strengths of **humans** and **computers**

Enables **insight**

communicates

VISUALIZATION

"Visualization is really about external cognition, that is, how resources outside the mind can be used to boost the cognitive capabilities of the mind."

Stuart Card

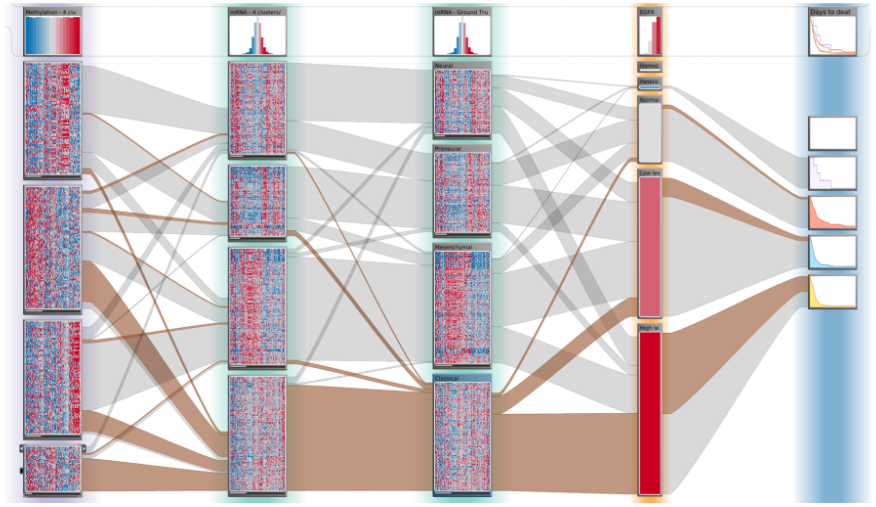
one of the pioneers of applying human factors in human–computer interaction

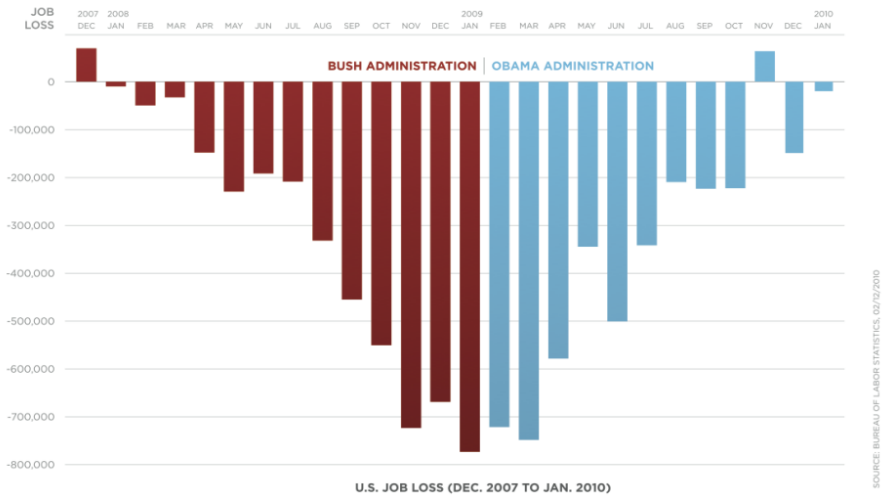
WHY VISUALIZE?

- To inform humans: communication

- When questions are not well defined: exploration
  - *What is the structure of positive comments in social network?*
  - *What is the movement pattern of customer in a supermarket?*
  - *Which drug can help patient X ?*

PURPOSE OF VISUALIZATION





Open Exploration — Confirmation — Communication

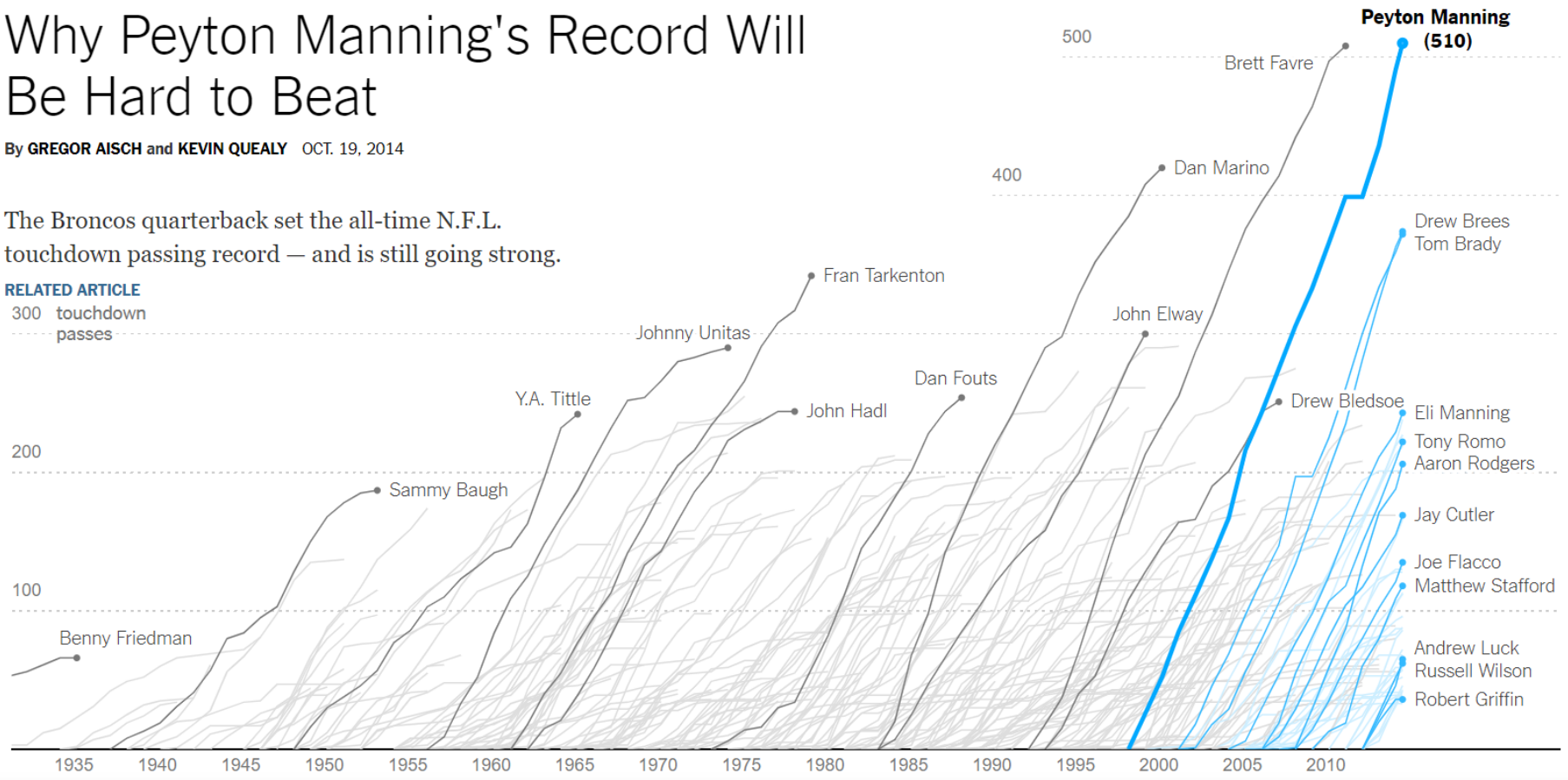
COMMUNICATION

Why Peyton Manning's Record Will Be Hard to Beat

By **GREGOR AISCH** and **KEVIN QUEALY** OCT. 19, 2014

The Broncos quarterback set the all-time N.F.L. touchdown passing record — and is still going strong.

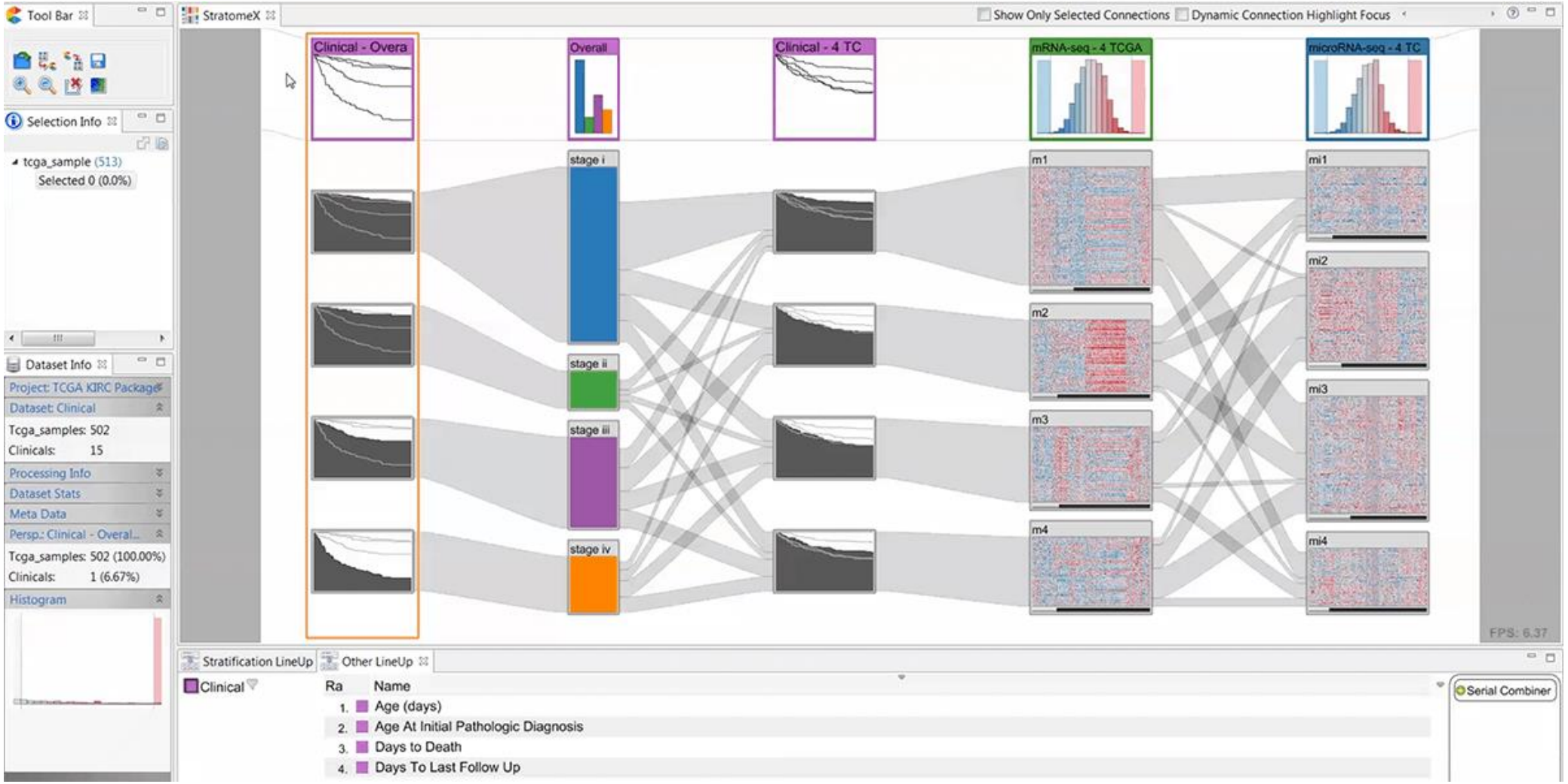
RELATED ARTICLE



https://www.nytimes.com/interactive/2014/10/19/upshot/peyton-manning-breaks-touchdown-passing-record.html?abt=0002&abg=1&smid=tw-upshotnyt

EXAMPLE OF EXPLORATION

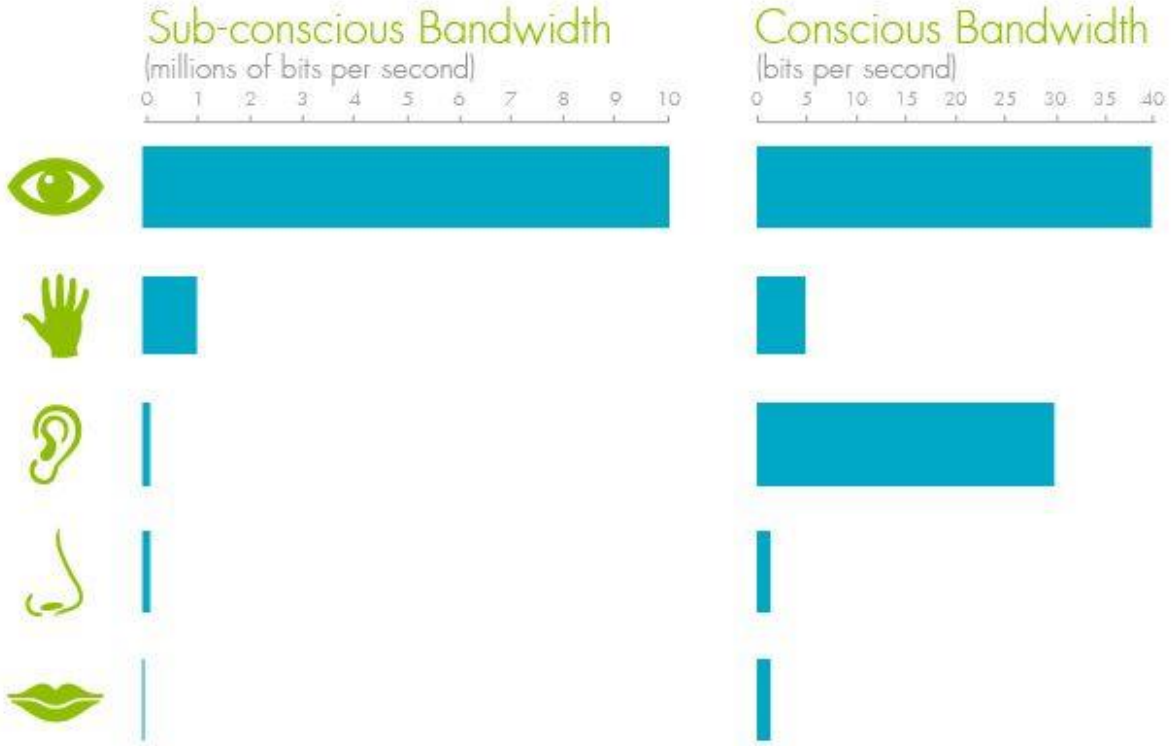
Tool to explore Genomic Stratifications in Cancer



WHY GRAPHICS?

Figures

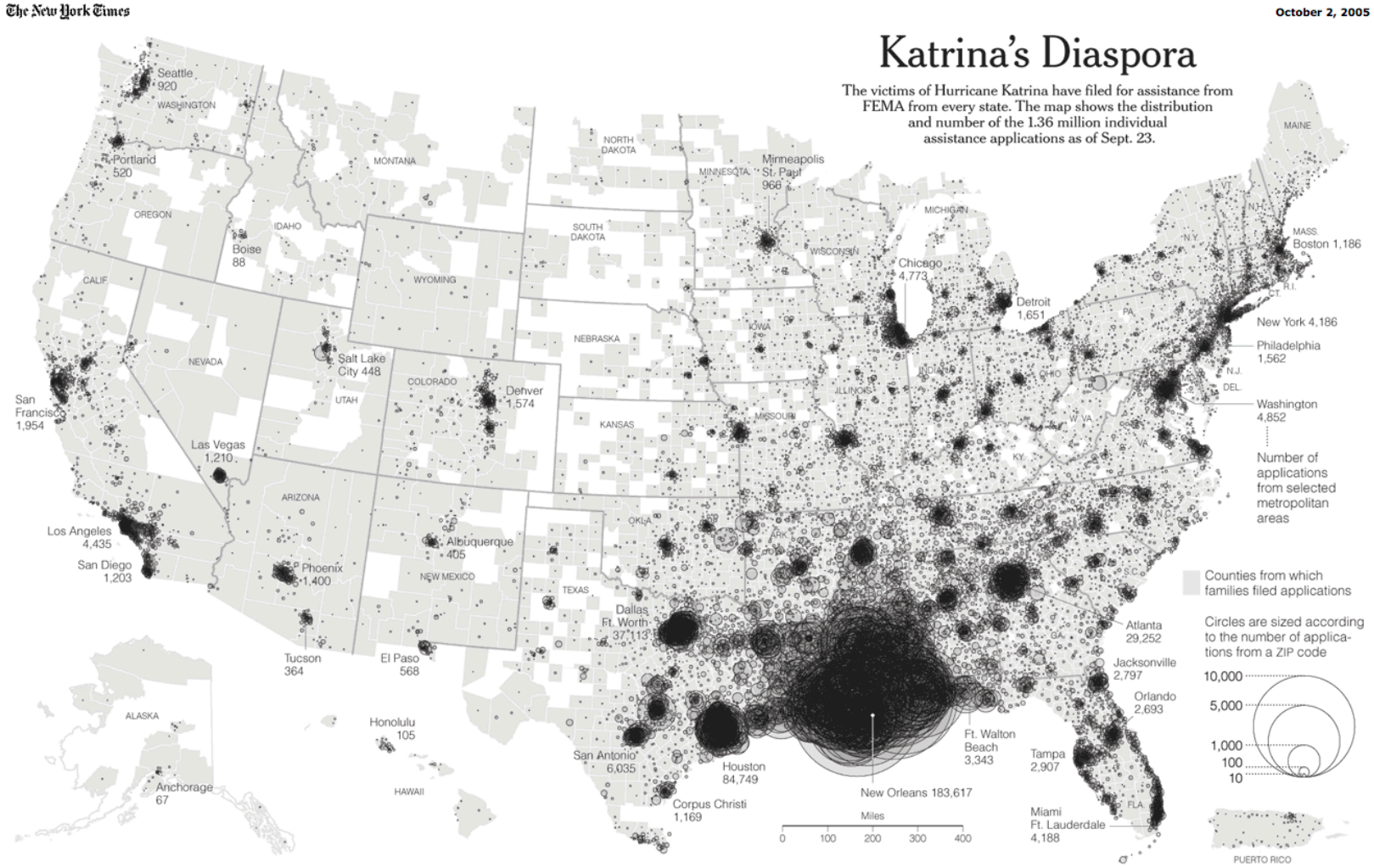
- Are **richer**; provide more info with less clutter and in less space
- Provide the *gestalt effect*: give an overview, **make structure more visible**
- Are more **accessible**, easier to understand, faster to grasp, more comprehensible, more memorable, more fun, and less formal



city's main public hospital was a wreck, and the city's public-housing projects were shuttered.

Campanella then switched to an identically constructed map, only this time based on 2010 census data, and in bits and pieces on the screen there was a simple and arresting picture of what Katrina meant. In the neighborhoods that were once a dense black, many of the little squares had thinned and turned gray. The sharp lines that once separated the teapot from Central City were now blurry: the white areas of the city were pushing north, into the vacuum left by the exodus. The Bywater was graying, as it gentrified still further. "Before Katrina, an American Community Survey estimate of New Orleans Parish population was four hundred and fifty-five thousand, and about sixty-eight per cent black," Campanella said. "Now the latest estimate is three hundred and eighty-four thousand, and it's about

TEXTUAL VS IMAGE



WHEN NOT TO VISUALIZE? WHEN TO AUTOMATE?

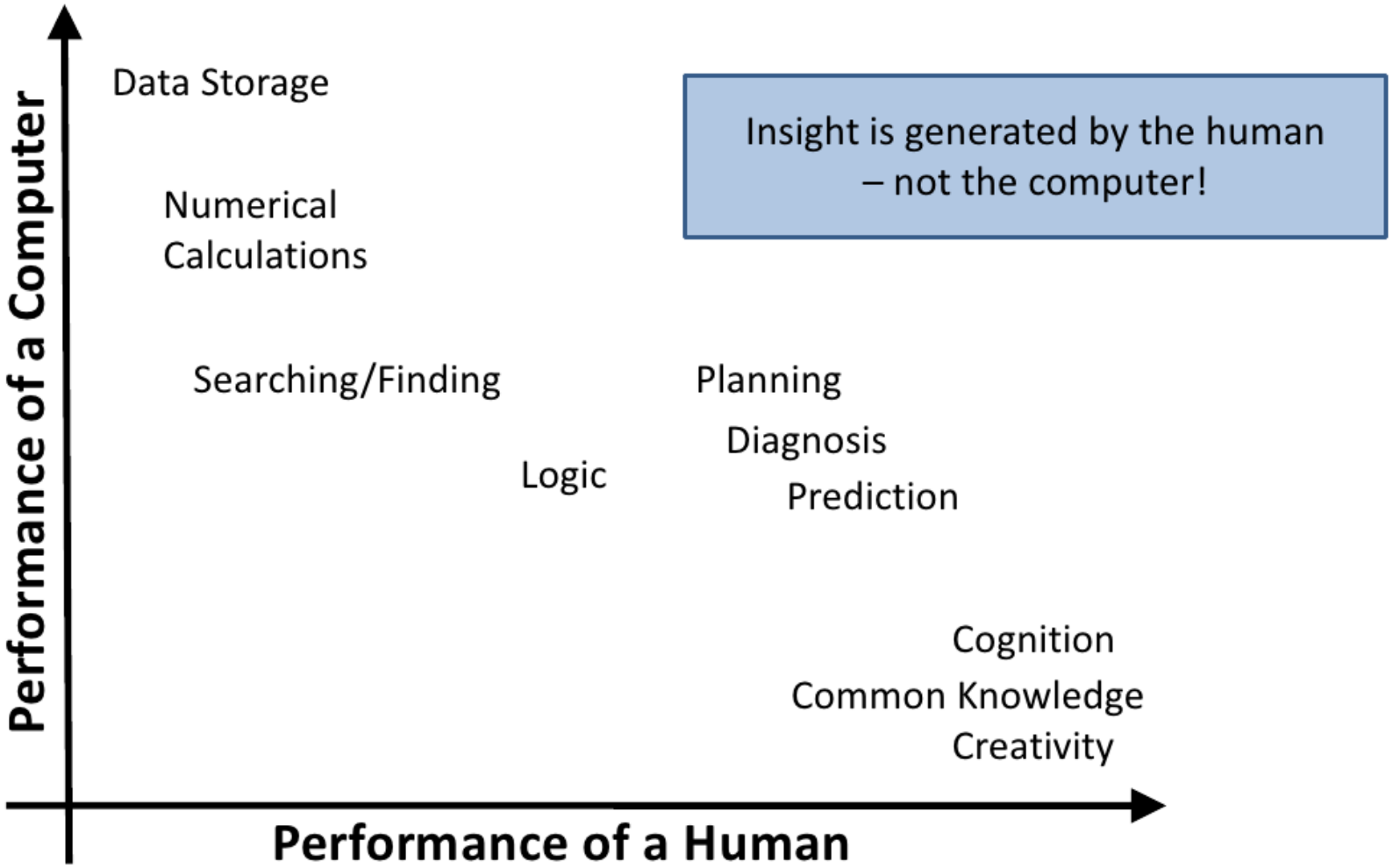
Well defined question on well-defined dataset

- *What is the current unemployment rate?*
- *Which city has the biggest contribution to the GDP of a country?*

No human intervention possible/necessary

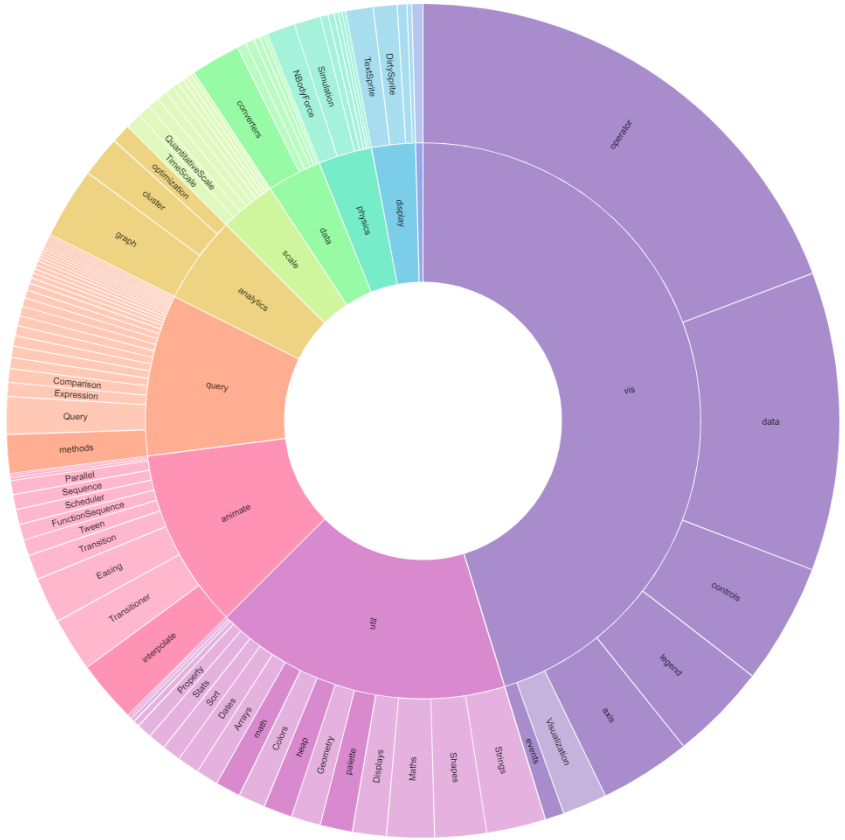
- Decision needed in minimal time
  - *High frequency stock market trading: which stock to buy/sell?*
  - *Manufacturing: is bottle broken?*
- Impractical for human to be involved
  - *Automatic data products*

COMPUTER VS HUMAN



WHY USE COMPUTER?

- Interaction
  - "drill down" into data
- Integration
  - Integrate with algorithms
  - Make visualization part of a data analysis pipeline



[Sunburst by John Stasko, Implementation in Caleydo by Christian Partl]
https://observablehq.com/@d3/zoomable-sunburst

WHY USE COMPUTER?

- Efficiency
  Re-use charts / methods for different datasets
- Quality
  Precise data driven rendering
- Storytelling
  - Use time

TELL STORIES

WHY NOT JUST USE STATISTICS

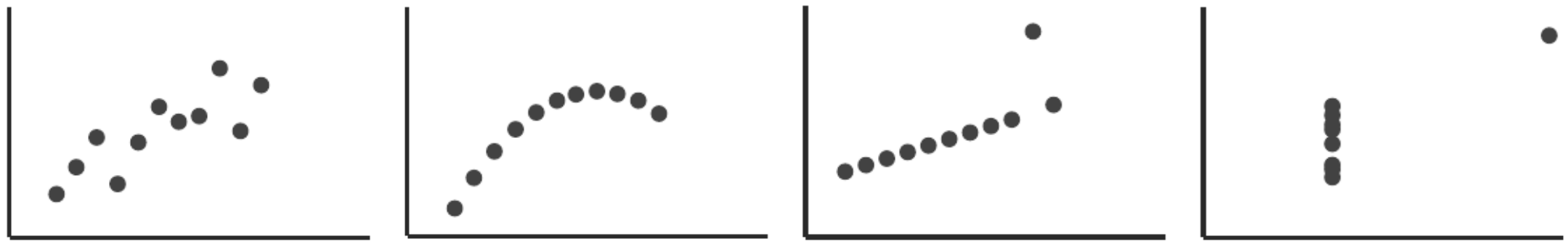
Anscombe's quartet

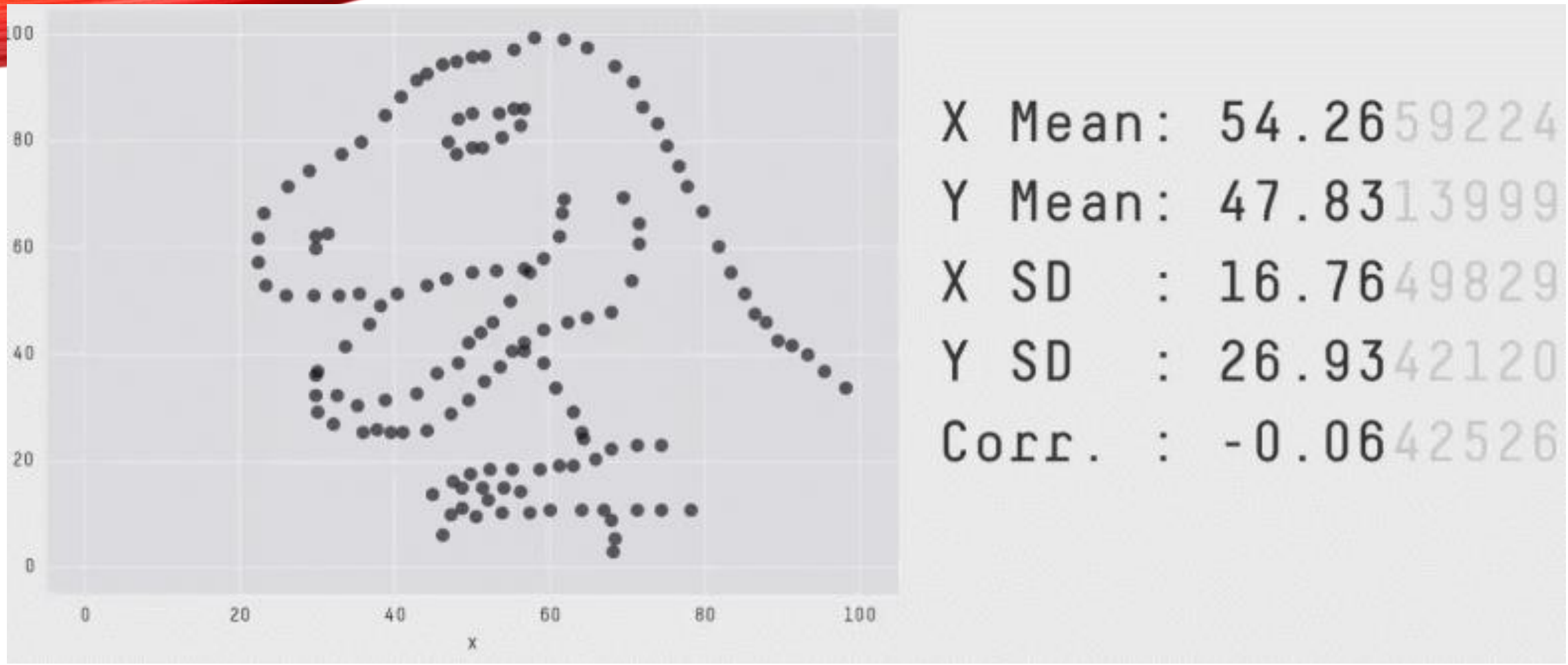
I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

For all four datasets

Property	Value
Mean of *x*	9
Sample variance of *x* : σ^2	11
Mean of *y*	7.50
Sample variance of *y* : σ^2	4.125
Correlation between *x* and *y*	0.816
Linear regression line	*y* = 3.00 + 0.500*x*
Coefficient of determination of the linear regression : R^2	0.67

ANSCOMBE'S QUARTET





Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing, CHI 2017, Justin Matejka, George Fitzmaurice

VISUALIZATION = **HUMAN** **DATA** **INTERACTION**

DATA

VISUALIZATION IN DATA SCIENCE

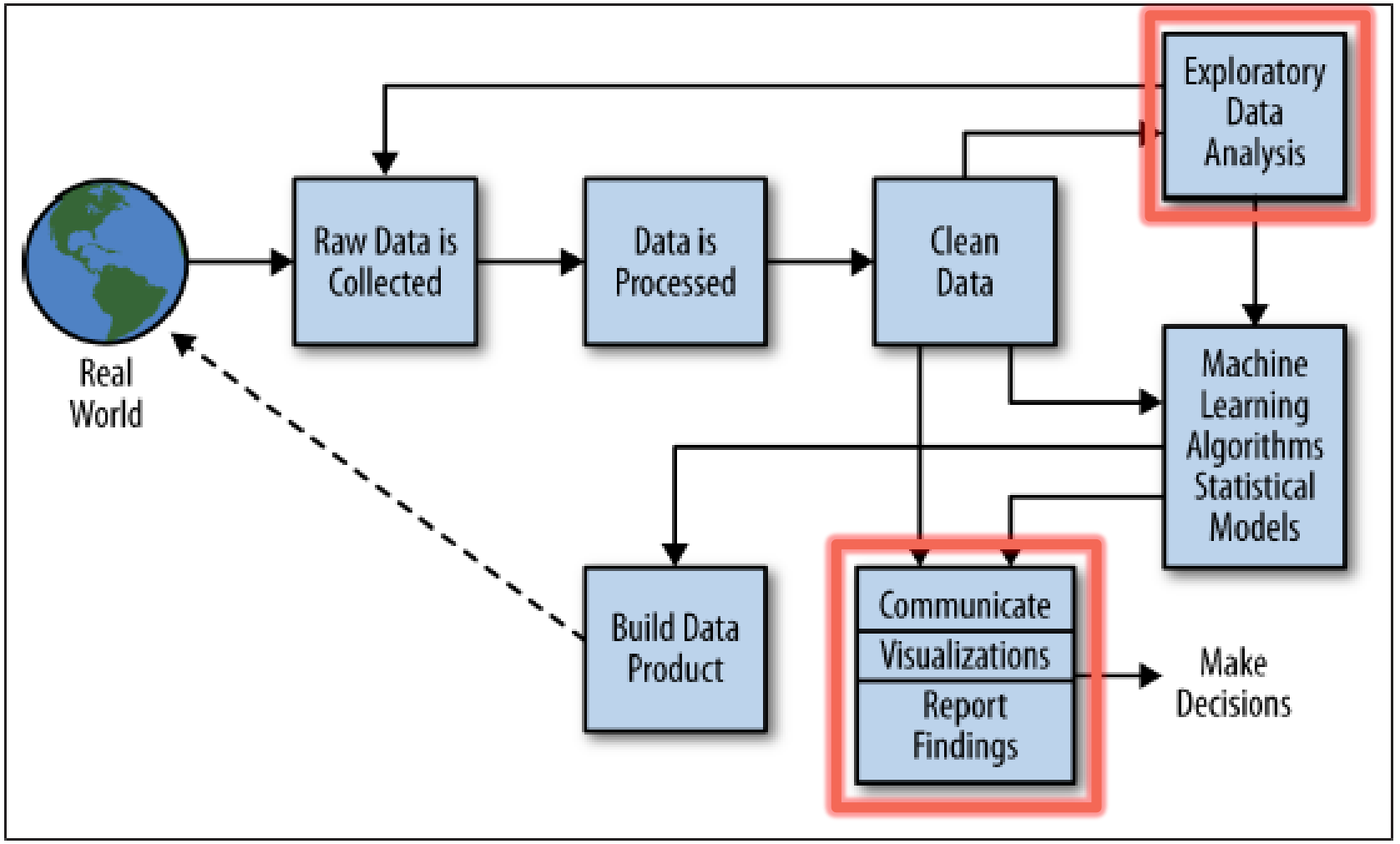


Figure 2-2. The data science process

BIG DATA

- Big Data

  2017: 2.5 exabytes (quintillion bytes)of data per day, largely unstructured

  2020: expected to reach 44 zettabytes

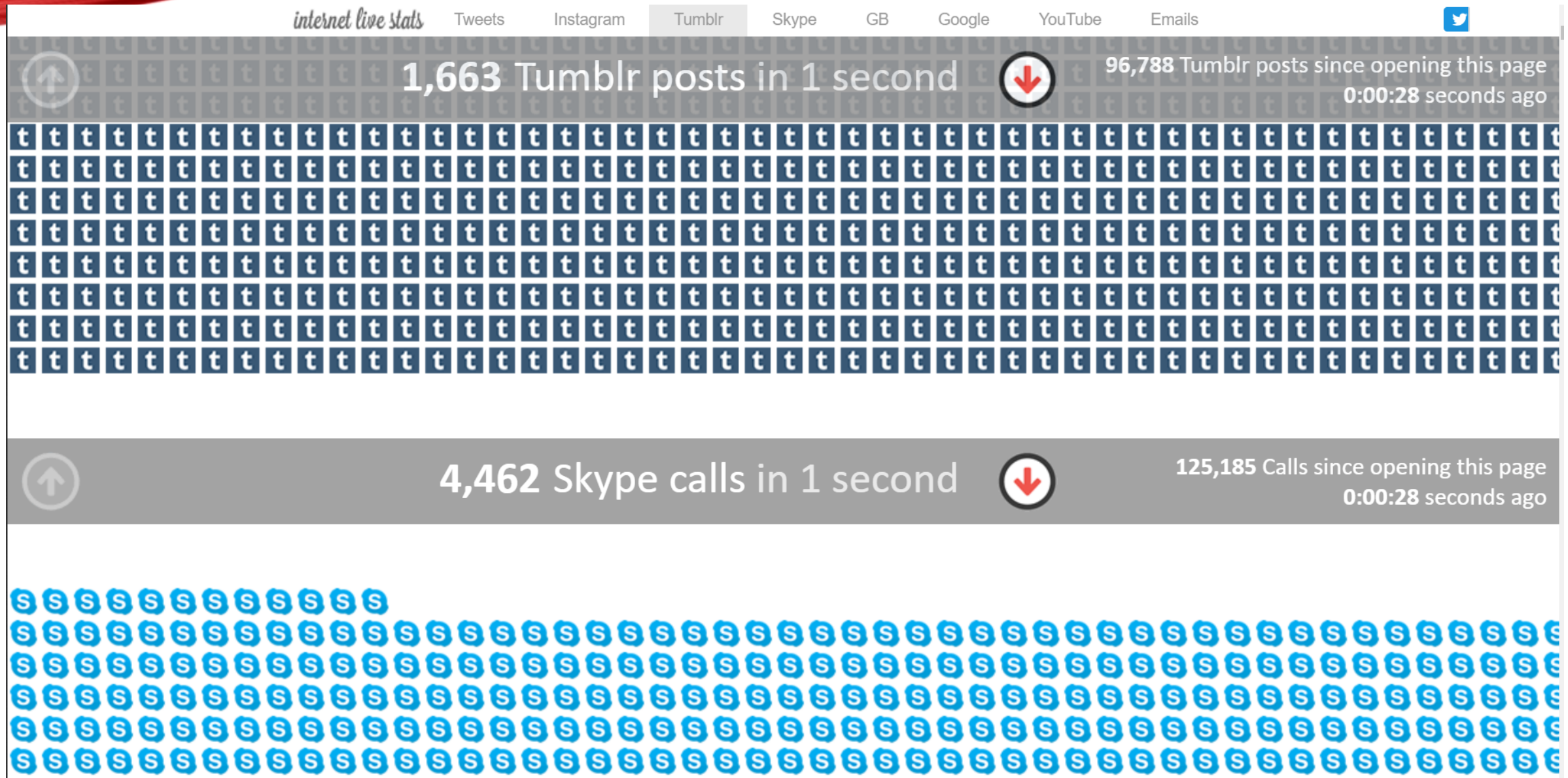
- 90% of the data created in last two years

Abbreviation	Unit	Value	Size (in bytes)
b	bit	0 or 1	1/8 of a byte
B	bytes	8 bits	1 byte
KB	kilobytes	1,000 bytes	1,000 bytes
MB	megabyte	$1{,}000^2$ bytes	1,000,000 bytes
GB	gigabyte	$1{,}000^3$ bytes	1,000,000,000 bytes
TB	terabyte	$1{,}000^4$ bytes	1,000,000,000,000 bytes
PB	petabyte	$1{,}000^5$ bytes	1,000,000,000,000,000 bytes
EB	exabyte	$1{,}000^6$ bytes	1,000,000,000,000,000,000 bytes
ZB	zettabyte	$1{,}000^7$ bytes	1,000,000,000,000,000,000,000 bytes
YB	yottabyte	$1{,}000^8$ bytes	1,000,000,000,000,000,000,000,000 bytes

1 million TB →

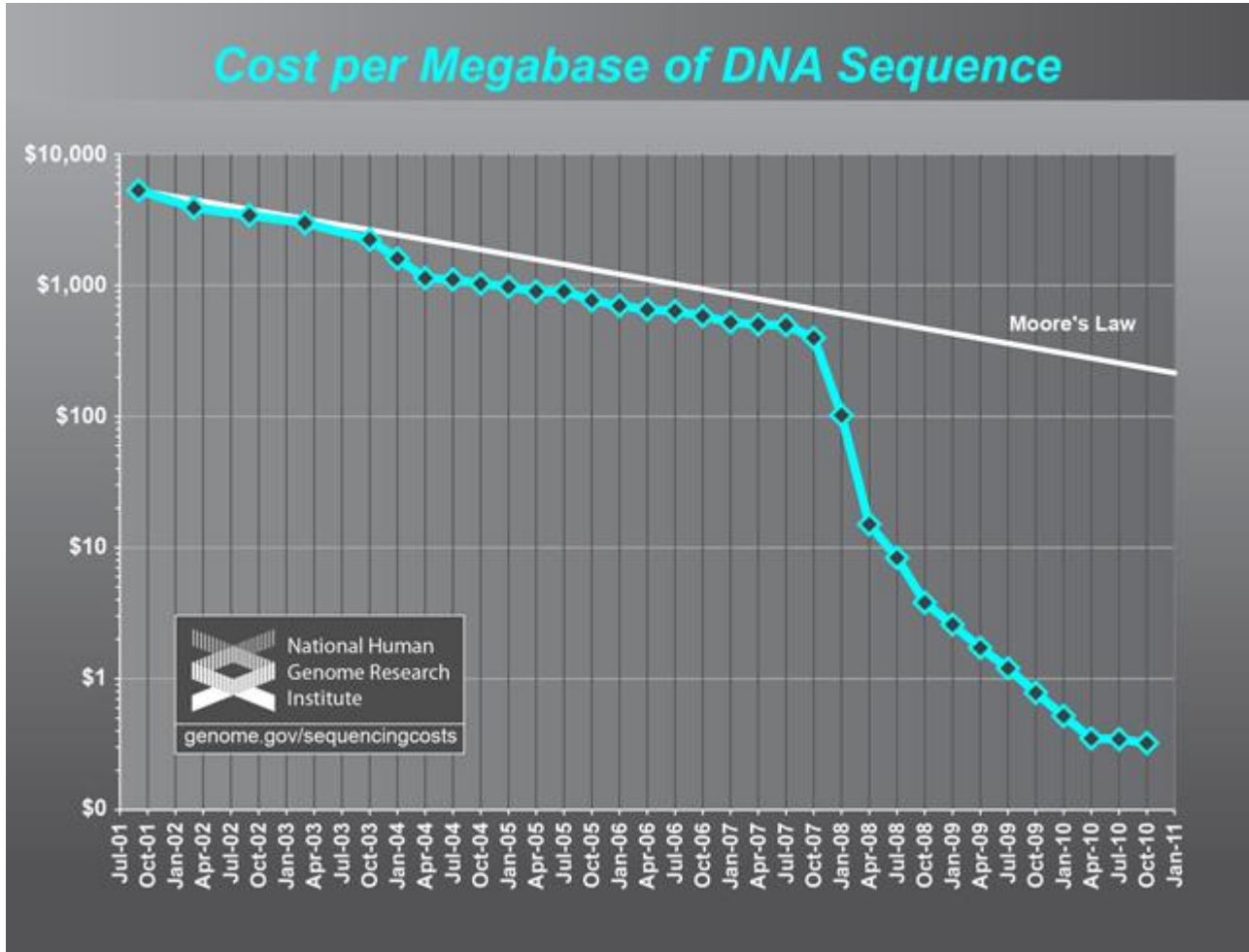
1000 billion TB →

SOURCE OF DATA



BIG DATA IN SCIENCE

- Big data is transforming science and engineering
  - Internet of things
  - Sensor
  - Camera
- In chemistry
- In medicine
- In smart building/smart city



GET A SENSE OF BIG DATA

- Example: CERN Large Hadron Collider Data
  CERN has publicly released over 300TB of data: CERN Open Data Portal

How much is that?

- A **DVD-R** holds 4.7 GB. You'd need **63,830** of them to hold 300 TB.
- It takes Pandora about a day and a half to burn through a gig of mobile data. So if the CERN data was an album, you could **stream it in just over 1,230 years**.
- At 350 MB per hour for 4K video streaming, so if the CERN data was a 4K movie it'd probably be about 857,142 hours, or about **98 years** long.
- But it ain't no thing compared to what the National Security Agency works with. Going by 2013 figures the agency released, the NSA's various activities **"touch" 300 TB of data every 15 minutes** or so
(Popular Mechanics Article

BIG DATA IN DATA CENTER

- NSA Utah Data Center (Bluffdale, Utah)
  Storage Capacity?
  estimates vary, but Forbes magazine estimates 12 exabytes (12,000 petabytes or 12 **million** terabytes)

"The ability to take data—to be able to **understand** it, to **process** it, to **extract value** from it, to **visualize** it, to **communicate** it—that's going to be a hugely important skill in the next decades, … because now we really do have **essentially free and ubiquitous data**."

Hal Varian, Google's Chief Economist
The McKinsey Quarterly, Jan 2009

HUMAN

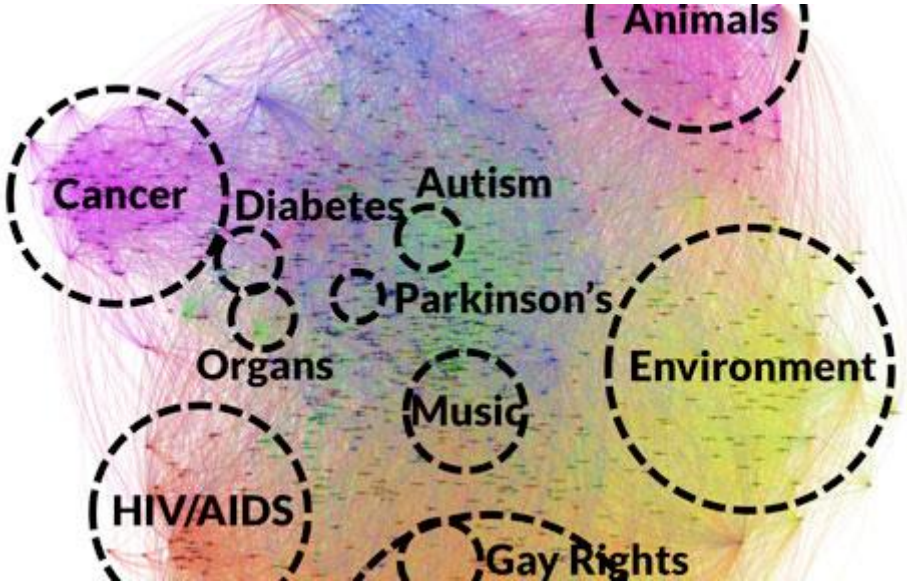
In human-data interaction

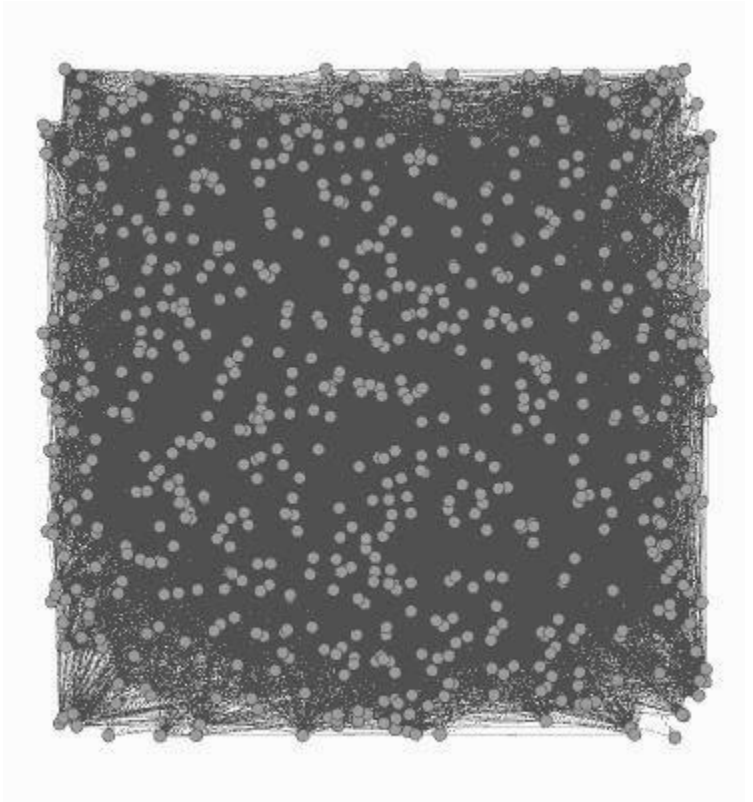
WHY HUMAN?

- Leveraging human capabilities
  Pattern discovery: clusters, outliers, trends
  Contextual knowledge: expectations for datasets, explanations for patterns
  Action: humans learn and take action

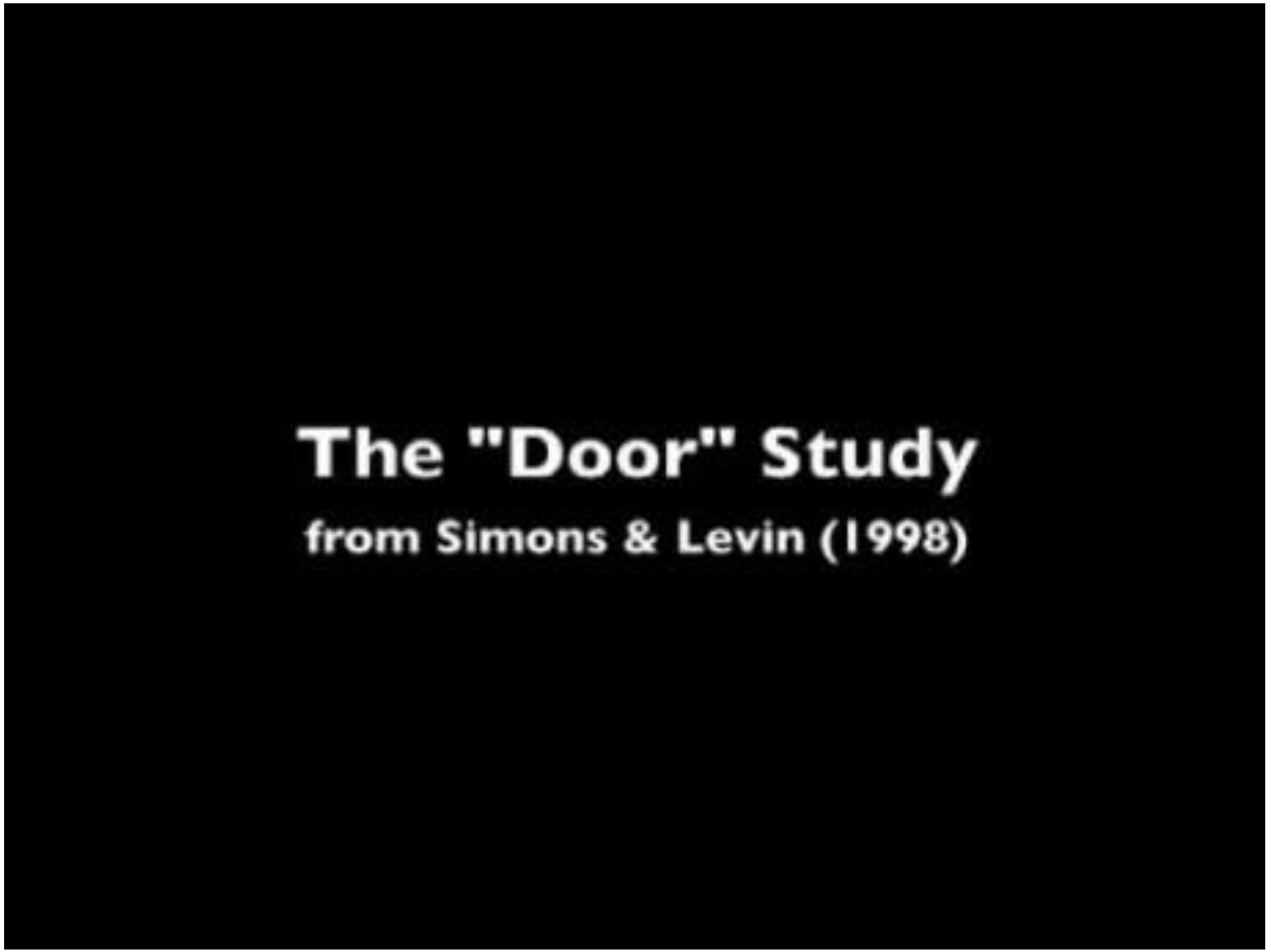
But humans have **limitation**, we have to design for them.

CAN BE DRAWN BUT CAN WE READ?



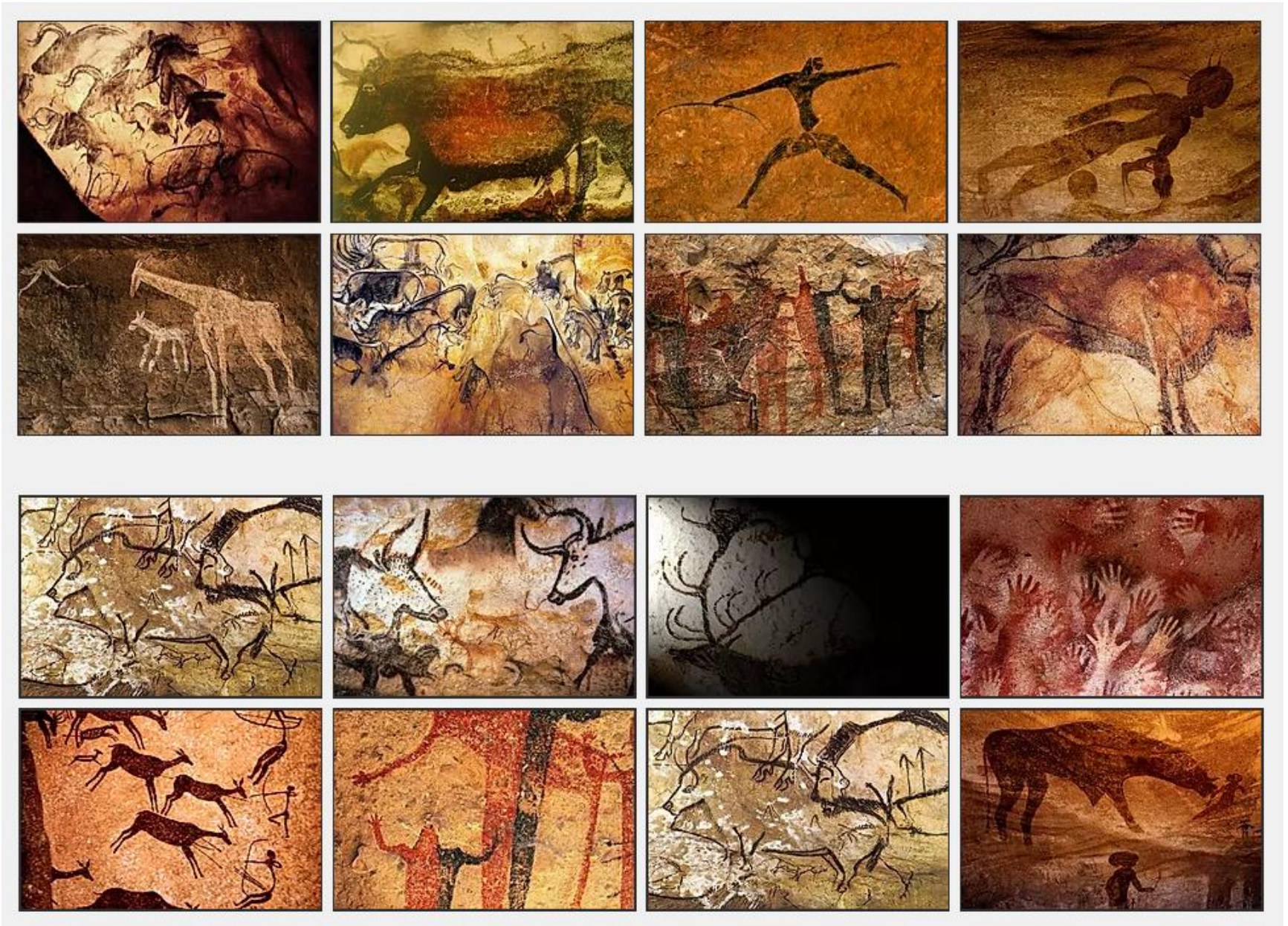


LIMITS OF COGNITION



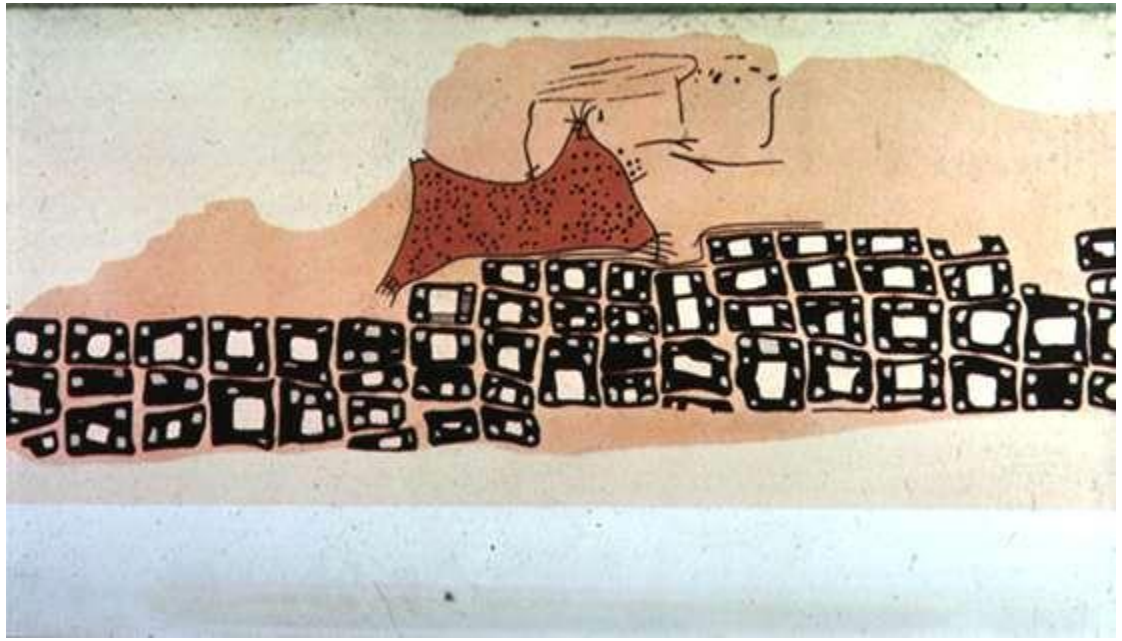
HISTORY OF VISUALIZATION

ROCK AND CAVE



The History of Visual Communication

RECORD

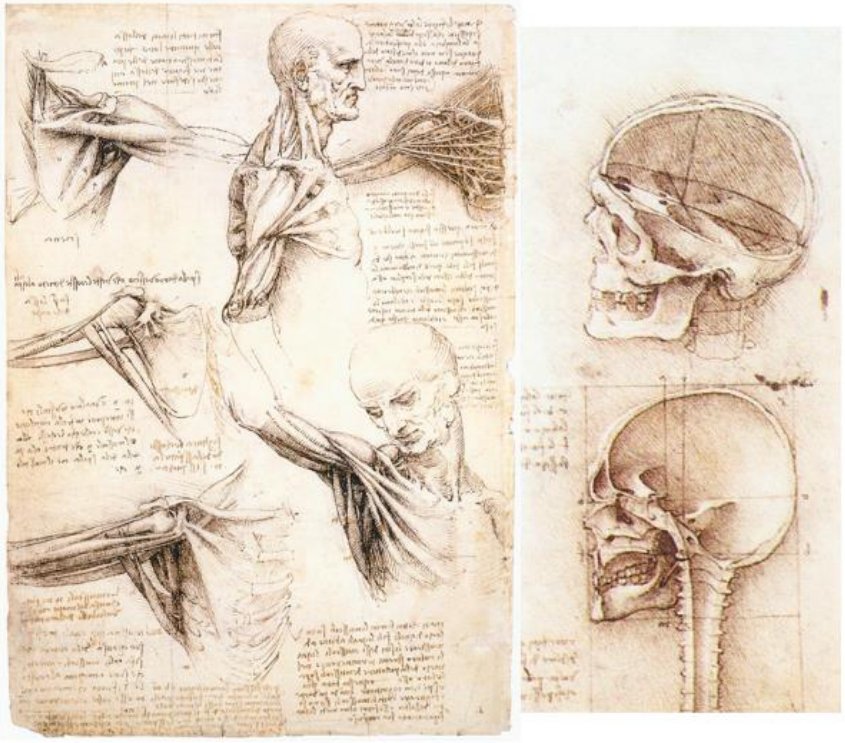


Konya town map, Turkey, c. 6200 BC

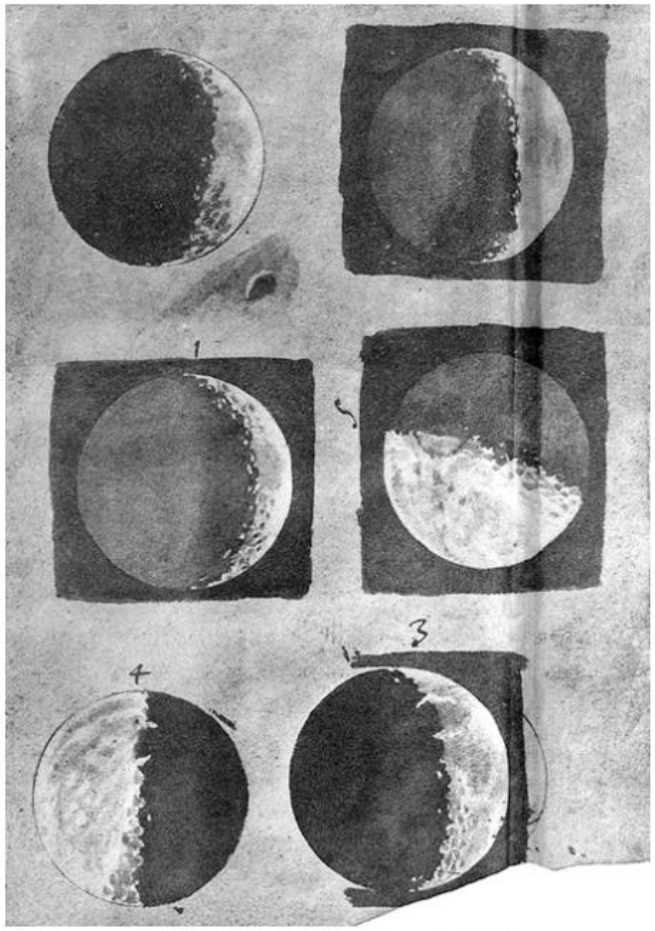
Anaximander's Map of the World

Anaximander of Miletus, c. 550 BC

RECORD



Leonardo Da Vinci, ca. 1500



Galileo Galilei, 1616

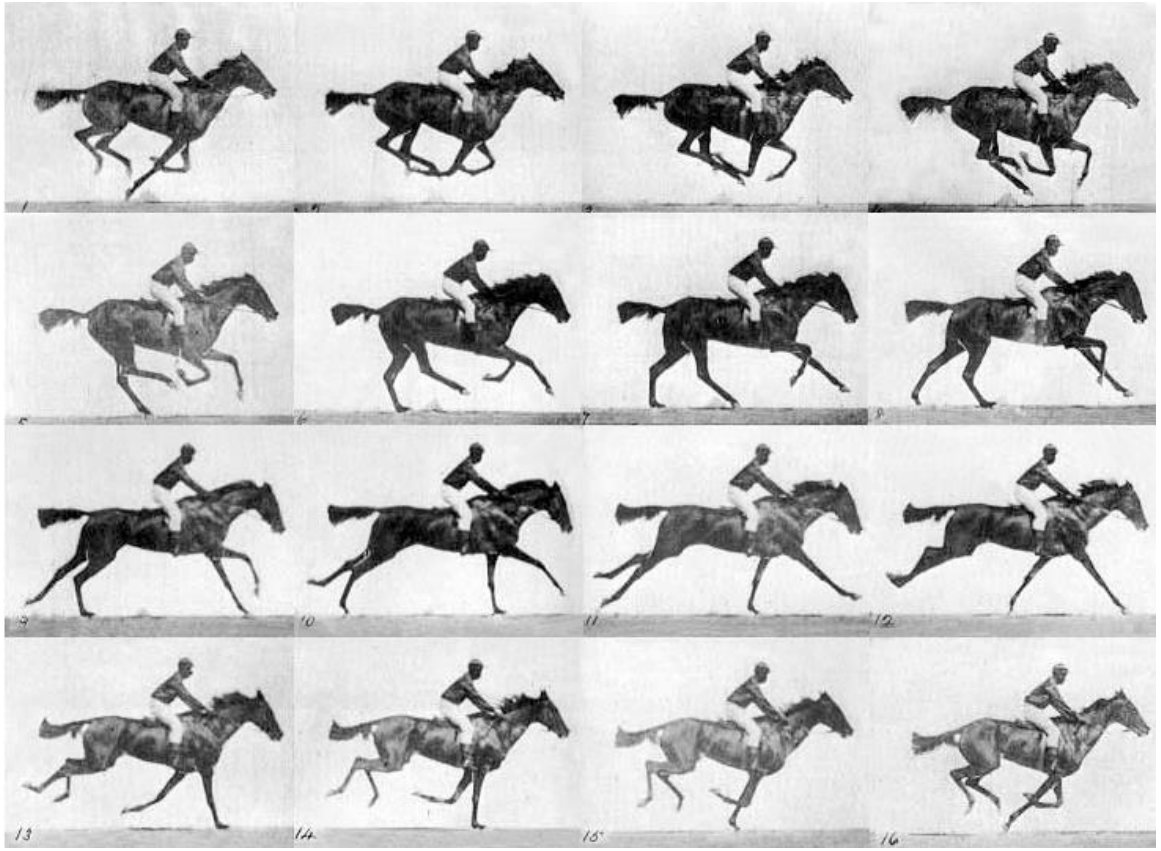
William Curtis (1746-1799)

Donald Norman

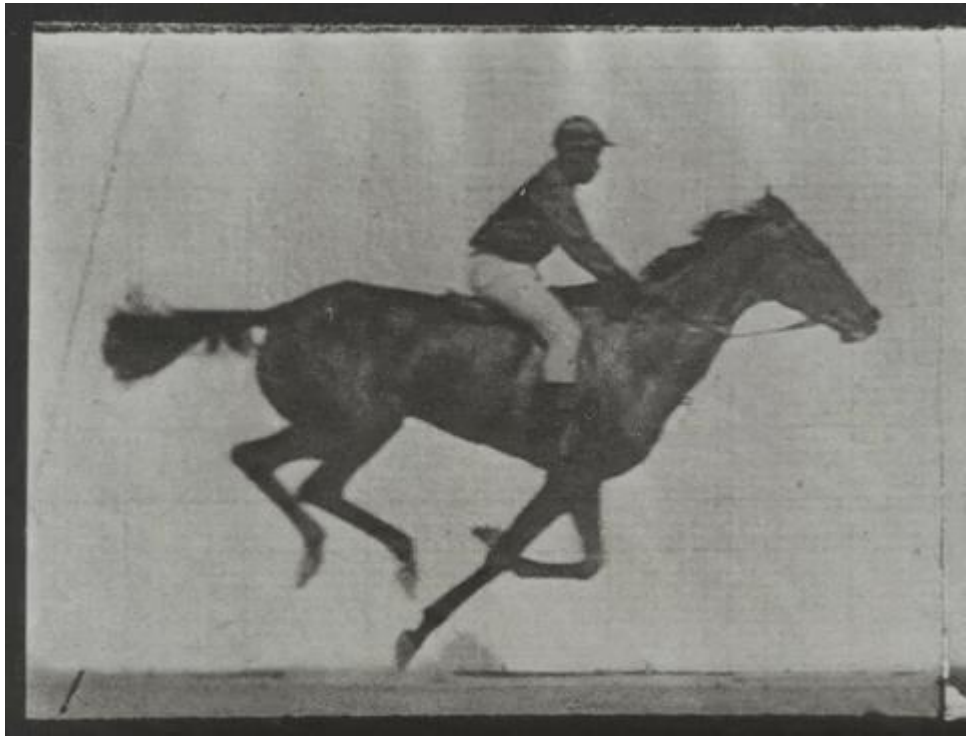
The History of Visual Communication

The Galileo Project, Rice University

RECORD



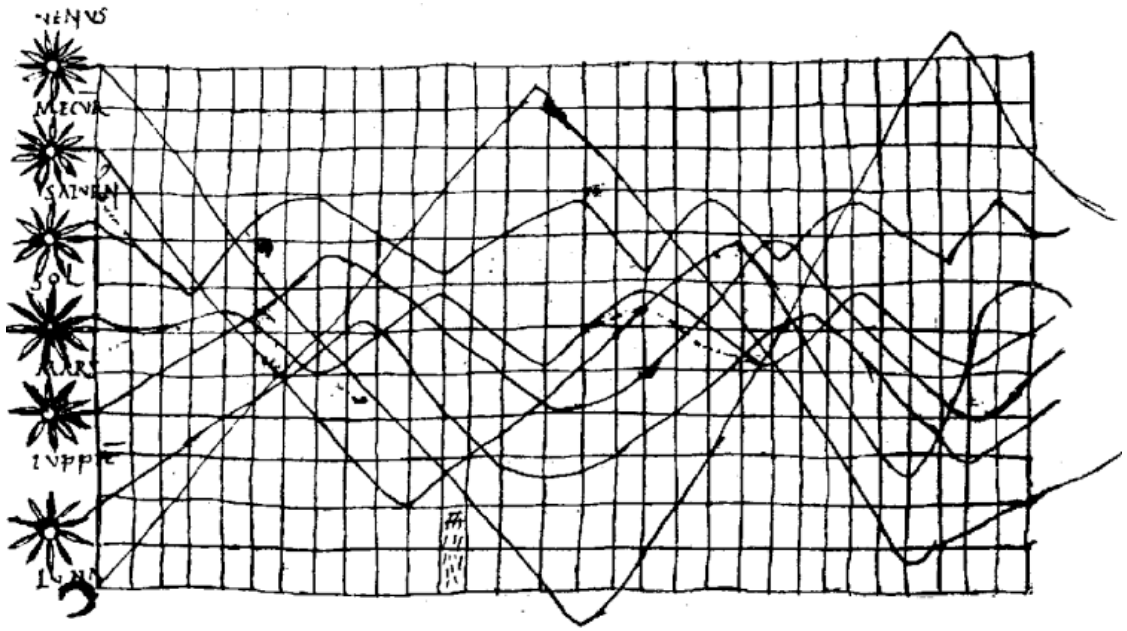
Eadweard J. Muybridge, 1878.
A Galloping horse



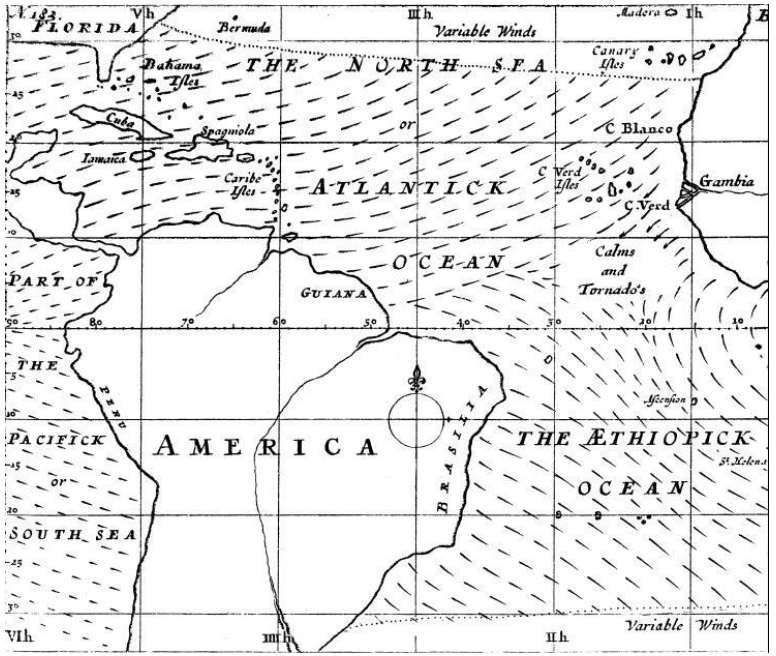
https://en.wikipedia.org/wiki/Eadweard_Muybridge

ANALYZE – TIME SERIES

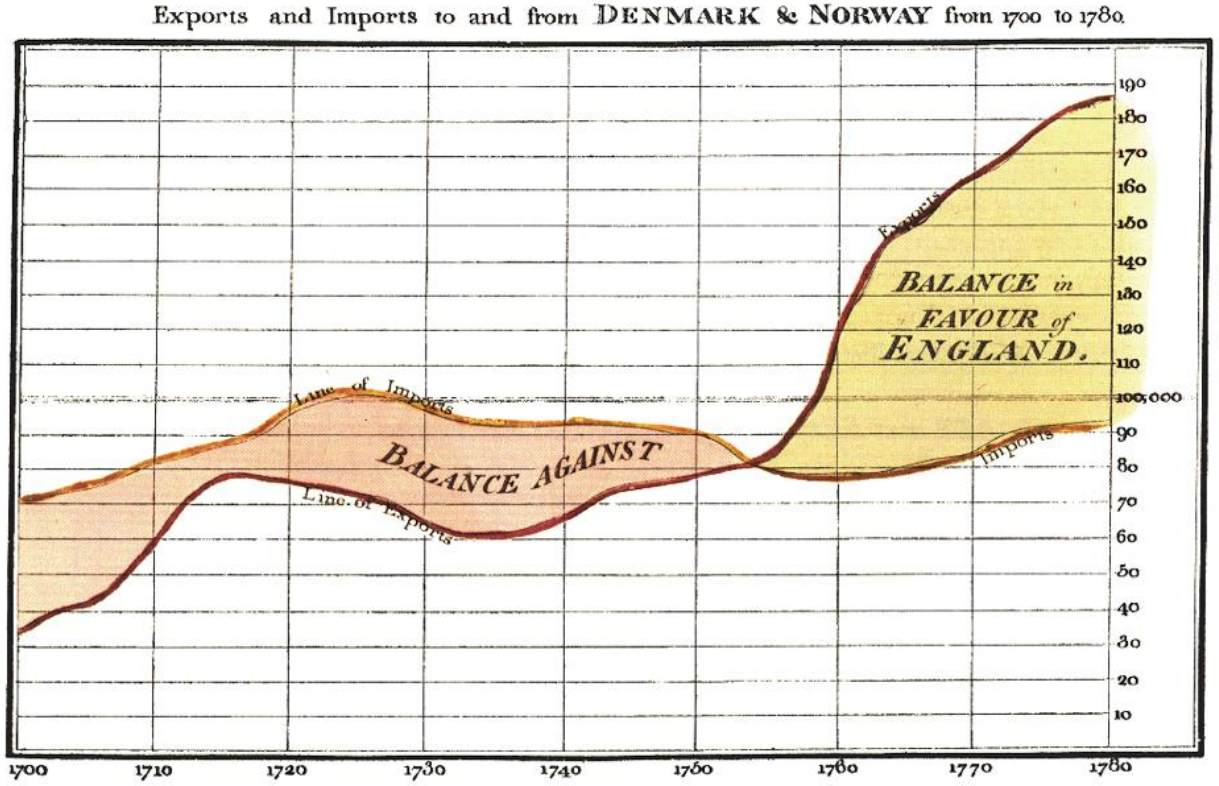
Planetary Movement Diagram, c. 950



Halley's Wind Map, 1686



ANALYZE


Exports and Imports to and from DENMARK & NORWAY from 1700 to 1780
BALANCE in FAVOUR of ENGLAND.
BALANCE AGAINST
Line of Imports
Line of Exports
Exports
Imports
190
180
170
160
150
140
130
120
110
100,000
90
80
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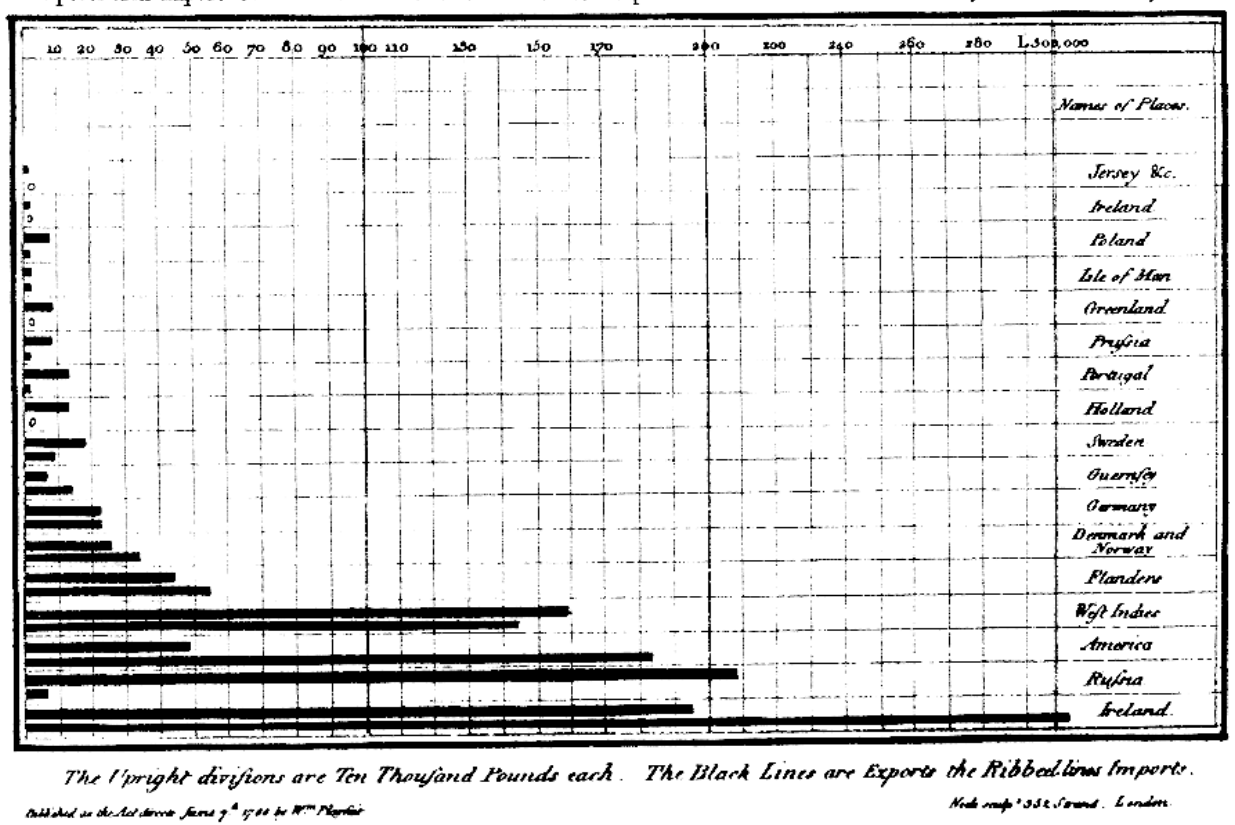
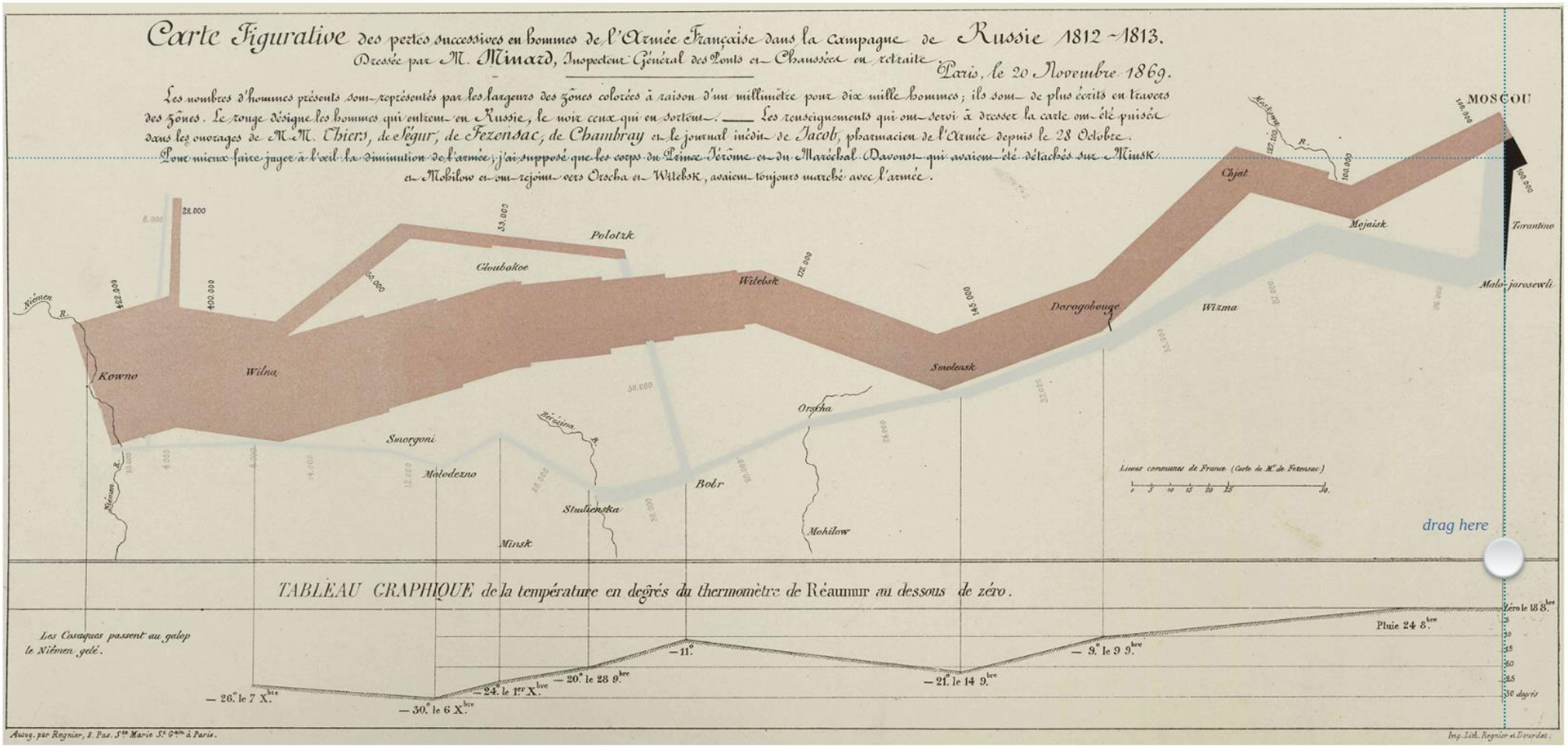

Exports and Imports of SCOTLAND to and from different parts for one Year from Christmas 1780 to Christmas 1781.
10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250 260 270 280 L.300,000
Names of Places.
Jersey &c.
Ireland
Poland
Isle of Man
Greenland
Prussia
Portugal
Holland
Sweden
Guernsey
Germany
Denmark and Norway
Flanders
West Indies
America
Russia
Ireland
The Upright divisions are Ten Thousand Pounds each. The Black Lines are Exports the Ribbed lines Imports.

FIND PATTERN

John Snow, 1854
Cholera outbreak

Spread by airbone ?
Map to show it is waterbone

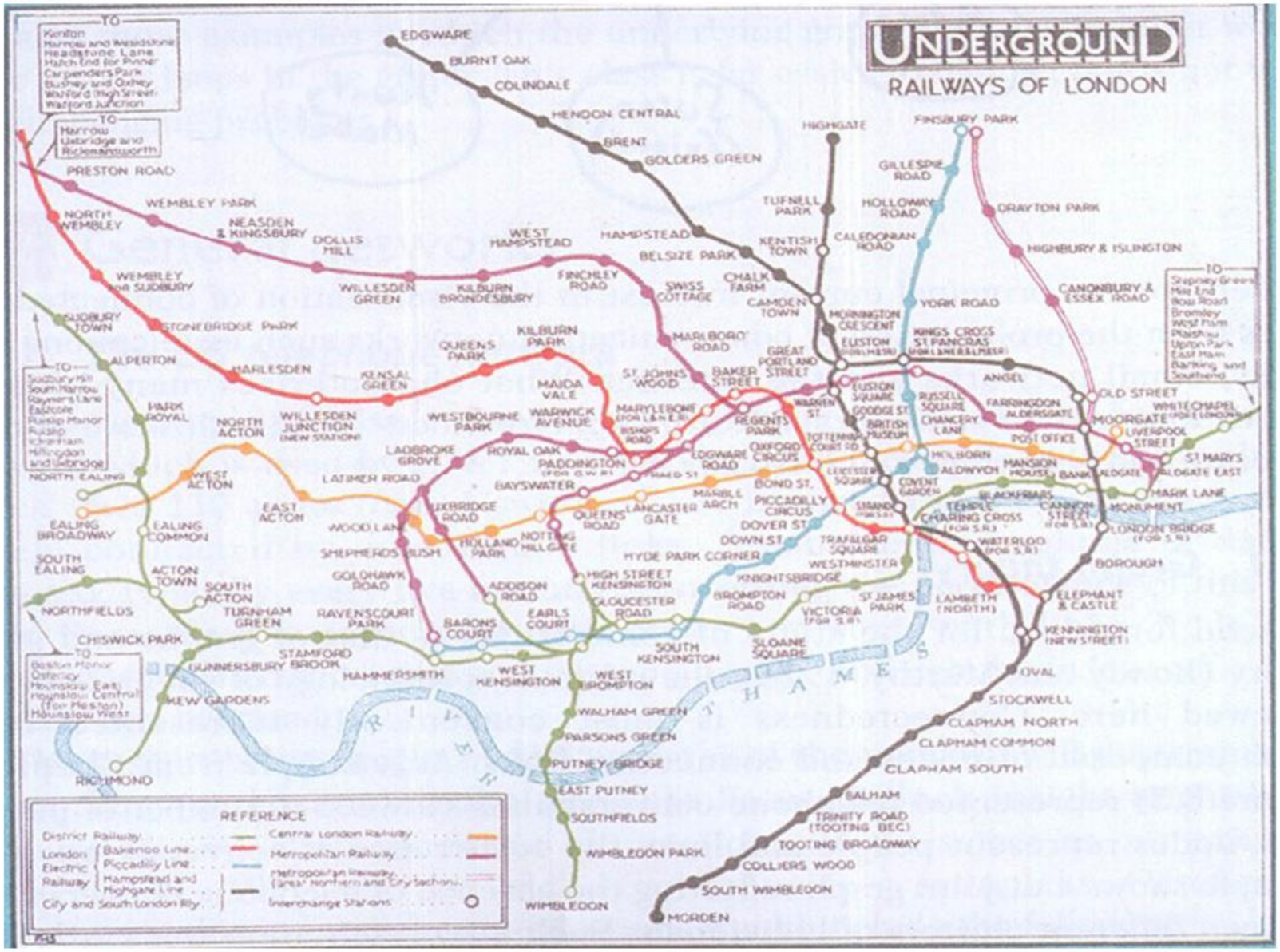
COMMUNICATE



https://www.masswerk.at/minard/ - Napoleon march - C.J. Minard, 1869

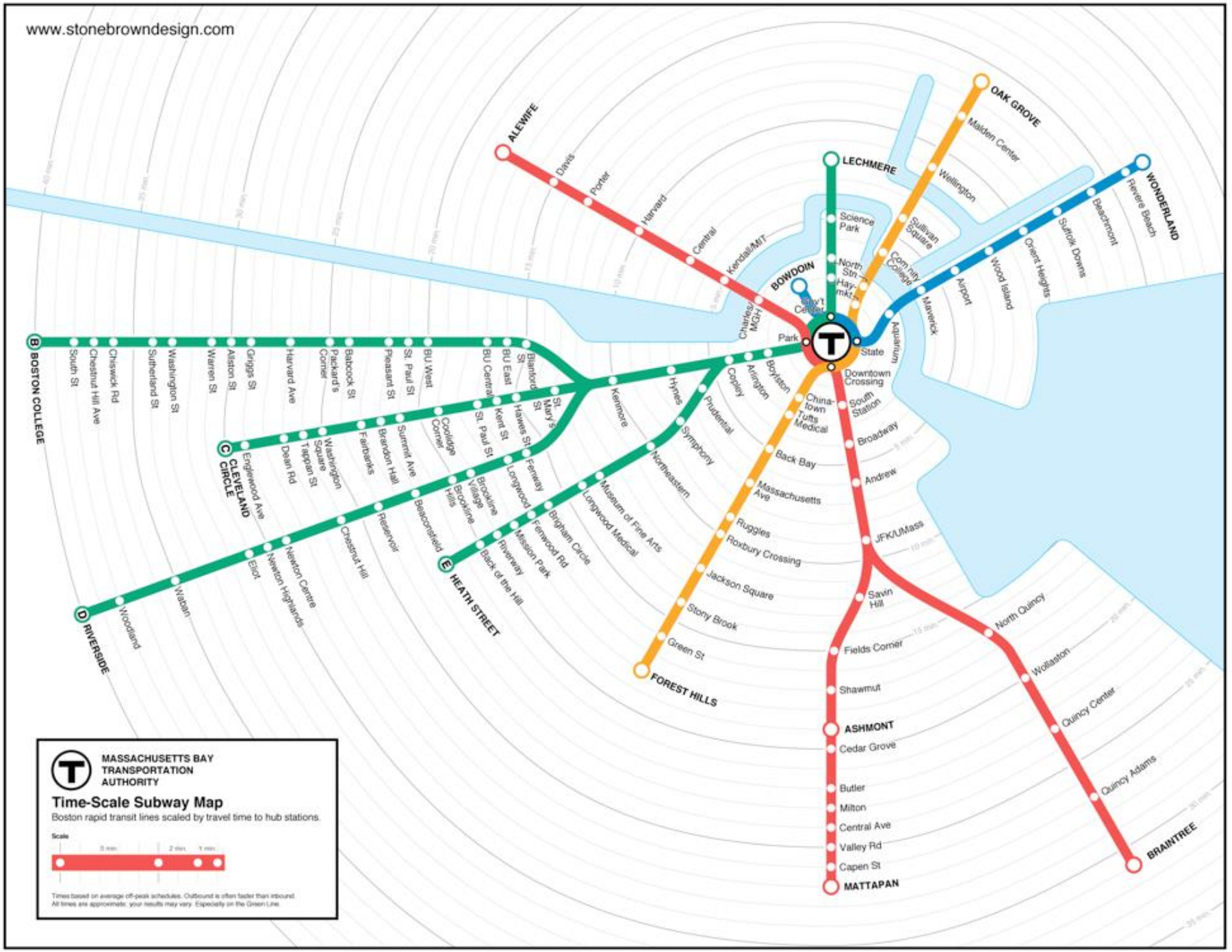
http://infowetrust.com/scroll/

COMMUNICATE



London Subway Map, 1927

SUBWAY MAP


www.stonebrowndesign.com
ALEWIFE
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Science Park
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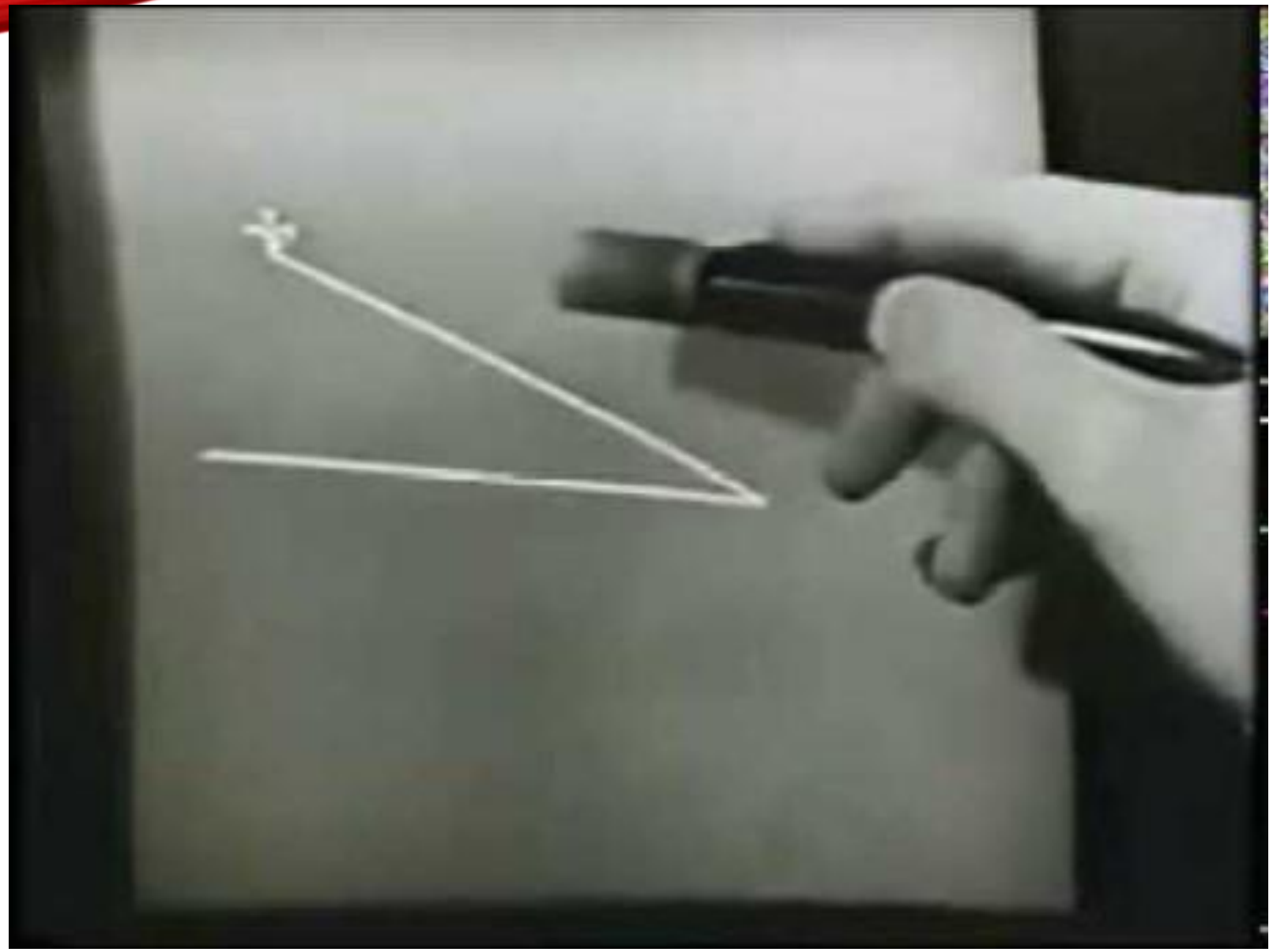
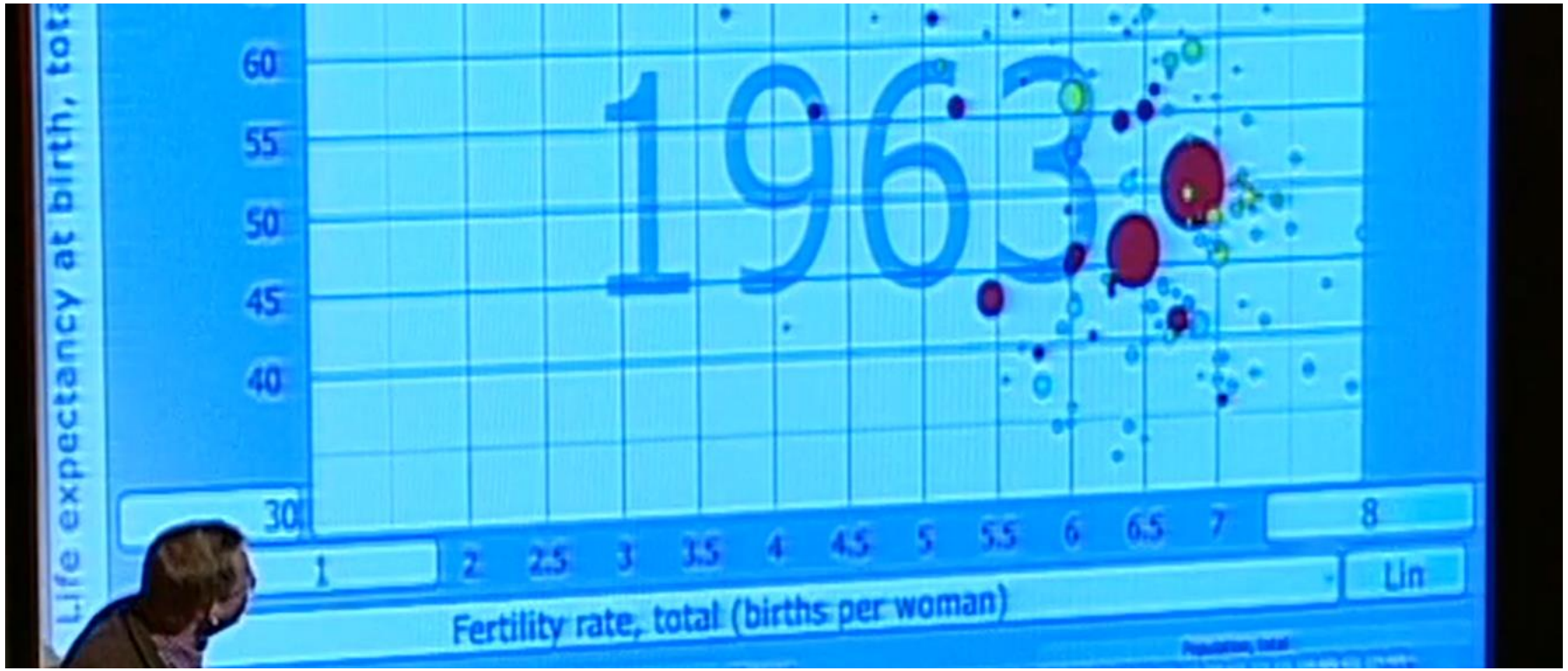
BOWDOIN
Gov't Center
Park
State
Downtown Crossing
OAK GROVE
Malden Center
Wellington
Sullivan Square
Comm'nty College
Maverick
Airport
Wood Island
Orient Heights
Suffolk Downs
Beachmont
Revere Beach
WONDERLAND
Aquarium
Boylston
Arlington
Copley
Hynes
Kenmore
B BOSTON COLLEGE
South St
Chestnut Hill Ave
Chiswick Rd
Sutherland St
Washington St
Warren St
Alston St
Griggs St
Harvard Ave
Packard's Corner
Babcock St
Pleasant St
St. Paul St
BU West
BU Central
BU East
Blanford St
C CLEVELAND CIRCLE
Englewood Ave
Dean Rd
Tappan St
Washington Square
Fairbanks
Brandon Hall
Summit Ave
Coolidge Corner
St. Paul St
Kent St
Hawes St
St. Mary's St
Fenway
Longwood
Brookline Village
Brookline Hills
Beaconsfield
D RIVERSIDE
Woodland
Waban
Eliot
Newton Highlands
Newton Centre
Chestnut Hill
Reservoir
E HEATH STREET
Back of the Hill
Riverway
Mission Park
Fenwood Rd
Longwood Medical
Museum of Fine Arts
Northeastern
Symphony
Prudential
China-town
Tufts Medical
Back Bay
Massachusetts Ave
Ruggles
Roxbury Crossing
Jackson Square
Stony Brook
Green St
FOREST HILLS
South Station
Broadway
Andrew
JFK/UMass
Savin Hill
Fields Corner
Shawmut
ASHMONT
Cedar Grove
Butler
Milton
Central Ave
Valley Rd
Capen St
MATTAPAN
North Quincy
Wollaston
Quincy Center
Quincy Adams
BRAINTREE
MASSACHUSETTS BAY TRANSPORTATION AUTHORITY
Time-Scale Subway Map
Boston rapid transit lines scaled by travel time to hub stations.
Scale
Times based on average off-peak schedules. Outbound is often faster than inbound.
All times are approximate; your results may vary. Especially on the Green Line.

INTERACT



Ivan Sutherland, 1963

MODERN EXAMPLES



Hans Rosling, TED 2006

https://www.ted.com/talks/hans_rosling_the_best_stats_you_ve_ever_seen

COURSE OBJECTIVES

How to **efficiently visualize data**

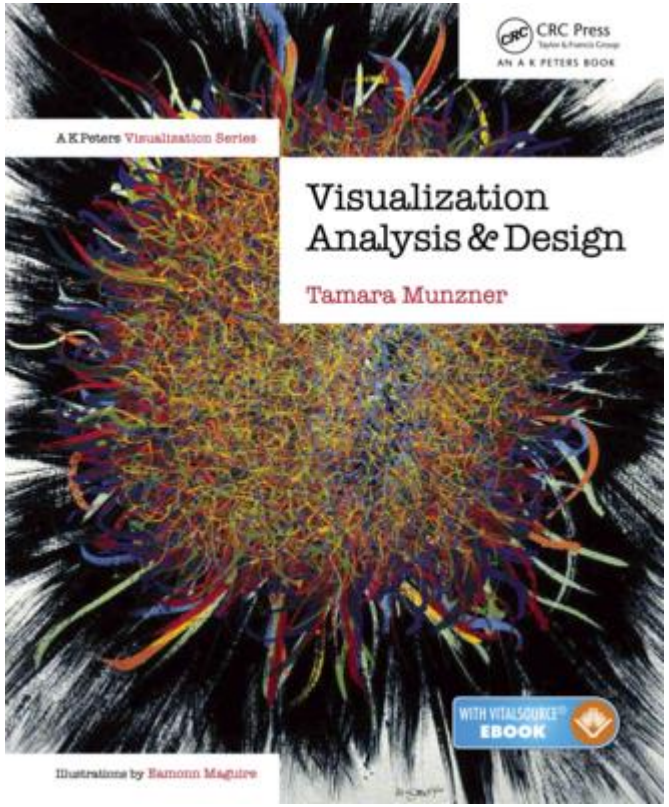
Evaluate and **critique** visualization designs

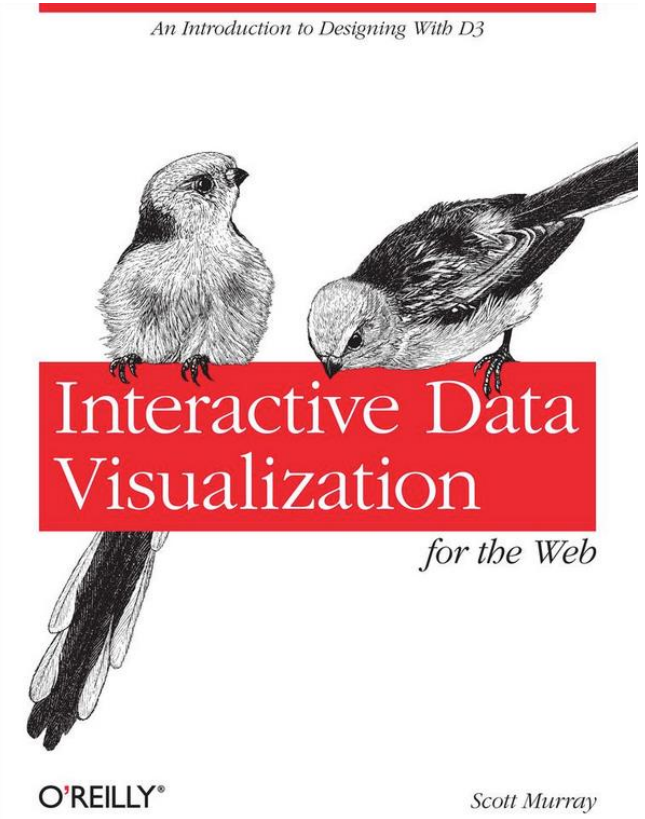
Apply fundamental principles & techniques

Design visual data analysis solutions

Implement interactive data visualizations (Web development skills)

TEXTBOOK


CRC Press
AN A K PETERS BOOK
A K Peters Visualization Series
Visualization Analysis & Design
Tamara Munzner
WITH VITALSOURCE EBOOK
Illustrations by Eamonn Maguire


An Introduction to Designing With D3
Interactive Data Visualization
for the Web
O'REILLY
Scott Murray

COURSE SCHEDULE

1. Introduction
2. Perception
3. Data abstraction, data types
4. Marks and channels
5. Design guidelines
6. Visualizing Interaction
7. Views, focus and context
8. Tabular Data
9. Storytelling with visualization
10. Visualizing networks and trees
11. Visualizing multivariate networks
12. Tasks Analysis, Designing and Evaluating Visualizations
13. Maps
14. Text visualization
15. Filtering and aggregation

GRADING

- Lab/Assignments/Quiz: 40%
- Midterm exam: 30%
- Final exam: 30%

PROJECT

- Design and implement a web-based **interactive** visualization to answer questions you have about some topic of your own choosing
- Task

1. Acquire data
2. Design a visualization
3. Implement it
4. Evaluate the result

- Team: 3-4 students
- To SUBMIT

1. A project proposal (by 16/03/2020)
2. A functional project prototype (The week after the midterm exam)
3. A final project submission (The week before the final exam)

PROJECT PROPOSAL

- Project title, link to github repo
- Background and motivation
- Objectives: the primary questions to answer
- Data: where and how to collect
- Data processing:
- Visualization design: 3 alternative prototype designs
- Must-have features
- Optional features
- Project schedule

TO SUBMIT:

3-4 pages of text, 5-6 pages of sketches of the prototype design

SciVis, InfoVis – bridging the community divide?!

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1 Introduction

Scientific Visualization (*SciVis*) and *Information Visualization* (*InfoVis*) are well-established and often used terms in the research field of visualization. But instead of intuitively illustrating two research fields – with disjunctive goals, challenges, and approaches, which they are not –, this terminology of SciVis vs. InfoVis rather represents a manifested community divide: there are *SciVis researchers* and there are *InfoVis researchers* (and there are only a few who appear on both sides), there are *SciVis conferences/journals* and there are *InfoVis conferences/journals*, etc.

In this panel, we boil up a discussion about the pros and cons of this community divide, we identify the good reasons for staying apart from each other (yes, there seem to be some of these...) as well as the good reasons for getting together a bit more. We also delineate to what extent this divide is due to historic, social, and organizational reasons (as compared to significant and inherent differences between the fields of InfoVis and SciVis).

It is important for us to outline opportunities for benefitting from each other (information visualization of scientific data as well as SciVis goodies for InfoVis). Considering future research challenges for both SciVis and InfoVis, e.g., in the context of *visual analytics*, we call for the educated "Think big!" and propose to consider visualization research as one joint field, hosting disjunctive parts, but also offering opportunities for promising and challenging joint activities, not at the least in the context of joint applications.

2 Position Statements

In the following, the panelists and the moderator contribute their points of view on the panel subject.

2.1 Breadth vs. Depth, and the Usefulness of Interdisciplinarity (Daniel Weiskopf)

As discussed earlier (e.g. in the IEEE Visualization Panel 2004 on InfoVis and SciVis [5]), InfoVis and SciVis are historically distinct fields, with largely different groups of researchers. Content-wise InfoVis and SciVis can be distinguished according to the data with which they deal. SciVis is typically applied to data with an intrinsic spatial layout (e.g., a flow simulation in 3D space), whereas InfoVis deals with data that has no pre-defined spatialization (e.g., graphs of web links). Similarly, SciVis data tends to be continuous, whereas InfoVis data tends to be discrete in nature.

The different problem settings and required background knowledge are good reasons to keep SciVis and InfoVis separate. In general, there is a trend in all areas of research toward more specialization because more advanced research questions require more in-depth knowledge and methodologies. This generic observation is also true for both SciVis and InfoVis, which have matured as research fields. Researchers need to have a thorough background in their respective areas, which is hard to achieve for the greatly differing types of methodologies typically employed in InfoVis and SciVis. Similarly, the issue of expertise affects the organizational form: the chairs, program committees, and reviewers of conferences or workshops should be experts, which is easier to achieve for narrower topic areas. Likewise, a specialized conference tends to be more appealing to an expert audience.

On the other hand, InfoVis and SciVis share many goals and basic principles, e.g., visual perception and interaction models. Therefore, a strong interaction between both fields can be productive; and, this interaction is already happening as illustrated by researchers that actively publish in both communities. It should be pointed out that visualization, in general, has started as an interdisciplinary endeavor, and it still is. Therefore, it is more than natural to maintain "interdisciplinary" links between InfoVis and SciVis. However, the advantages of interdisciplinary research may be even more pronounced for collaborations with outside fields. For example, feature extraction in 3D unsteady flow might benefit most from joint work with CFD engineers [10]; or color design issues and visual attention for interactive multi-variate data visualization might be investigated together with perceptual psychologists. Therefore, the "Think big!" should especially address the outreach to outside fields.

The current model of the separation in InfoVis and SciVis and the co-location of the IEEE Infovis Symposium and the IEEE Visualization Conference can be considered a good solution for the time being: it allows for separate organizational forms for InfoVis and SciVis and, at the same time, it facilitates interdisciplinary communication and discussion between both communities. Furthermore, additional co-located workshops provide enough flexibility to accommodate new trends (e.g. VAST and visual analytics) or venues for specialized interdisciplinary research topics.

2.2 In the InfoVis Search for Scientific Insight (Kwan-Liu Ma)

SciVis and InfoVis have each developed into distinct fields of study. I started my career as a SciVis researcher but I find myself now being actively engaged in both fields. Why? In the early days, for example, my work, and the one of many others, placed a strong focus on high-performance rendering of 3D physical phenomena or structures. However, if we look at the complete tasks of data understanding in the scientific discovery process, we can find many examples showing the need of information visualization. There are

some inherent InfoVis aspects of the scientific data analysis problem which were not looked at. We need to either develop new InfoVis techniques or make creative use of known techniques for addressing these aspects.

In volume rendering, a critical task is to classify the different materials in a volume according to the purpose of visualization. Usually, classification is done by defining, through an interactive graphics editor, a transfer function mapping voxel values to color and opacity. The conventional 1D transfer function is of limited effectiveness in performing the actual classification. To obtain better classification results, higher-dimensional transfer functions that take into account more properties of the data, such as gradient, neighboring texture, and position, are needed. Similar challenges are found in the feature extraction tasks for 4D volume data from physics simulations, from which a data set typically consists of from hundreds to thousands of time steps. A single transfer function would not be able to accommodate the often varying dynamic range of data values over time. Furthermore, some of the features are defined by multiple scalar and vector quantities, suggesting that the feature extraction must be done in a high-dimensional space. The complexity of the conventional user interface, however, rises with the dimensionality of transfer functions. Higher dimensional transfer functions are too confusing, if not impossible, for the user to define directly. In this case, we can make use of information visualization techniques, many of which have been designed for understanding high-dimensional data.

If we generalize this transfer function problem and consider that the user's task is essentially a search for insightful visualization within a space defined by a set of data properties, visualization representing the user's search pattern can allow the user to easily switch back and forth between different points in the search space. How to represent this search space is more an InfoVis problem, which appears in many scientific applications. The solution that we derive must be integrated with both the SciVis and quantitative analysis steps through linked or superimposed views of the data.

The complete data understanding process involves multiple steps including quantitative analyses at different scales, visualization of the analysis results in both the data space and some derived space, and examining uncertainty inherently in the data or introduced in the process of visualization. In fact, visualization, if appropriately designed, can also be used to effectively direct the overall data analysis process. These needs demand new information visualization capabilities. Furthermore, modern large-scale scientific investigations are collaborative in nature. Next generation visualization technologies must help users keep track of their visualization experience, use it to generate new visualizations, and share it with others. An ideal visualization system should be therefore capable of displaying not only different aspects of the data but also information about how a picture is made and related to other pictures, as well as the knowledge derived from the pictures [3]. This is again an information visualization problem.

Over the past 15 years, we have made a lot of progress in advancing the state of the art in scientific visualization [4]. To develop a complete visualization solution for large-scale scientific investigation, however, I anticipate to see even greater exchange between researchers in SciVis and InfoVis. This annual meeting provides the best venue to facilitate such very much needed exchange.

2.3 Different Data and Different People (Jarke J. van Wijk)

In my view, there are two major differences between InfoVis and SciVis. Firstly, the kind of data studied differs. Generalizing, InfoVis is concerned with abstract and discrete data, such as multivariate data, trees and graphs, whereas SciVis deals with spatial and continuous data, such as volume and flow data. Secondly, the supporting disciplines and the background of the members of the communities are different. A member from the InfoVis community has typically a background in human computer interaction, psychology, statistics, or data mining, while an SciVis researcher will often have a background in for instance computer graphics, geometric modeling, numerical computing, and parallel processing. Is there a problem? I do not think so. Both communities are active, flourishing, and produce many interesting results.

Nevertheless, real innovation comes from out-of-the-box thinking and crossing boundaries, and I think there is much potential here. As a member of the SciVis community who later joined the InfoVis community also, I will focus on what SciVis can contribute to InfoVis, using the two differences mentioned as a guide.

One central problem in visualization is the scale of data sets. If really large data sets have to be handled, they have to be transformed. One approach is to convert discrete, abstract data into continuous, spatial data. A prime example is the use of landscape metaphors where discrete objects (articles, news messages, etc.) are positioned in 2D or 3D space, and next density distributions are generated and shown. Using such conversions, the InfoVis community can take advantage of exisiting SciVis methods and techniques.

Secondly, all disciplines mentioned that are used in SciVis can also be applied for InfoVis. As an example, I have a background in computer graphics and geometric modeling, and I used this to provide extra cues for the visualization of hierarchical data. This has led to for instance the cushion treemaps [9]. I believe that realism can be used as a means to visualize abstract data more effectively, and graphics people know how to do this. As another example, I have shown how differential geometry can be used to attack zooming and panning issues [8].

In short, I think there are many interesting opportunities for SciVis methods and SciVis researchers to contribute to InfoVis. And obviously, also the opposite case can be made.

2.4 InfoVis is from Venus, SciVis from Mars (Robert Kosara)

How different are the InfoVis and SciVis communities? Let me count the ways:

Technical soundness. While SciVis papers are usually very technical and go into great depth of how a particular method is implemented and why, InfoVis papers tend to be light on implementation details, and consider implementation a minor problem. This is of course problematic on the InfoVis side, where dataset sizes are growing, and we have little existing work that can deal with that. SciVis, on the other hand, easily gets lost in implementation issues that sometimes only work on a particular (brand of) graphics card, and whose novelty and useful life are therefore limited.

Theory. While building new things is certainly a big part of our life as computer scientists, we also need to understand the theoretical basics of our field. Most model papers have so far been published at InfoVis [6], or by truly interdisciplinary people at Vis [7]. But in order to become a proper field of research, we need a foundational theory of visualization, and a much deeper understanding of the things we do [2].

Process. Understanding the processes of design and implementation of visualizations is extremely important, especially for a discipline as young as visualization. InfoVis has a history of publishing artistic papers as well as design studies, both of which put a lot of emphasis on how the results were achieved,

not just on the outcome. This is very valuable especially to beginners and outsiders, and needs to be emphasized a lot more.

Abstraction. While scientific visualization deals with things that are real, or that could be seen if they were, InfoVis usually does not. What does a bank account look like? How do you depict the results of a survey so that you can understand them and draw conclusions? The question of coming up with a visual representation from nothing in InfoVis demands a different approach than the quasi-photographic view of SciVis.

Interdisciplinary work. Visualization is inherently interdisciplinary. People who mostly come from a computer science background try to produce images that are perceived easily, and that communicate data effectively to the user. We draw on perceptual psychology, cognition, illustration, visual communication, etc. Yet this is hardly acknowledged especially in SciVis, while InfoVis researchers started conducting user studies and basing their work on literature in other fields long ago. The number of artists and other non-computer scientists is also much larger than in scientific visualization, which tends to scare them off with the level of technical knowledge that is required for entry.

Women. Look around you, how many women do you see? If there are quite a few, you stumbled into InfoVis or an InfoVis session at Vis. If there are mostly men, you're at Vis. Does this bother you? It certainly should, we need a lot more variety and color in our field, and InfoVis is doing a lot better here.

Some of the points in the above list are connected, and most of them present strengths as well as weaknesses. We can certainly act as if InfoVis and SciVis were the same, and hope that by doing so the differences will simply disappear – but that is not going to help. In fact, we can profit much more from realizing and understanding the differences, and trying to leverage them. Who says an InfoVis paper can't be extremely technical, or a SciVis technique can't be documented from start to end and thoroughly evaluated?

Like it or not, the historical division between InfoVis and SciVis is here to stay. But this should be taken as an advantage, as something that can drive innovation, rather than an obstacle.

2.5 Communities, Divides and Bridges (Helwig Hauser)

It is not in the nature of panel discussions to significantly change the course of history. If at all, panels can stimulate a broader discussion amongst scientists about relevant issues. But this, at least, is very important. And the community divide between SciVis and InfoVis definitely is a relevant issue, and even though already previously, e.g., at Vis 2003 [5], panelists discussed this issue, it is still necessary to explicitly discuss what possibly divides or integrates SciVis and InfoVis. Thinking about it and discussing the issue, we can try to delineate what we just think from what actually is true. This, hopefully, then leads to a more informed (and a less emotional) point of view.

When being open to new points of view, when being ready to expand the horizon of our usual considerations, we can identify a lot of useful input from outside worlds [1]. Many InfoVis solutions proof very useful when it comes to the visualization of multivariate, scientific data (see, for example, Fig. 1 for a InfoVis-SciVis-mixed analysis of scientific data). Increasingly, also results from the SciVis world, such as the general-purpose use of GPUs, for example, or the use of semitransparency for visualization, inspire the InfoVis world. Also, we have to face the fact that research topics change and evolve over time, both also in the fields of SciVis and InfoVis, and we are departing from known grounds and enter new ones [1]. SciVis data, for example, becomes more versatile, i.e., multi-dimensional (e.g., time-dependent and 3D), multivariate, multi-modal, multi-typed (e.g., also including discrete, categorical, nominal data, etc.), aso. InfoVis, on the other hand, faces new challenges when dealing with really large data, when going 3D, or when being confronted with dense data distributions which tightly relate to the scenario of continuous data as in SciVis.

Another point of view is the one of practitioners. For many of those who actually need visualization, it really makes no point of whether something is SciVis or InfoVis – visualization solutions are needed – and if they stem from different fields, that's just fine, but not really relevant (in the first place). In many application cases, we anyhow see that the visualization part only makes up a limited part of the complete solution – other aspects like data acquisition and management, documentation and reporting, ergonomics of the GUI, automation, etc., make up important parts, as well. From this point of view, for example, it not really makes a whole lot of sense to finely distinguish between SciVis and InfoVis. A similar point of view, for example, also is the one of funding organizations – it might pay off to sell our technology as Vis (not as SciVis or InfoVis). The same also holds for application people. Do the life sciences need SciVis or InfoVis? They probably need both.

Finally, there is yet another issue, which shows up when we take a step backwards and consider the panel topic from a slightly larger distance: in addition to SciVis and InfoVis, there is a lot more which also is visualization, or at least tightly relates to it. Knowledge visualization, for example, which much more focusses on presentation than on exploration and analysis, and cartographic/geographic visualization, which – in principle – has important aspects in common with InfoVis (but with a strong and central relation to geographical, i.e., spatial references), make up just two related fields (amongst others more), which also should be considered. Accordingly, it is well possible that in future, we not only talk about SciVis and InfoVis, but about *x*Vis, with *x* being a lot more, or, dropping the *x*, just about Vis.

3 Short Biographies

Daniel Weiskopf is an assistant professor of computing science and a co-director of the Graphics, Usability, and Visualization Lab (GrUVi) at Simon Fraser University. His research interests include scientific visualization, GPU methods, real-time computer graphics, mixed realities, as well as special and general relativity. He received a diploma degree (MSc) in physics and a PhD in physics, both from the University of Tübingen, Germany, and he did his Habilitation in computer science at the Univ. of Stuttgart, Germany.

Kwan-Liu Ma is a professor of computer science at the University of California at Davis. He has a PhD in computer science from the University of Utah. Before joining UC Davis in 1999, he was with ICASE/NASA LaRC as a research scientist. His research spans the fields of visualization, user interface design, and high-performance computing. Presently, he has projects in terascale visualization, intelligent visualization interfaces, security visualization, and large graph visualization. More information about his work can be found at http://www.cs.ucdavis.edu/~ma/.

Jarke J. van Wijk obtained a MSc degree in Industrial Design Engineering in 1982 and a PhD degree in computer science in 1986 from Delft University of Technology, both with honors. He has worked at a software company and at the Netherlands Energy Research Foundation ECN before he joined the Technische Univer-

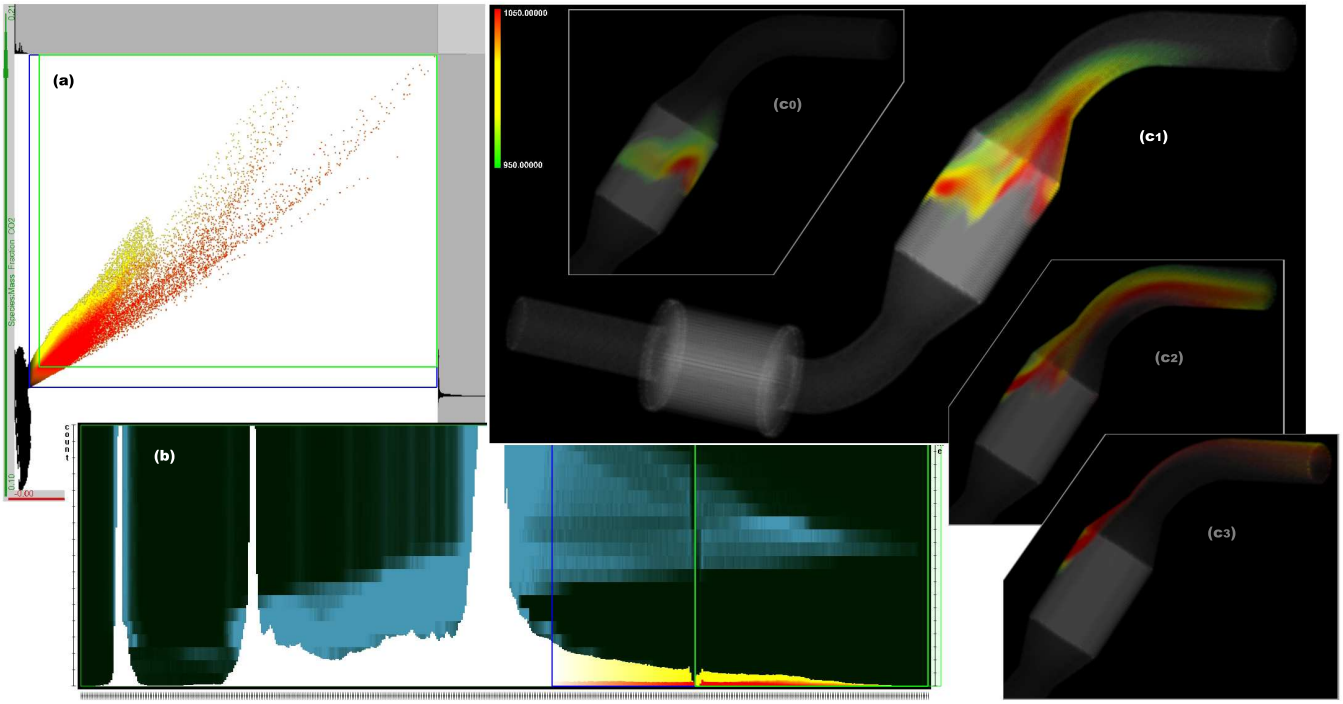

Figure 1: Visualization of the oxidation during the regeneration of a Diesel particulate filter (SciVis & InfoVis integrated). In the scatterplot (a) the CO-CO_2 distribution of the data is visualized (larger amounts of CO/CO_2 have been selected through brushing) and in the histogram (b) the temperature distribution is shown (highest values brushed). A 3D focus+context visualization (c1) shows the data according to the selections in (a) & (b), i.e., highlighting the oxidation front (at time $t = 35s$, temperatures color-coded). Dimmed views (c0), (c2), and (c3) show the data at time steps $t = 30s$, $t = 40s$, and $t = 45s$ after the start of the regeneration, respectively.

siteit Eindhoven in 1998, where he became a full professor in Visualization in 2001. He is a member of IEEE, ACM SIGGRAPH, and Eurographics. He has been paper cochair for IEEE Visualization in 2003 and 2004, and is paper cochair for IEEE InfoVis in 2006. His main research interests are information visualization and flow visualization, both with a focus on the development of new visual representations.

Robert Kosara is an Assistant Professor of Computer Science at the University of North Carolina at Charlotte (UNCC). He received his M.Sc. and Ph.D. degrees in Computer Science from Vienna University of Technology in Austria. His interest in information visualization is based to a large part on the specific problems he has to solve in this field, and the way it forces him to work in a truly interdisciplinary way.

Helwig Hauser graduated in 1995 from Vienna Univ. of Techn., Austria. In 1998, he received the degree of a Dr.techn. (PhD), also from TU Wien. From 1994 until mid-2000, Helwig Hauser worked as a Univ.-Ass. (assistant professor) at the Inst. of Computer Graphics at TU Wien. Since 2000, he works for the VRVis Research Center in Vienna, Austria, as a key researcher in the field of visualization. In 2003, Helwig Hauser became the scientific director of VRVis (`http://www.VRVis.at/`). His main interests include volume, flow, and information visualization, and the combination thereof. Helwig Hauser is member of IEEE and EUROGRAPHICS.

Acknowledgements – Daniel Weiskopf acknowledges support from NSERC (Discovery Grant). Helwig Hauser acknowledges the K*plus* funding of VRVis.

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PERCEPTION

PERCEPTION - COGNITION

RED	GREEN	BLUE	YELLOW	PINK
ORANGE	BLUE	GREEN	BLUE	WHITE
GREEN	YELLOW	ORANGE	BLUE	WHITE
BROWN	RED	BLUE	YELLOW	GREEN
PINK	YELLOW	GREEN	BLUE	RED

Is there something wrong?

PERCEPTION

Perception

- Identification and interpretation of sensory information
- From the physical stimulus to recognizing information
- Shaped by learning, memory, expectation

Cognition

- The processing of information, applying knowledge

Hear someone speak: Perception

Understand the language and the words: Cognition

PERCEPTION VS. COGNITION

PERCEPTION

Eye, optical nerve, visual cortex

Basic perception

First processing

Not conscious

Reflexes

COGNITION

Recognizing objects

Relations between objects

Conclusion drawing

Problem solving

Learning, …

PERCEPTION - COGNITION

RED	GREEN	BLUE	YELLOW	PINK
ORANGE	BLUE	GREEN	BLUE	WHITE
GREEN	YELLOW	ORANGE	BLUE	WHITE
BROWN	RED	BLUE	YELLOW	GREEN
PINK	YELLOW	GREEN	BLUE	RED

WHAT IS THERE
VS
WHAT DO WE SEE

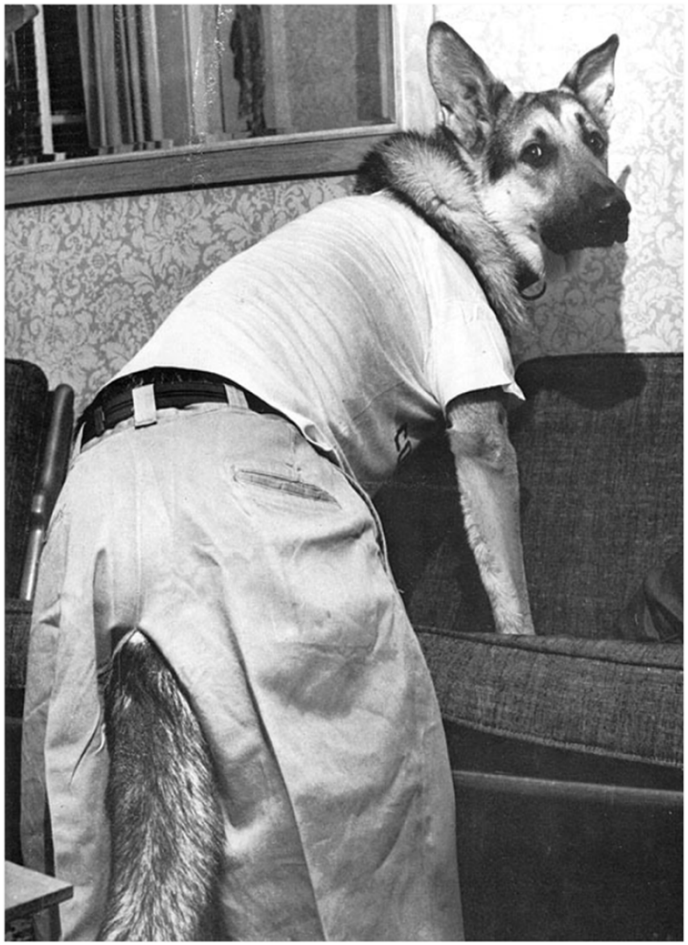
Emergence Images

http://graphics.stanford.edu/~niloy/research/emergence/emergence_image_siga_09.html

OUR PERCEPTION IS BASED ON PRIORS

We have a model of the world

We try to fit what we see into this model

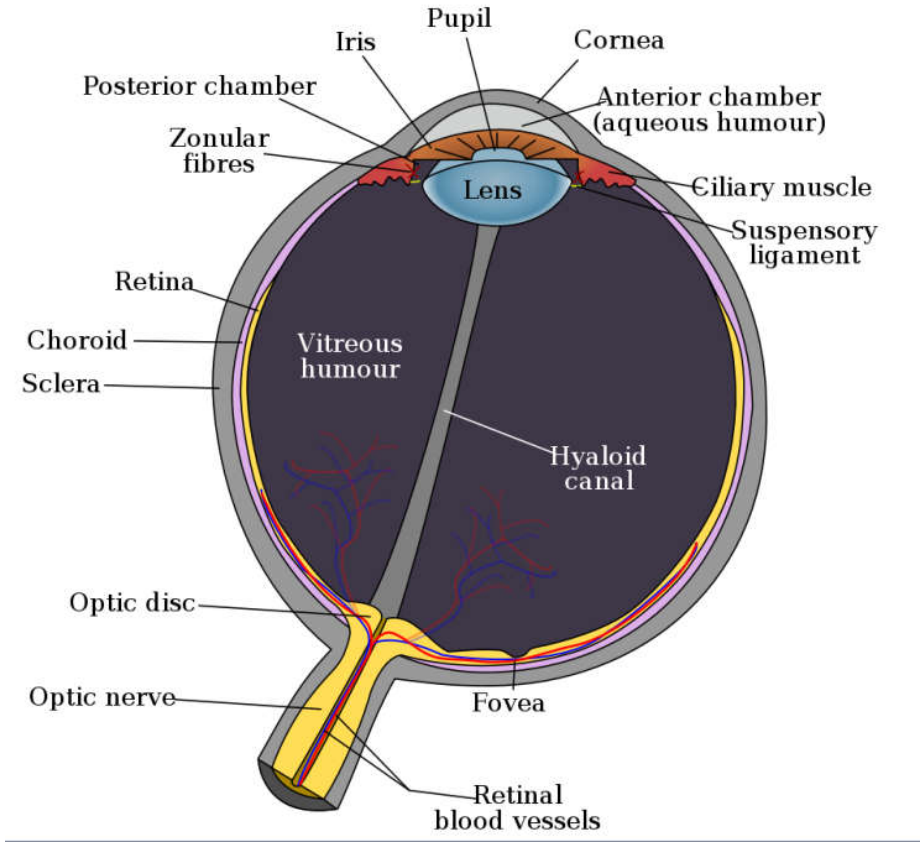


dribbleglass.com

WHO DO YOU SEE?

OUR EYE

HUMAN VISUAL SYSTEM



Cone cells: for color vision.

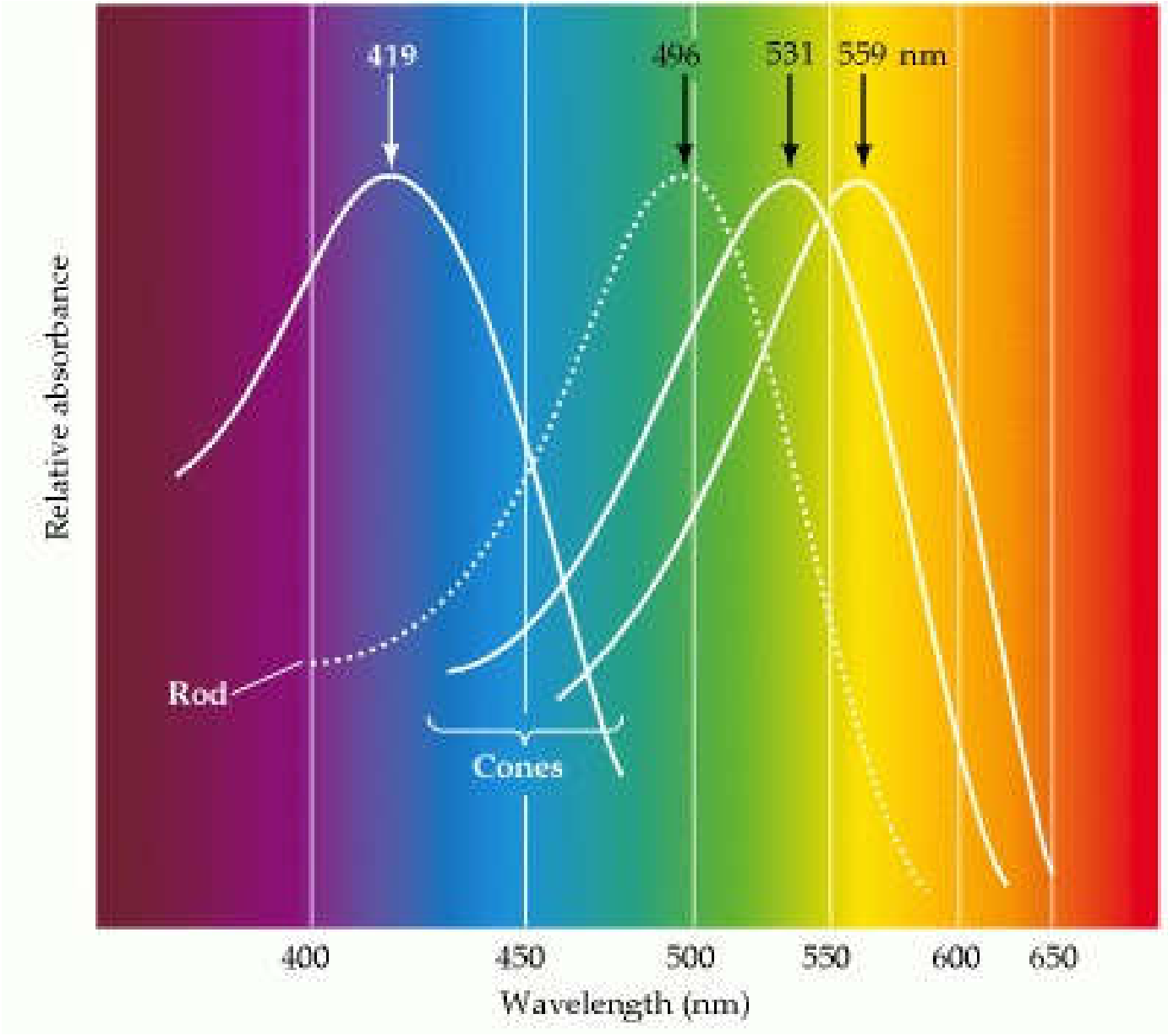
- Dense in the center. Fovea has only cone cells

Rods cells: for black/white

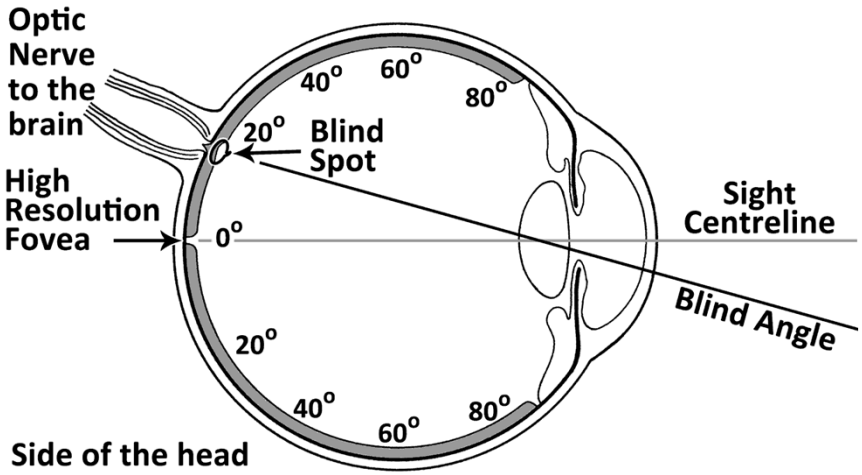
COLOR VISION

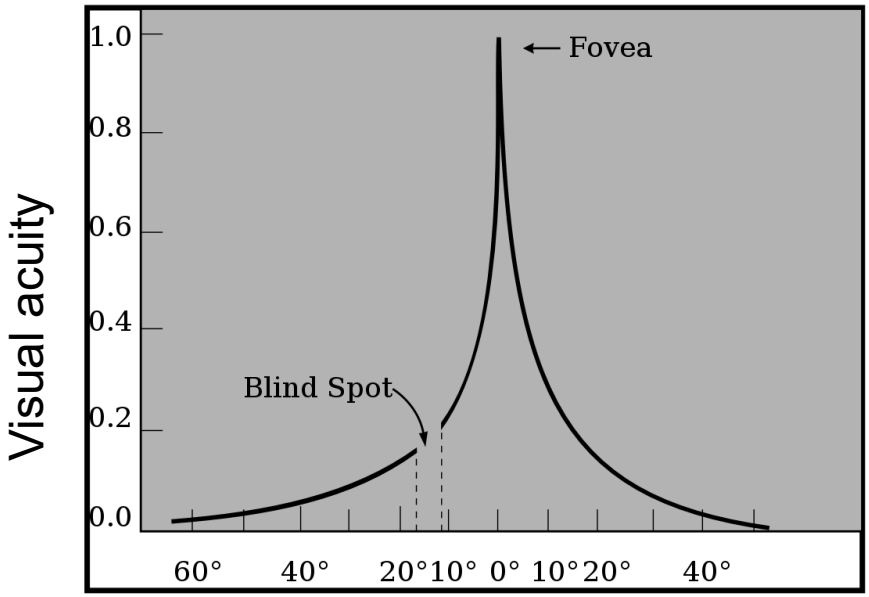
3 types of cone with different sensitivity to light:

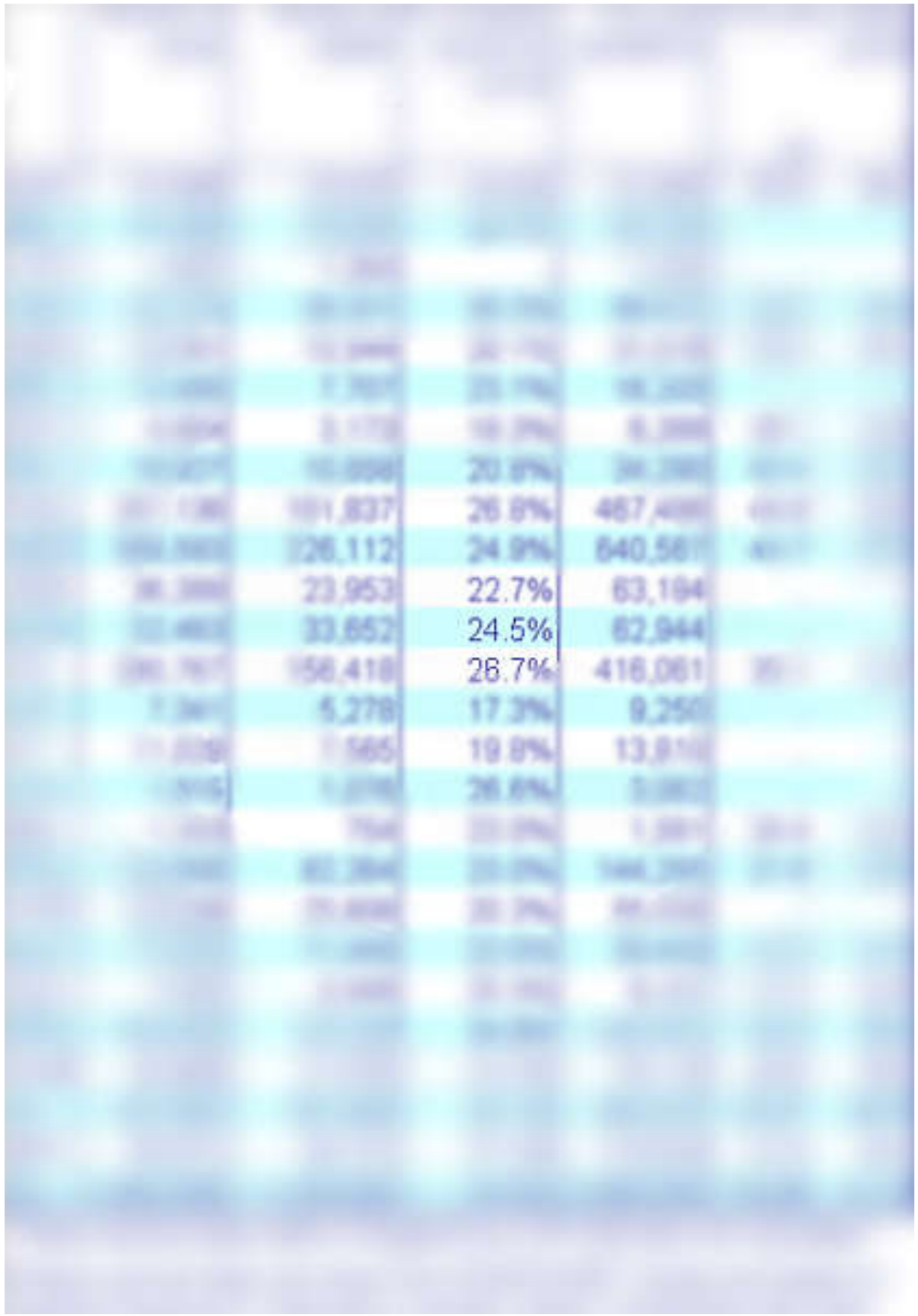
- Red (63%)
- Green (31%)
- Blue (6%)



VISUAL ACUITY





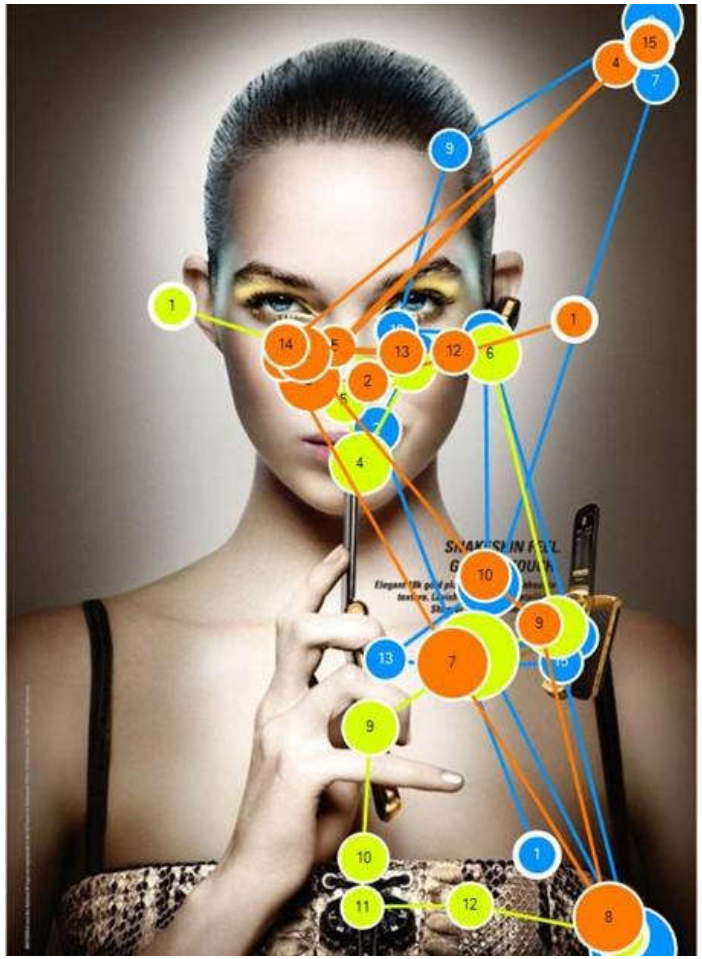

20.8%
26.8%
24.9%
22.7%
24.5%
26.7%
17.3%
19.8%

HUMAN VISUAL SYSTEM

Not similar to a camera

Vision works as sequence of

- **Fixation**: maintaining gaze on single location (200-600 ms)
- **Saccades:** moving between different locations (20-100 ms)



HUMAN VISUAL SYSTEM

No general purpose vision

What we see depends on our goals and expectations

Relative judgements: strong

Absolute judgments: weak

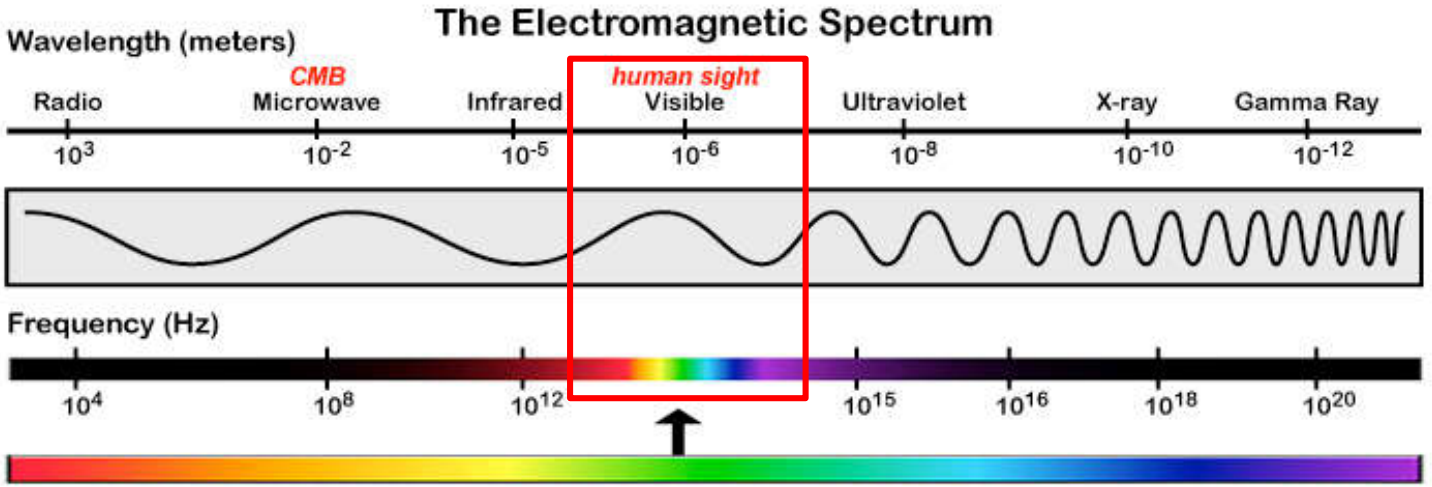
Ames Room

AMES ROOM

COLOR

COLOR BASICS

The electromagnetic spectrum

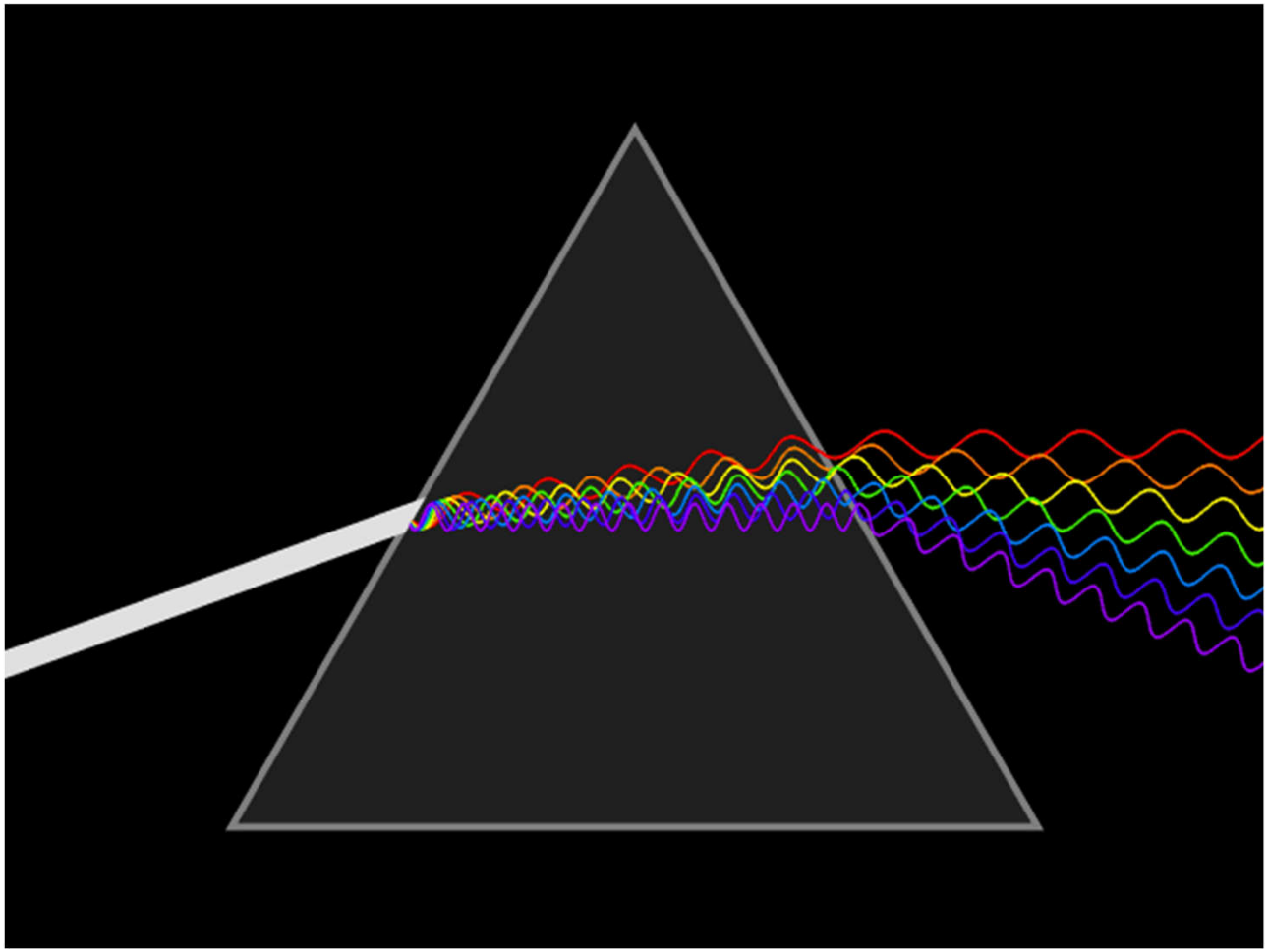


- Visible part: 390 – 750 nm

- Spectral colors: evoked by a single wavelength
(red, orange, yellow, green, blue, and violet**)**

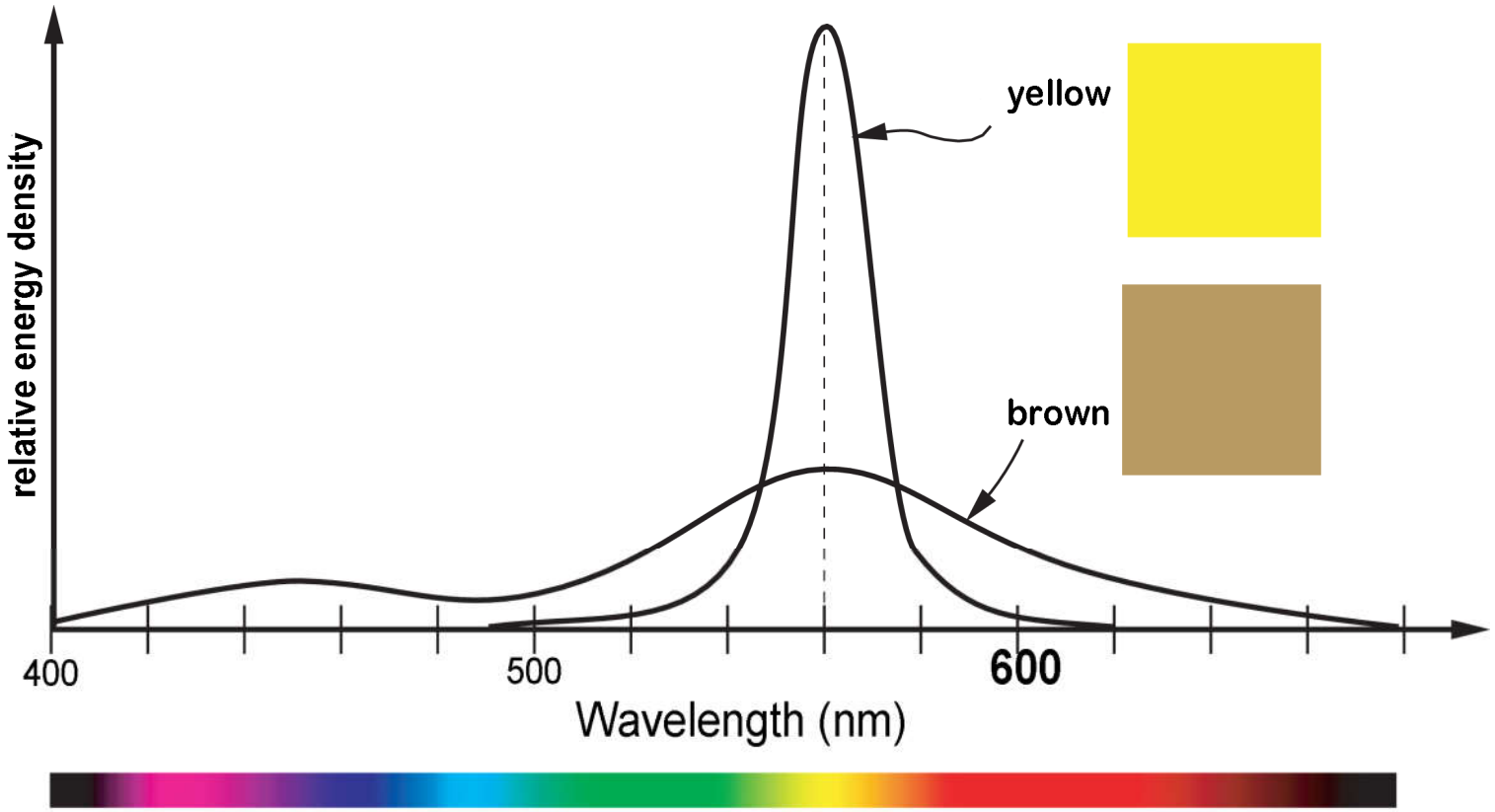
- Other colors: unsaturated colors: mix of multiple wavelengths
(purple, magenta, … and white**)**

DISPERSING LIGHT WITH PRISM



COLOR != WAVELENGTH

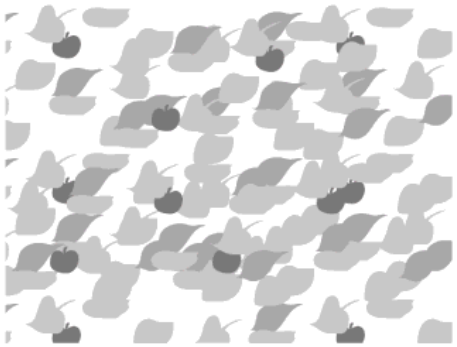
color = wavelength + energy

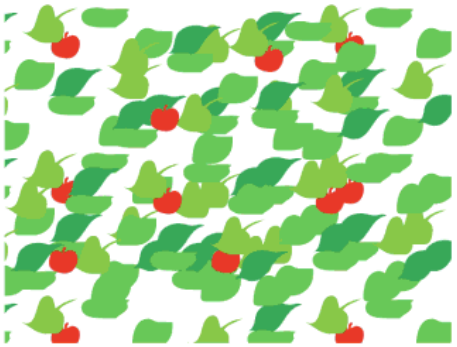


COLOR

Color is irrelevant to much of normal vision.
Does not help to perceive

- layout of objects
- how they are moving
- what shape they are

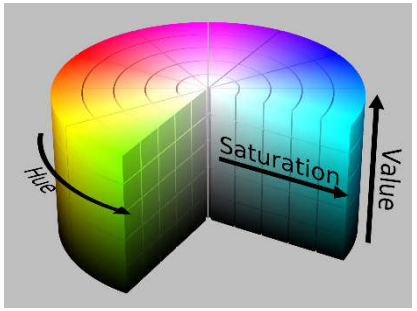




Color breaks camouflage

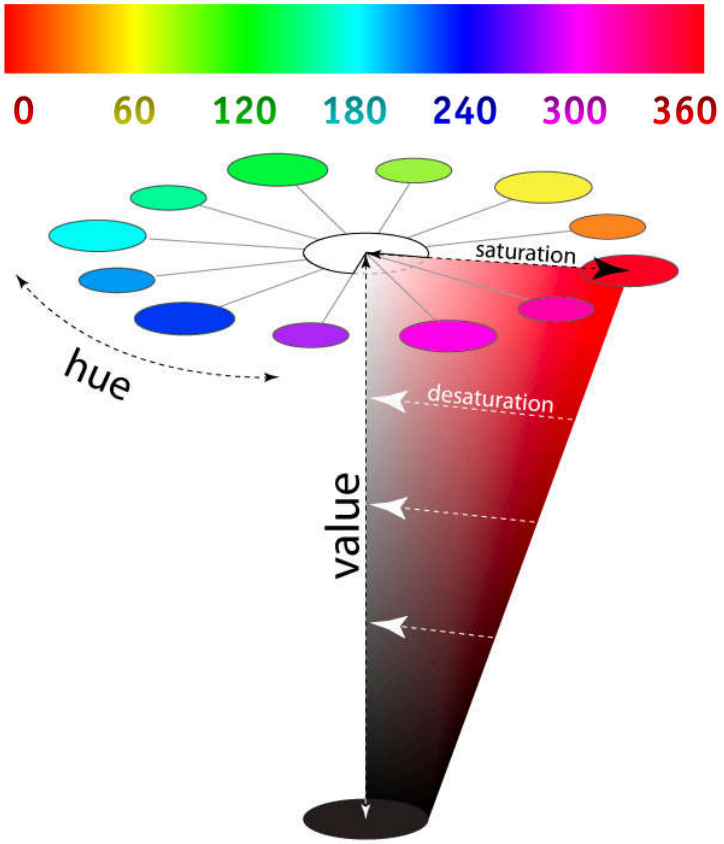
Tells about material properties (e.g. quality of food)

DIMENSIONS OF COLOR



HSV model

Hue
color value expressed as a number from 0 to 360



Saturation

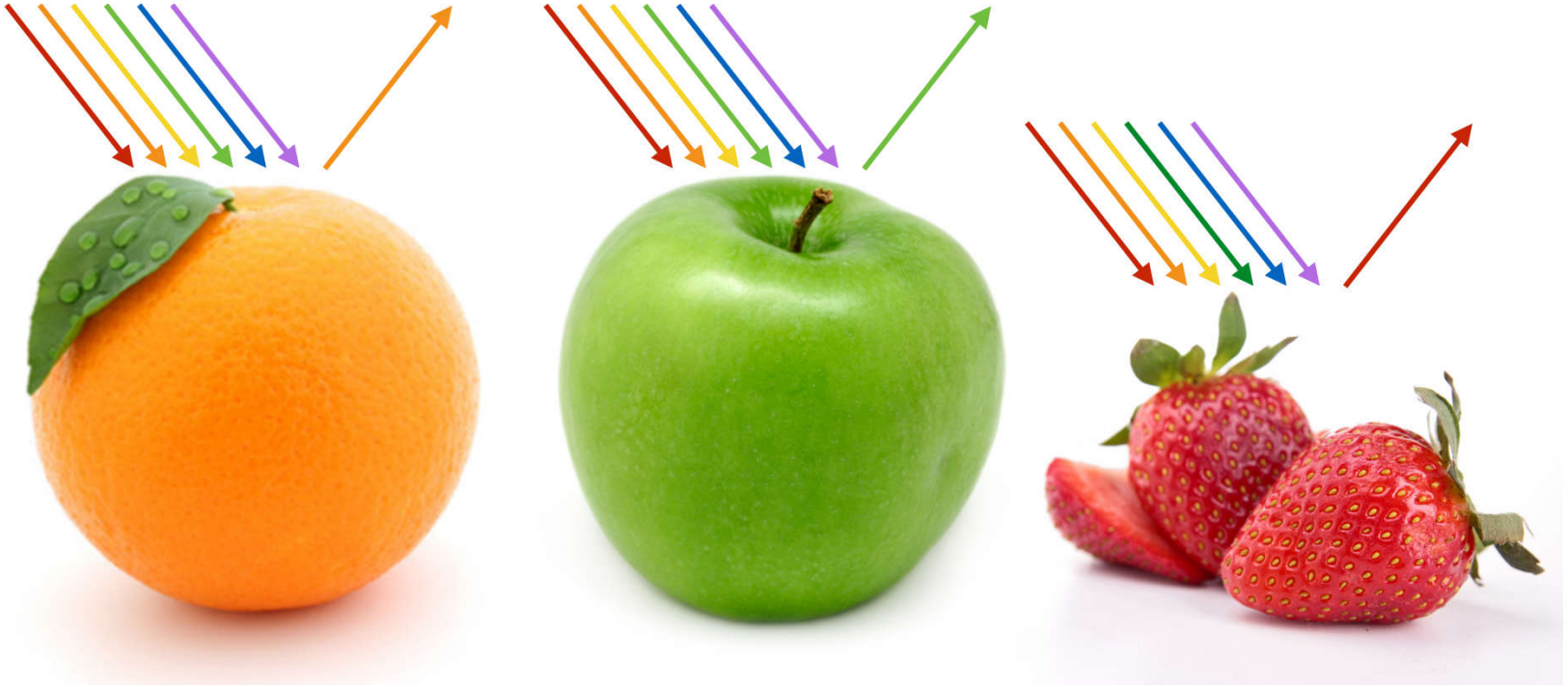
- the purity of a color
- pigments: no white/black is added
- light: what is the ratio of dominant wavelength to others

Value

Lightness or darkness of a color
the intensity of light

Other color models: RGB, HSL, …

COLOR IN REAL LIFE



PAINT MIXING

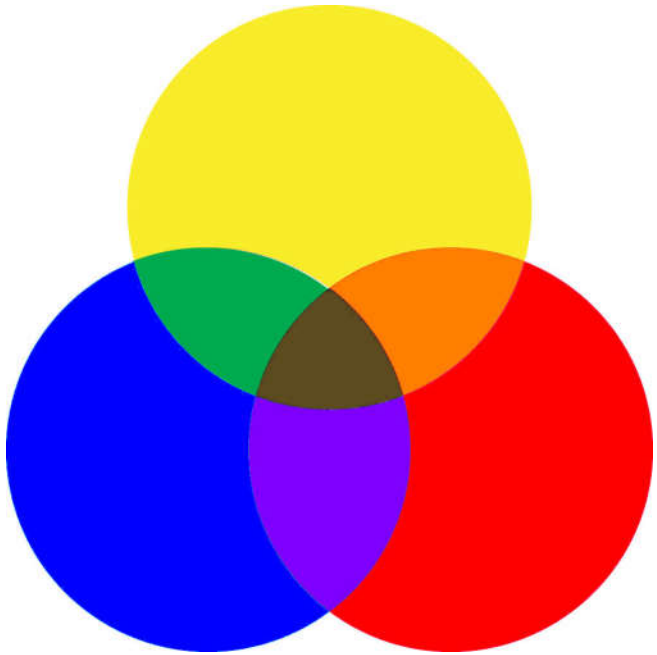
Physical mixing of opaque paints

Primary: red, yellow, blue

Secondary (mixed): green, orange, purple

Subtractive model

- absorb wavelength



INK MIXING

Subtractive mix of transparent inks

Primary: cyan, magenta, yellow

Secondary: RGB

Approx. black = C+M+Y

True black = C+M+Y+K

Subtractive model

LIGHT MIXING

Additive mix of colored lights

Primary: Red, Green, Blue

Secondary: Cyan, Magenta, Yellow

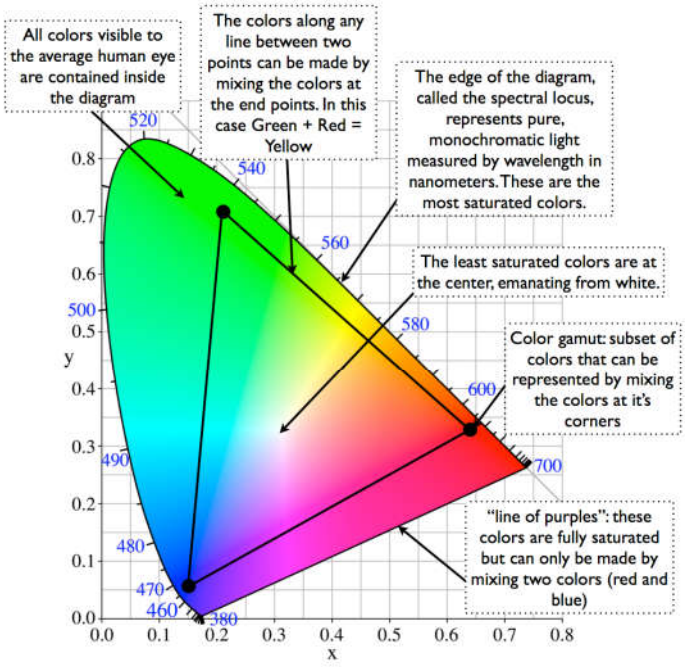
Additive model

COLOR GAMUT

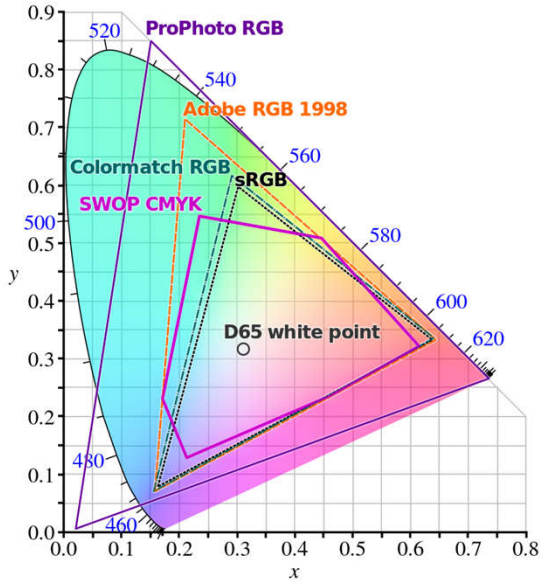
The curve shows all visible colors

Polygon: Set of all colors that can be produced by a device

Primaries are arbitrary



Anatomy of a CIE Chromaticity Diagram

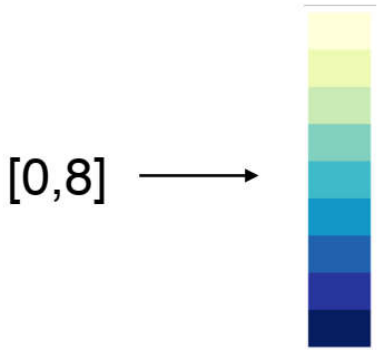


COLOR IN VISUALIZATION

COLORMAP

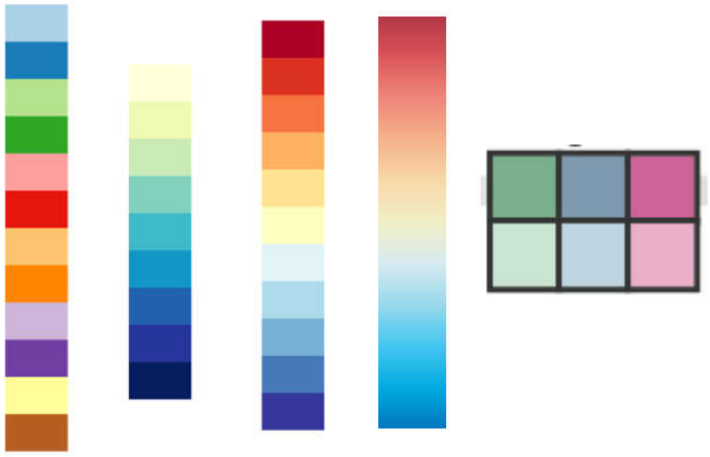
Specifies a mapping between color and values

- Categorical vs ordered
- Sequential vs diverging
- Segmented vs continuous
- Univariate vs bivariate

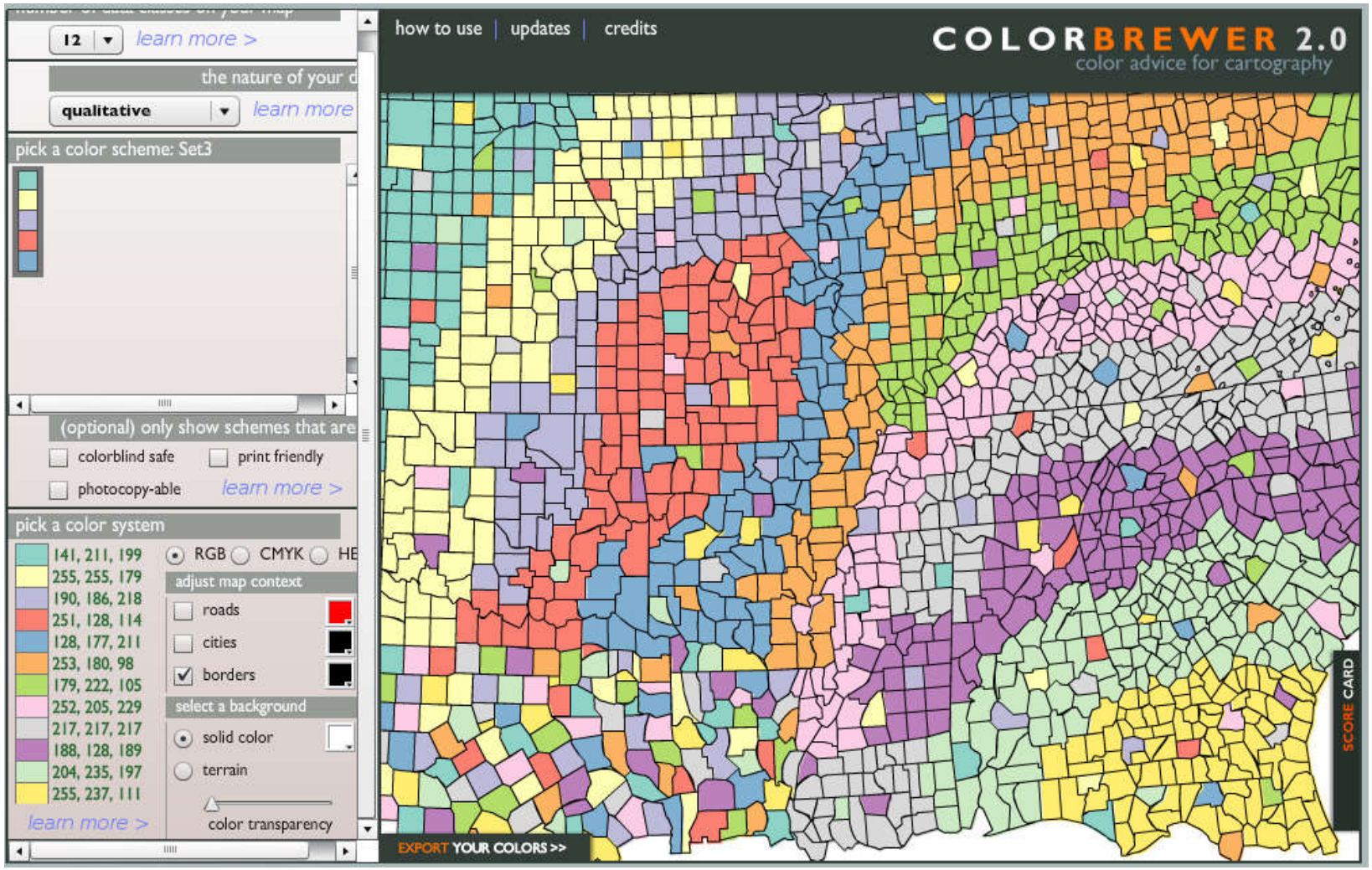


Expressiveness

Match colormap to attribute characteristics



APPLICATION: LABELLING

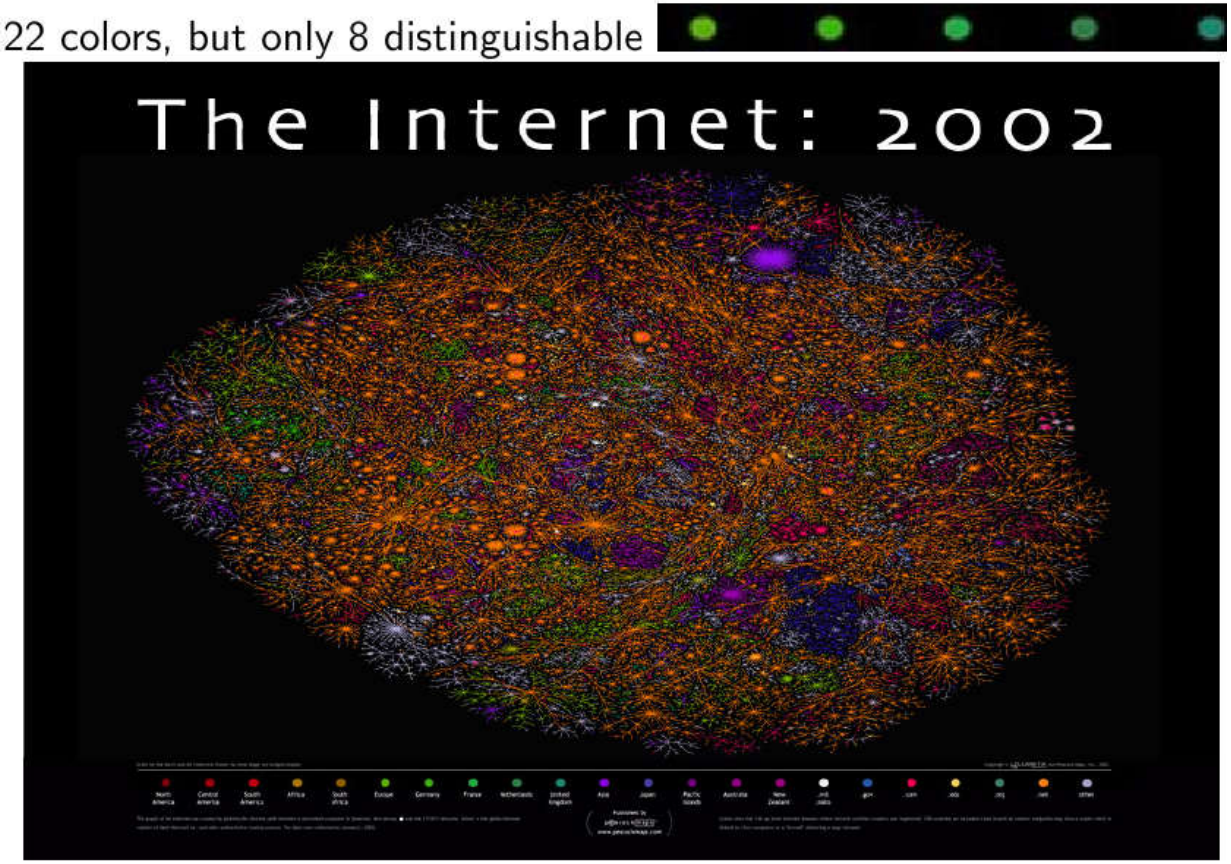


Carefully designed color scheme for 12 colors [colorbrewer]

COLOR FOR QUALITATIVE DATA

Color labeling (nominal information coding)
recommended: about 6, no more than 10

22 colors, but only 8 distinguishable



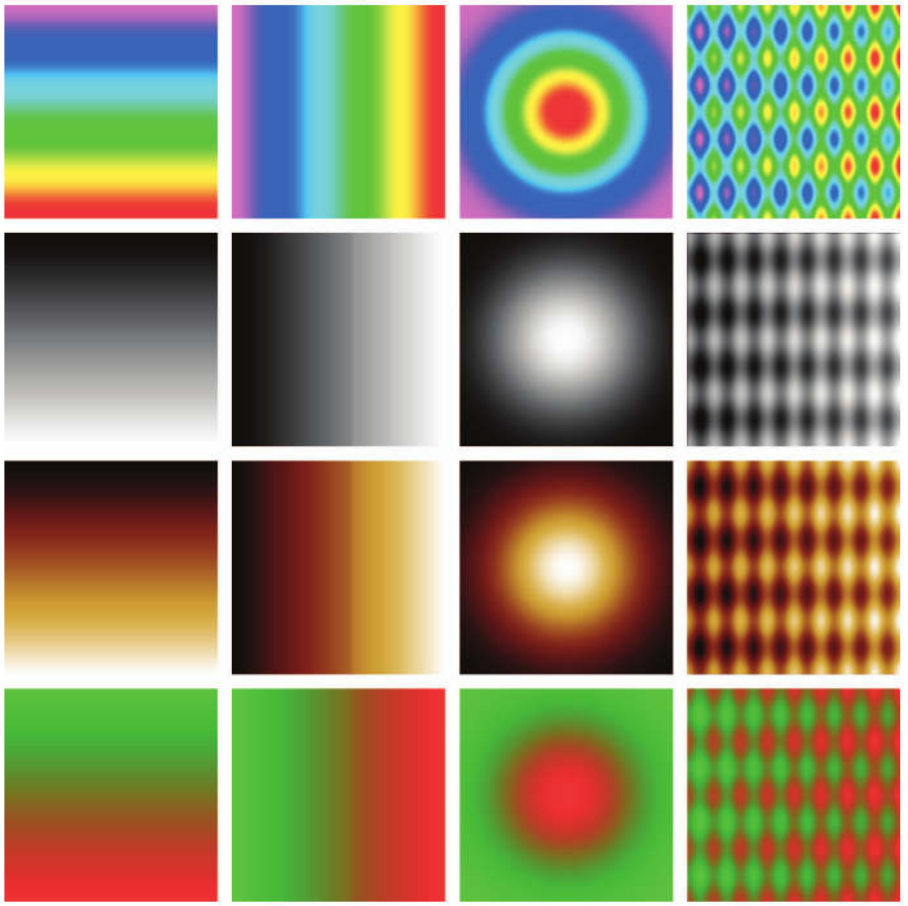
COLOR FOR QUANTITATIVE DATA

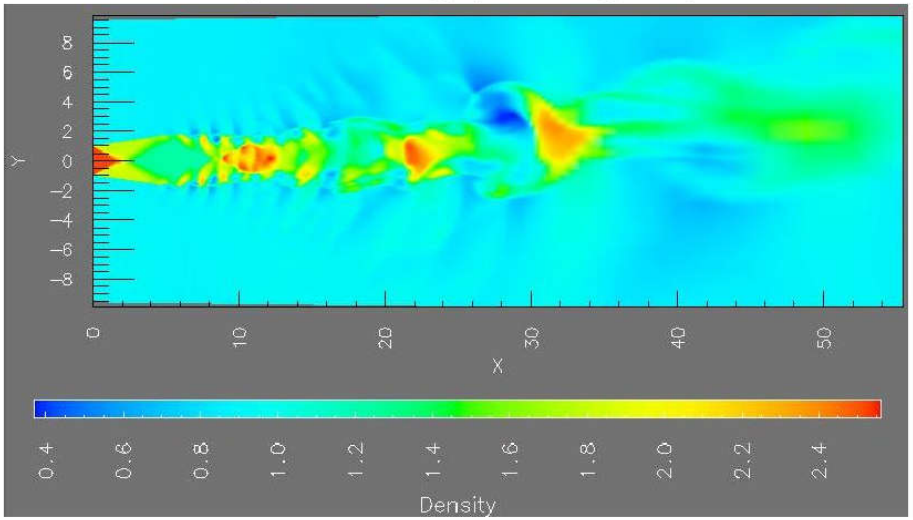
use **value**

saturation works but not as good

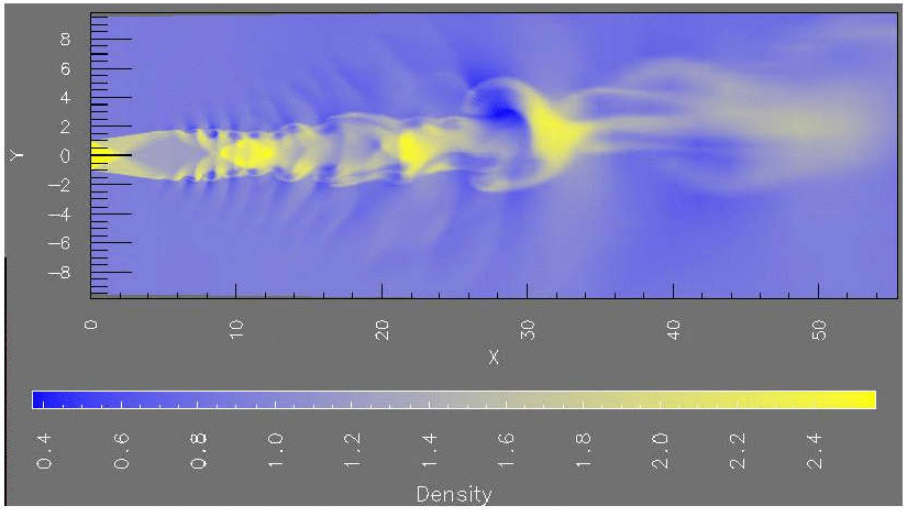
don‘t use **hue**!

Danger: rainbow color map





[Rogowitz and Treinish, Why Should Engineers and Scientists Be Worried About Color? http://www.research.ibm.com/people/l/lloydt/color/color.HTM]



[Rogowitz and Treinish, How NOT to Lie with Visualization, www.research.ibm.com/dx/proceedings/pravda/truevis.htm]

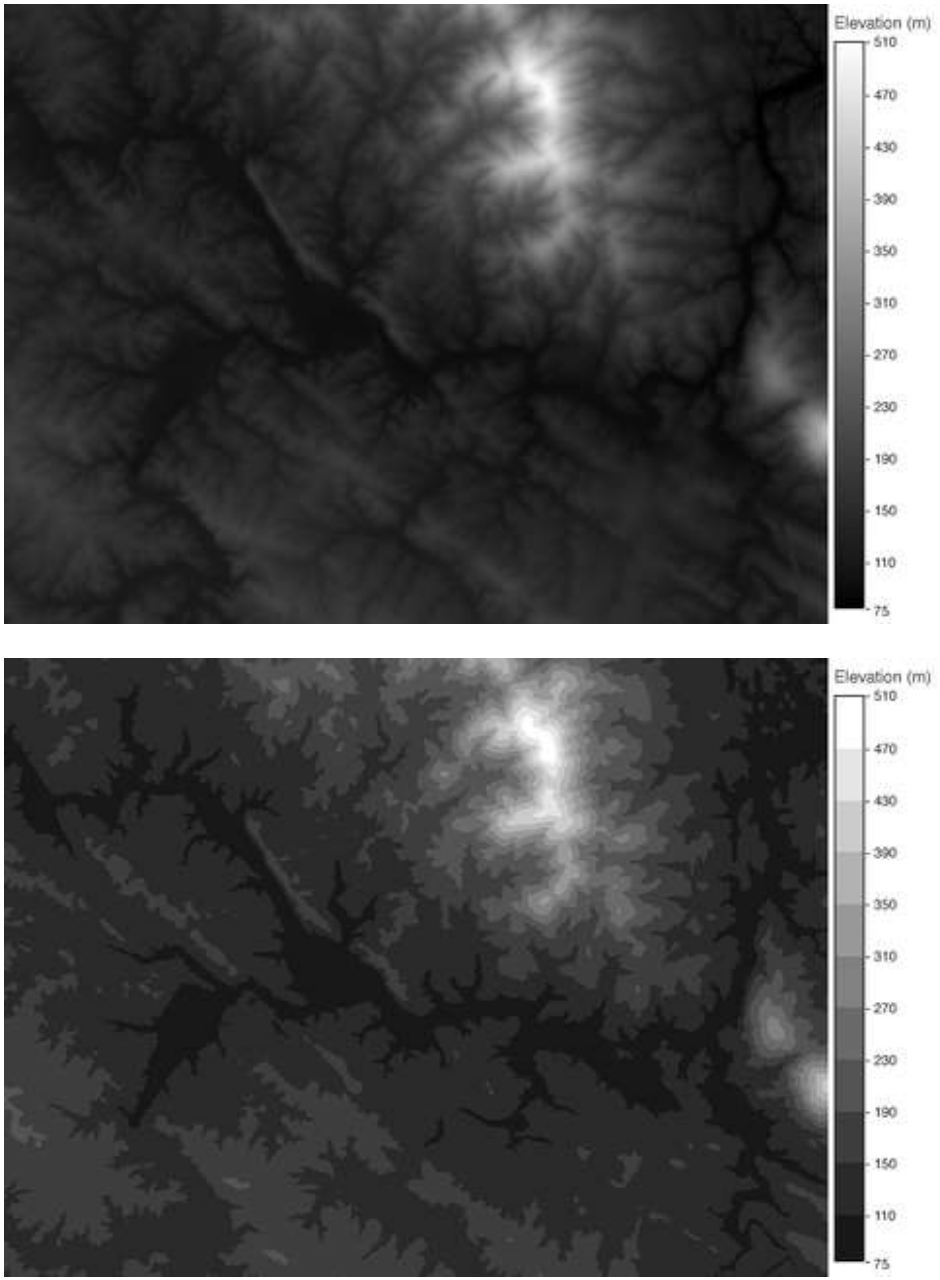
TO BIN OR NOT TO BIN?

What is faster?

What is more accurate?

Continuous was faster

Binned was often more accurate



Evaluating the Impact of Binning 2D Scalar Fields
Lace Padilla, P. Samuel Quinan, Miriah Meyer, Sarah Creem-Regehr
Proceedings of InfoVis16

COLOR BLINDNESS

NOTION

10% of males, 1% of females

Most common: red-green weakness / blindness

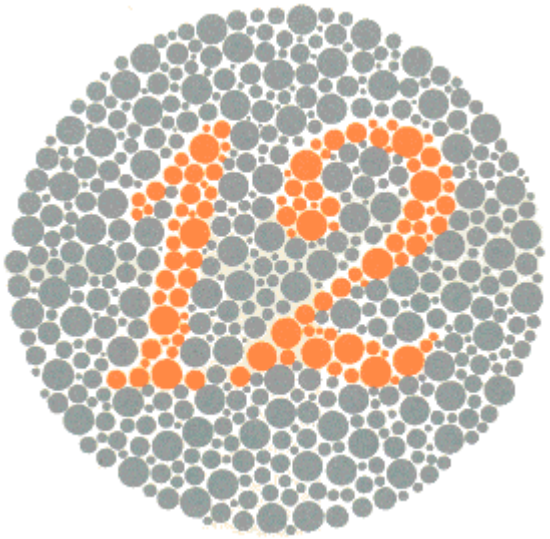
Reason: lack of medium or long wavelength receptors, or altered spectral sensitivity (most common: green shift)

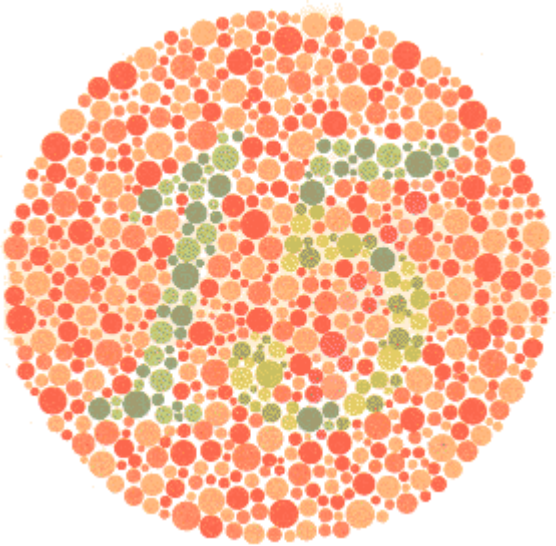
Normal

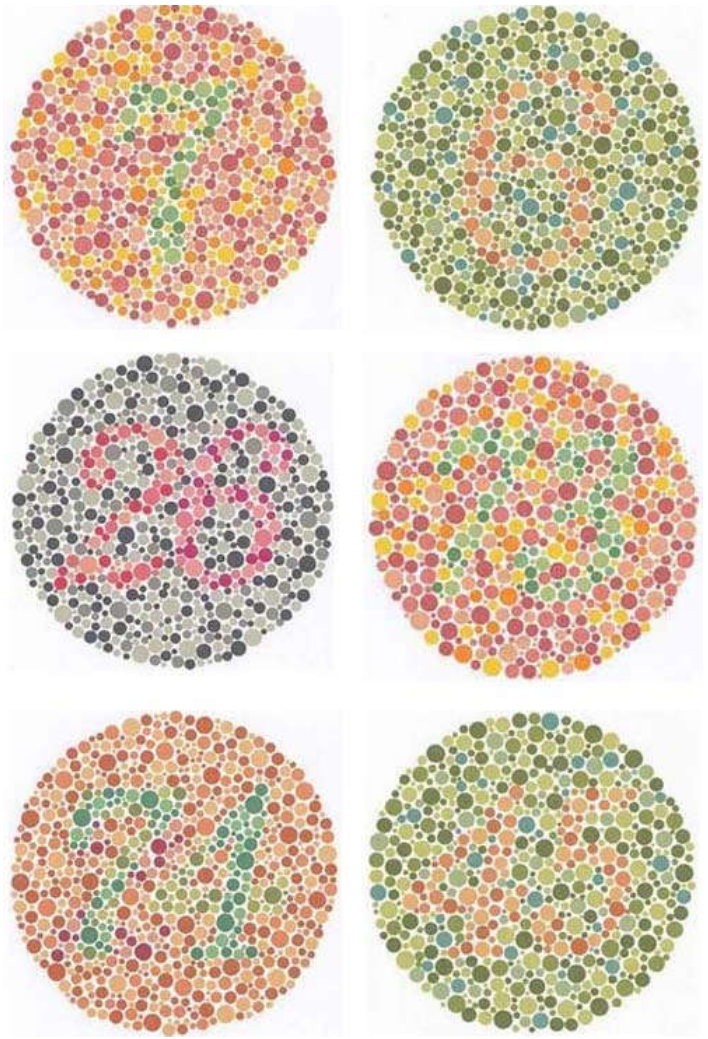
Deuteranopia (no green receptors)

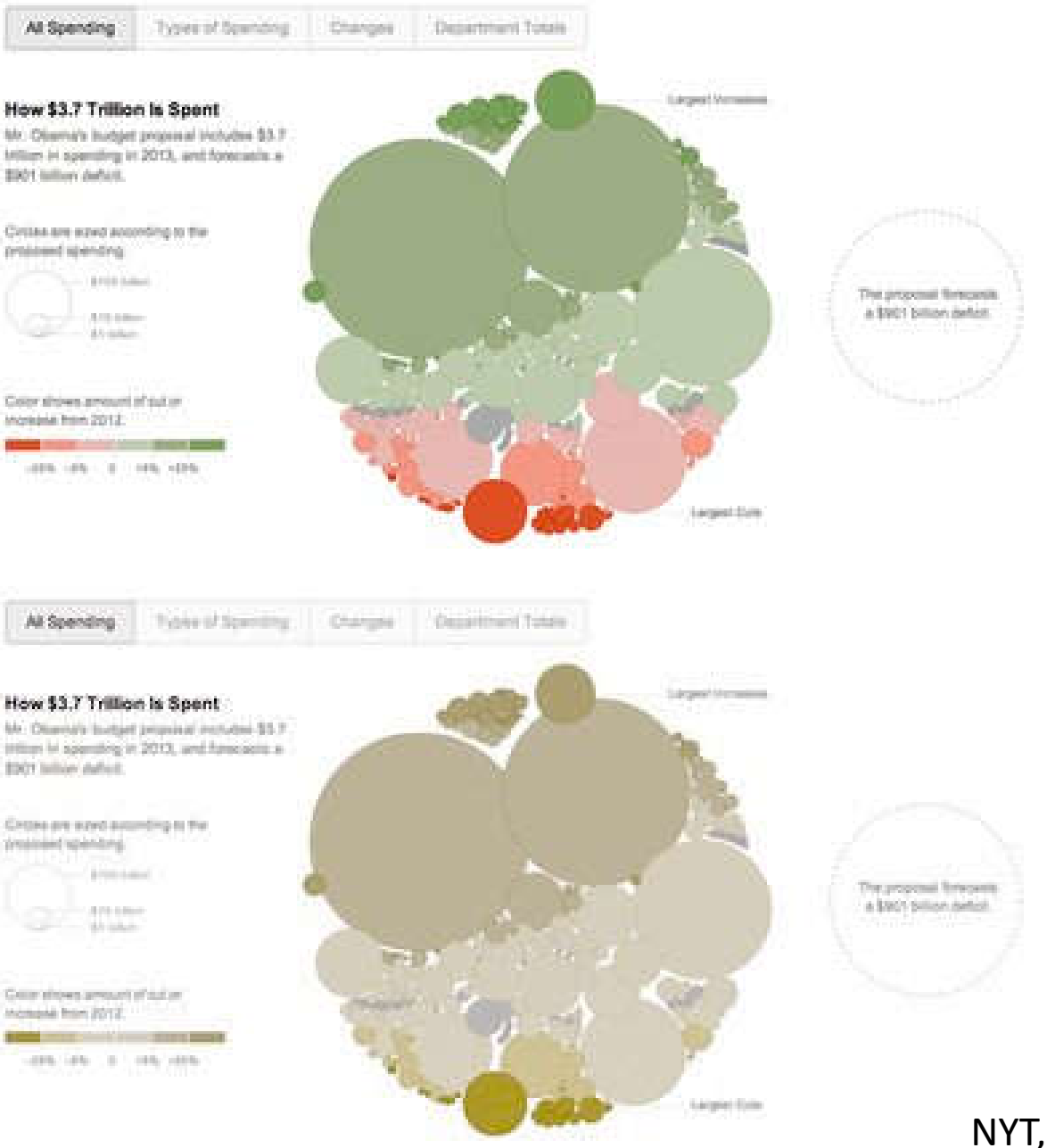
Protanopia (no red receptors)

COLOR BLINDNESS TEST



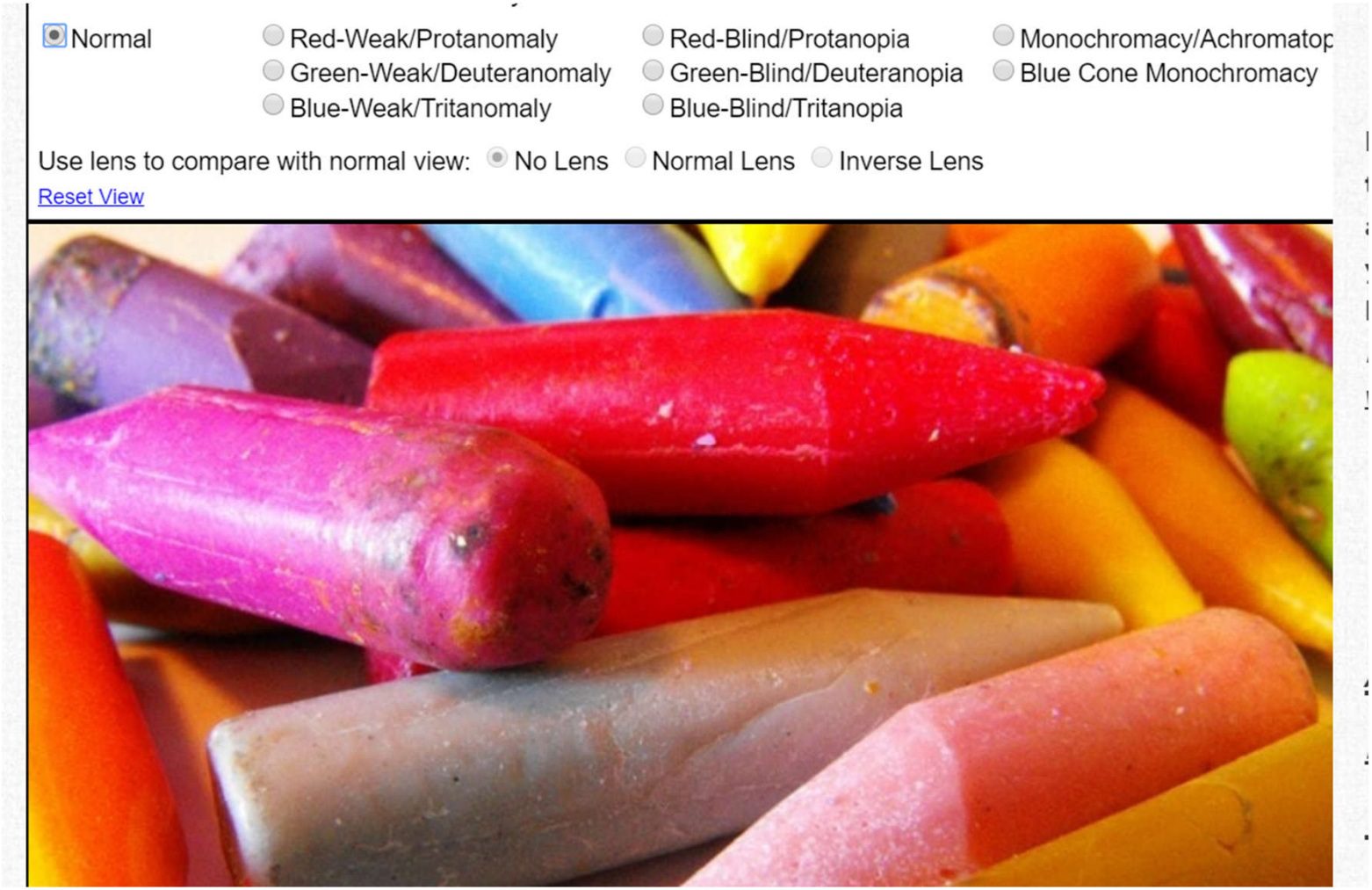






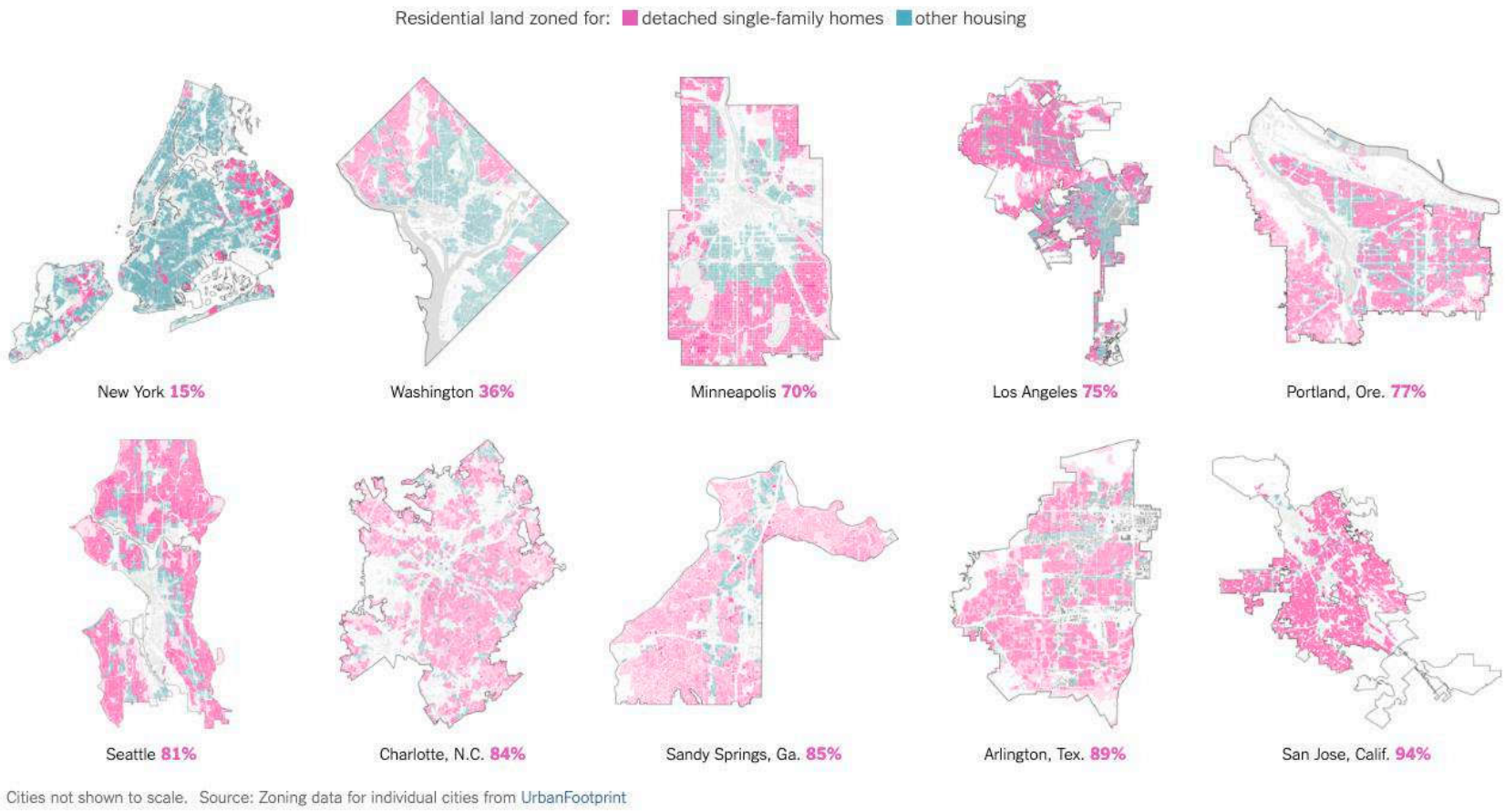
NYT, Feb 12, 2012

SIMULATE COLOR VISION DEFICIENCIES



https://www.color-blindness.com/coblis-color-blindness-simulator/

COLOR BLINDNESS CHECKING



NY Times Zoning Maps: http://nyti.ms/2XVAuie

Drag and drop or paste your file in the area below or: Choose File D9WlOrwXsAA4Z07.jpg

Trichromatic view:	*Anomalous Trichromacy:*	*Dichromatic view:*	*Monochromatic view:*
○ Normal	○ Red-Weak/Protanomaly	○ Red-Blind/Protanopia	○ Monochromacy/Achromatopsia
	○ Green-Weak/Deuteranomaly	◉ Green-Blind/Deuteranopia	○ Blue Cone Monochromacy
	○ Blue-Weak/Tritanomaly	○ Blue-Blind/Tritanopia	

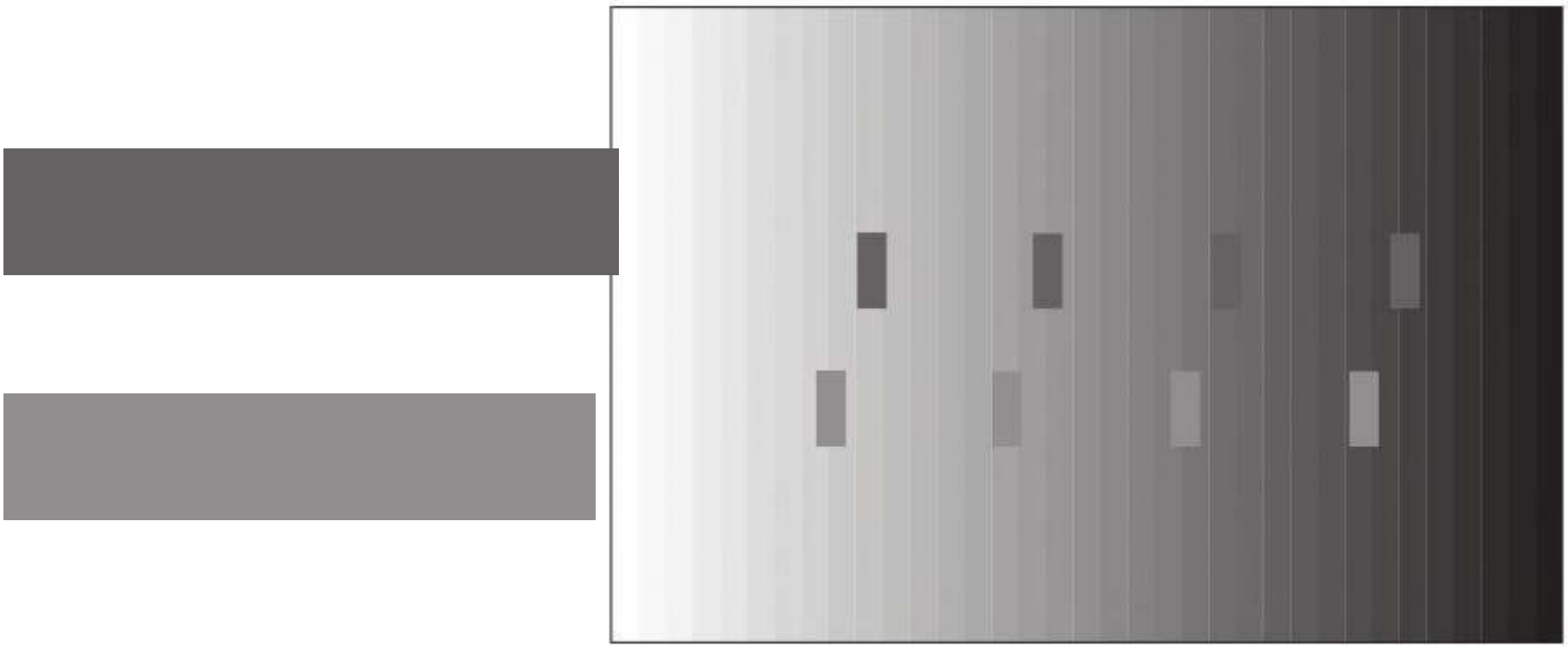
Use lens to compare with normal view: ◉ No Lens ○ Normal Lens ○ Inverse Lens

Reset View Open simulated image in new window

COLOR IS RELATIVE

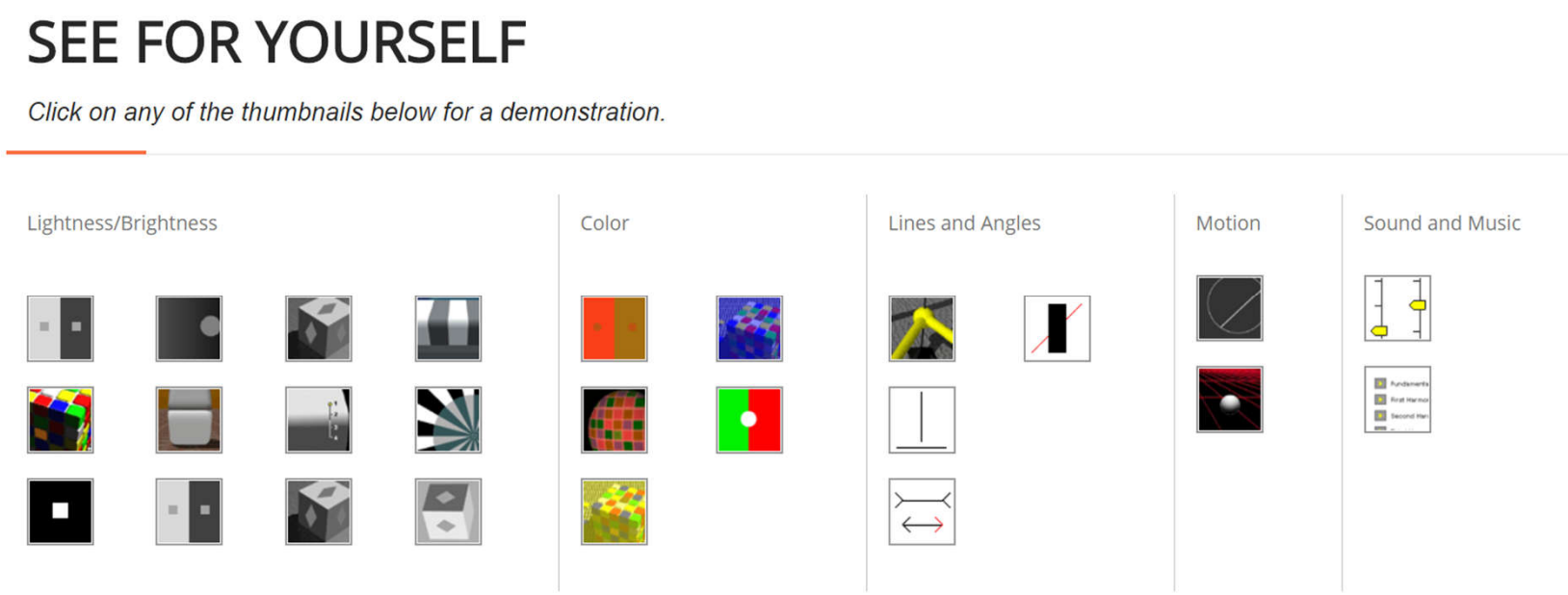
SIMULTANEOUS BRIGHTNESS CONTRAST

The perceived brightness of an object is relative to it‘s background



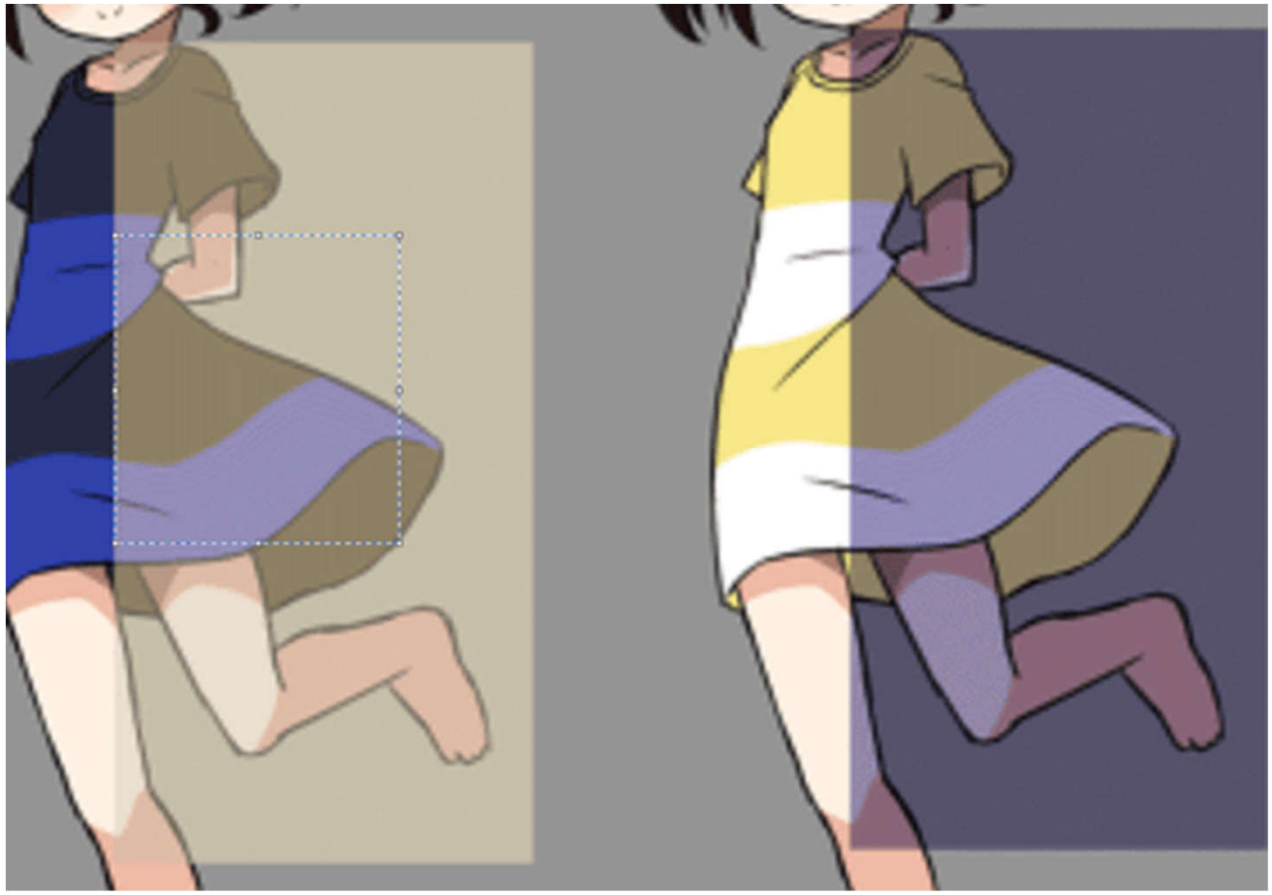
FOR MORE DEMO

http://purveslab.net/see-for-yourself/

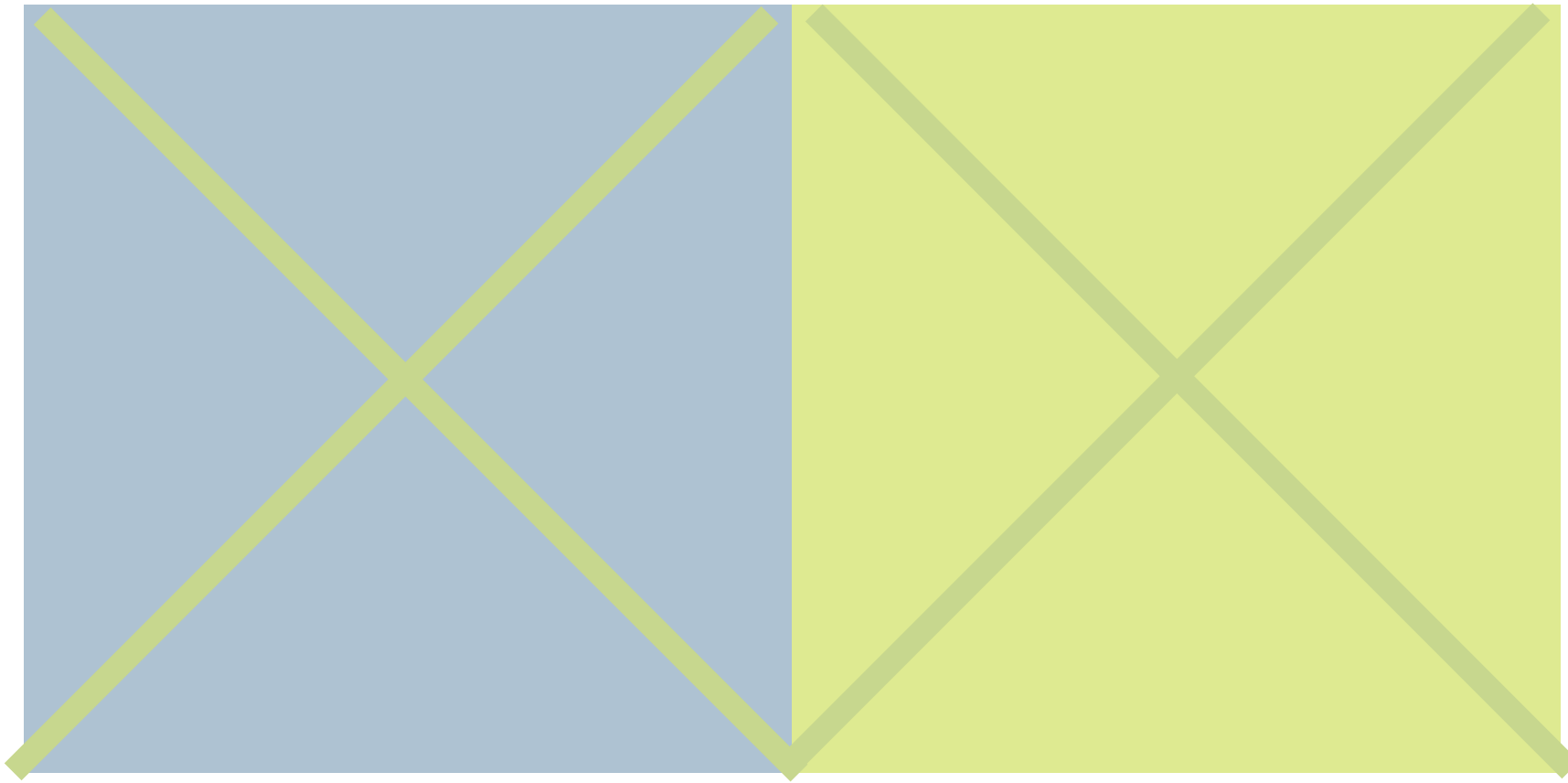


BLACK AND BLUE OR WHITE AND GOLD

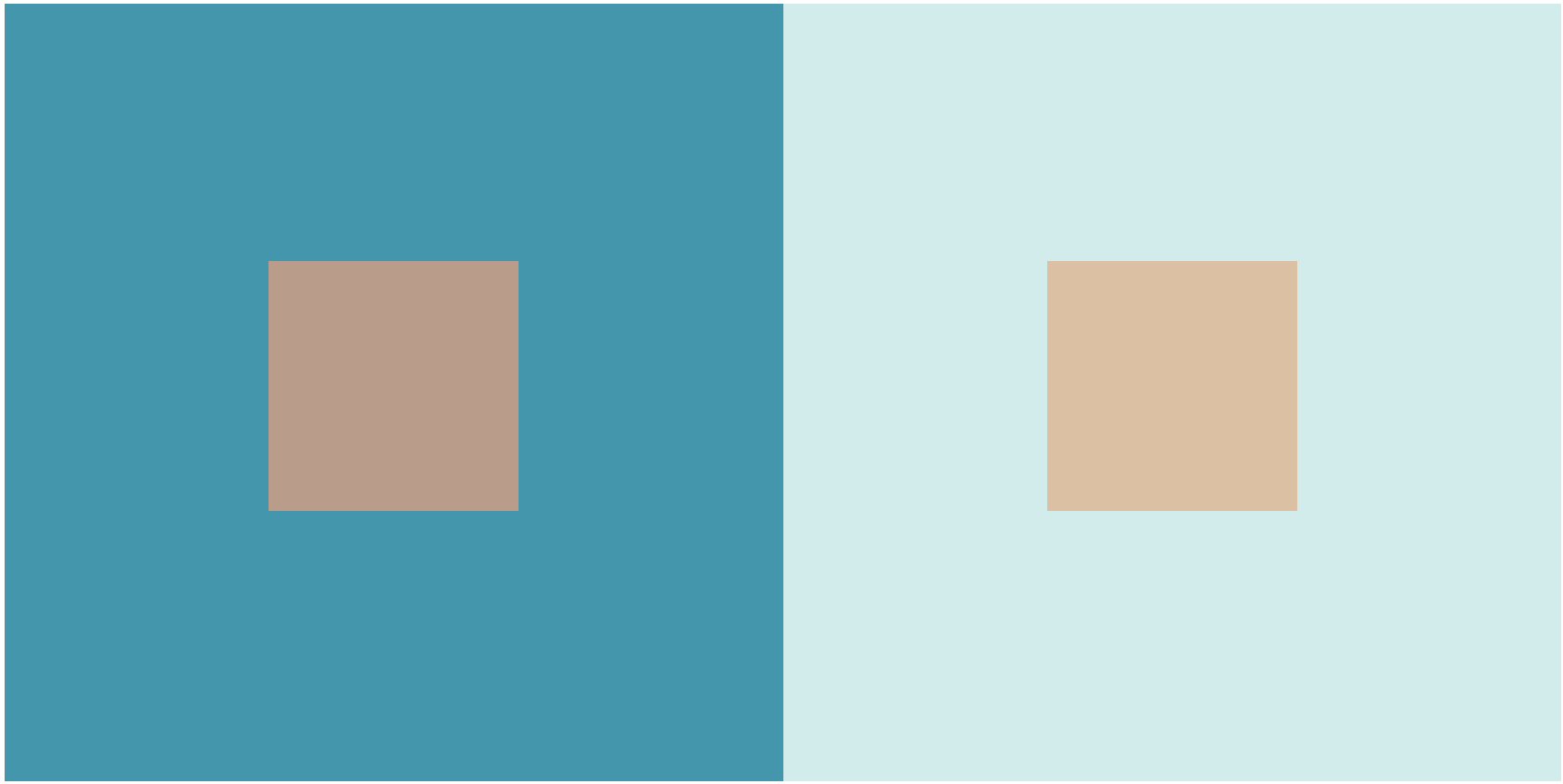
https://imgur.com/hxJjUQB



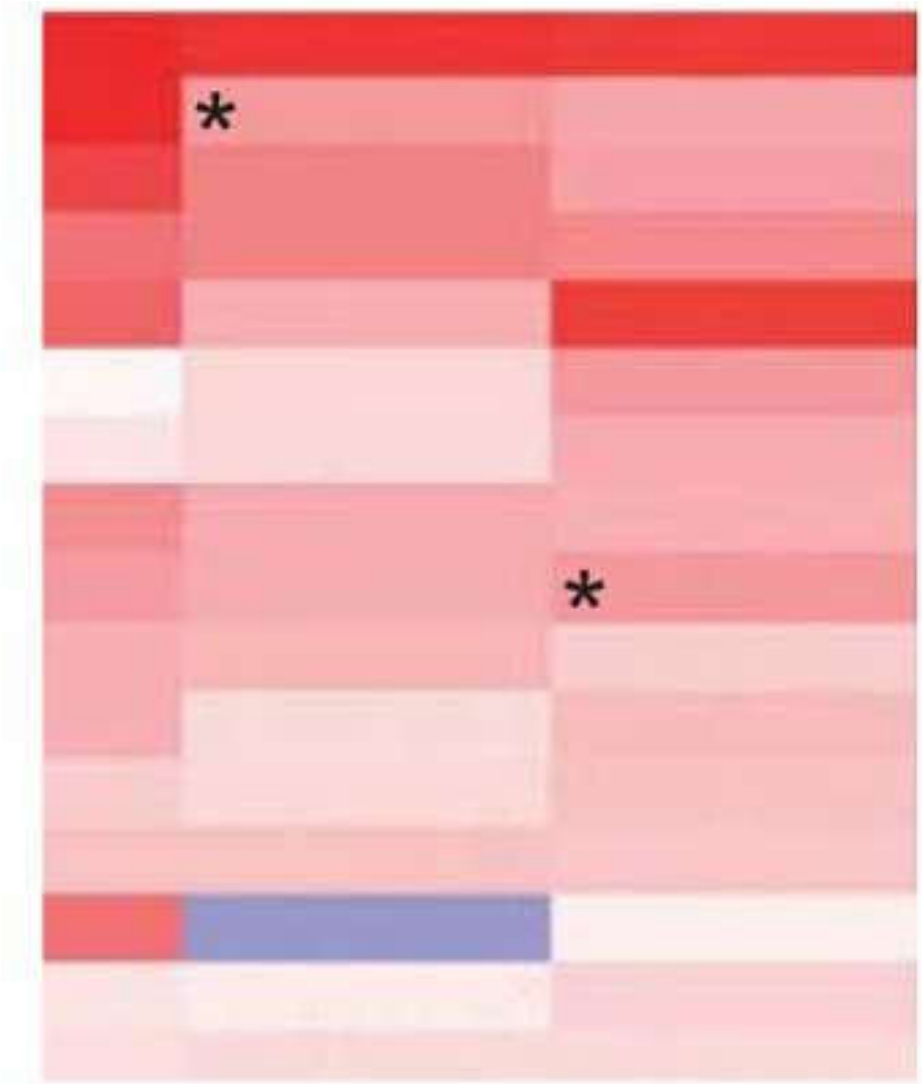
INTERACTION OF COLOR



INTERACTION OF COLOR

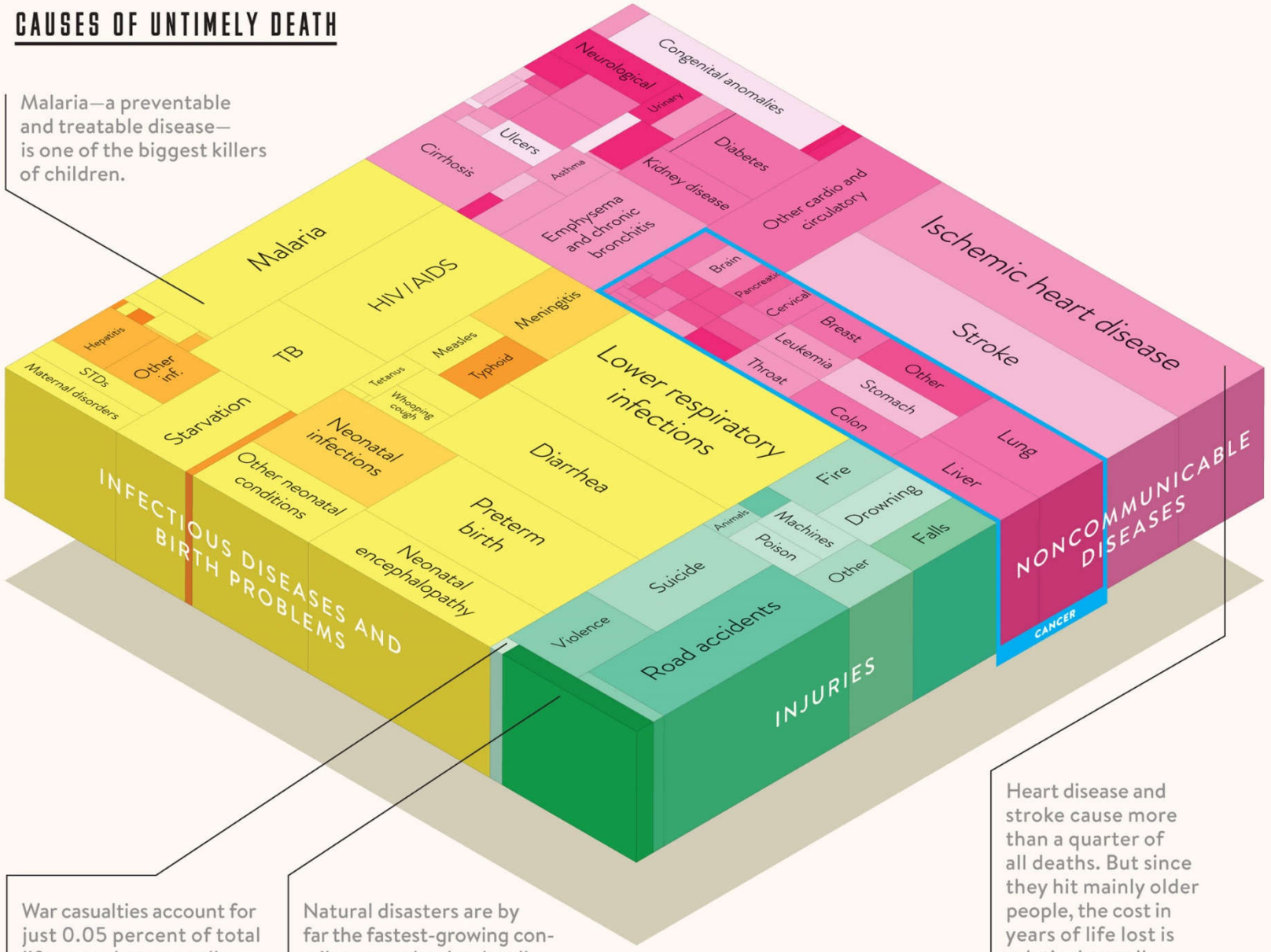


INTERACTION OF COLOR



[Wong 2010]

CAUSES OF UNTIMELY DEATH



I love this graph because it shows that while the number of people dying from communicable diseases is still far too high, those numbers continue to come down [..] But there remains much to do to cut down the deaths in that yellow block even more dramatically.[…]

Bill Gate

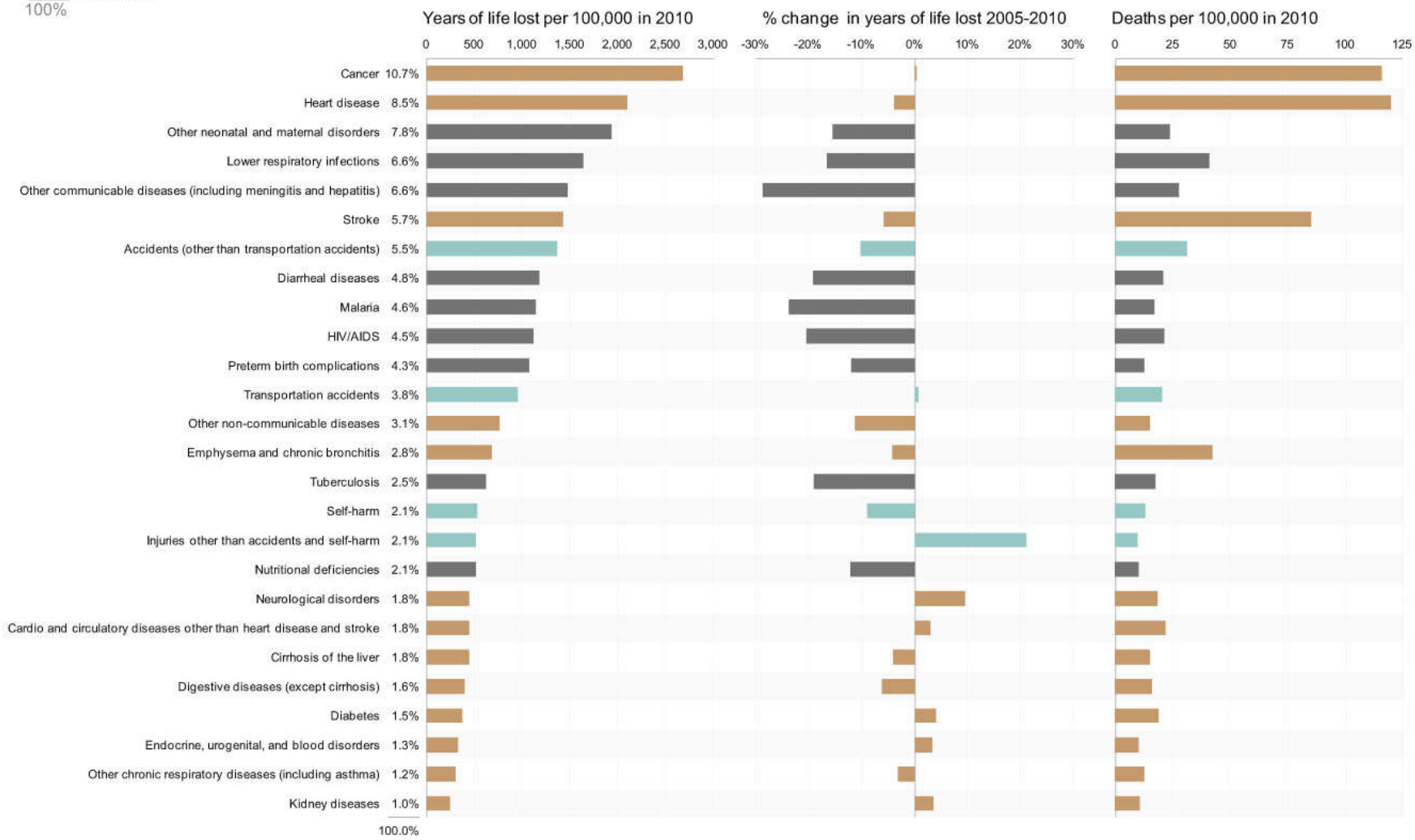
EVALUATE THE GRAPH

how well does the graph above tell this story?

Global Causes of Lost Life

44% Communicable, maternal, neonatal, and nutritional disorders
43% Non-communicable diseases
13% Injuries
100%

Comparing the number of deaths alone, as shown in the right-most graph below, doesn't tell the entire story. Some causes of death have a greater effect on the young, which can be seen when comparing years of life lost in the leftmost graph.



Some causes of death contribute disproportionately to years of life lost because of their effect on the young. For example, malaria, while not huge in the number of deaths, is much more signficant in the number of years that are lost.

Two interesting changes reside in "Injuries other than accidents and self-harm." War, which accounted for only 0.05% of years of life lost, decreased since 2005 by 31.5% in years of life lost per 100,000 people. Natural disasters, which accounted for 0.65% of years of life lost, increased by 217% in years of life lost per 100,000.

Communicable, maternal, neonatal, and nutritional disorders (the gray bars) are often easier to prevent through healthcare than other causes of death. This reveals itself in the graph above by the fact that all of these disorders have decreased during this five year period.

The five forms of cancer that cause the most deaths are trachea/bronchus/lung (2.9%), stomach (1.4%), liver (1.4%), colon/rectum (1.4%), and breast (0.8%).

All cardiovascular and circulatory diseases combined account for 30% of deaths.

Redesigned by Perceptual Edge

POPOUT

POPOUT

Properties detected by the low-level visual system

very rapid - 200-250 milliseconds

very accurate

processed in parallel

Happens before focused attention → pre-attentive

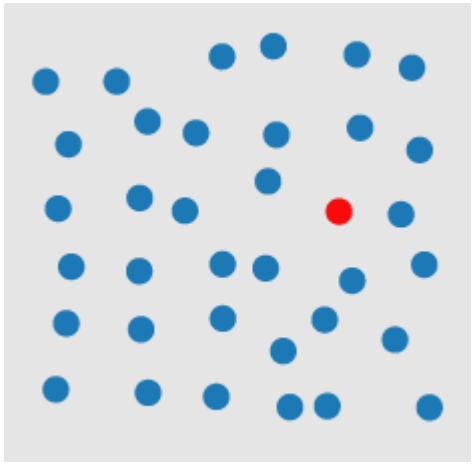
Attention is very important for cognition

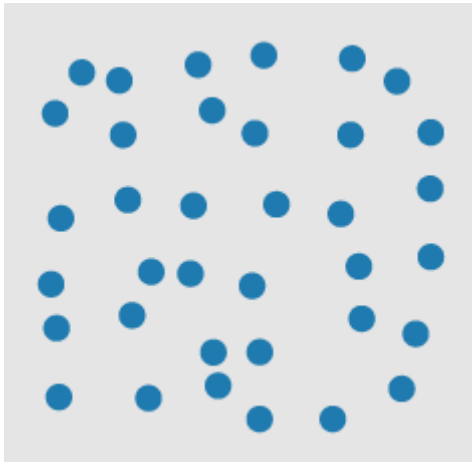
Independent of the number of distractors

Opposite: sequential search (processed serially)

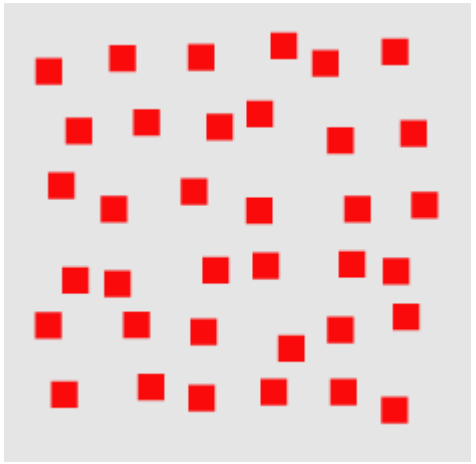
Examples based on Healey,
https://www.csc2.ncsu.edu/faculty/healey/PP/index.html

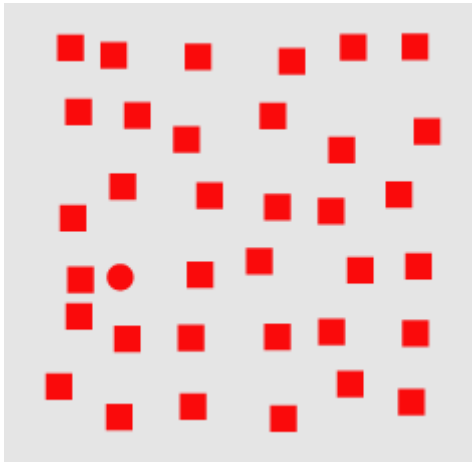
DIFFERENCE IN HUE

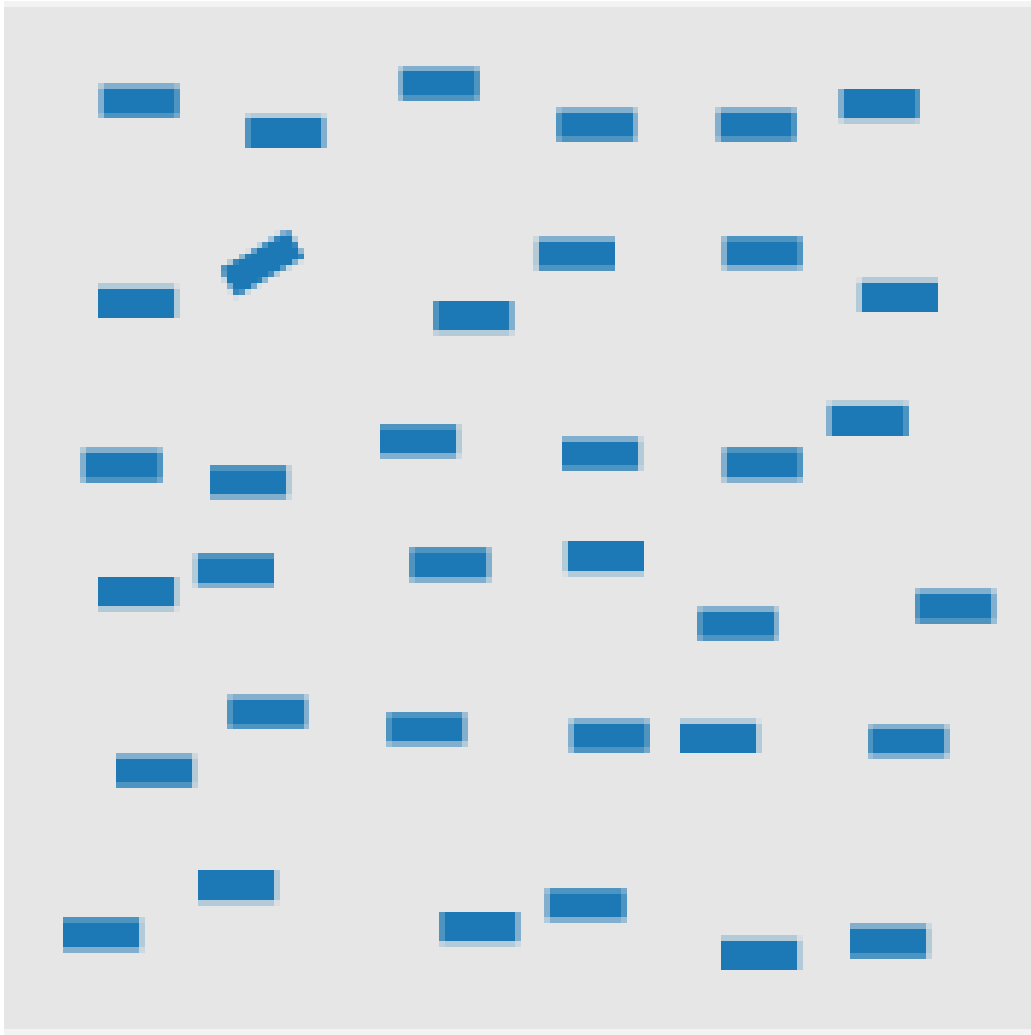




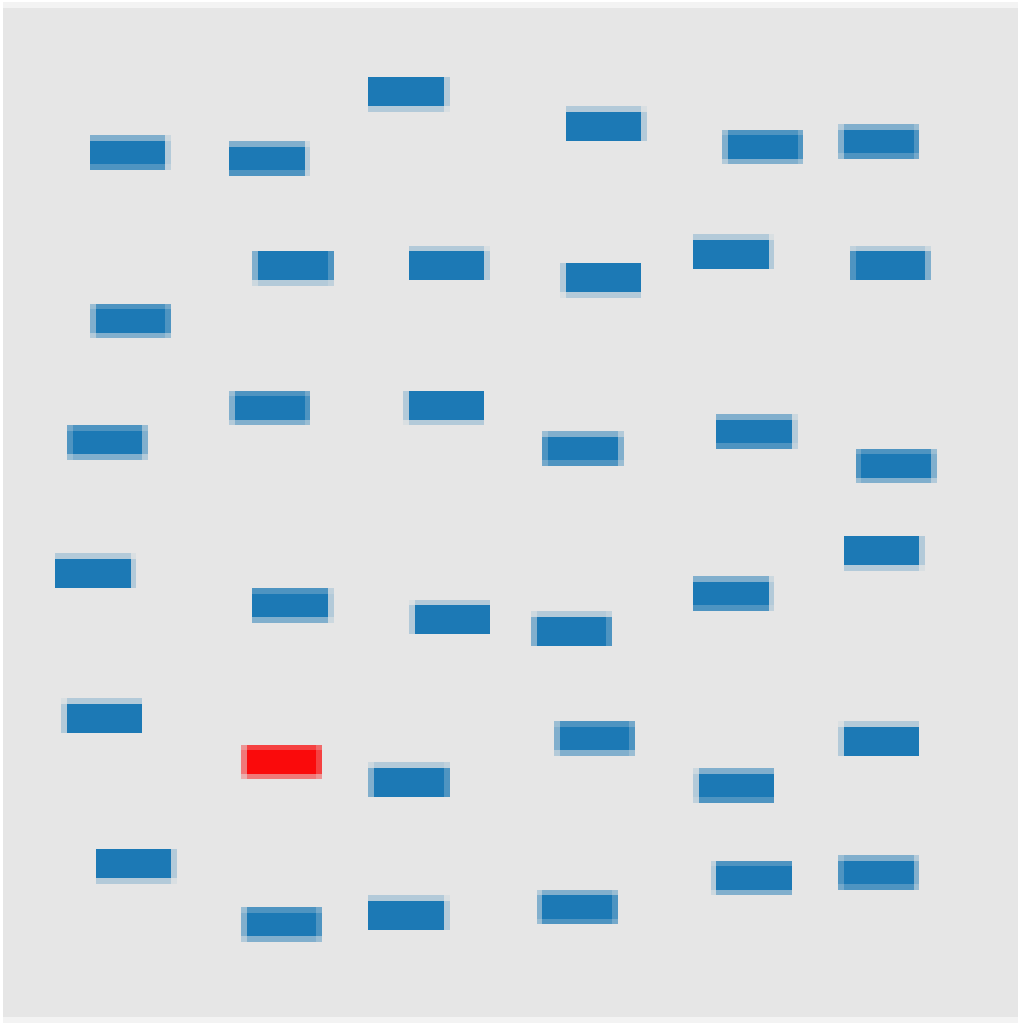
DIFFERENCE IN CURVATURE / FORM



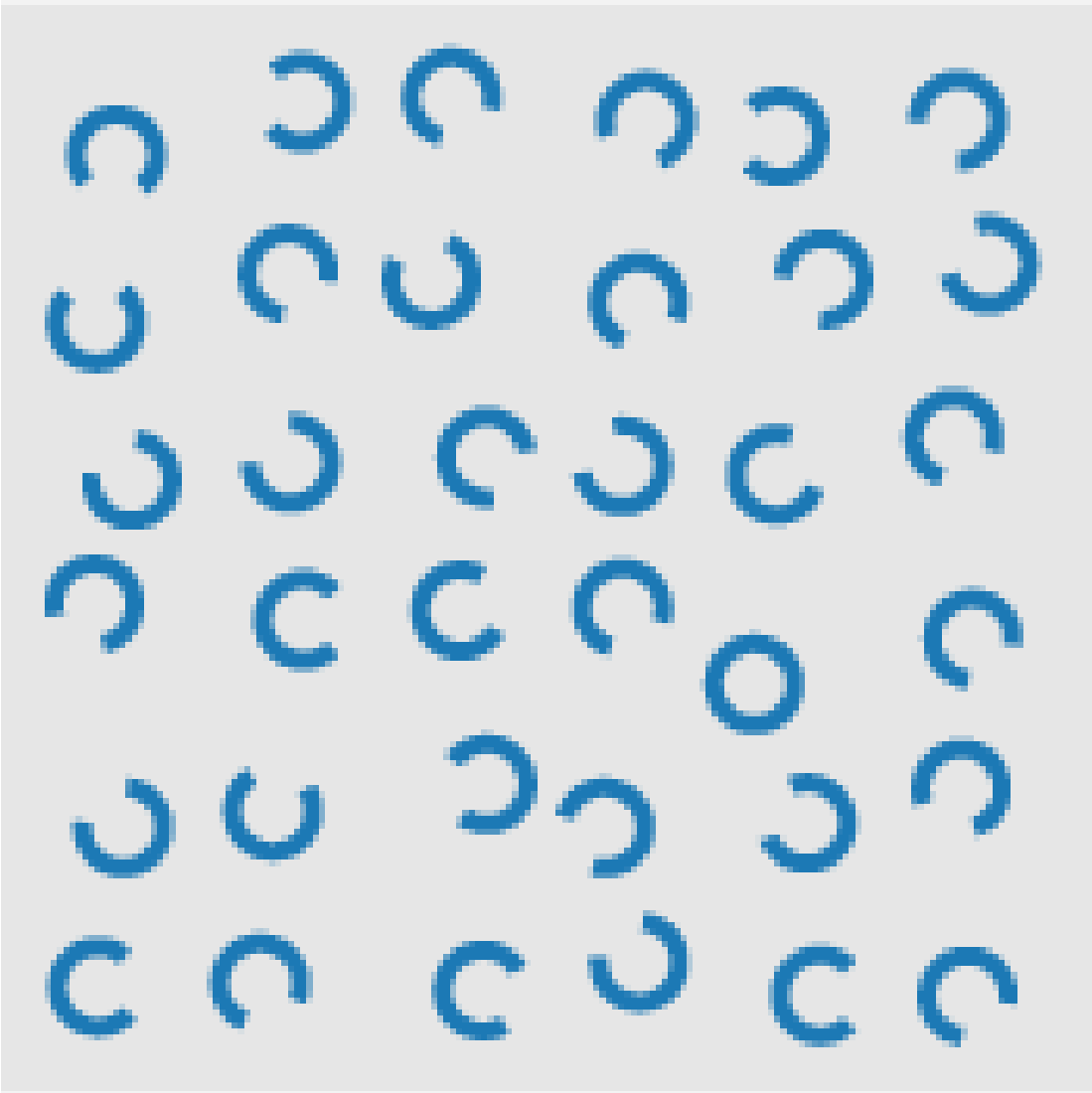




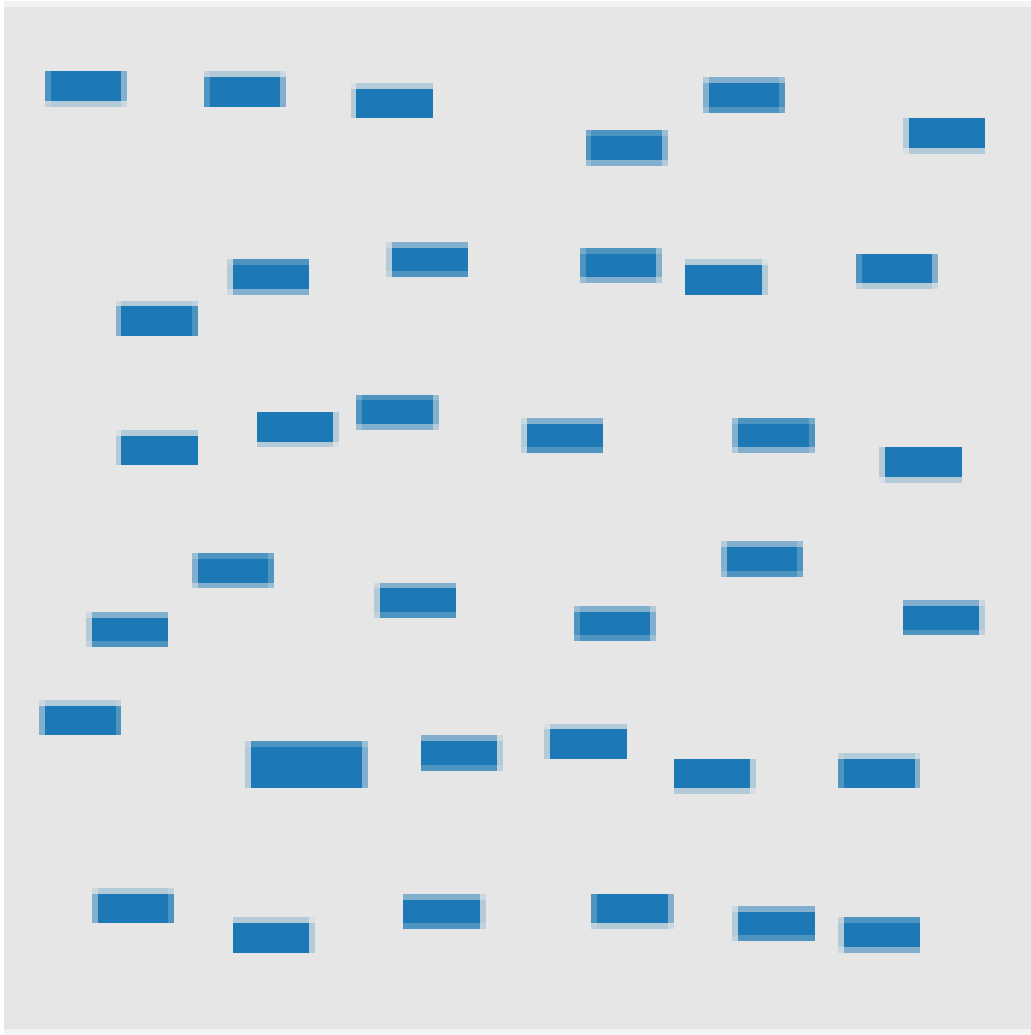
orientation



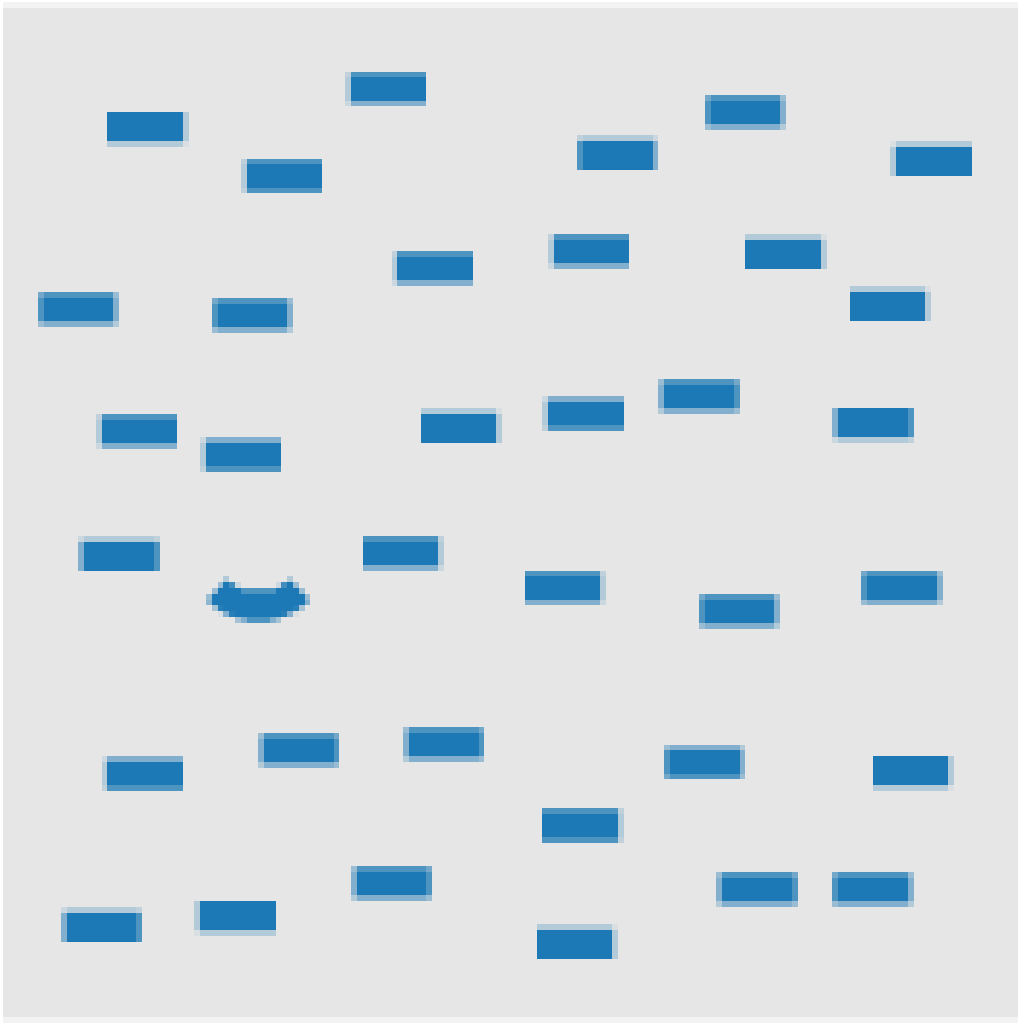
Hue



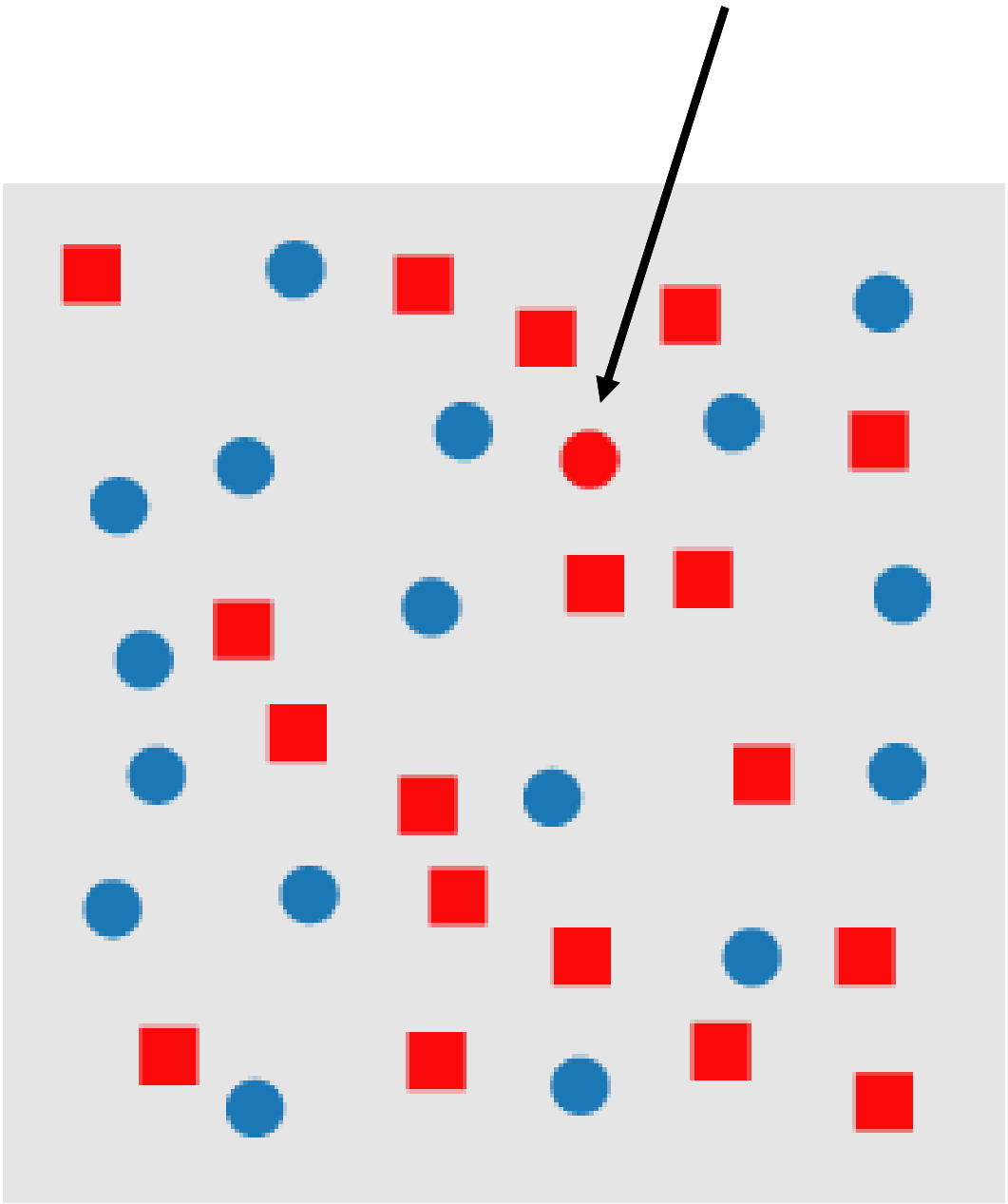
Closure



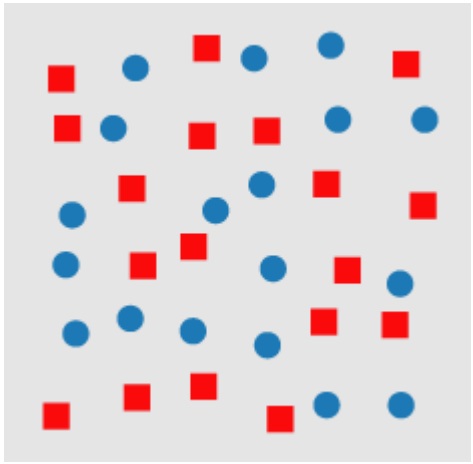
size

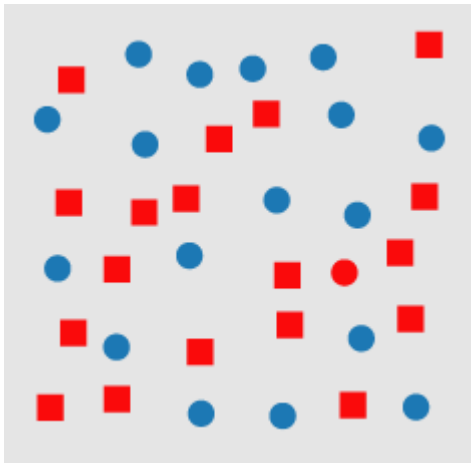


curvature



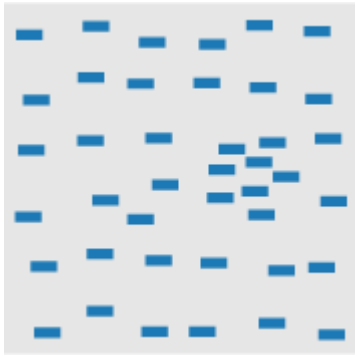
NOT PRESERVED ON COMBINATIONS

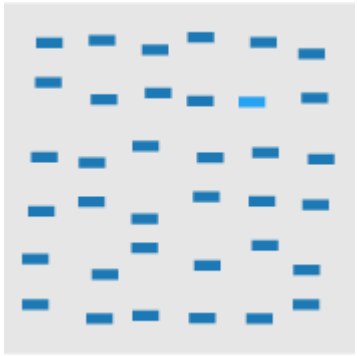


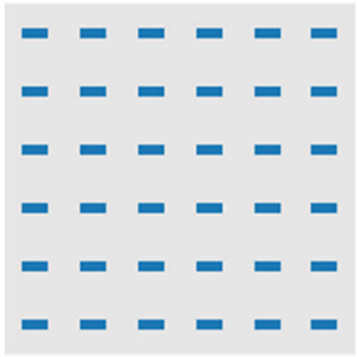


Conjunction Targets – no unique visual property
target: red, circle
distractor objects have both properties

SOME MORE PRE-ATTENTIVE PROPERTIES







TASK

- **Target detection**

  Detect the presence or absence of a target
- **Boundary detection**

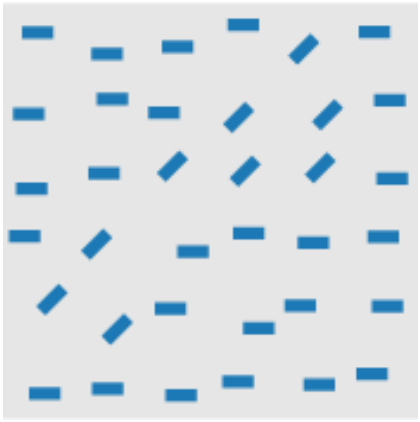
  Detect a texture boundary between two groups of elements, where are of the elements in each group have a common visual property
- **Region tracking**

  Track one or more elements with a unique visual feature as they move in time and space
- **Counting and estimation**

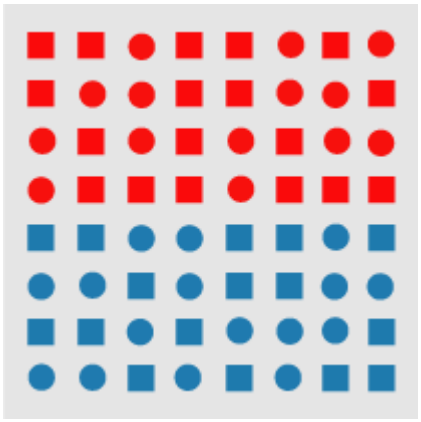
  Users count or estimate the number of elements with a unique visual feature

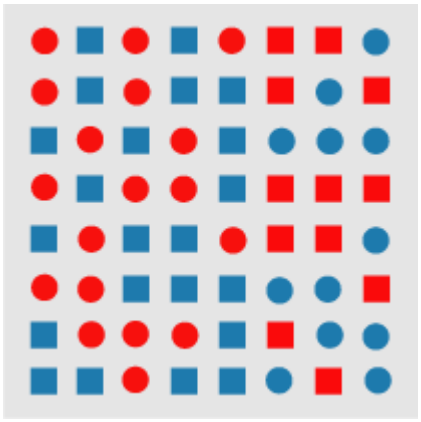
TASK

Number Estimation

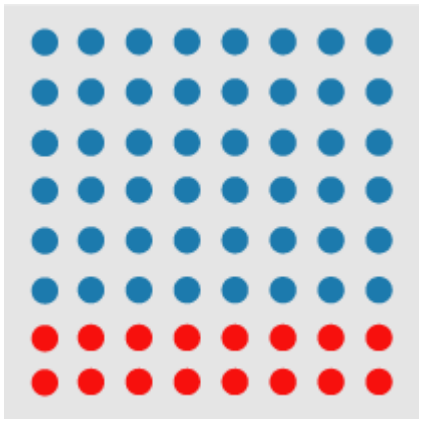


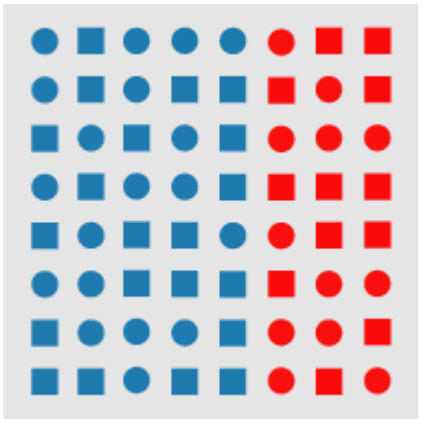
Boundary Detection

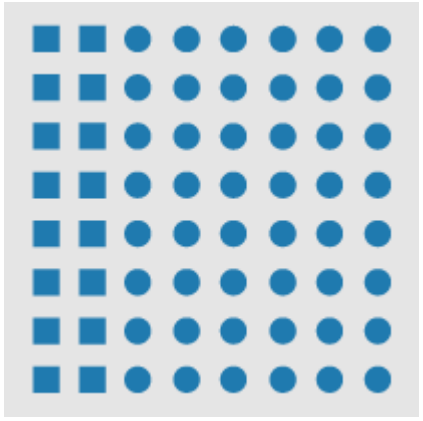


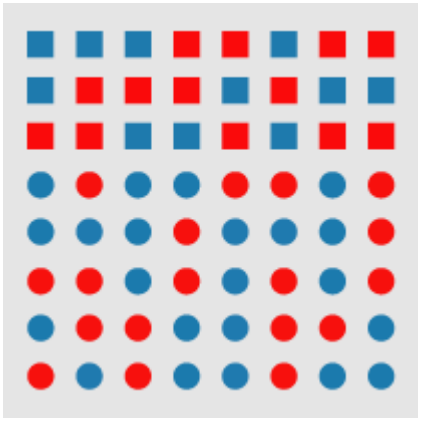


HIERARCHY OF PREATTENTIVE FEATURE









THEORIES OF PREATTENTIVE PROCESSING

For more theories

http://www.csc.ncsu.edu/faculty/healey/PP/index.html

PREATTENTIVE PROCESSING IN VIS

Can be used to draw attention to areas of interest

Can be used to express similarity/group memberships

Visual features must be carefully designed

Conjunctions must be avoided

colorful vis vs highlight

CHANGE BLINDNESS

CHANGE BLINDNESS

Detail of an image cannot be remembered across separate scenes

except in areas with focused attention

Interruption (e.g. a blink, eye saccade or blank screen) amplifies this effect

Not failure of vision system

Failure due to inappropriate attentional guidance

CHANGE BLINDNESS

Various theories about causes

- Overwriting: information that was not abstracted is lost
- First impression: only initial view is abstracted
- Nothing is stored: only abstract concepts are committed to memory
- Everything is stored, nothing is compared: we compare only when we are forced to
- Feature combination: scenes are combined as long as they make sense

Influencing factors

- Attention
- Expectation (knowing something will change)
- Semantic importance of changed object
- Low level object properties overlooked more easily

ATTENTION BLINDNESS

WHODUNNIT?

KEY POINTS

To find meaning in what we see **we must selectively pay attention** to what is important

Low-level vision is driven by object features rather than a conscious effort where to look (e.g., pre-attentive processing)

Attention is driven by preexisting knowledge, expectations, and goals stored in long-term memory

GESTALT PRINCIPLES

Wertheimer, 1923, and recent extensions

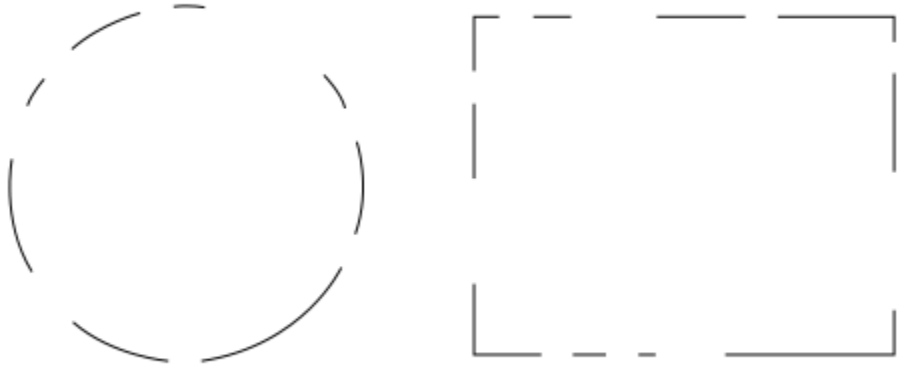
GESTALT PRINCIPLES

Patterns that transcend the visual stimuli that produced them

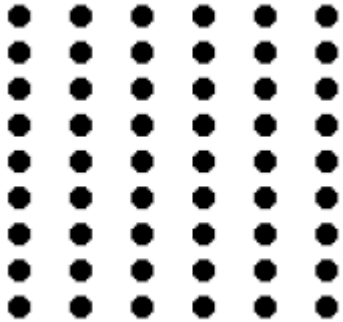
- The whole is something else than the sum of its parts - Kurt Koffka

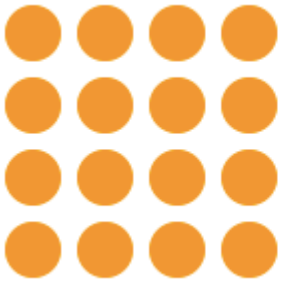
Vs

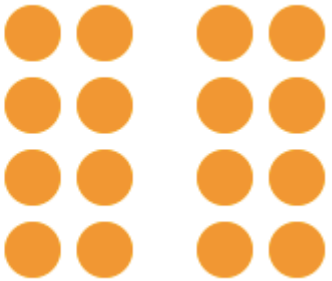
- The whole is greater than the sum of the parts

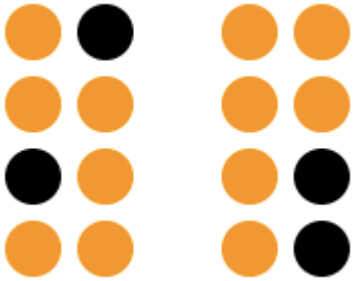


PROXIMITY

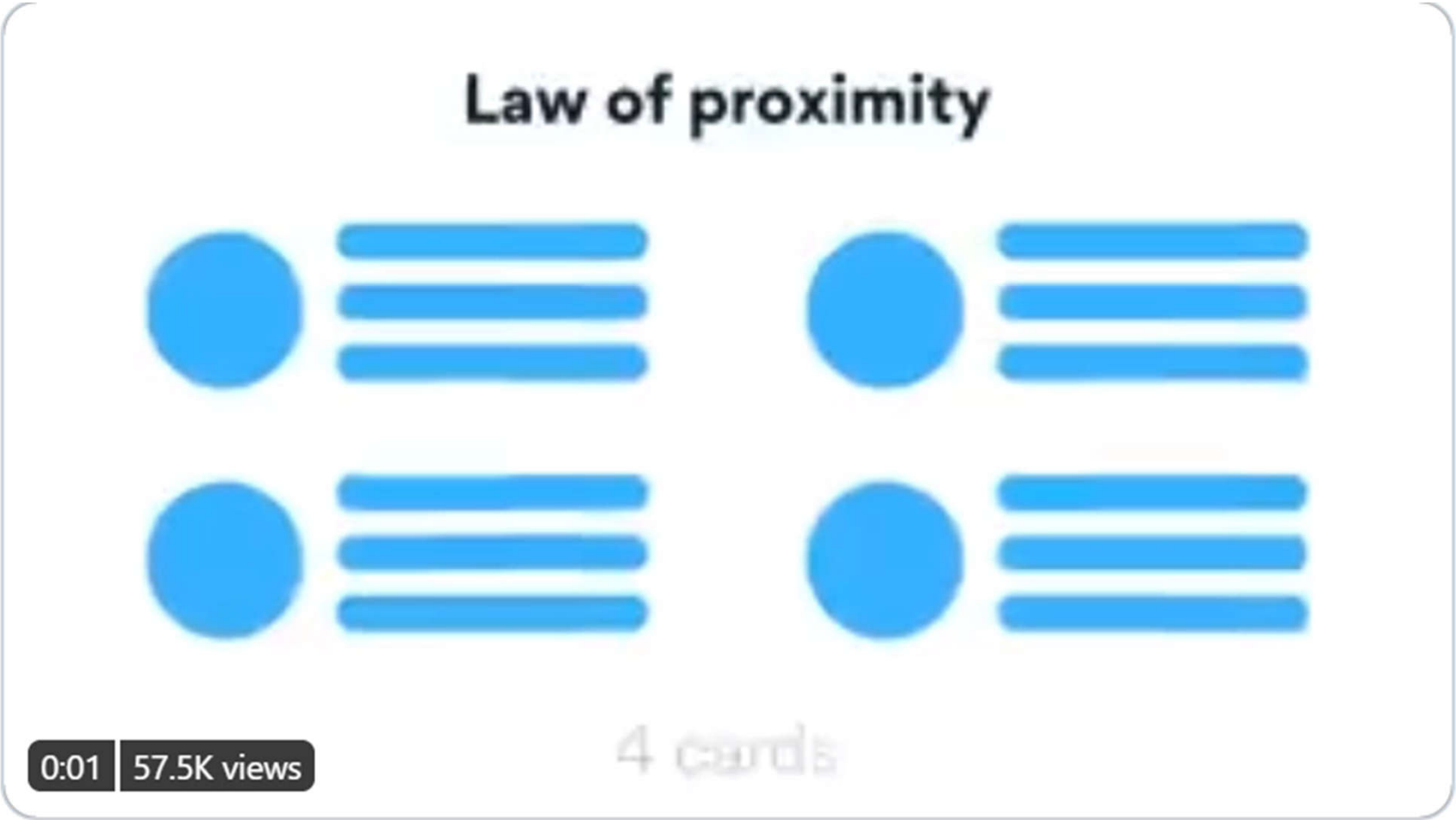








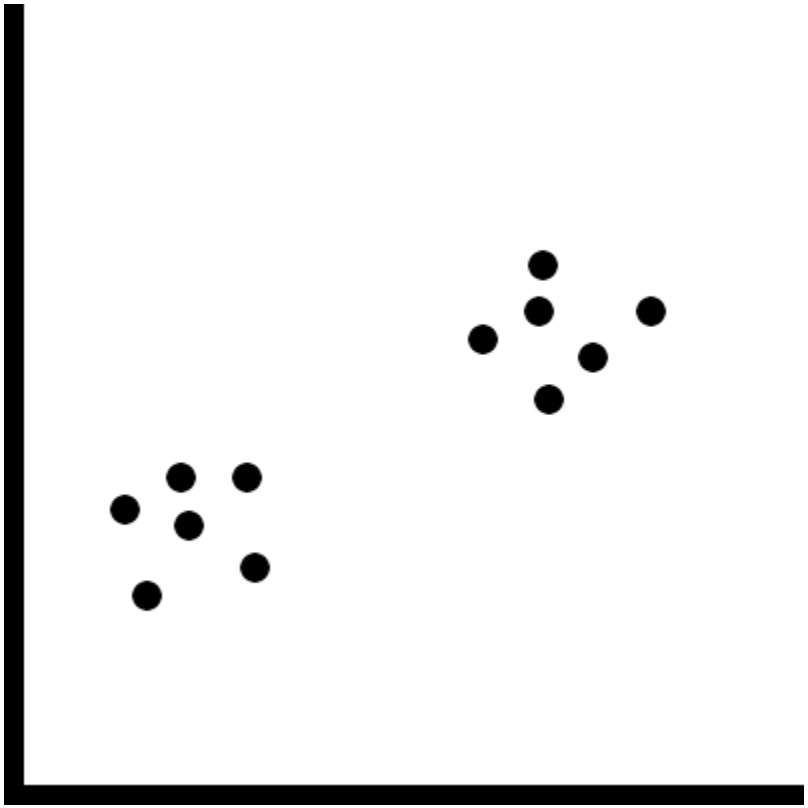
Andy Rutledge, “Gestalt Principles of Perception”



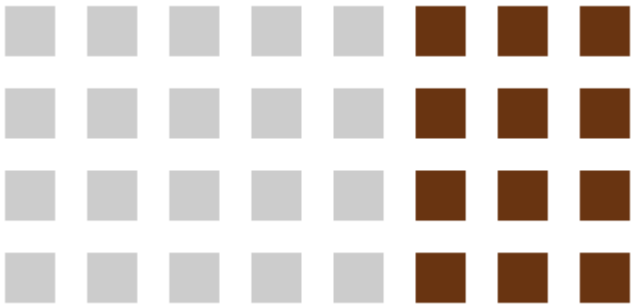
https://twitter.com/pablostanley/status/974303621092225024

PROXIMITY

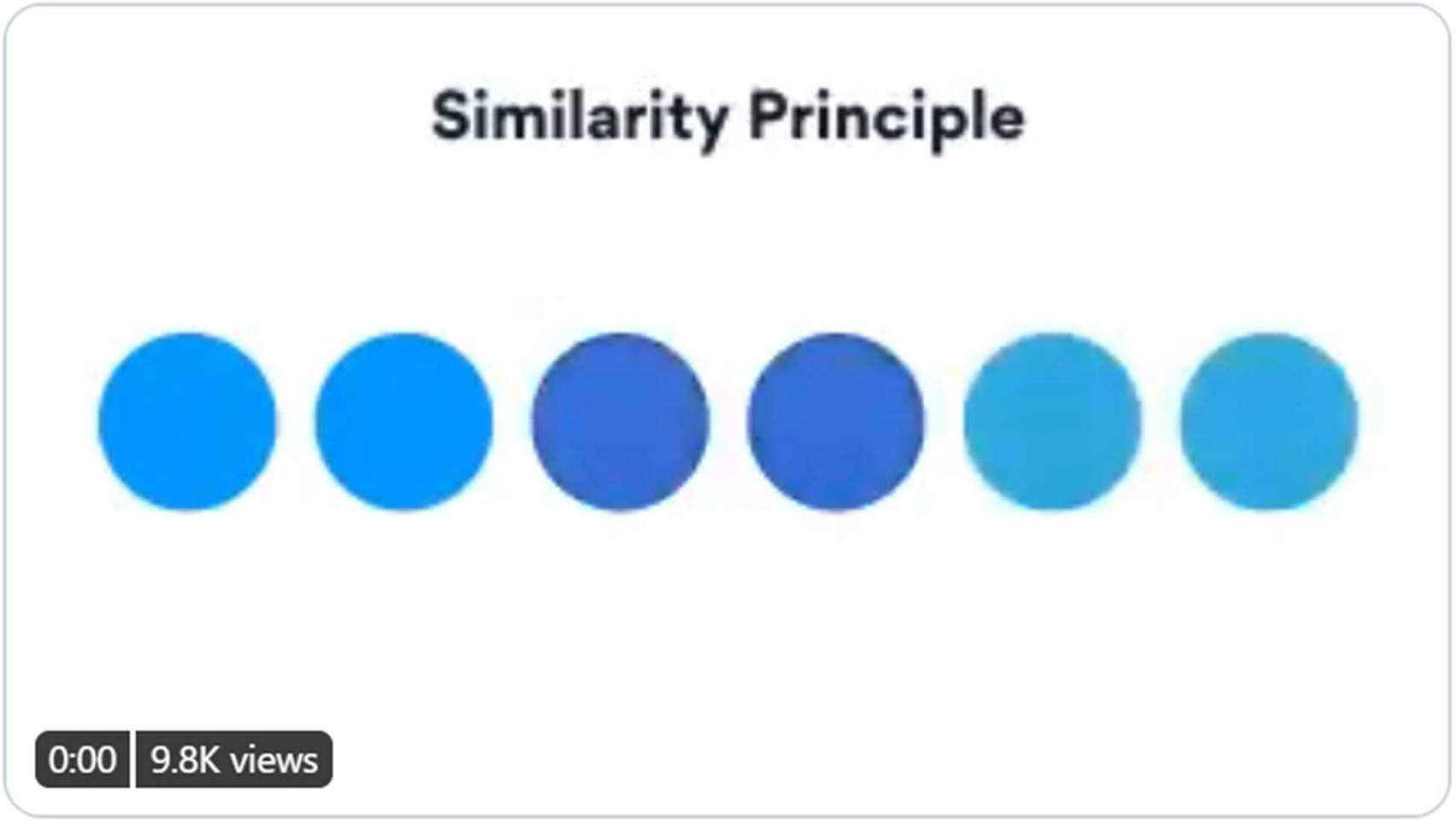
Grouping/linking by placing entities in close proximity

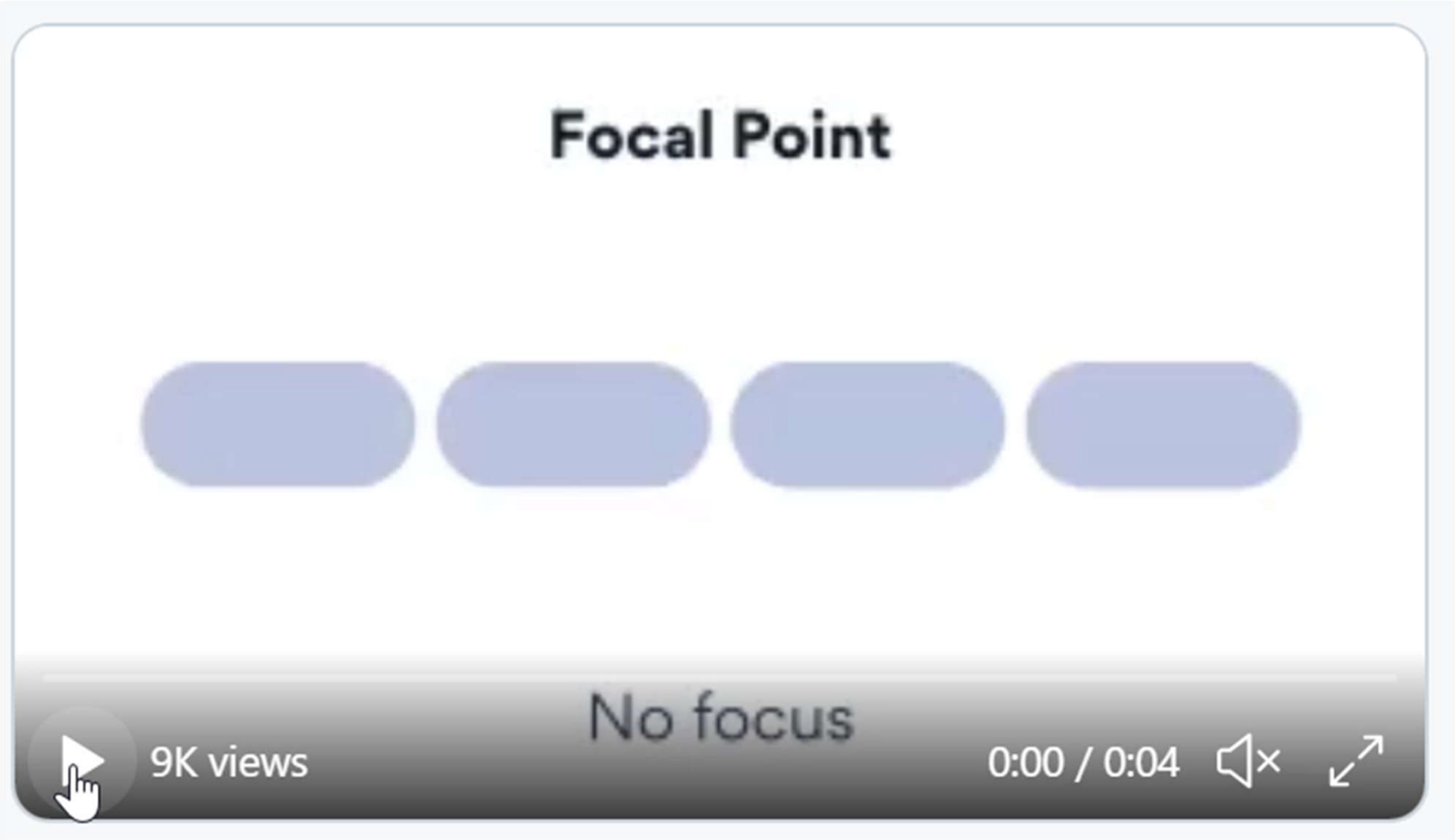


SIMILARITY



Andy Rutledge, “Gestalt Principles of Perception “


Similarity Principle
0:00
9.8K views

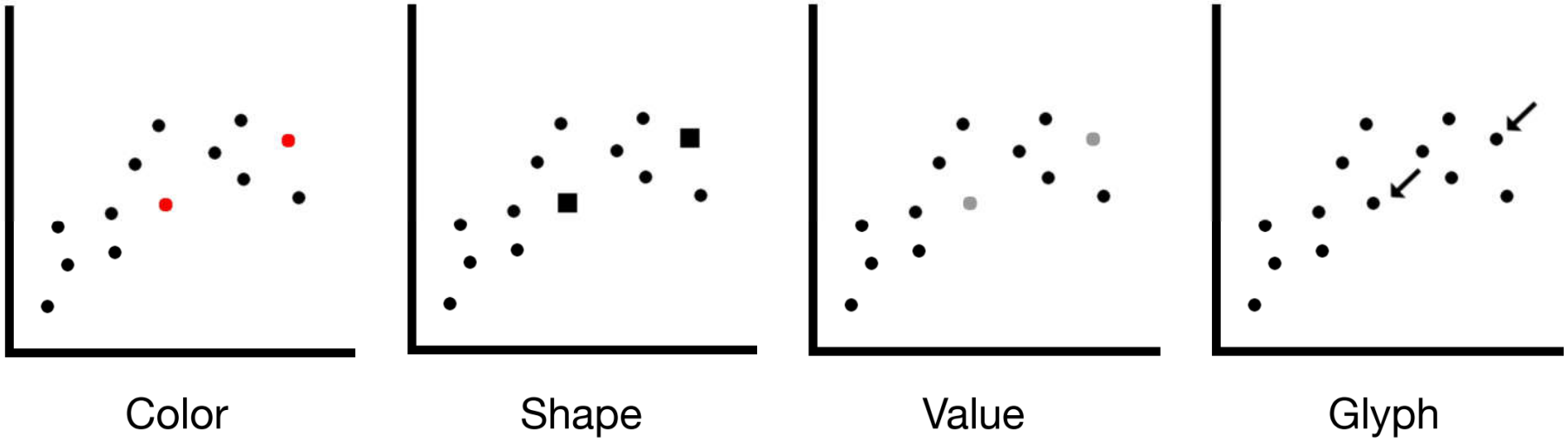

Focal Point
No focus
9K views
0:00 / 0:04

SIMILARITY

Co-modulation of a chanel

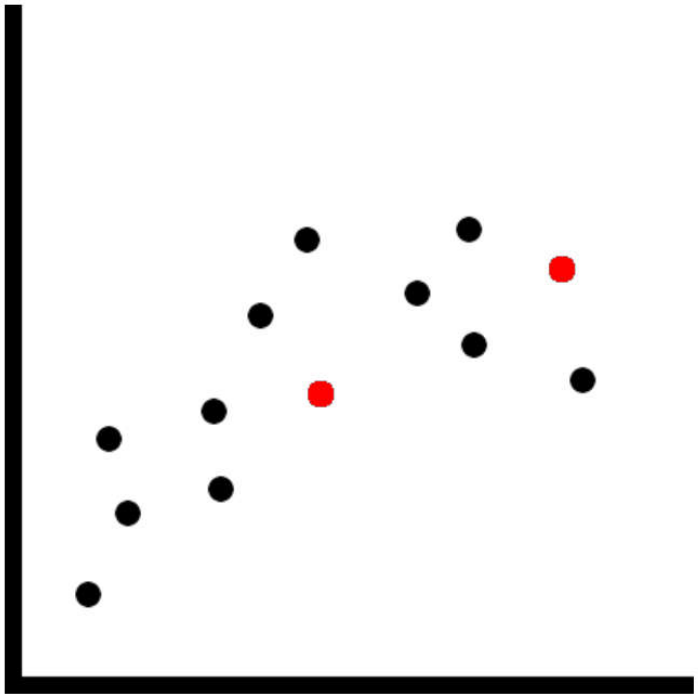
color, shape, size, value, orientation, texture…

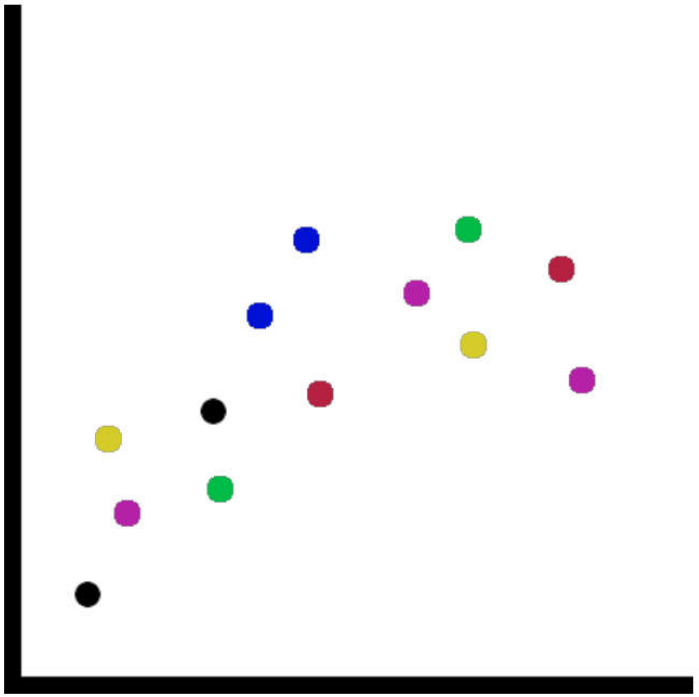
Adding a glyph, label, frame, background



COLOR – PERCEPTION ISSUES

Popout: works for 1-2 simultaneous, but not for more

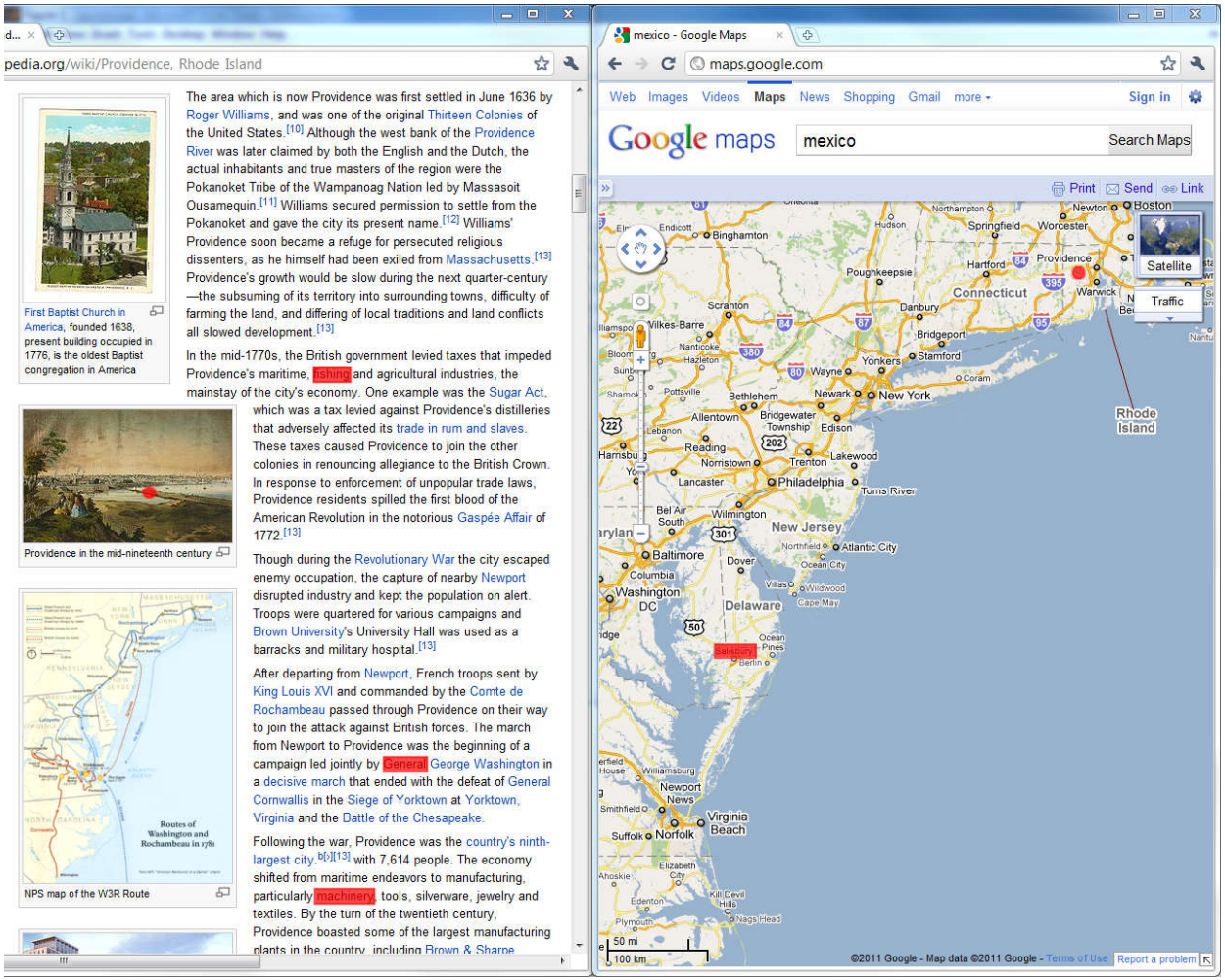




COLOR – PERCEPTION ISSUES

Slower in a cluttered environment

Size of colored object relevant

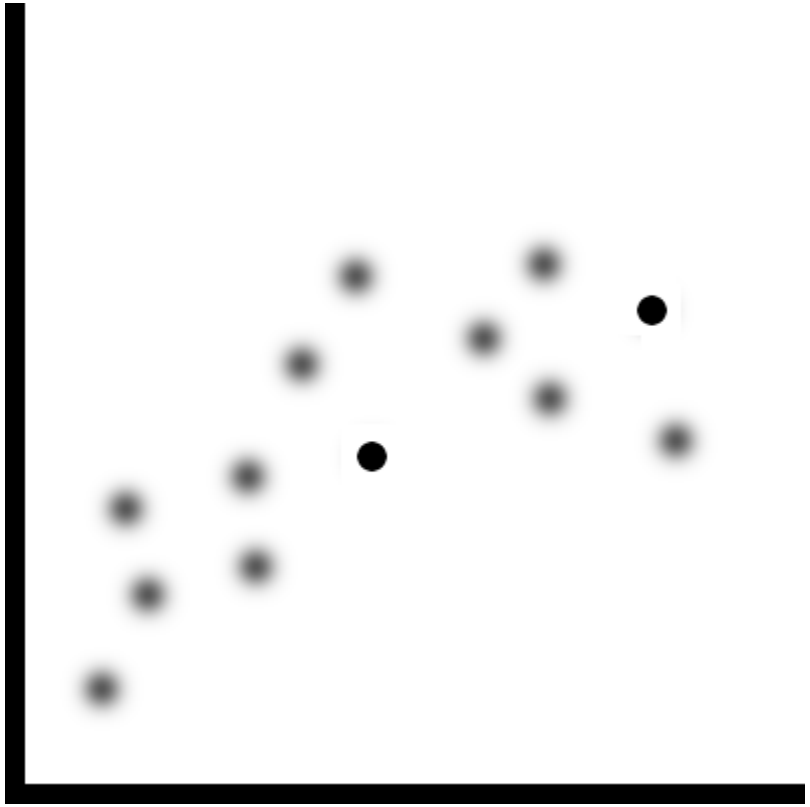


SIMILARITY

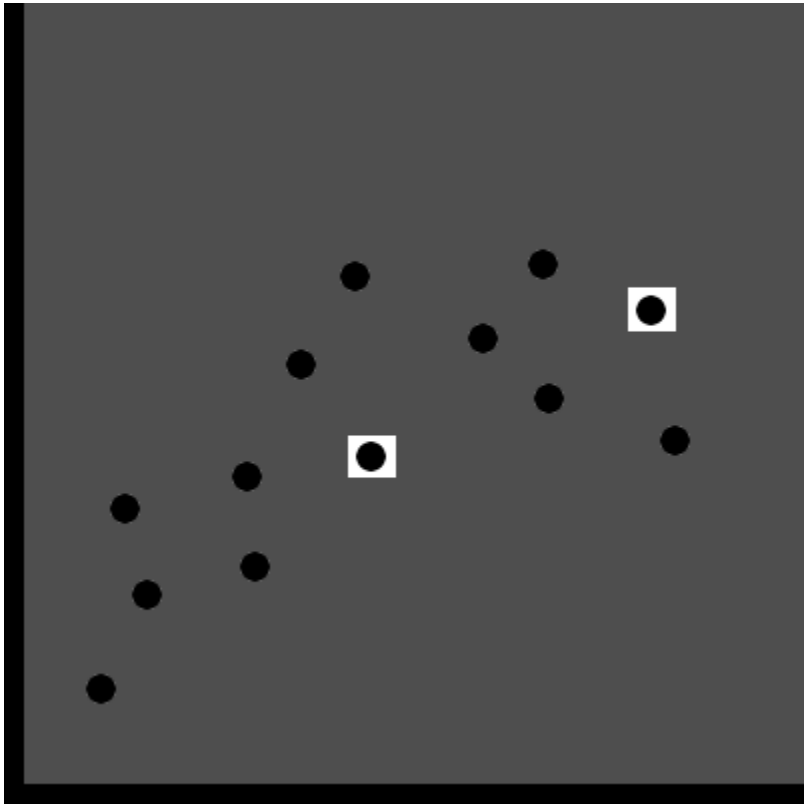
Modulate everything else

blurring, darkening, desaturating,

Blur

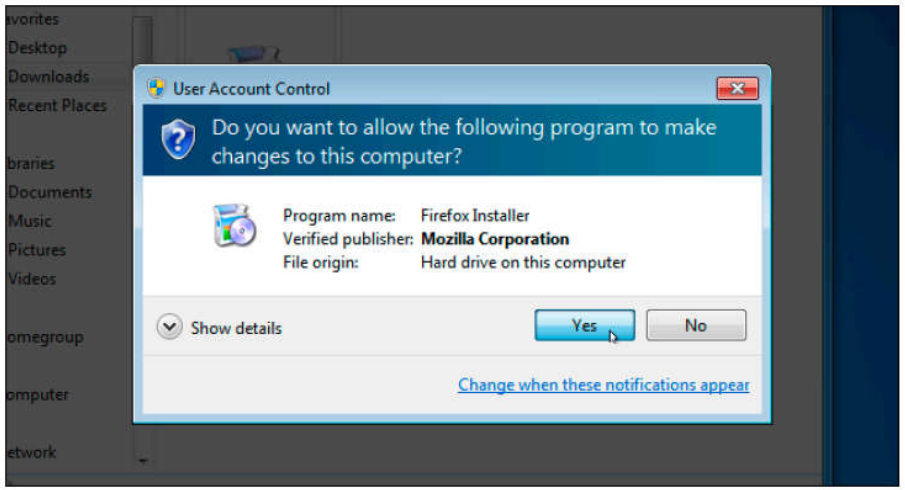


Darkening

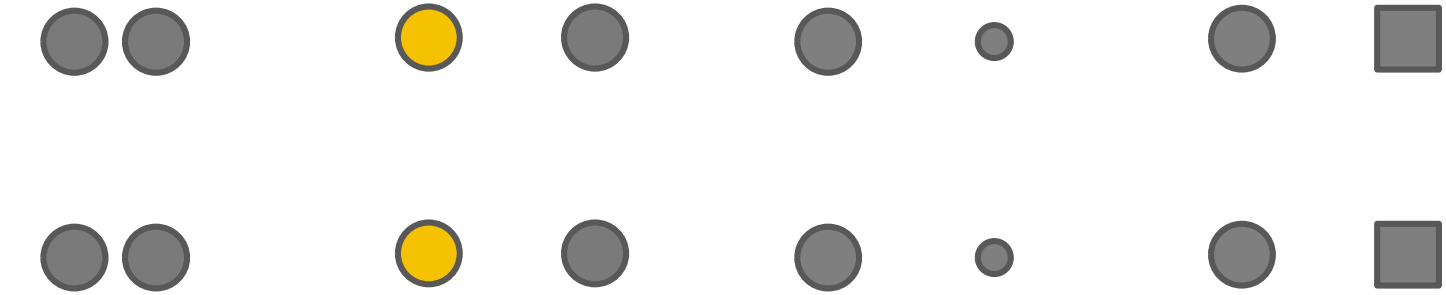


MODULATING EVERYTHING ELSE

Don’t use unless the sole objectives is to guide attention toward one (set of) items

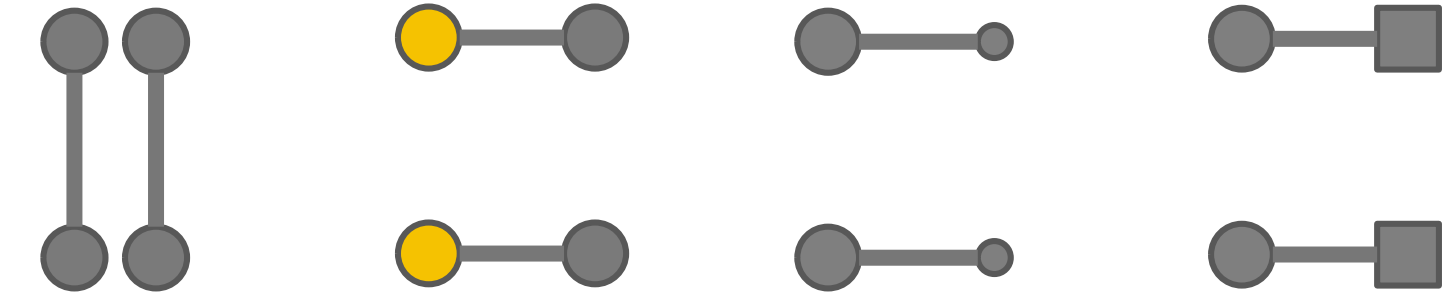


GROUPING



Proximity Color Size Shape

GROUPING

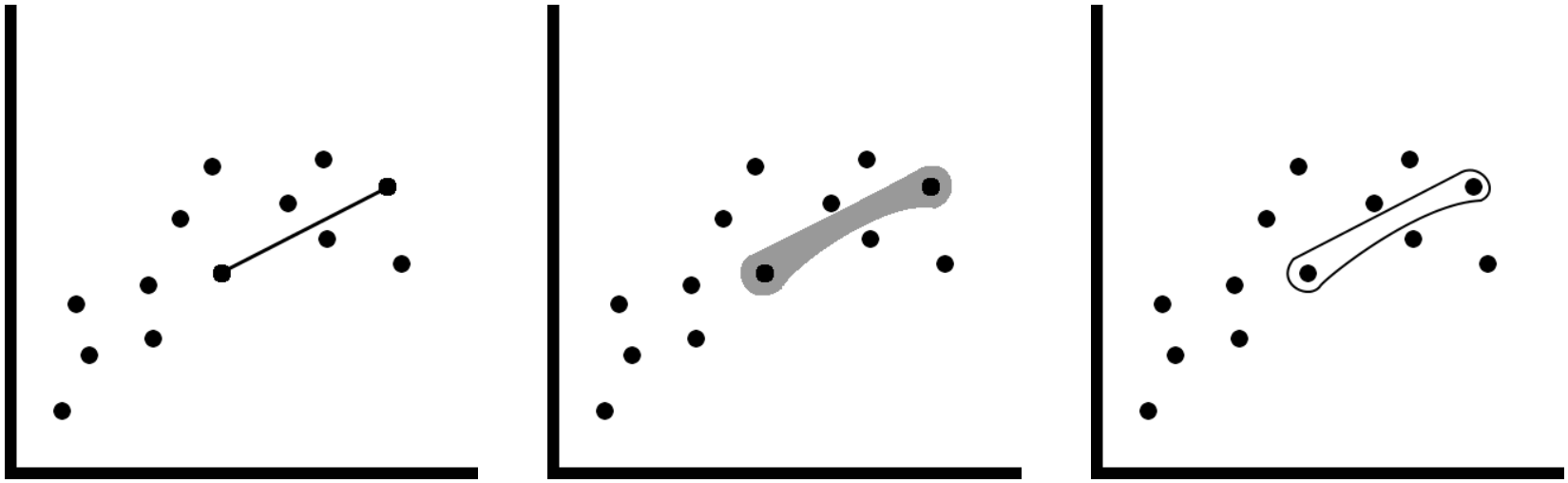


Proximity Color Size Shape

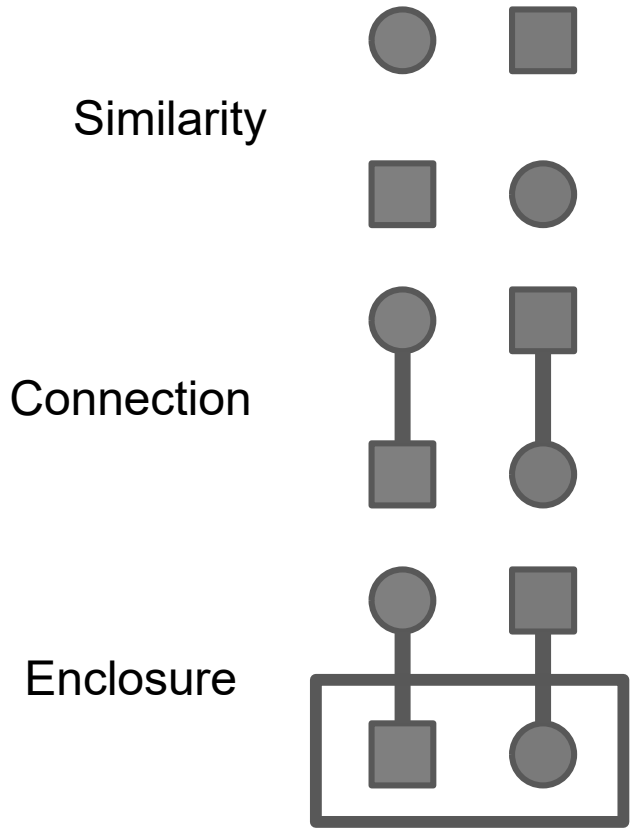
CONNECTEDNESS AND COMMON REGION

Connected items with a line or curve

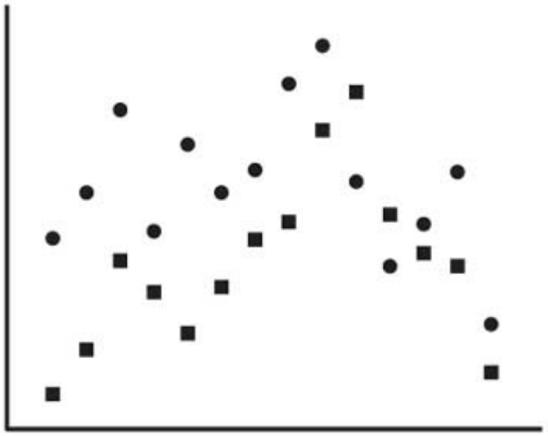
Surround items with a outline, surface, volume

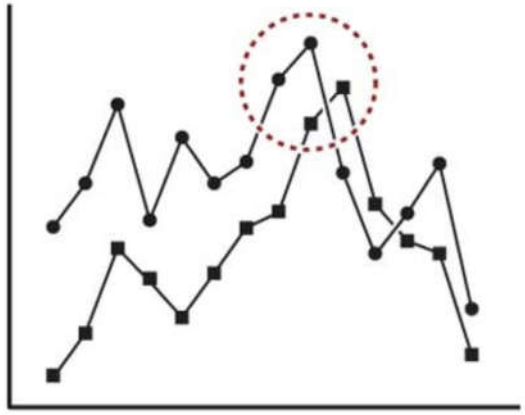


ENCLOSURE / COMMON REGION

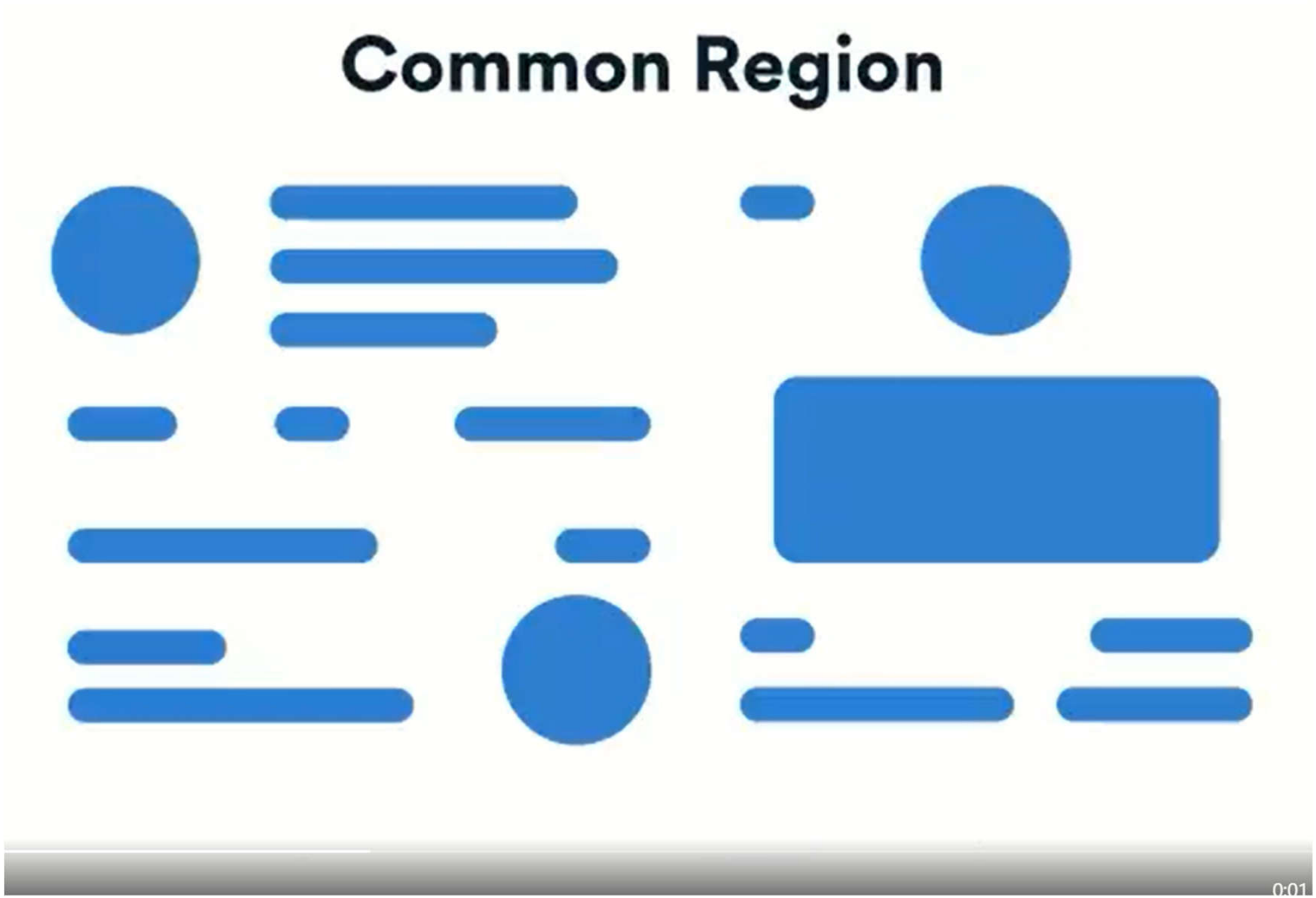


CONNECTEDNESS AND ENCLOSURE





Bang Wong, “Gestalt Principles, 1”, Nature Methods


Common Region
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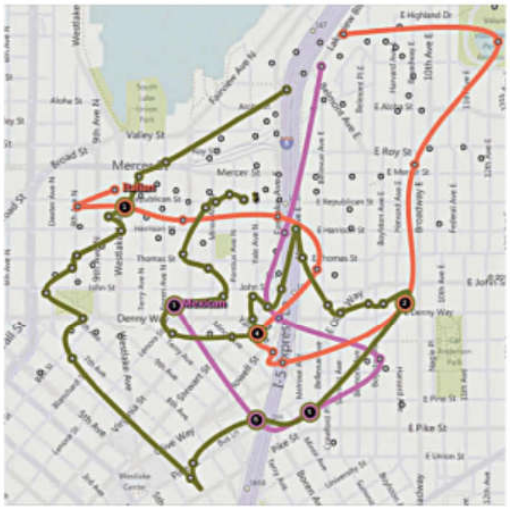
CONNECTEDNESS VARIETIES

Bubble Sets

Image by [Dinkla et al., 2011]
Technique by [Collins et al.,2009]

Line Sets

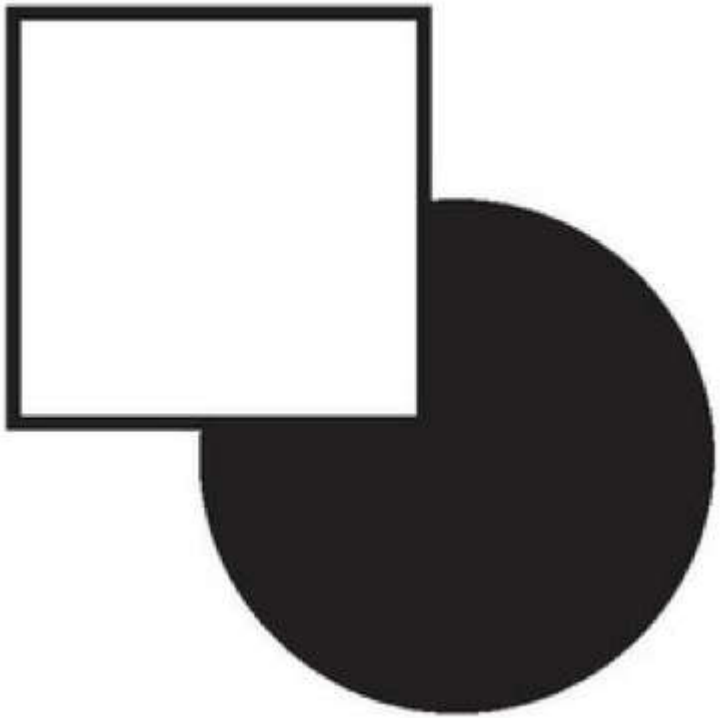


[Alper et al., 2011]

Kelp Diagrams

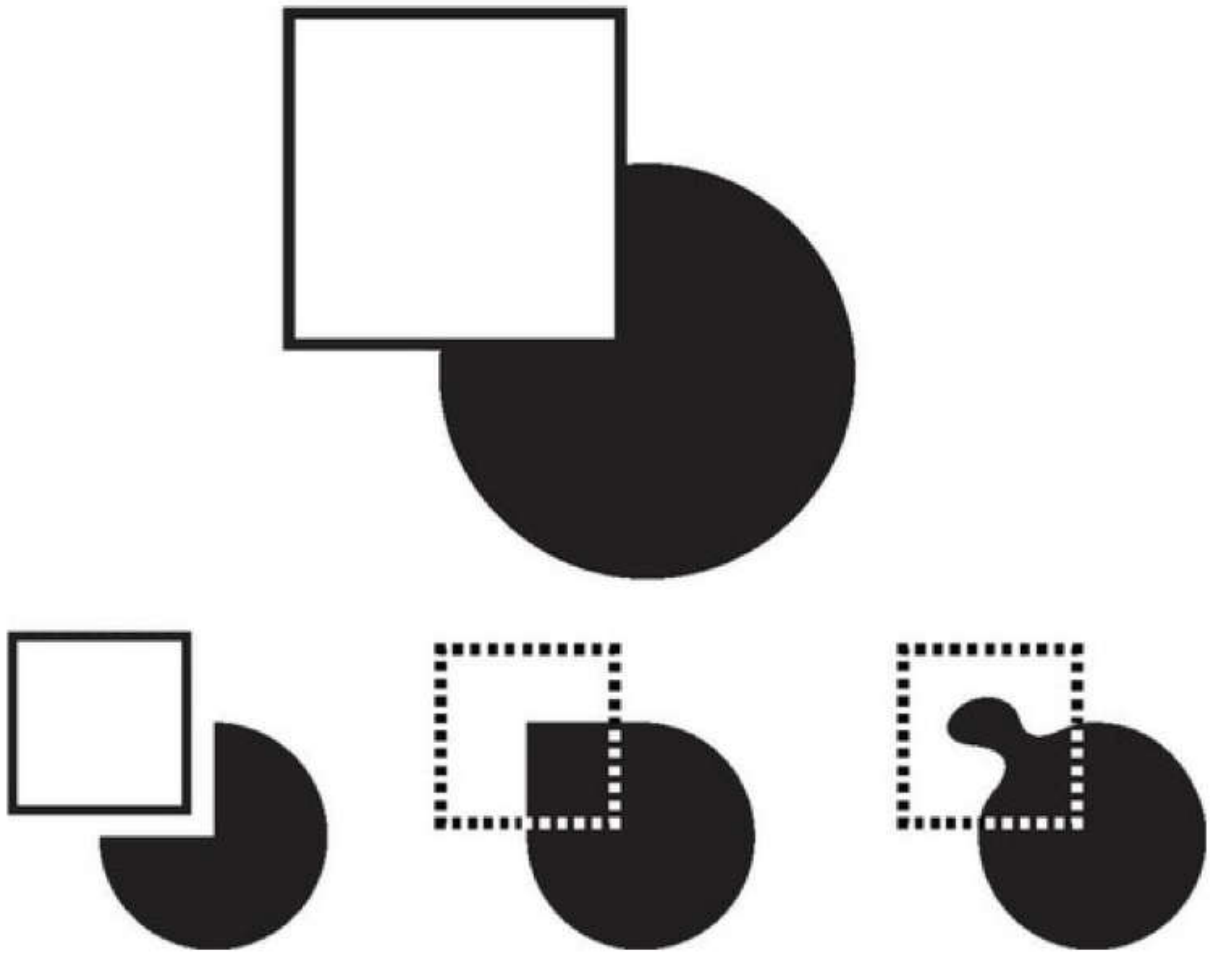
[Dinkla et al., 2012]

CONTINUITY

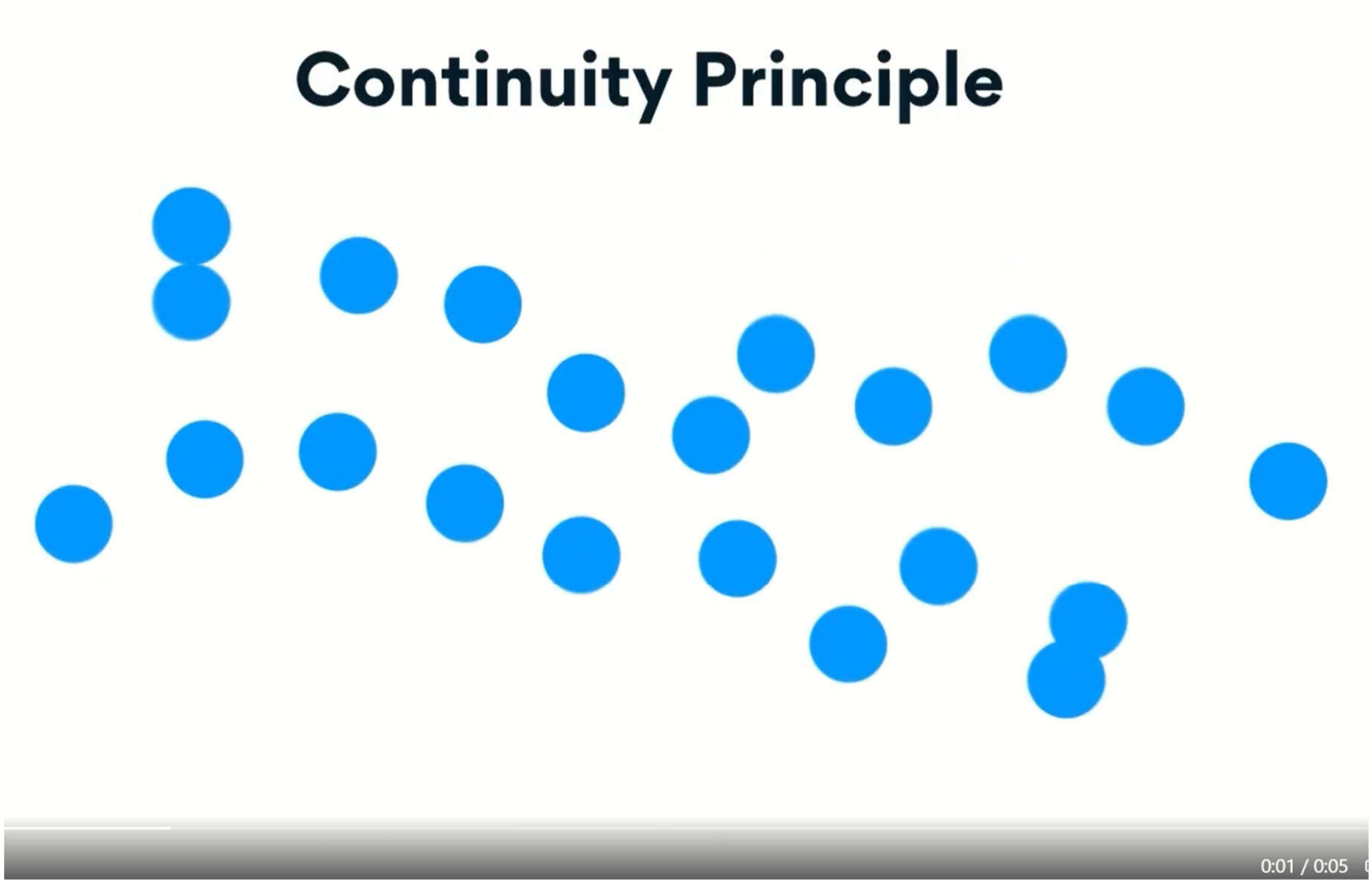


Bang Wong, “Gestalt Principles, II”, Nature Methods

CONTINUITY

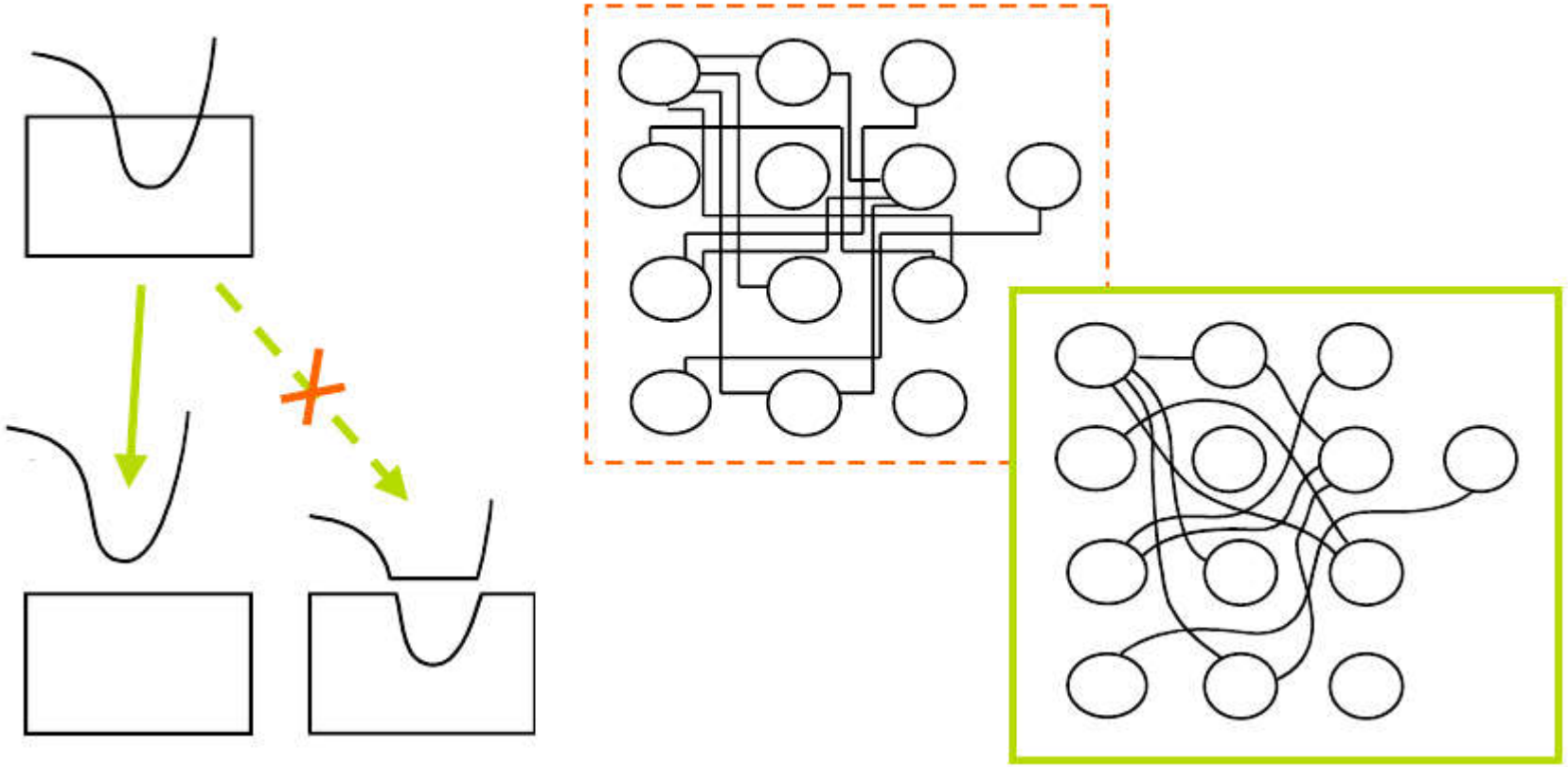


CONTINUITY PRINCIPLE



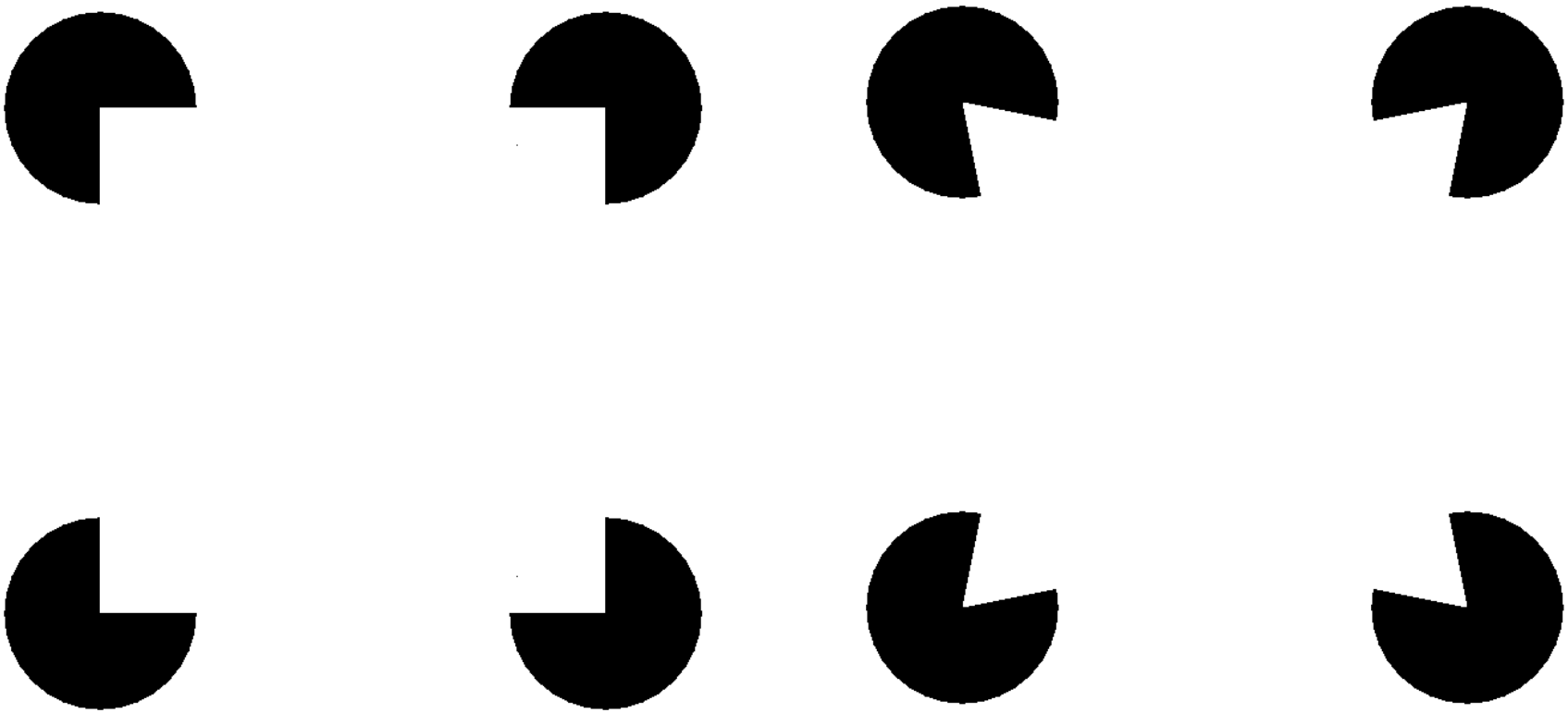
CONTINUITY

Things: smooth & continuous



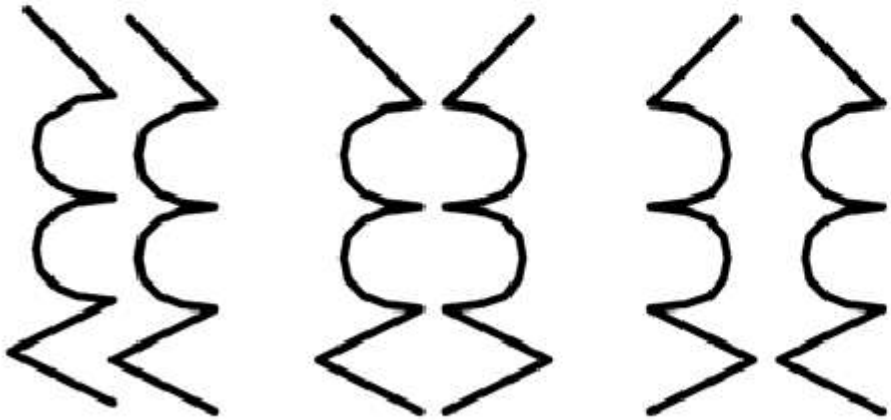
CLOSURE / COMPLETION

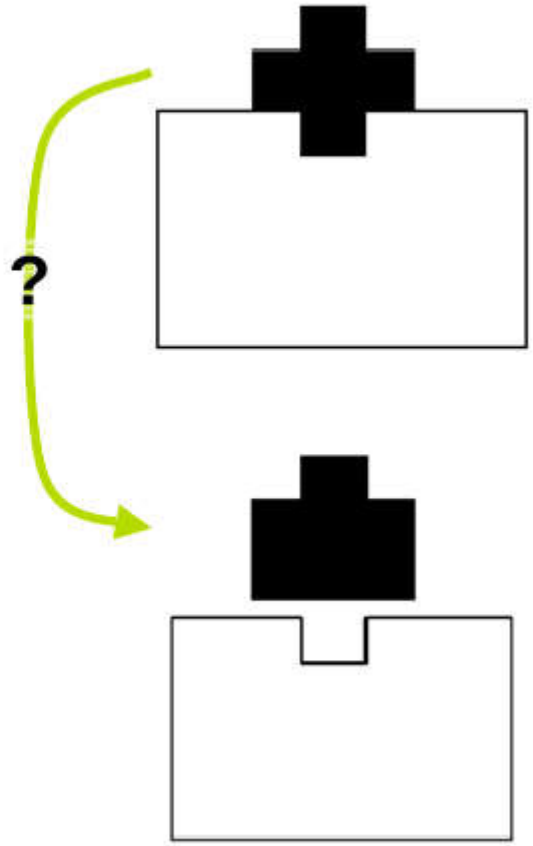
CLOSURE



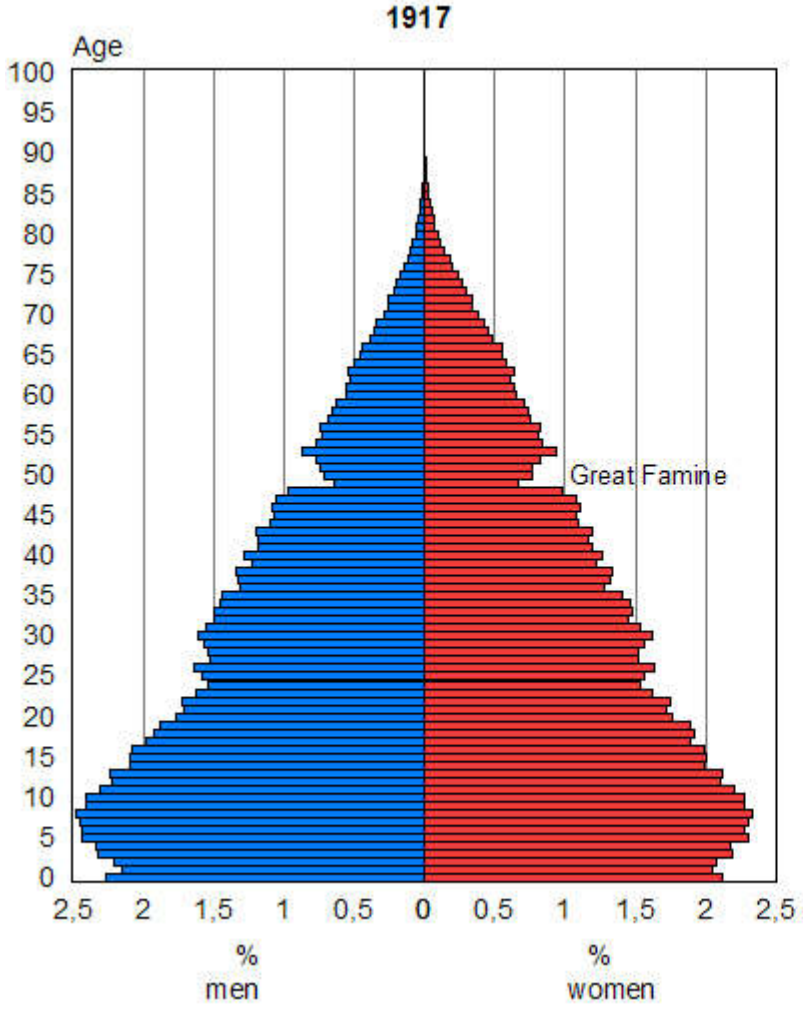
SYMMETRY

Things: symmetric





POPULATION PYRAMID



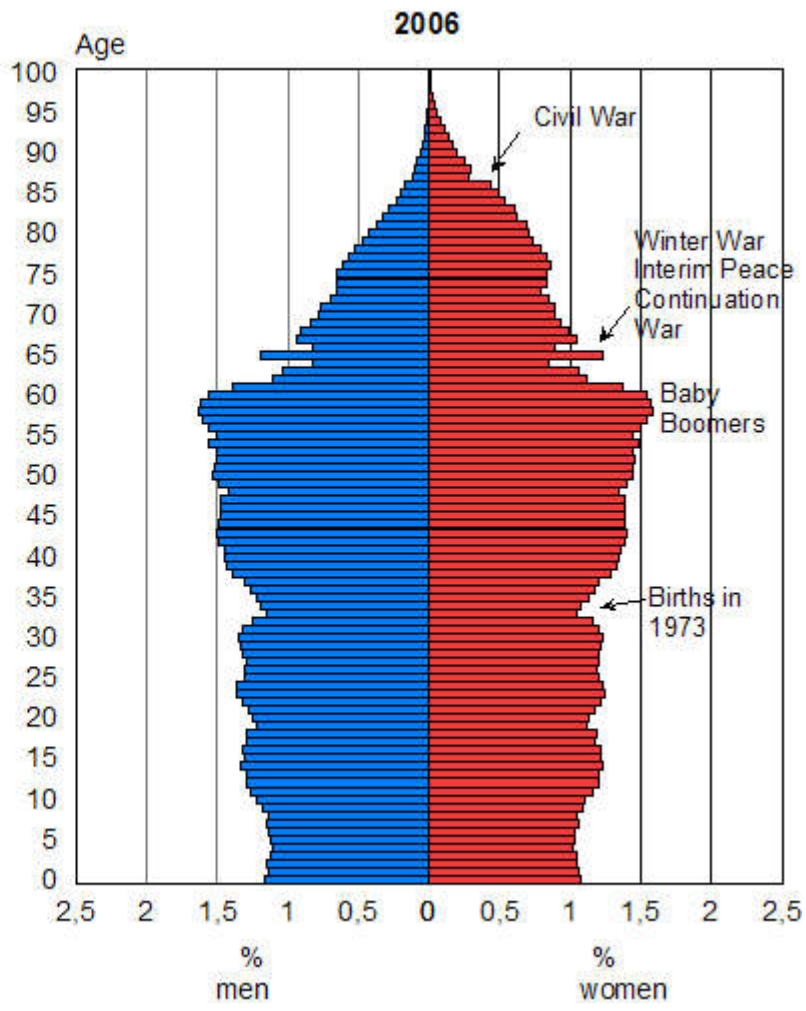
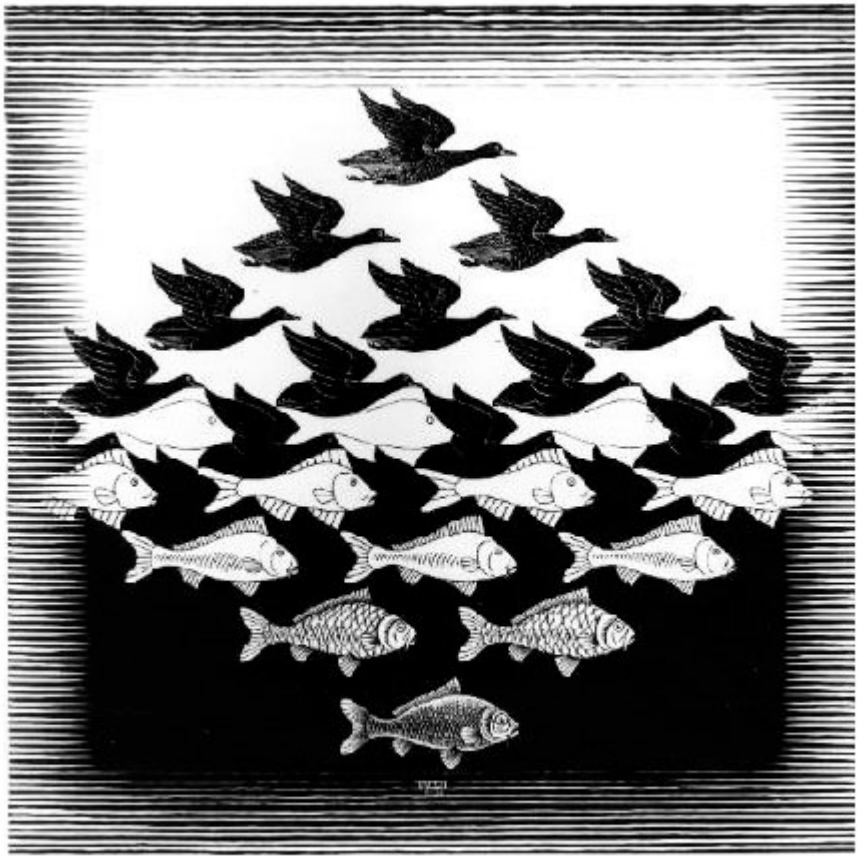


FIGURE / GROUND



M.C. Escher: *Sky and Water I* 1938 woodcut

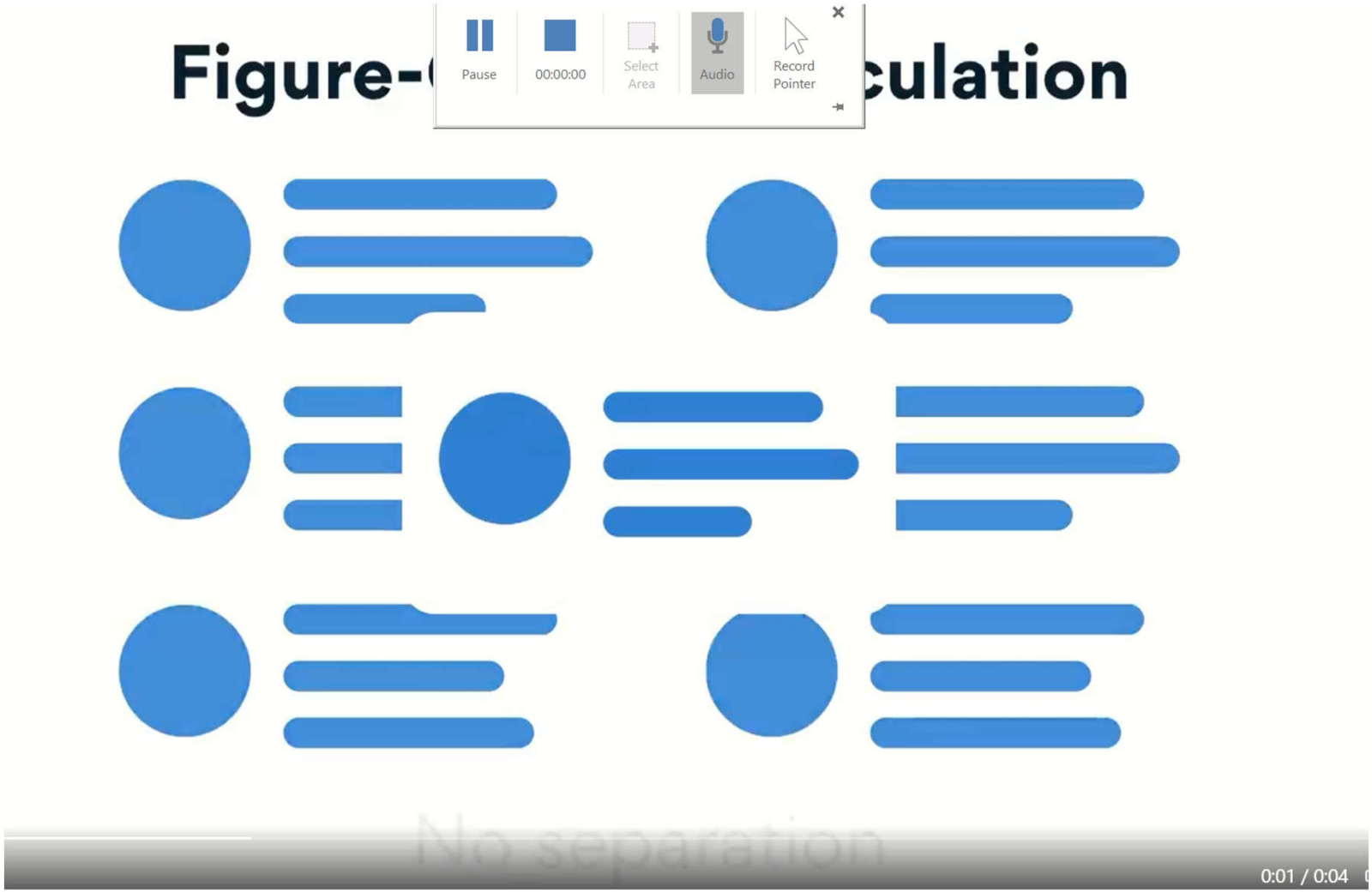


Figure-
culation
Pause
00:00:00
Select Area
Audio
Record Pointer
No separation
0:01 / 0:04

APPLICATION

pedia.org/wiki/Providence,_Rhode_Island

First Baptist Church in America, founded 1638, present building occupied in 1776, is the oldest Baptist congregation in America

The area which is now Providence was first settled in June 1636 by Roger Williams, and was one of the original Thirteen Colonies of the United States.[10] Although the west bank of the Providence River was later claimed by both the English and the Dutch, the actual inhabitants and true masters of the region were the Pokanoket Tribe of the Wampanoag Nation led by Massasoit Ousamequin.[11] Williams secured permission to settle from the Pokanoket and gave the city its present name.[12] Williams' Providence soon became a refuge for persecuted religious dissenters, as he himself had been exiled from Massachusetts.[13] Providence's growth would be slow during the next quarter-century —the subsuming of its territory into surrounding towns, difficulty of farming the land, and differing of local traditions and land conflicts all slowed development.[13]

In the mid-1770s, the British government levied taxes that impeded Providence's maritime, fishing and agricultural industries, the mainstay of the city's economy. One example was the Sugar Act, which was a tax levied against Providence's distilleries that adversely affected its trade in rum and slaves. These taxes caused Providence to join the other colonies in renouncing allegiance to the British Crown. In response to enforcement of unpopular trade laws, Providence residents spilled the first blood of the American Revolution in the notorious Gaspée Affair of 1772.[13]

Providence in the mid-nineteenth century

Though during the Revolutionary War the city escaped enemy occupation, the capture of nearby Newport disrupted industry and kept the population on alert. Troops were quartered for various campaigns and Brown University's University Hall was used as a barracks and military hospital.[13]

NPS map of the W3R Route

After departing from Newport, French troops sent by King Louis XVI and commanded by the Comte de Rochambeau passed through Providence on their way to join the attack against British forces. The march from Newport to Providence was the beginning of a campaign led jointly by General George Washington in a decisive march that ended with the defeat of General Cornwallis in the Siege of Yorktown at Yorktown, Virginia and the Battle of the Chesapeake.

Following the war, Providence was the country's ninth-largest city.[b][13] with 7,614 people. The economy shifted from maritime endeavors to manufacturing, particularly machinery, tools, silverware, jewelry and textiles. By the turn of the twentieth century, Providence boasted some of the largest manufacturing plants in the country, including Brown & Sharpe

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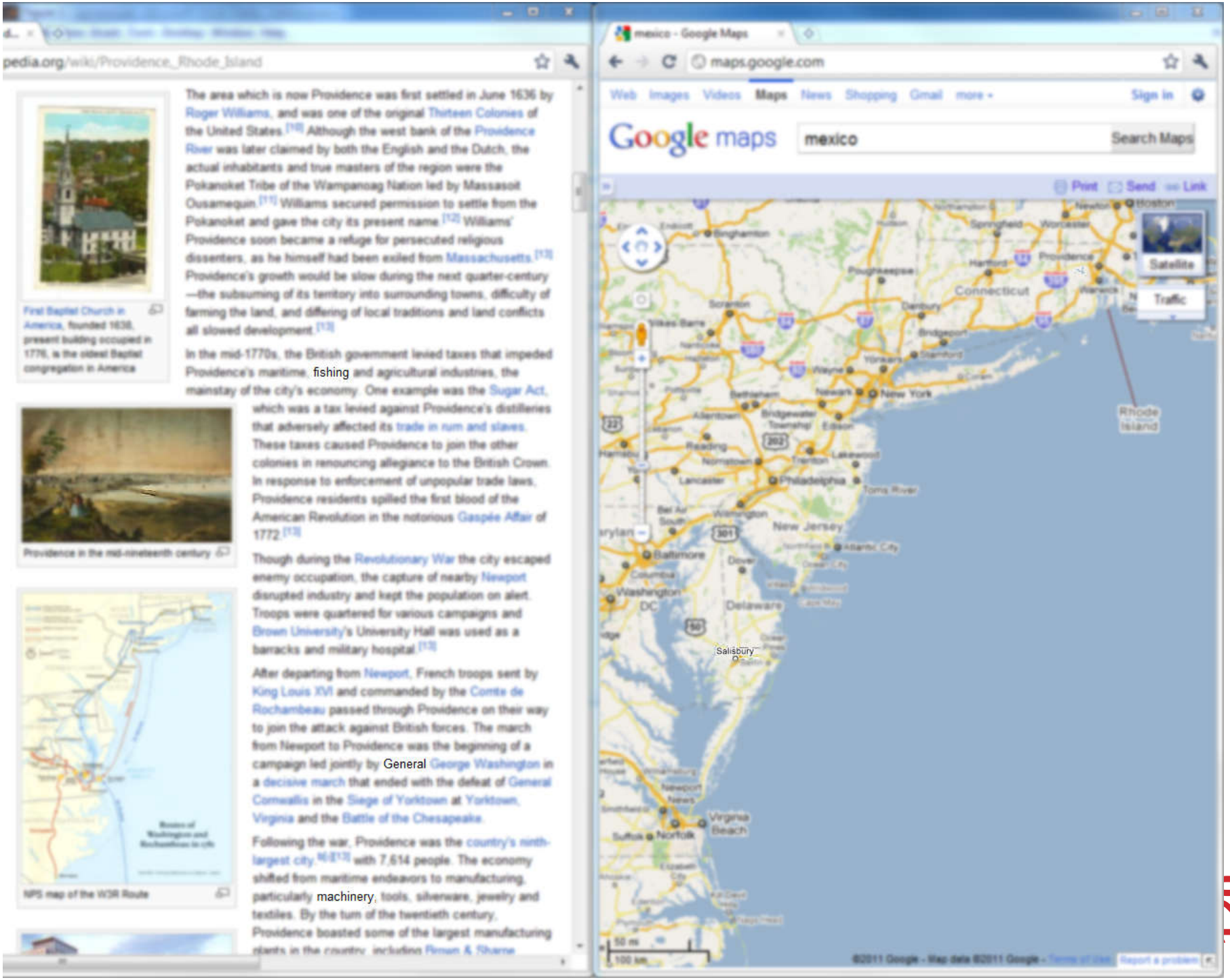
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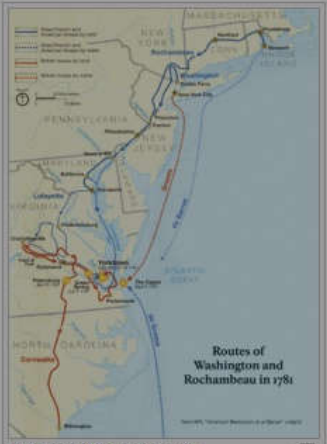

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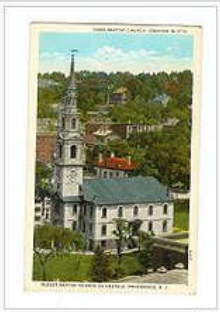

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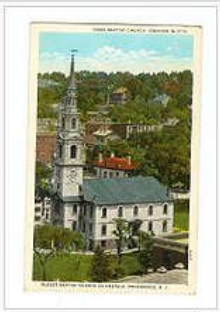

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TAKEAWAYS

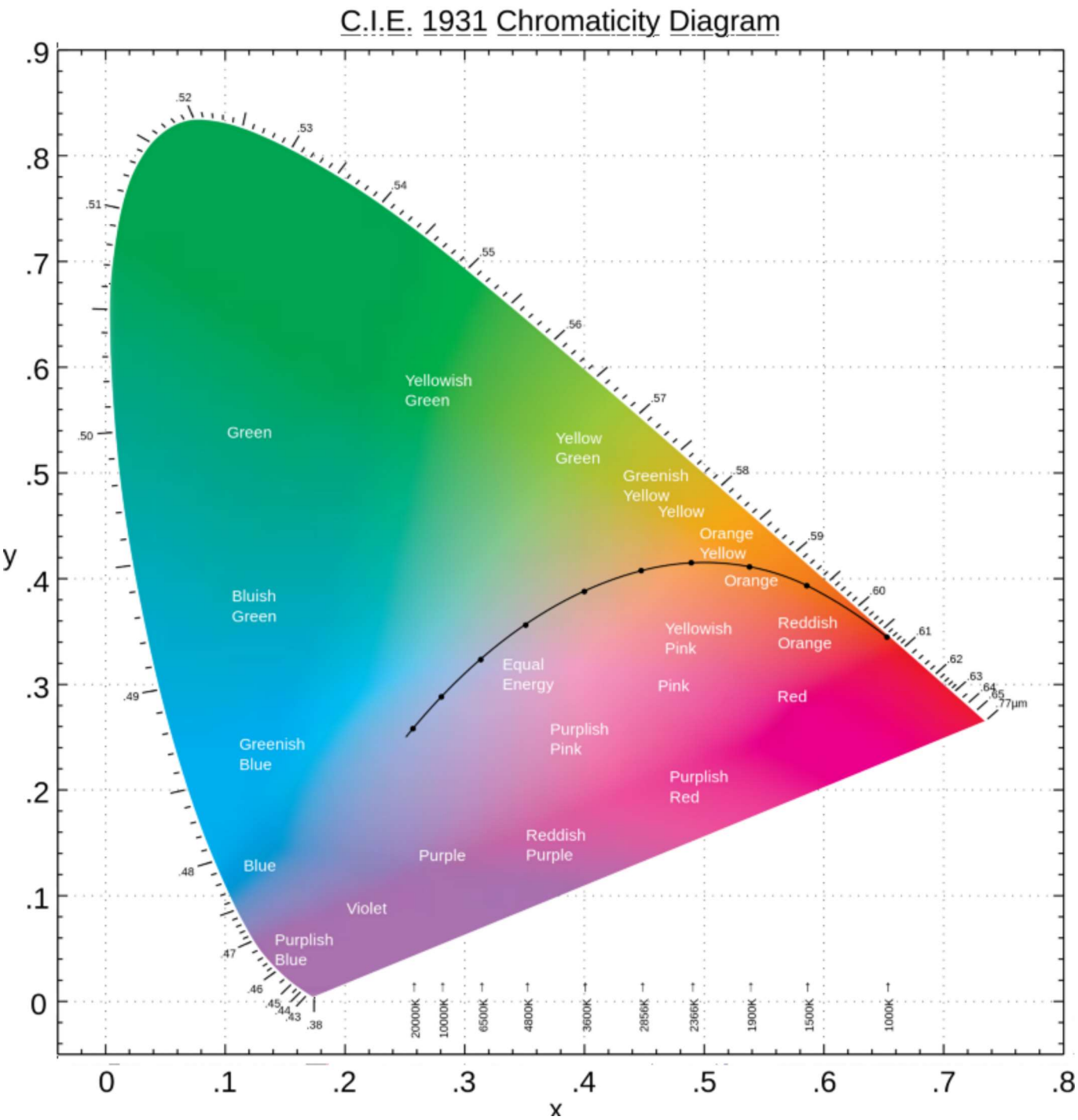
Knowing about Perception is important for Vis and UI design

- How to choose your colors
- How to show relationships
- How to draw attention
- How to minimize risk of overlooking

A Beginner's Guide to (CIE) Colorimetry

Chandler Abraham Follow

Sep 11, 2016 · 18 min read



Edit 2017: This was the first thing I ever wrote about color. I'm older and wiser and I wanted to issue a few disclaimers. Understanding the foundation of CIE colorimetry is very practical but here are some things to remember.

- **CIE colorimetry isn't even half the story of color science, it's a tiny piece of the huge puzzle of human perception and engineering that is color science. Don't confuse understanding it with understanding color perception. Colorimetry is essentially linear algebra, human perception is much more complex.**
- **CIE colorimetry isn't the whole story of colorimetry. It's the modern standard but the theoretical work was done by James Maxwell in the 1850s.**
- **Differentiating theoretical colorimetry from the CIE's implementation of it will be a milestone in your understanding. I recommend Jan Koenderink's Color For the Sciences:** https://mitpress.mit.edu/books/color-sciences

by @cba

Feedback welcome: cbabraham@gmail.com

This blog post is an introduction to the CIE 1931 color system, an international standard model of human color vision that is used all around you. Do you see any color prints around you? Are you reading this on a color display? All of that is made possible by CIE 1931.

Commission internationale de l'éclairage (CIE) is a 100 year old organization that creates international standards related to light and color.

CIE 1931 is a Color Matching System. Color matching does not attempt to describe how colors appear to humans, color matching tells us how to numerically specify a measured color, and then later accurately reproduce that measured color (e.g. in print or digital displays).

To reiterate that point, Color Matching Systems are not focused on describing color with qualities like hue or saturation, they just tell us what combinations of light appear to be the same color to most people (they "match"). Color matching allows gives us a basic framework for color reproduction.

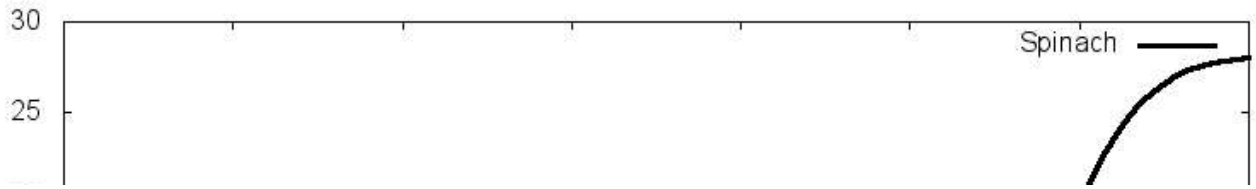
You can think of CIE 1931 like you would Newtonian physics. It's a mathematical generalization of human color vision, that allows us to define, and accurately reproduce colors in most situations. It does not fully describe the subjective and complicated process of human color vision, it's not without edge cases, but it is relatively easy to work with.

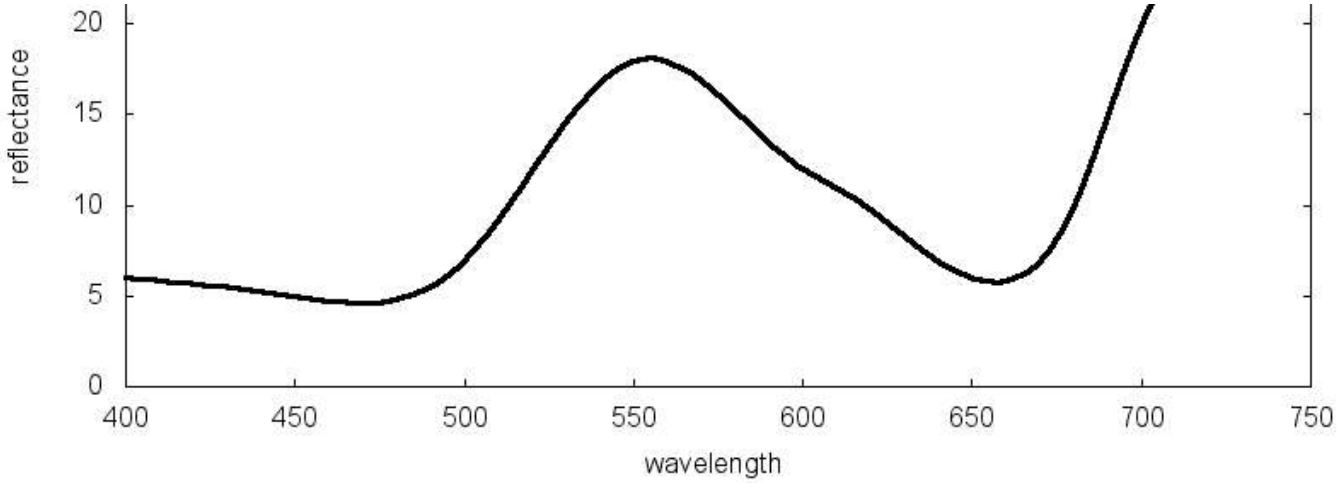
1. Trichromatic Color Theory
2. 1924 Luminous Efficiency Function, V(λ)
3. 1931 RGB Color Matching Functions
4. Wright Guild Color Matching Experiments
5. RGB Chromaticity Coordinates
6. Non-Spectral Colors
7. CIE XYZ Color Space
8. Real World Color Spaces
9. Beyond Color Matching
10. Acknowledgements, Further Reading

1) Trichromatic Color Theory

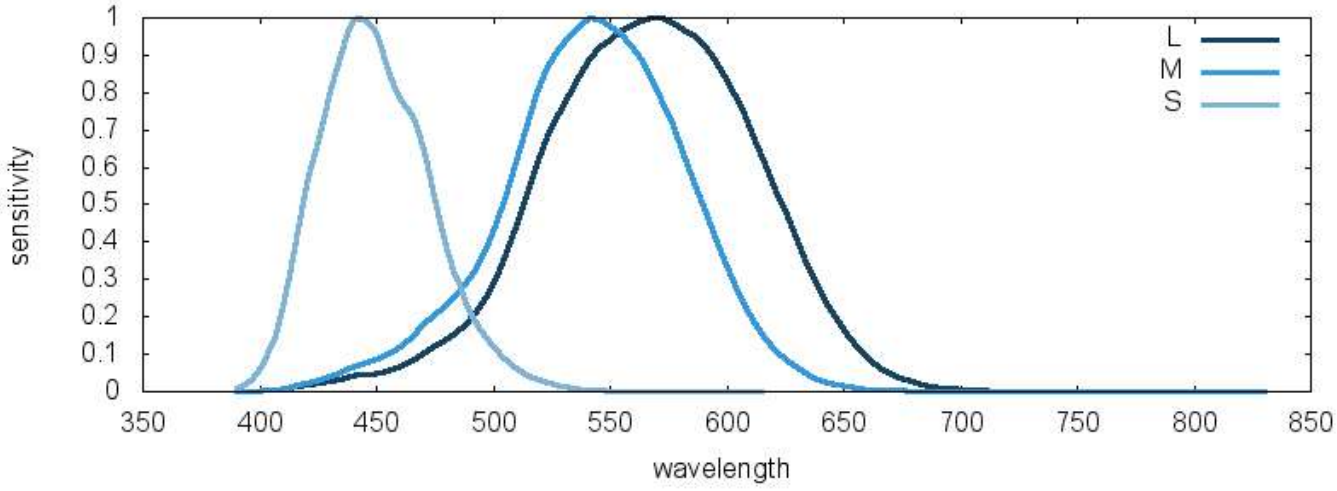
A principal quality of light is its wavelength. Humans can perceive light with waves as small as ~380nm and as large as ~750nm.

The sun emits light across many wavelengths and objects around us reflect some wavelengths and absorb some wavelengths. Below is the spectral distribution of a spinach leaf illuminated by sunlight. The leaf absorbs most wavelengths as heat, but reflects visible light around 550nm. We see this particular reflected spectral composition as green.

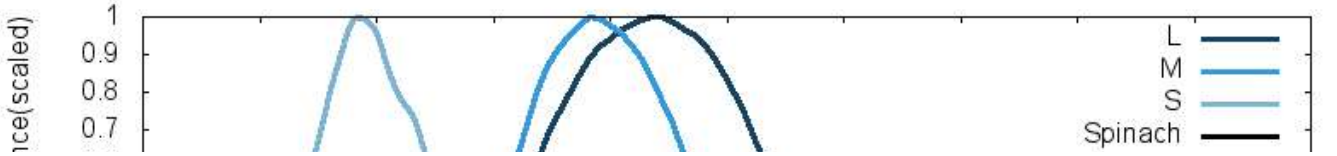


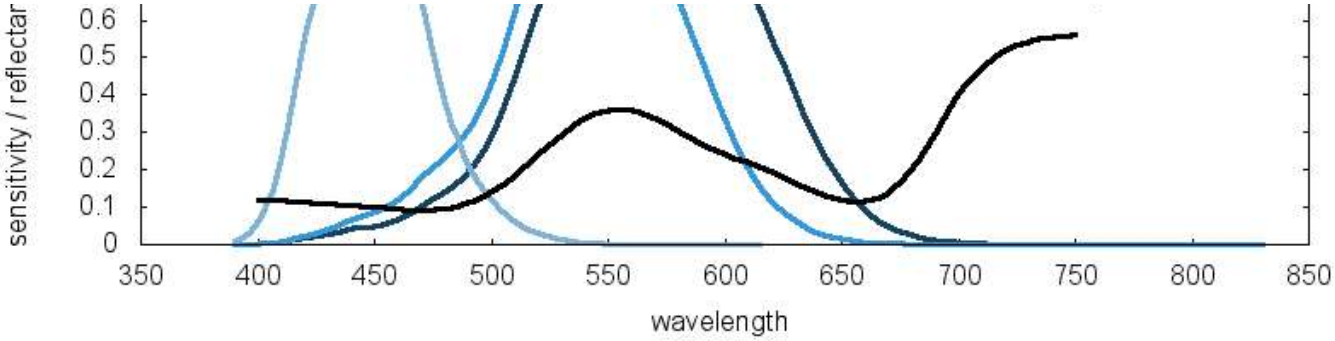


Inside the eye there are three types of cone photoreceptors called Long, Medium and Short that contribute to color discrimination. They are all sensitive to different, yet overlapping, wavelengths of light. They are commonly associated with the color they are most sensitive too, L = red, M = green, S = blue.



When you view the charts overlaid, you can see that the spinach mostly reflects light outside of the eye's visual range, and inside our range it mostly reflects light centered around our M cone.

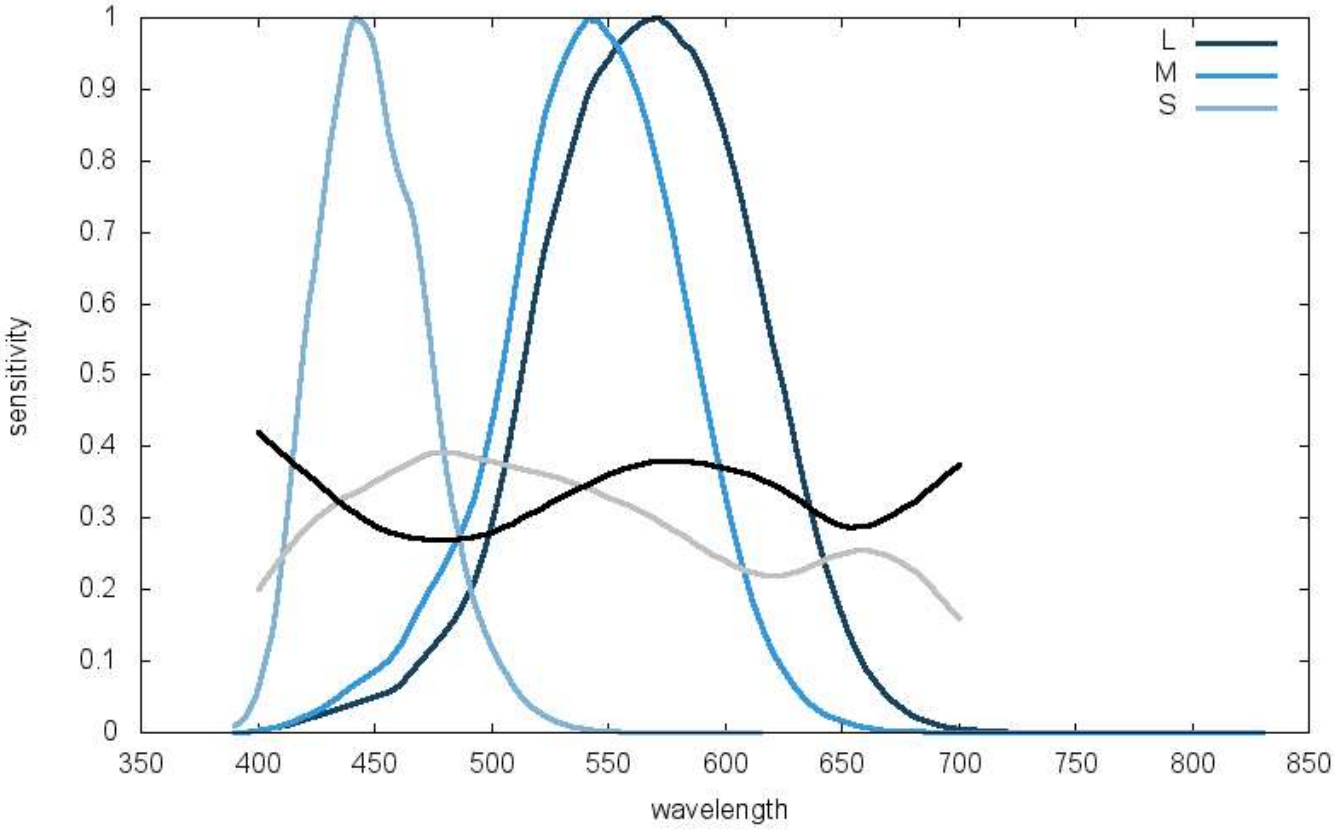




The eye's cone system takes a complex spectral distribution (like the spinach) and reduces it down to three numerical values, each representing how much the three cones were stimulated. These cone values are fed into the next part of the brain's visual processing pipeline, which we won't go into (see: Opponent Color Theory).

This is important: The trichromatic process is lossy, you can't go from the output of the cones back to the original spectral distribution. **This is because different spectral distributions can stimulate the cones in the exact same way.**

Shown here are two different spectral distributions that are indistinguishable to the human eye. You can see that the M cone is stimulated by both distributions equally, just on opposite ends of the cone's sensitivity range.



You could imagine a leaf and a green car that look the same to you, but physically have different reflectance properties. It turns out every color (or, unique cone output) can be created from many different spectral distributions. Color science started to make a lot more sense when I finally understood this.

This phenomenon is called metamerism and it has huge ramifications for color reproduction. It means we don't need the original light to reproduce an observed color. We can reproduce a color if we can create a spectral distribution of light with the same cone response as the original distribution.

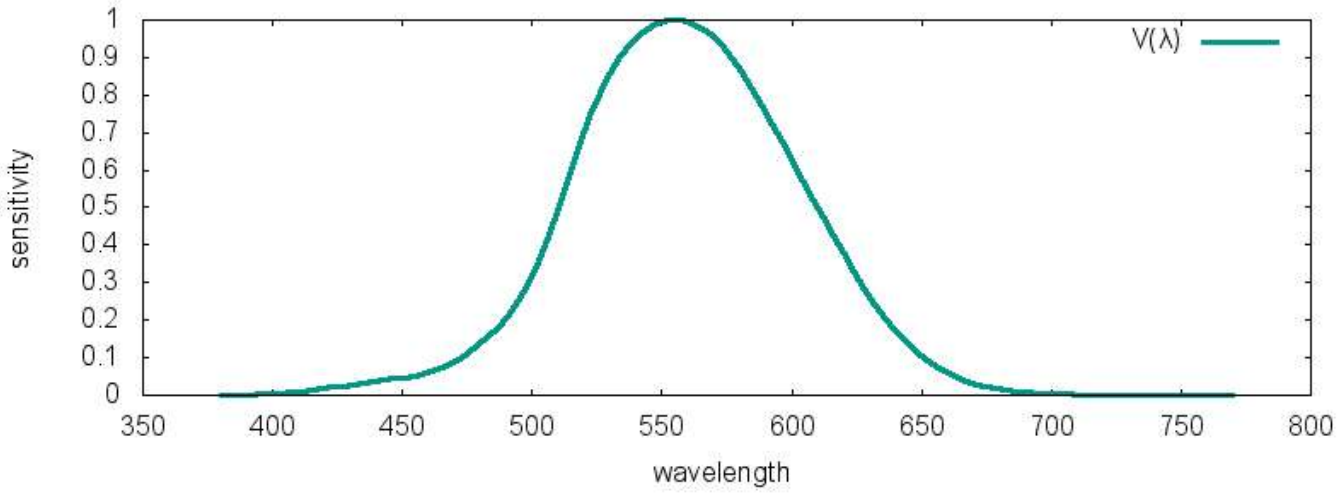
This is why you can create a color that looks like spectral yellow on a LCD display that has no yellow lights. The LCD can combine red and green light in the right amounts to mimic the human cone response of spectral yellow light.

CIE 1931 is a model that tell us how to create these matches. Excited yet?

2) 1924 Luminous Efficiency Function, V(λ)

> tip: λ means wavelength

Before the fundamental work of colorimetry occurred CIE published a function, V(λ), that describes the human eye's sensitivity to light at different wavelengths in daylight.



What this tell us is that two lights, 550nm and 400nm can have the same radiance (an objective unit) but appear to have different brightnesses.

Or to put it another way, if you have a green and blue light that appear to be equally bright, then you know that the blue light is more luminous.

Note: The 1924 V(λ) was eventually shown to under represent our sensitivity at the blue end of the spectrum.

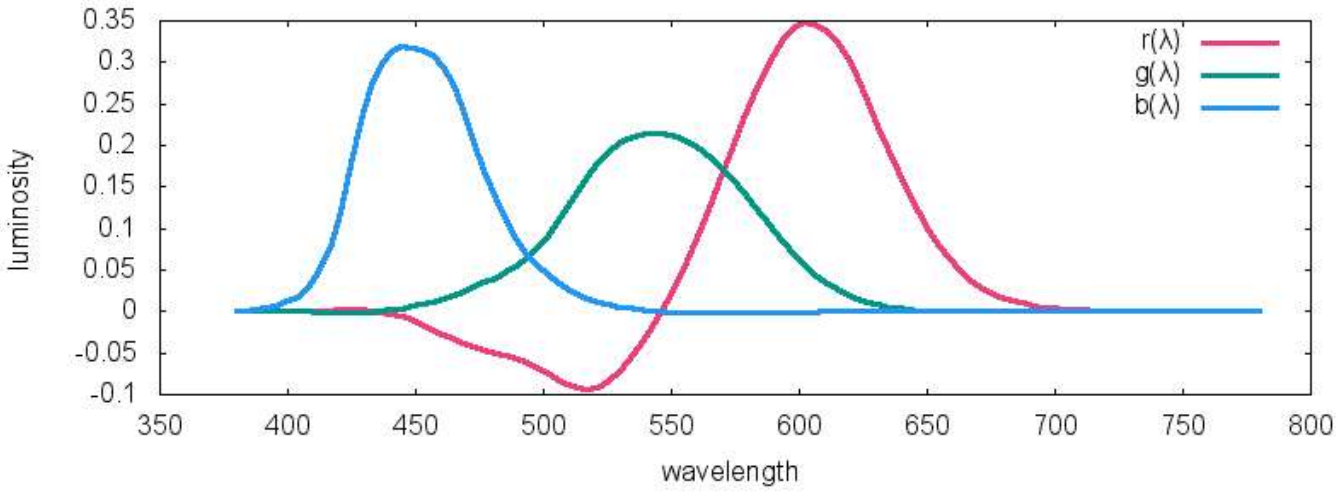
3) 1931 RGB Color Matching Functions

CIE 1931 contains 3 functions called the RGB color matching functions.

Let's say you have three lights, a red and green and a blue with precisely known single wavelengths. If you point these lights at the same spot on the wall is it possible to adjust the power of each light until the spot had the exactly same color as 540nm light?

This is exactly the kind of question color matching functions can answer. Color Matching functions have three fixed primary colors and each function outputs the amount of primary needed to create a desired color when all three are mixed.

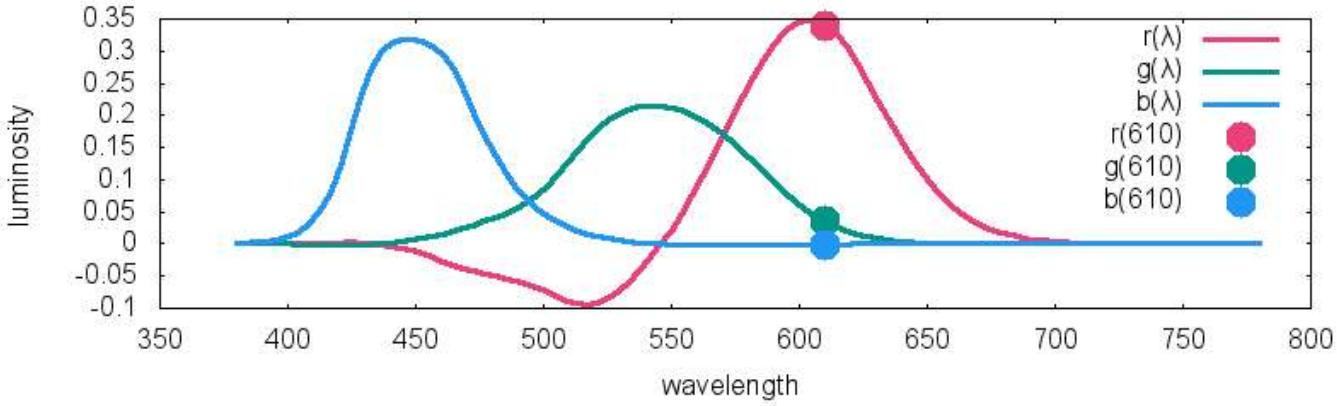
700nm, 546.1nm and 435.8nm are the primaries of the 1931 RGB color matching functions: r(λ), g(λ), b(λ). Plotted below.



Let's use this to make some matches in our hypothetical light setup. Note, this scenario is going to seem very specific and contrived but later we will show how it can be generalized to explain all color matches.

We'll pick a target wavelength of 610nm, the goal is to reproduce the color of this light with our primary lights.

```
λ = 610nm r(λ) = 0.34756 g(λ) = 0.04776 b(λ) = -0.00038
```

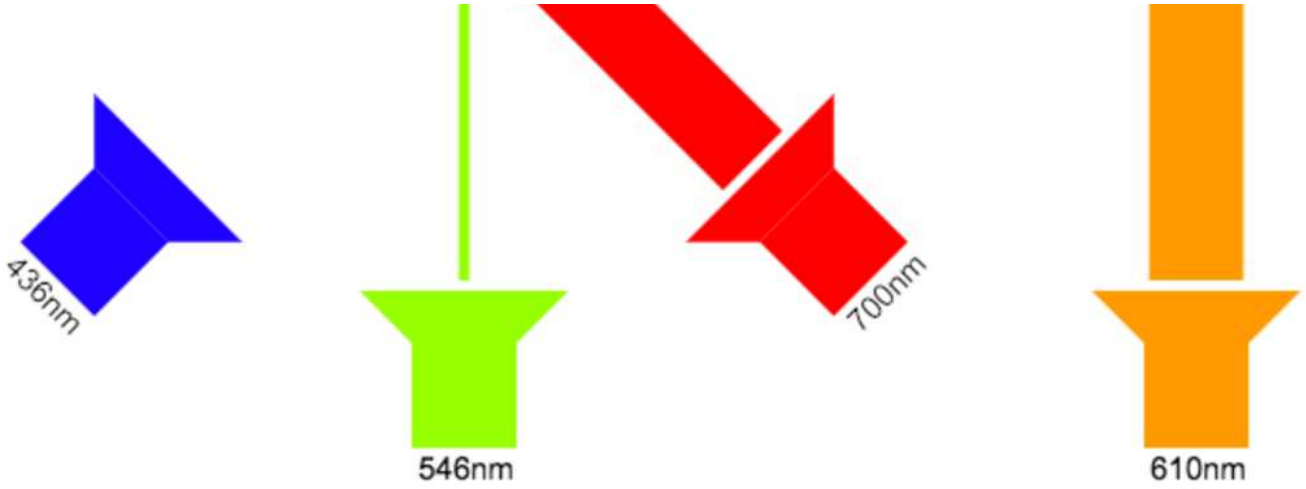


One quirk about these functions, they are scaled as if all the primary lights are equally bright. As mentioned earlier, a blue light must be more luminous than an equally bright green light. So in order to get absolute luminance needed for our match, we must un-scale the values using a set of relative luminance values provided to us along with the function.

```
Lr = 1 # r luminance scale
Lg = 4.5907 # g luminance scale
Lb = 0.0601 # b luminance scale
λ = 610nm
R = r(λ)/Lr = 0.34756
G = g(λ)/Lg = 0.04776
B = b(λ)/Lb = -0.00038
```

The resulting values R, G, and B are called **tristimulus values**, they are the required luminosity of the test lights to achieve a color match.

Instead of subjecting you to the colorimetric equations that describes this match, here's a diagram of the scenario.



Since blue is negative, we leave the light off because negative light isn't possible. What this means is that we actually can't create a perfect match of 610nm light with these primaries, but by using the positive amounts of red and green we can create something very close. Negative values are explored more in the next section.

The exercise we've described is not a day-to-day use of color matching functions, but as we will see arranging these lights is similar to how you might vary the intensity of rgb sub-pixels to create an LCD pixel of a desired color.

4) Wright Guild Color Matching Experiments

I used the lighting arrangement in the previous section as a demonstration of a physical application of the color matching functions, but it actually has its origins in the creation of those functions.

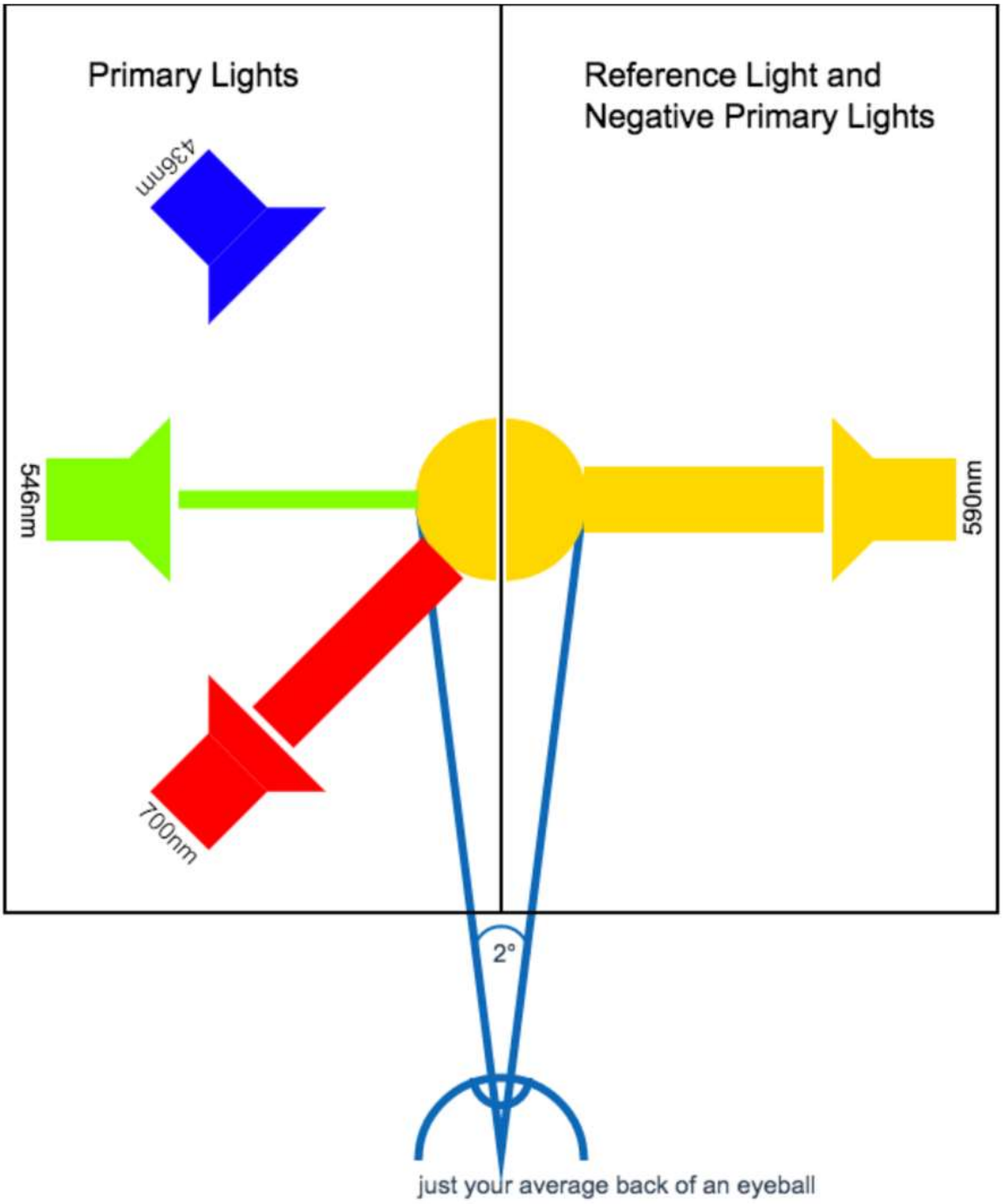
In the 1920s two color scientists, W. D. Wright and J. Guild, each performed similar color vision experiments. Wright performed his experiment on 10 subjects, Guild used 7. Their results agreed with each other so well that they were combined by CIE to create the RGB color matching functions we've been discussing.

Color vision can sometimes vary, so it was important to find test subjects with no color vision deficiencies or yellowing of the lens (happens with old age) in order to create functions that worked for the average person. The combination of the Wright Guild subject data is known as the 1931 Standard Observer.

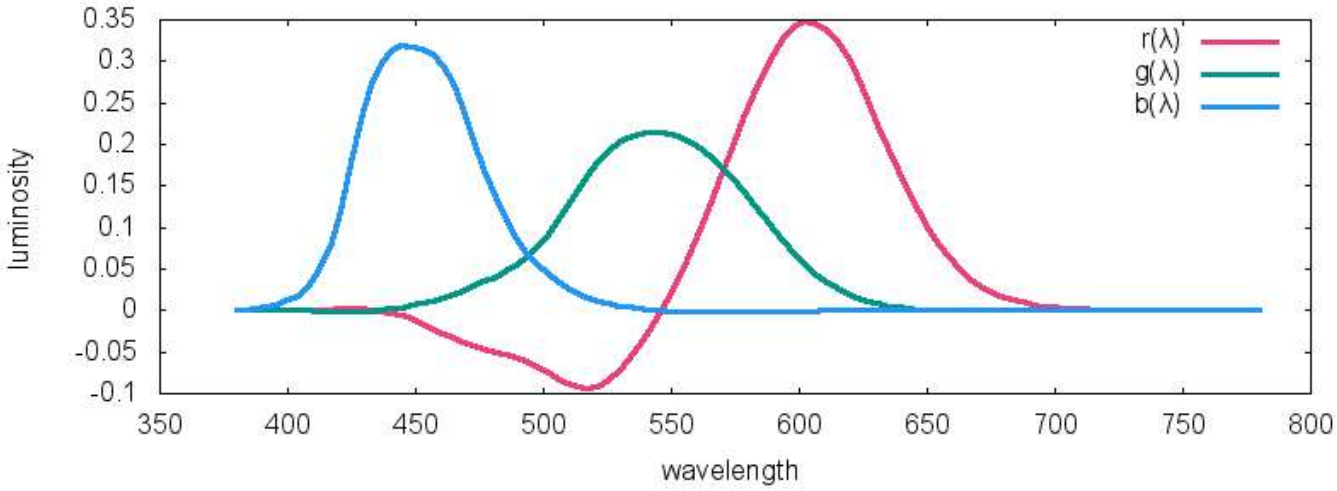
The experiment setup asks a subject to adjust the values of 3 primary lights until they produce a color indistinguishable from a reference light. This is repeated using a

reference light for every visible wavelength (at some increment). When this is done you have the values of your color matching functions.

Here's an example. On the right is a monochromatic (single wavelength) yellow, on the left the test subject has adjusted the primary lights so that their mixture is identical to the monochromatic light.



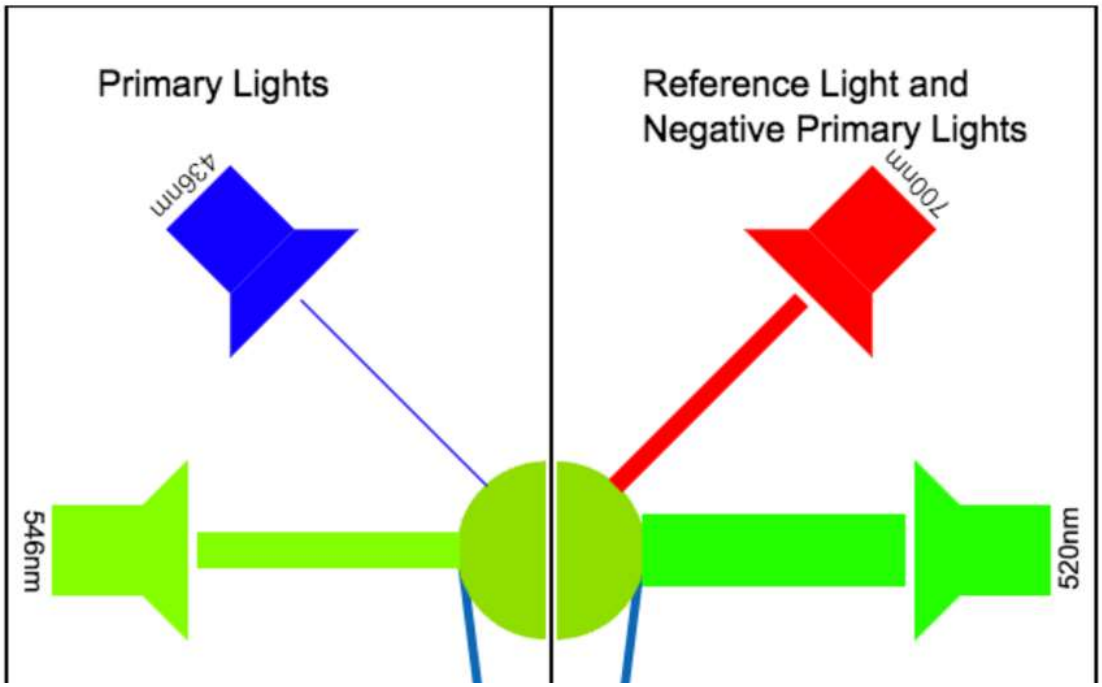
About those negative values

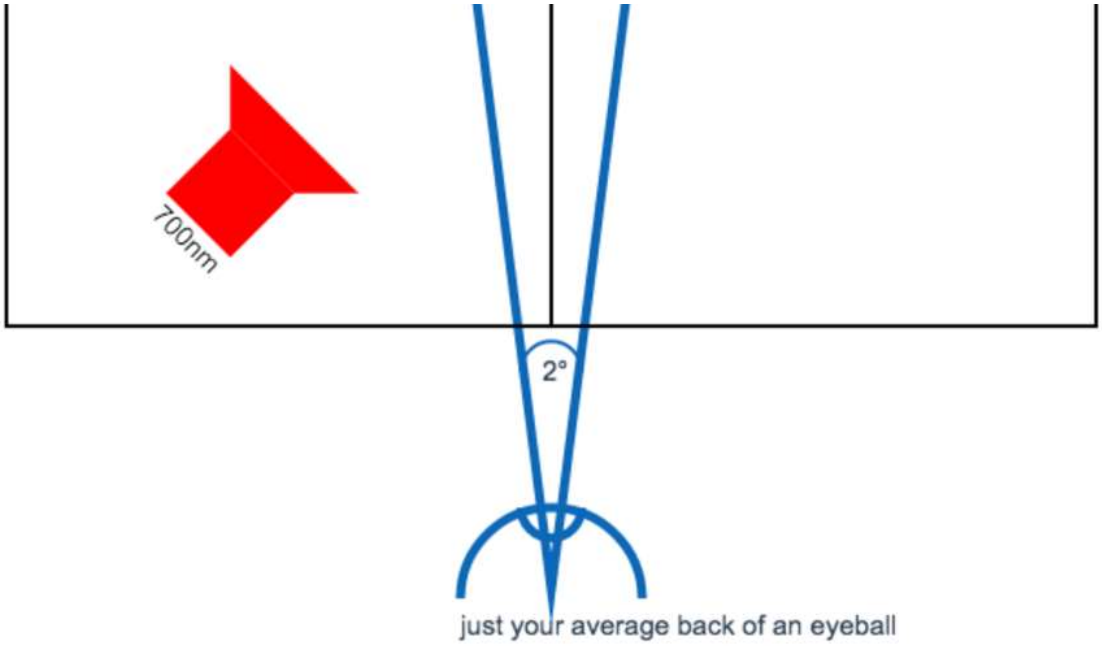


You might have noticed in the color matching functions that sometimes they call for a negative amount of light to achieve a match, such as 520nm. What this means is that the test subject was unable to achieve a match using positive values of the primary lights. To address this, some of the primary lights were mixed in on the opposite side of the screen with the reference light, until a color match was able to be made.

As it turns out, no 3 primaries can create every spectral color. By using the negative light trick researchers were able to quantify a color match in spectral ranges not matchable by their chosen primaries.

520nm is an example of a bright green that wasn't achievable with the test primaries unless a negative amount of red was used.





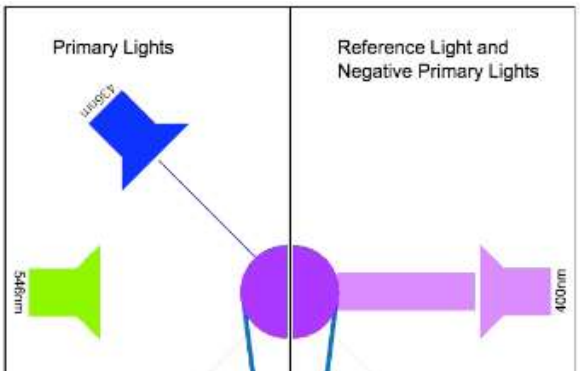
Restrictions on viewing conditions

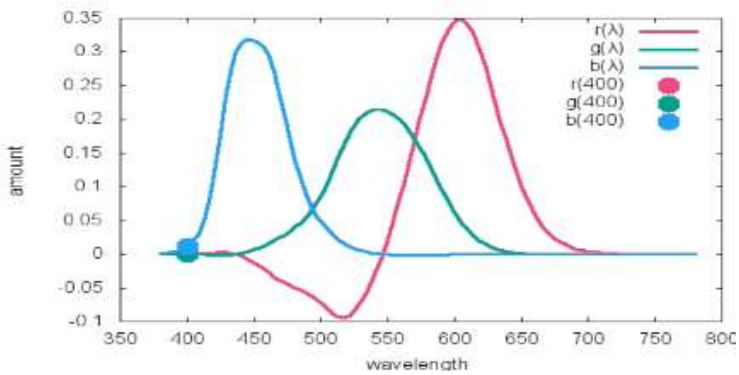
If you hold out your thumb in front of you, it takes up a 2° field of view. If you make a fist, that's 10°.

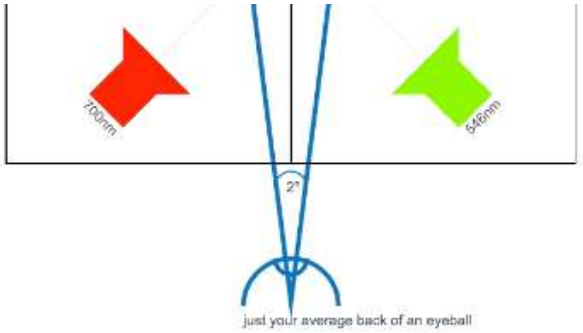
The 1931 Standard Observer is only valid for colors viewed at a 2° field of view. This places the light on a spot on the back of your eye called the Fovea. This is a spot with high cone density, giving you maximum color discrimination and limited rod interference.

Color researchers realized that a 10° color matching function would be more representative of day to day color perception so color matching experiments were repeated at 10° and published as the 1964 10° Supplementary Standard Observer.

This might be more fun than useful but here is an animation cycling through all of the configurations of the 1931 2° tests.







5) RGB Chromaticity Coordinates

Colors specified as three tristimulus values, like in the RGB, are difficult to visualize. It's not easy to imagine what a color looks like at a specific coordinate inside a three dimensional space.

One way we can simplify this problem is to remove the intensity dimension from the data, so that dark red and light red are the same value. This concept of color minus intensity is called Chromaticity.

This is what's happening when you adjust the brightness on your monitor, you are changing the colors of the pixels but the chromaticity of pixel remains the same. This shows why we don't always care about the original intensity information, and why Chromaticity is a useful concept.

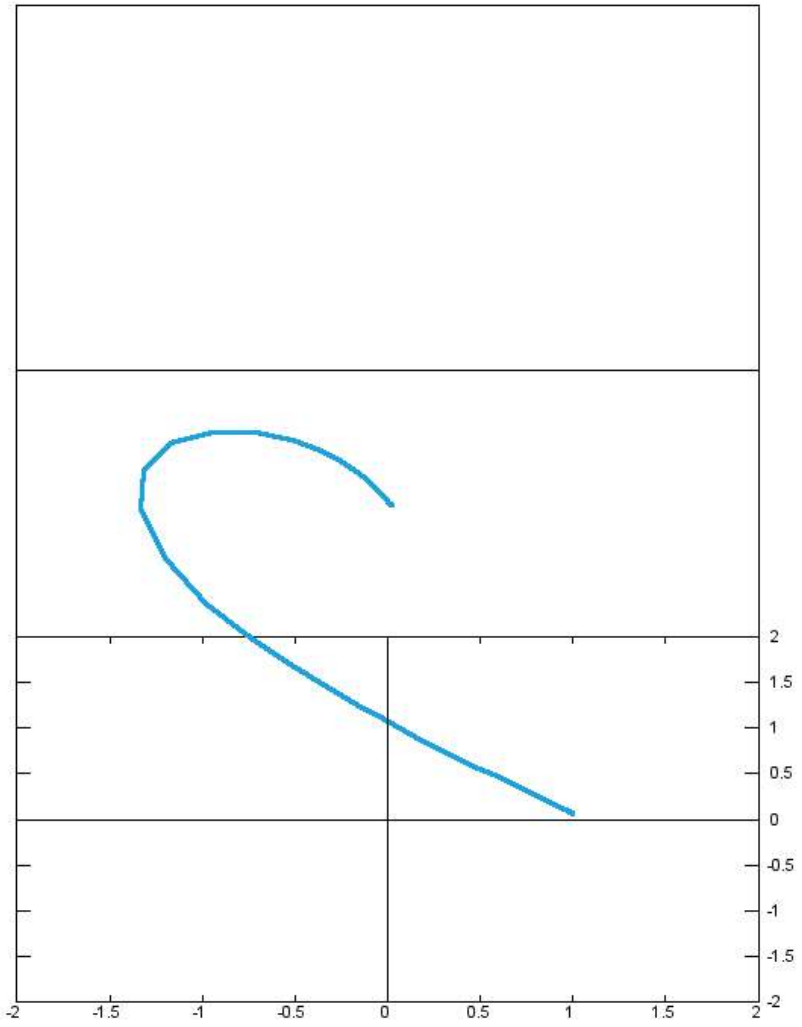
This is the equation to convert tristimulus values into chromaticity coordinates.

```
RGB are tristimulus values
r = R/(R+G+B)
g = G/(R+G+B)
b = B/(R+G+B)
rgb are chromaticity coordinates and r + g + b = 1
```

rgb is about the ratio, it tells you "To match 400nm light, much more blue is needed than red or green"

RGB is about absolute amounts, it tells you "To match 400nm light, very little of any primary light is needed"

We can plot every value of rgb as a point in three dimensions. This gives us a curve that represents the chromaticity of each spectral wavelength of light, also called a spectral locus.



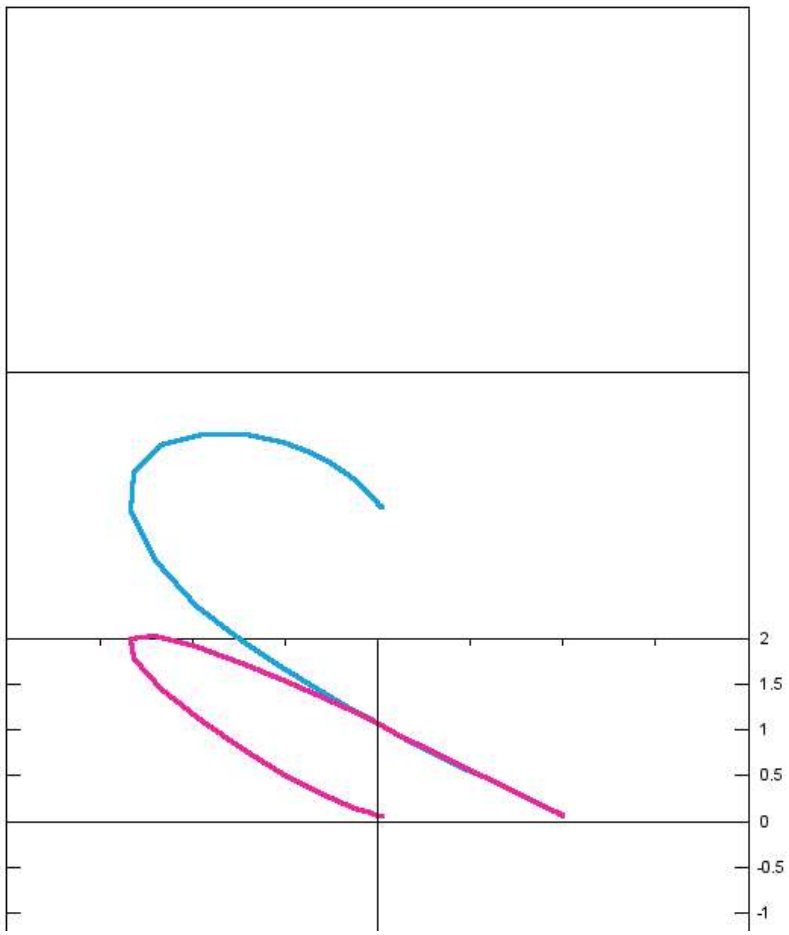
One Weird Trick

Even though we still have three values (rgb), they all sum to one, which means we can always determine the third value from the first two. Like so:

```
r = -1.5 #known
g = 2 #known
b = ? #unknown

if r + g + b must equal 1

then b = 1 - g - r

so b must be 0.5
```

By removing intensity information from the data (in the conversion from RGB to rgb), we have actually converted a three dimensional space into two dimensions, which we can take advantage of for simplified visualizations.

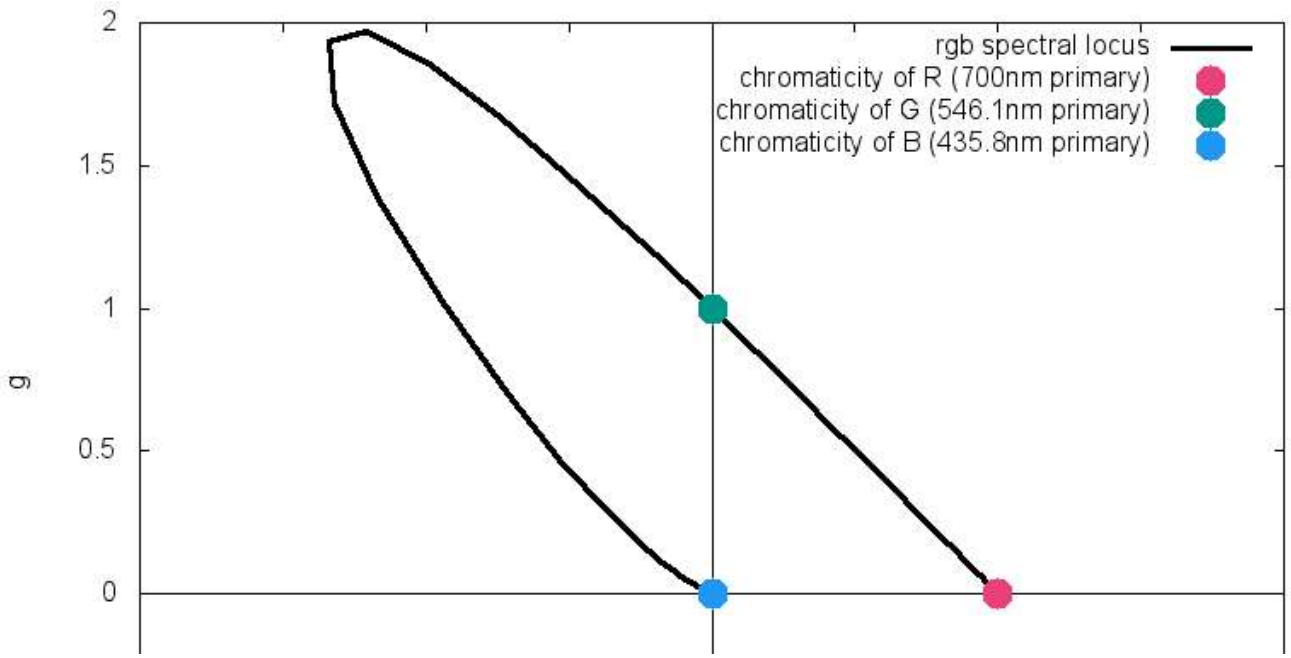
Since we can safely drop a dimension without losing information, let's convert the b axis to all zeroes and just plot rg in two dimensions. This is called a projection to the rg plane.

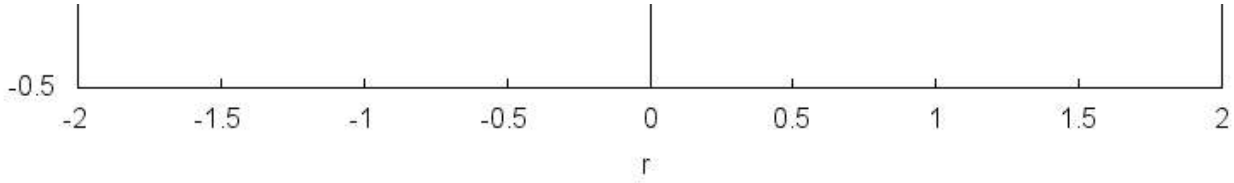


Dropping the 3D plot we end up with our 2D projection of the rg chromaticity space, our first chromaticity diagram!

Let's break down what we're looking at

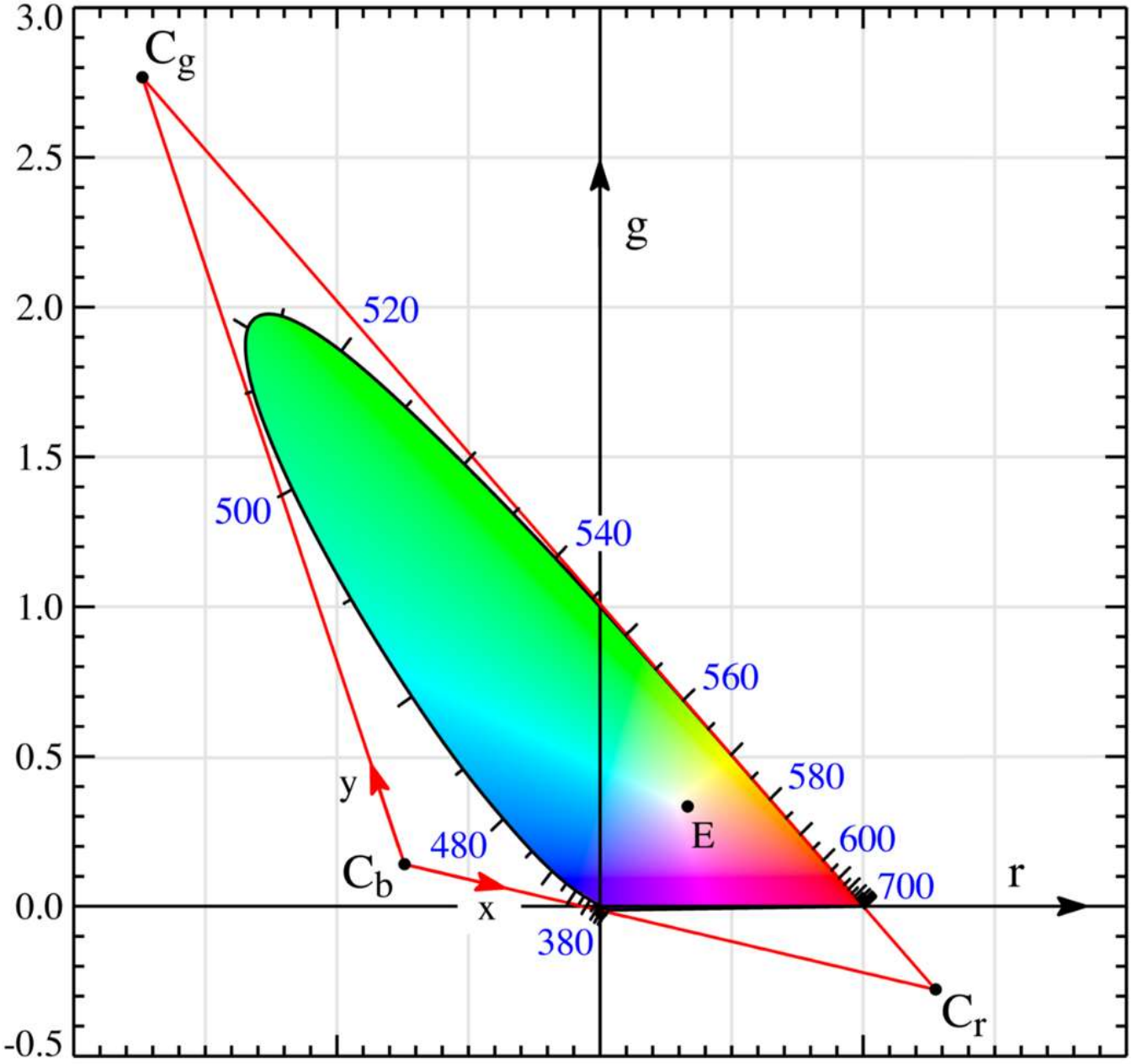
- Every point on the outer curve is the chromaticity coordinate of a spectral color.
- Every point inside the curve is a chromaticity of a non-spectral color.
- Every point outside the curve is an imaginary chromaticity that is meaningless and has no realizable color.
- Our original RGB primaries, are spectral colors, so they fall on the curve.
- Every point inside the triangle formed by the primaries is a chromaticity that can be created with those lights.
- Points inside the curve but not the triangle are real chromaticities but you would need different primary lights to create them.





My plotting abilities are not quite up to the task of rendering a colored chromaticity diagram so here is the rg diagram from wikipedia, ignore the red Cr, Cg and Cb triangle for now. When looking at chromaticity renderings remember that real colors have luminosity, chromaticities do not.

This diagram shows colors that have the appropriate chromaticities rendered with a visually interesting luminosity.



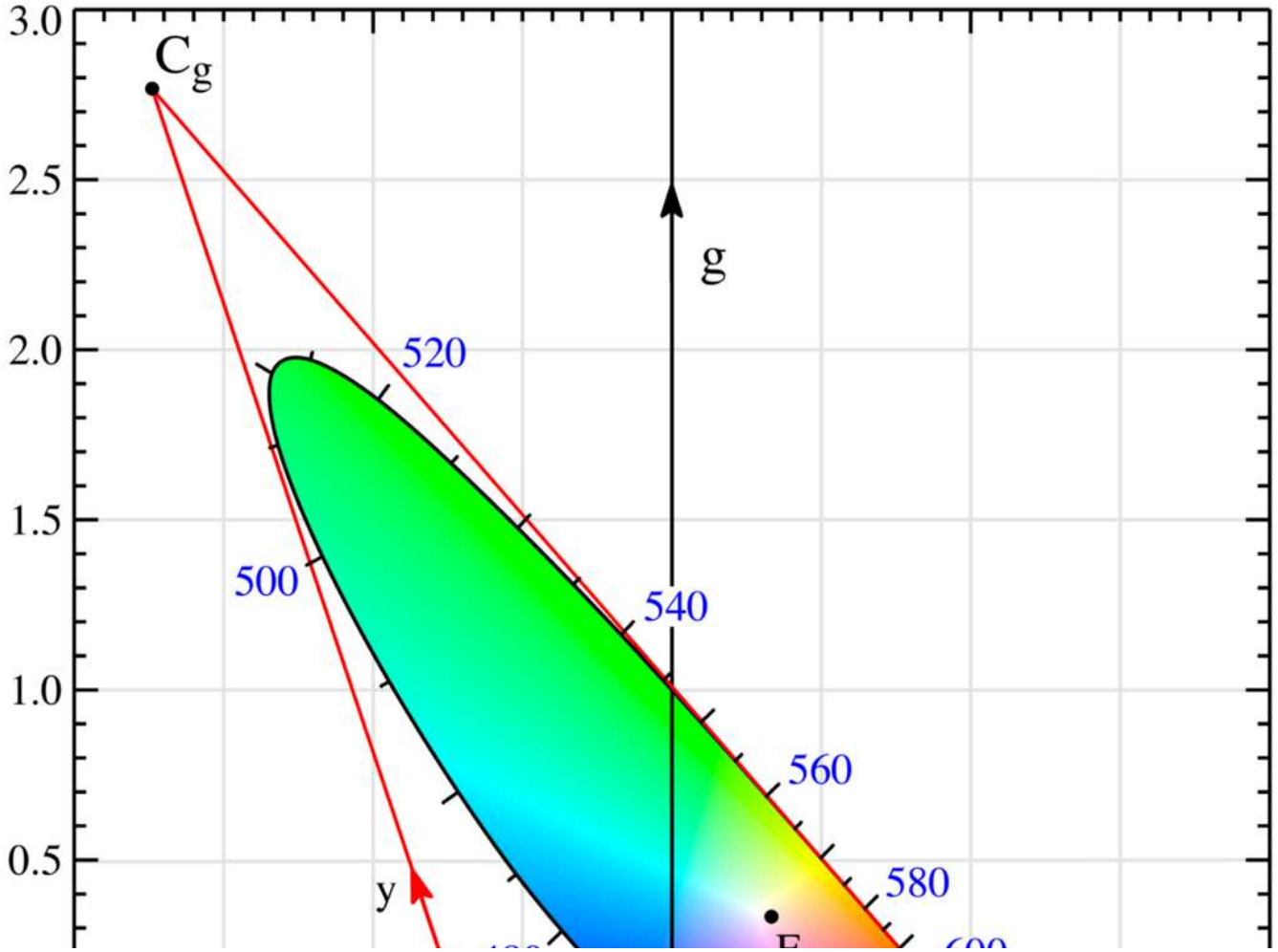
2.0 1.0 0.0 1.0 2.0

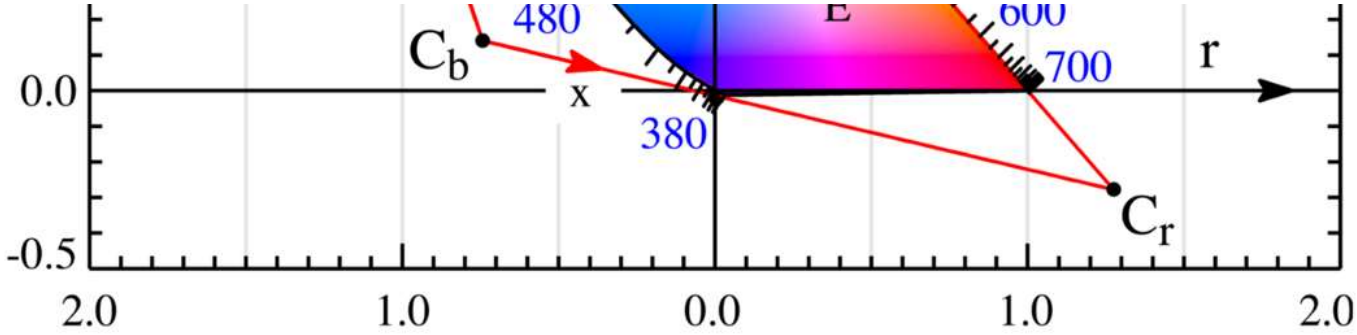
6) Non Spectral Colors

We know the chromaticities of the spectrum, so it makes sense that we could color the boundary of the curve, but what does it mean to color the inside of the curve?

The space inside the curve turns out to represent non spectral chromaticities, chromaticities that only can be created from mixtures of light. If you draw a line between any two points on the curve, the midpoint is the chromaticities of mixing those two spectral chromaticities.

The spectral colors are featured prominently in colorimetry as the spectral locus but you almost never run into these colors them in the real world. Narrow band spectral colors can be generated with lasers but the colors of every day objects are wide broadband spectral power distributions that well inside the boundary of the chromaticity diagram, like the spinach leaf at the start of this post.





7) CIE XYZ Color Space

If you're still following along, I commend you. We're in the home stretch now.

The 1931 XYZ Color Matching Functions are a linear transformation of the 1931 RGB Color Matching Functions in order to give them some mathematically convenient properties. In fact, the RGB CMFs we've been discussing are mostly unused in practical colorimetry. Modern color spaces are defined in terms of XYZ.

Keep in mind that XYZ and RGB are different transformations of the same data, anything you can do with one you can do with the other, and colors can be converted back and forth between them.

The big idea of XYZ is that one of the three functions could be transformed so that it aligned very closely to 1924 Luminosity Function, V(λ). This would mean that luminosity of a color could be determined entirely from looking at one of that color's primary values. Additionally, before computers the manual math involved in using the CMFs was difficult and there was a desire to have CMFs that didn't contain any negative values.

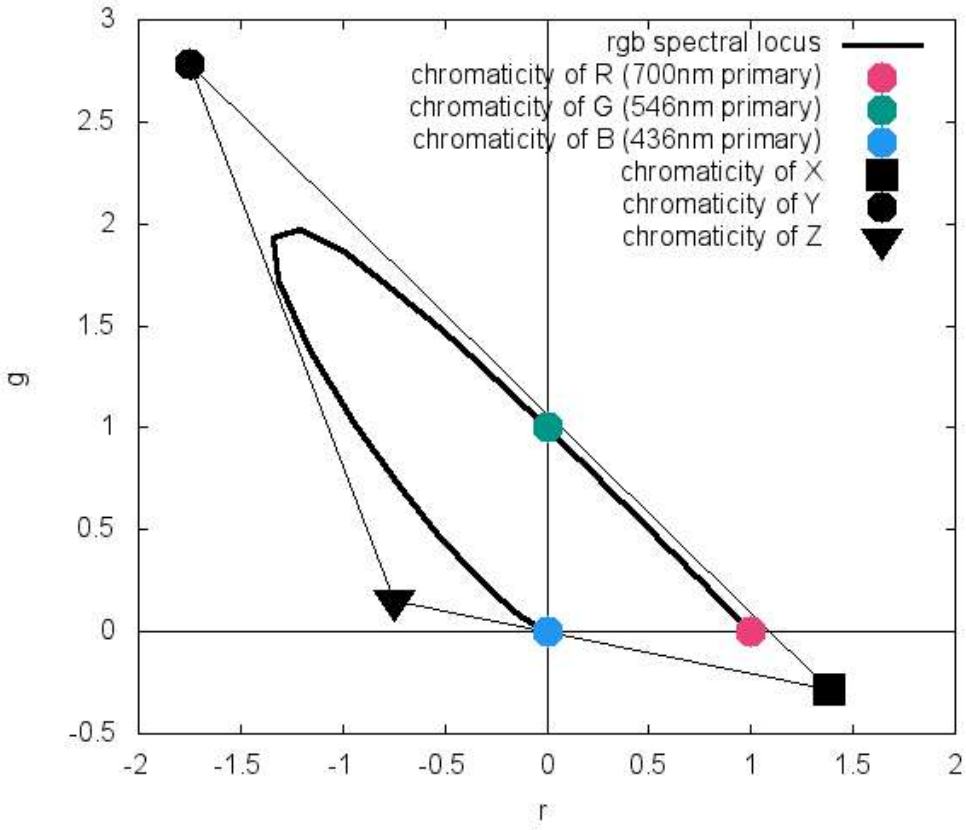
There is a slight problem however, in order to create an all positive transformation of the RGB CMFs you have to use imaginary primaries that don't correspond to any real color.

You can see this in the rg chromaticity diagram from the previous section, the only way to form a triangle around all of the real colors is to choose primary coordinates outside of the real colors. This is exactly what XYZ does.

Three new primaries, X Y and Z were picked such that when the RGB data was transformed to have XYZ as the new axes the Y primary mapped to the V curve and all values were positive.

Let's check it out.

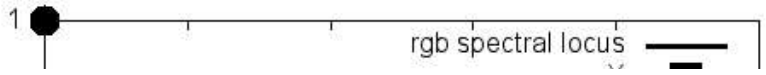
Before, in rg chromaticity space:

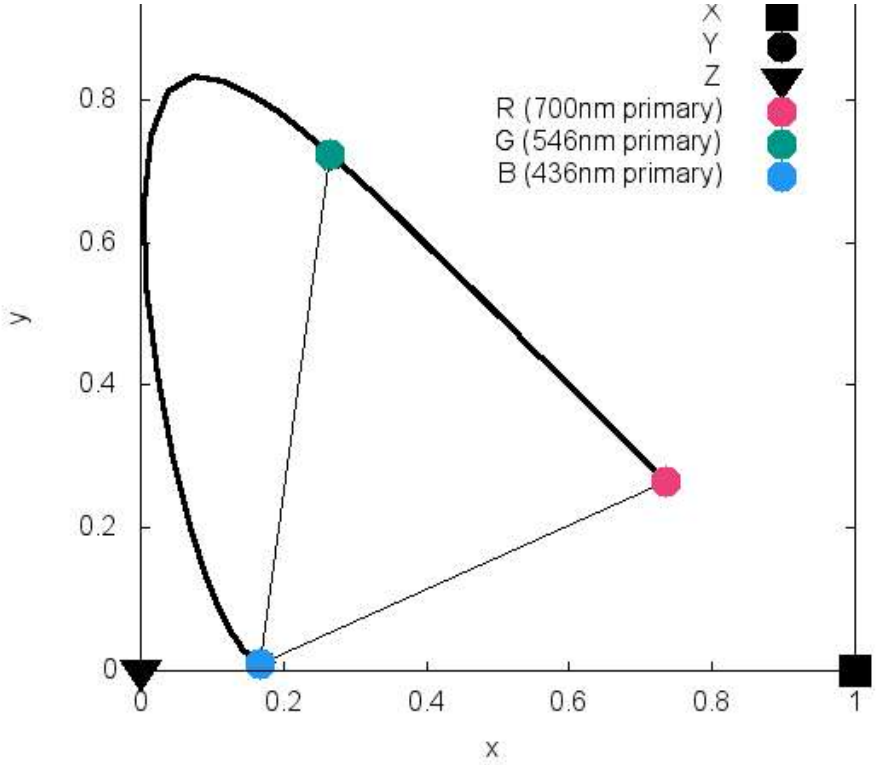



```
Matrix coefficents that convert the RGB space into one where the
carefully chosen XYZ primaries exist at coordinates (1,0) (0,1) and
(0,0) in the new chromaticity space.

[X]   | 2.768 1.751 1.130|   [R]
[Y] = | 1.000 4.590 0.060| * [G]
[Z]   | 0     0.056 5.594|   [B]


source: Colorimetry by Janos Schanda, page 30
```

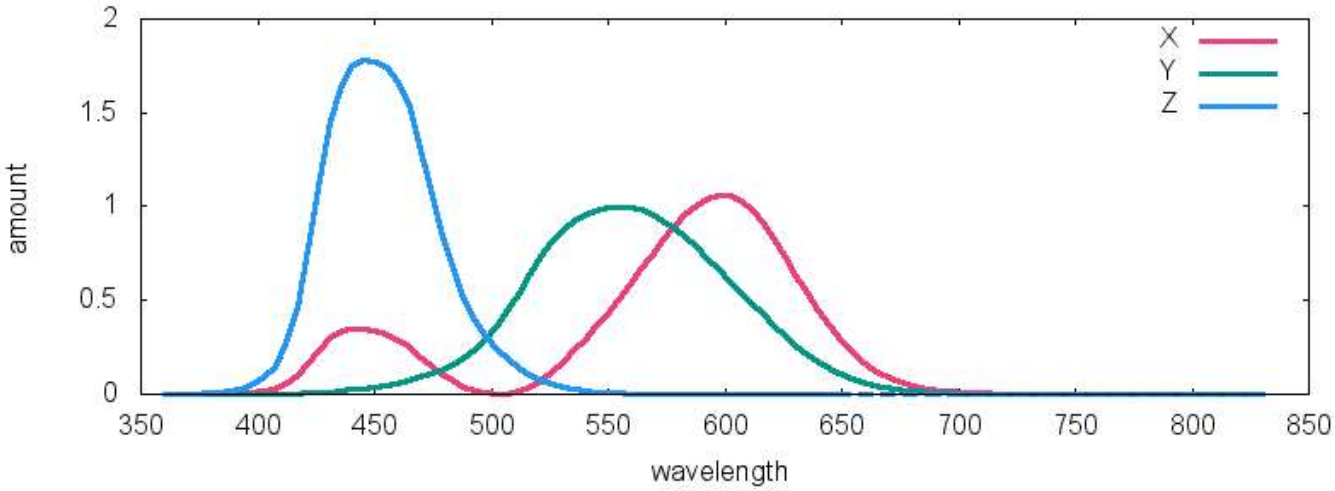
~ waves linear algebra wand ~

After, in xy chromaticity space. Look, all positive values!

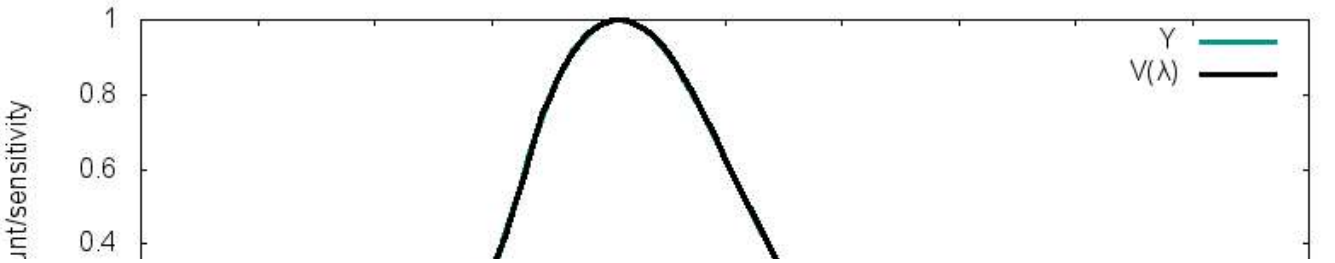


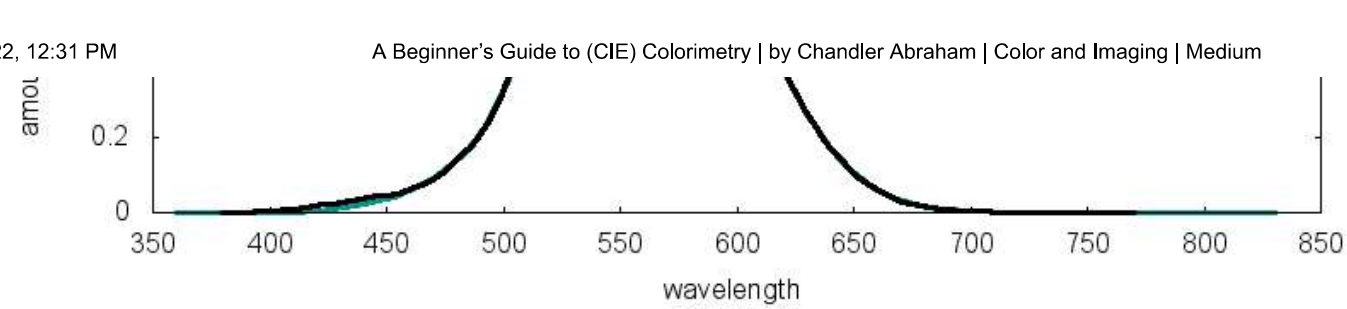


Similarly, our old friend the RGB Color Matching Functions becomes the all positive XYZ Color Matching Functions:



Plotting the Y CMF against the luminosity from earlier shows the near perfect alignment.





The spectral locus still defines the coordinates of every spectral color, only now it's in terms of primaries that don't exist.

Coordinates specified in terms of imaginary colors might not seem very useful, but when it comes time to create physical colors you will basically do another linear transformation to convert the coordinates into your desired color space. If your XYZ color endd up having all positive values in the new color space, and if your new color space has non-imaginary primaries, then you have a recipe for real color creation. This is the subject of this 1996 blogpost by the founder of Autodesk, John Walker

There are probably thousands of color spaces specific to individual digital displays, but each one hypothetically only needs to know two. They need to know about XYZ and they need to know how to convert XYZ into their supported color space. XYZ is the common language used to communicate color.

8) Real World Color Spaces

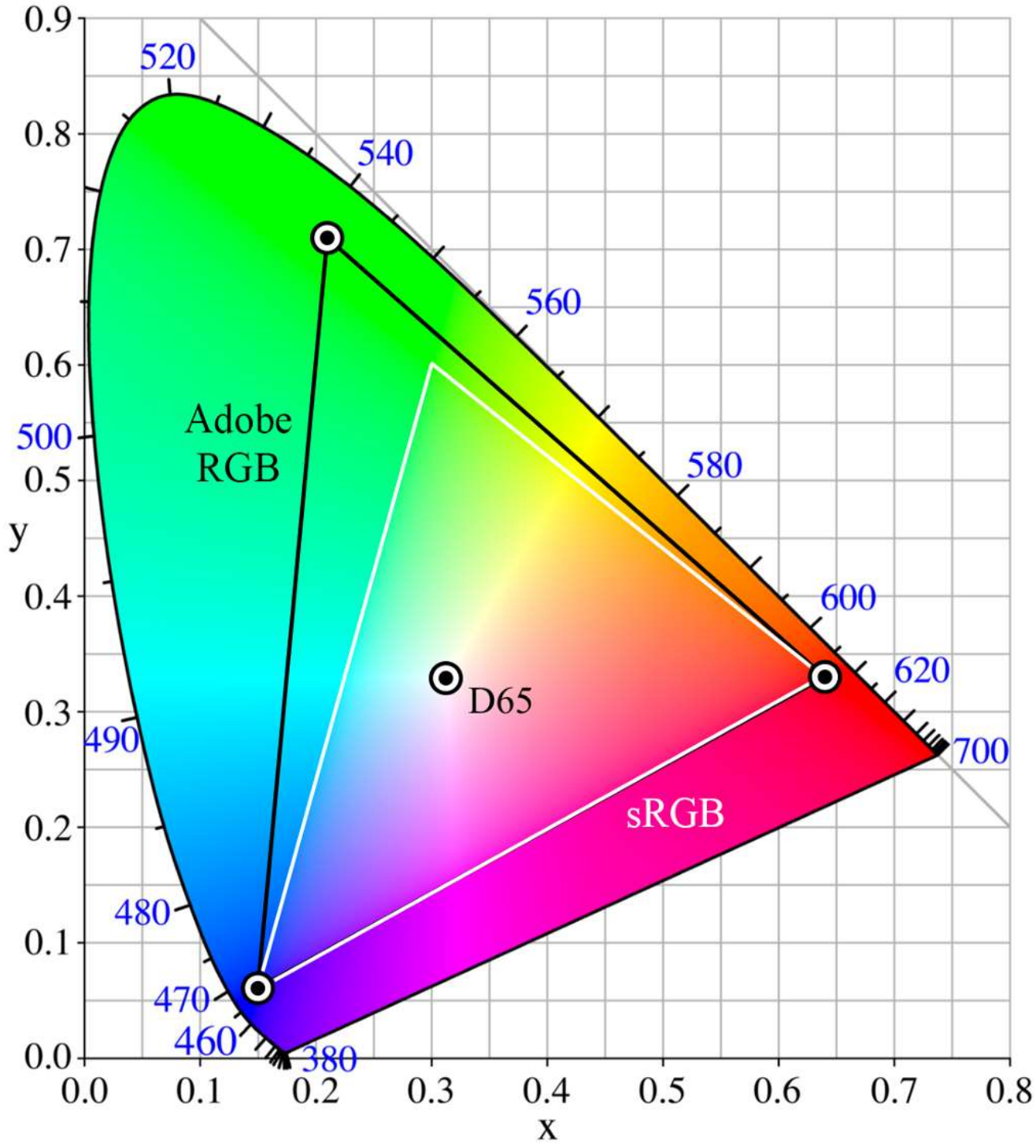
The release of the 1931 XYZ CMFs kick started the practical applications of accurate color reproduction in science and media.

> The first request that I received to specify a colour on the CIE system was from our Ministry of Agriculture and Fisheries, who wished to define the colour of forced rhubarb in connection with a National Mark scheme they were introducing. This showed surprising enterprise on the part of our civil servants and it also provoked an amusing article in our humorous mazagine, Punch.
>
> -W. D. Wright.

If you have ever dealt with colors in web development, you've probably seen colors specified as rgb values like 'rgb(31,157,167)'. These aren't referring to the 1931 RGB color space. They mostly commonly refer to the sRGB color space, created by Microsoft in 1996.

sRGB, and another popular web color space Adobe RGB are each defined by their own three primary colors. Those primaries are defined by specifying their XYZ values.

You can plot the primaries of other color spaces onto the XYZ chromaticity diagram to see the triangles they form. Every chromaticity inside the triangle is a chromaticity that can be specified in this that color space. The area inside a primary triangle is known as a color gamut.



You can see that neither sRGB or Adobe RGB can express the entire gamut of human color perception. If you photograph something with a color outside the range of the color space used by your camera, it will get usually get clipped to the closest expressible color.

Armed with the knowledge of an sRGB color, like rgb(184, 44, 161). You know how much you would need to vary the three sRGB primaries to create that color. If you had a digital display whose sub pixels were exactly the color of sRGB primaries you would be set! But that's usually not the case.

This is an array of 9 pixels, each made up of 3 sub pixels. The three colors of the sub pixels are the primary colors of this display.

If you have an sRGB value, you can do a linear transformation back into XYZ space. Then you can do another linear transformation to make the value in terms of the primaries of the display device.

Now you can display that color in real life, by varying the intensities of the sub pixels of the display accordingly.

9) Beyond Color Matching

As I mentioned in the introduction, Color Matching deals with the accurate reproducibility of color, it tells us if one color looks like another but not what they look like. In fact colors that match can retain that match even as their appearance changes across different lighting conditions.

Color Appearance Models seek to solve the much more difficult problem of quantifying colors in dimensions that are meaningful and intuitive to humans. Dimensions like, hue, saturation and brightness.

Another advanced topic is Color Difference Models, quantifying how different two colors seem from each other so that you can create colors spaces with perceptual uniformity. In these, the distance between any two colors in the coordinate space mirrors the perceived visual difference in the colors.

I like this quote from Wright.

> Stability of matching also means that a colour specification cannot take account of the changes in appearance of a colour inuduced by adaptation, and it is for this reason that we must regard a tristimulus specification as a measure of the quality of the stimulus rather than of the quality of the sensation.
>
> This may be regarded as a limitation of colour measurement and it must be admitted that we are at present a long way from establishing a system on which colour appearance can be specified exactly, yet experience shows that colorimetry as at present practised has many applications and is, indeed, an esssential state towards the specficiation of colour apperance.
>
> — W. D. Wright, The Measurement of Color, 1944

10) Acknowledgments, Further Reading

Last summer two excellent articles and a tweet storm on color plunged me into a year long quest to understand some basic color science.

All three of them were kind enough to answer my questions.

Additionally, a huge thanks to Professors Susan Farnand, Michael J. Murdoch and Mark Fairchild of the Munsell Color Science Laboratory for pointing me in the right directions.

Here are some materials I found particularly interesting

books

The Measurement of Colour, 4th edition by William David Wright

Colorimetry: Understanding the CIE System by Janos Schanda. (this includes a fascinating essay by Wright about the history of how 1931 played out)

papers

How the CIE 1931 Color-Matching Functions Were Derived from Wright–Guild Data

Calculation from the original experimental data of the CIE 1931 RGB standard observer spectral chromaticity co-ordinates and color matching functions

Colour Rendering of Spectra

misc

Obituary: W. David. Wright 1906–1997

I got most of the source data for the XYZ color matching functions and chromaticity coordinates from this handy database http://www.cvrl.org

The less well known RGB color matching functions I got from the official CIE publication.
CIE 15: Technical Report: Colorimetry

Credit for the cover photo chromaticity rendering goes to wikipedia user Fuzzypeg.

Credit for the RGB vs sRGB comparison goes to wikipedia user Mbearnstein37

Colors

About Write Help Legal

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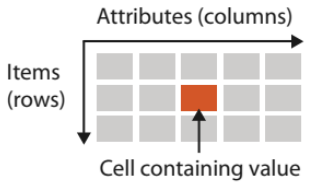
DATA

TERMS

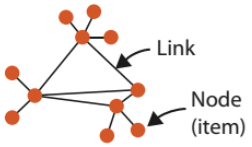
Dataset types: to be visualized

➔ **Dataset Types**

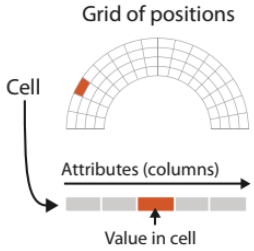
➔ Tables



➔ Networks

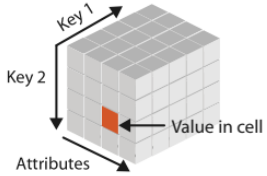


➔ Fields (Continuous)



➔ Geometry (Spatial)

➔ *Multidimensional Table*



➔ *Trees*

Data types: fundamental units

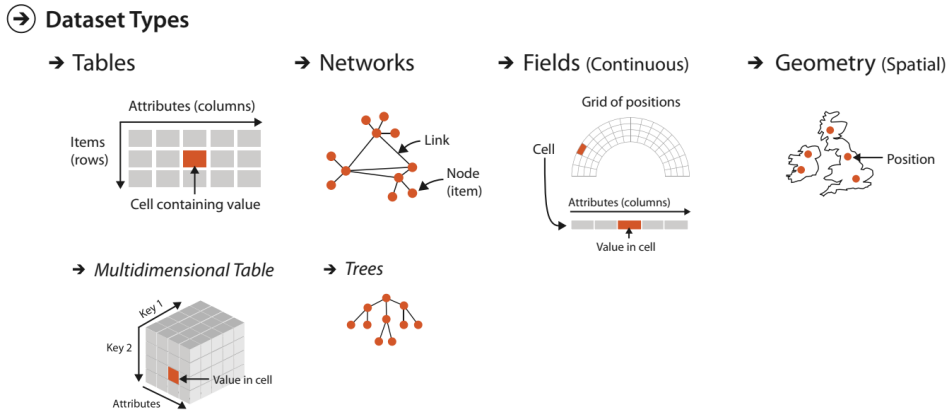
➔ **Data Types**

➔ Items ➔ Attributes ➔ Links ➔ Positions ➔ Grids

STRUCTURE

Structured

- Know data types, semantics



Unstructured

- No predefined data model
- Text, interspersed with facts (dates, times, locations)
- Video, images

Translate into structured data

- Natural Language Processing, text mining
- Object recognition, tracking

TEXT EXAMPLE: PHRASE NET

Pattern in text: “X’s Y” in novels

[van Ham, InfoVis 2009]

DATA SEMANTICS

Basil, 7, S, Pear

What does it mean?

Semantics: real world meaning

Name? city? Fruit? Height? Age? Day of month?

→ Metadata

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

DATA TYPES

Structural or mathematical interpretation of data

Item, link, attribute, position, grid

Different from data types in programming !

ITEMS & ATTRIBUTES

Item: individual entity, discrete

e.g., Patient, Car, Stock, city

“independent variable”

Attribute: measured, observed, logged property

e.g., Patient: height, blood pressure

Car: horsepower, make

“dependent variable”

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

OTHER DATA TYPES

Links

- Express relationship between two items
- Friendship on Facebook

Positions

- Spatial data -> location in 2D or 3D
- Pixels in photo, Voxels in MRI scan, latitude / longtitude

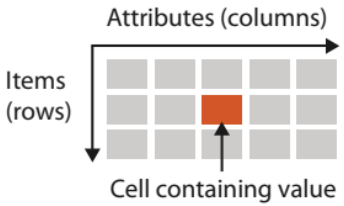
Grids (Not grids in MS Excel)

- Sampling strategy for continuous data
- How many Voxels in MRI scan, positions of weather stations in a city

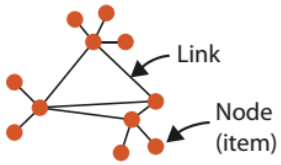
DATASET TYPES

Dataset Types

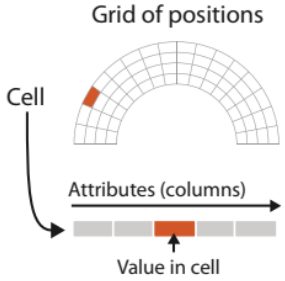
Tables



Networks

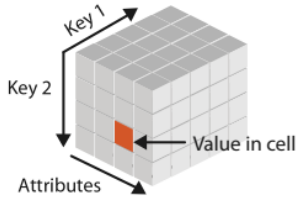


Fields (Continuous)



Geometry (Spatial)

Multidimensional Table



Trees

TABLES

Flat table

- one item per row
- Each column is attribute
- Unique (implicit) key
- **No duplicates**

Multidimensional table

- Indexing based on multiple keys

MULTIDIMENSIONAL TABLES

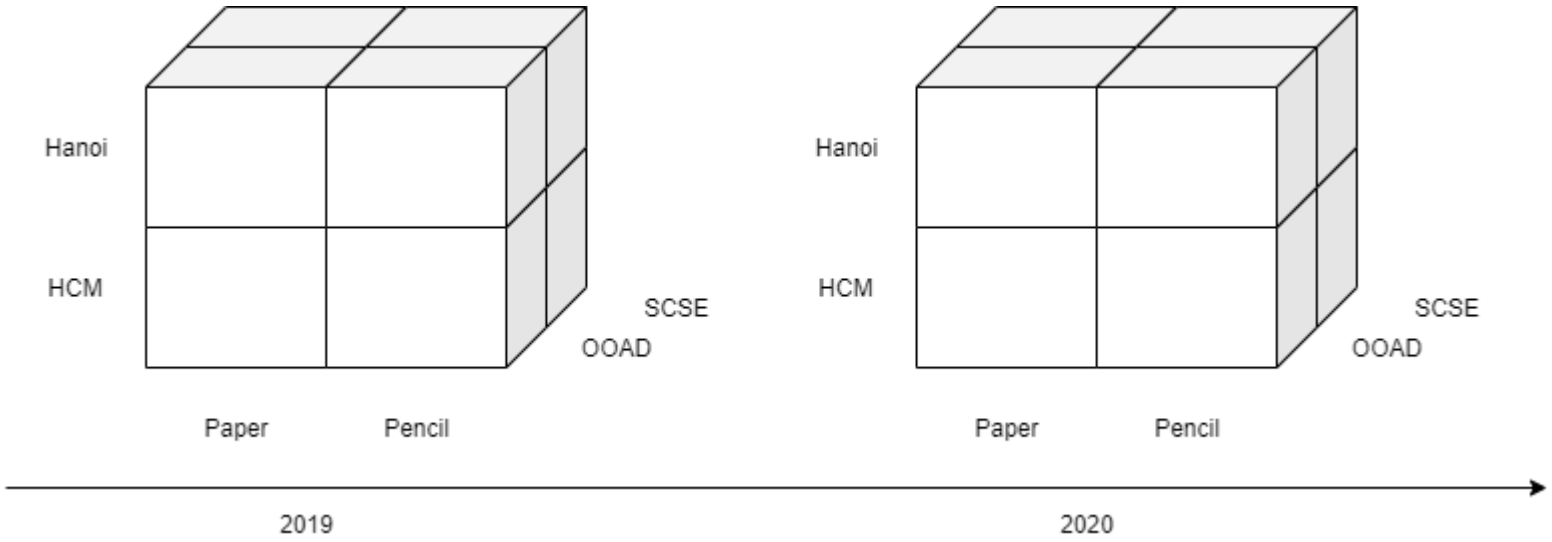
Multiple keys

- Item, Store location, Customer, Period

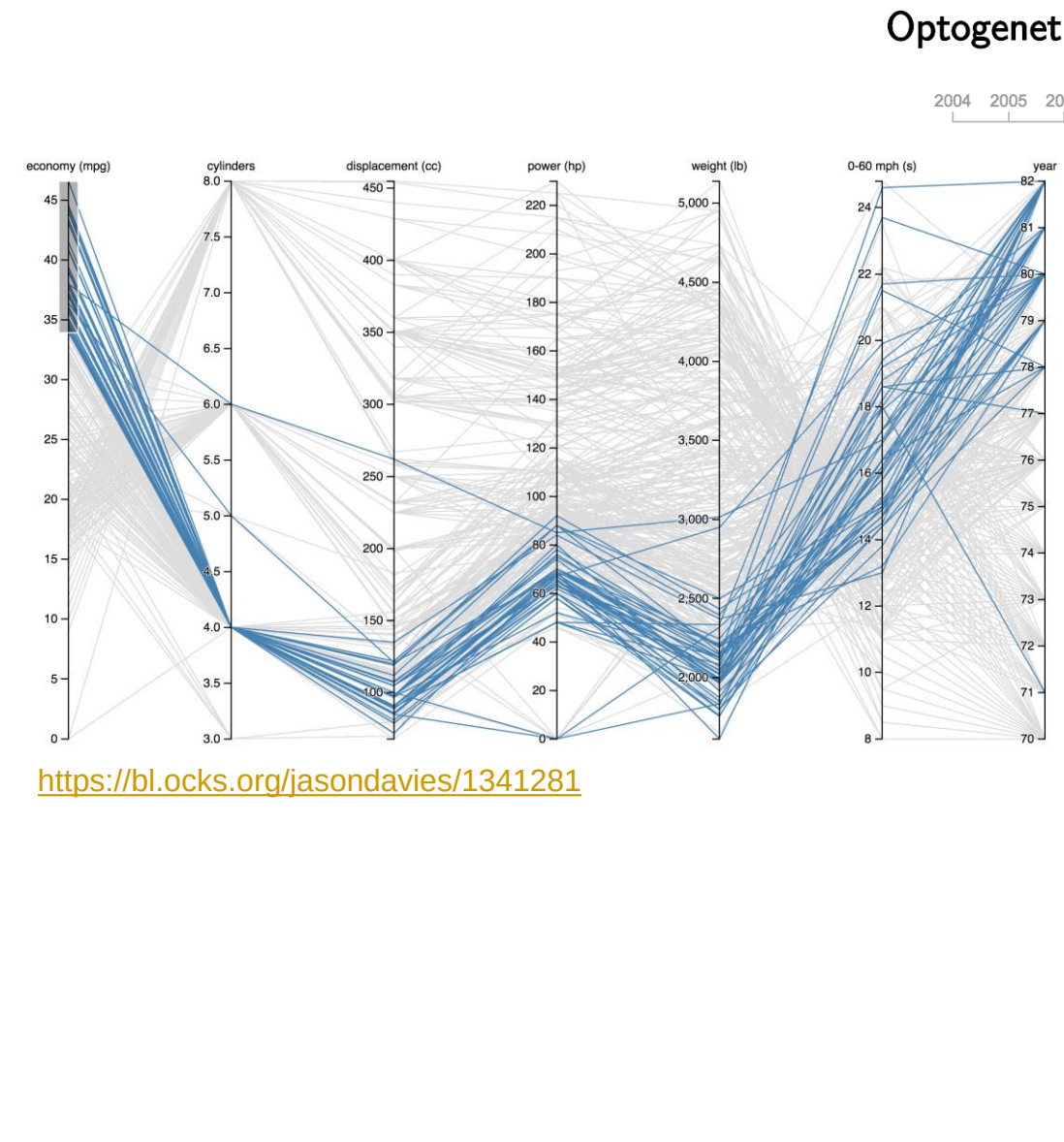
Attribute

- Quantity

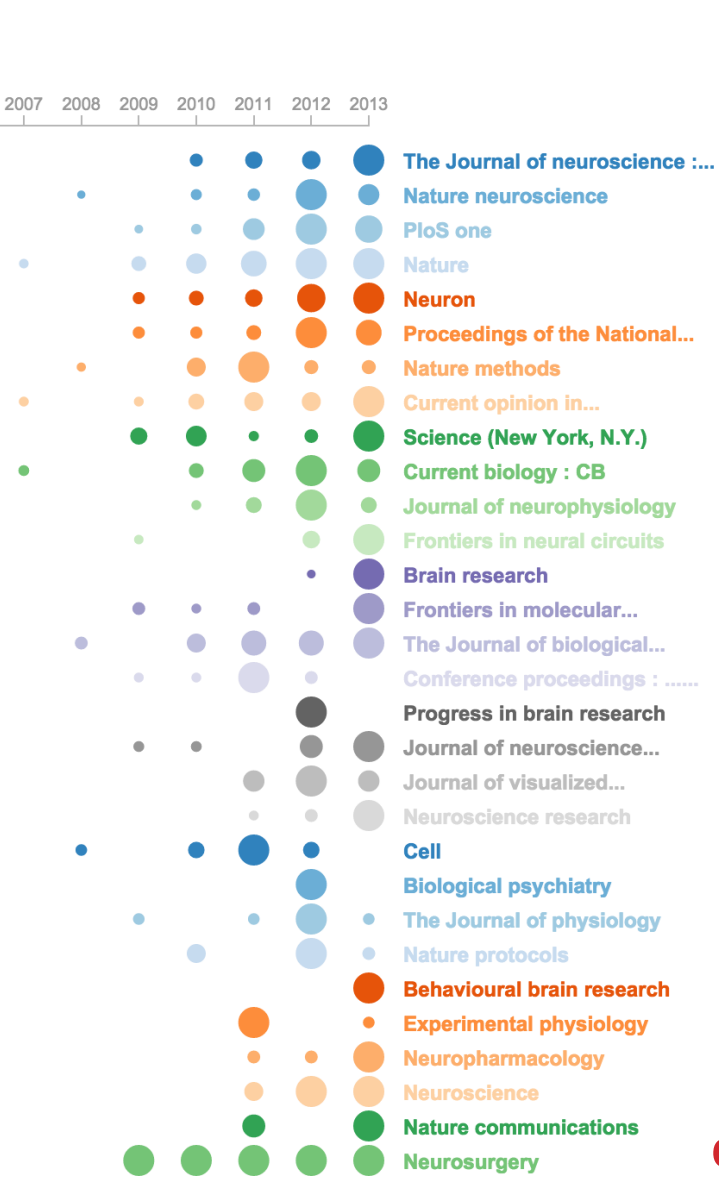
Item	Store location	Customer	Quantity
Paper	HCM	SCSE	400
Pencil	HCM	OOAD	50
Paper	Hanoi	OOAD	100
Pencil	Hanoi	SCSE	20



VISUALIZE TABLES



https://bl.ocks.org/jasondavies/1341281



COLLECTIONS

How we groups items

Sets

- Unique items, unordered

List

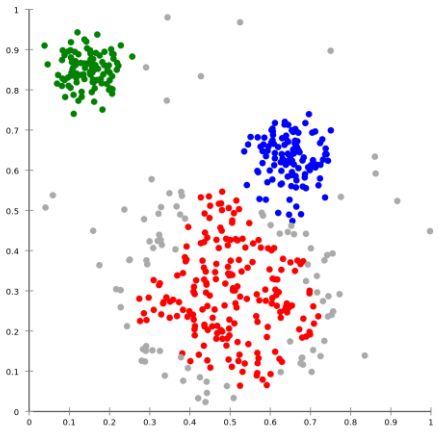
- Ordered, duplicates allowed

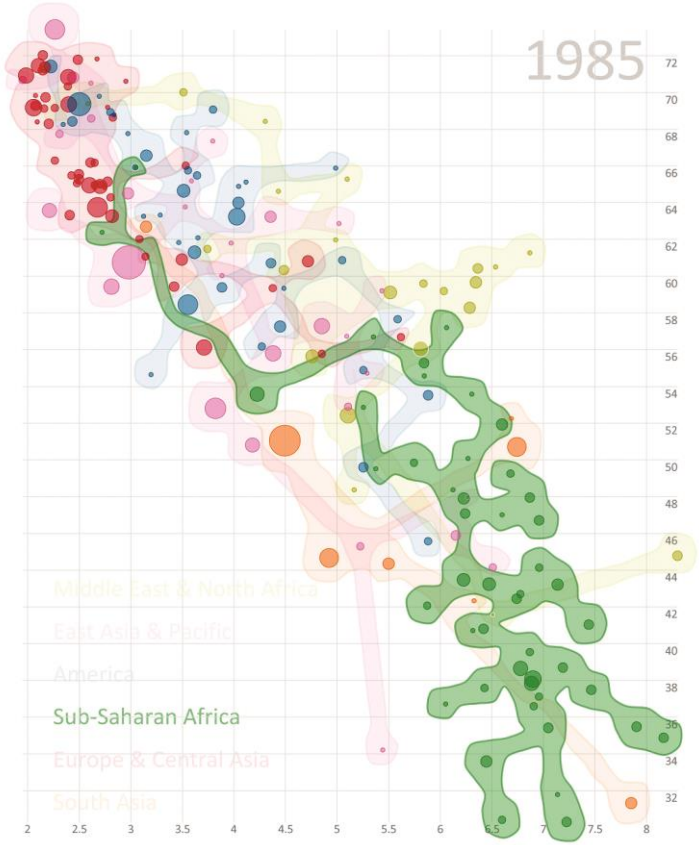
Clusters

- Groups of similar items

https://caleydo.org/tools/lineup/

Rank	School Name Filter: <None>	Academic repu	E	Facult	Citatio	I	I
1.	Massachusetts Inst						
2.	University of Camb						
3.	Harvard University	100 (1)			100 (1)		
4.	UCL (University Co						
5.	University of Oxfor						
6.	Imperial College L						
7.	Yale University						
8.	University of Chic						





GRAPHS/NETWORKS

Set of nodes, set of edges

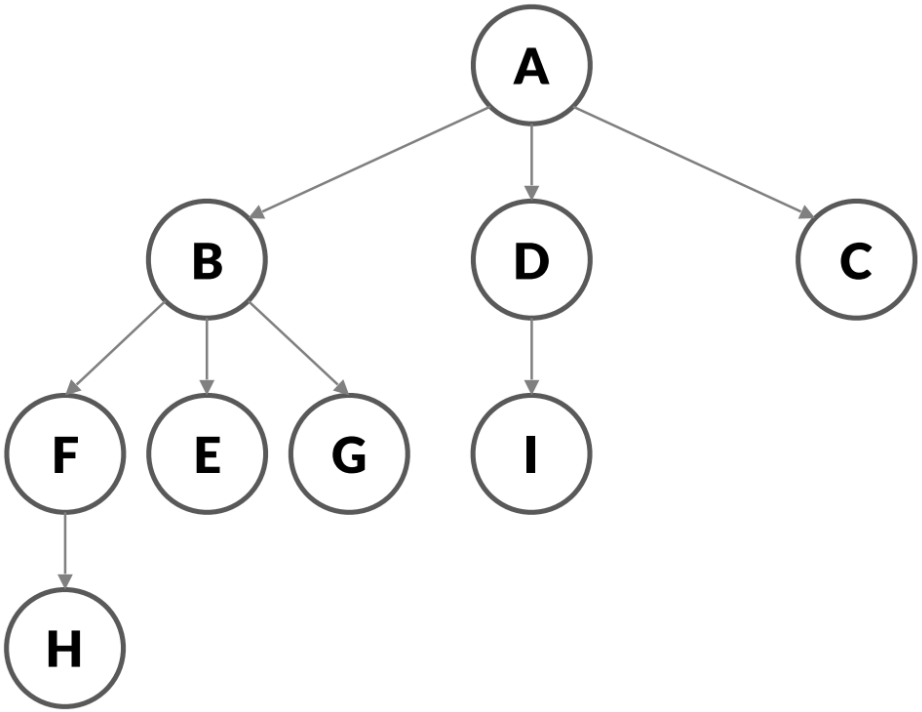
Connecting these vertices

Diagrammatic Example

TREES

A tree is a graph with no cycles

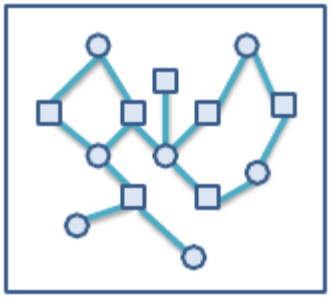
Tree



SPECIAL GRAPHS

A bipartite graph

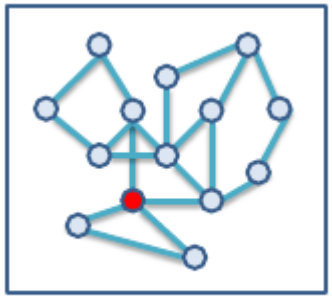
- Vertices can be partitioned into two independent sets



Bipartite Graph

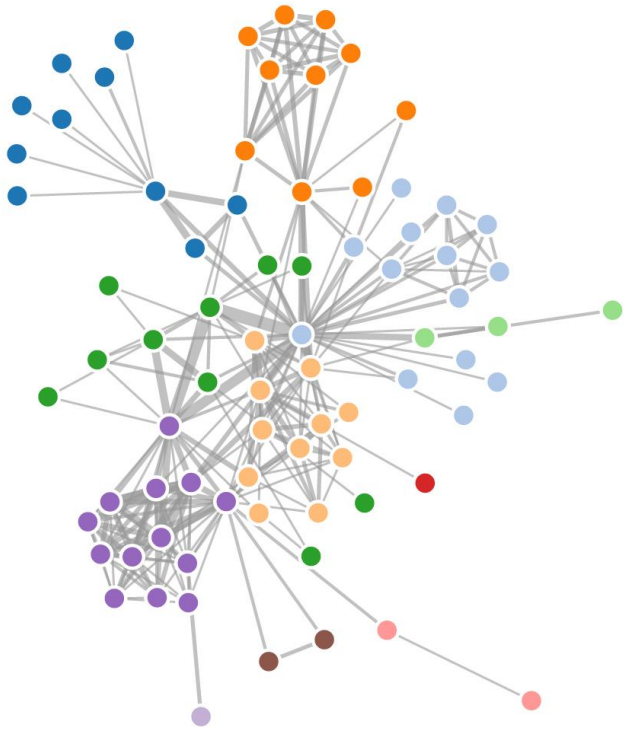
An articulation point

- Is a vertex which if deleted would break up a connected graph



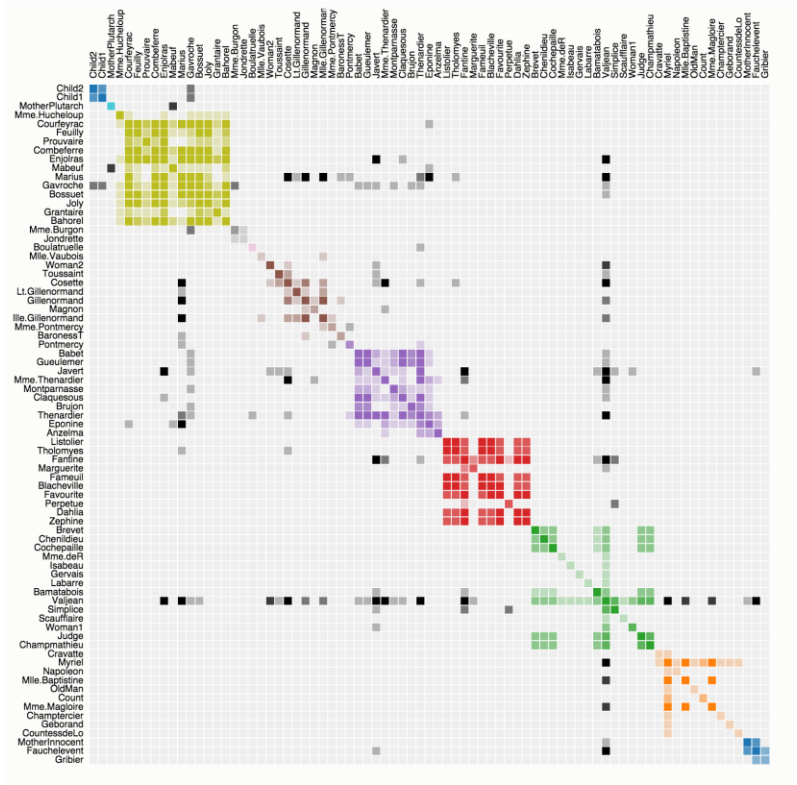
Articulation Point (red)

VISUALIZING GRAPHS



Node-Link diagram

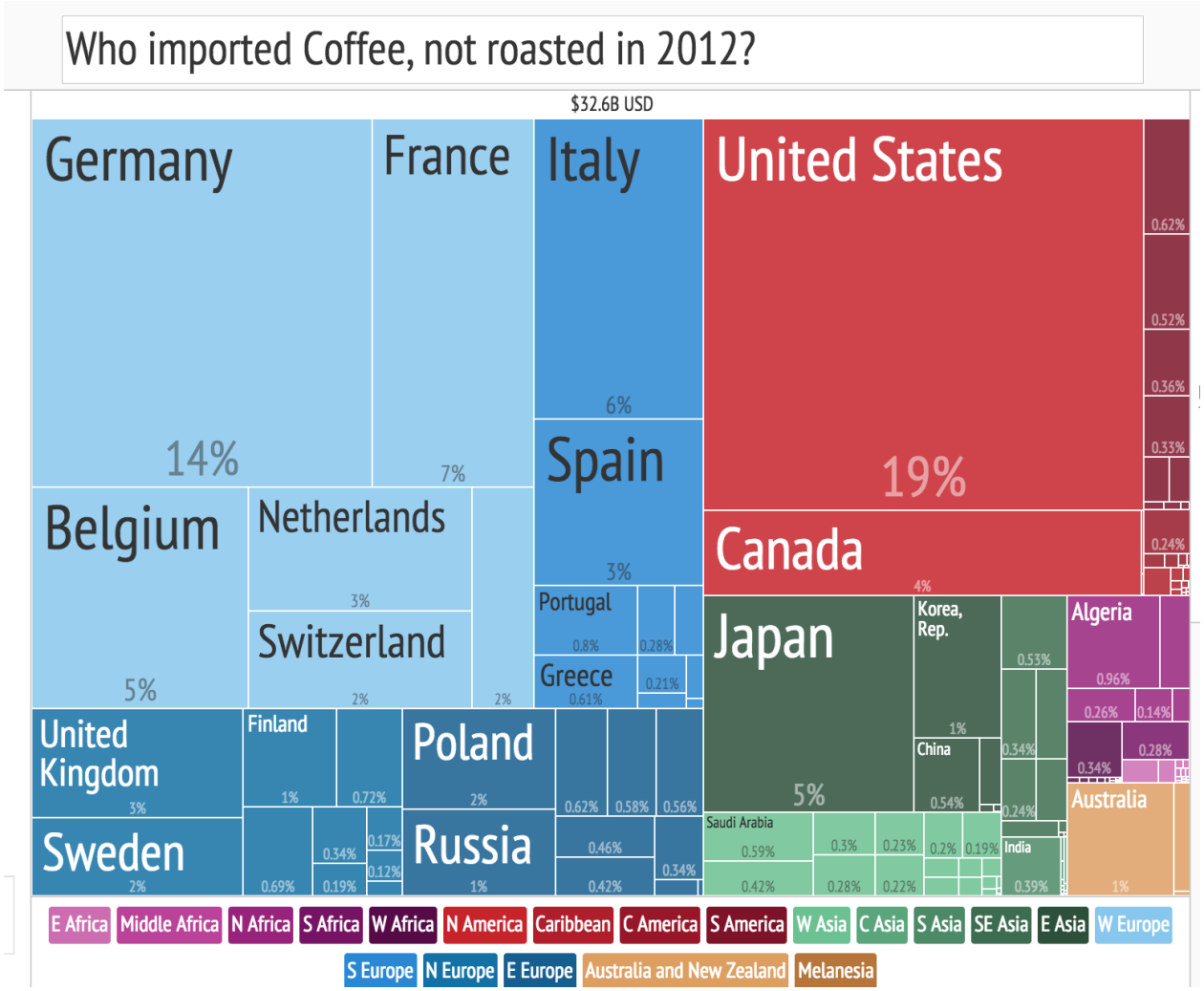
https://observablehq.com/@d3/force-directed-graph



Matrix

https://bost.ocks.org/mike/miserables/

VISUALIZING GRAPHS



Treemap (Implicit Tree Visualization)

https://atlas.cid.harvard.edu/explore/?tradeDirection=import&year=2012&product=726&country=undefined&redirected=true

FIELDS

Sets of attributes values associated with cells

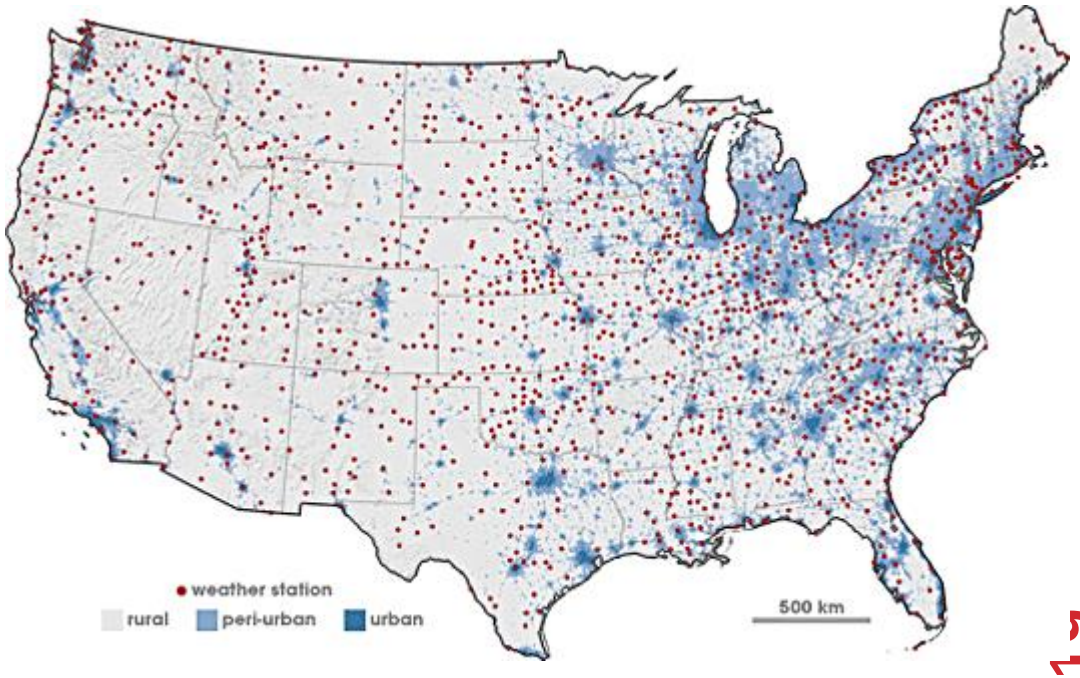
Cell contains data from **continuous domain**

- Temperature, pressure, wind velocity

Measured or simulated

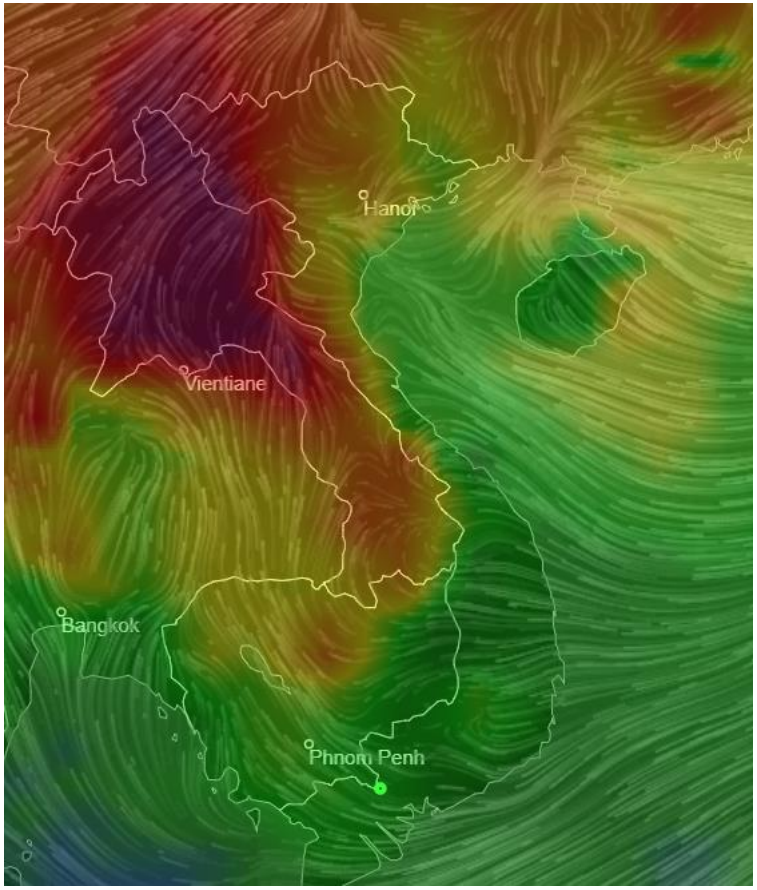
Sampling & **Interpolation**

- Signal processing & stats



https://www.giss.nasa.gov/research/features/200711_temptracker/page2.html

FIELD EXAMPLES: AIR QUALITY



Air visual

https://www.iqair.com/earth?nav

FIELDS: GRID TYPES

Uniform grid

Geometry & topology can be computed

Rectilinear Grid

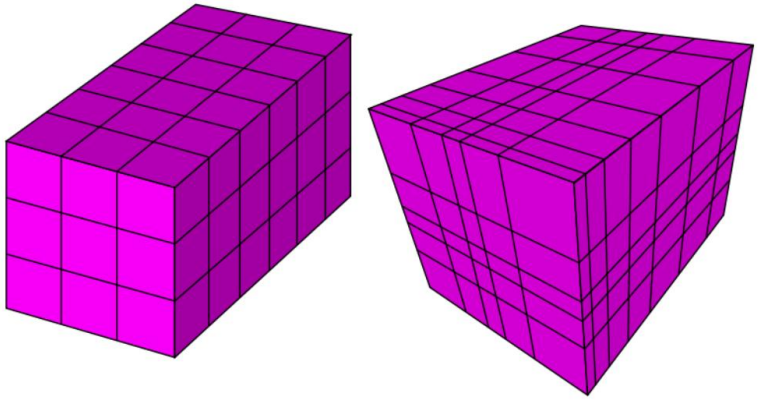
Nonuniform sampling

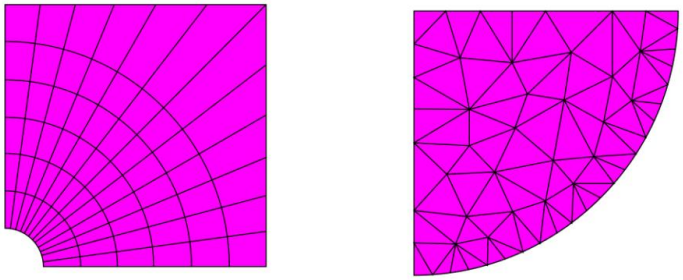
Structured grid

Allow curvilinear grids

Unstructured grid

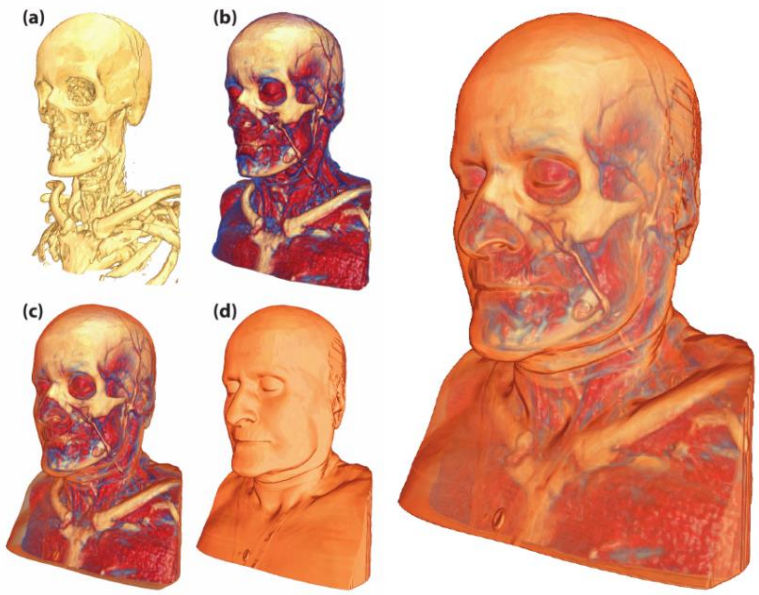
full flexibility, store position and connection



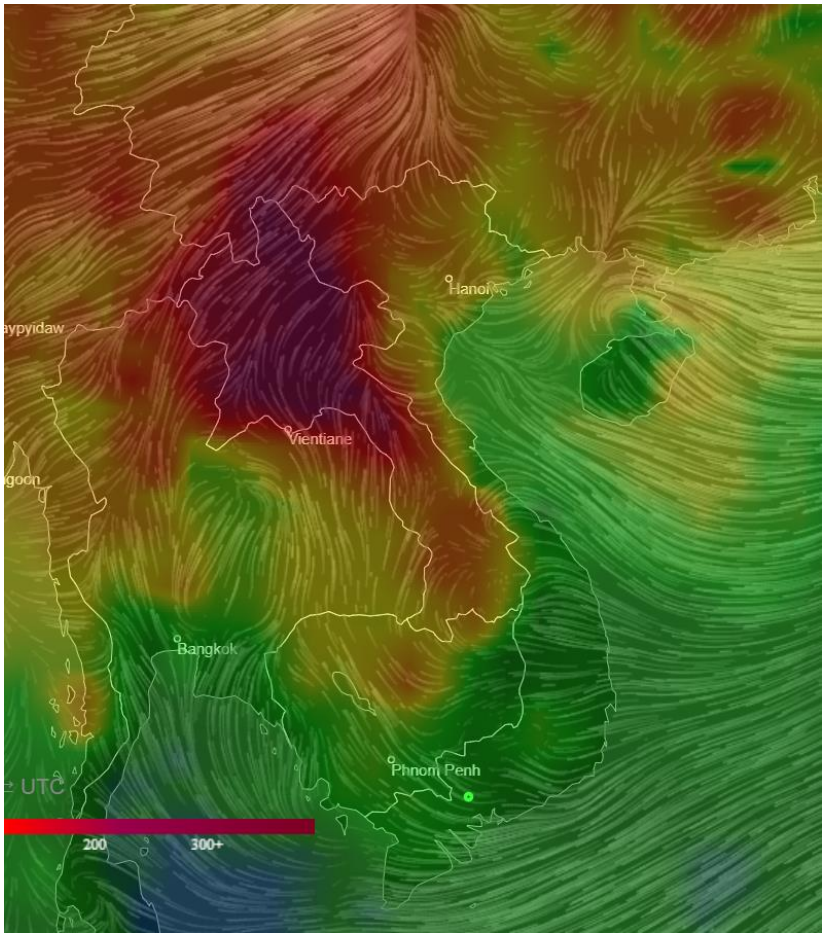


https://en.wikipedia.org/wiki/Regular_grid

VISUALIZING FIELDS



[Bruckner 2007]



READING IN CLASS

INFOVIS
VS
SCIVIS

ACADEMIC SUBFIELDS

Information Vis

"Abstract Data"

- Tables, Graphs, Maps

Free to choose spatial layout

Perception Research

Visual Analytics

InfoVis + Stats + Machine learning

Applied Work

Systems

Funding buzzword

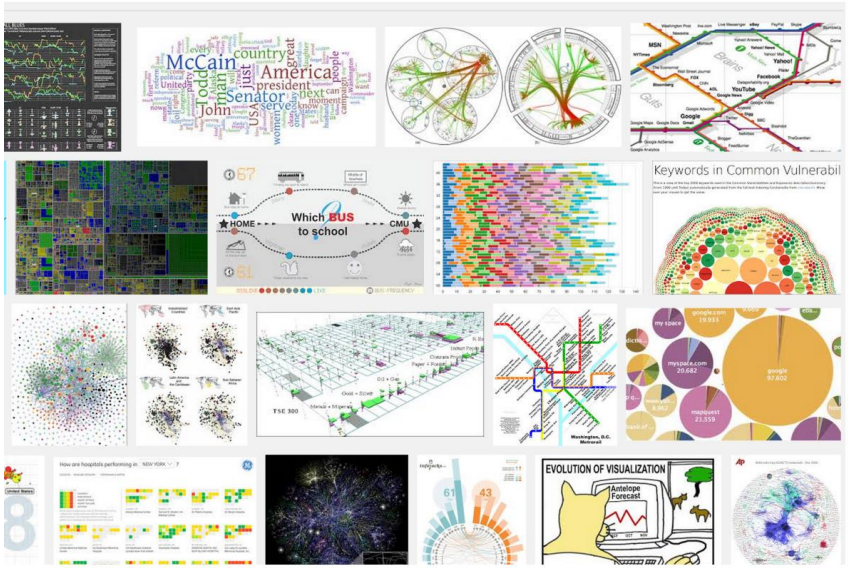
Scientific Vis

"Spatial Data" (Fields)

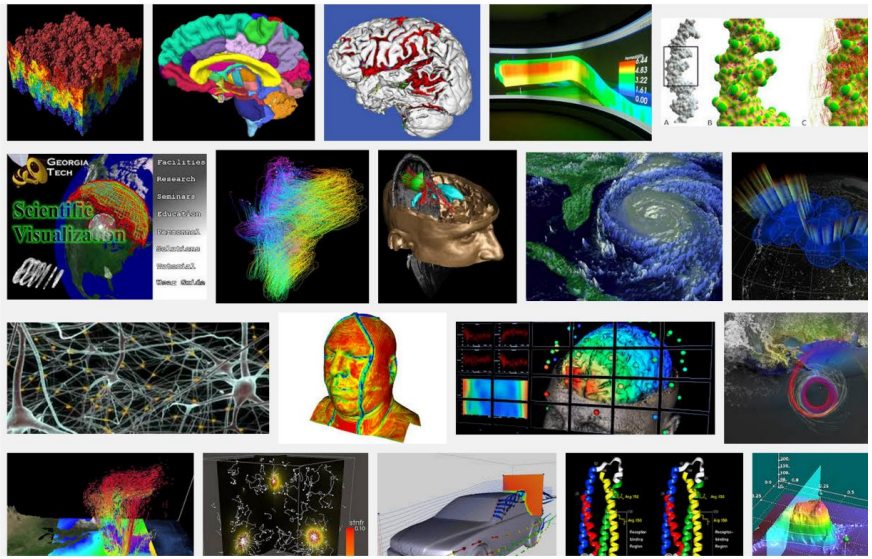
Not free to choose spatial layout

Find best way to depict reality

INFOVIS VS SCIVIS



InfoVis: White Background



SciVis: Black Background

GEOMETRY

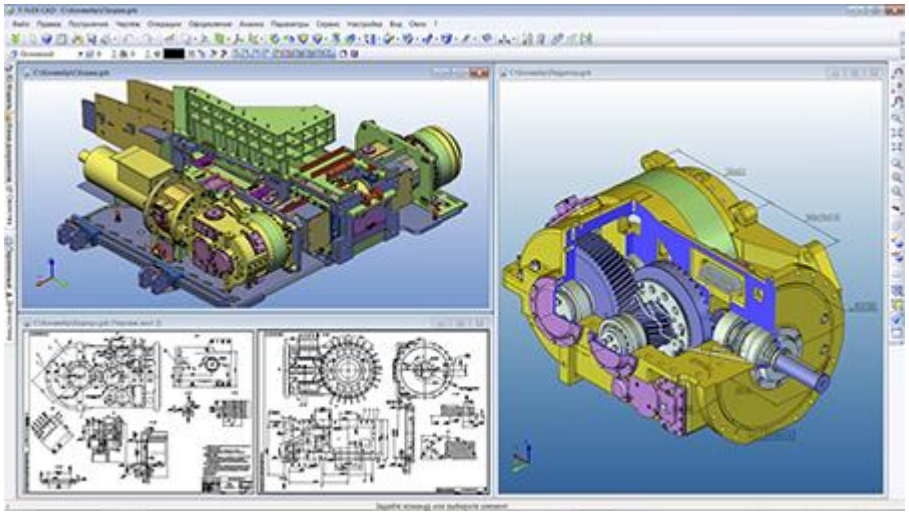
Shape of items

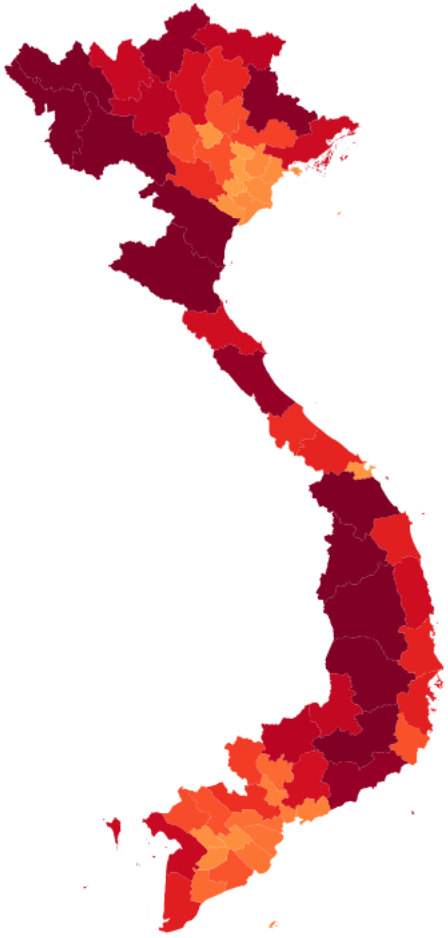
Explicit spatial positions

Points, lines, curves, surfaces, regions, volumnes

Important in Computer Graphics, CAD

Not a core visualization topic





ATTRIBUTE TYPES

Categorical (nominal)

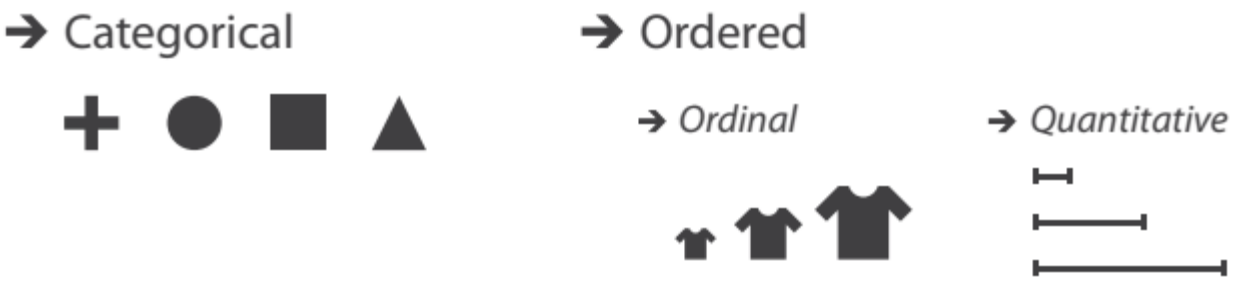
- Compare equality

Ordered

- Ordinal
  - Great/Less than defined

Quantitative

- Arithmetic possible
  - e.g., length, weight



QUANTITATIVE: INTERVAL

May have **equal differences** between successive points on the scale

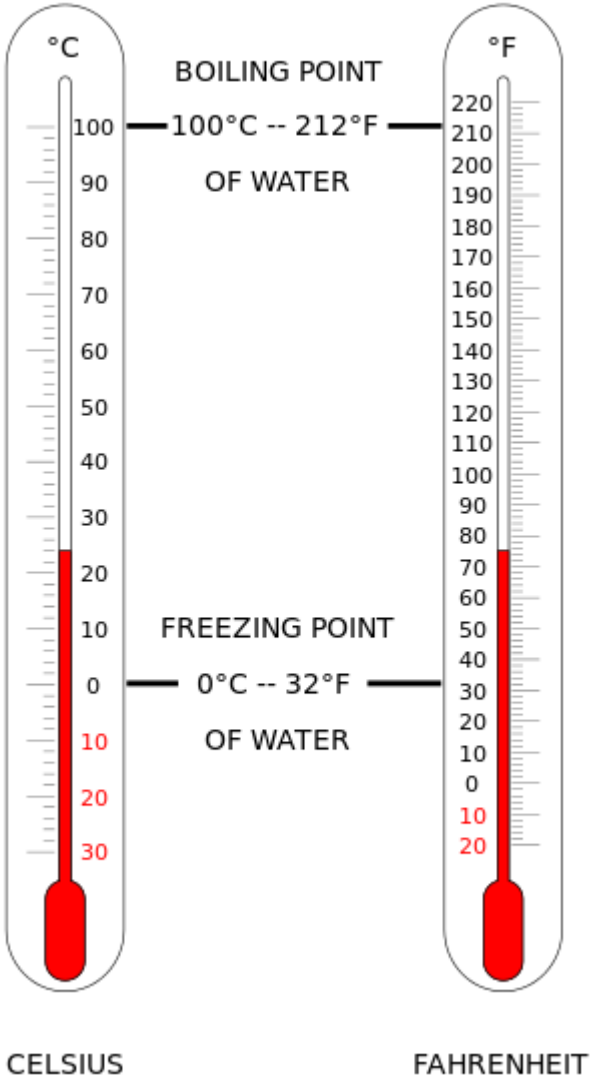
But the position of zero is arbitrary

Does zero mean none?

- Dates; Location;

Cannot compare directly (Fahrenheit vs Celsius)

Only differences can be compared (but ratios)



QUANTITATIVE: RATIO

The relative magnitudes of scores and the differences between them matter. The position of zero is fixed

Zero: there is nothing of the measured entity observed

Measurements: length, mass, age, weight, speed

Can measure ratios & proportions

e.g., Can say 2 times faster than the other

DATA TYPES

Nominal (categories, labels)

Operations: =, ≠

Ordinal (ordered)

Operations: =, ≠, >, <

Interval (location of zero arbitrary)

Operations: =, ≠, >, <, +, - (distance)

Ratio (zero fixed)

Operations: =, ≠, >, <, +, -, ×, ÷ (proportions)

QUIZ

What type of variable (nominal, ordinal, interval, or ratio) are the following:

1. Marathon race times
2. Major
3. Product rating
4. IQ score
5. Product name

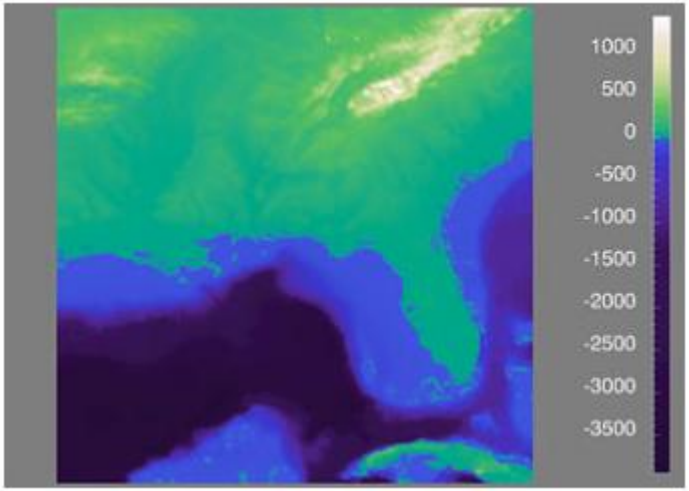
SEQUENTIAL & DIVERGING DATA

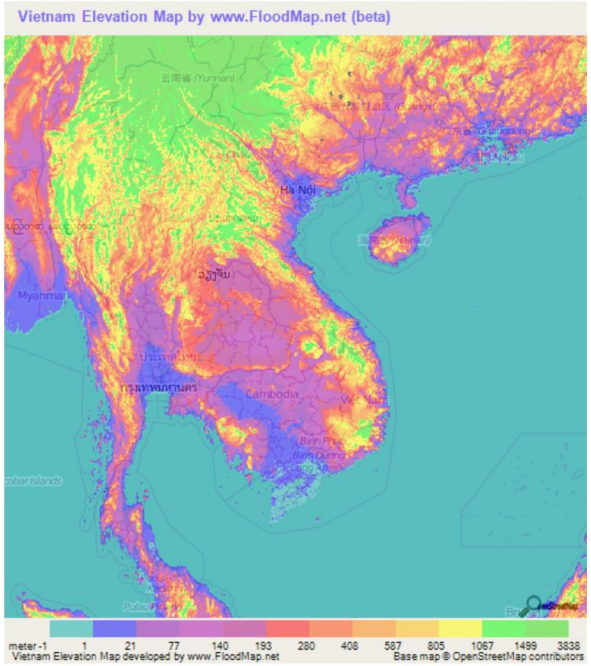
Sequential

- Homogeneous from min to max

Diverging

- Two or multiple sequences that meet
- Elevation dataset: above sea level & below sea level
- Temperature of water: below or above freezing / boiling





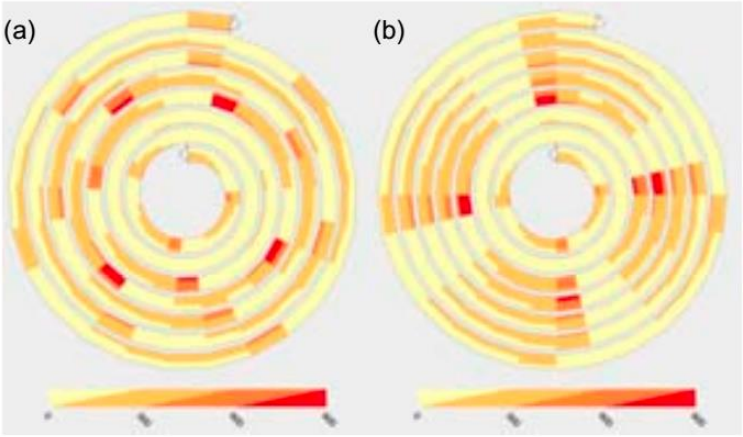
OTHER STRUCTURE

Cyclic data

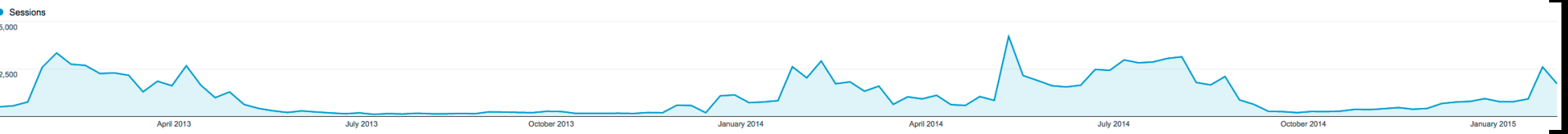
- Time (hours, week, month, year)

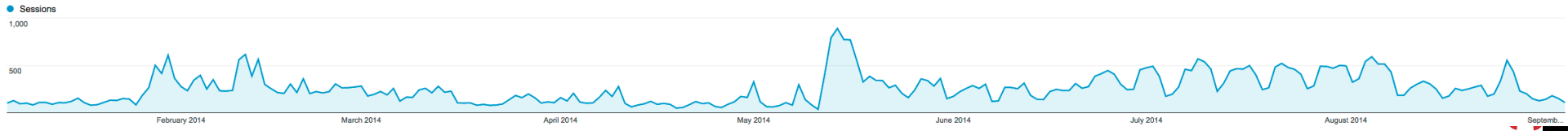
Aggregation

- Might be patterns on multiple levels



Respiratory disease cases.
Left: 25 day pattern
Right: 28 day pattern
[Tominski 2008]





DATA VS CONCEPTUAL MODEL

Data model: low-level description of the data

Set with operations

Conceptual model: mental construction

Includes semantics, support reasoning

Data	Conceptual
1D floats	temperature
3D vector of floats	space

DATA VS CONCEPTUAL MODEL

From data model

19.5, 29.0, -1 (floats)

Using conceptual model

Temperature

To data type

Continuous to 4 significant digits (Q)

Hot, warm, cold (O)

Burned vs Not burned (N)

COMBINATIONS, DERIVED DATA

Networks can have attributes

Attributes have hierarchies

Data types can be transformed

Chapter 2

What: Data Abstraction

2.1 The Big Picture

Figure 2.1 shows the abstract types of *what* can be visualized. The four basic dataset types are tables, networks, fields, and geometry; other possible collections of items include clusters, sets, and lists. These datasets are made up of different combinations of the five data types: items, attributes, links, positions, and grids. For any of these dataset types, the full dataset could be available immediately in the form of a static file, or it might be dynamic data processed gradually in the form of a stream. The type of an attribute can be categorical or ordered, with a further split into ordinal and quantitative. The ordering direction of attributes can be sequential, diverging, or cyclic.

2.2 Why Do Data Semantics and Types Matter?

Many aspects of vis design are driven by the kind of data that you have at your disposal. What kind of data are you given? What information can you figure out from the data, versus the meanings that you must be told explicitly? What high-level concepts will allow you to split datasets apart into general and useful pieces?

Suppose that you see the following data:

```
14, 2.6, 30, 30, 15, 100001
```

What does this sequence of six numbers mean? You can't possibly know yet, without more information about how to interpret each number. Is it locations for two points far from each other in three-dimensional space, $14, 2.6, 30$ and $30, 15, 100001$? Is it two points closer to each other in two-dimensional space, $14, 2.6$ and

30, 30, with the fifth number meaning that there are 15 links between these two points, and the sixth number assigning the weight of '100001' to that link?

Similarly, suppose that you see the following data:

```
Basil, 7, S, Pear
```

These numbers and words could have many possible meanings. Maybe a food shipment of produce has arrived in satisfactory condition on the 7th day of the month, containing basil and pears. Maybe the Basil Point neighborhood of the city has had 7 inches of snow cleared by the Pear Creek Limited snow removal service. Maybe the lab rat named Basil has made seven attempts to find his way through the south section of the maze, lured by scent of the reward food for this trial, a pear.

To move beyond guesses, you need to know two crosscutting pieces of information about these terms: their semantics and their types. The **semantics** of the data is its real-world meaning. For instance, does a word represent a human first name, or is it the shortened version of a company name where the full name can be looked up in an external list, or is it a city, or is it a fruit? Does a number represent a day of the month, or an age, or a measurement of height, or a unique code for a specific person, or a postal code for a neighborhood, or a position in space?

The **type** of the data is its structural or mathematical interpretation. At the data level, what kind of thing is it: an item, a link, an attribute? At the dataset level, how are these data types combined into a larger structure: a table, a tree, a field of sampled values? At the attribute level, what kinds of mathematical operations are meaningful for it? For example, if a number represents a count of boxes of detergent, then its type is a quantity, and adding two such numbers together makes sense. If the number represents a postal code, then its type is a *code* rather than a *quantity*—it is simply the name for a *category* that happens to be a number rather than a textual name. Adding two of these numbers together does not make sense.

Table 2.1 shows several more lines of the same dataset. This simple example table is tiny, with only nine rows and four columns. The exact semantics should be provided by the creator of the dataset; I give it with the column titles. In this case, each person has a unique identifier, a name, an age, a shirt size, and a favorite fruit.

Sometimes types and semantics can be correctly inferred simply by observing the syntax of a data file or the names of variables

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

Table 2.1. A full table with column titles that provide the intended semantics of the attributes.

within it, but often they must be provided along with the dataset in order for it to be interpreted correctly. Sometimes this kind of additional information is called **metadata**; the line between data and metadata is not clear, especially given that the original data is often derived and transformed. In this book, I don't distinguish between them, and refer to everything as *data*.

► Deriving data is discussed in Section 3.4.2.3.

The classification below presents a way to think about dataset and attribute types and semantics in a way that is general enough to cover the cases interesting in vis, yet specific enough to be helpful for guiding design choices at the abstraction and idiom levels.

2.3 Data Types

Figure 2.2 shows the five basic **data types** discussed in this book: items, attributes, links, positions, and grids. An **attribute** is some specific property that can be measured, observed, or logged.* For example, attributes could be salary, price, number of sales, protein expression levels, or temperature. An **item** is an individual entity that is discrete, such as a row in a simple table or a node

★ Synonyms for *attribute* are **variable** and **data dimension**, or just **dimension** for short. Since *dimension* has many meanings, in this book it is reserved for the visual channels of spatial position as discussed in Section 6.3.

Figure 2.2. The five basic data types: items, attributes, links, positions, and grids.

in a network. For example, items may be people, stocks, coffee shops, genes, or cities. A **link** is a relationship between items, typically within a network. A **grid** specifies the strategy for sampling continuous data in terms of both geometric and topological relationships between its cells. A **position** is spatial data, providing a location in two-dimensional (2D) or three-dimensional (3D) space. For example, a position might be a latitude–longitude pair describing a location on the Earth's surface or three numbers specifying a location within the region of space measured by a medical scanner.

2.4 Dataset Types

A **dataset** is any collection of information that is the target of analysis.* The four basic **dataset types** are tables, networks, fields, and geometry. Other ways to group items together include clusters, sets, and lists. In real-world situations, complex combinations of these basic types are common.

★ The word *dataset* is singular. In vis the word **data** is commonly used as a singular mass noun as well, in contrast to the traditional usage in the natural sciences where *data* is plural.

Figure 2.3 shows that these basic dataset types arise from combinations of the data types of items, attributes, links, positions, and grids.

Figure 2.4 shows the internal structure of the four basic dataset types in detail. Tables have cells indexed by items and attributes, for either the simple flat case or the more complex multidimensional case. In a network, items are usually called nodes, and they are connected with links; a special case of networks is trees. Continuous fields have grids based on spatial positions where cells contain attributes. Spatial geometry has only position information.

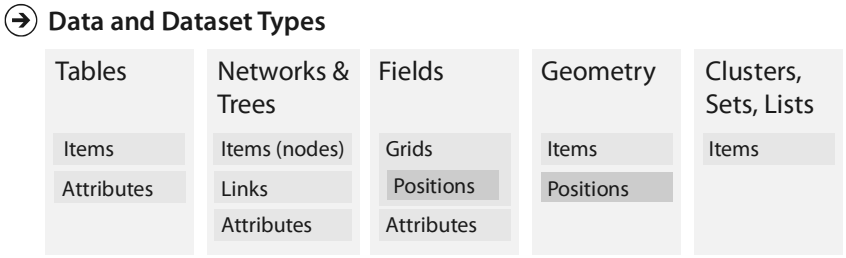


Figure 2.3. The four basic dataset types are tables, networks, fields, and geometry; other possible collections of items are clusters, sets, and lists. These datasets are made up of five core data types: items, attributes, links, positions, and grids.

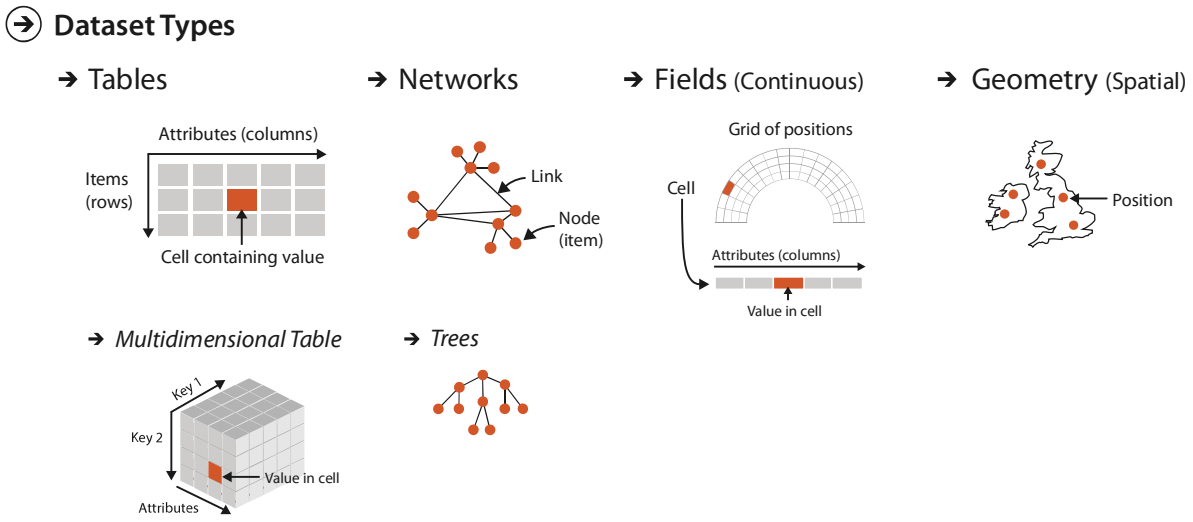


Figure 2.4. The detailed structure of the four basic dataset types.

2.4.1 Tables

Many datasets come in the form of **tables** that are made up of rows and columns, a familiar form to anybody who has used a spreadsheet. In this chapter, I focus on the concept of a table as simply a type of dataset that is independent of any particular visual representation; later chapters address the question of what visual representations are appropriate for the different types of datasets.

▶ Chapter 7 covers how to arrange tables spatially.

For a simple **flat table**, the terms used in this book are that each row represents an **item** of data, and each column is an **attribute** of the dataset. Each **cell** in the table is fully specified by the combination of a row and a column—an item and an attribute—and contains a **value** for that pair. Figure 2.5 shows an example of the first few dozen items in a table of orders, where the attributes are order ID, order date, order priority, product container, product base margin, and ship date.

A **multidimensional table** has a more complex structure for indexing into a cell, with multiple keys.

▶ Keys and values are discussed further in Section 2.6.1.

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box		7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box		7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	[illegible]	5-Low	Wrap Bag	0.56	1/20/05
69	[illegible]5	4-Not Specified	Small Pack	0.44	6/6/05
69	[illegible]5	4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05
97	1/29/06	3-Medium	Small Box	0.38	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

Figure 2.5. In a simple table of orders, a row represents an *item*, a column represents an *attribute*, and their intersection is the *cell* containing the value for that pairwise combination.

★ A synonym for *networks* is **graphs**. The word *graph* is also deeply overloaded in vis. Sometimes it is used to mean *network* as we discuss here, for instance in the vis subfield called *graph drawing* or the mathematical subfield called *graph theory*. Sometimes it is used in the field of statistical graphics to mean **chart**, as in bar graphs and line graphs.

★ A synonym for *node* is **vertex**.

★ A synonym for *link* is **edge**.

2.4.2 Networks and Trees

The dataset type of **networks** is well suited for specifying that there is some kind of relationship between two or more items.* An item in a network is often called a **node**.* A **link** is a relation between two items.* For example, in an articulated social network the nodes are people, and links mean friendship. In a gene interaction network, the nodes are genes, and links between them mean that these genes have been observed to interact with each other. In a computer network, the nodes are computers, and the links represent the ability to send messages directly between two computers using physical cables or a wireless connection.

Network nodes can have associated attributes, just like items in a table. In addition, the links themselves could also be considered to have attributes associated with them; these may be partly or wholly disjoint from the node attributes.

It is again important to distinguish between the abstract concept of a network and any particular visual layout of that network where the nodes and edges have particular spatial positions. This chapter concentrates on the former.

▶ The spatial arrangement of networks is covered in Chapter 9.

2.4.2.1 Trees

Networks with hierarchical structure are more specifically called **trees**. In contrast to a general network, trees do not have cycles: each child node has only one parent node pointing to it. One example of a tree is the organization chart of a company, showing who reports to whom; another example is a tree showing the evolutionary relationships between species in the biological tree of life, where the child nodes of humans and monkeys both share the same parent node of primates.

2.4.3 Fields

The **field** dataset type also contains attribute values associated with cells.[1] Each **cell** in a field contains measurements or calculations from a **continuous** domain: there are conceptually infinitely many values that you might measure, so you could always take a new measurement between any two existing ones. Continuous phenomena that might be measured in the physical world or simulated in software include temperature, pressure, speed, force, and density; mathematical functions can also be continuous.

For example, consider a field dataset representing a medical scan of a human body containing measurements indicating the density of tissue at many sample points, spread regularly throughout a volume of 3D space. A low-resolution scan would have 262,144 cells, providing information about a cubical volume of space with 64 bins in each direction. Each cell is associated with a specific region in 3D space. The density measurements could be taken closer together with a higher resolution grid of cells, or further apart for a coarser grid.

Continuous data requires careful treatment that takes into account the mathematical questions of **sampling**, how frequently to

[1] My use of the term *field* is related to but not identical to its use in the mathematics literature, where it denotes a mapping from a domain to a range. In this case, the domain is a Euclidean space of one, two, or three dimensions, and the adjective modifying *field* is a statement about the range: **scalars**, **vectors**, or **tensors**. Although the term *field* by itself is not commonly found in the literature, when I use it without an adjective I'm emphasizing the continuous nature of the domain, rather than specifics of the ranges of scalars, vectors, or tensors.

take the measurements, and **interpolation**, how to show values in between the sampled points in a way that does not mislead. Interpolating appropriately between the measurements allows you to **reconstruct** a new view of the data from an arbitrary viewpoint that's faithful to what you measured. These general mathematical problems are studied in areas such as signal processing and statistics. Visualizing fields requires grappling extensively with these concerns.

In contrast, the table and network datatypes discussed above are an example of **discrete** data where a finite number of individual items exist, and interpolation between them is not a meaningful concept. In the cases where a mathematical framework is necessary, areas such as graph theory and combinatorics provide relevant ideas.[2]

2.4.3.1 Spatial Fields

Continuous data is often found in the form of a **spatial field**, where the cell structure of the field is based on sampling at spatial positions. Most datasets that contain inherently spatial data occur in the context of tasks that require understanding aspects of its spatial structure, especially shape.

For example, with a spatial field dataset that is generated with a medical imaging instrument, the user's task could be to locate suspected tumors that can be recognized through distinctive shapes or densities. An obvious choice for visual encoding would be to show something that spatially looks like an X-ray image of the human body and to use color coding to highlight suspected tumors. Another example is measurements made in a real or simulated wind tunnel of the temperature and pressure of air flowing over airplane wings at many points in 3D space, where the task is to compare the flow patterns in different regions. One possible visual encoding would use the geometry of the wing as the spatial substrate, showing the temperature and pressure using size-coded arrows.

The likely tasks faced by users who have spatial field data constrains many of the choices about the use of space when designing visual encoding idioms. Many of the choices for **nonspatial data**, where no information about spatial position is provided with the dataset, are unsuitable in this case.[⋆]

★ A synonym for *nonspatial data* is **abstract data**.

[2] Technically, all data stored within a computer is discrete rather than continuous; however, the interesting question is whether the underlying semantics of the bits that are stored represents samples of a continuous phenomenon or intrinsically discrete data.

Thus, the question of whether a dataset has the type of a spatial field or a nonspatial table has extensive and far-reaching implications for idiom design. Historically, vis diverged into areas of specialization based on this very differentiation. The subfield of **scientific visualization**, or **scivis** for short, is concerned with situations where spatial position is *given* with the dataset. A central concern in scivis is handling continuous data appropriately within the mathematical framework of signal processing. The subfield of **information visualization**, or **infovis** for short, is concerned with situations where the use of space in a visual encoding is *chosen* by the designer. A central concern in infovis is determining whether the chosen idiom is suitable for the combination of data and task, leading to the use of methods from human–computer interaction and design.

2.4.3.2 Grid Types

When a field contains data created by sampling at completely regular intervals, as in the previous example, the cells form a **uniform grid**. There is no need to explicitly store the **grid geometry** in terms of its location in space, or the **grid topology** in terms of how each cell connects with its neighboring cells. More complicated examples require storing different amounts of geometric and topological information about the underlying grid. A **rectilinear grid** supports nonuniform sampling, allowing efficient storage of information that has high complexity in some areas and low complexity in others, at the cost of storing some information about the geometric location of each each row. A **structured grid** allows curvilinear shapes, where the geometric location of each cell needs to be specified. Finally, **unstructured grids** provide complete flexibility, but the topological information about how the cells connect to each other must be stored explicitly in addition to their spatial positions.

2.4.4 Geometry

The **geometry** dataset type specifies information about the shape of items with explicit spatial positions. The items could be points, or one-dimensional lines or curves, or 2D surfaces or regions, or 3D volumes.

Geometry datasets are intrinsically spatial, and like spatial fields they typically occur in the context of tasks that require shape understanding. Spatial data often includes hierarchical structure at multiple scales. Sometimes this structure is provided intrinsically

with the dataset, or a hierarchy may be derived from the original data.

Geometry datasets do not necessarily have attributes, in contrast to the other three basic dataset types. Many of the design issues in vis pertain to questions about how to encode attributes. Purely geometric data is interesting in a vis context only when it is derived or transformed in a way that requires consideration of design choices. One classic example is when contours are derived from a spatial field. Another is when shapes are generated at an appropriate level of detail for the task at hand from raw geographic data, such as the boundaries of a forest or a city or a country, or the curve of a road. The problem of how to create images from a geometric description of a scene falls into another domain: computer graphics. While vis draws on algorithms from computer graphics, it has different concerns from that domain. Simply showing a geometric dataset is not an interesting problem from the point of view of a vis designer.

► Section 3.4.2.3 covers deriving data.

► Section 8.4 covers generating contours from scalar fields.

Geometric data is sometimes shown alone, particularly when shape understanding is the primary task. In other cases, it is the backdrop against which additional information is overlaid.

2.4.5 Other Combinations

Beyond tables, there are many ways to group multiple *items* together, including sets, lists, and clusters. A **set** is simply an unordered group of items. A group of items with a specified ordering could be called a **list**.* A **cluster** is a grouping based on attribute similarity, where items within a cluster are more similar to each other than to ones in another cluster.

★ In computer science, **array** is often used as a synonym for *list*.

There are also more complex structures built on top of the basic network type. A **path** through a network is an ordered set of segments formed by links connecting nodes. A **compound network** is a network with an associated tree: all of the nodes in the network are the leaves of the tree, and interior nodes in the tree provide a hierarchical structure for the nodes that is different from network links between them.

Many other kinds of data either fit into one of the previous categories or do so after transformations to create derived attributes. Complex and hybrid combinations, where the complete dataset contains multiple basic types, are common in real-world applications.

The set of basic types presented above is a starting point for describing the *what* part of an analysis instance that pertains to

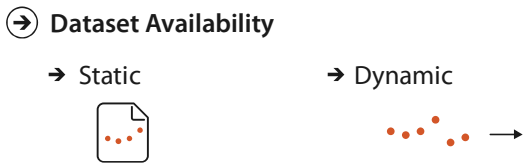


Figure 2.6. Dataset availability can be either static or dynamic, for any dataset type.

data; that is, the **data abstraction**. In simple cases, it may be possible to describe your data abstraction using only that set of terms. In complex cases, you may need additional description as well. If so, your goal should be to translate domain-specific terms into words that are as generic as possible.

2.4.6 Dataset Availability

Figure 2.6 shows the two kinds of dataset availability: *static* or *dynamic*.

The default approach to vis assumes that the entire dataset is available all at once, as a **static file**. However, some datasets are instead **dynamic streams**, where the dataset information trickles in over the course of the vis session.* One kind of dynamic change is to add new items or delete previous items. Another is to change the values of existing items.

★ A synonym for *dynamic* is **online**, and a synonym for *static* is **offline**.

This distinction in availability crosscuts the basic dataset types: any of them can be static or dynamic. Designing for streaming data adds complexity to many aspects of the vis process that are straightforward when there is complete dataset availability up front.

2.5 Attribute Types

Figure 2.7 shows the attribute types. The major disinction is between categorical versus ordered. Within the ordered type is a further differentiation between ordinal versus quantitative. Ordered data might range sequentially from a minimum to a maximum value, or it might diverge in both directions from a zero point in the middle of a range, or the values may wrap around in a cycle. Also, attributes may have hierarchical structure.

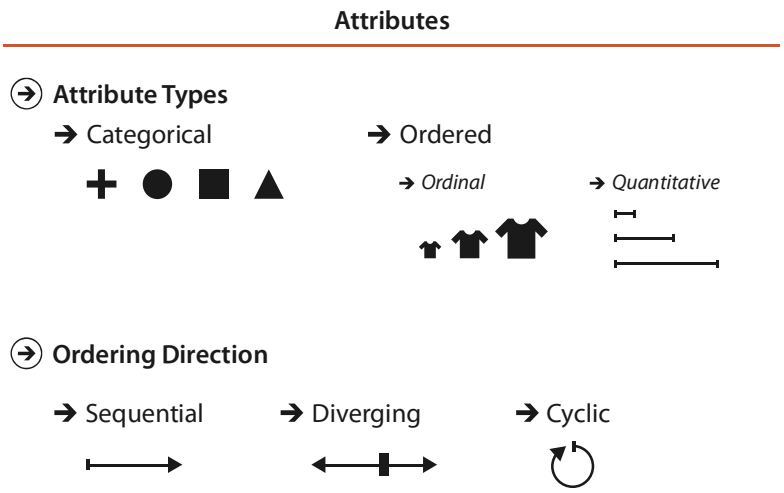


Figure 2.7. Attribute types are categorical, ordinal, or quantitative. The direction of attribute ordering can be sequential, diverging, or cyclic.

2.5.1 Categorical

★ A synonym for *categorical* is **nominal**.

The first distinction is between categorical and ordered data. The type of **categorical** data, such as favorite fruit or names, does not have an implicit ordering, but it often has hierarchical structure.* Categories can only distinguish whether two things are the same (apples) or different (apples versus oranges). Of course, any arbitrary external ordering can be imposed upon categorical data. Fruit could be ordered alphabetically according to its name, or by its price—but only if that auxiliary information were available. However, these orderings are not implicit in the attribute itself, the way they are with quantitative or ordered data. Other examples of categorical attributes are movie genres, file types, and city names.

2.5.2 Ordered: Ordinal and Quantitative

All **ordered** data does have an implicit ordering, as opposed to unordered *categorical* data. This type can be further subdivided. With **ordinal** data, such as shirt size, we cannot do full-fledged arithmetic, but there is a well-defined ordering. For example, large minus medium is not a meaningful concept, but we know that medium falls between small and large. Rankings are another kind

of ordinal data; some examples of ordered data are top-ten lists of movies or initial lineups for sports tournaments depending on past performance.

A subset of ordered data is **quantitative** data, namely, a measurement of magnitude that supports arithmetic comparison. For example, the quantity of 68 inches minus 42 inches is a meaningful concept, and the answer of 26 inches can be calculated. Other examples of quantitative data are height, weight, temperature, stock price, number of calling functions in a program, and number of drinks sold at a coffee shop in a day. Both integers and real numbers are quantitative data.[3]

In this book, the *ordered* type is used often; the ordinal type is only occasionally mentioned, when the distinction between it and the quantitative type matters.

2.5.2.1 Sequential versus Diverging

Ordered data can be either **sequential**, where there is a homogeneous range from a minimum to a maximum value, or **diverging**, which can be deconstructed into two sequences pointing in opposite directions that meet at a common zero point. For instance, a mountain *height* dataset is sequential, when measured from a minimum point of sea level to a maximum point of Mount Everest. A *bathymetric* dataset is also sequential, with sea level on one end and the lowest point on the ocean floor at the other. A full *elevation* dataset would be diverging, where the values go up for mountains on land and down for undersea valleys, with the zero value of sea level being the common point joining the two sequential datasets.

2.5.2.2 Cyclic

Ordered data may be **cyclic**, where the values wrap around back to a starting point rather than continuing to increase indefinitely. Many kinds of time measurements are cyclic, including the hour of the day, the day of the week, and the month of the year.

2.5.3 Hierarchical Attributes

There may be hierarchical structure within an attribute or between multiple attributes. The daily stock prices of companies collected

[3] In some domains the quantitative category is further split into **interval** versus **ratio** data [Stevens 46]; this distinction is typically not useful when designing a visual encoding, so in this book these types remain collapsed together into this single category.

over the course of a decade is an example of a time-series dataset, where one of the attributes is time. In this case, time can be aggregated hierarchically, from individual days up to weeks, up to months, up to years. There may be interesting patterns at multiple temporal scales, such as very strong weekly variations for weekday versus weekend, or more subtle yearly patterns showing seasonal variations in summer versus winter. Many kinds of attributes might have this sort of hierarchical structure: for example, the geographic attribute of a postal code can be aggregated up to the level of cities or states or entire countries.

► Section 13.4 covers hierarchical aggregation in more detail, and Section 7.5 covers the visual encoding of attribute hierarchies.

2.6 Semantics

Knowing the type of an attribute does not tell us about its semantics, because these two questions are crosscutting: one does not dictate the other. Different approaches to considering the semantics of attributes that have been proposed across the many fields where these semantics are studied. The classification in this book is heavily focused on the semantics of keys versus values, and the related questions of spatial and continuous data versus nonspatial and discrete data, to match up with the idiom design choice analysis framework. One additional consideration is whether an attribute is temporal.

2.6.1 Key versus Value Semantics

A **key** attribute acts as an index that is used to look up **value** attributes.* The distinction between key and value attributes is important for the dataset types of tables and fields, as shown in Figure 2.8.

★ A synonym for *key attribute* is **independent attribute**. A synonym for *value attribute* is **dependent attribute**. The language of independent and dependent is common in statistics. In the language of data warehouses, a synonym for *independent* is **dimension**, and a synonym for *dependent* is **measure**.

2.6.1.1 Flat Tables

A simple **flat table** has only one key, where each item corresponds to a row in the table, and any number of value attributes. In this case, the key might be completely implicit, where it's simply the index of the row. It might be explicit, where it is contained within the table as an attribute. In this case, there must not be any duplicate values within that attribute. In tables, keys may be categorical or ordinal attributes, but quantititive attributes are typically unsuitable as keys because there is nothing to prevent them from having the same values for multiple items.

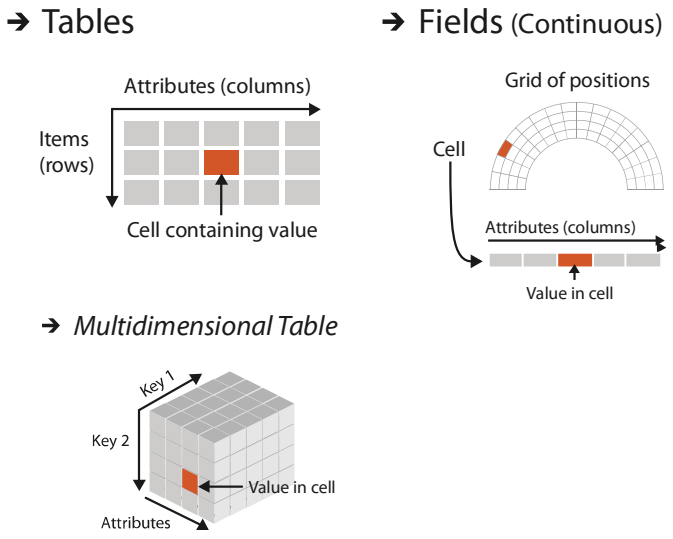


Figure 2.8. Key and value semantics for tables and fields.

For example, in Table 2.1, *Name* is a categorical attribute that might appear to be a reasonable key at first, but the last line shows that two people have the same name, so it is not a good choice. *Favorite Fruit* is clearly not a key, despite being categorical, because *Pear* appears in two different rows. The quantitative attribute of *Age* has many duplicate values, as does the ordinal attribute of *Shirt Size*. The first attribute in this flat table has an explicit unique identifier that acts as the key.[4] This key attribute could either be ordinal, for example if the order that the rows were entered into the table captures interesting temporal information, or categorical, if it's simply treated as a unique code.

Figure 2.9 shows the order table from Figure 2.5 where each attribute is colored according to its type. There is no explicit key: even the *Order ID* attribute has duplicates, because orders consist of multiple items with different container sizes, so it does not act as a unique identifier. This table is an example of using an implicit key that is the row number within the table.

[4] It's common to store the key attribute in the first column, for understandability by people and ease of building data structures by computers.

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
69	6/4/05	4-Not Spec		0.6	6/6/05
70	12/18/06	5-Low		0.59	12/23/06
70	12/18/06	5-Low		0.82	12/23/06
96	4/17/05	2-High		0.55	4/19/05
97	1/29/06	3-Medium		0.38	1/30/06
129	11/19/08	5-Low		0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

quantitative
ordinal
categorical

Figure 2.9. The order table with the attribute columns colored by their type; none of them is a key.

2.6.1.2 Multidimensional Tables

The more complex case is a **multidimensional table**, where multiple keys are required to look up an item. The combination of all keys must be unique for each item, even though an individual key attribute may contain duplicates. For example, a common multidimensional table from the biology domain has a gene as one key and time as another key, so that the value in each cell is the activity level of a gene at a particular time.

The information about which attributes are keys and which are values may not be available; in many instances determining which attributes are independent keys versus dependent values is the goal of the vis process, rather than its starting point. In this case, the successful outcome of analysis using vis might be to recast a flat table into a more semantically meaningful multidimensional table.

2.6.1.3 Fields

Although fields differ from tables a fundamental way because they represent continuous rather than discrete data, keys and values are still central concerns. (Different vocabulary for the same basic idea is more common with spatial field data, where the term *independent variable* is used instead of *key*, and *dependent variable* instead of *value*.)

Fields are structured by sampling in a systematic way so that each grid cell is spanned by a unique range from a continuous domain. In spatial fields, spatial position acts as a quantitative key, in contrast to a nonspatial attribute in the case of a table that is categorical or ordinal. The crucial difference between fields and tables is that useful answers for attribute values are returned for locations throughout the sampled range, not just the exact points where data was recorded.

Fields are typically characterized in terms of the number of keys versus values. Their **multivariate** structure depends on the number of value attributes, and their **multidimensional** structure depends on the number of keys. The standard multidimensional cases are 2D and 3D fields for static measurements taken in two or three spatial dimensions,[5] and fields with three or four keys, in the case where these measurements are time-varying. A field can be both multidimensional and multivariate if it has multiple keys and multiple values. The standard classification according to multivariate structure is that a **scalar field** has one attribute per cell, a **vector field** has two or more attributes per cell, and a **tensor field** has many attributes per cell.⋆

★ These definitions of *scalar*, *vector*, and *tensor* follow the common usage in vis. In a strict mathematical sense, these distinctions are not technically correct, since scalars and vectors are included as a degenerate case of tensors. Mapping the mathematical usage to the vis usage, **scalars** mean mathematical tensors of order 0, **vectors** mean mathematical tensors of order 1, and **tensors** mean mathematical tensors of order 2 or more.

2.6.1.4 Scalar Fields

A **scalar field** is univariate, with a single value attribute at each point in space. One example of a 3D scalar field is the time-varying medical scan above; another is the temperature in a room at each point in 3D space. The geometric intuition is that each point in a scalar field has a single value. A point in space can have several different numbers associated with it; if there is no underlying connection between them then they are simply multiple separate scalar fields.

2.6.1.5 Vector Fields

A **vector field** is multivariate, with a list of multiple attribute values at each point. The geometric intuition is that each point in a vector

[5] It's also possible for a spatial field to have just one key.

field has a direction and magnitude, like an arrow that can point in any direction and that can be any length. The length might mean the speed of a motion or the strength of a force. A concrete example of a 3D vector field is the velocity of air in the room at a specific time point, where there is a direction and speed for each item. The dimensionality of the field determines the number of components in the direction vector; its length can be computed directly from these components, using the standard Euclidean distance formula. The standard cases are two, three, or four components, as above.

2.6.1.6 Tensor Fields

A **tensor field** has an array of attributes at each point, representing a more complex multivariate mathematical structure than the list of numbers in a vector. A physical example is stress, which in the case of a 3D field can be defined by nine numbers that represent forces acting in three orthogonal directions. The geometric intution is that the full information at each point in a tensor field cannot be represented by just an arrow and would require a more complex shape such as an ellipsoid.

2.6.1.7 Field Semantics

This categorization of spatial fields requires knowledge of the attribute semantics and cannot be determined from type information alone. If you are given a field with multiple measured values at each point and no further information, there is no sure way to know its structure. For example, nine values could represent many things: nine separate scalar fields, or a mix of multiple vector fields and scalar fields, or a single tensor field.

2.6.2 Temporal Semantics

A **temporal** attribute is simply any kind of information that relates to time. Data about time is complicated to handle because of the rich hierarchical structure that we use to reason about time, and the potential for periodic structure. The time hierarchy is deeply multiscale: the scale of interest could range anywhere from nanoseconds to hours to decades to millennia. Even the common words *time* and *date* are a way to partially specify the scale of temporal interest. Temporal analysis tasks often involve finding or verifying periodicity either at a predetermined scale or at some scale not known in advance. Moreover, the temporal scales of interest do not all fit into a strict hierarchy; for instance, weeks do not fit

cleanly into months. Thus, the generic vis problems of transformation and aggregation are often particularly complex when dealing with temporal data. One important idea is that even though the dataset semantics involves change over time, there are many approaches to visually encoding that data—and only one of them is to show it changing over time in the form of an animation.

Temporal attributes can have either value or key semantics. Examples of temporal attributes with dependent value semantics are a duration of elapsed time or the date on which a transaction occurred. In both spatial fields and abstract tables, time can be an independent key. For example, a time-varying medical scan can have the independent keys of x, y, z, t to cover spatial position and time, with the dependent value attribute of density for each combination of four indices to look up position and time. A temporal key attribute is usually considered to have a quantitative type, although it's possible to consider it as ordinal data if the duration between events is not interesting.

▶ Section 3.4.2.3 introduces the problem of data transformation. Section 13.4 discusses the question of aggregation in detail.

▶ Vision versus memory is discussed further in Section 6.5.

2.6.2.1 Time-Varying Data

A dataset has **time-varying** semantics when time is one of the key attributes, as opposed to when the temporal attribute is a value rather than a key. As with other decisions about semantics, the question of whether time has key or value semantics requires external knowledge about the nature of the dataset and cannot be made purely from type information. An example of a dataset with time-varying semantics is one created with a sensor network that tracks the location of each animal within a herd by taking new measurements every second. Each animal will have new location data at every time point, so the temporal attribute is an independent key and is likely to be a central aspect of understanding the dataset. In contrast, a horse-racing dataset covering a year's worth of races could have temporal value attributes such as the race start time and the duration of each horse's run. These attributes do indeed deal with temporal information, but the dataset is not time-varying.

A common case of temporal data occurs in a **time-series** dataset, namely, an ordered sequence of time–value pairs. These datasets are a special case of tables, where time is the key. These time-value pairs are often but not always spaced at uniform temporal intervals. Typical time-series analysis tasks involve finding trends, correlations, and variations at multiple time scales such as hourly, daily, weekly, and seasonal.

The word **dynamic** is often used ambiguously to mean one of two very different things. Some use it to mean a dataset has *time-varying* semantics, in contrast to a dataset where time is not a key attribute, as discussed here. Others use it to mean a dataset has *stream* type, in contrast to an unchanging file that can be loaded all at once. In this latter sense, items and attributes can be added or deleted and their values may change during a running session of a vis tool. I carefully distinguish between these two meanings here.

► The dataset types of dynamic streams versus static files are discussed in Section 2.4.6.

2.7 Further Reading

The Big Picture The framework presented here was inspired in part by the many taxonomies of data that have been previously proposed, including the synthesis chapter at the beginning of an early collection of infovis readings [Card et al. 99], a taxonomy that emphasizes the division between continuous and discrete data [Tory and Möller 04a], and one that emphasizes both data and tasks [Shneiderman 96].

Field Datasets Several books discuss the spatial field dataset type in far more detail, including two textbooks [Telea 07, Ward et al. 10], a voluminous handbook [Hansen and Johnson 05], and the *vtk* book [Schroeder et al. 06].

Attribute Types The attribute types of categorical, ordered, and quantitative were proposed in the seminal work on scales of measurement from the psychophysics literature [Stevens 46]. Scales of measurement are also discussed extensively in the book *The Grammar of Graphics* [Wilkinson 05] and are used as the foundational axes of an influential vis design space taxonomy [Card and Mackinlay 97].

Key and Value Semantics The Polaris vis system, which has been commercialized as Tableau, is built around the distinction between key attributes (independent dimensions) and value attributes (dependent measures) [Stolte et al. 02].

Temporal Semantics A good resource for time-oriented data vis is a recent book, *Visualization of Time-Oriented Data* [Aigner et al. 11].

MARKS & CHANNELS

How can I visually represent two numbers

e.g., 4 and 8

MARKS & CHANNELS

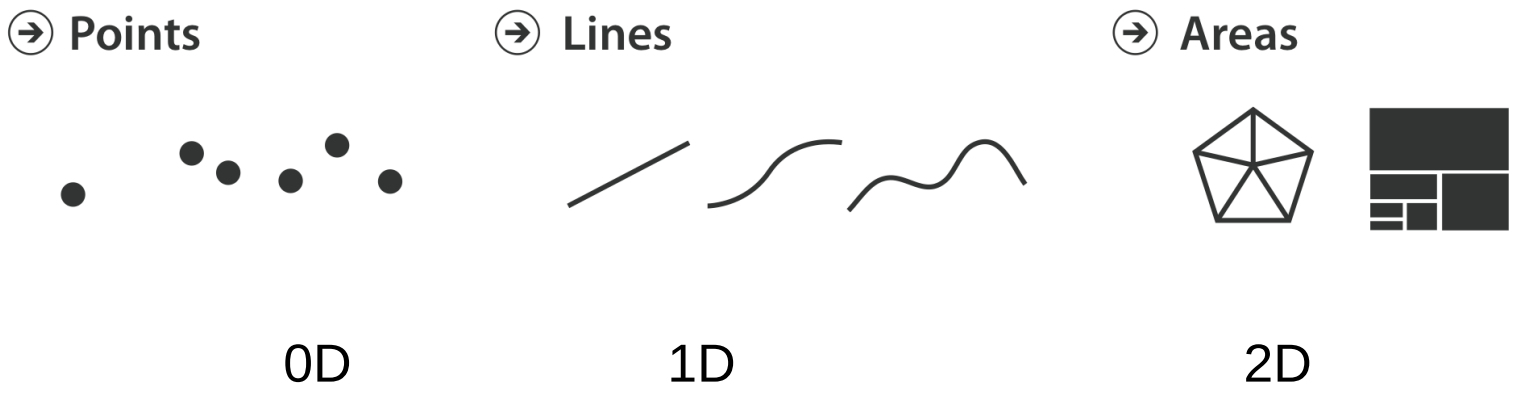
Marks: represent **items** or **links**

Channels: change **appearance** based on **attribute**

Channel = visual variable

MARKS FOR ITEMS

Basic geometric elements

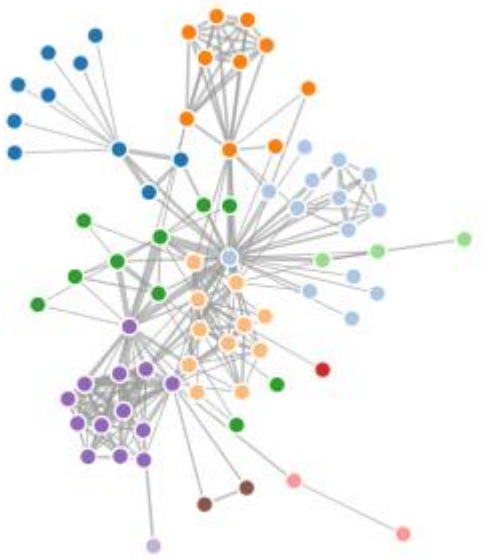


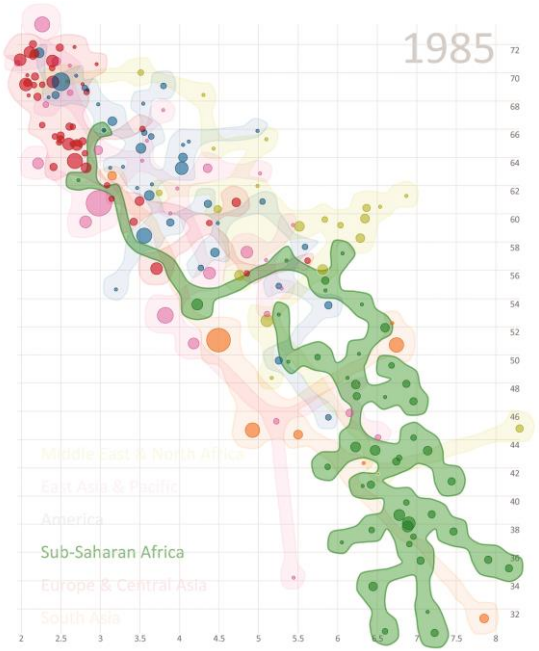
3D mark: Volume, but rarely used

MARKS FOR LINKS

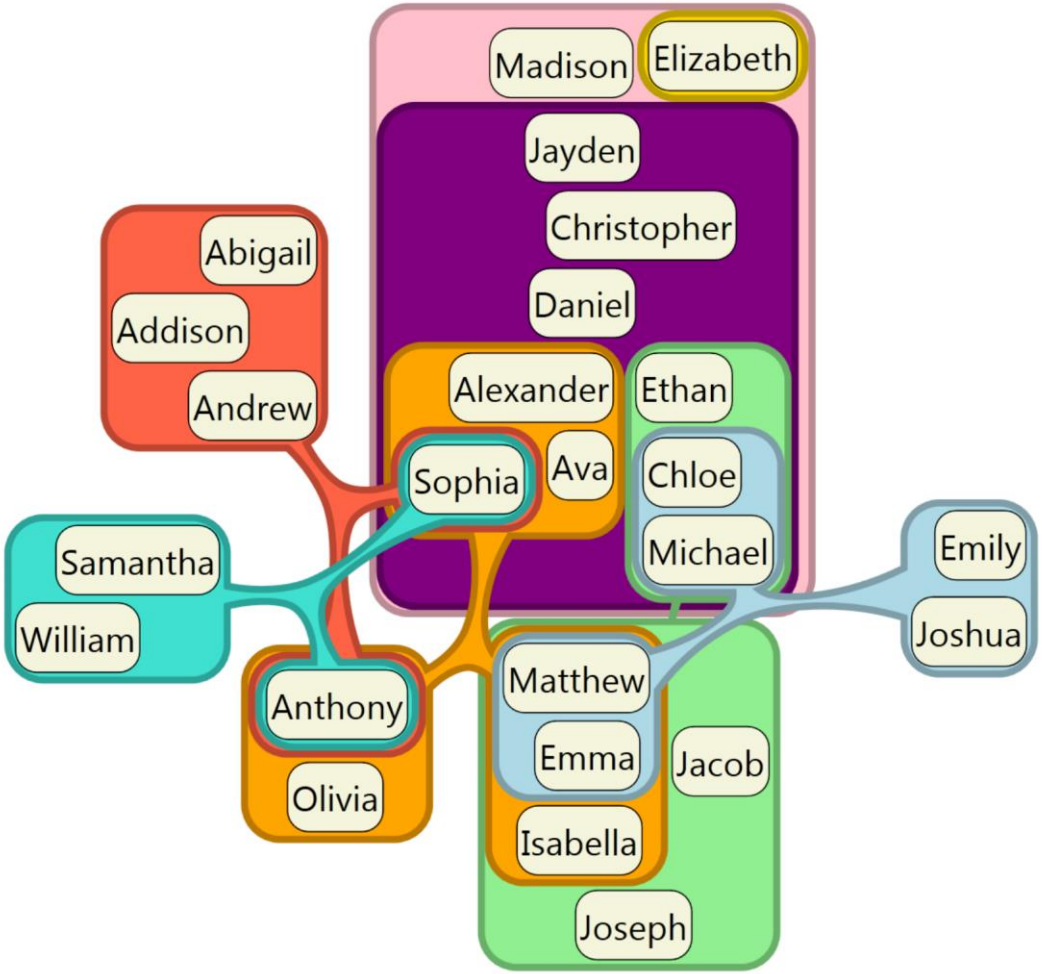
Containment

Connection





NESTED CONTAINMENT



CHANNELS

Change appearance proportional to or based on attributes

- **Position**
  - Horizontal
  - Vertical
  - Both

- **Color**

- **Shape**

- **Tilt**

- **Size**
  - Length
  - Area
  - Volume

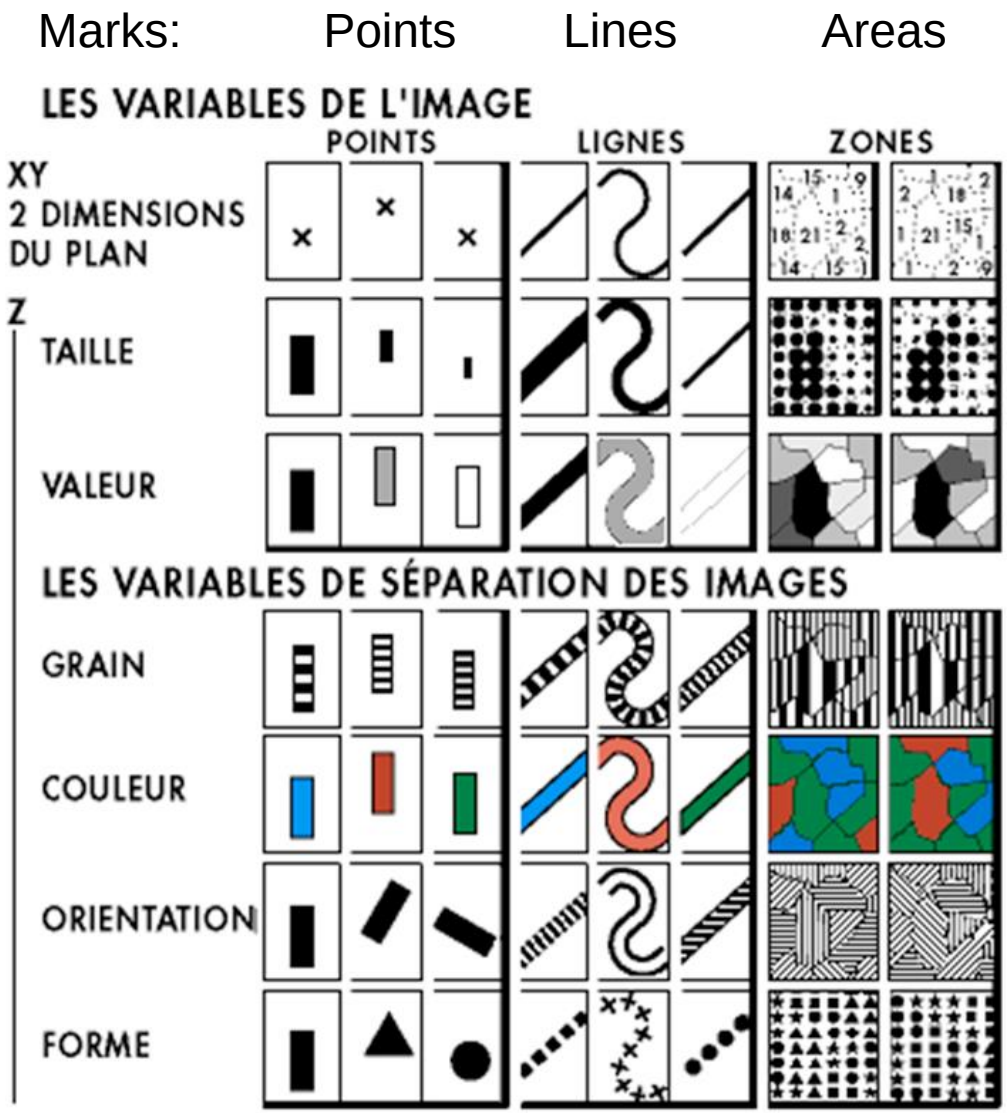
JACQUES BERTIN

French cartographer

who makes or draws maps

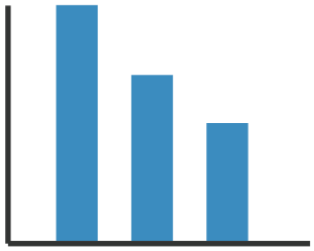
Theoretical principles for visual encodings

BERTIN'S VISUAL VARIABLES



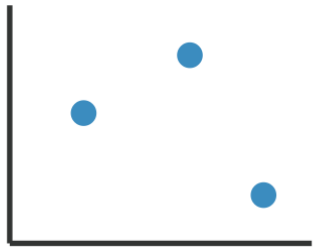
Semiology of Graphics [J. Bertin, 67]

USING MARKS AND CHANNELS



Mark: Line

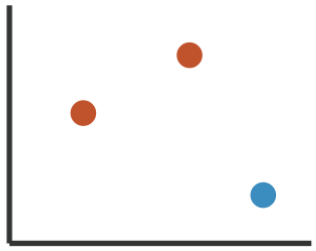
Channel:
Length, Position
1 quantitative attribute
1 categorical attribute



Mark: Point

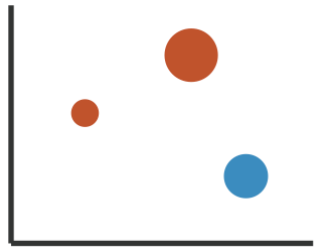
Channel: Position

1 quantitative attr.



Adding Hue

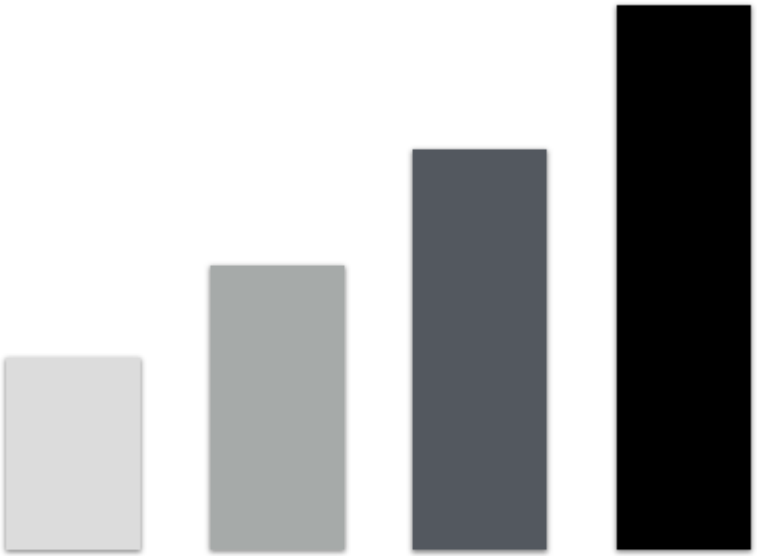
+1 categorical attr.



Adding Size

+1 quantitative attr.

REDUNDANT ENCODING



Length, Position and Value

GOOD BAR CHART?

Rule: Use channel proportional to data!

https://twitter.com/ChaseThomason/status/1118478036507164672?s=19

TYPES OF CHANNELS

Magnitude channels

How much? Which rank?

Position

Length

Saturation …

Identity channels

What?

Shape

Color (hue)

Spatial region...

Ordinal & Quantitative data

Categorical Data

PRINCIPLES OF EXPRESSIVENESS AND EFFECTIVENESS

Expressiveness principle

The visual encoding should express **all of, and only**, the information in the dataset attributes

Effectiveness principle:

The importance of attribute should match the **salience** of the channel

Means:

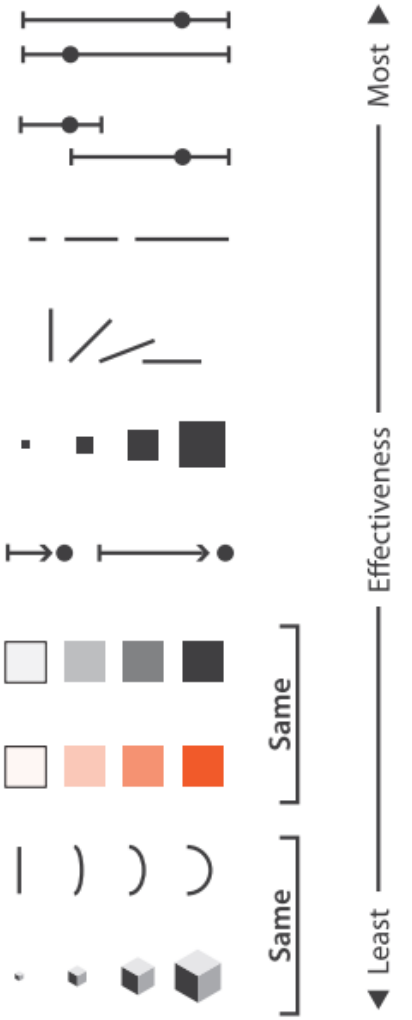
- The most important attributes should be encoded with the most effective channels in order to be most noticeable
- Then the following important attributes match with less effective channels

RANK OF CHANNELS

Channels: Expressiveness Types and Effectiveness Ranks

Magnitude Channels: Ordered Attributes

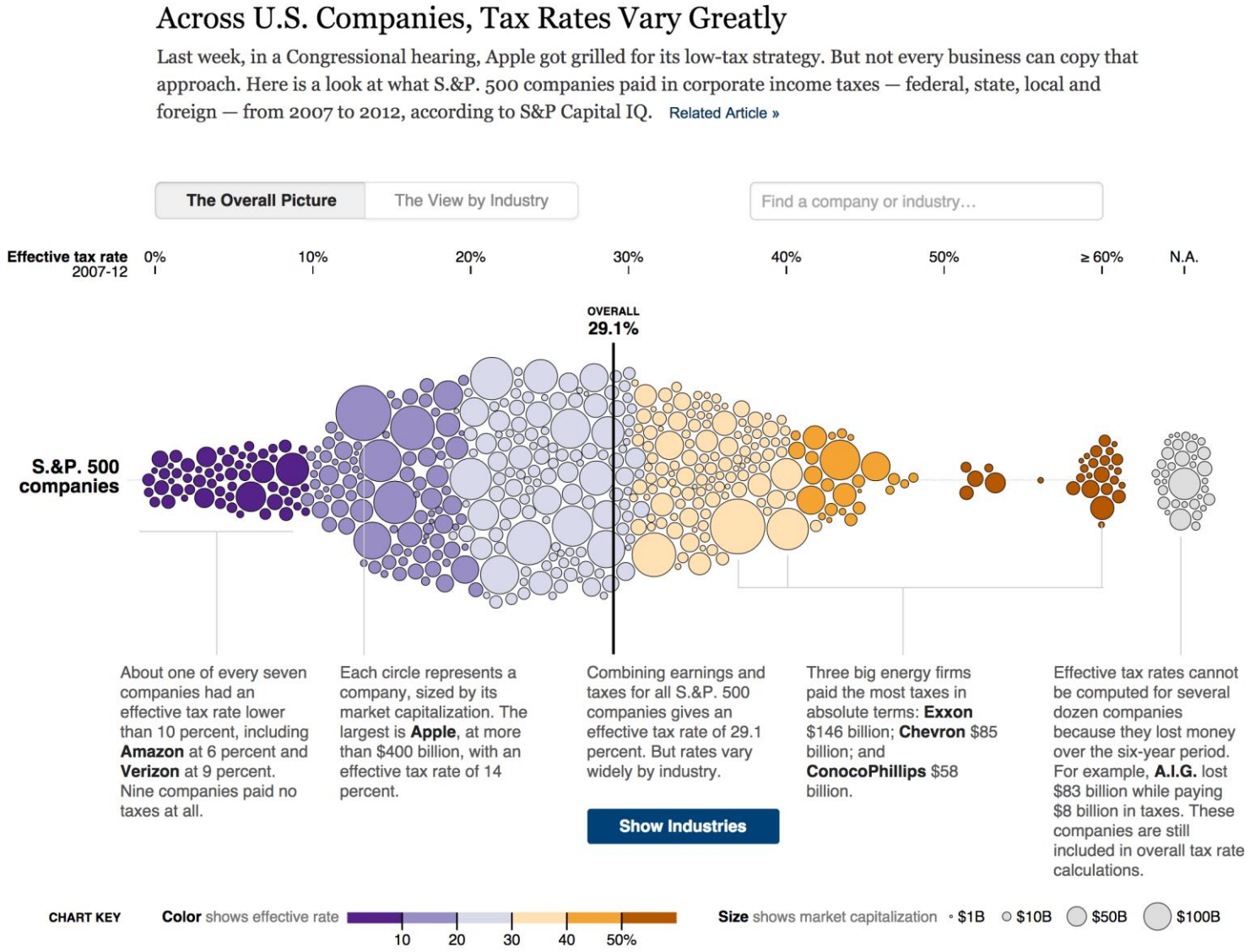
- Position on common scale
- Position on unaligned scale
- Length (1D size)
- Tilt/angle
- Area (2D size)
- Depth (3D position)
- Color luminance
- Color saturation
- Curvature
- Volume (3D size)



Identity Channels: Categorical Attributes

- Spatial region
- Color hue
- Motion
- Shape

WHAT VISUAL VARIABLES ARE USED?



http://www.nytimes.com/interactive/2013/05/25/sunday-review/corporate-taxes.html

CHARACTERISTICS OF CHANNELS

Selective

- Is a mark distinct from other marks?
- Can we make out the difference between two marks?

Associative

- Does it support grouping?

Quantitative (Magnitude vs identity channels)

- Can we quantify the difference between two marks?

CHARACTERISTICS OF CHANNELS

Order (Magnitude vs Identity)

Can we see a change in order?

Length

How many unique marks can we make?

POSITION

Strongest visual variables

Suitable for all data types

Problems:

- Sometimes not available (spatial data, map)
- Cluttering (many items overlapped)

Selective: Yes

Associative: Yes

Quantitative: Yes

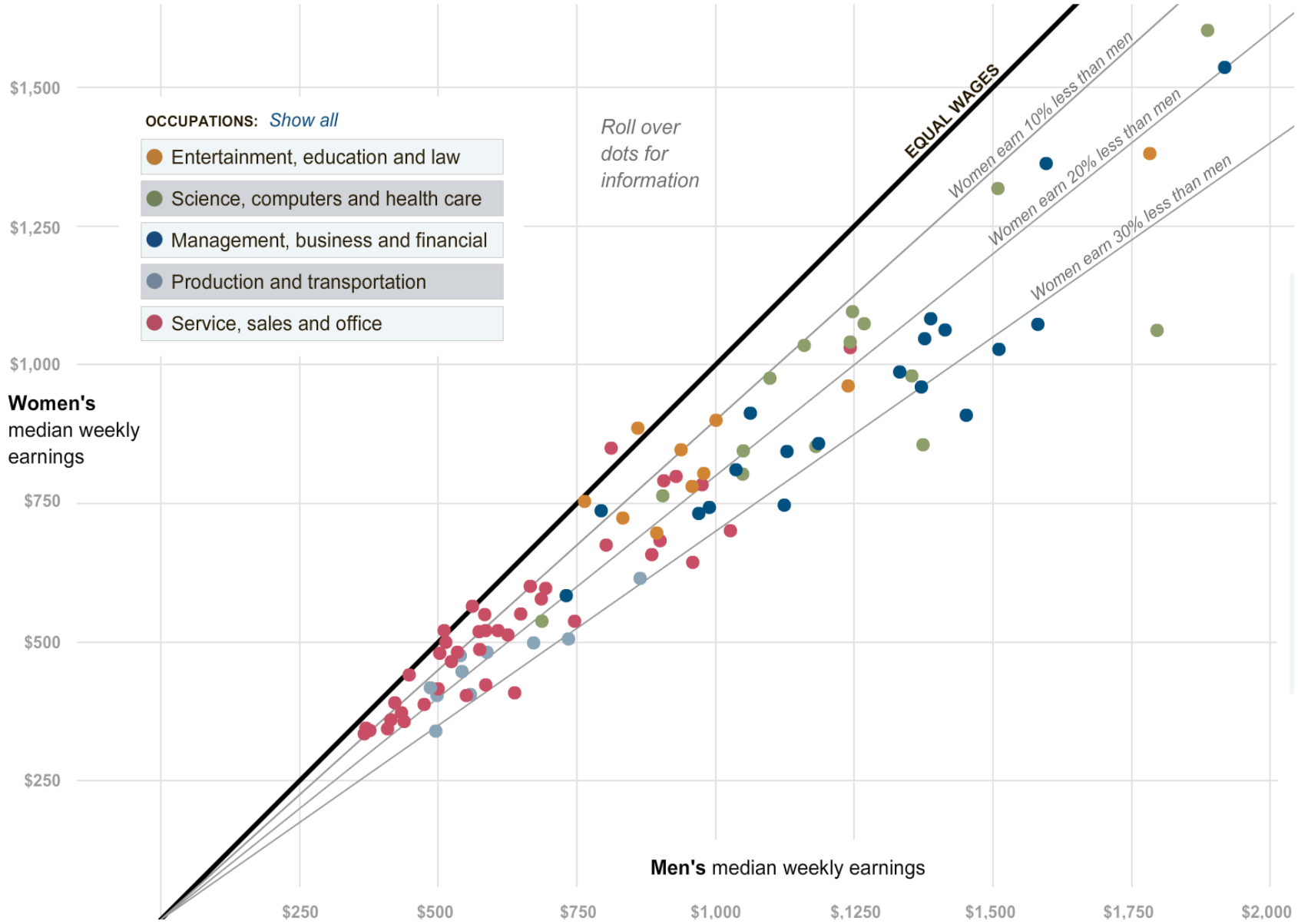
Order: Yes

Length: Fairly big

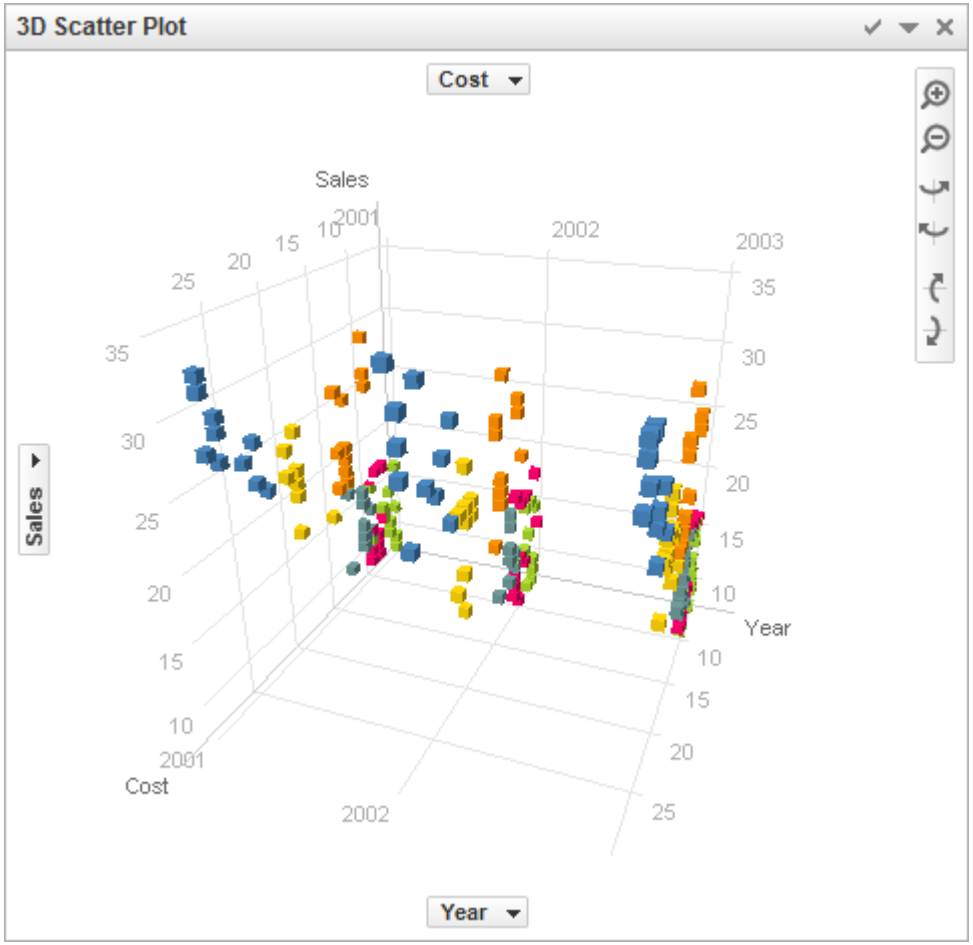
(options)

→ Good channel

EXAMPLE: SCATTERPLOT



POSITION IN 3D? NOT SO GOOD



LENGTH & SIZE

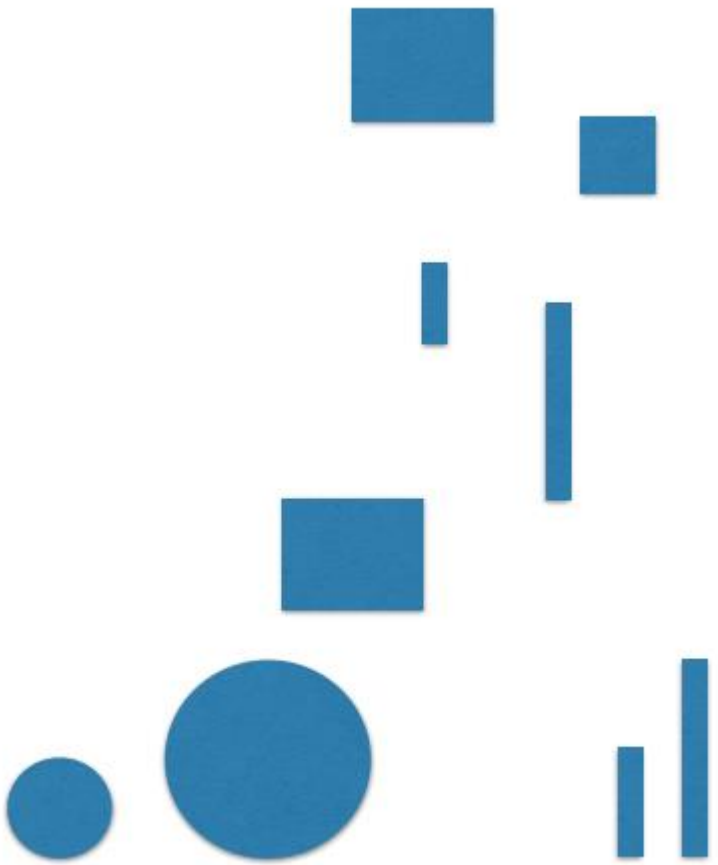
Good for 1D, OK for 2D, Bad for 3D

Easy to see which one is bigger

Aligned bars use position redundantly

1D length:

Selective:	yes
Associative:	yes
Quantitative:	yes
Order:	yes
Length:	yes



EXAMPLE 2D SIZE: BUBBLES

Four Ways to Slice Obama's 2013 Budget Proposal

Explore every nook and cranny of President Obama's federal budget proposal.

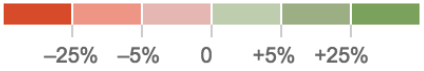
All Spending | Types of Spending | Changes | Department Totals

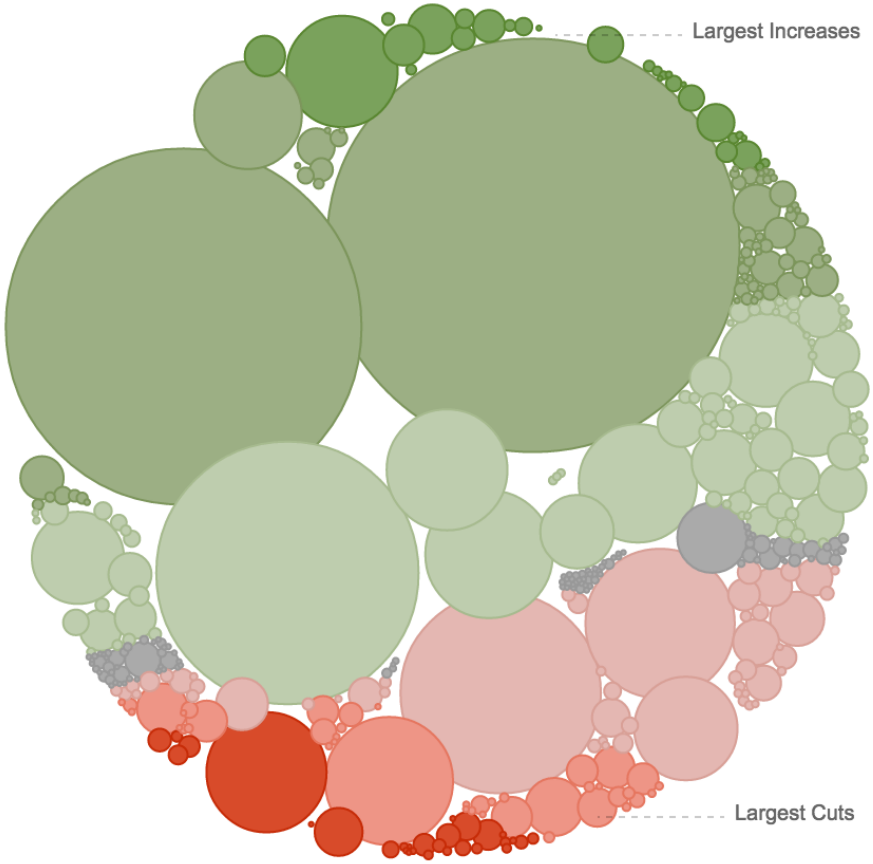
How $3.7 Trillion Is Spent

Mr. Obama's budget proposal includes $3.7 trillion in spending in 2013, and forecasts a $901 billion deficit.

Circles are sized according to the proposed spending.

Color shows amount of cut or increase from 2012.





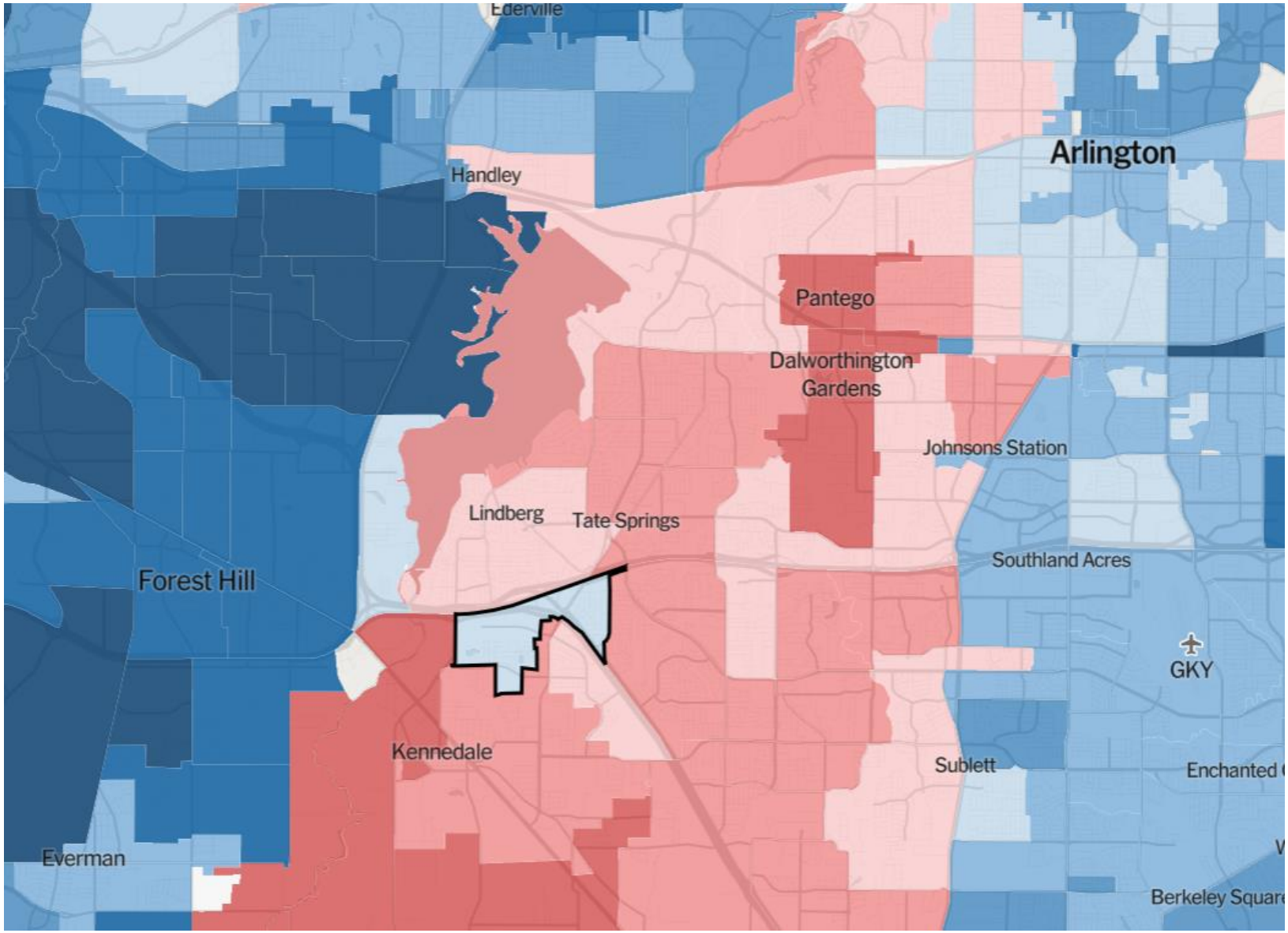
VALUE/LUMINANCE/SATURATION

OK for quantitative data when length & size are used

Not very many shades recognizable

Selective:	yes
Associative:	yes
Quantitative:	somewhat
Order:	yes
Length:	limited (around 7 – 8)

EXAMPLE: DIVERGING VALUE-SCALE



https://www.nytimes.com/interactive/2018/upshot/election-2016-voting-precinct-maps.html#11.08/32.700/-97.197/115304

COLOR

Good for qualitative data (identity channel)

Limited number of classes/length (7-10)

Do not work for quantitative data

Lots of pitfalls !

Good practice: minimize use of color for encoding data

Selective:	yes
Associative:	yes
Quantitative:	no
Order:	no
Length:	limited

COLOR: BAD EXAMPLE

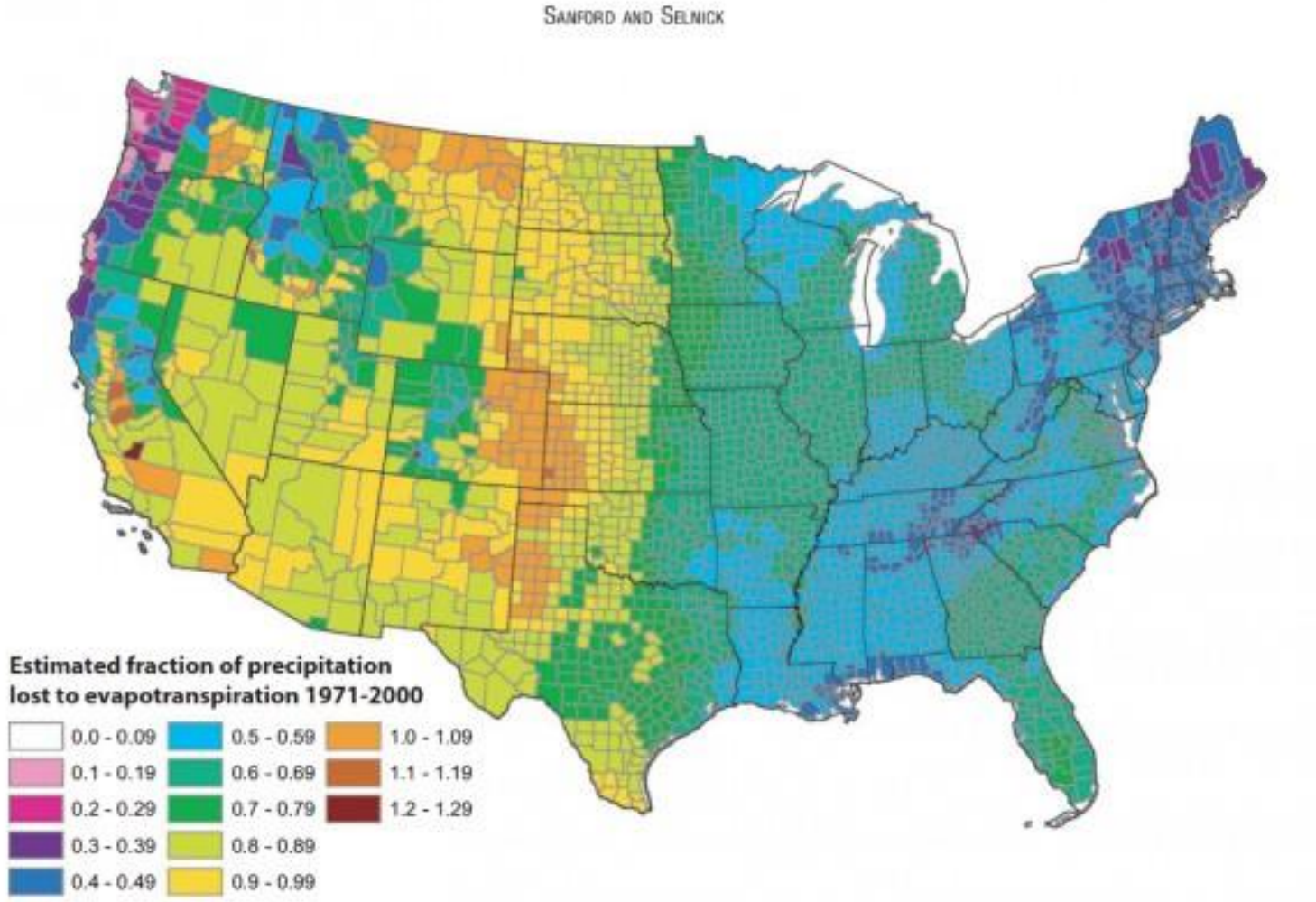
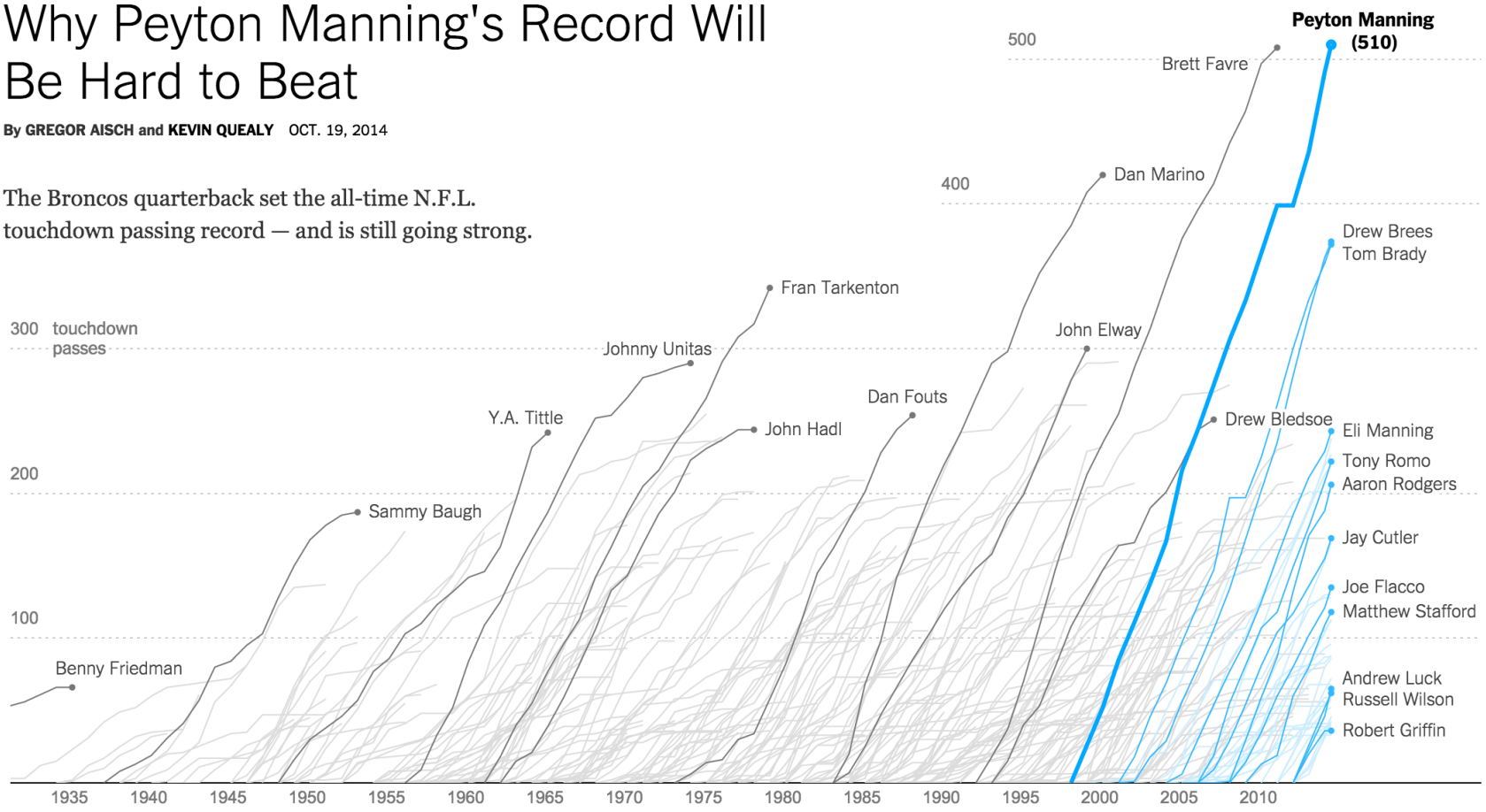


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

COLOR: GOOD EXAMPLE

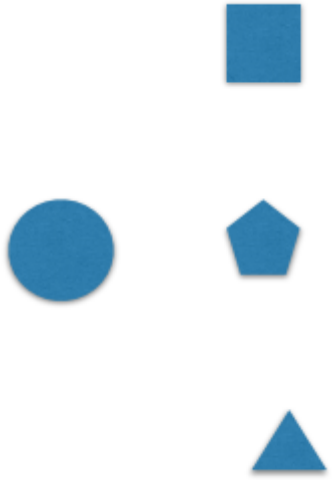


SHAPE

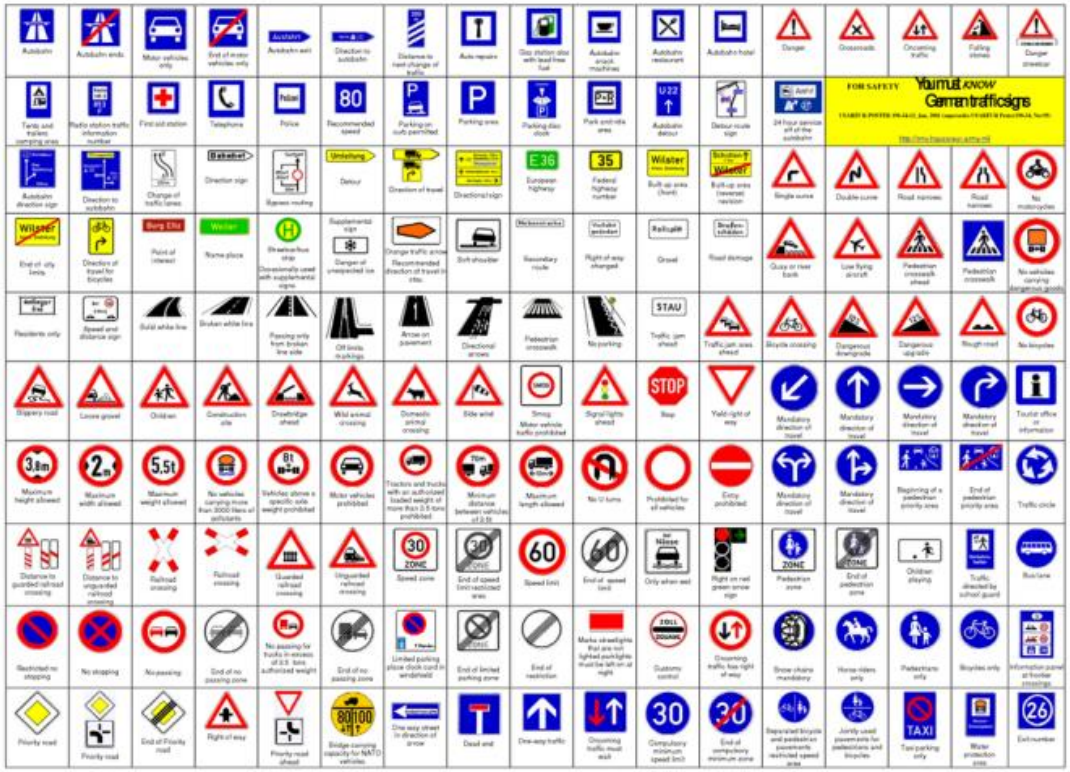
Great to recognize many classes

No grouping, ordering

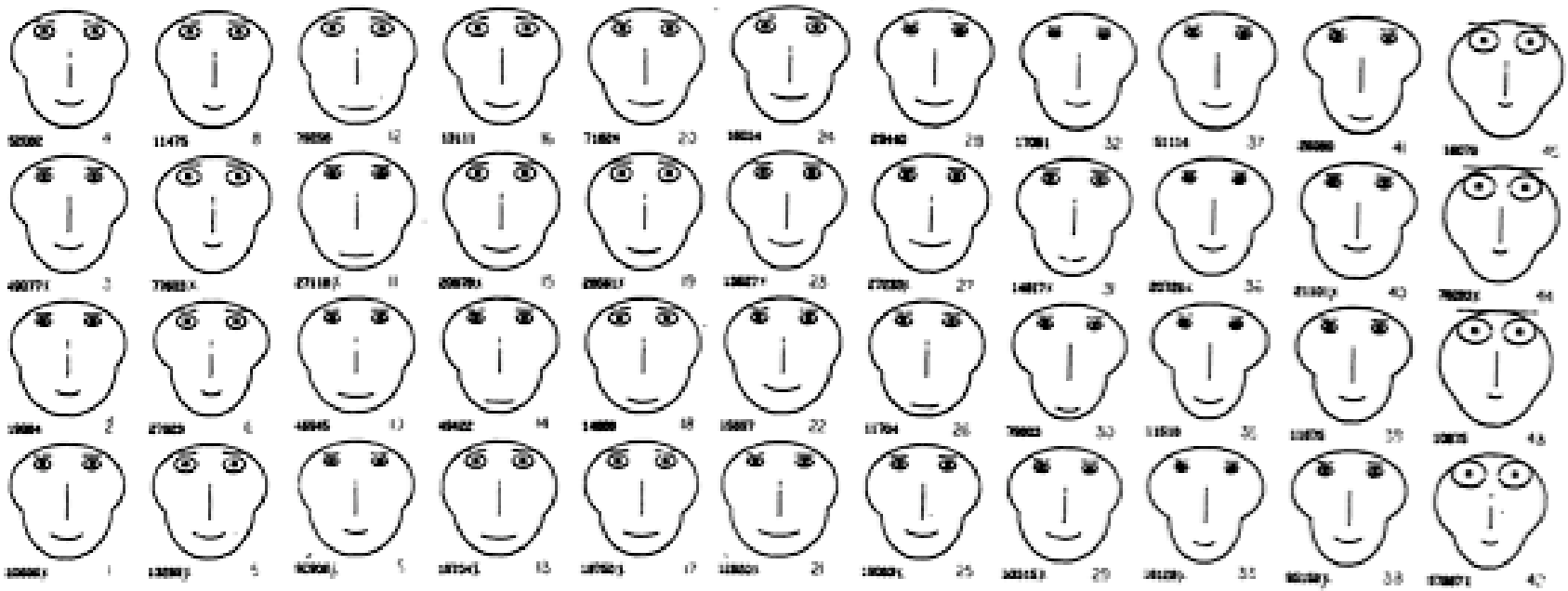
Selective:	yes
Associative:	limited
Quantitative:	no
Order:	no
Length:	vast



ALFA ROMEO
ACURA
ASTON MARTIN
ASTON MARTIN
Audi
BENTLEY
BMW
MITSUBISHI MOTORS
Ferrari
Photoshop plugins



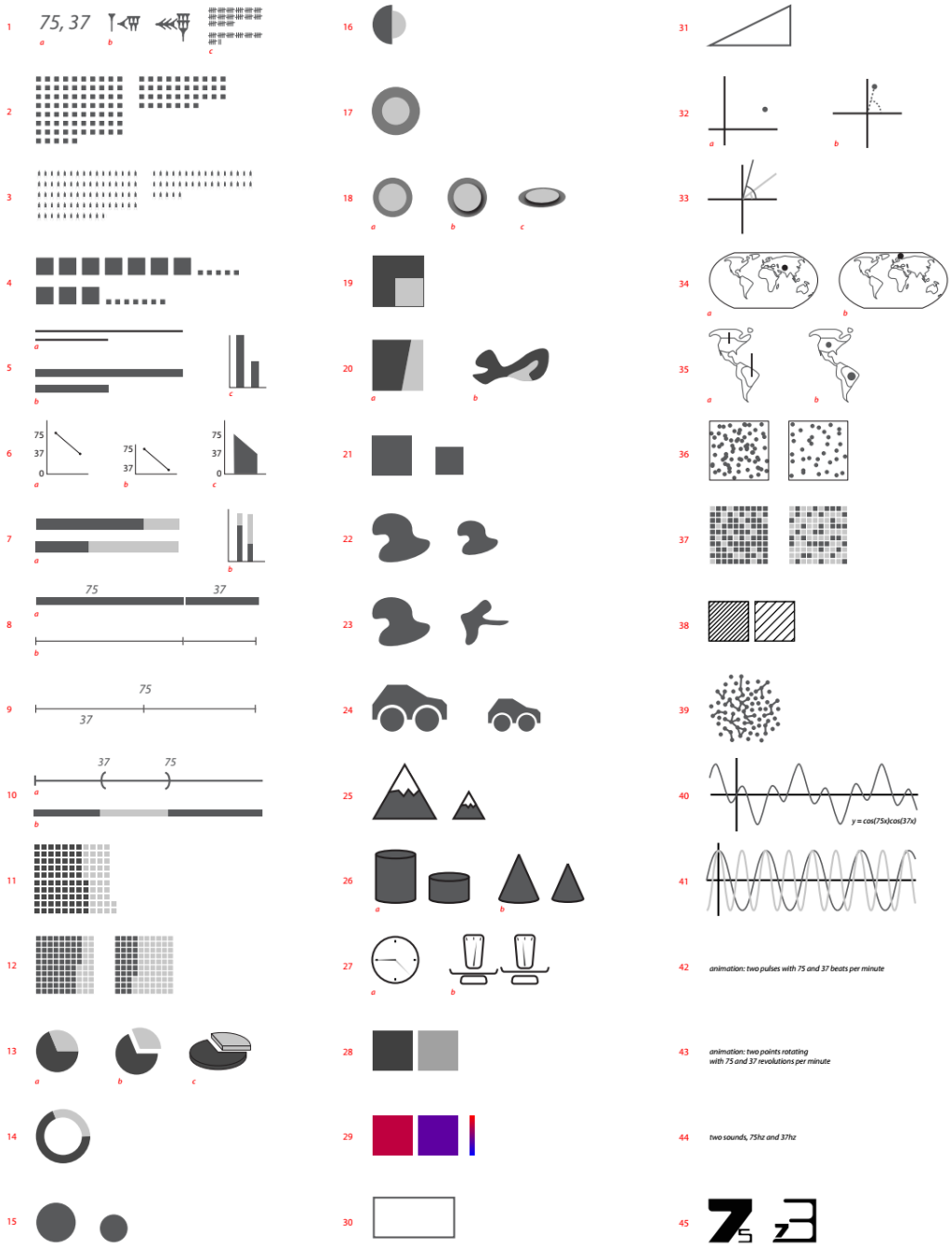
Idea: use facial parameters to map quantitative data



Does it work? Not really

Critique: https://eagereyes.org/criticism/chernoff-faces

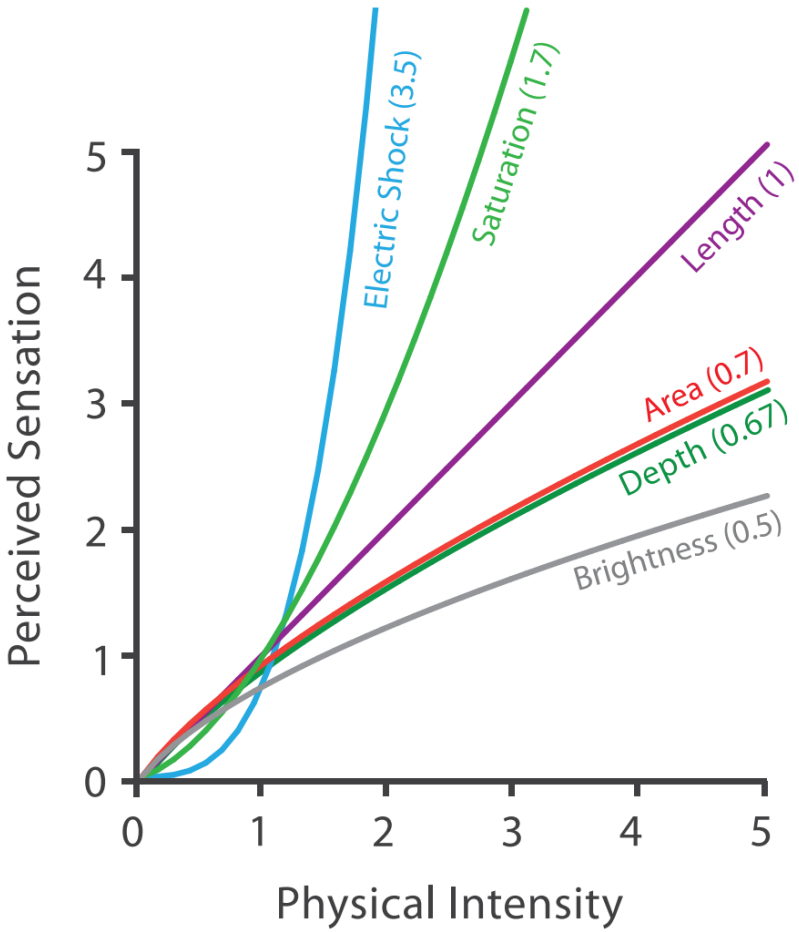
MORE CHANNELS



https://en.rockcontent.com/blog/2012/07/27/45-ways-to-communicate-two-quantities/

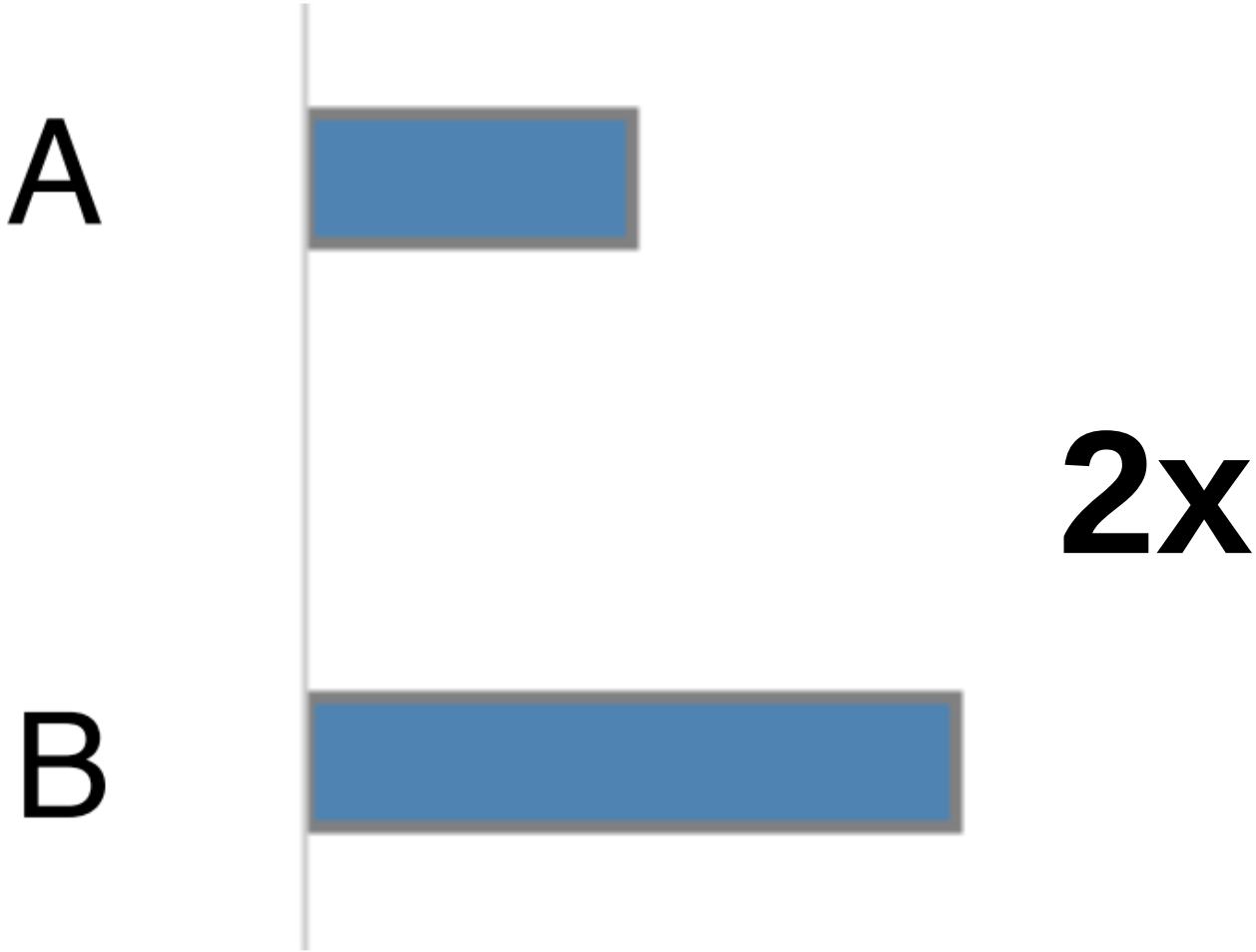
ACCURACY OF CHANNELS

Steven's Psychophysical Power Law: $S = I^N$

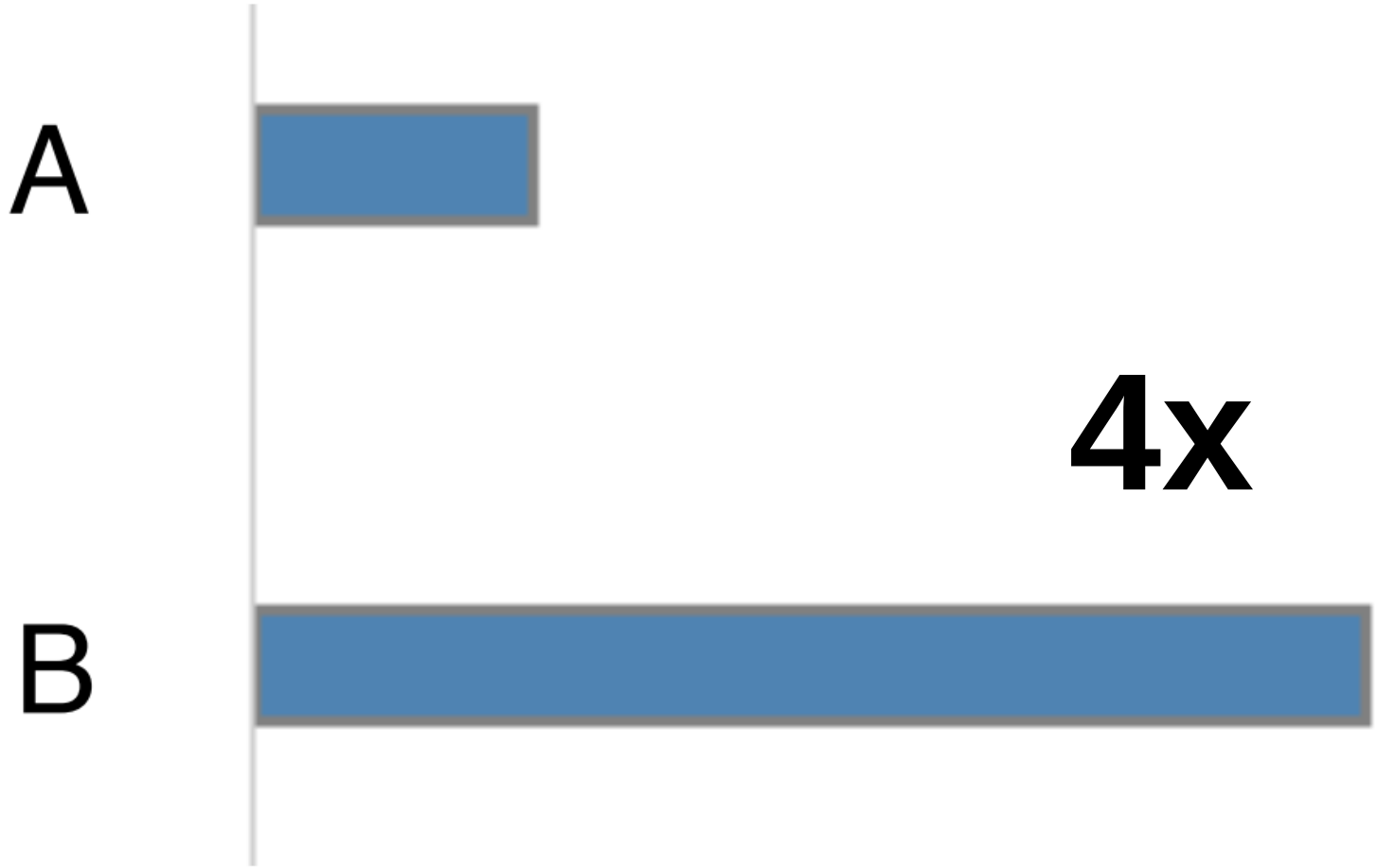


S = sensation
I = intensity

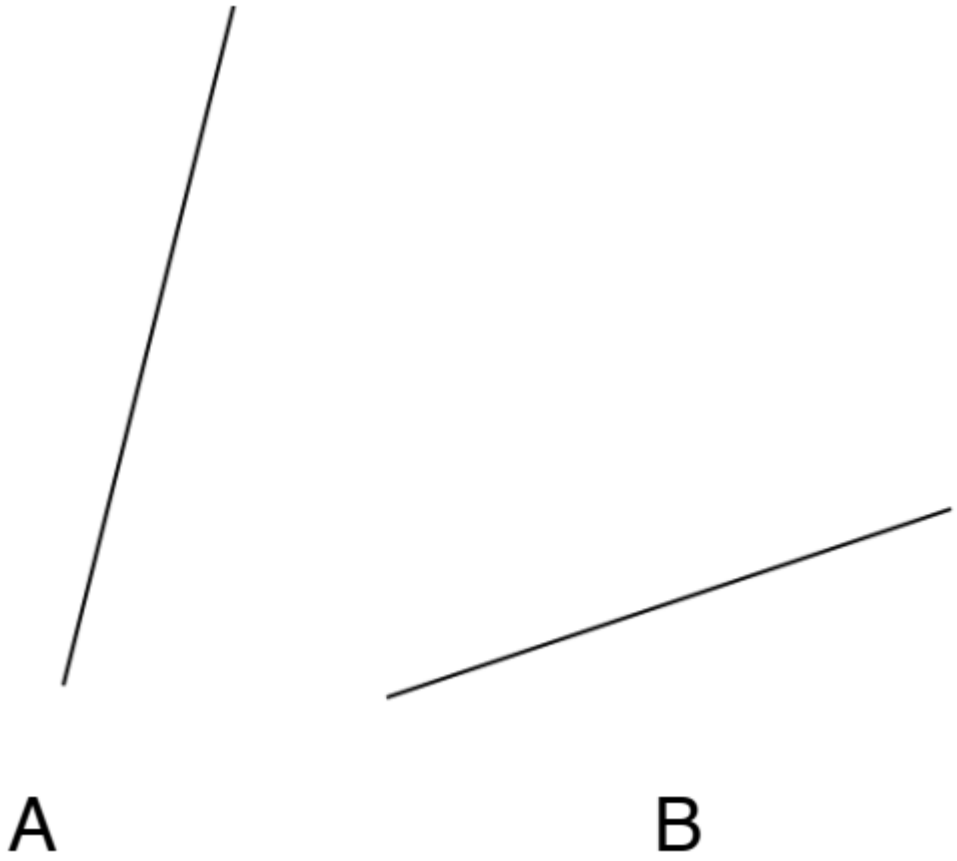
HOW MUCH LONGER?



HOW MUCH LONGER

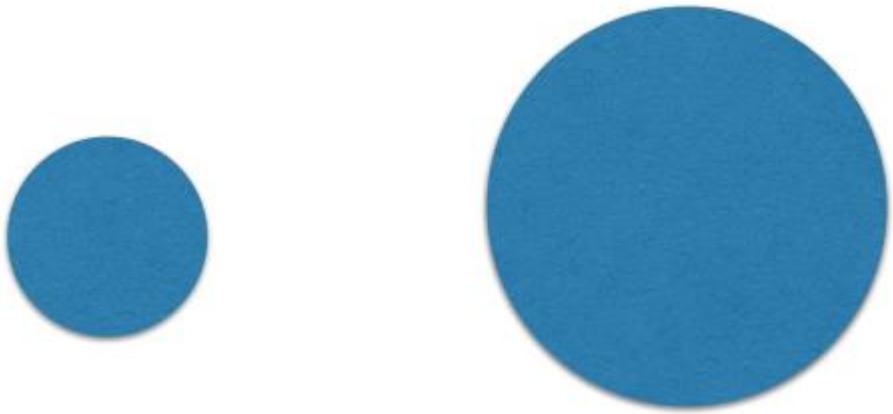


HOW MUCH STEEPER?



~4x

HOW MUCH LARGER?



A B

2x diameter
4x area

HOW MUCH LARGER?

A

B

5x

HOW MUCH LARGER (AREA)?

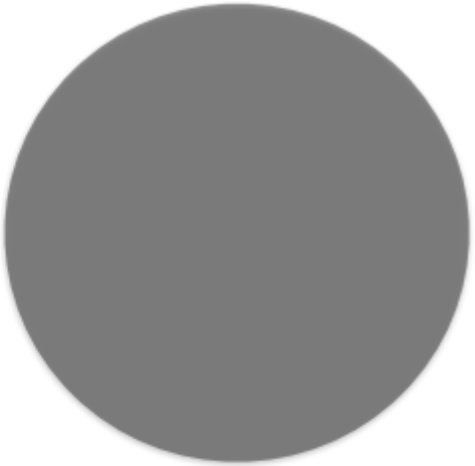
A

B

3x

HOW MUCH DARKER?

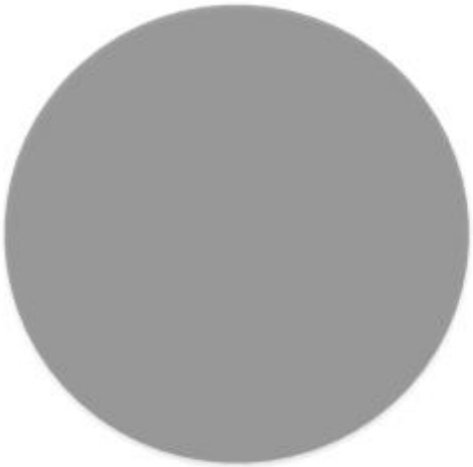
A



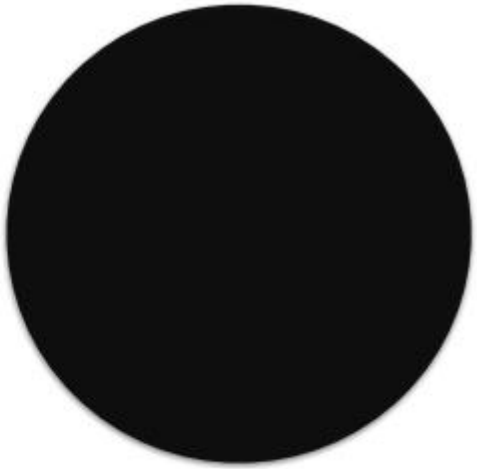
B

2x

HOW MUCH DARKER?



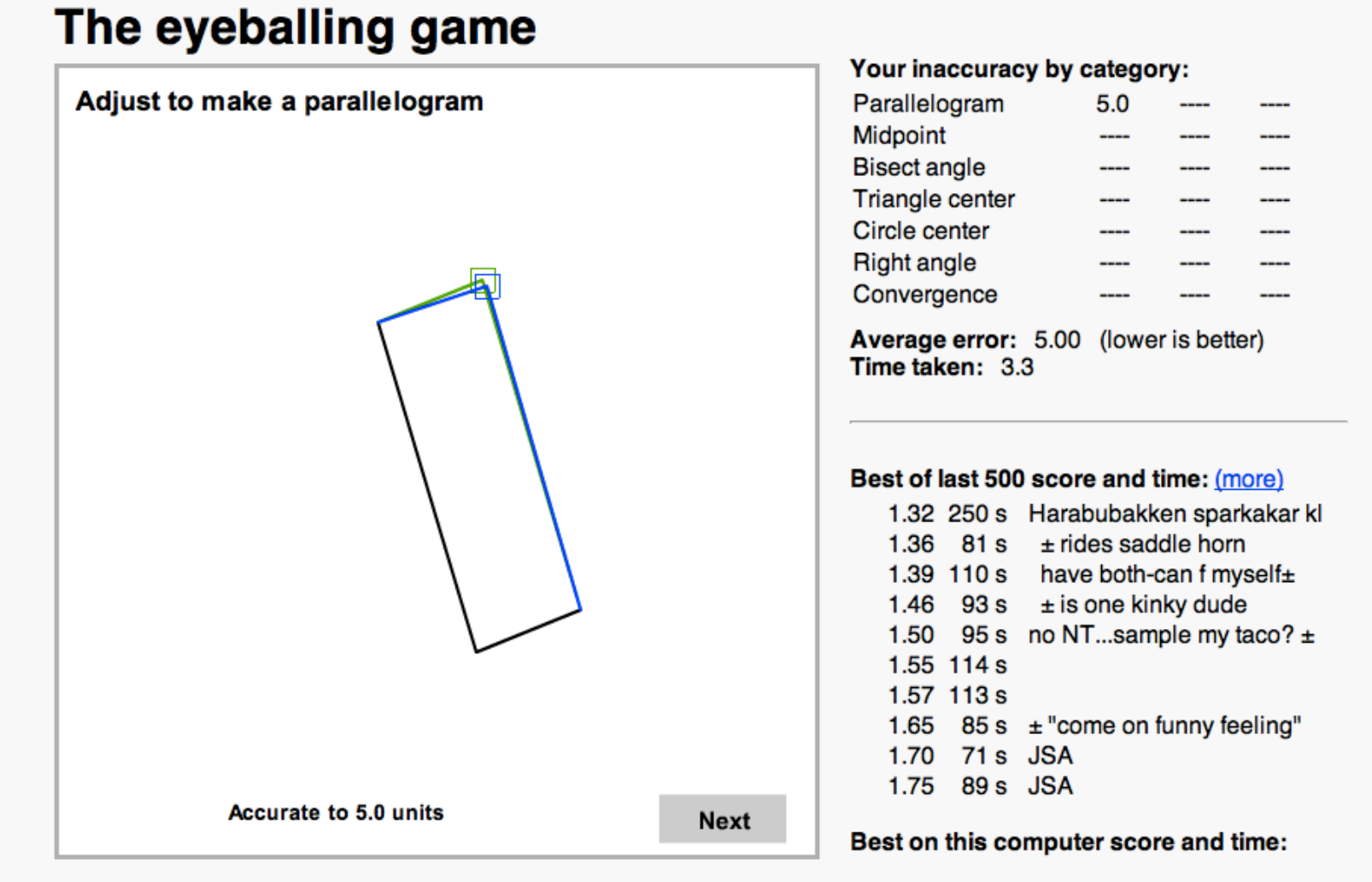
A



B

3x

POSITION, LENGTH & ANGLE



https://woodgears.ca/eyeball/index.html

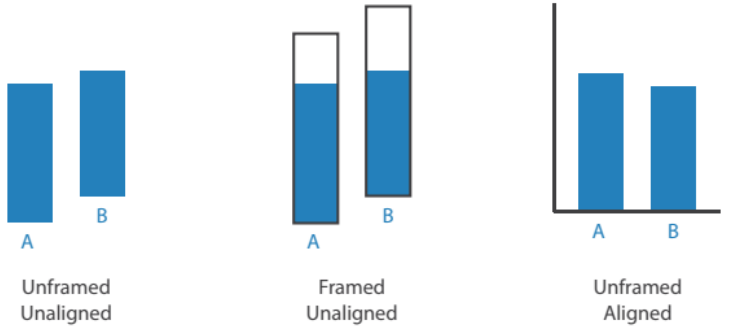
OTHER FACTORS AFFECTING ACCURACY

Alignment

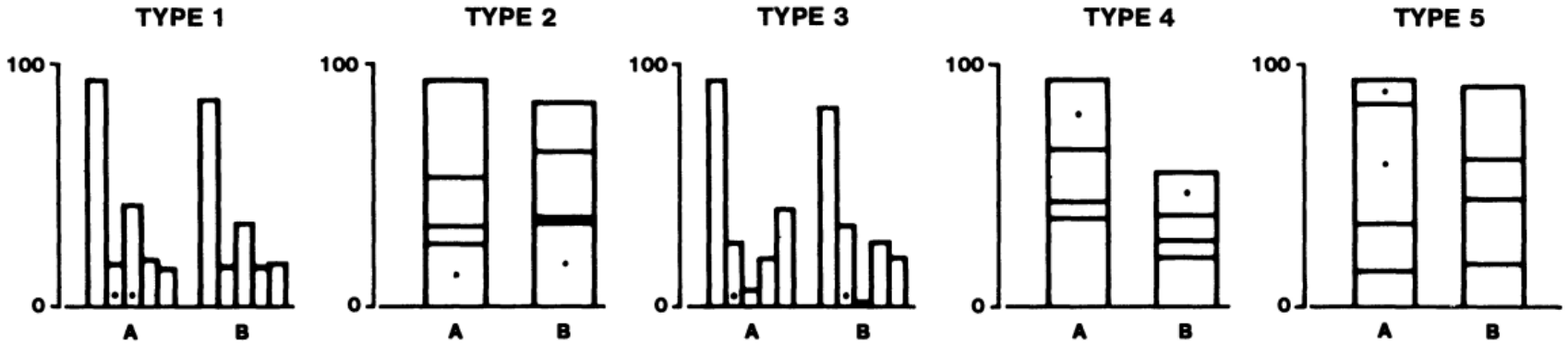
Distractors

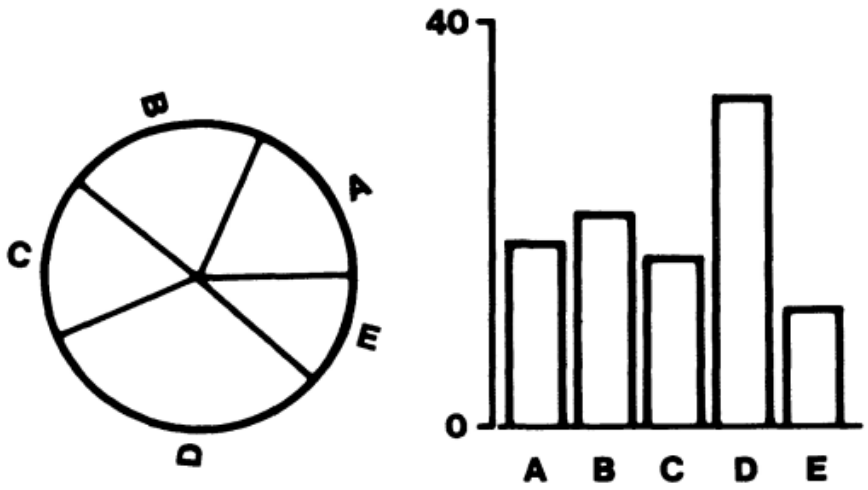
Distance

Common scale

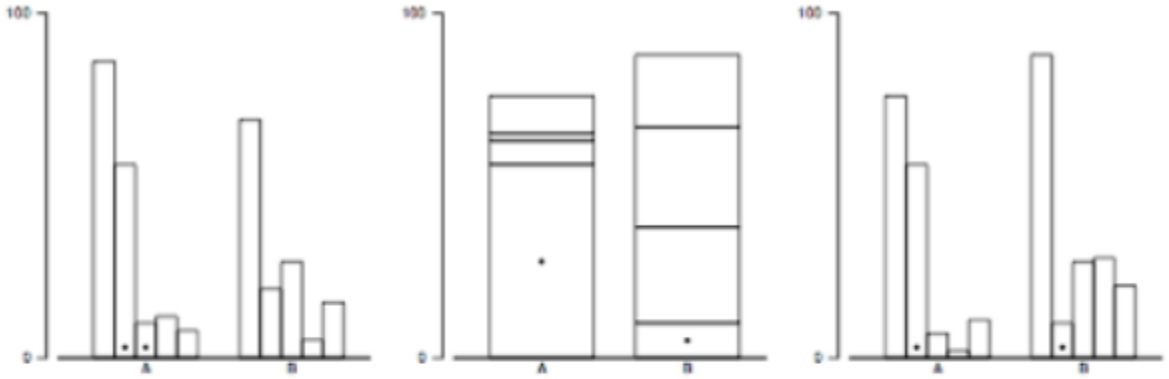


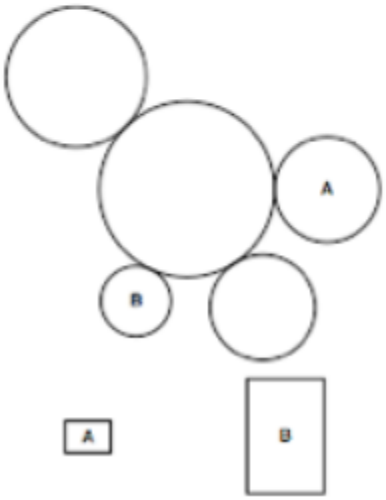
RESEARCH BY CLEVELAND 1984

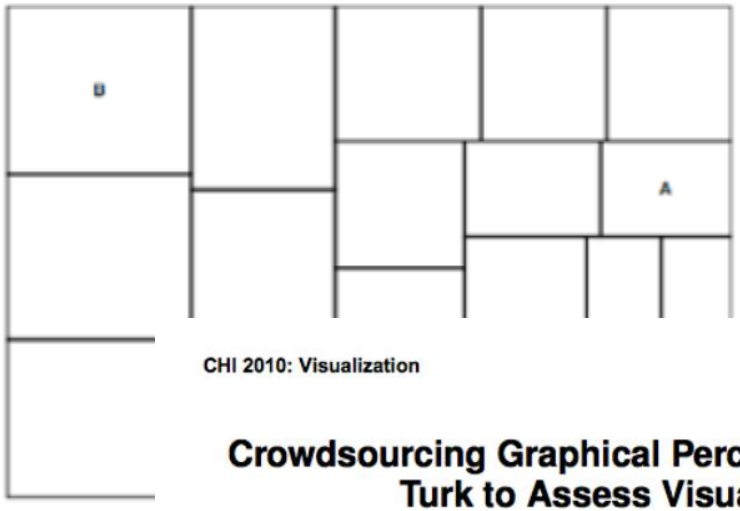




HEER & BOSTOCK, 2010







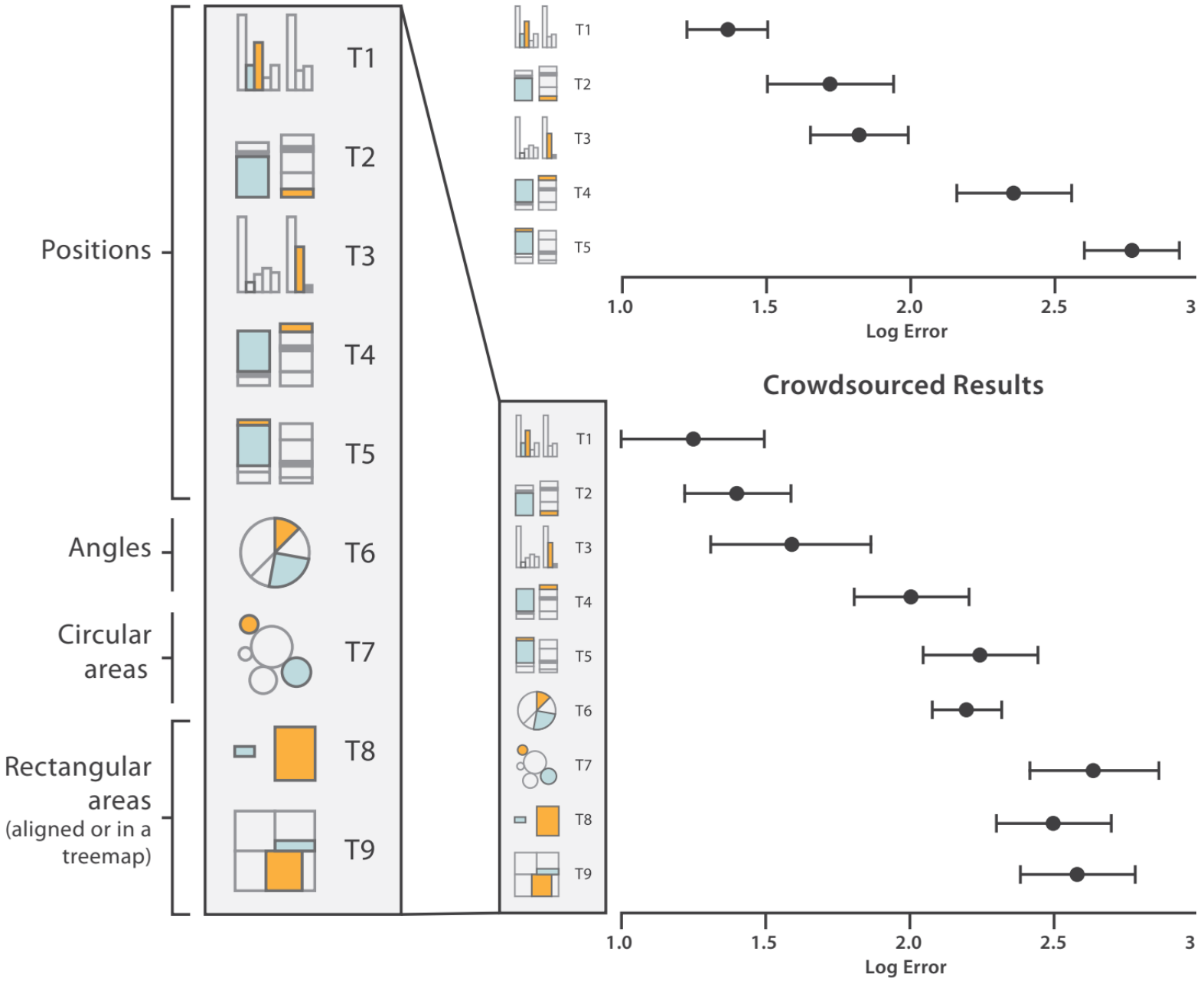
CHI 2010: Visualization

April 10–15, 2010, Atlanta, GA, USA

Crowdsourcing Graphical Perception: Using Mechanical Turk to Assess Visualization Design

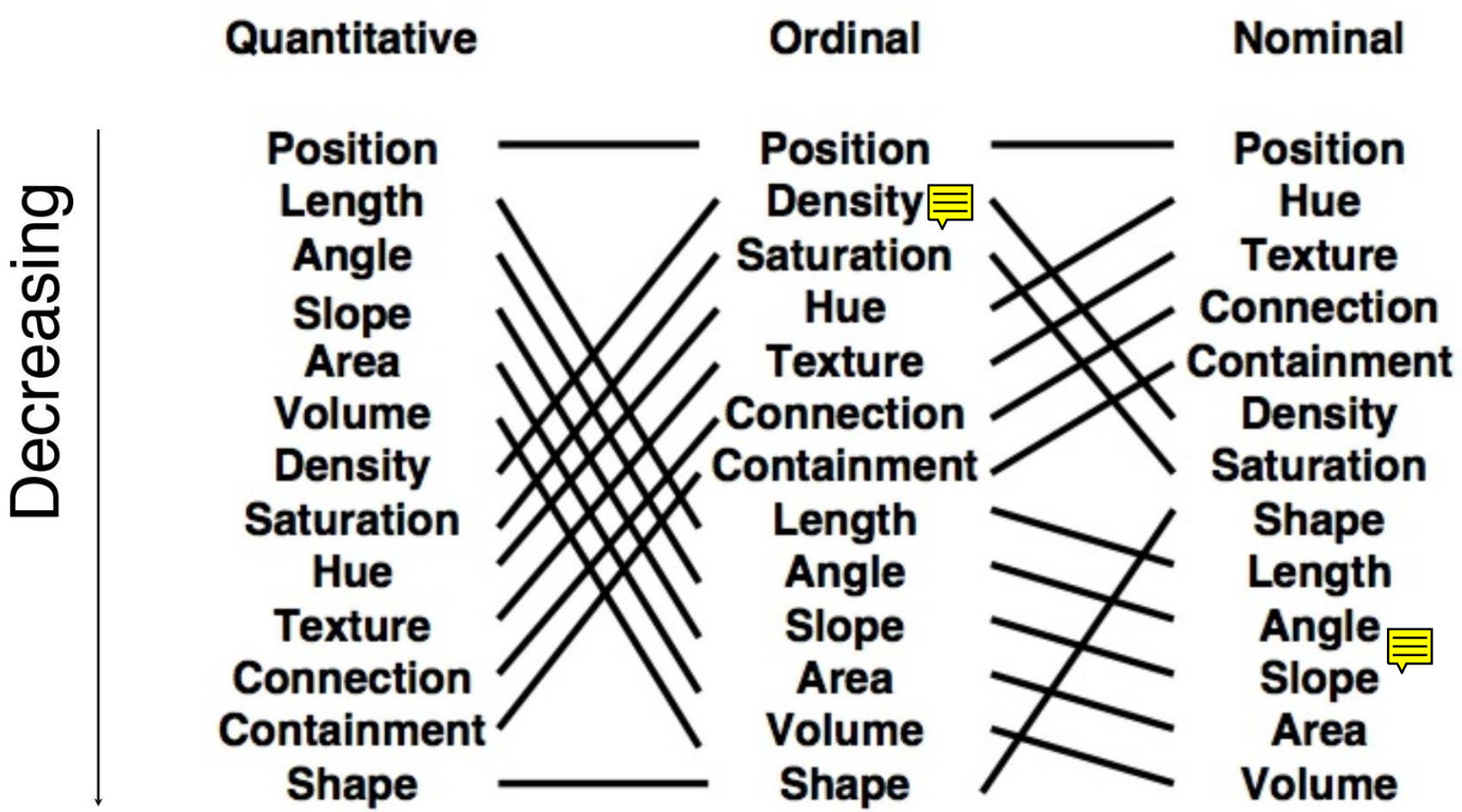
Jeffrey Heer and Michael Bostock
Computer Science Department
Stanford University
{jheer, mbostock}@cs.stanford.edu

https://www.mturk.com/worker



Error = log2(judged percent - true percent + 1/8)

JOCK MACKINLAY, 1986

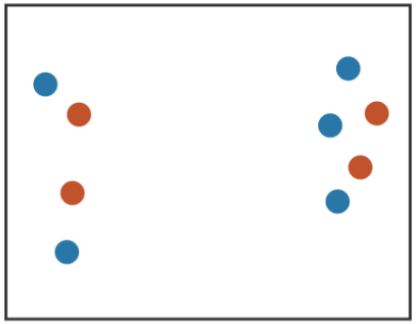


[Mackinlay, Automating the Design of Graphical Presentations of Relational Information, 1986]

SEPARABILITY OF ATTRIBUTES

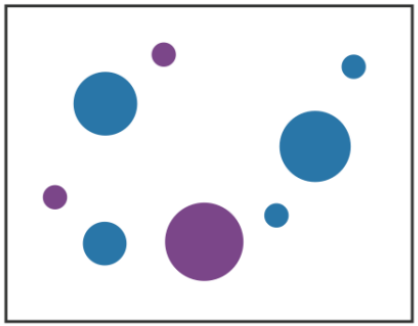
Can we combine multiple visual variables?

Position
+ Hue (Color)



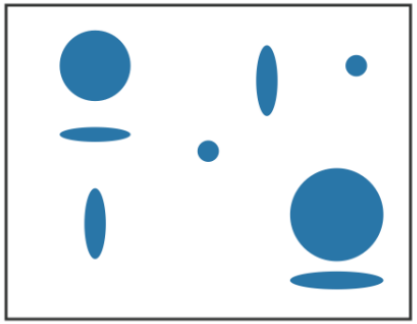
Fully separable

Size
+ Hue (Color)



Some interference

Width
+ Height



Some/significant interference

Red
+ Green

Major interference

T. Munzner, Visualization Analysis and Design, 2014

DESIGN GUIDELINE

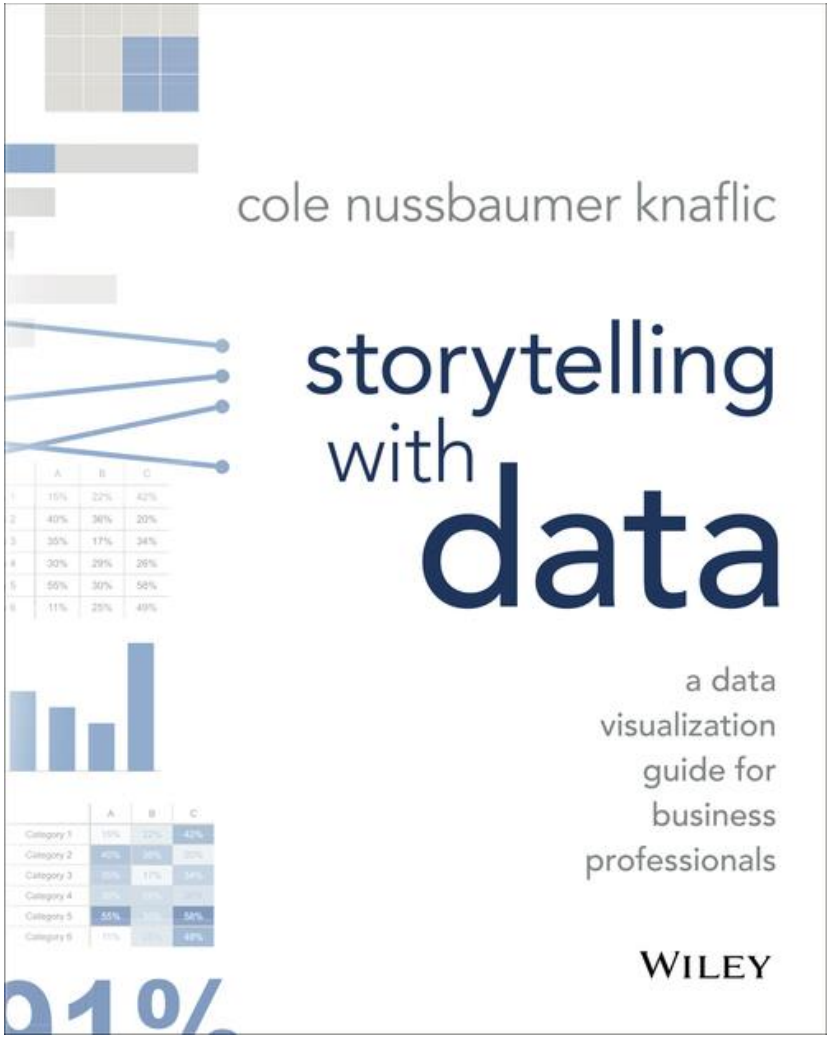
PRINCIPLES

#1: The visualization should show all of the data and only the data

PRINCIPLES

#2: Use the best visual channel available for the most important aspect of the data

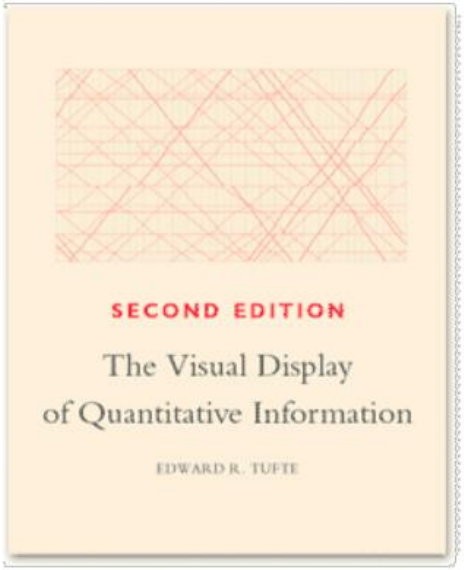
MATERIALS

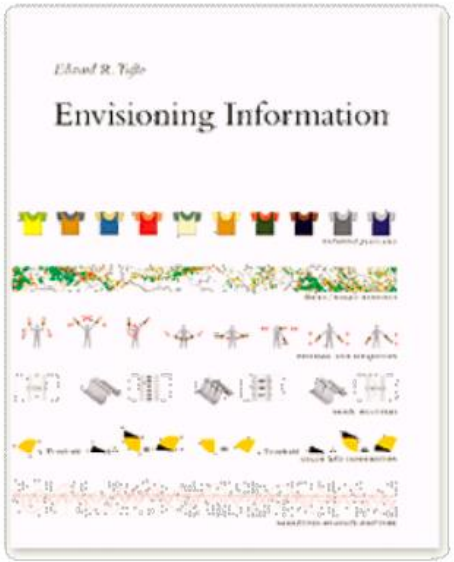

cole nussbaumer knaflic
storytelling with data
a data visualization guide for business professionals
WILEY

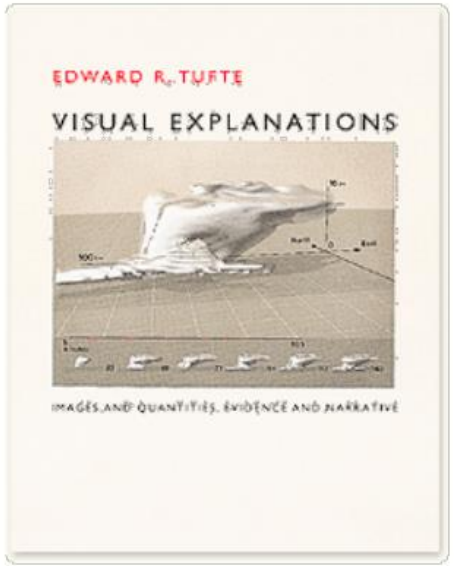
MATERIALS

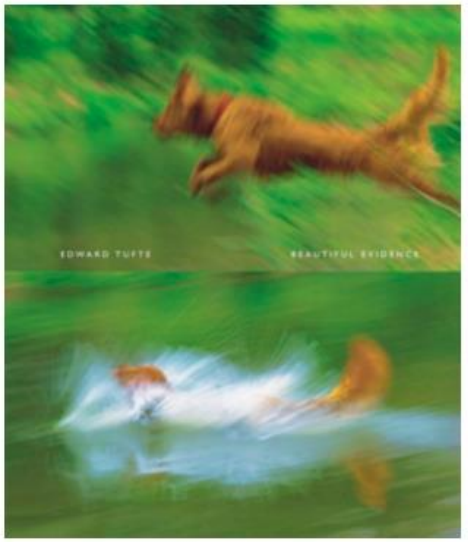
Edward Tufte

Author of fundamental books on visualization









https://en.wikipedia.org/wiki/Edward_Tufte

THE VISUAL DISPLAY OF QUANTITATIVE INFORMATION

Graphical excellence is the well-designed presentation of interesting data – a matter of substance, of statistics, and of design

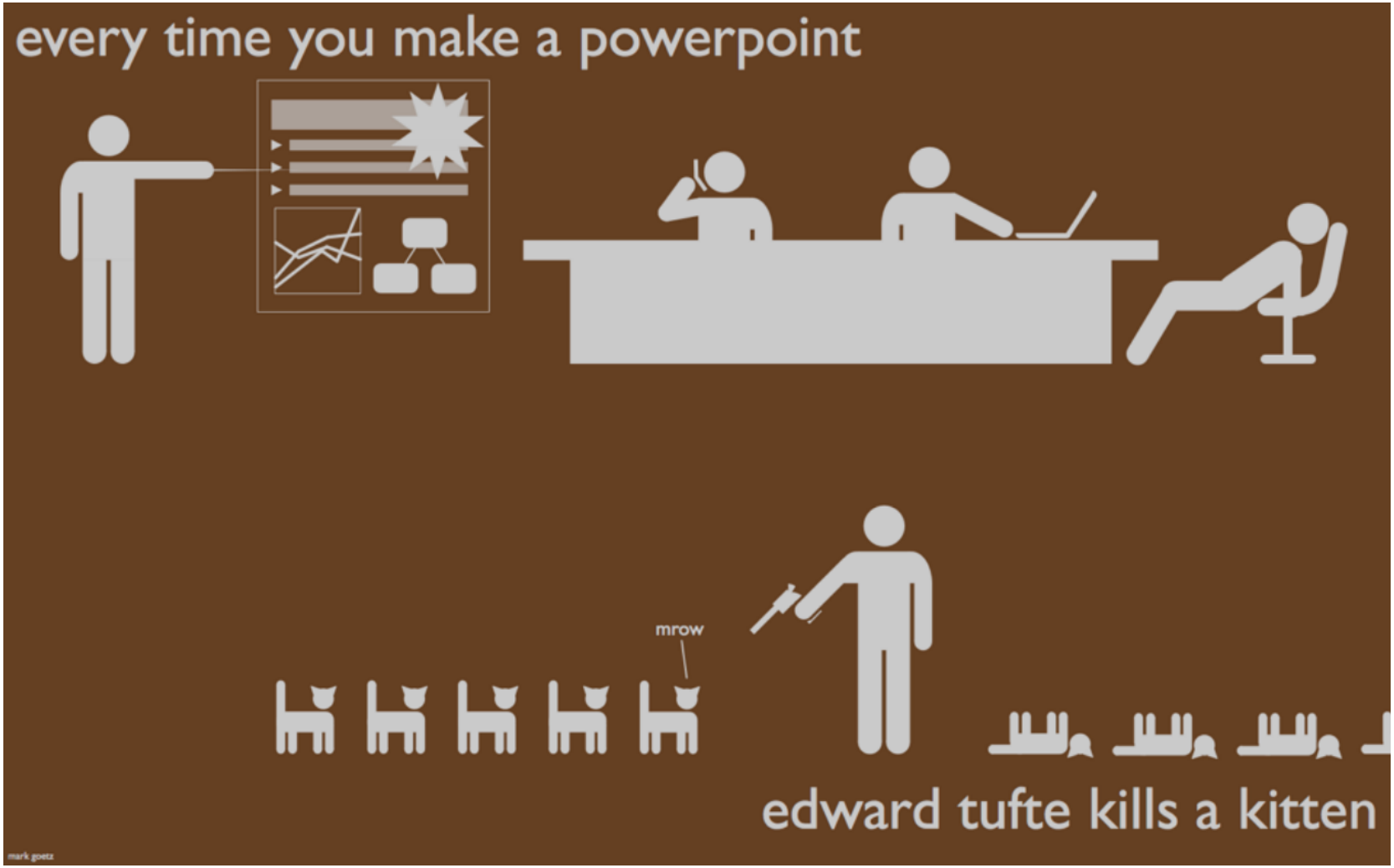
Edward Tufte

MORE GUIDELINES

- Graphical excellence consists of complex ideas communicated with clarity, precision, and efficiency.

- Graphical excellence is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space.

And many more in **The Visual Display of Quantitative Information**


every time you make a powerpoint
mrow
edward tufte kills a kitten
mark goetz

TUFTE'S INTEGRITY PRINCIPLES

Show **data variation,** not design variation

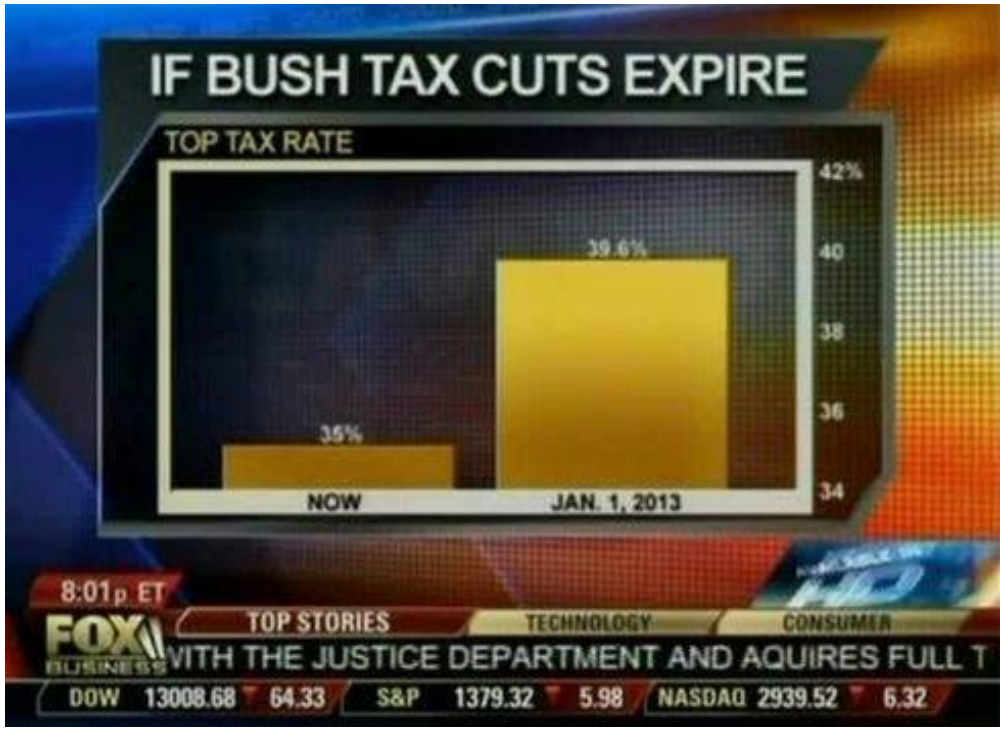
Clear, detailed, and thorough **labeling** and **appropriate scales**

Size of the **graphic effect** should be **directly proportional to the numerical quantities** ("lie factor")

THE LIE FACTOR

$$\frac{\text{Size of effect shown in graphic}}{\text{Size of effect in data}}$$

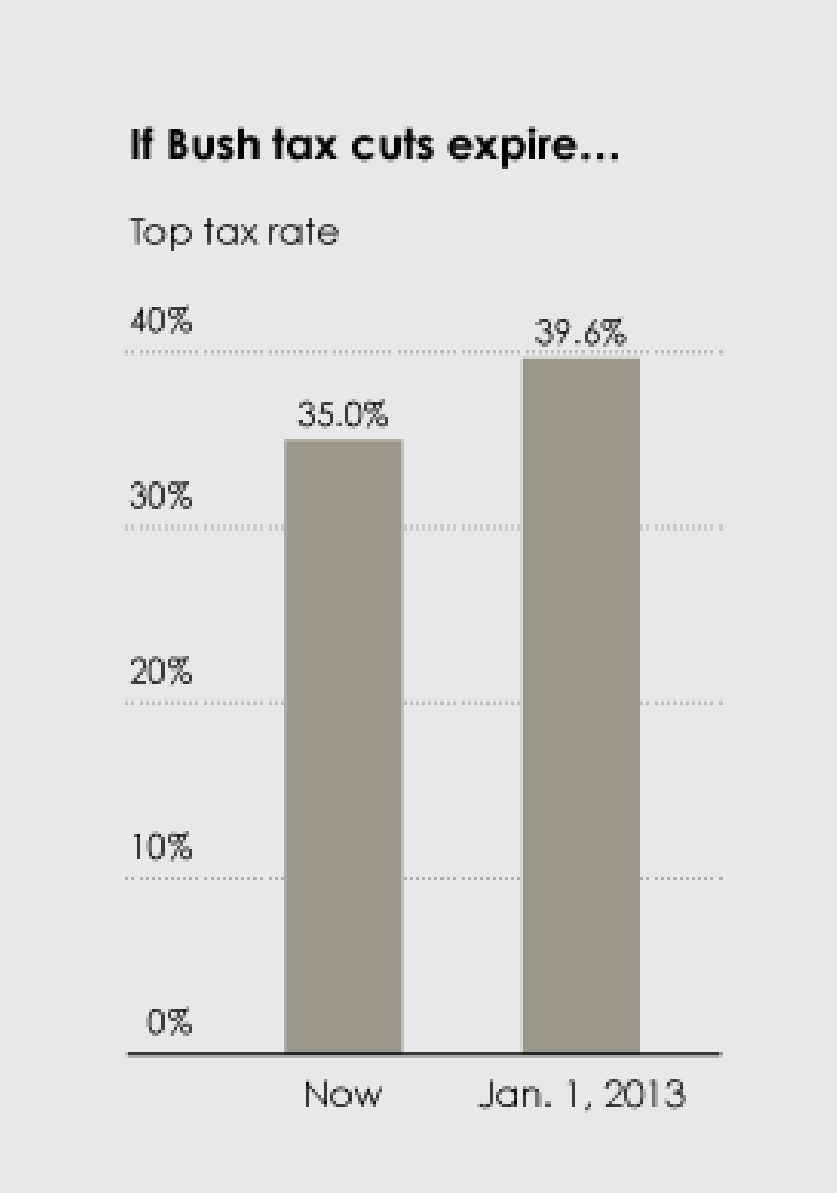
LIE FACTOR - GRAPHICAL INTEGRITY



Magnitude in data must correspond to magnitude of mark

Effect in Data: factor 1.14
Effect in Graphic: factor 5
Lie Factor: 5/1.14 = 4.38

SCALE DISTORTIONS



WHAT'S WRONG?

Viele Bezieher mit "ungeklärter Staatsbürgerschaft"

Die größte Gruppe in der Liste der Mindestsicherungsbezieher ist aber jene der "ungeklärten Staatsbürgerschaft". Dass es sich bei den 16.712 Personen um

WHAT'S WRONG?

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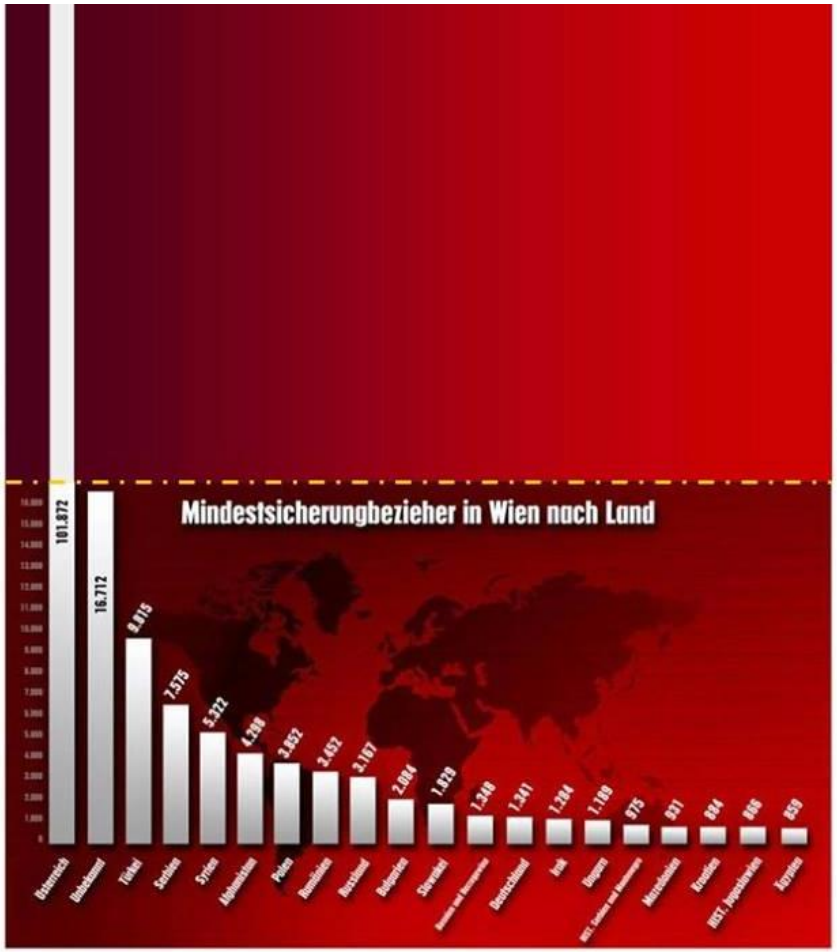

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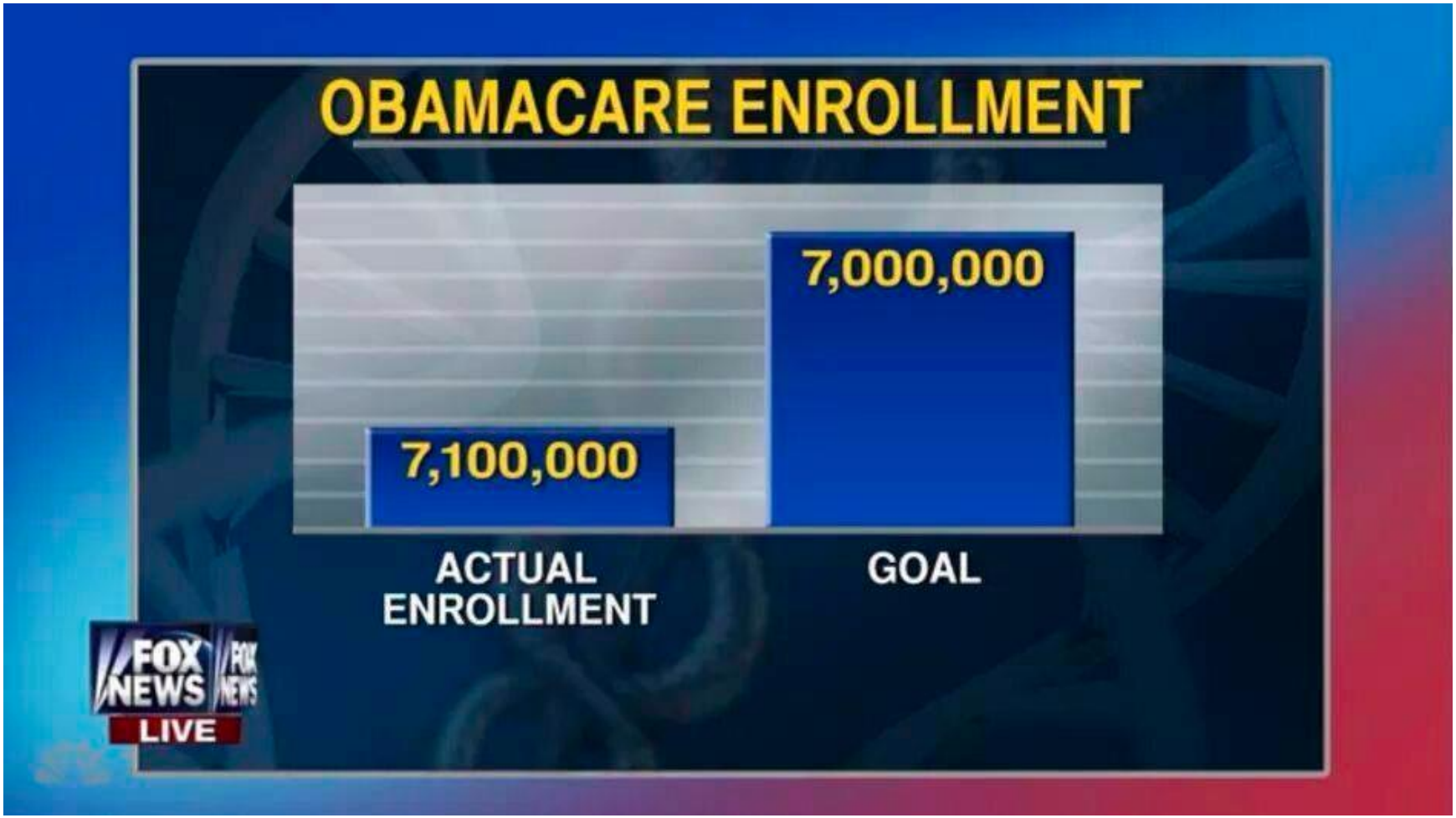
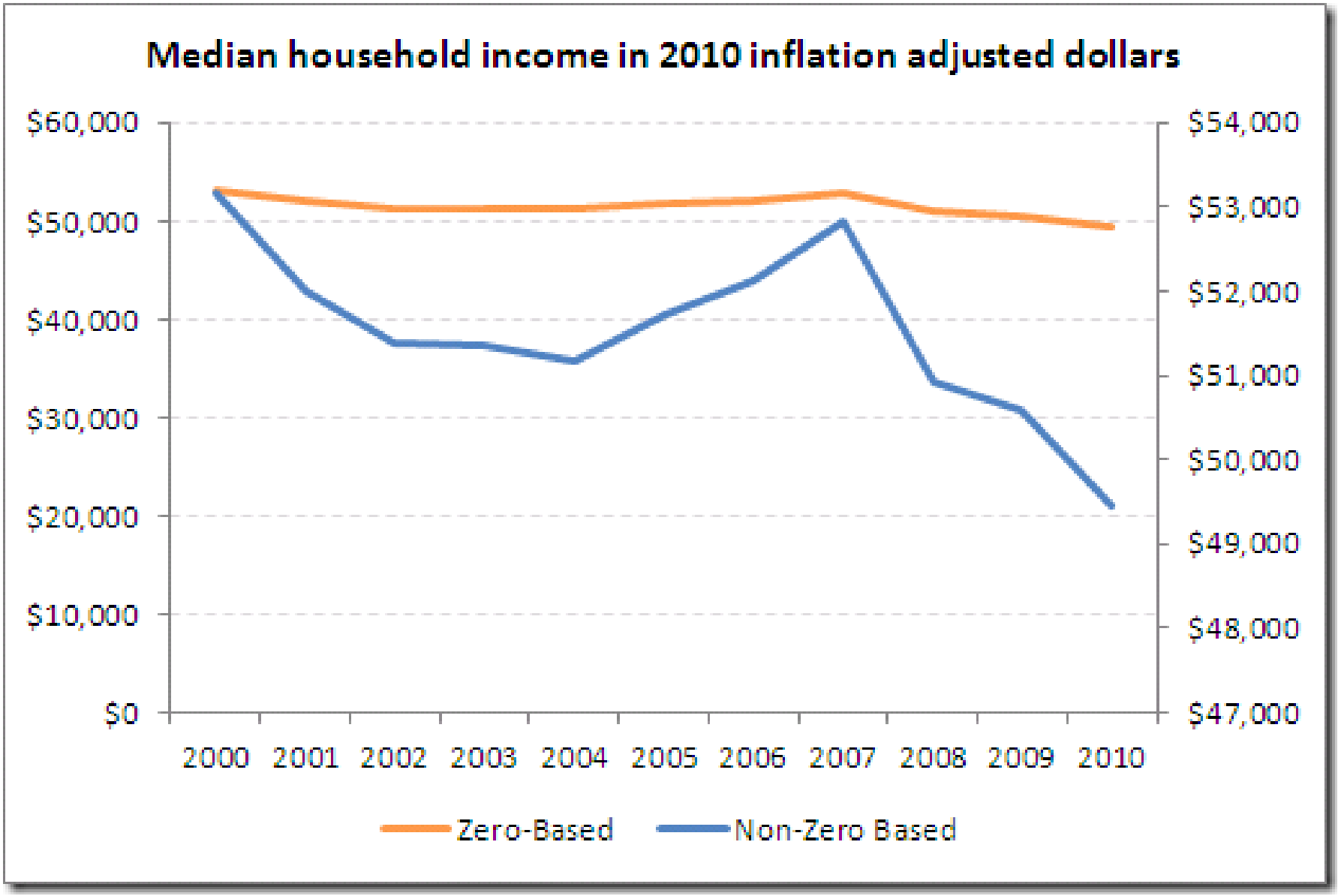

OBAMACARE ENROLLMENT
7,000,000
7,100,000
ACTUAL ENROLLMENT
GOAL
FOX NEWS
FOX NEWS
LIVE

START AT 0?


Median household income in 2010 inflation adjusted dollars
$60,000
$50,000
$40,000
$30,000
$20,000
$10,000
$0
$54,000
$53,000
$52,000
$51,000
$50,000
$49,000
$48,000
$47,000
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
Zero-Based
Non-Zero Based

Use a baseline that shows the data, not the zero-point

Think about: what is a meaningful baseline?

SCALE AT 0

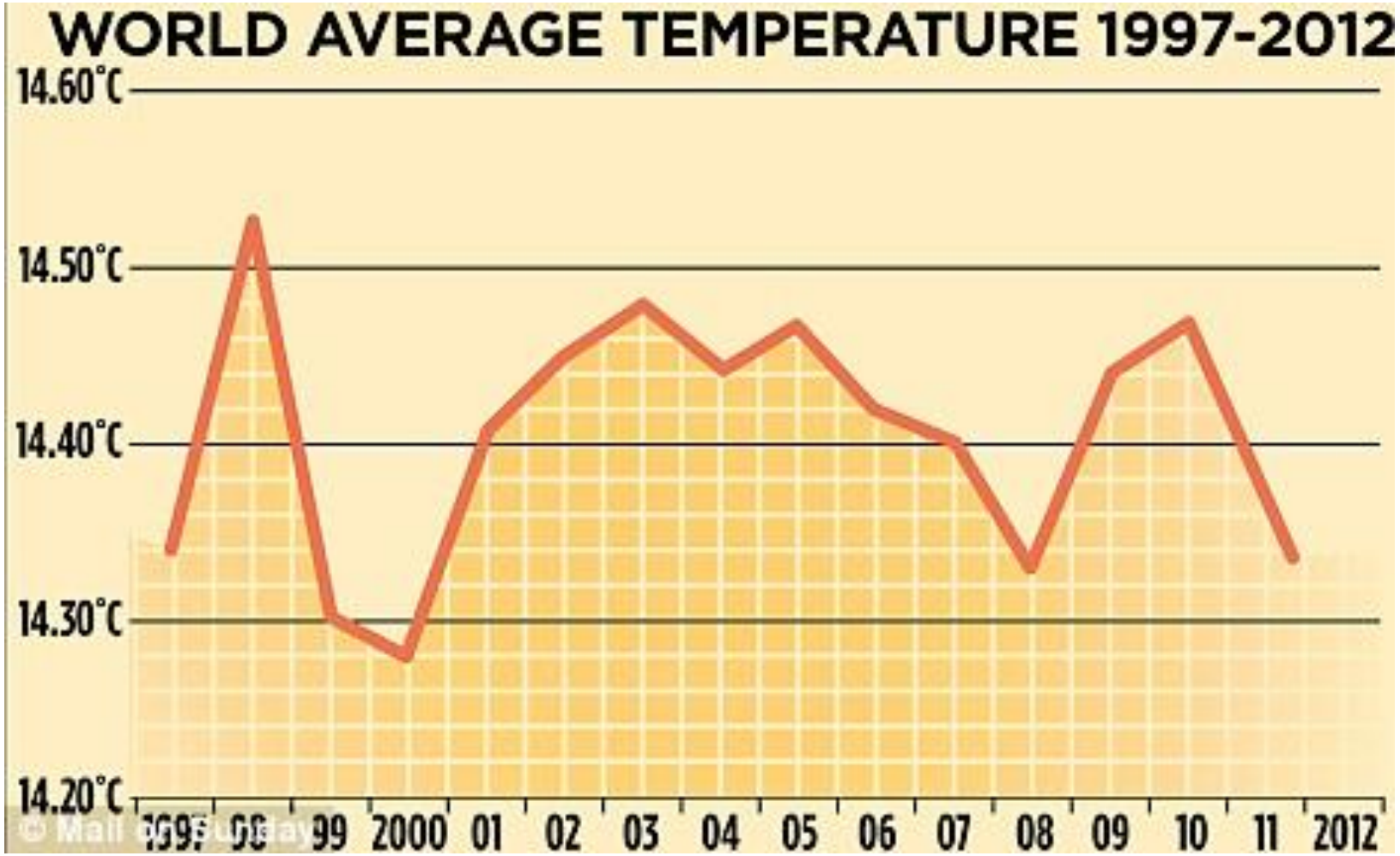
FRAMING

Vis can be used to lie

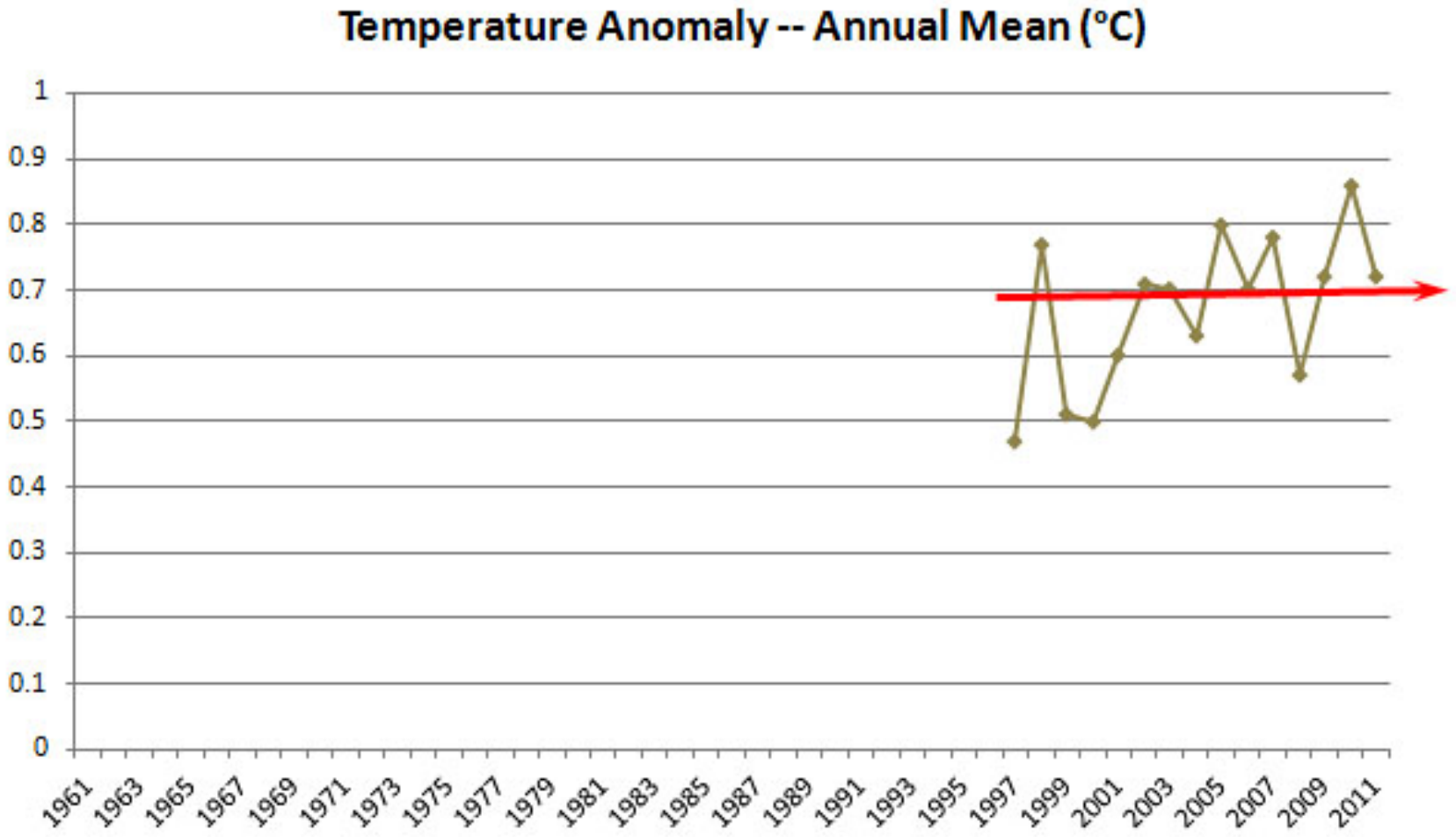
as language or statistics

When showing something, make sure that you're faithful to the data

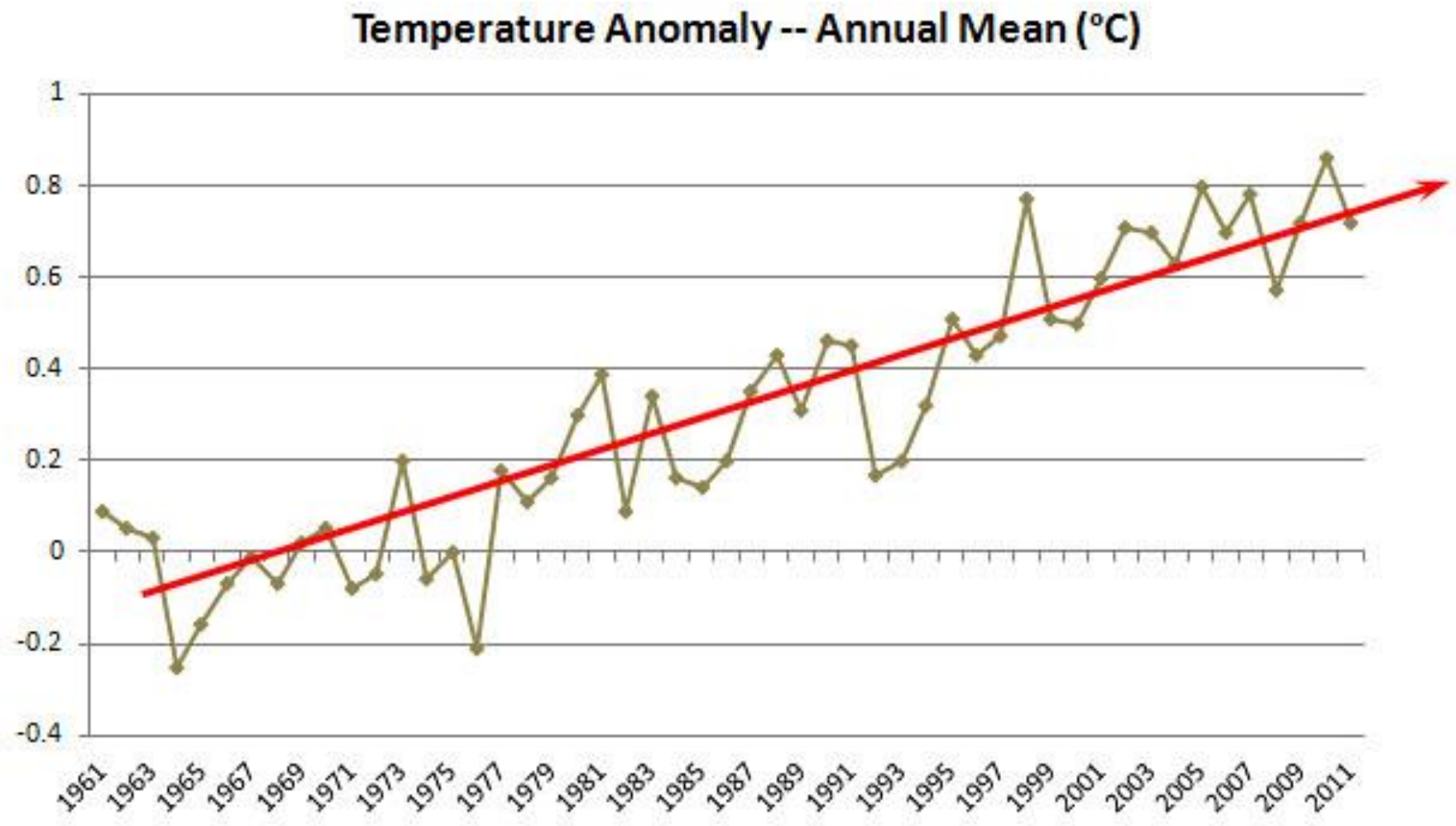
GLOBAL WARMING?



The Daily Mail, UK, Jan 2012


Temperature Anomaly -- Annual Mean (°C)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1961
1963
1965
1967
1969
1971
1973
1975
1977
1979
1981
1983
1985
1987
1989
1991
1993
1995
1997
1999
2001
2003
2005
2007
2009
2011

GLOBAL WARNING – FRAME THE DATA



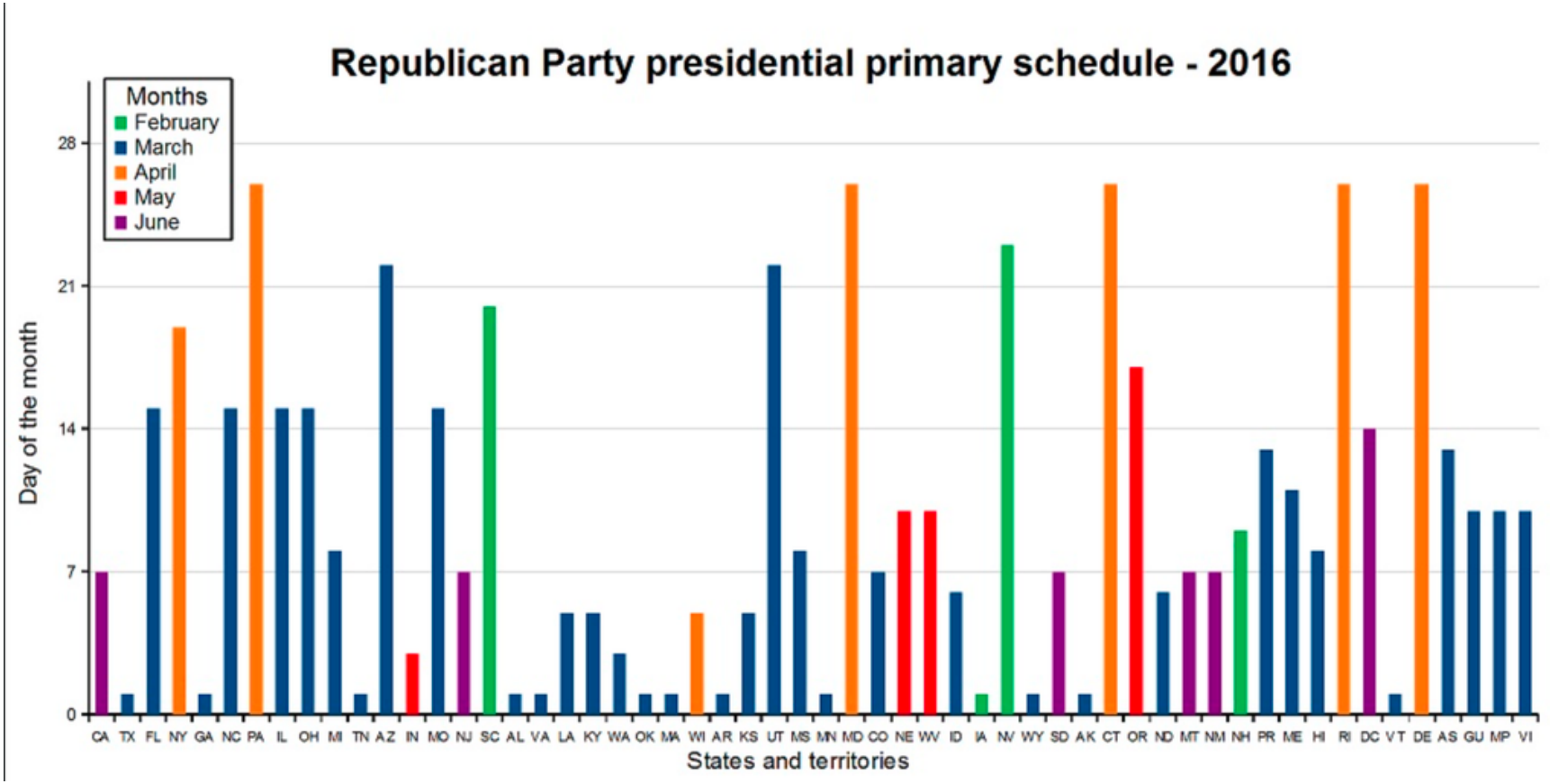
WHAT'S WRONG

How 2012 Stacks Up
The Warmest Years on Record
Contiguous U.S.
55.3
54.31
54.32
54.14
53.93
53.81
55.5°
55.0°
54.5°
54.0°
53.5°
53.0°
1921
1999
1934
2006
1998
2012
Source: NOAA's National Climatic Data Center - State of the Climate National Overview
CLIMATE CENTRAL

SCALE DISTORTIONS IN TEMPORAL DATA

SCALE DISTORTIONS IN TEMPORAL DATA

WHAT'S WRONG?

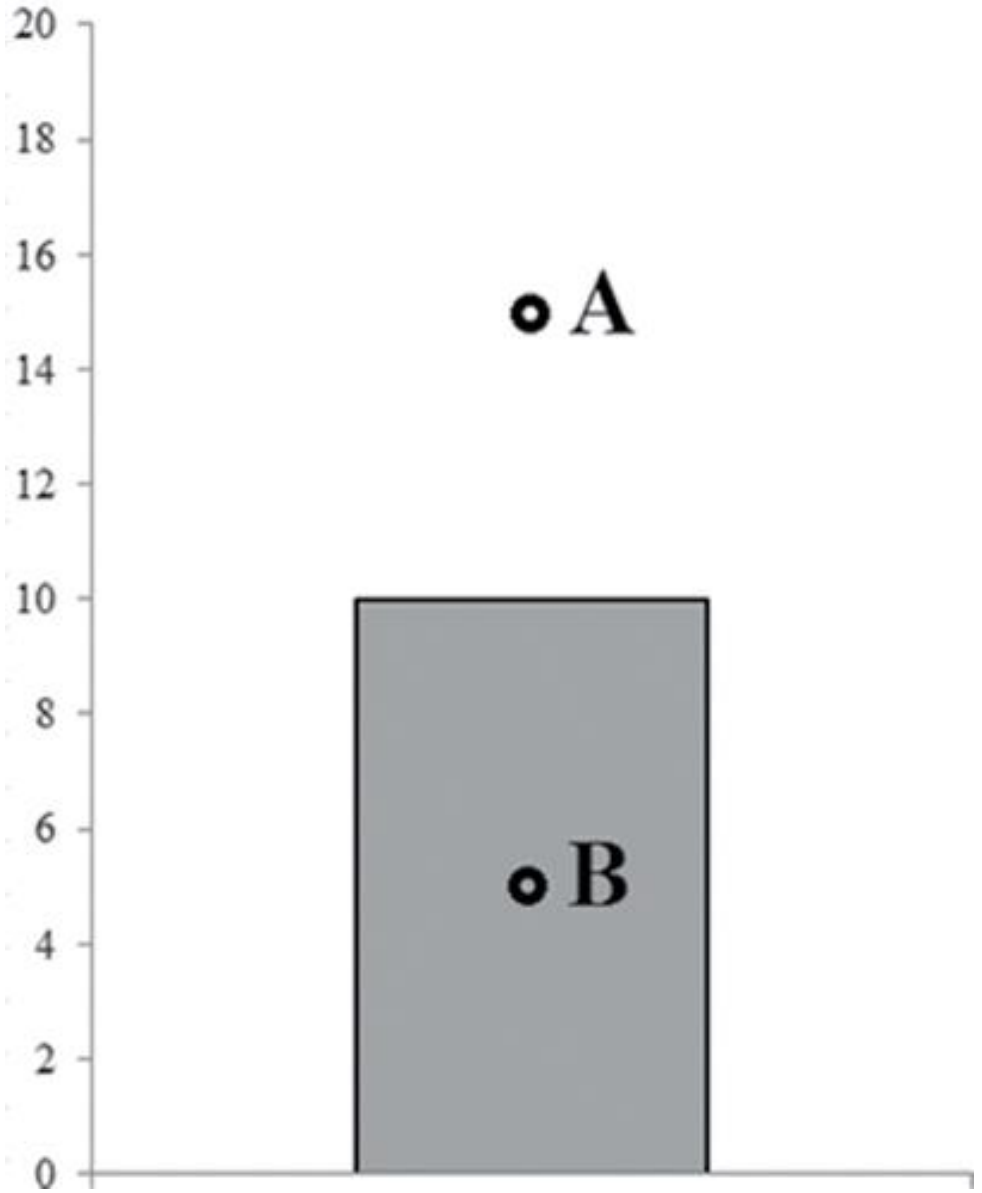

Republican Party presidential primary schedule - 2016
Months
February
March
April
May
June
Day of the month
0
7
14
21
28
CA TX FL NY GA NC PA IL OH MI TN AZ IN MO NJ SC AL VA LA KY WA OK MA WI AR KS UT MS MN MD CO NE WV ID IA NV WY SD AK CT OR ND MT NM NH PR ME HI RI DC VT DE AS GU MP VI
States and territories

BIASES

Height of the Bar encodes mean of a distribution

Which value is more likely to belong to the distribution?

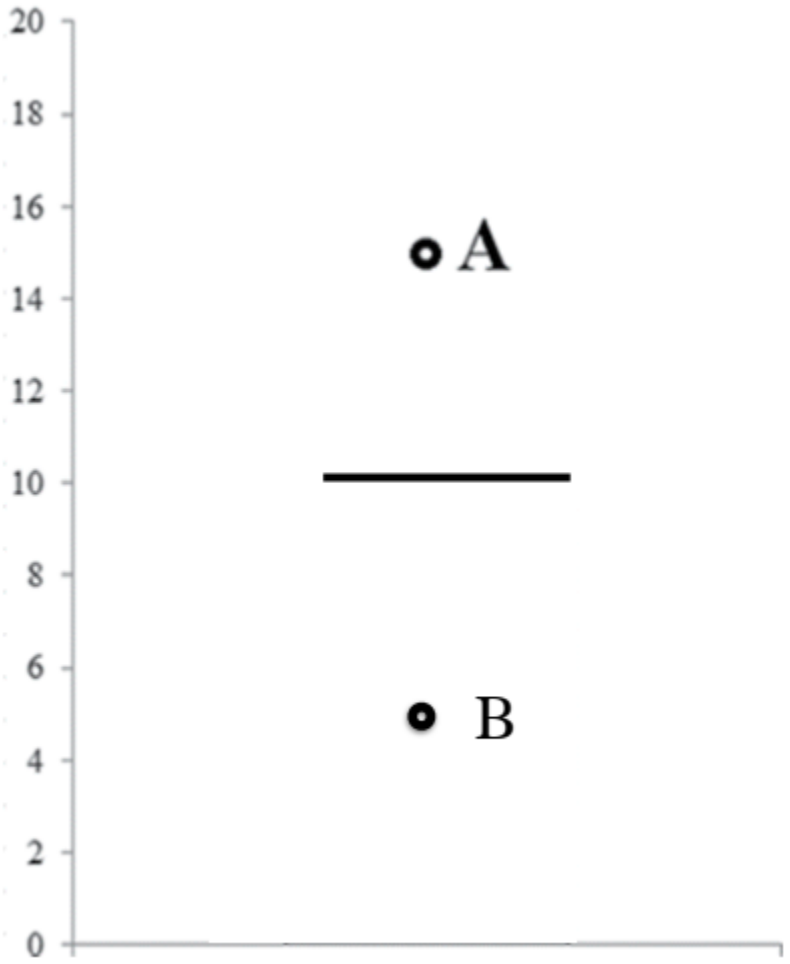
A or B?



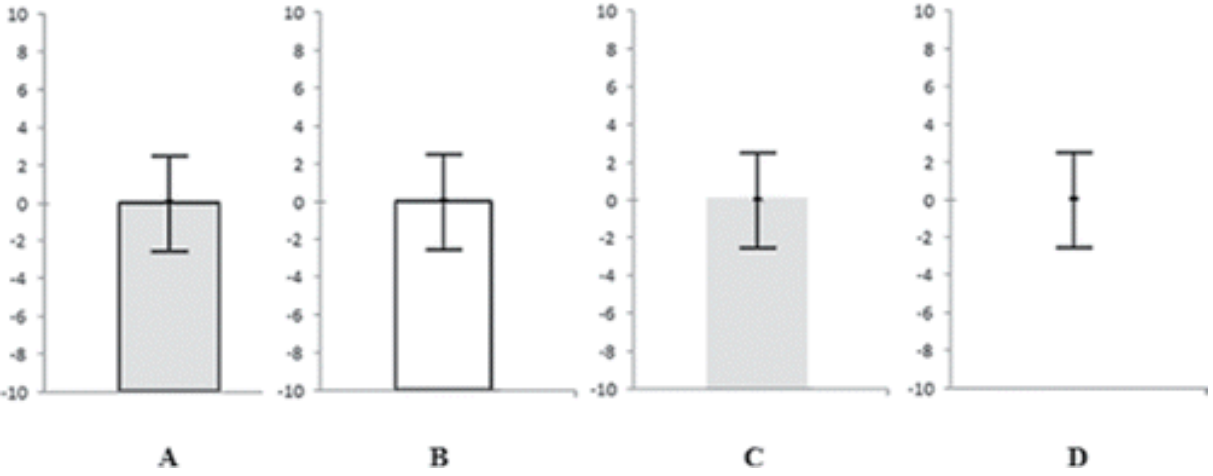
BIASES

We can plot the data faithfully, but still perceive it wrongly!

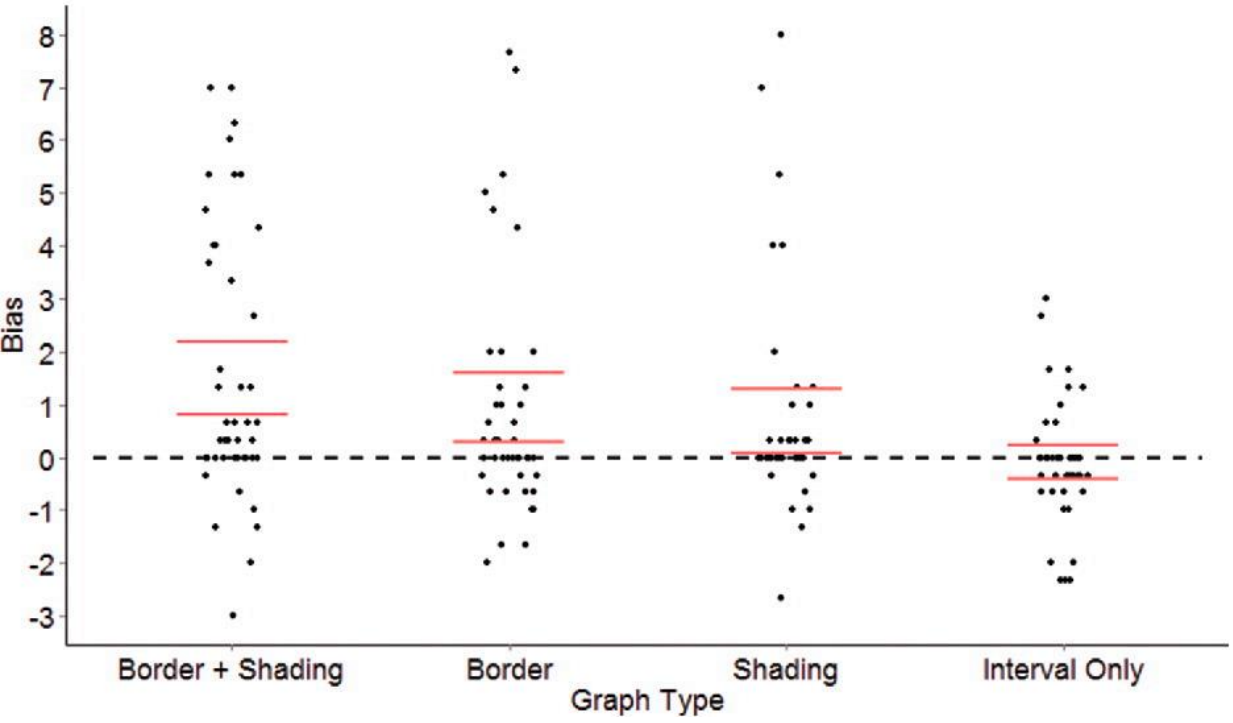
Mean of a distribution with a bar



WITHIN THE BAR BIAS



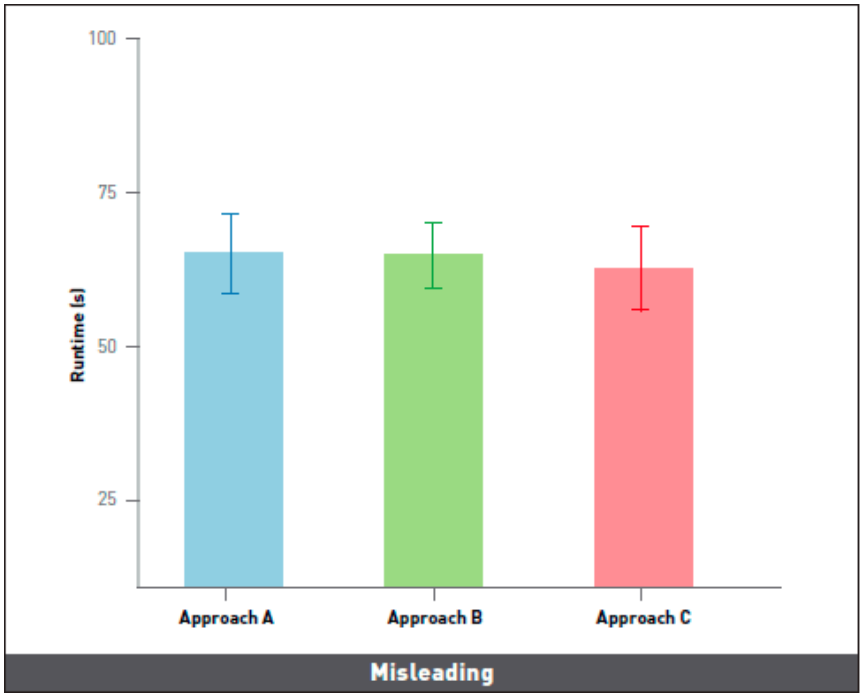
Experimental Conditions

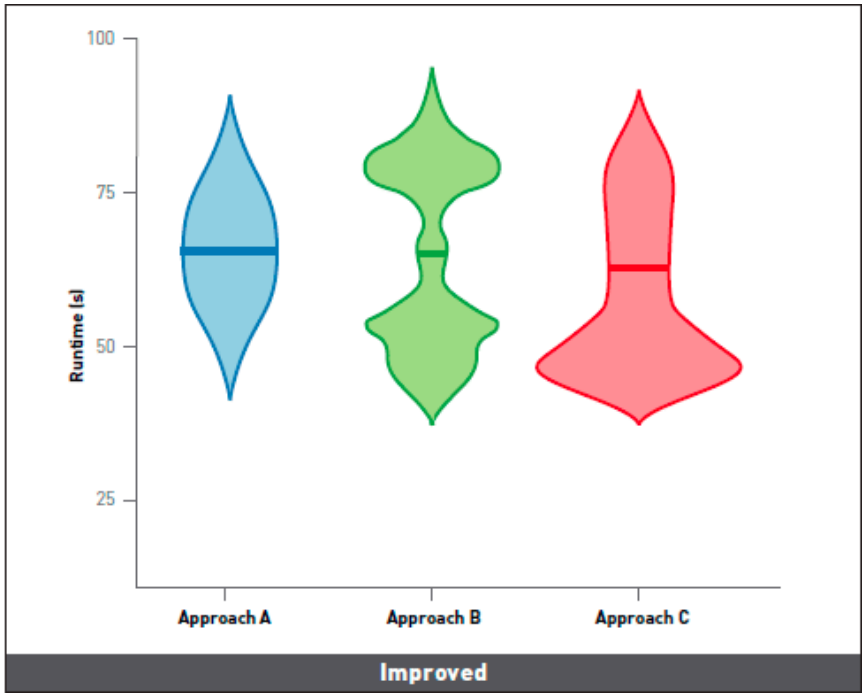


Results

Christopher S. Pentoney & Dale E. Berger (2016) Confidence Intervals and the Within-the-Bar Bias, The American Statistician, 70:2, 215-22

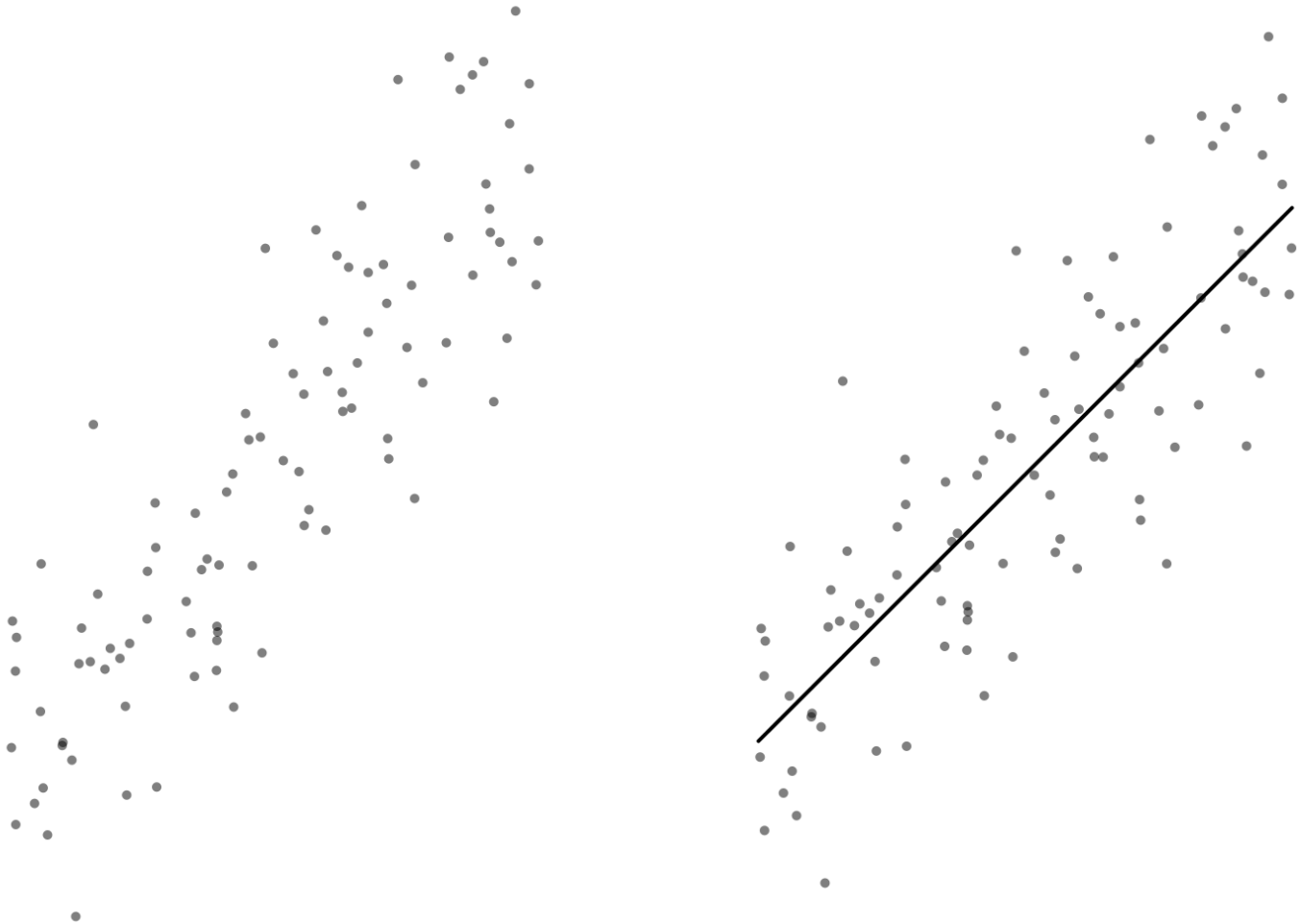
CAREFUL WHEN DESIGNING AGGREGATED CHARTS



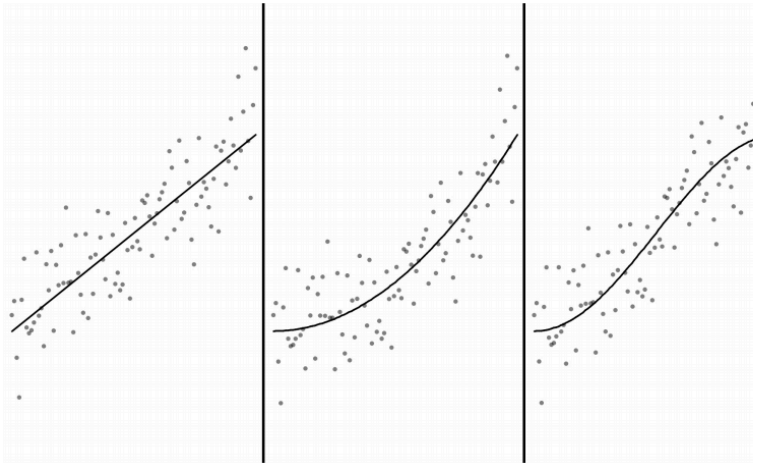


Should use median and confidence interval only when the data is normal distribution

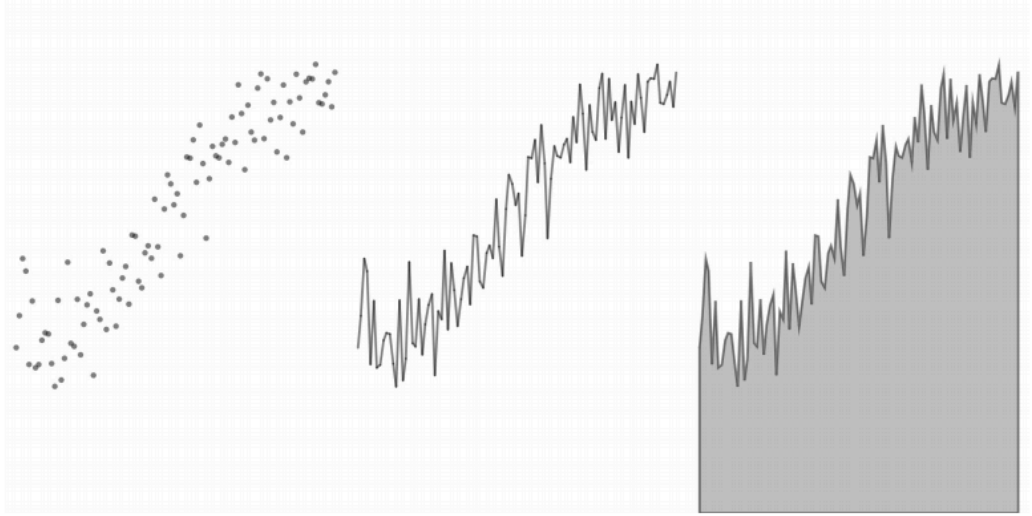
WHAT'S THE TRENDLINE



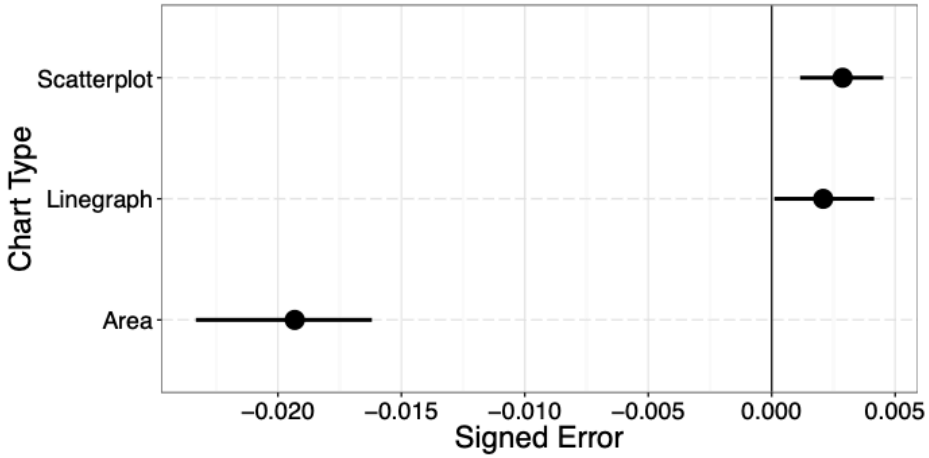
REGRESSION BY EYE



We’re good at spotting trends

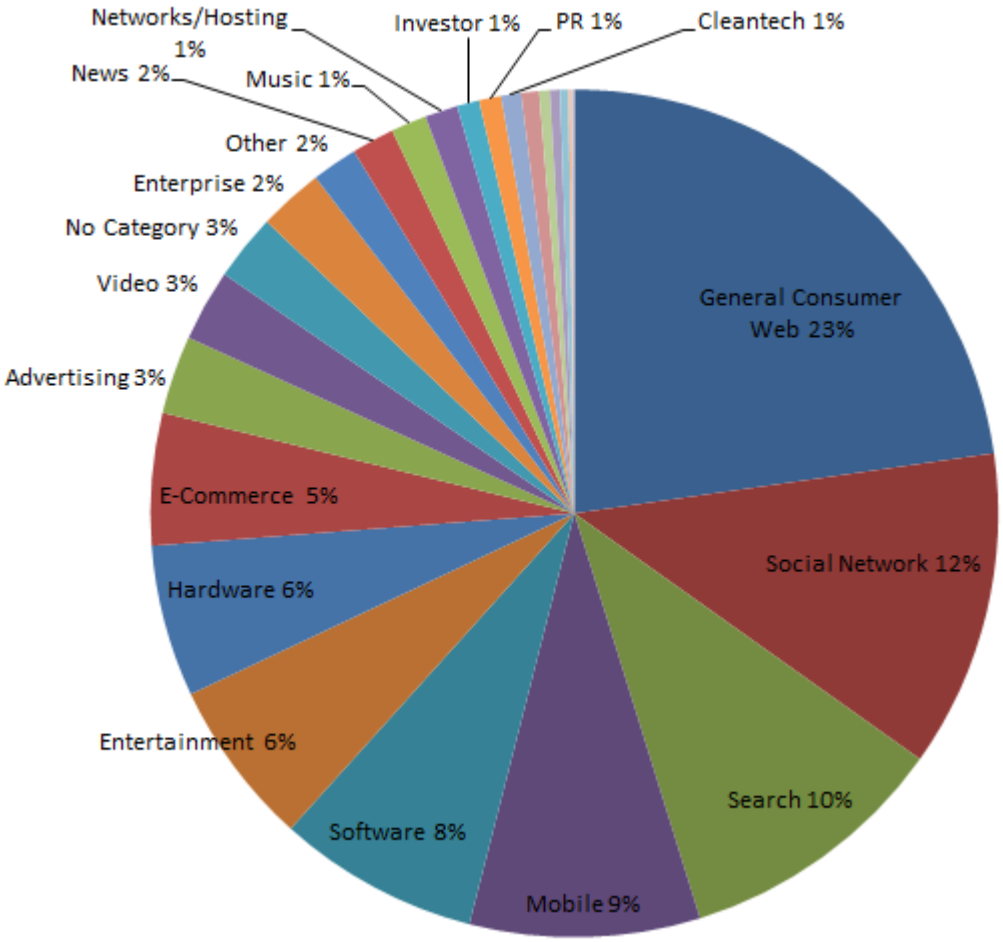


But the wrong vis technique can deceive us



http://idl.cs.washington.edu/files/2017-RegressionByEye-CHI.pdf

DEATH TO PIE CHARTS

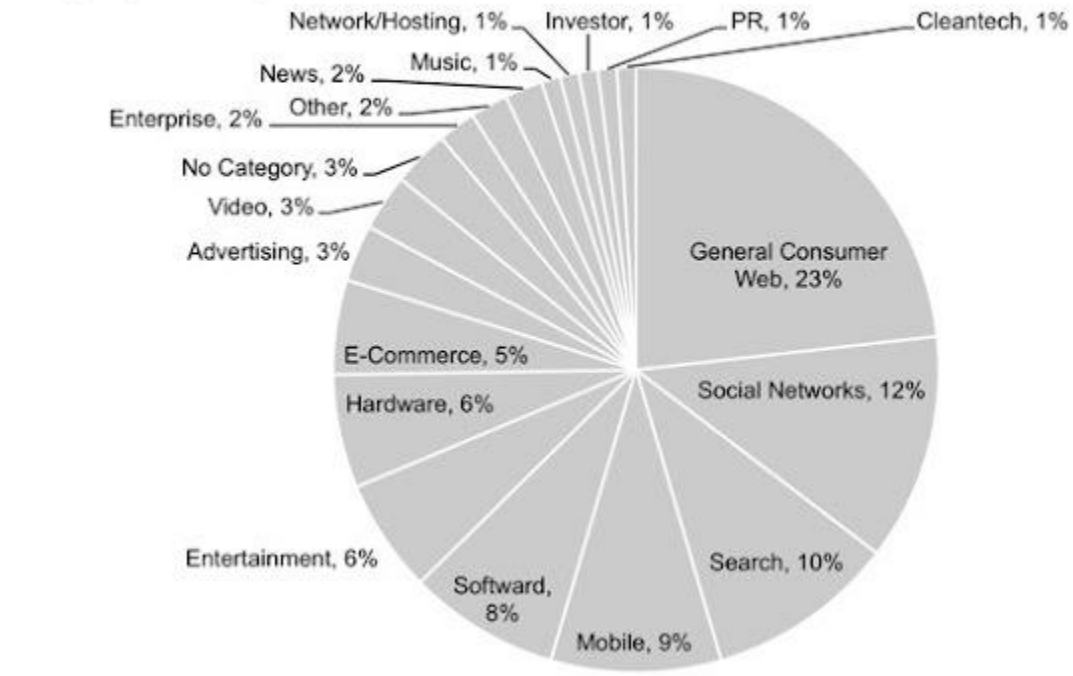


Share of coverage on TechCrunch

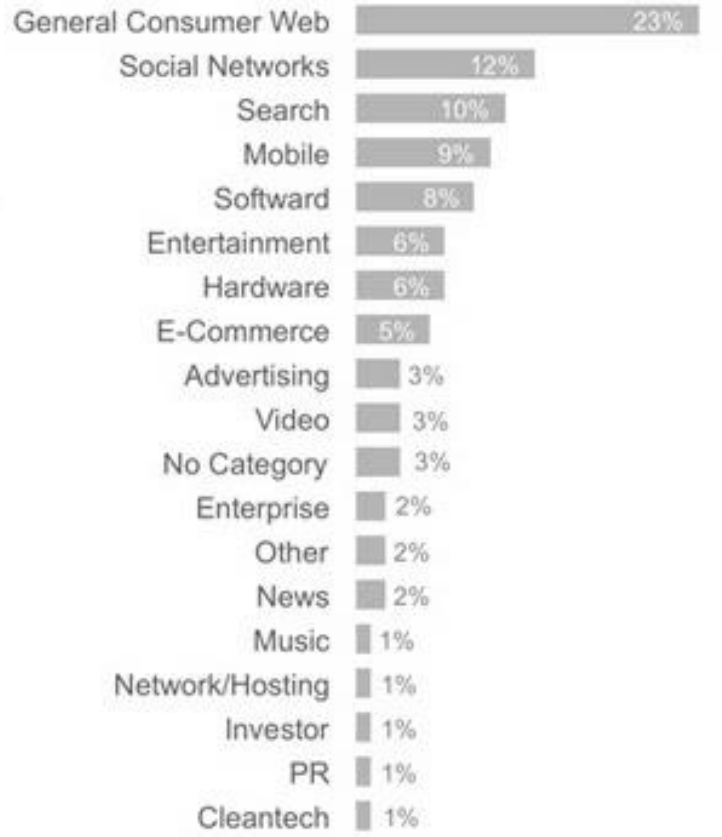
www.storytellingwithdata.com/2011/07/death-to-pie-charts.html
Cole Nussbaume

REDESIGN

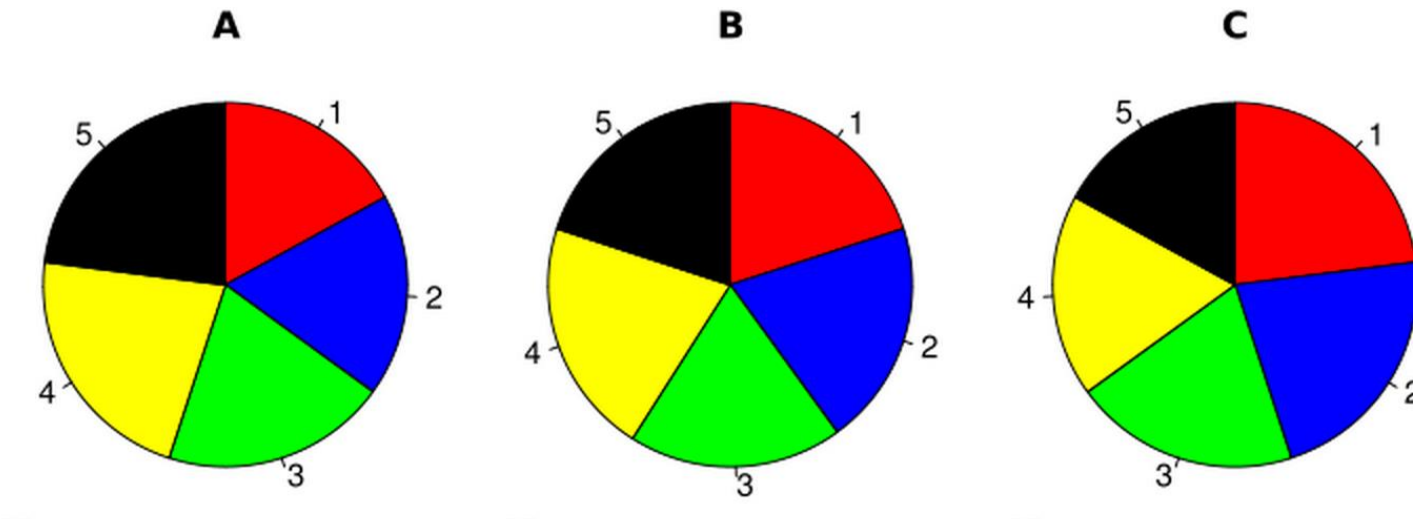
TechCrunch Coverage: 2005 - 2011
A slightly better pie?



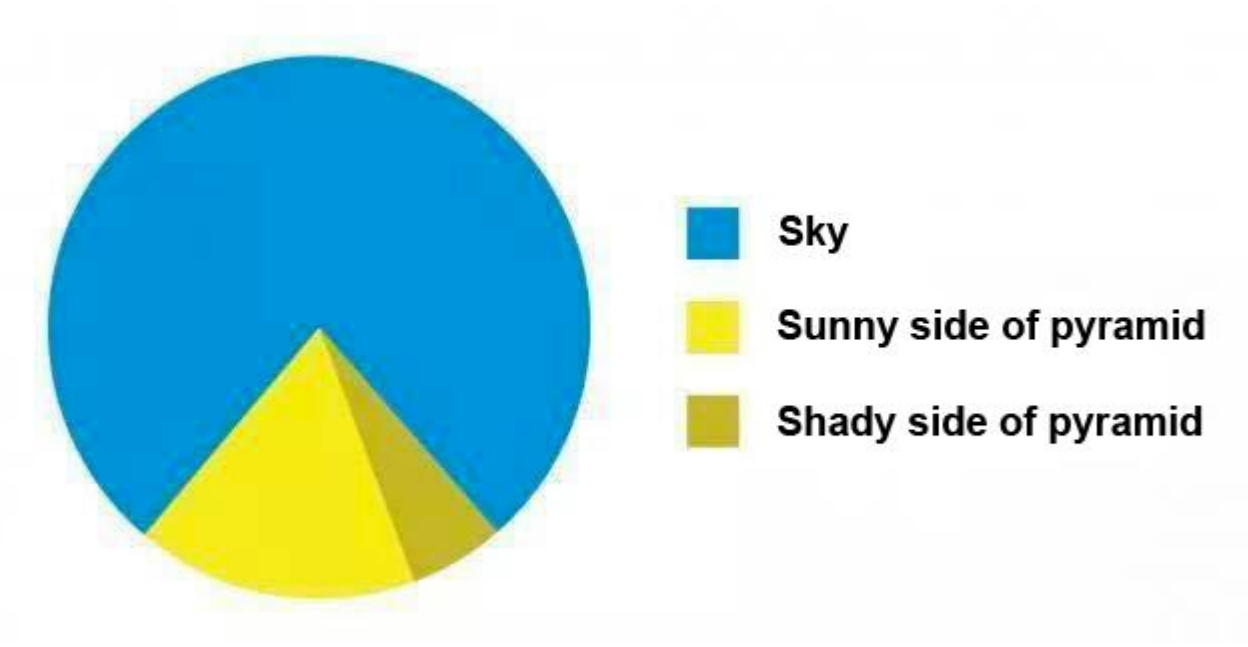
TechCrunch Coverage: 2005 - 2011
Bars are best!



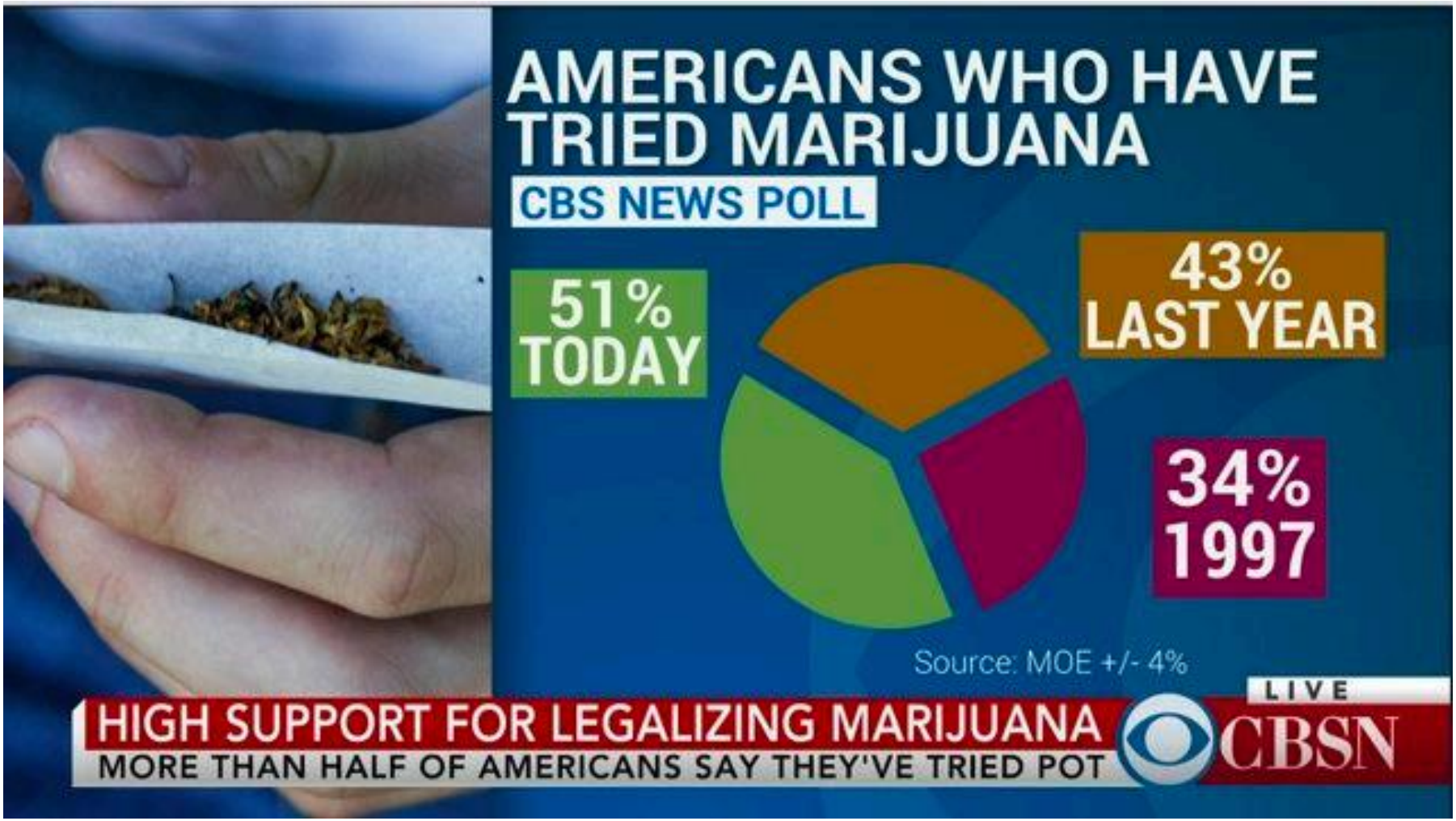
CAN YOU SPOT THE DIFFERENCES?



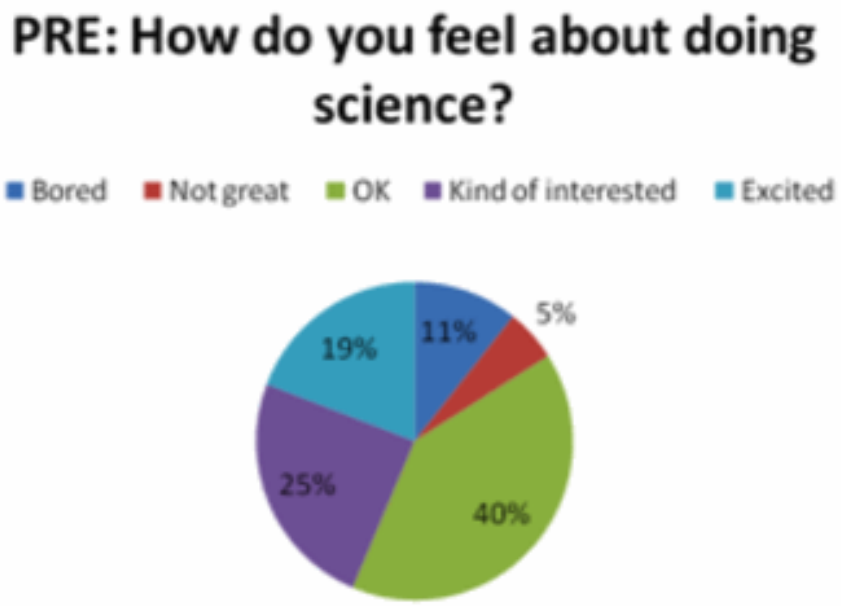
A GOOD PIE CHART?



BAD EXAMPLE AGAIN



SO, WHAT TO USE INSTEAD



http://www.storytellingwithdata.com/blog/2014/06/alternatives-to-pies

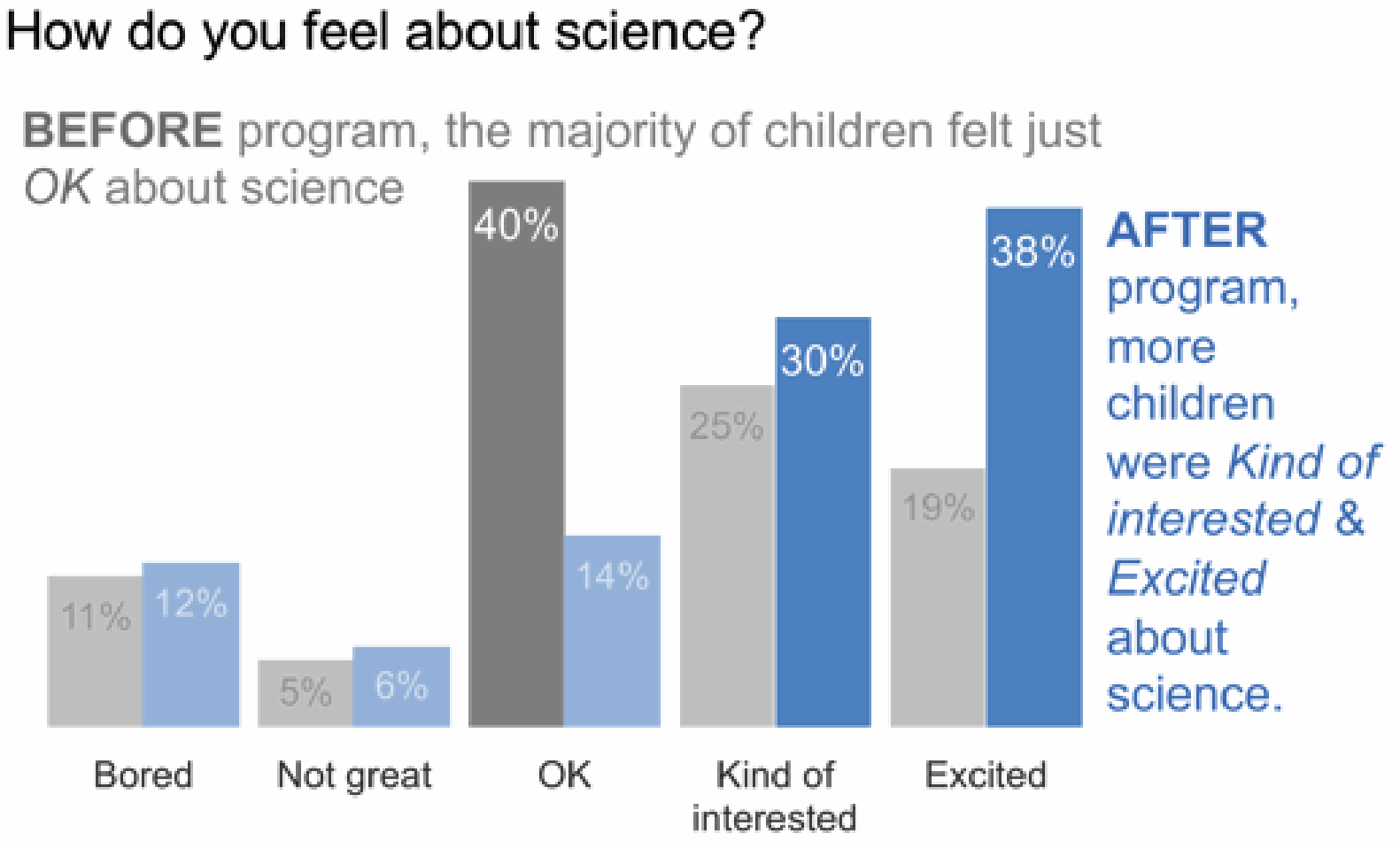
ALTERNATIVE #1. NUMBER DIRECTLY

After the pilot program,

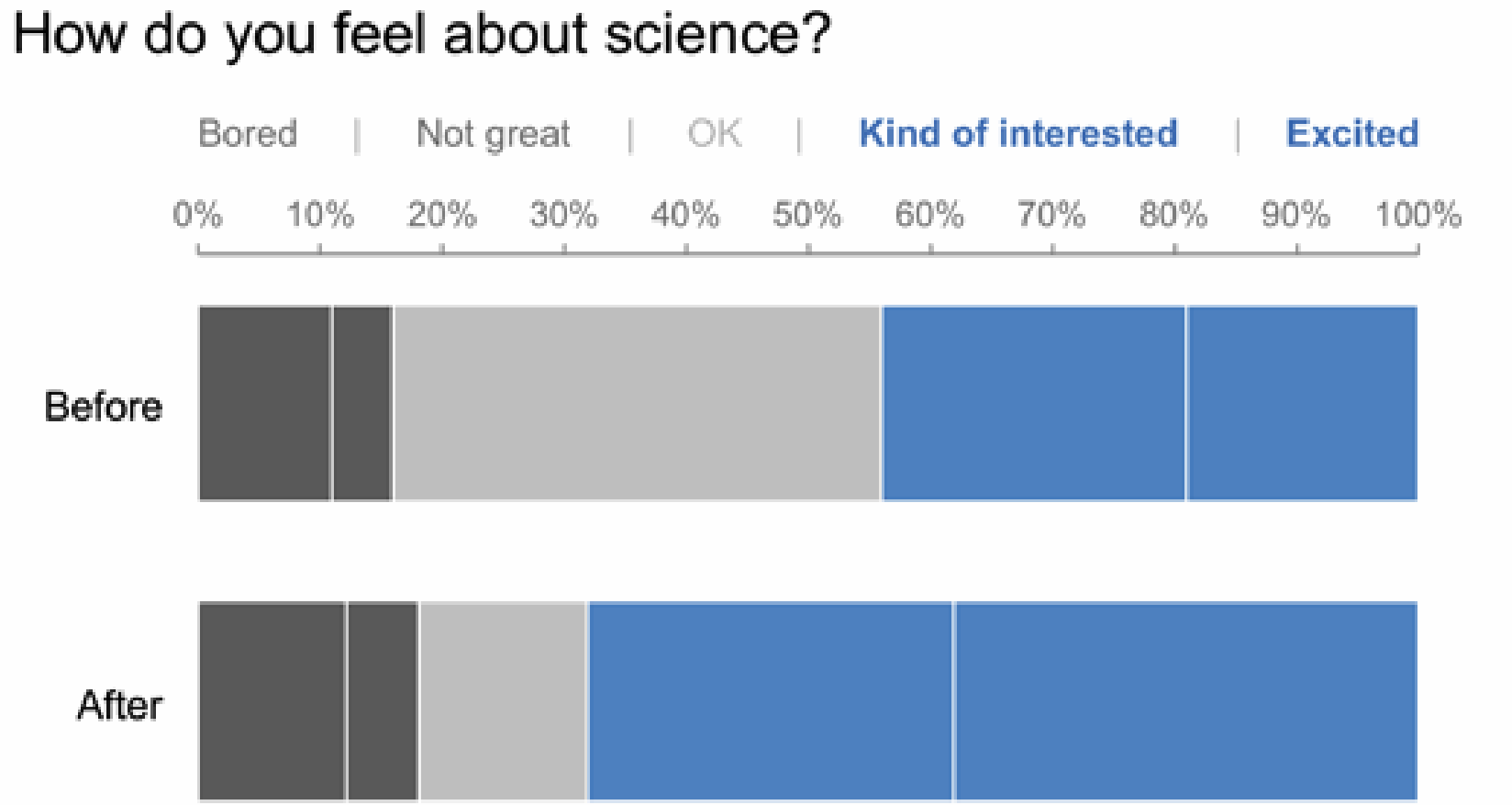
68%

of kids expressed interest towards science,
compared to 44% going into the program.

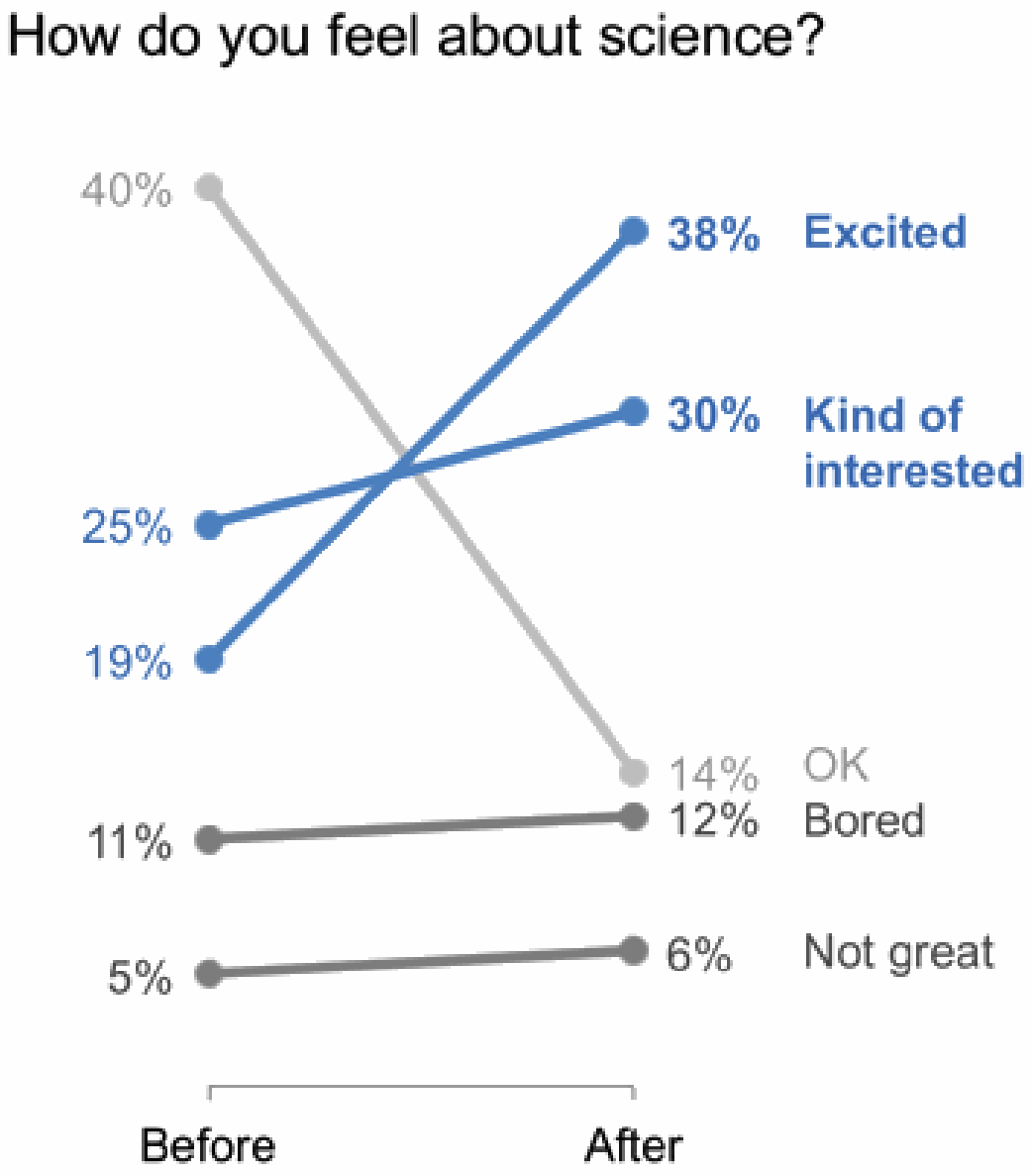
ALTERNATIVE #2: SIMPLE BAR GRAPH



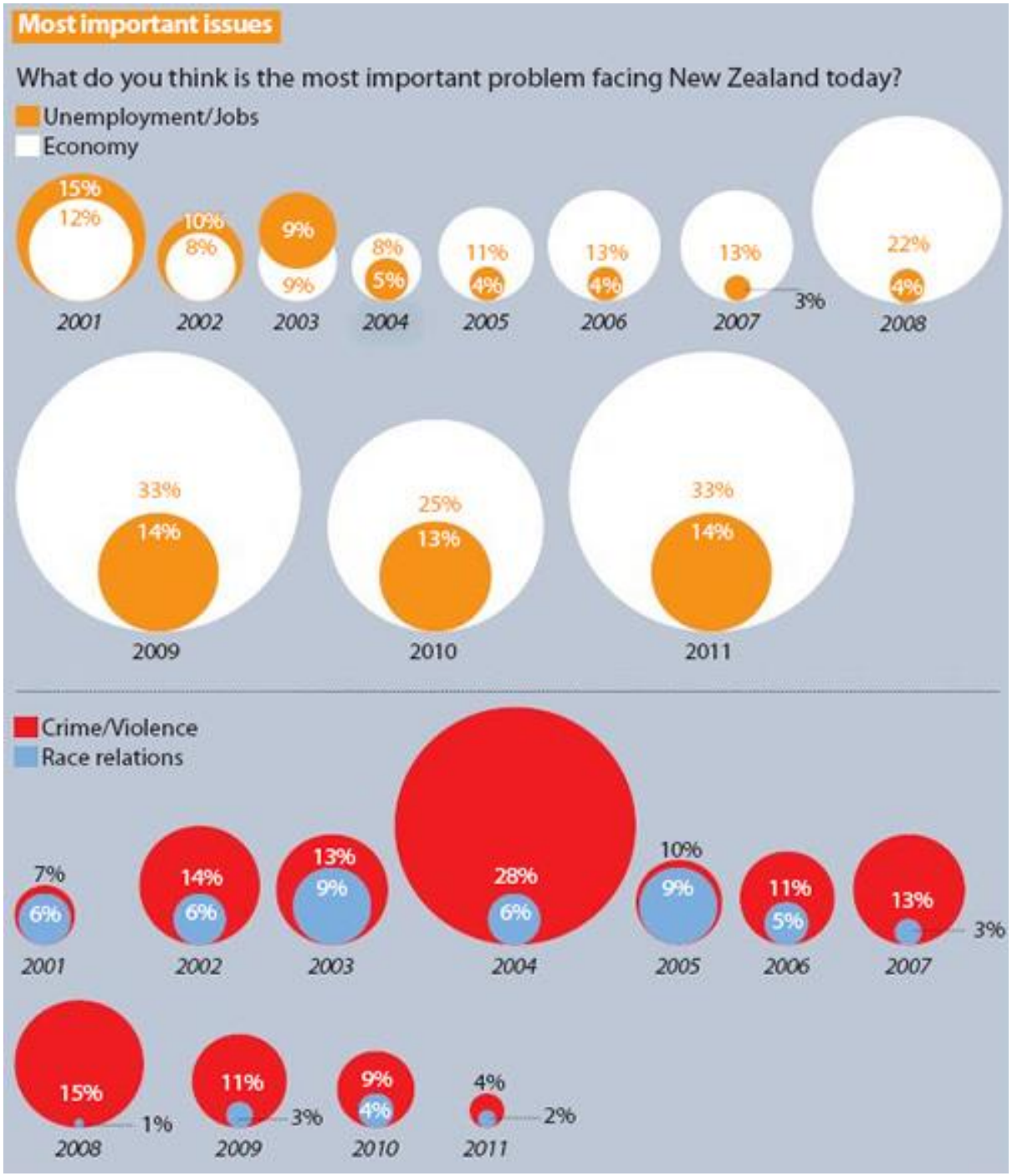
ALTERNATIVE #3: 100% STACKED HORIZONTAL BAR GRAPH

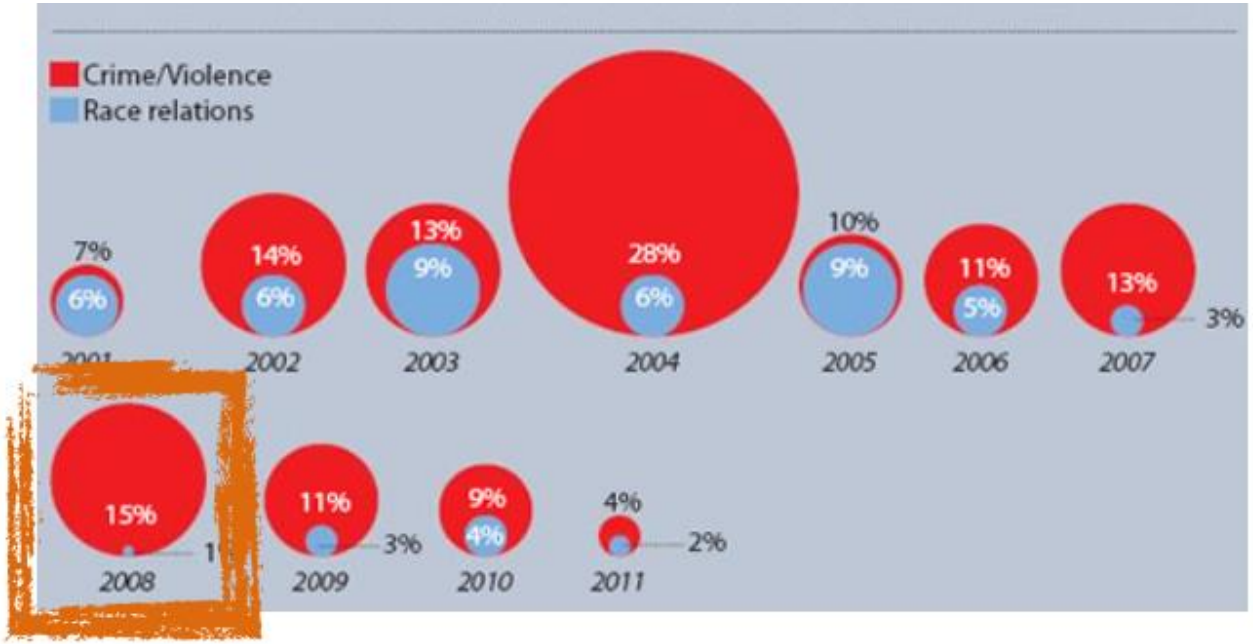


ALTERNATIVE #4: SLOPEGRAPH



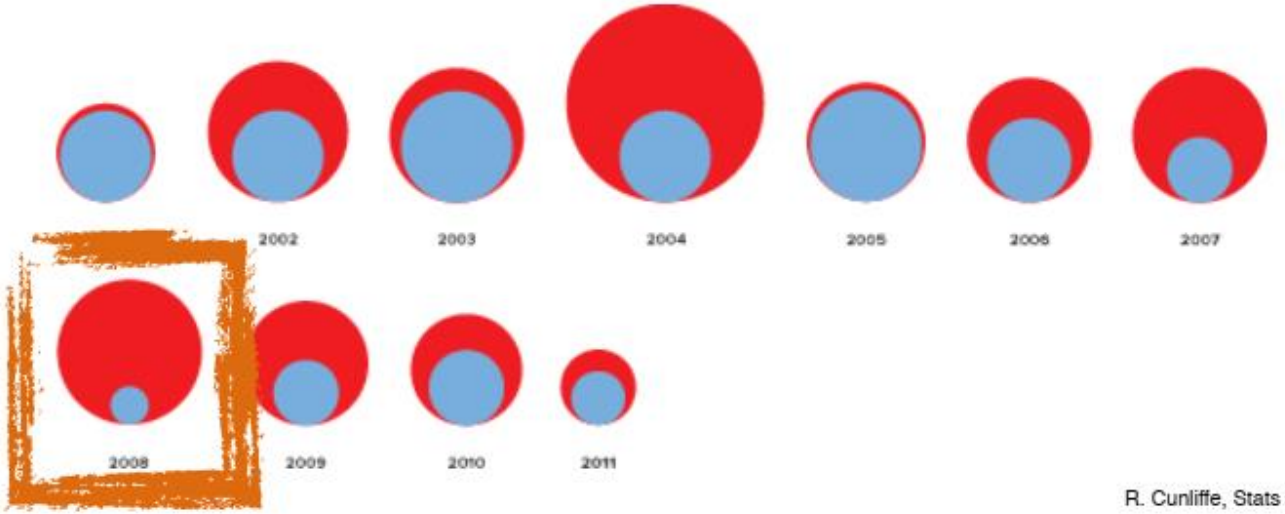
DESIGN CRITIQUE/ REDESIGN


Most important issues
What do you think is the most important problem facing New Zealand today?
Unemployment/Jobs
Economy
15%
12%
2001
10%
8%
2002
9%
9%
2003
8%
5%
2004
11%
4%
2005
13%
4%
2006
13%
3%
2007
22%
4%
2008
33%
14%
2009
25%
13%
2010
33%
14%
2011
Crime/Violence
Race relations
7%
6%
2001
14%
6%
2002
13%
9%
2003
28%
6%
2004
10%
9%
2005
11%
5%
2006
13%
3%
2007
15%
1%
2008
11%
3%
2009
9%
4%
2010
4%
2%
2011

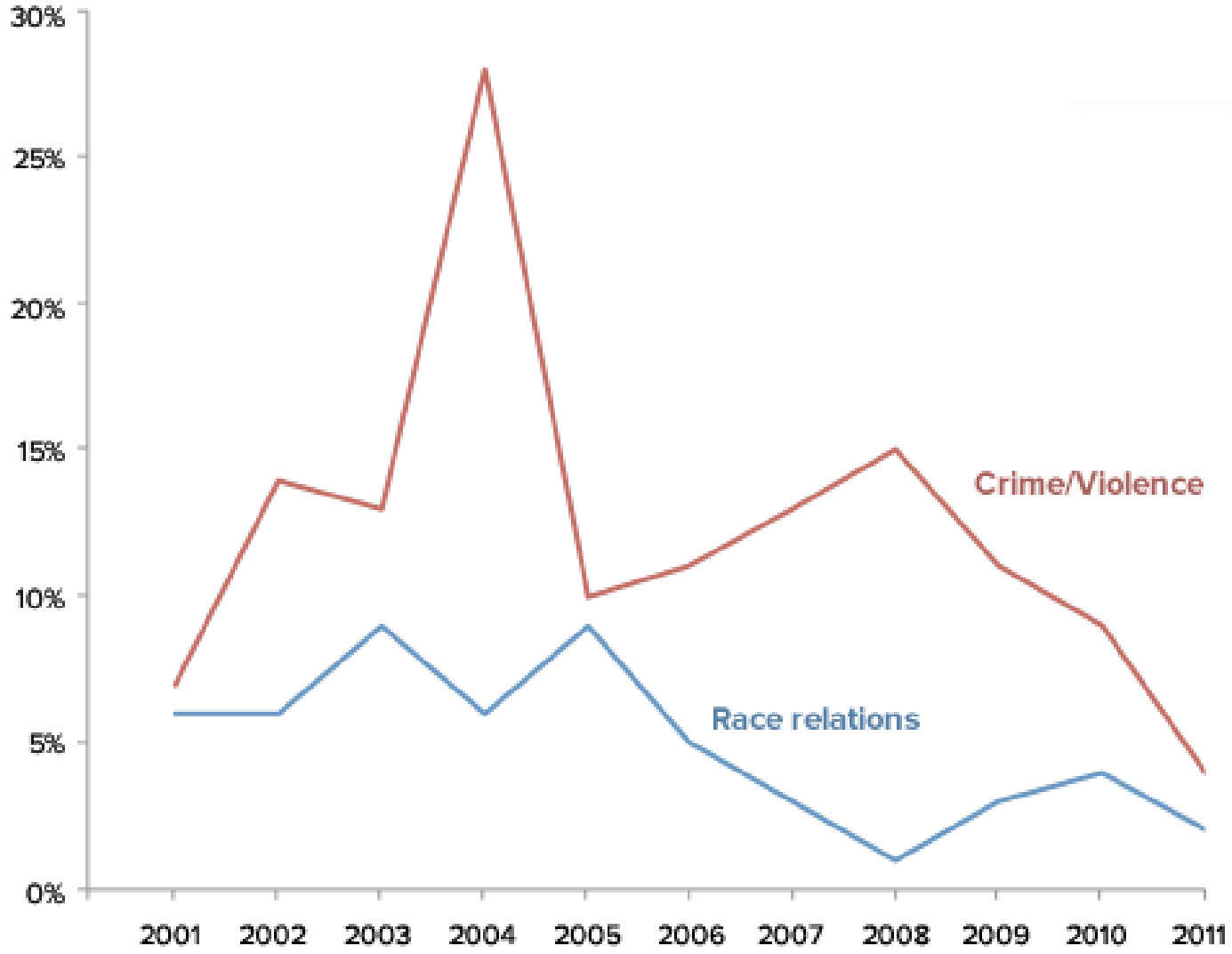


Quantity encoded by diameter, not area!

Fixing that:

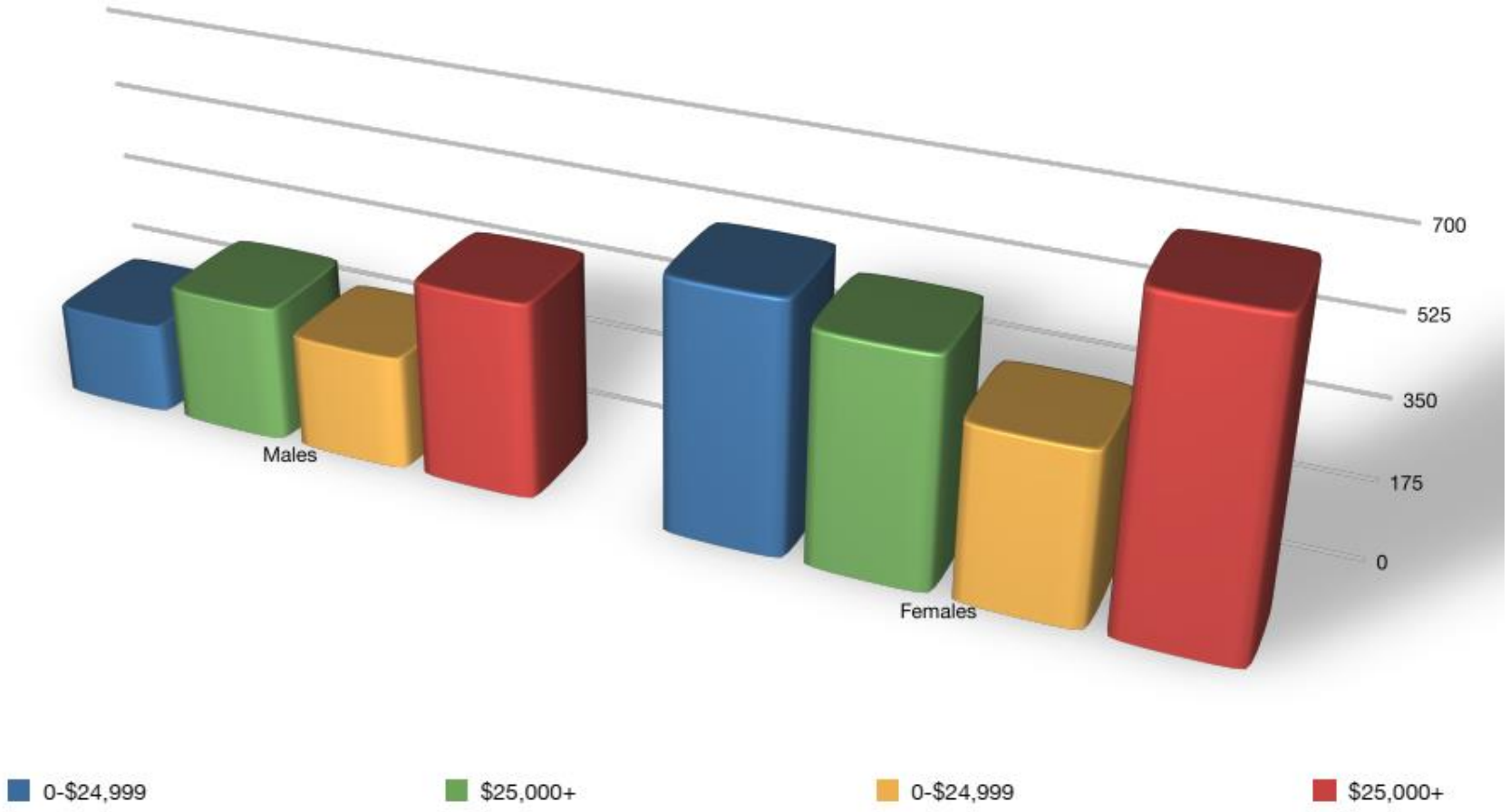


R. Cunliffe, Stats Chat

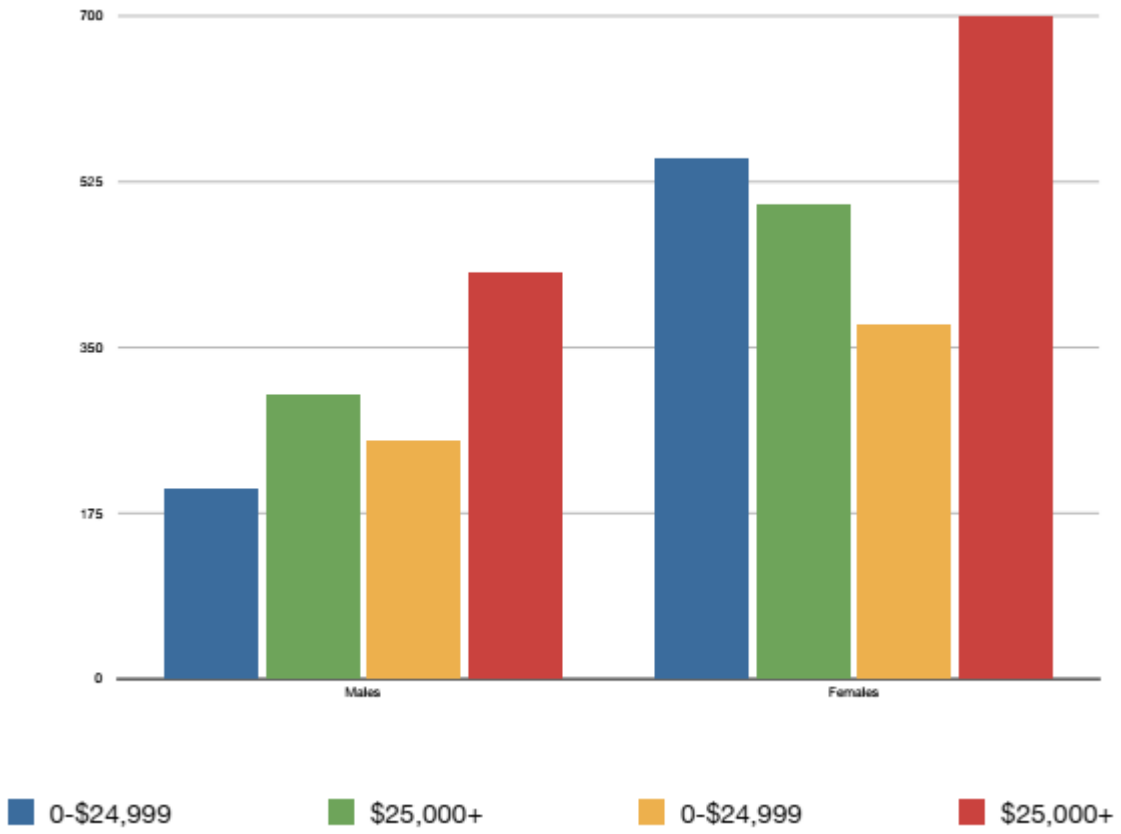

30%
25%
20%
15%
10%
5%
0%
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
Crime/Violence
Race relations

VISUALIZATION DESIGN PRINCIPLES

MAXIMIZE DATA-INK RATIO

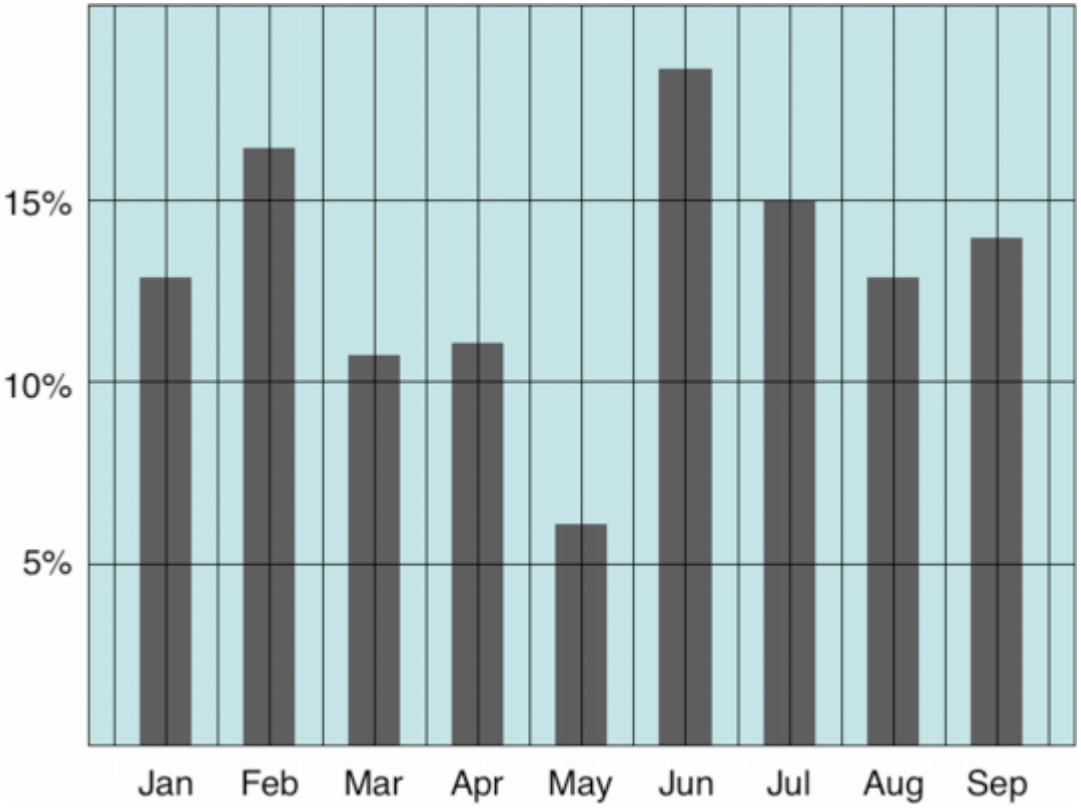


MAXIMIZE DATA-INK RATIO



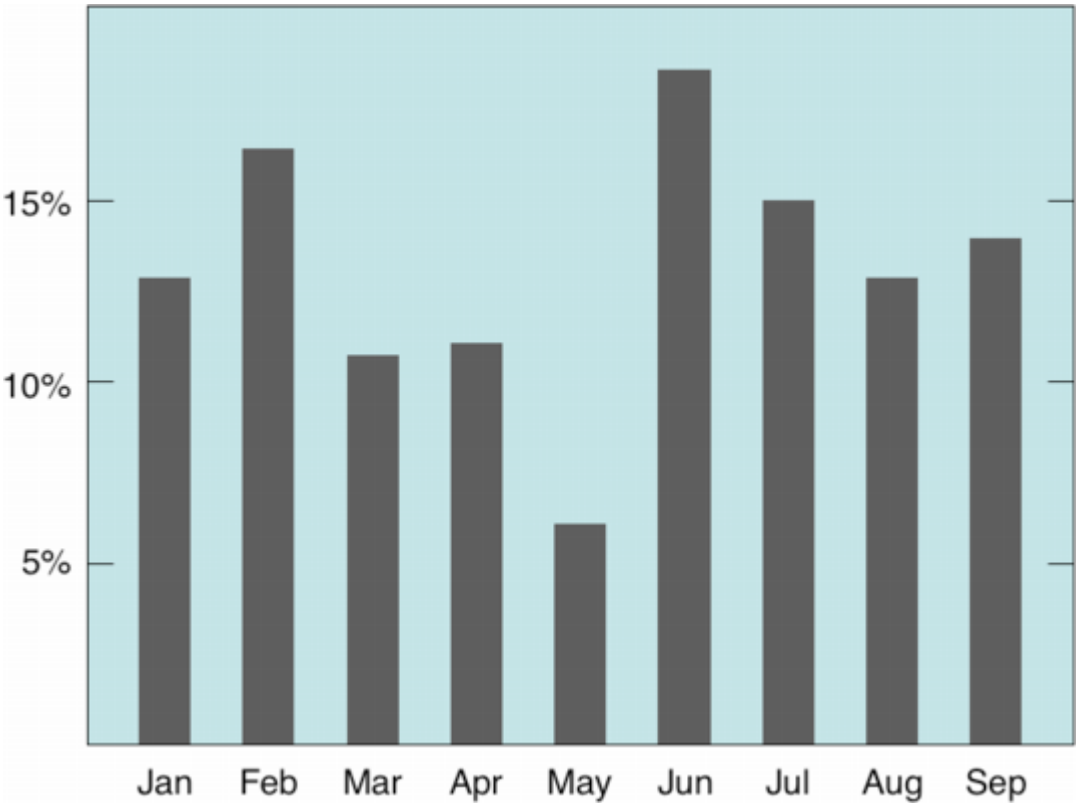
AVOID CHART JUNK

Extraneous visual elements that distract from the message

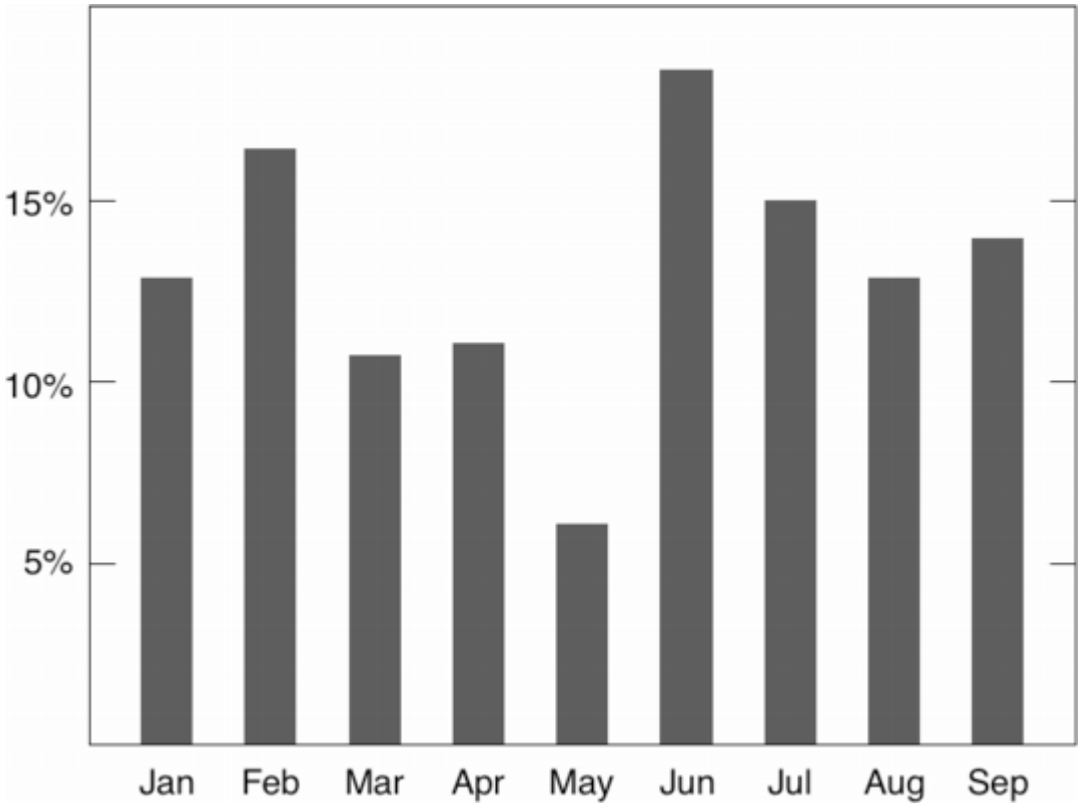


ongoing, Tim Brey

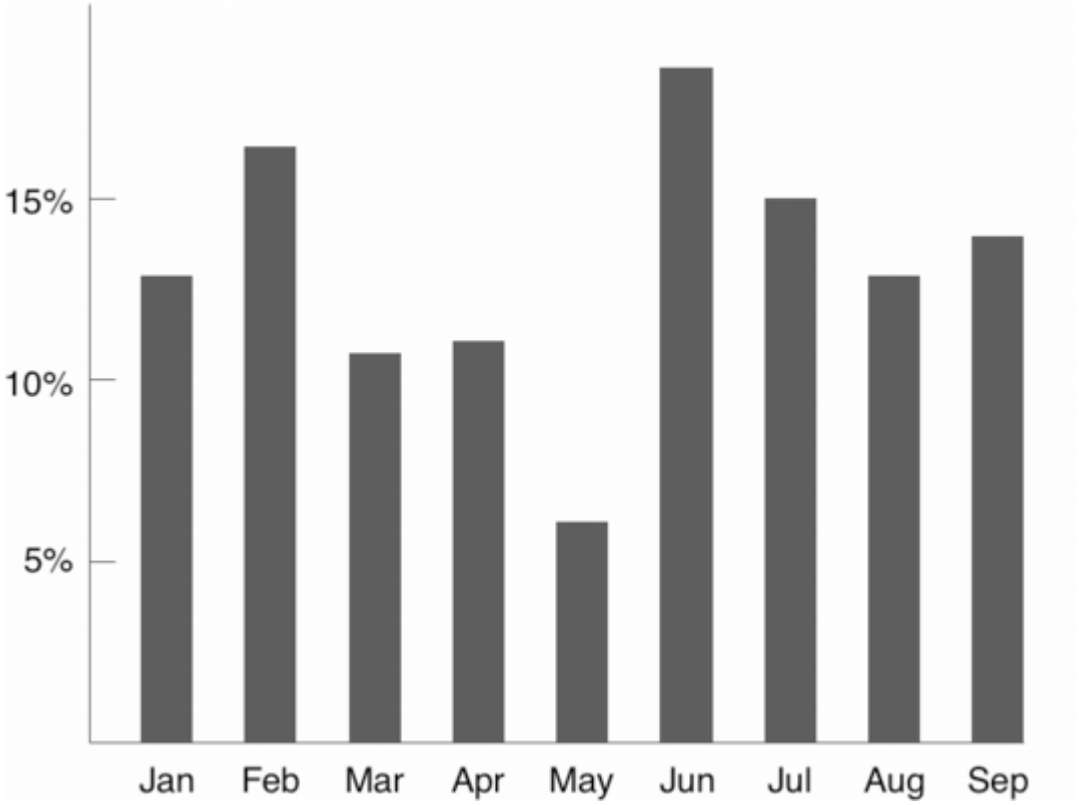
AVOID CHART JUNK



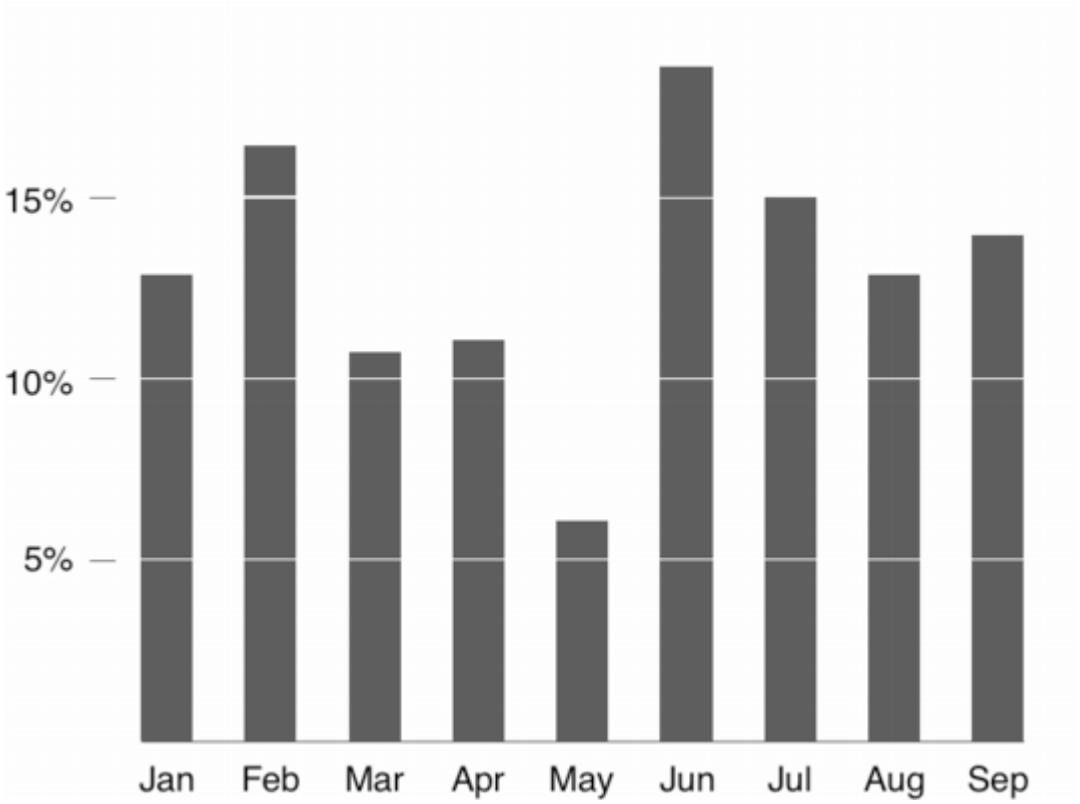
AVOID CHART JUNK



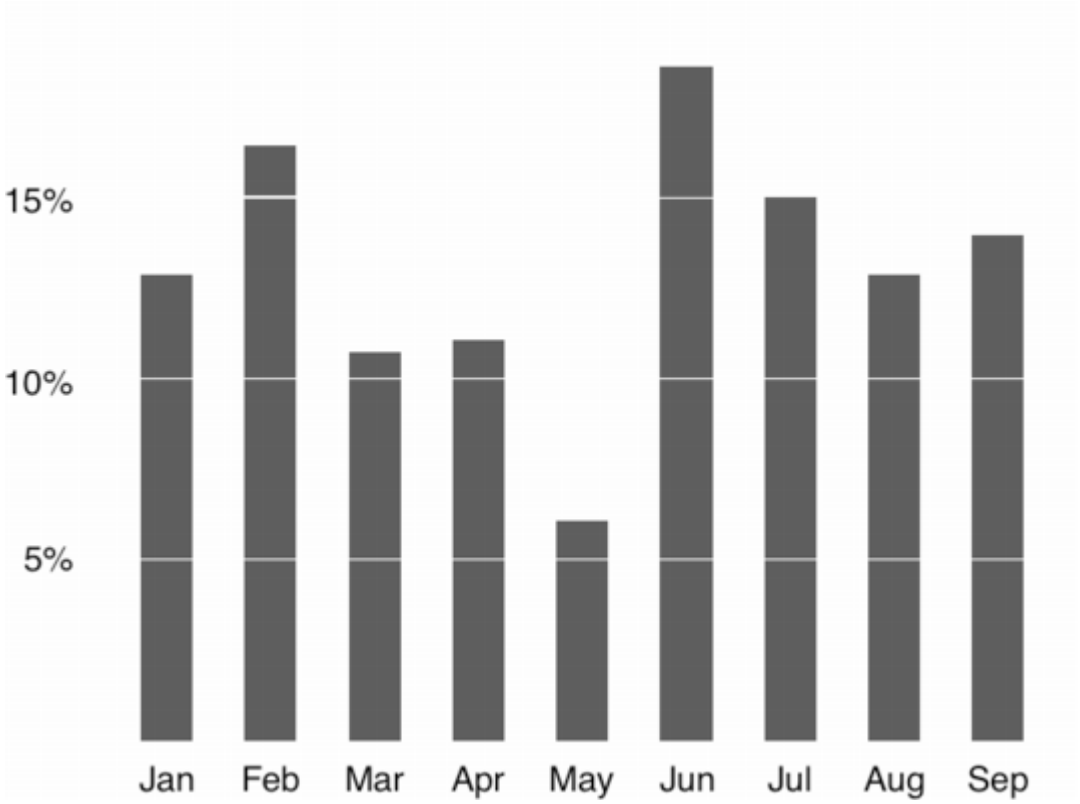
AVOID CHART JUNK



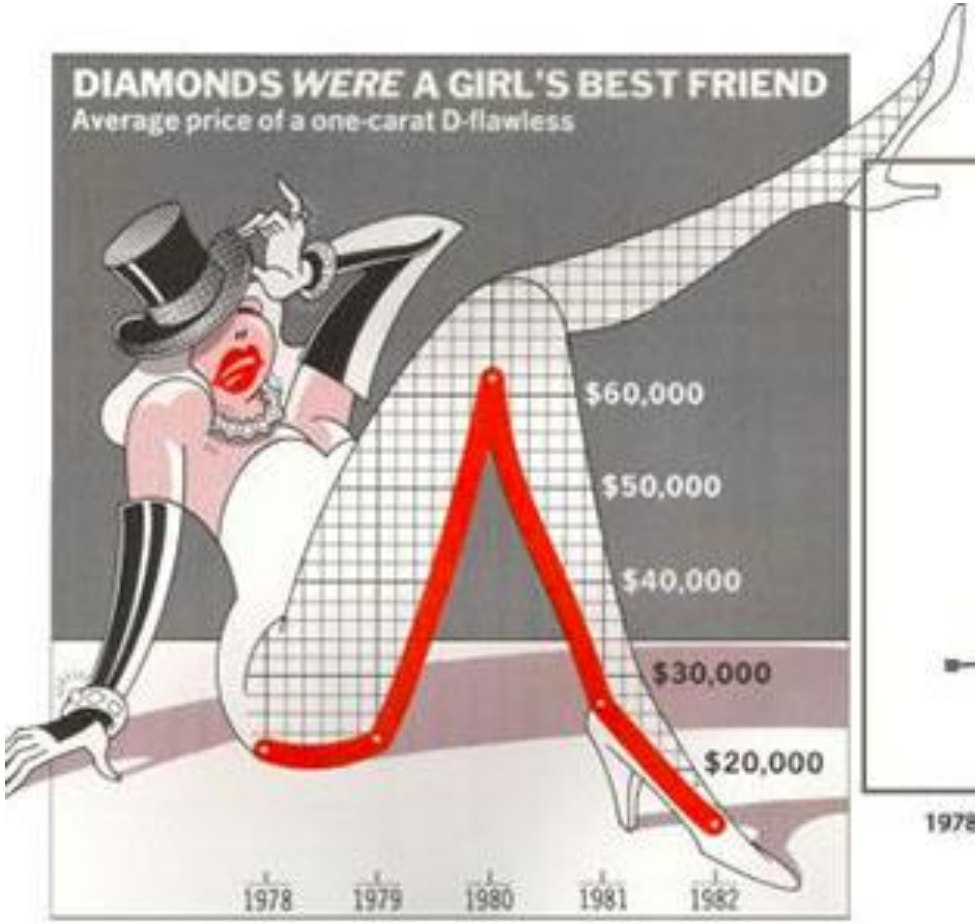
AVOID CHART JUNK

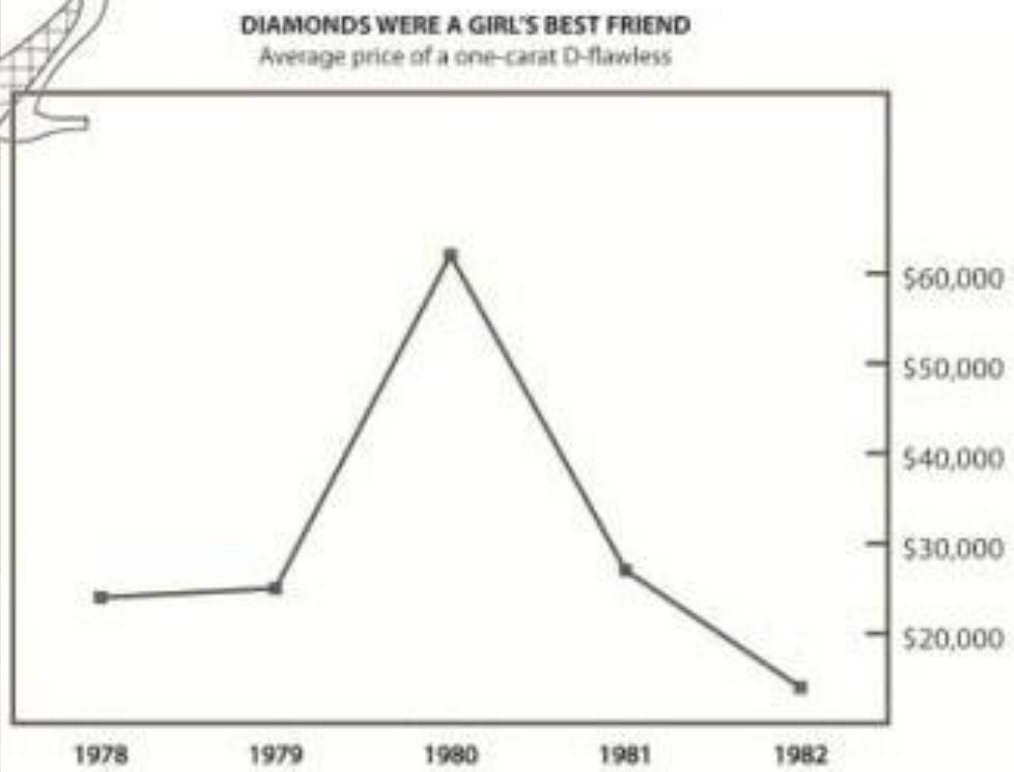


AVOID CHART JUNK



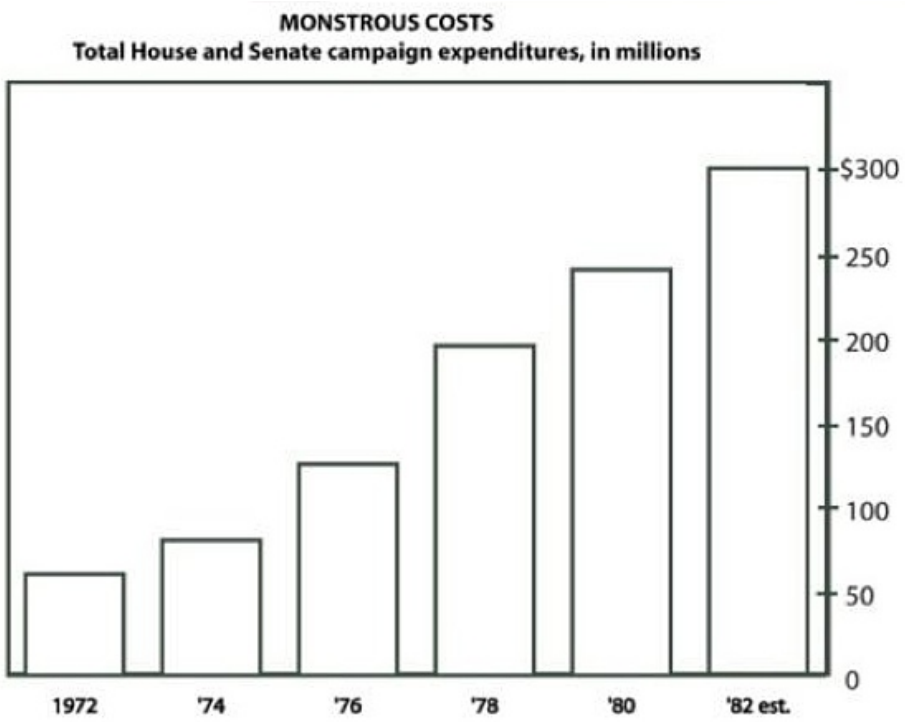
WHICH IS BETTER?





Bateman et al. 2010

WHICH IS BETTER?



Useful Junk? The Effects of Visual Embellishment on Comprehension and Memorability of Charts

Scott Bateman, Regan L. Mandryk, Carl Gutwin,
Aaron Genest, David McDine, Christopher Brooks
Department of Computer Science, University of Saskatchewan, Saskatoon, Saskatchewan, Canada
scott.bateman@usask.ca, regan@cs.usask.ca, gutwin@cs.usask.ca,
aaron.genest@usask.ca, dam085@mail.usask.ca, cab938@mail.usask.ca

ABSTRACT
Guidelines for designing information charts often state that the presentation should reduce 'chart junk' – visual embellishments that are not essential to understanding the data. In contrast, some popular chart designers wrap the presented data in detailed and elaborate imagery, raising the questions of whether this imagery is really as detrimental to understanding as has been proposed, and whether the visual embellishment may have other benefits. To investigate these issues, we conducted an experiment that compared embellished charts with plain ones, and measured both interpretation accuracy and long-term recall. We found that people's accuracy in describing the embellished charts was no worse than for plain charts, and that their recall after a two-to-three-week gap was significantly better. Although we are cautious about recommending that all charts be produced in this style, our results question some of the premises of the minimalist approach to chart design.

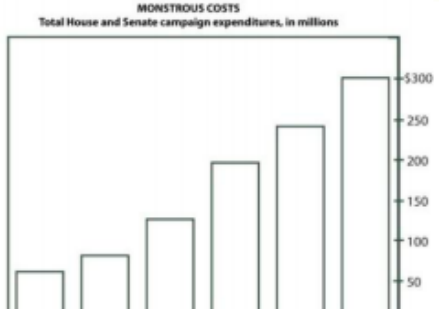
Author Keywords
Charts, information visualization, imagery, memorability.

ACM Classification Keywords
H5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

General Terms
Design, Human Factors

INTRODUCTION
Many experts in the area of chart design, such as Edward Tufte, criticize the inclusion of visual embellishment in charts and graphs; their guidelines for good chart design often suggest that the addition of *chart junk,* decorations and other kinds of non-essential imagery, to a chart can make interpretation more difficult and can distract readers from the data [22]. This *minimalist* perspective advocates *data-ink* – or the ink in the chart used to represent data.

Despite these minimalist guidelines, many designers include a wide variety of visual embellishments in their charts, from small decorations to large images and visual backgrounds. One well-known proponent of visual embellishment in charts is the graphic artist Nigel Holmes, whose work regularly incorporates strong visual imagery into the fabric of the chart [7] (e.g., Figure 1).



EXPERIMENTAL RESULTS

1. No difference for **interpretation accuracy**
2. No difference in **recall accuracy after a five-minute gap**
3. Significantly **better recall for Holmes charts** of both the chart topic and the details (categories and trend) **after long-term gap** (2-3 weeks).
4. Participants **saw value messages** in the Holmes charts significantly more often than in the plain charts.
5. Participants found the Holmes charts **more attractive, most enjoyed** them, and found that they were **easiest and fastest to remember**.

USE CHART JUNK? IT DEPENDS!

PROS

Persuasion

Memorability

Engagement

CONS

Biased analysis

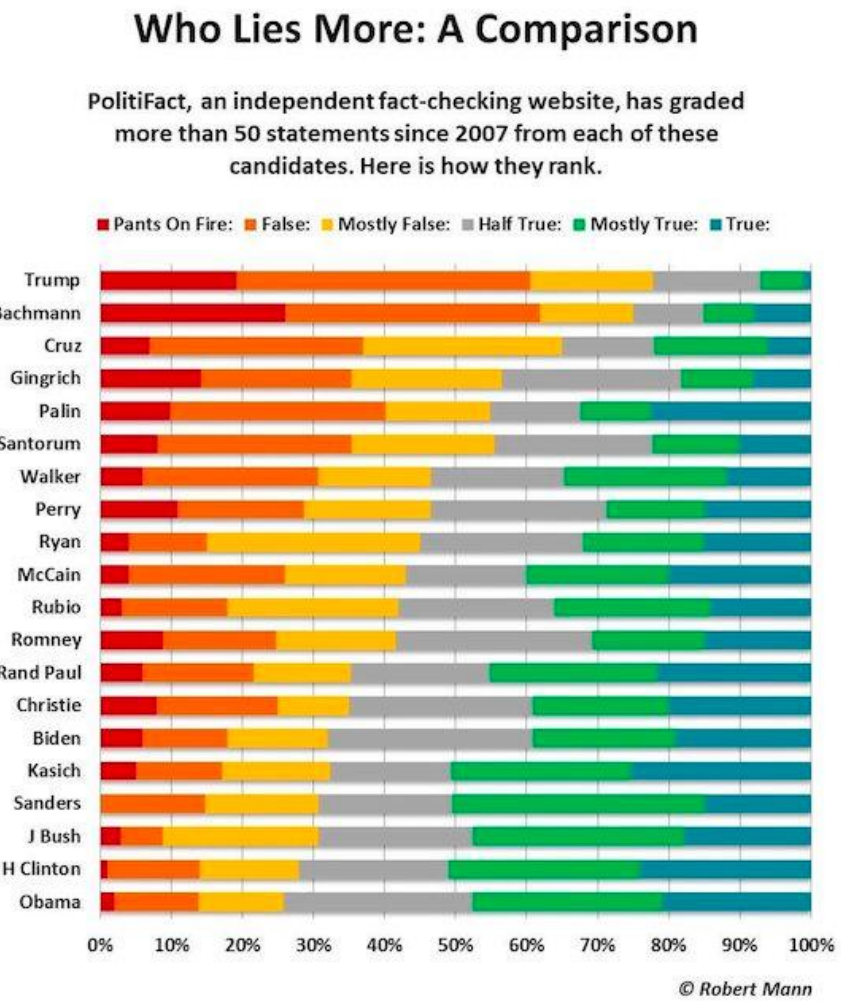
Trustworthiness

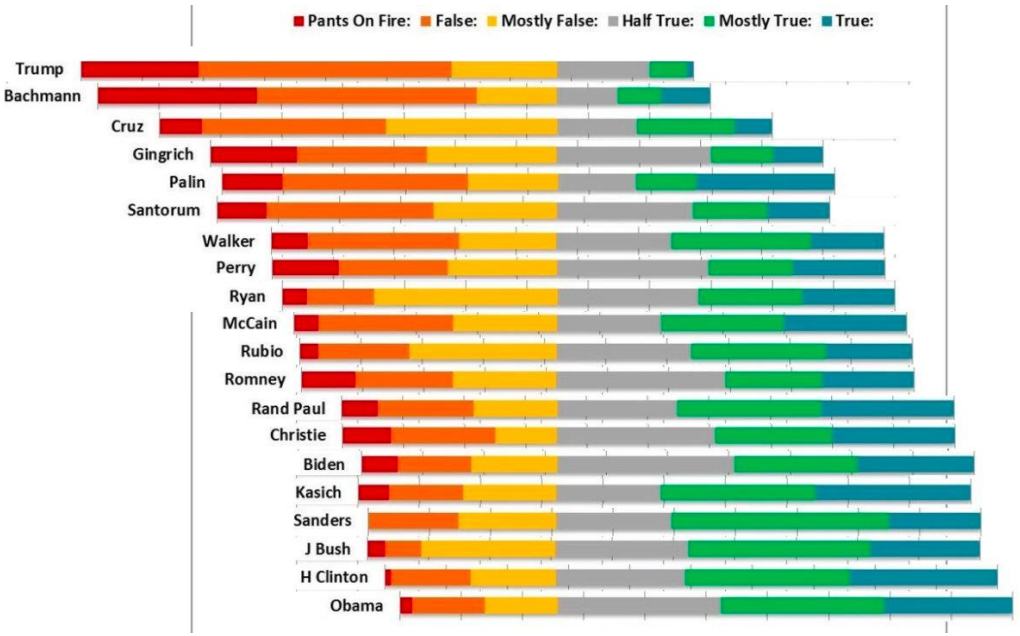
Interpretability

Space efficiency

Effort

ALIGNMENT MATTERS





https://twitter.com/infowetrust/status/760521739092627457

http://www.visualisingdata.com/2016/08/little-visualisation-design-part-21/

NO UNJUSTIFIED 3D

Depth judgment is bad

N = 0.67

Occlusion

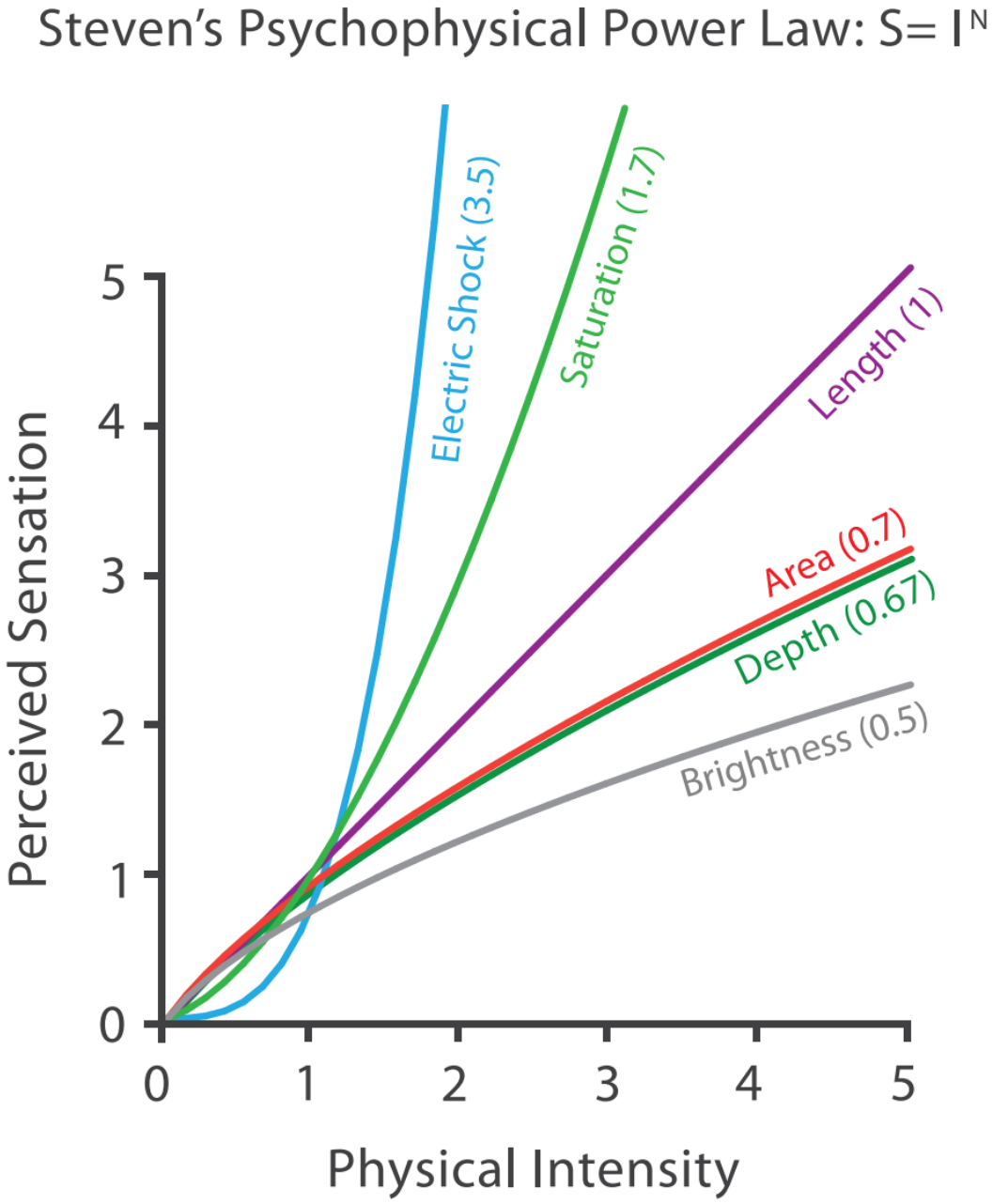
Perspective distortion

Color: Lighting /

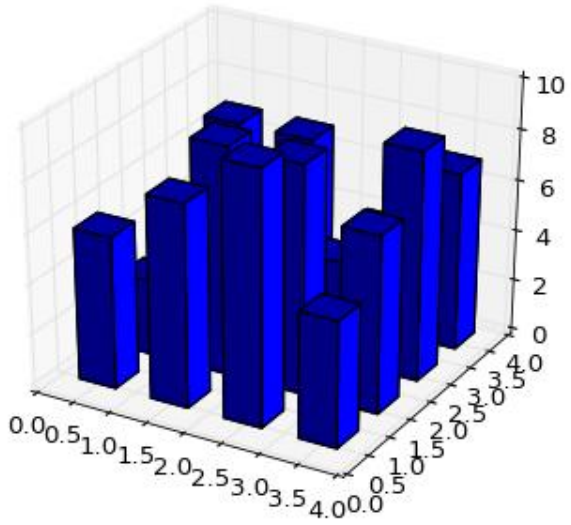
Shadows /

Shading

Title text illegible



DON'T


0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
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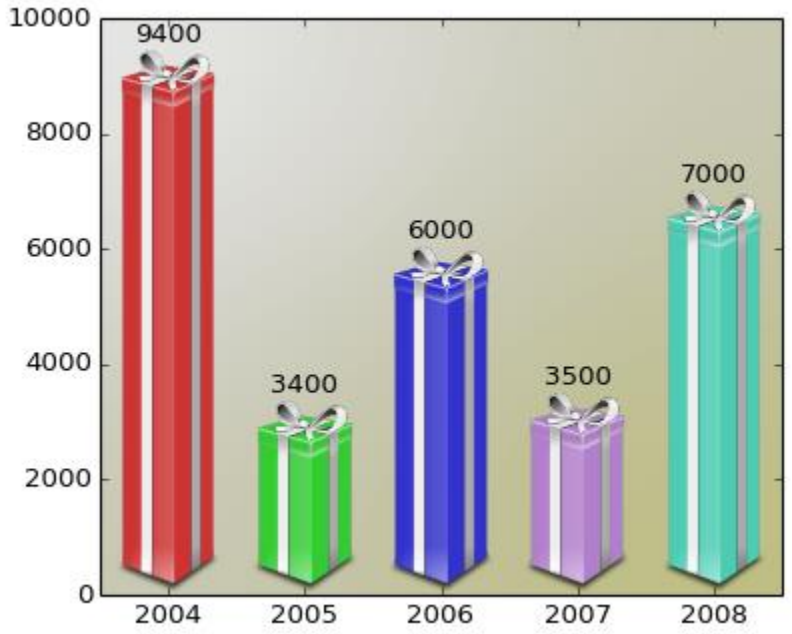
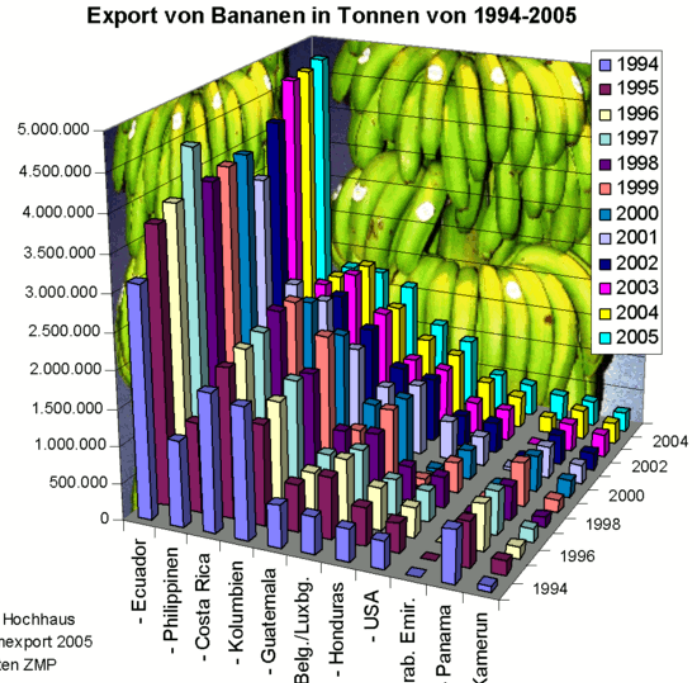
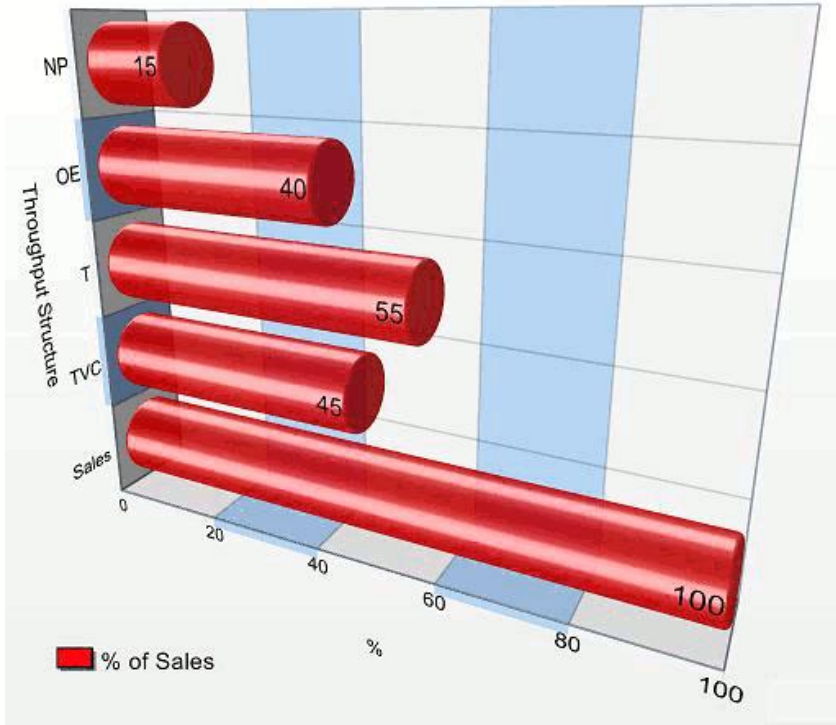
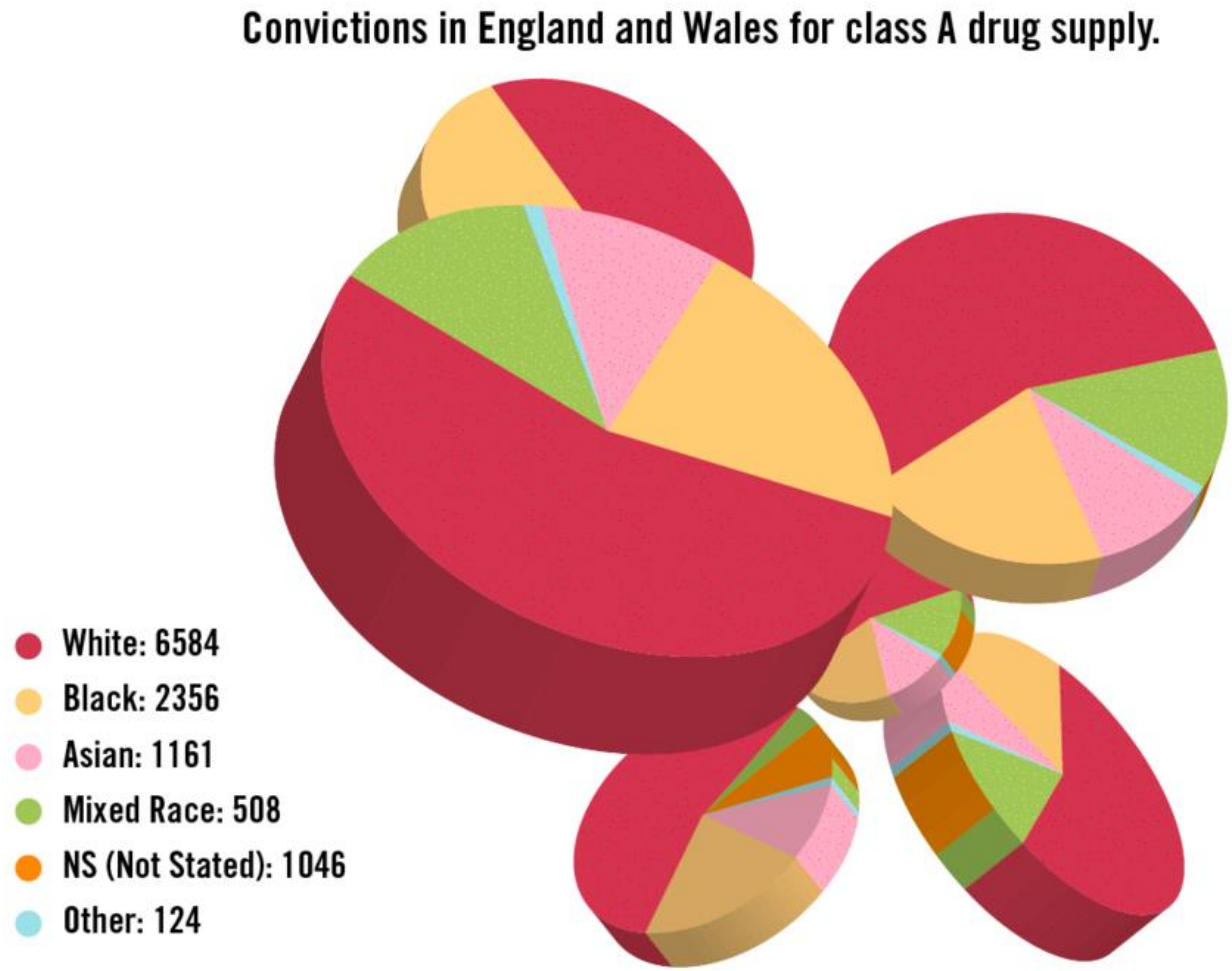
0
2
4
6
8
10


10000
8000
6000
4000
2000
0
9400
3400
6000
3500
7000
2004
2005
2006
2007
2008

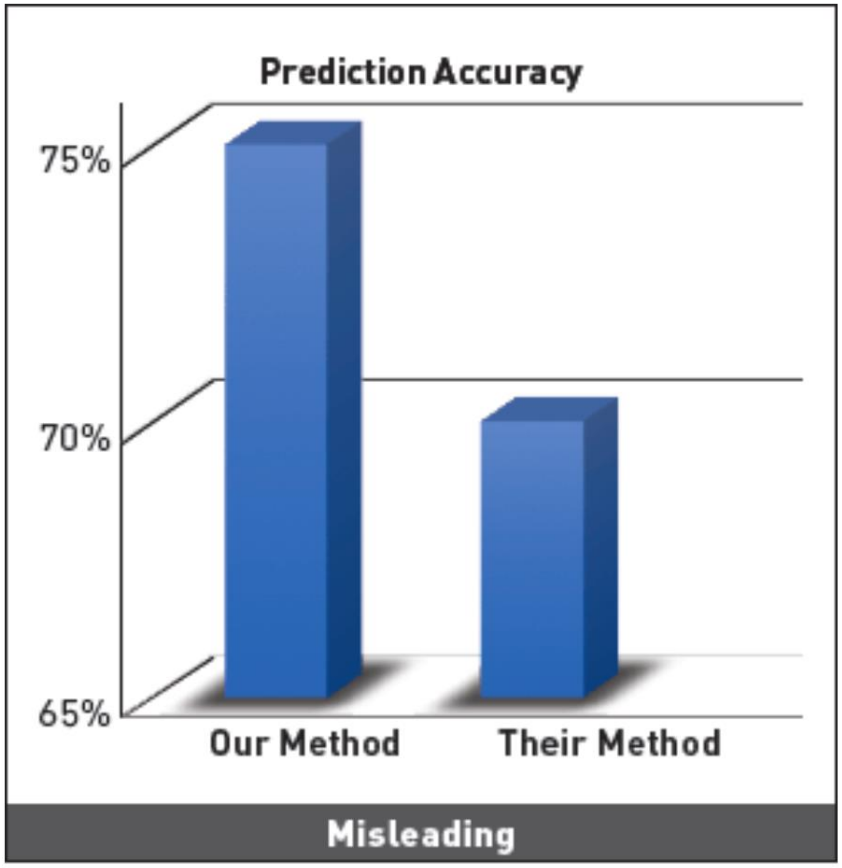

Export von Bananen in Tonnen von 1994-2005
5.000.000
4.500.000
4.000.000
3.500.000
3.000.000
2.500.000
2.000.000
1.500.000
1.000.000
500.000
0
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2004
2002
2000
1998
1996
1994
- Ecuador
- Philippinen
- Costa Rica
- Kolumbien
- Guatemala
- Belg./Luxbg.
- Honduras
- USA
rab. Emir.
Panama
Kamerun
Dr. Hochhaus
Banexport 2005
Daten ZMP

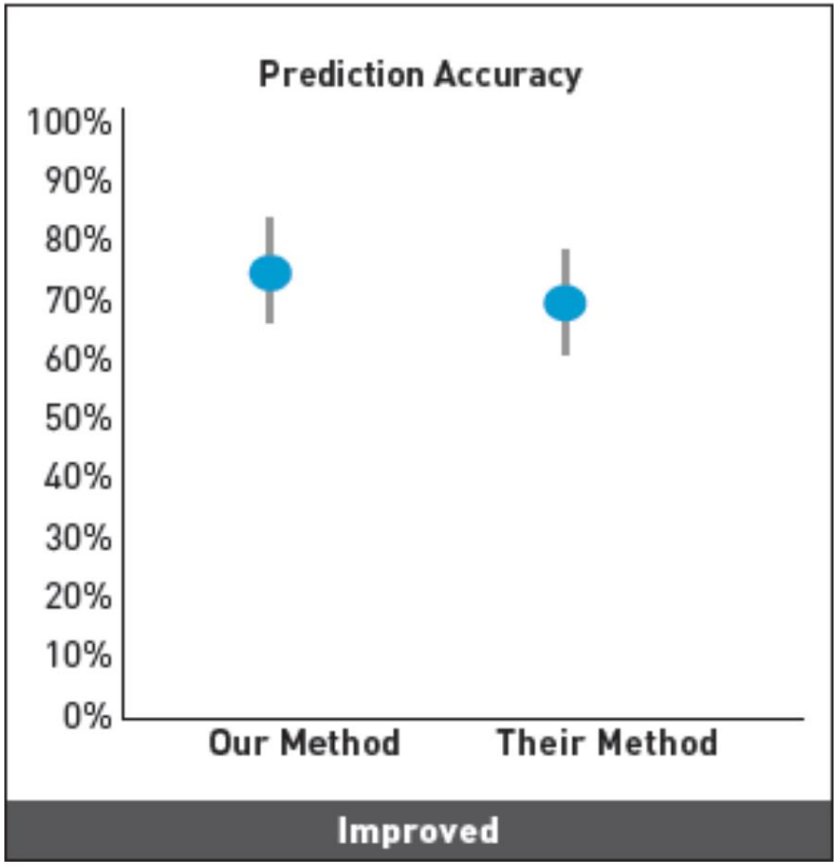

NP
15
OE
40
T
55
TVC
45
Sales
100
Throughput Structure
0
20
40
60
80
100
%
% of Sales

DON'T


Convictions in England and Wales for class A drug supply.
White: 6584
Black: 2356
Asian: 1161
Mixed Race: 508
NS (Not Stated): 1046
Other: 124

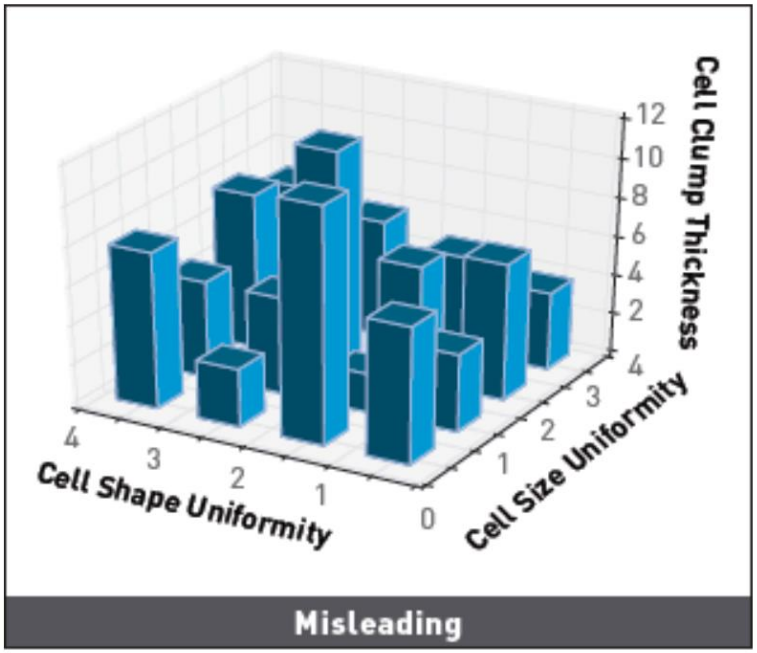
3D DESIGN ALTERNATIVE



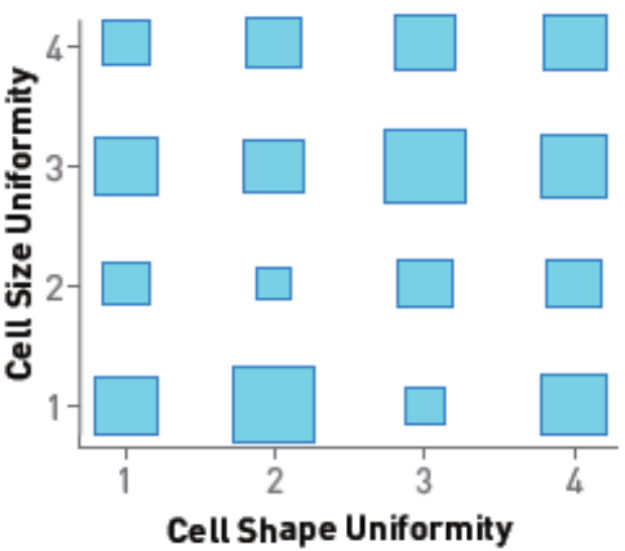


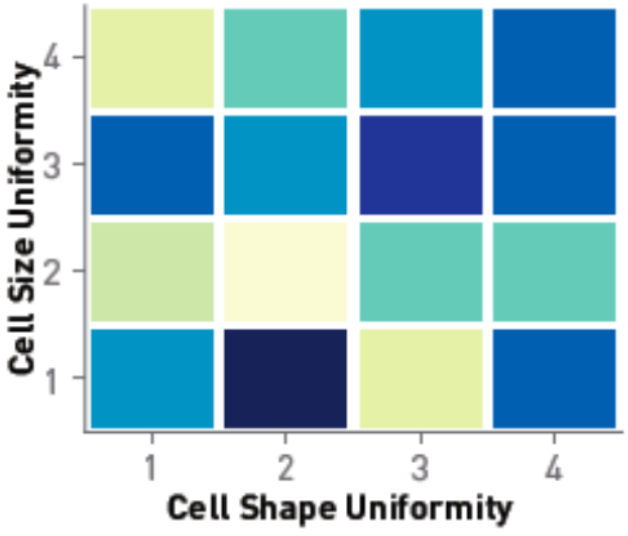
http://interactions.acm.org/archive/view/july-august-2018/the-good-the-bad-and-the-biased

3D DESIGN ALTERNATIVE



Misleading



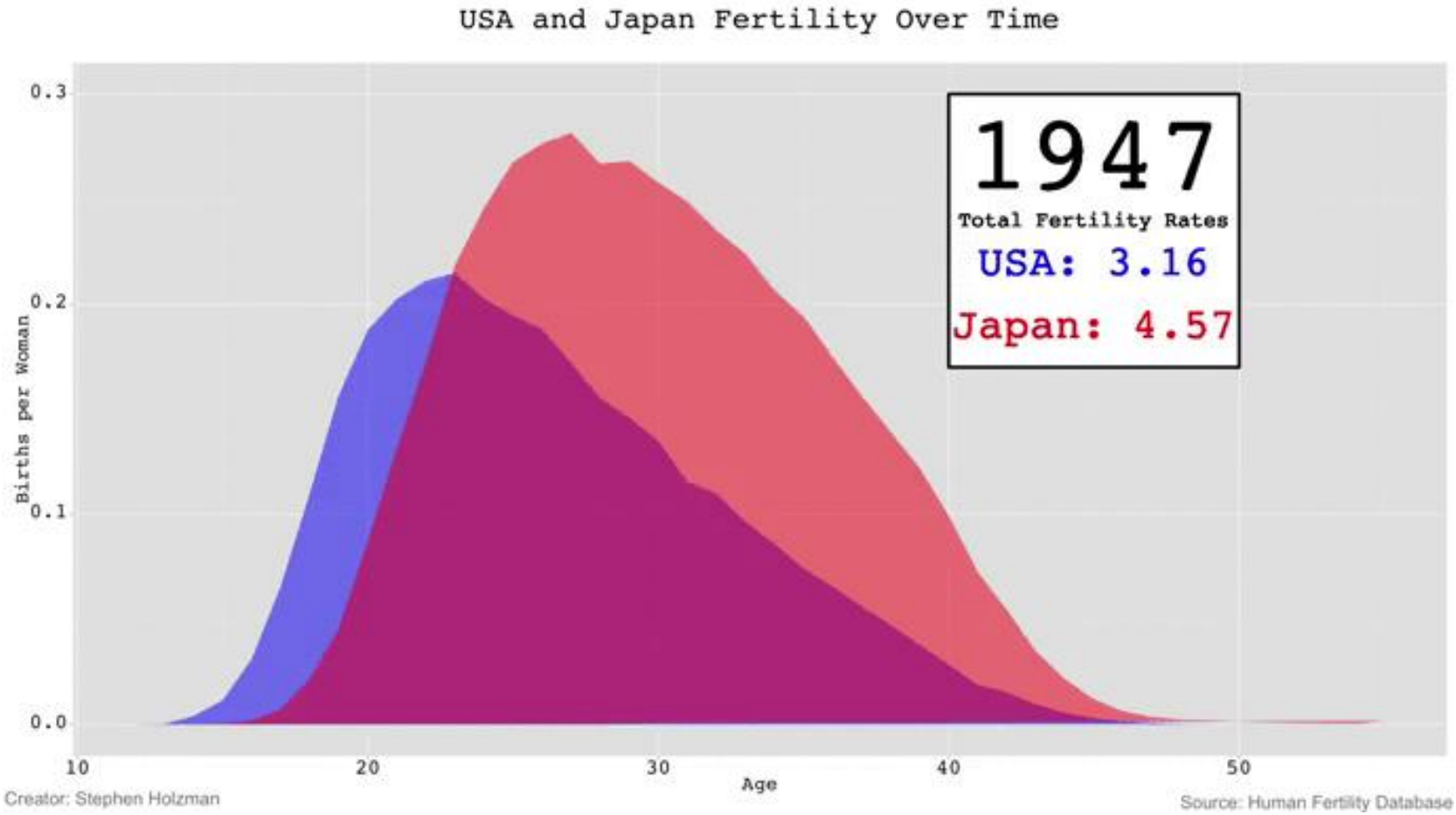


Improved

http://interactions.acm.org/archive/view/july-august-2018/the-good-the-bad-and-the-biased

EYES BEAT MEMORY

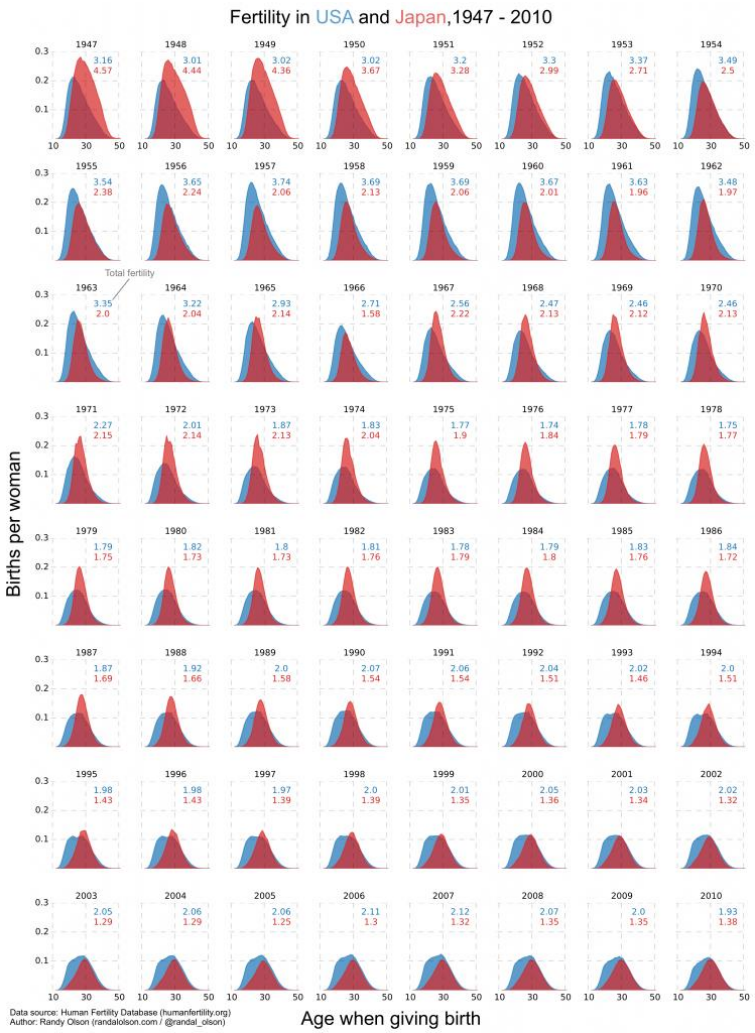
Don’t make people memory, show them



http://www.randalolson.com/2015/08/23/small-multiples-vs-animated-gifs-for-showing-changes-in-fertility-rates-over-time/

What can we do differently?

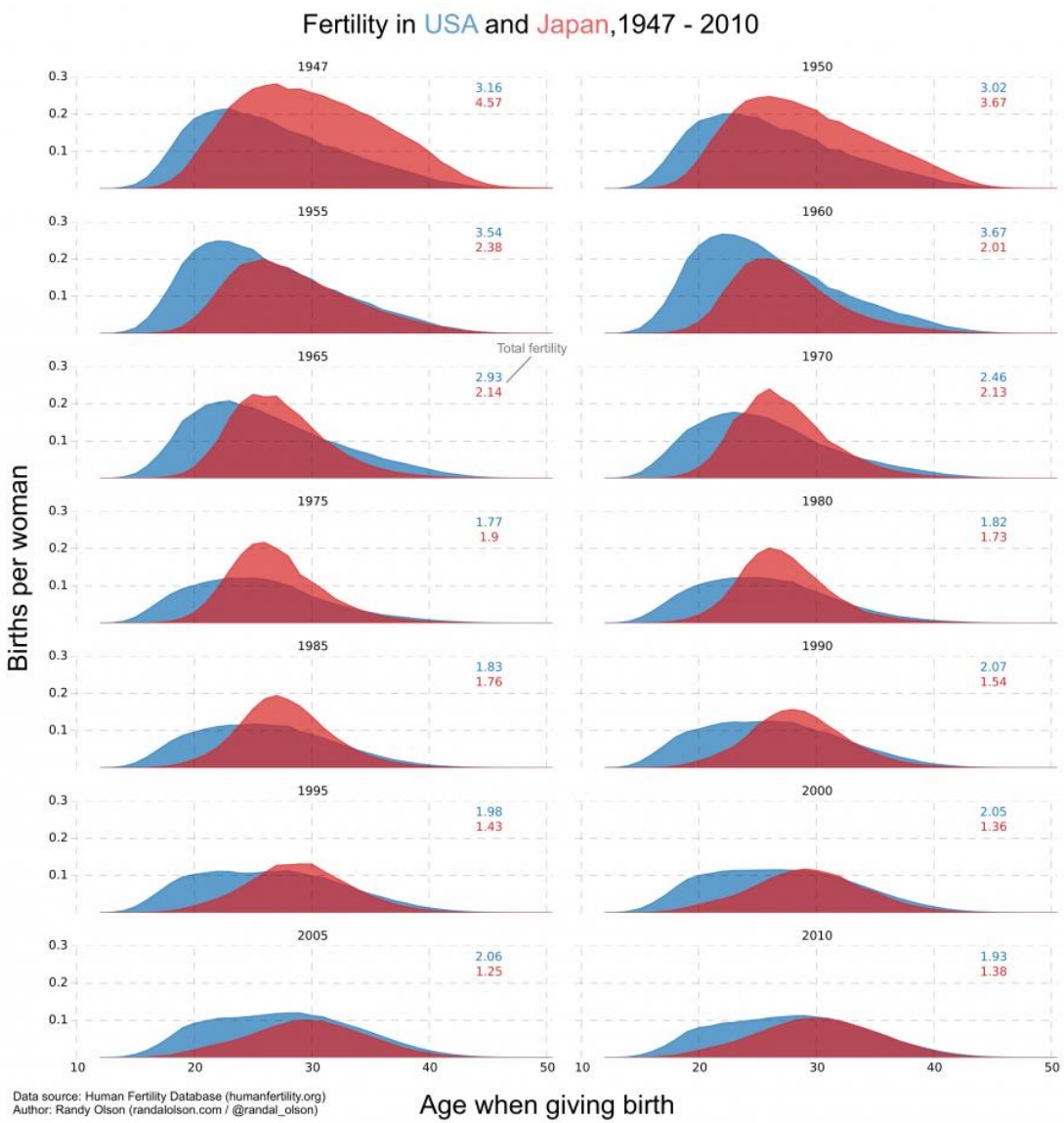
EYES BEAT MEMORY: SMALL MULTIPLES



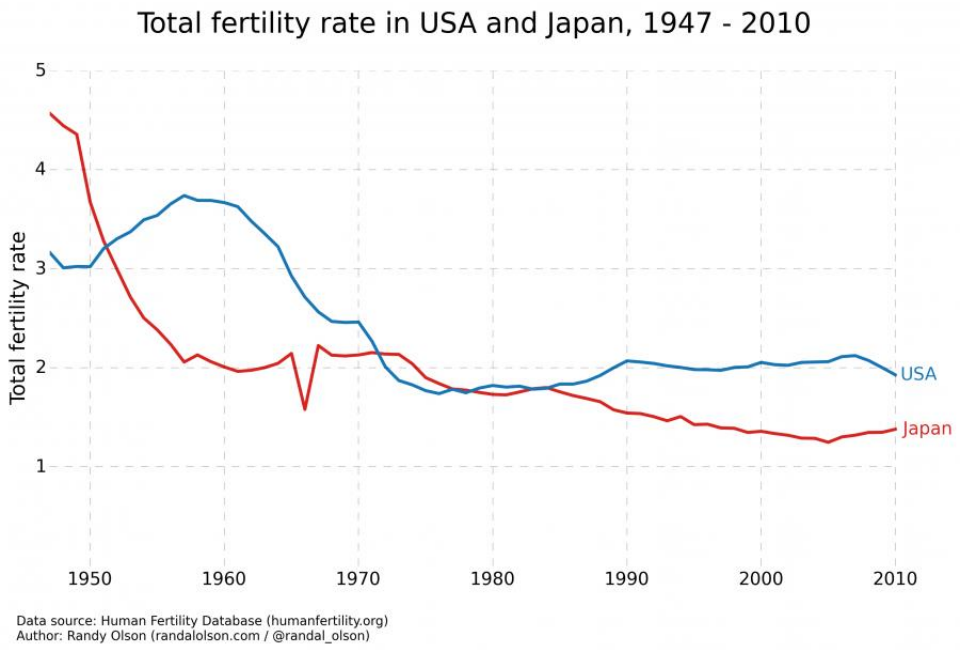
A lot of charts

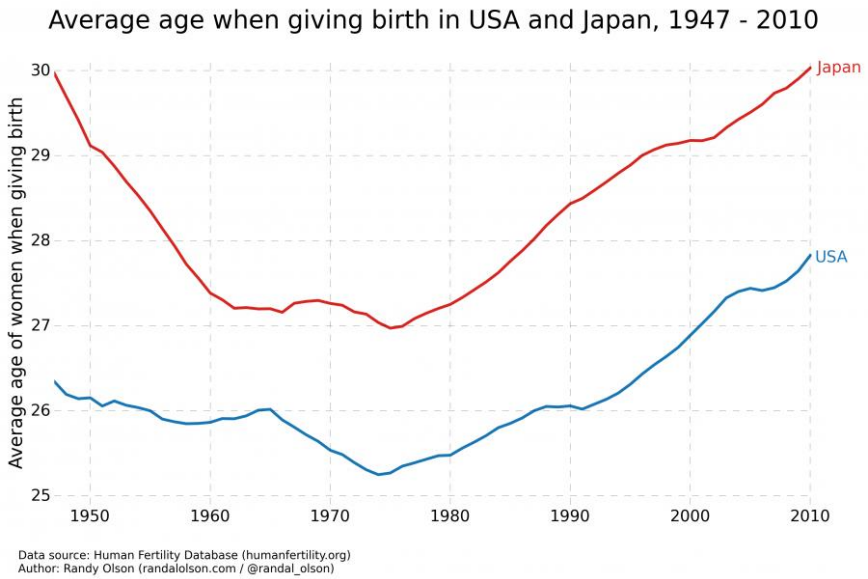
Do we need all of them?

EYES BEAT MEMORY: SMALL MULTIPLES

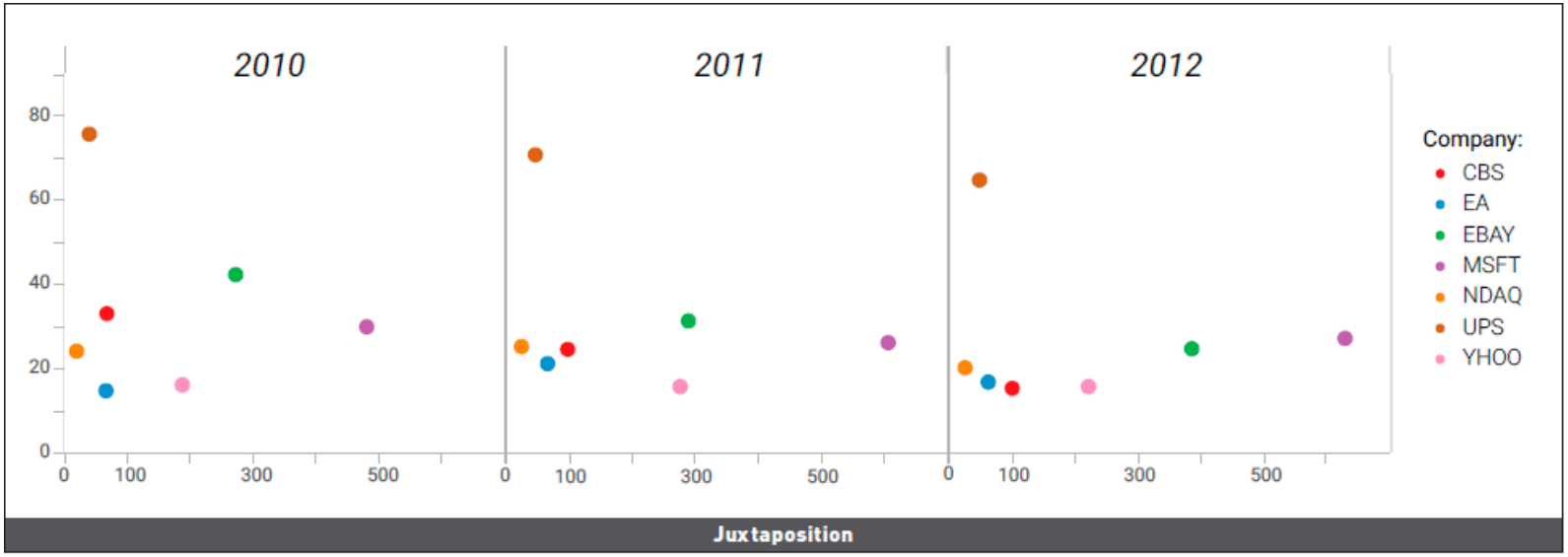


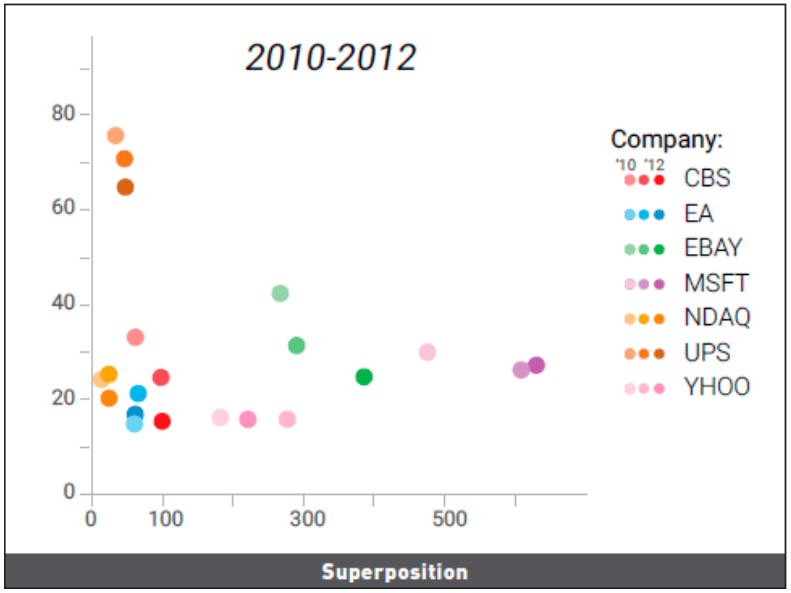
SIMPLIFY

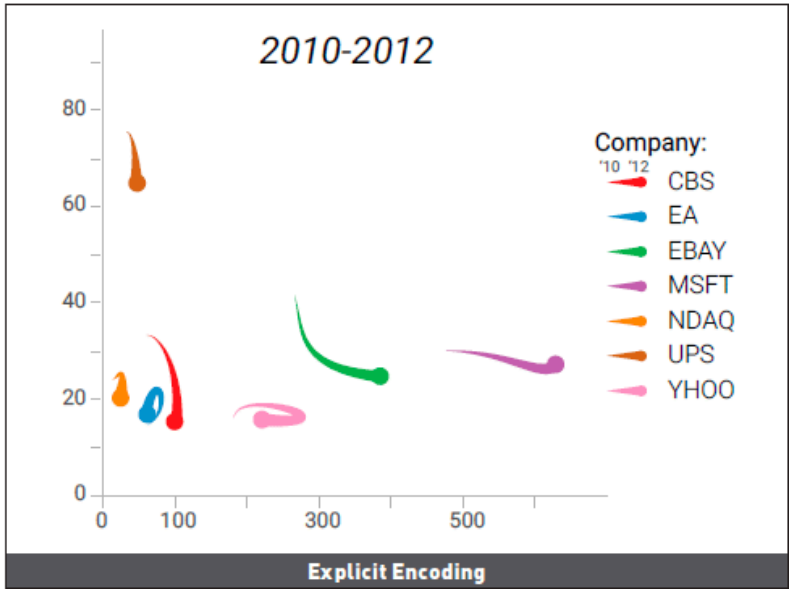

Total fertility rate in USA and Japan, 1947 - 2010
Total fertility rate
5
4
3
2
1
1950
1960
1970
1980
1990
2000
2010
USA
Japan
Data source: Human Fertility Database (humanfertility.org)
Author: Randy Olson (randalolson.com / @randal_olson)


Average age when giving birth in USA and Japan, 1947 - 2010
Average age of women when giving birth
30
29
28
27
26
25
1950
1960
1970
1980
1990
2000
2010
Japan
USA
Data source: Human Fertility Database (humanfertility.org)
Author: Randy Olson (randalolson.com / @randal_olson)

SMALL MULTIPLE DESIGN ALTERNATIVES







http://interactions.acm.org/archive/view/july-august-2018/the-good-the-bad-and-the-biased

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Scott Bateman, Regan L. Mandryk, Carl Gutwin,
Aaron Genest, David McDine, Christopher Brooks
Department of Computer Science, University of Saskatchewan, Saskatoon, Saskatchewan, Canada
scott.bateman@usask.ca, regan@cs.usask.ca, gutwin@cs.usask.ca,
aaron.genest@usask.ca, dam085@mail.usask.ca, cab938@mail.usask.ca

ABSTRACT
Guidelines for designing information charts often state that the presentation should reduce 'chart junk' – visual embellishments that are not essential to understanding the data. In contrast, some popular chart designers wrap the presented data in detailed and elaborate imagery, raising the questions of whether this imagery is really as detrimental to understanding as has been proposed, and whether the visual embellishment may have other benefits. To investigate these issues, we conducted an experiment that compared embellished charts with plain ones, and measured both interpretation accuracy and long-term recall. We found that people's accuracy in describing the embellished charts was no worse than for plain charts, and that their recall after a two-to-three-week gap was significantly better. Although we are cautious about recommending that all charts be produced in this style, our results question some of the premises of the minimalist approach to chart design.

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ACM Classification Keywords
H5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

General Terms
Design, Human Factors

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CHI 2010, April 10–15, 2010, Atlanta, Georgia, USA.
Copyright 2010 ACM 978-1-60558-929-9/10/04....$10.00.

Despite these minimalist guidelines, many designers include a wide variety of visual embellishments in their charts, from small decorations to large images and visual backgrounds. One well-known proponent of visual embellishment in charts is the graphic artist Nigel Holmes, whose work regularly incorporates strong visual imagery into the fabric of the chart [7] (e.g., Figure 1).

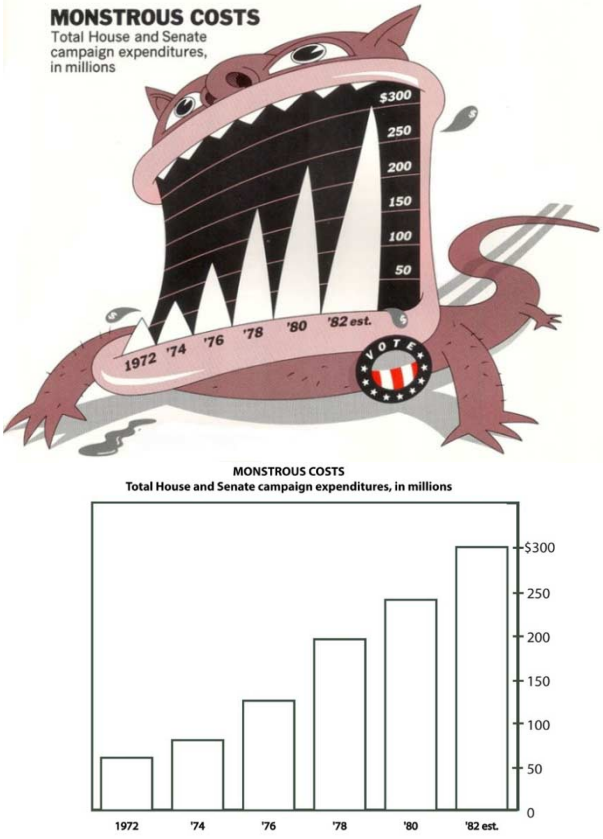

Figure 1. A chart by Holmes [7] (above), and a 'plain' version.

These kinds of charts appear regularly in many mass-media publications, and the widespread use of embellished designs raises questions about whether the minimalist position on chart design is really the better approach. Two issues in particular are raised: first, whether visual embellishments do in fact cause comprehension problems; and second, whether the embellishments may provide additional information that is valuable for the reader. For example, the added visual imagery in a Holmes-style chart could draw

the reader's eye, could help to convey a specific message, or could make the chart more memorable.

There is very little evidence from prior research to answer these questions. Previous studies have suggested that minor decoration in charts may not hamper interpretation [1,11,20], and work in psychology has shown that the use of imagery can affect memorability [4,6,14,17], but there is very little work that looks at how chart imagery can affect the way people view information charts.

To answer these questions, we carried out an investigation that tested the effects of visual embellishments on two aspects of interaction with charts: interpretation accuracy, and memorability. We conducted a study in which participants looked at several different minimalist and embellished charts (taken directly from Holmes's work) and answered questions about each chart's topic and details. Participants then answered the same questions again, after a gap of either a few minutes or of around 2 weeks.

Our study showed two main results. First, we did not find any difference in people's interpretation accuracy between the embellished and minimalist charts, despite the fact that the embellished charts contained a much larger amount of non-data-ink. Second, participants were able to remember significantly more about the embellished charts after a gap longer than 12 days, suggesting that the addition of strong visual images does help people to remember both the topic and details of the chart.

We followed up these performance results with an analysis of eye tracking data, which showed that people did spend time looking at the non-data ink in the embellished charts. These results support the idea that people do notice and process the visual imagery, which may be helping to encode the chart in their memory. Further, in their subjective ratings of which type of charts they preferred for reading and remembering participants overwhelmingly chose Holmes-style charts over plain charts.

Our study showed that if memorability is important, visual imagery can help to fix a chart in a viewer's memory. However, we do not advocate this strategy as a general principle, because the use of strong images in charts is contentious – for example, images convey messages that may intentionally or unintentionally bias the viewer's interpretation of the data. More importantly, then, our work shows that there *can* be strong effects from the inclusion of visual embellishments – something that has not been reported before – and that this phenomenon should be better understood by researchers and chart designers.

PREVIOUS WORK

The Debate over Visual Embellishment

In the field of information design there has been a long-standing debate concerning the use of imagery and 'unnecessary' visual embellishments. While some of these arguments are aesthetic in nature, considerable focus is also placed on what presentation style best conveys information to a viewer in graphic form. We divide this work into two sides: those that discourage the use of visual embellishments, and those that encourage their use.

The most well-known proponent of eliminating visual embellishments is Edward Tufte. Tufte takes the stance that "…it's wrong to distort the data measures—the ink locating values of numbers—in order to make an editorial comment or fit a decorative scheme" (p.59, [22]). Tufte proposes the *data-ink ratio*, where he argues that all ink that is not used to present data should be removed [22]. The data-ink ratio can be calculated by dividing the ink used for displaying data (*data-ink)* by the total ink used in the graphic, with the goal of having the ratio as close to 1 as possible. Cleveland [3] similarly proposed a series of recommendations for graph design, based on his research into graphical perception, and emphasized a minimalist approach to presenting data graphically. These minimalist views aim to reduce interpretation effort and increase interpretation accuracy when a user examines an information chart.

Although Tufte's and Cleveland's rules of thumb are well known, designers and publications continue to produce charts with a wide variety of chart junk, decorations, low data-ink ratios, and 3D representations. In a survey of journal publications, magazines, and newspapers from 1985 to 1994, Zacks *et al.* [24] found that there had been little change in the prevalence of graph features that are identified by Tufte, Cleveland, and others as undesirable. This suggests that there may be other reasons for including non-data-ink in graphs.

The other side of the debate, where embellishment of data graphics is encouraged, is exemplified by the work of Nigel Holmes. Holmes states that a data graphic "must engage the reader's interest" ([7], p. 60), and he demonstrates how this can be achieved using graphic imagery. He also states that "the purpose for making a chart is to clarify or make visible the facts that otherwise would lie buried in a mass of written materials" ([7], p. 21).

Interpretation Time and Errors

The minimalist perspective argues that the inclusion of visual embellishments and correspondingly lower data-ink ratios in graphs decreases the interpretability of data. There is some support for this assertion when interpretability is defined as accuracy and speed in answering questions based on graphs. Schonlau and Peters [18] did find a small but significant decrease in accuracy when a third dimension was added to pie charts. Siegrist [19] found that 3D bar and pie charts usually take more time to evaluate, although the accuracy of those evaluations was on par with 2D versions. Gillan and Richman [5] showed that high data-ink ratios were correlated positively with faster response times and greater accuracy, although further investigation demonstrated that varying the particular location and function of the additional ink changed results. For example, background images made interpretation harder, but lines for

X and Y axes improved response time. Their findings support Tufte's assertion that data-ink should be maximized, but only "within reason."

However, other research differs, finding no significant differences in interpretability between varying ratios of data to ink. Kulla-Mader [11] found no differences in interpretation errors between three ratios of data-ink; however, ratios varied only in the presence and colour of gridlines, and the existence of chart borders. Blasio and Bisantz [1] found no significant difference in interpretation time between low data-ink ratios and medium ratios in real-time monitoring tasks. They did find a significant difference between their high data-ink condition and the other conditions, but the high data-ink data was presented as a table, while other ratio conditions were charts. In addition, visual psychophysics research into graphing elements has found that graph elements with irrelevant extra dimensions are processed just as quickly, and in some cases more quickly, than simpler elements [20].

Aesthetics and Preferences

Tufte has suggested that even well-known chart designs, such as the rectangular bars of bar charts, contain redundant data. One of his redesigns suggest removing the left vertical line of the bar, leaving only an L-shaped indicator of a data point; he asserts that the left vertical bar is redundant ink. However, some researchers have suggested that moderate increases in visual embellishment such as fully-drawn graph elements (e.g. rectangular bars for values rather than L-shaped representations) or small levels of chart junk provide a valuable aesthetic improvement to the graphic. Inbar et al. [9] found that users preferred non-minimalist graphs; although, they suggested that the participant's unfamiliarity with the minimalist design may have been a confounding factor in the study. They conclude that the problem is complicated, a conclusion supported by research into 3D graphs: Levy et al. [12] found that users have a preference for using 3D graphs for showing information to others and enhancing memorability, but 2D graphs for immediate personal use.

These findings make sense within the context of Tractinsky and Meyer [21], who found that graphs were more likely to be designed with chart junk when designers had persuasion or impressiveness as an objective, particularly when the data in the graph reflected poorly on the graph designer.

Chart Memorability

Ware identifies 'memory extension' as one of the ways in which visualizations extend human cognition [23]. Charts, especially when they are used within the context of persuasion or presentation, are designed to aid in the memorability of the presented data [10], and existing research has supported this fact.

In both recognition and free-recall situations, research suggests that pictures lead to better performance than words alone for experimental learning tasks [6,17]. The impact of pictorial stimuli on written retention has been shown with young children and college students (e.g., [4,14]). A recent study examined the effects of pictorial illustrations on younger and older adults' recall of the content of short sentences, and found that when pictorial stimuli is combined with verbal stimuli, content recall was improved for both groups [2].

The emotional tone of pictures also appears to affect memory. Mather and Nesmith [13] found that participants were more likely to remember the location of positive and negative pictures than non-arousing pictures during an incidental-coding task. Since enhancement of memory occurred for both positive and negative pictures, it can be noted that arousal (rather than valence) is the critical factor.

Blasio and Bisantz [1] argue that in monitoring tasks, such as situational awareness of a control board, interpretability is paramount. Reductions in interpretation in these settings can significantly increase costs for corporations or make typical tasks much more mentally demanding. The majority of charts are generated for static media, but in these situations memorability of a graph may be as, or more, important than the interpretability; as having the static chart media on-hand when it is needed may not be possible. There have been very few studies of memorability in graphs. Kelly [10] found no differences in memorability between high and low data-ink conditions. Kelly's findings may have been confounded by the questions asked in the study, which included estimation-based components that could have biased the results.

COMPARISON OF PLAIN AND EMBELLISHED CHARTS

We designed an experiment to investigate differences in comprehension and recall between charts with two dramatically different levels of visual imagery. We maximized the differences in visual embellishment between two sets of charts while remaining as realistic as possible. For our embellished charts, we selected fourteen charts from Nigel Holmes' book *Designer's Guide to Creating Charts and Diagrams* [7] (see Figure 2). All of the charts incorporated some degree of chart junk, had relatively low data-ink ratios, and often contained images and cartoons that were woven into the presentation of the data. These charts were designed to attract the eye, engage the reader, and sometimes provide a particular value message over and above the presentation of the data itself.

For each of the Holmes charts we created a plain version of the chart (Figure 2), taking care to include exactly the same data in the chart (including type of chart, titles, axis labels, and axis values). In creating the plain charts we did not focus on specifically maximizing the data-ink ratio, but developed charts that reflected Inbar's conjecture [9] about the 'sweet spot' of chart design where high data-ink ratios and familiarity intersect to create comfortably readable graphs. As a result, the plain charts preserved axis ticks, the overall chart style (e.g. bar, line, or pie), and the positioning of axis values and labels as they existed in the original

Holmes version. However, axis lines (both for X and Y) were removed, and the axis titles and chart title were repositioned if necessary to conform to more traditional title positions. Bar and line charts were framed by a black box, and all used grayscale.

Participants and Apparatus

Twenty participants (9 male, 11 female), aged between 18 and 40 were recruited from a local university (7 graduate, 13 undergraduate). Of the 20 participants, 17 said they at least occasionally created charts and 19 said they at least occasionally read or interpreted charts.

The charts were digitized and formatted to be displayable on a computer monitor as full-screen slides, yielding 14 charts, each with a Holmes version and a plain version. A standard Windows PC was used to present the charts on a 24-inch widescreen monitor; we used a Tobii eye tracker to capture eye-gaze data.

Each slideshow presented the fourteen charts, alternating between Holmes and plain versions. Each participant saw only one version of each chart, either Holmes or plain. Presentation order for the versions was counterbalanced, but chart order was not. We treated the first two slides as training. A fixation-cross slide was displayed between each chart to provide the participants with a visual break.

Procedure

The experiment had two parts: a chart reading and description phase, and a recall phase. To prevent intentional learning, participants were not told about the recall phase – they were told that there were two parts to the study and that the first part was unrelated to the second part. Participants were assigned to one of two order conditions, with ten in each condition.

Part 1: Reading and Description Task

Participants were seated approximately 24 inches from the computer monitor, and the eye-tracker was calibrated for each participant. The experimenter instructed the participants that they would be shown a series of charts that they could examine for as long as they needed while answering four questions related to the chart. Participants were informed that they could answer the questions without prompting or ask for assistance in the form of a reminder. The answers were stated verbally to the experimenter. If at any time, the participant strayed from either the prescribed question ordering or subject matter, the experimenter would redirect the conversation to focus on the question.

The experimenter asked the four questions verbally; the list of questions is shown below, followed by extra instructions (in italics) provided if the participant did not understand. Questions represent components of high-level chart comprehension.

Q1–Subject: ‘What is the chart is about?’ *‘Tell me about the basic subject of the chart.’*

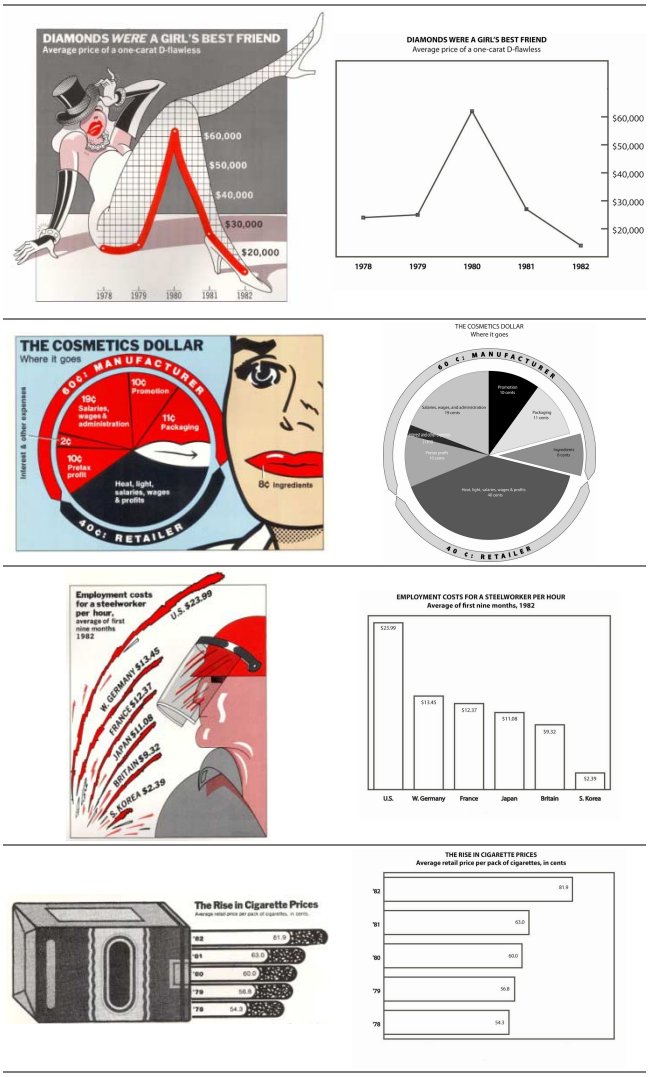

Figure 2. Example charts used in the study: Holmes versions (left) and equivalent plain versions (right).

Q2–Values: ‘What are the displayed categories and values?’ *‘Tell me how the chart is organized and any relevant values.’*

Q3–Trend: ‘What is the basic trend of the graph?’ ‘*Tell me whether the chart shows any changes and describe these changes.’* (Note that this question was not relevant for pie charts.)

Q4–Value Message: ‘Is the author trying to communicate some message through the chart?’ *‘Is the author trying to get across a specific point or is he or she merely presenting objective information?’*

If participants failed to provide a complete answer to any question, the experimenter went through a series of increasingly specific prompts until either the participant sufficiently answered the question or the list of prompts was exhausted. The prompts were as follows:

1. ‘Can you be more specific?’ This prompt was provided as generic reminder to elaborate.

2. 'Can you be more specific about *X*?', where *X* was a question-specific prompt. e.g. 'the basic subject'; 'how they've categorized the data'; 'any changes the chart shows'; or 'whether there is a message'.
3. A specific prompt directly related to the component of the participant's answer currently missing. (E.g. 'What were the values on the x-axis of the graph?')

Part 2: Recall Task

After the reading and description task, participants were placed in one of two recall conditions (immediate recall or long-term recall), with ten participants in each. The long-term participants scheduled their second part of the study for two to three weeks later. Participants in both conditions had previously been informed that the second portion of the experiment was unrelated to the first and therefore remained unaware of the recall task.

Participants in the immediate recall condition played a five-minute game after describing the charts to clear their visual and linguistic memory before starting the recall task.

At the beginning of the recall task, whether in the immediate or long-term recall condition, participants were reminded of the four component questions they had been asked for each chart during the description task (subject, values, trend, and message) then instructed to recall as many of the charts as possible. The experimenter recorded any correctly recalled charts (e.g. "I remember one about the price of diamonds"). After the participant had recalled as many charts as possible, the experimenter asked the participants to describe the charts as completely as they could, based on the four component questions. The descriptive recall was performed in the order of chart recall, not experimental presentation. If at any point the participant recalled a new chart, it was added to the end of the recall list, but the participant was redirected to the current chart's descriptive recall stage. Finally, the experimenter prompted the participant for each un-recalled chart (e.g. "Do you remember the chart called 'Monstrous Costs'?") and continued with a descriptive recall stage for each in turn.

Measures

We collected three types of data from the experiment: experimenter-coded response scores, eye-gaze data, and a preference questionnaire.

Response Scores

A single experimenter coded the participant responses. The same scale was used in both description and recall parts of the study. The experimenter updated a checklist as the participant described each graph. For each question in both the description and long-term recall the experimenter scored the participant's answer in two parts: first, whether a prompt was required (scored as either "Yes" or "No", regardless of what level of prompting was required; second, the quality of their response (independent of prompting) on the following four point scale.

3 points (all correct): The participant provided all of the relevant information in a given category.

2 points (mostly correct): The participant provided most of the relevant information in a given category, but omitted one or two important details.

1 point (mostly incorrect): The participant provided some correct information in a given category, but omitted most relevant details.

0 points (all incorrect): The participant provided no correct or relevant information in a given category.

0 points (I don't know): The participant stated that they did not know any of the relevant information in a given category. This was scored as completely incorrect.

An example of scoring: to a participant looking at the Holmes 'Monstrous Costs' chart, we would ask question Q3: 'What is the basic trend of the chart?' If the participant responded, 'I don't understand,' we would elaborate: 'Tell me whether the chart shows any changes and describe these changes.' The participant might answer 'The teeth get bigger every year.' This answer would score 1 point, as it is not a complete answer (with incorrect information about the period of the data reported) but provides at least some information that the bars increase. The experimenter would then provide additional prompts starting with 'Can you be more specific?' A complete answer scoring four points might be 'The chart shows that campaign expenditures by the house increased by about 50 million dollars every two years, starting in 1972 and ending in 1982.'

Early in the data collection stage, a second researcher reviewed the primary coder's results by independently coding a single participant through a review of the interview video. The primary coder also performed an additional coding of the same interview, based only on the video capture. All three versions, the original coding, the second coding by the primary coder, and the alternate coding by the second researcher differed only a few times and then never more than by one point.

Preference Questionnaire

After the recall portion of the experiment, participants were asked to choose their preferred chart format for a number of questions related to both the description of the charts and the recall of the charts. For participants in the immediate recall group, this questionnaire was administered on the same day as the description of the charts; for participants in the long-term recall group, this questionnaire was administered 12-22 days after the initial chart descriptions.

Gaze Data

We used the eye tracker throughout the description task to get an idea at what people looked at on a chart. For each chart, we used the Tobii 'area of interest' analysis software to designate each type of content that was in a chart. Chart content was coded by a single experimenter as one of the following (see Figure 3):

- Data: gazes in the region would be for primarily data

- Embellishment: gazes in the region would be for primarily non-data imagery
- Dual coded (Data and Embellishment): it could not reasonably be assumed if gazes in the region would be for data or imagery
- Other: any gaze not falling in a specified region

We removed gaze data for charts where gazes did not register correctly, which would have resulted from the participant shifting or moving out of range of the eyetracker during the study.

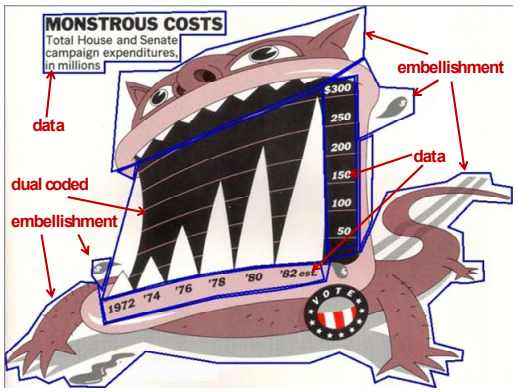

Figure 3. Area of interest analysis: gaze falling in any of the areas defined by the blue borders were labeled as data, embellishment, dual encoded, or other.

RESULTS

We first present the results related to describing the charts, followed by the recall results and the user preferences, and then analyze the participants' gaze data. In all cases we used a t-test for analysis; all of our hypotheses and questions related to a comparison of two groups (e.g. long-term vs. immediate recall). We applied a 1-tailed test where our stated hypothesis was that one group was significantly greater or less for a particular measure.

Description

Description scores were calculated from the coded responses. After removing data for the two training charts, and summing description scores over the remaining charts for each participant, we conducted pairwise t-tests to determine whether the presence of visual embellishments altered the quality of participant verbal chart descriptions. There were no differences in the scores of chart descriptions for the subject of the chart (t_{19}=0.84, p=.412), the categories in the charts (t_{19}=1.38, p=.185), or the trend of the charts (t_{19}=0.23, p=.818), depending on whether or not visual embellishments were present (see Figure 4). There was a difference in the description of a value message of the charts (t_{19}=3.37, p=.003), with better descriptions of a value message for the Holmes charts than the plain charts (see Figure 6).

We also looked at the total completion time for users in completing the description task – calculated from the total onscreen time spent with each chart. A pairwise t-test was used to determine if the presence of embellishments resulted in users spending more time to complete the description tasks. There was no difference in completion time for describing the charts (t_{19}=1.834, p=.082), depending on the presence of visual embellishments (Mean Holmes=2.60min, SD Holmes=0.67; Mean Plain=2.43min, SD Plain=0.68).

Recall

Recall scores were calculated from the coded responses. After removing data for the two training charts, and summing recall scores over the remaining charts for each participant, we conducted pairwise t-tests to determine if users were able to better recall charts that contained visual embellishments. Because our hypothesis was that visual embellishments would aid recall, we conducted 1-tailed t-tests with α=0.05.

For participants in the immediate recall group, there were no differences in recall scores for the subject of the chart (t_9=1.24, p=.124), the categories in the chart (t_9=1.21, p=.129), or the trend of the chart (t_9=0.35, p=.369) between charts that contained visual embellishments and charts that did not (see Figure 5). There was a difference in the recall score for the value message (t_9=2.24, p=.026), see Figure 6.

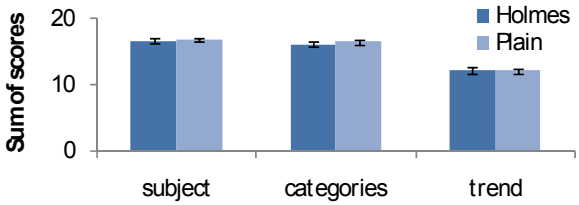

Figure 4. Means ± SE for description scores.

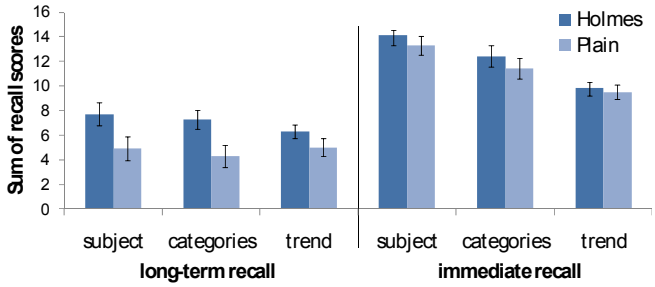

Figure 5. Means ±S E for recall scores for long-term and immediate recall.

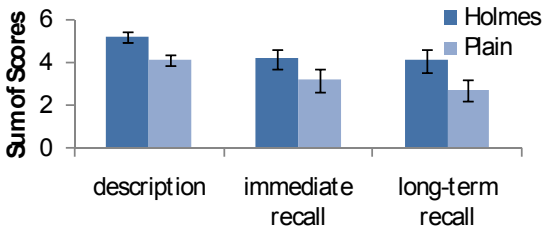

Figure 6. Means ± SE for sum of value message scores.

For participants in the long-term recall group, there were significant differences in recall scores for the subject of the chart (t_9=2.56, p=.015), the categories in the chart (t_9=5.03,

p≈.000), the trend of the chart (t_9=1.95, p=.042), and the value message (t_9=2.41, p=.020) between charts with visual embellishments and ones without (see Figure 5 & Figure 6).

In addition, participants in the long-term recall group required more prompting in order to recall subjects of the charts (t_9=2.67, p=.013), categories in the charts (t_9=2.81, p=.011), and trends of the charts (t_9=2.45, p=.018) for charts that did not contain visual embellishments. However, for participants in the immediate recall group, there were no differences in the required amount of prompting to recall subjects of the charts (t_9=1.41, p=.097), categories in the charts (t_9=0.32, p=.379), or trends of the charts (t_9=-0.32, p=.379) depending on whether or not charts contained visual embellishments (see Figure 7).

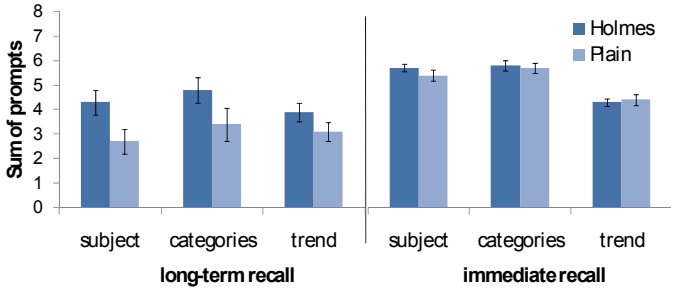

Figure 7. Means ±SE for the number of recall prompts needed for long-term and immediate recall.

User Preferences

After the recall portion of the experiment, participants were asked to choose their preferred chart format for a number of questions. Results are shown in Figure 8. Chi-squared tests showed that participants most enjoyed charts with visual embellishments (χ^2=8.9, p=.003) and found them to be the most attractive (χ^2=11.8, p=.001). Participants also found it easiest to remember the charts with visual embellishments (χ^2=15.2, p≈.000) and to remember the details of these charts (χ^2=8.9, p=.003). Finally, participants felt that it was both fastest to describe (χ^2=4.3, p=.039) and fastest to remember (χ^2=11.8, p=.001) charts containing visual embellishments. The remaining questions did not produce significant differences.

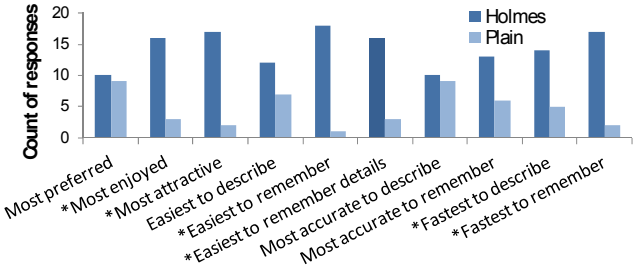

Figure 8. Count of user responses: *indicates significant difference between chart types from chi-squared test at α=0.05

Gaze Detection

During the recall portion of the experiment, we recorded users' gaze data to determine what proportion of time was spent looking at different aspects of the charts, including visual embellishments.

Holmes charts contained data, embellishments, and areas that were both data and embellishments (dual coded). In addition, areas of the chart that were not of interest (e.g., blank space) were categorized as 'other'. Plain charts only contained the data and other categories. Based on how long participants looked at the screen (sometimes they would look off screen or at the experimenter during the study), we calculated the proportion of time spent looking at the different chart elements. As Figure 9 shows, participants spent 67% of their on-screen time looking at data or data and embellishments for Holmes charts as compared to 78% for Plain charts. For Holmes charts, users spent 27% of their on-screen time looking at areas that contained both data and embellishments, and 13% of their time looking at embellishments that did not communicate any data. Although 40% of users' time was spent looking at embellishments for Holmes charts, this did not result in a longer time to describe the charts.

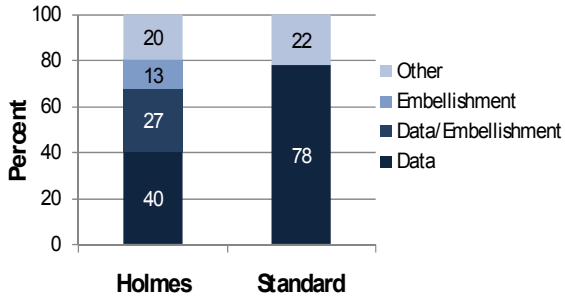

Figure 9. Percentage of on-screen time spent looking at different chart elements for Holmes and Plain charts.

DISCUSSION

The study showed five main findings:

- There was no significant difference between plain and image charts for interactive interpretation accuracy (i.e., when the charts were visible).
- There was also no significant difference in recall accuracy after a five-minute gap.
- After a long-term gap (2-3 weeks), recall of both the chart topic and the details (categories and trend) was significantly better for Holmes charts.
- Participants saw value messages in the Holmes charts significantly more often than in the plain charts.
- Participants found the Holmes charts more attractive, most enjoyed them, and found that they were easiest and fastest to remember.

In the next sections we propose explanations for these results, and then discuss several issues that these results raise for the practice of information presentation.

Why did the extra visual information in the image charts not lead to reduced interpretation and recall accuracy?

Some experts in chart design (and some previous research) suggest that the additional information in the Holmes-style

image charts should lead to greater difficulty in correctly interpreting the data in the chart. This did not happen in our study; we propose some possible reasons for our results.

First, the eye-tracker data suggests that people spent similar amounts of time looking at the data portions of the charts, regardless of their style, although with the Holmes charts the data was sometimes presented through imagery (dual coded). If participants' investigations of the data elements in both conditions were equal, it is reasonable that their interpretation of that data would be equally accurate.

It is important to note that we did not constrain the amount of time participants had to examine the charts. If viewing time was limited, it is possible that people could spend less time on the data elements (assuming that they look at both data and image in an interleaved fashion).

Second, the time that people spent looking at non-data elements in the image charts may not have hindered their interpretation as much as might be supposed. This is because the images in the Holmes charts are actually tightly connected to the subject and even the details of the chart. That is, both the overall message (which includes the topic) and some of the structural elements of the chart (such as the trend) are encoded in the image. For example, the image of the monster in Figure 1 evokes "Monstrous" in the chart's title, and the shape of the mouth and teeth show the increasing trend of the bars. the time spent examining the image itself may have helped people remember elements of the chart, possibly overcoming any detrimental effects caused by the increase in non-data information.

Nigel Holmes has in fact stated that one of the reasons for images in his charts is to allow faster pickup of messages:

> I think [Tufte] missed the point of much that I was trying to do: TIME magazine charts were aimed at lay readers, not unintelligent ones, but busy ones. I knew they'd get the point quicker if they were somehow attracted to the graphic. ([8], p. 76).

Whether this comes at the cost of interpretation richness will be discussed below – in terms of topics, categories, values, and trends, our study shows that the basic elements of the chart are correctly interpreted.

Why was long-term recall better for the image charts?
There are several possible reasons why participants were able to recall the Holmes charts better than the equivalent plain charts. First, as shown in previous research (e.g., [4,6,13,14]) the presence of an image along with other content provides an additional encoding in memory which can improve recall. The images in Holmes's charts are also closely associated with the subject of the chart, reinforcing people's memory.

Second, the Holmes images were all fairly different from one another, which may have helped people remember specific charts. The plain bar charts, in contrast, all had a similar visual appearance (as do all bar charts).

Third, the user's emotional response to the imagery in embellished charts may be a hidden factor. Whether from the subjective preference for the charts or the emotions evoked and associated with the chart imagery, it could be that the emotions, in combination with the visual imagery, help to anchor chart details in a viewer's memory.

Finally, it is intriguing that participants also remembered the details of the Holmes charts better than the plain charts. This may have occurred because of how closely the image is integrated into the chart data – for example, the monster's oddly shaped mouth and teeth show the rising line of the data. Remembering the picture, therefore, may also provide access to the details of the chart (e.g., the overall trend).

Why did participants see a value message in the images?
We believe that people saw value messages in the Holmes charts more often because the images do often convey values, and this was likely the intention of the designer. The image of a monster (and the word 'monstrous' in the title) conveys clear negative connotations that are part of the story. It is difficult to imagine using monster imagery with other more positive stories, such as rising vaccination rates or rising life expectancy ("monstrous life expectancy"?).

It is possible to include visual imagery in charts that reinforces the subject of the chart, but does not provide a value judgment. Although this was not true of the images used in our study, we present example image charts with no value message in Figure 10.

Design Questions and Implications

Should imagery be used in charts to improve recall?
The success of visual imagery in our study raises the question of whether designers should be encouraged to use images in their charts. This is a potentially contentious issue, and there is no way to make a clear recommendation either in favour or against visual imagery. In addition, further study is needed to consider other aspects of this practice before drawing strong conclusions; such as different tasks. For example, it could be that a task that required detailed analysis of charts is hampered more by embellishments, rather than the high-level description task that we asked of our participants. One could imagine that in some safety-critical systems, such as those used by flight control systems, limiting the presentation to the salient information would likely be the preferred course.

Perhaps the most important lesson for designers is that there may be more to the usability and utility of charts than is currently captured by minimalist design approaches.

One practical consideration arguing against the use of strong imagery in charts is the creativity, effort, and artistic ability required to do so. Finding an accompanying graphic that fits the story (and does not bring in any unwanted connotations), and finding a way to integrate the image into the representation of data, are likely to be difficult tasks that cannot be done well without a skilled designer. In addition,

some charts and some topics may not be amenable to Holmes-style visual presentations.

In the end, the purposes and traditions of the venue are likely to dictate the style of charts. As stated by Rock, publications can be oriented towards reading or viewing:

> Some things are designed for reading: scholarly journals, literary reviews, financial pages, and their ilk are fairly impenetrable to the casual page flipper. Other objects like USA Today, annual reports, fashion magazines, and so on are for looking. … Then there are the gray areas. These include newsmagazines and textbooks, which imply reading but are increasingly about looking. ([16] p. 84).

It is interesting to consider where the CHI proceedings fits into this continuum of reading and looking - we note that we shied away from presenting our experimental data in Holmes style (although partly from lack of artistic ability).

In our study we intentionally chose the most extreme type of visual embellishment that we could – namely, the full cartoon imagery used by Holmes. However, as stated earlier, there are several other types of visual embellishment that are less extreme, and it is possible that these other types will also help people to remember charts, without requiring such extensive imagery.

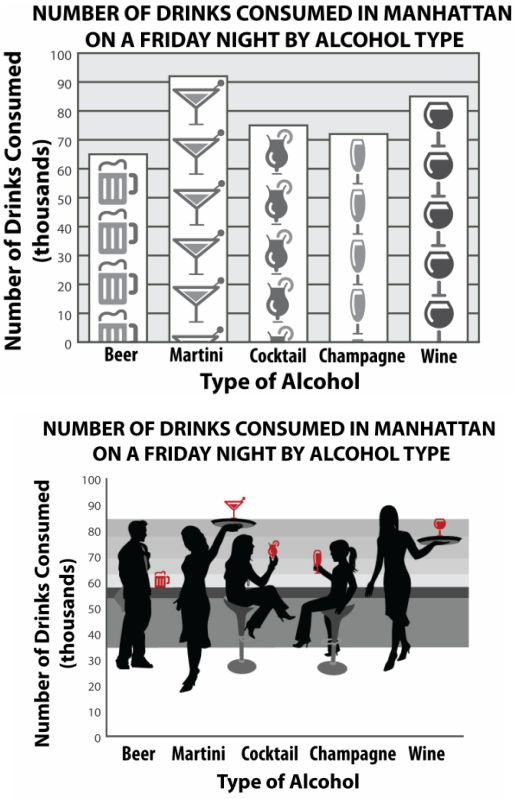

Figure 10. Less extreme visual imagery in charts.

Based on our results, we believe that there will be a difference between embellishments that provide images and those that do not. Psychology research suggests that pictures are useful in terms of memory and recall [4,6,14,17], and it is likely that a larger effect would be seen with charts that use pictures rather than non-image embellishments such as textures, colours, or abstract shapes. Figure 10 shows examples of possible designs that use imagery in less extreme ways.

The wider problem of bias in charts

Charts that include strong imagery have been criticized for introducing bias into the presentation, but it is worth noting that this is only a more explicit attempt to persuade than some of the other rhetorical devices that are used in written and visual communication. There is a tacit assumption in the minimalist approach that by removing embellishment and non-data ink, the presentation is made more objective, and the data 'is allowed to speak for itself.' However, despite their different visual style, the two charts in Figure 1 both tell the same story – that campaign expenditures are rising at a higher-than-linear rate – and both characterize this trend as 'monstrous.' There is therefore no guarantee that minimalist charts are free from bias.

Communication is, in most cases, oriented towards making an argument, and few if any pieces of writing are truly objective. There are many ways that even minimalist charts can be biased or used in support of a particular story – the type of chart, the selection of data, the ordering of bars, the scale of axes, the choice of titles and labels – all can be used to highlight or hide different aspects of the data.

Despite these potential adjustments behind the scenes, minimal charts still maintain the appearance of objectivity – but claiming objectivity is also a common rhetorical technique. As Rock [16] states,

> …images and charts seem to not imply an inherent point-of-view. They radiate a kind of false objectivity because the concept of the image-as-opinion is difficult for most people to grasp. (p. 86)

(Note that here 'image' implies the visual of even a minimalist chart, rather than the image of a Holmes chart).

The illusion of objectivity (as used in minimalist charts) and the use of evocative imagery (as used in Holmes charts) are perhaps just different approaches that work at different ends of the rhetorical spectrum. Designers and readers should remember that a Holmes chart is not necessarily more biased than its plain counterpart – but it may be more effective at conveying the value message that is part of the overall argument.

Stepping back another way, it can be noted that the use of charts altogether has been seen as a practice that takes away from the more expansive, judicious, and nuanced arguments that can be made through writing:

> Charts and diagrams are certainly useful for offering general, relational explications of an issue but they necessarily shave away the ambiguous, nuanced, or

> obscure aspects of any idea. The information has been preprocessed, prechewed; it can only lead to one conclusion. ([16], p. 85)

Although few practitioners would argue for dispensing with all charts, it is important for designers and readers to understand the wide variety of ways in which charts are used in support of stories and arguments, and to see chart imagery as just one point in this space.

CONCLUSION

Many experts suggest that visual embellishments and 'chart junk' cause interpretation problems and should be removed from information charts, but many examples exist of embellished charts that seem to be successful. To investigate some of the specific costs and benefits of these two approaches to chart design (minimalist and embellished), we compared interpretation accuracy and long-term recall for plain and 'Holmes-style' charts. We found that people's accuracy in describing the embellished charts was no worse than for plain charts, and that their recall after a two-to-three week gap was significantly better. In addition, participants preferred the embellished charts. Although we are cautious about proposing specific design recommendations, it seems clear that there is more to be learned about the effects of different types of visual embellishment in charts. Our results question some of the premises of the minimalist approach to chart design, and raise issues for designers about how charts are designed and used in different publications and different contexts.

ACKNOWLEDGEMENTS

We thank Nigel Holmes for granting us permission to use his images, and the CHI reviewers for their comments. This research was supported by NSERC.

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INTERACTION

SPECTRUM

Static content

Infographics

Books

Dynamic content

1. Animated content

   “Auto-play”, user is not in control

2. **Interactive content**

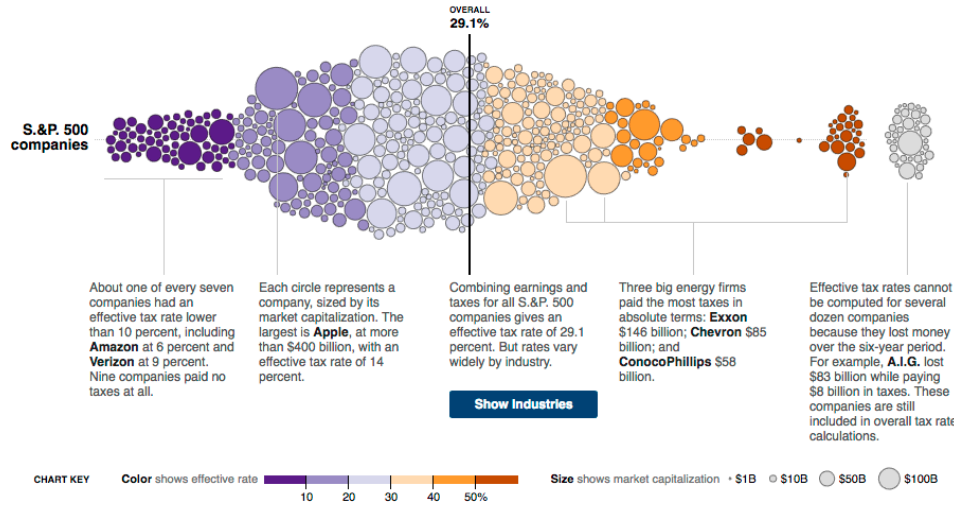
   Changes are a result of user actions

WHY INTERACT WITH VISUALIZATION?

Need to explore data that is big/complex

Too much data

Too many ways to show it

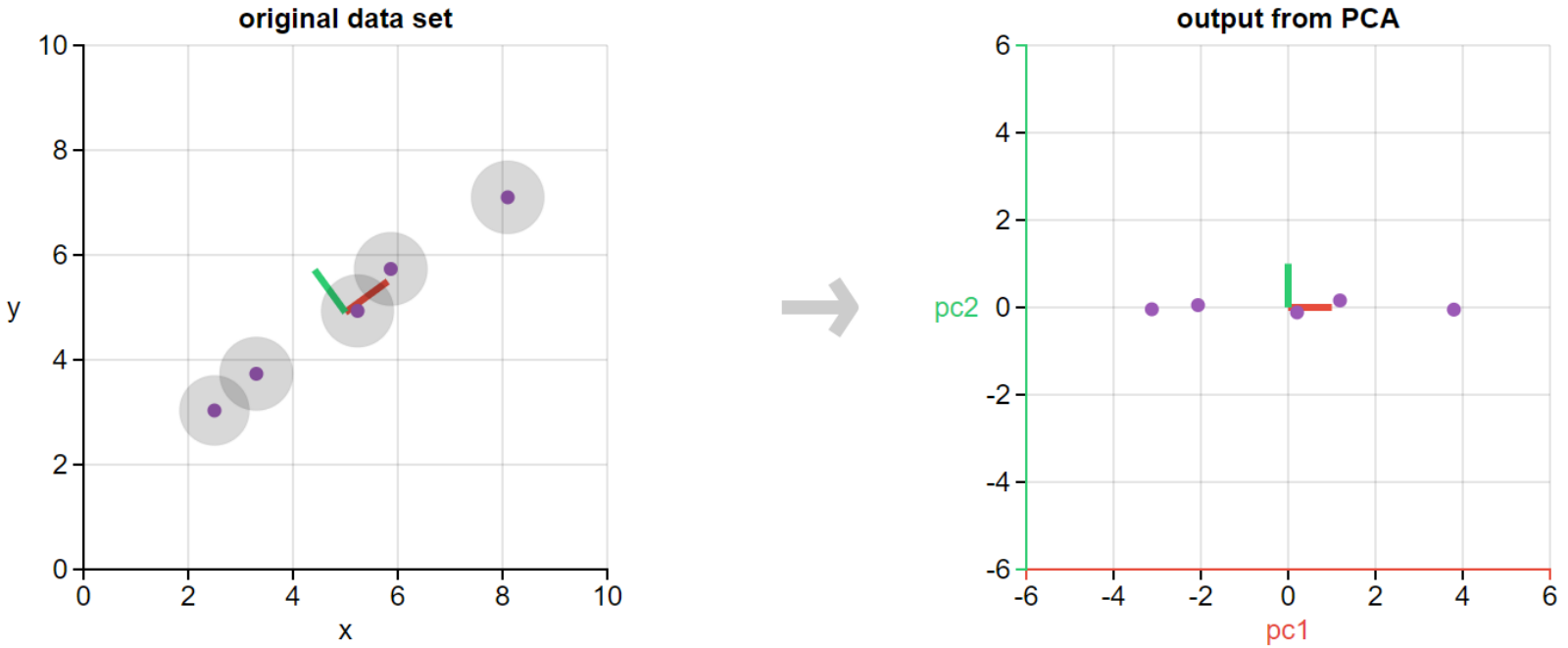


WHY INTERACTION WITH VIS?

Interaction amplifies cognition

Understand things better if we can touch them

When we can observe cause and effect



https://setosa.io/ev/principal-component-analysis/

INTERACTION METHODS

What do you design for?

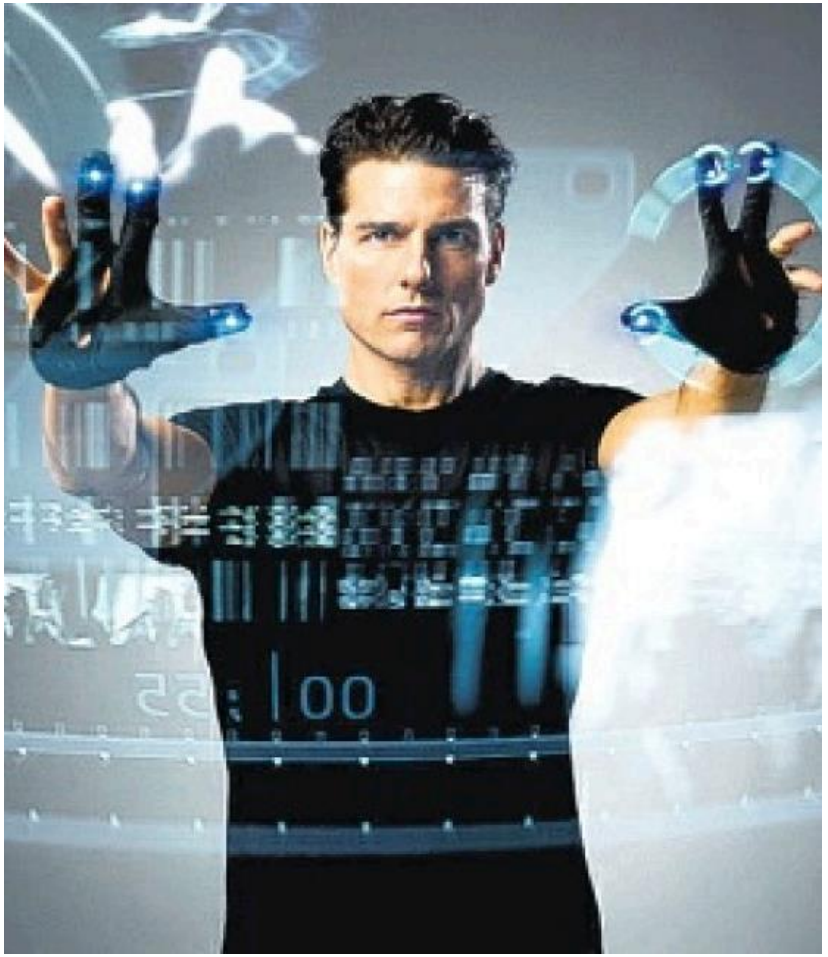
Mouse, keyboard?

Touch interaction / mobile?

Gestures?

Eye movement?

Speech?

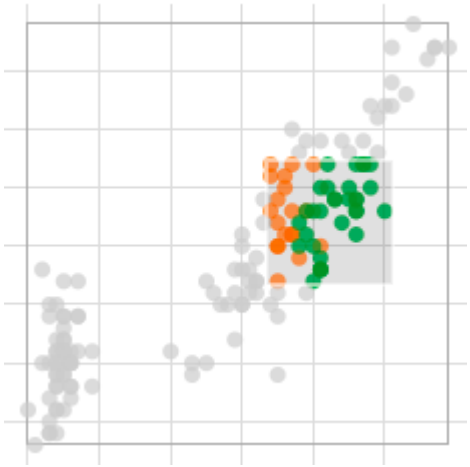


https://www.youtube.com/watch?v=QXLfT9sFcbc

DIRECT MANIPULATION

Interact directly with objects

Continuous feedback/updates



Indirect interact

Using a query, a slides, etc.

https://observablehq.com/@d3/brushable-scatterplot-matrix

TYPES OF INTERACTION

Single view

Change over time

Navigation

Semantic zooming

Filtering and Querying

Focus + Context

Multiple views

Selection (Details on demand)

Linking & brushing

Adapting representation

PURPOSES OF INTERACTION

DOI:10.1145/2133806.2133821

Article development led by acmqueue
queue.acm.org

A taxonomy of tools that support the fluent and flexible use of visualizations.

BY JEFFREY HEER AND BEN SHNEIDERMAN

Interactive Dynamics for Visual Analysis

TABLE 1: Taxonomy of interactive dynamics for visual analysis

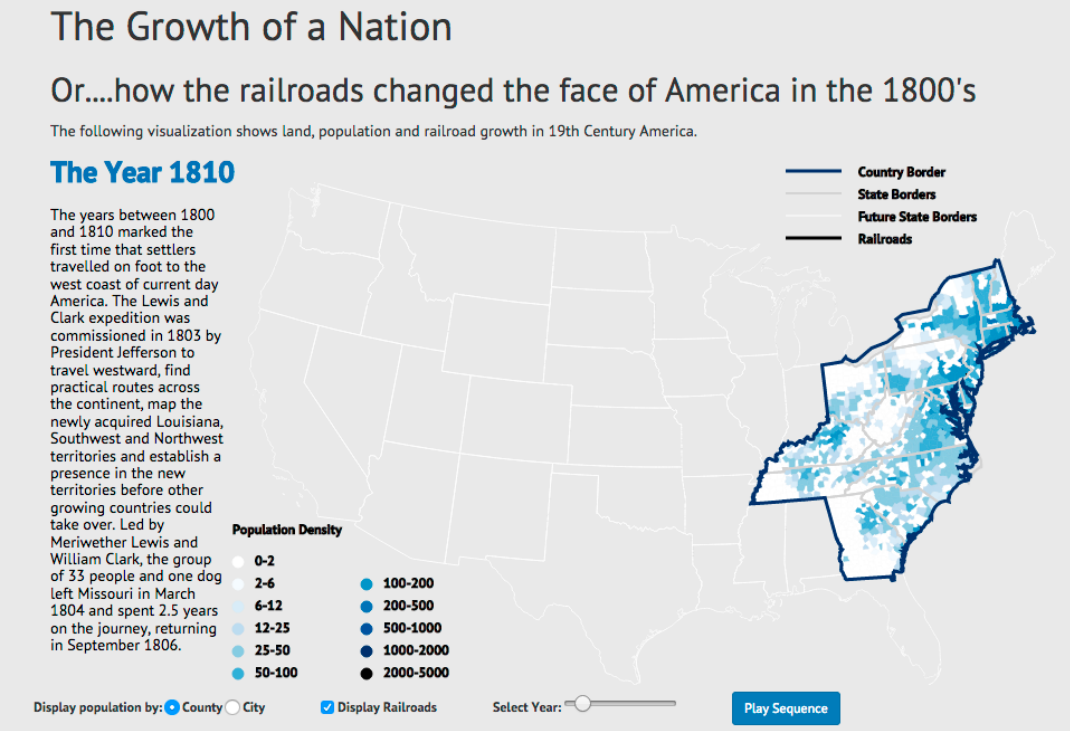
Data & View Specification	**Visualize** data by choosing visual encodings. **Filter** out data to focus on relevant items. **Sort** items to expose patterns. **Derive** values or models from source data.
View Manipulation	**Select** items to highlight, filter, or manipulate them. **Navigate** to examine high-level patterns and low-level detail. **Coordinate** views for linked, multi-dimensional exploration. **Organize** multiple windows and workspaces.
Process & Provenance	**Record** analysis histories for revisitation, review and sharing. **Annotate** patterns to document findings. **Share** views and annotations to enable collaboration. **Guide** users through analysis tasks or stories.

CHANGE OVER TIME/TRANSITIONS

CHANGE OVER TIME

Use slides to see view with data at different times

Sometimes better to show difference explicitly



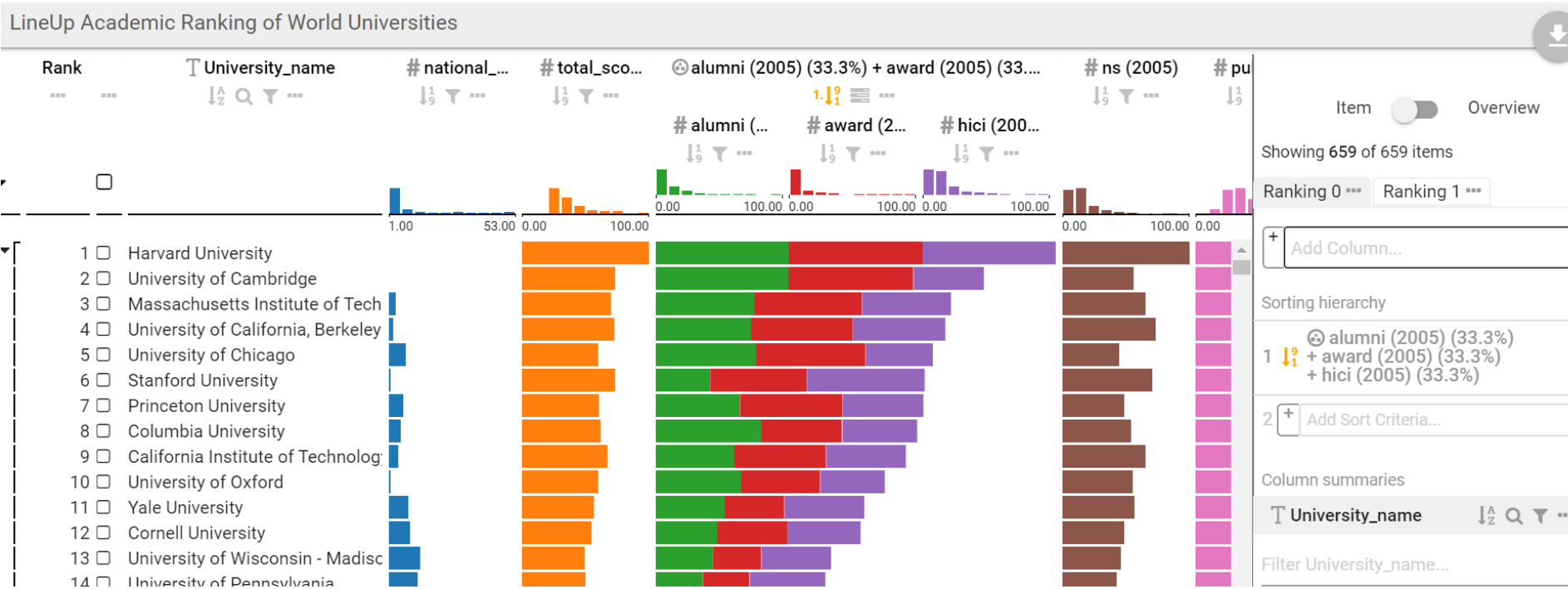
http://laurenwood.github.io/

CHANGE OVER TIME

Doesn't have to be literal time:

change as you go

as part of an analysis process



https://lineup.js.org/app/#shanghai

https://caleydo.org/tools/lineup/

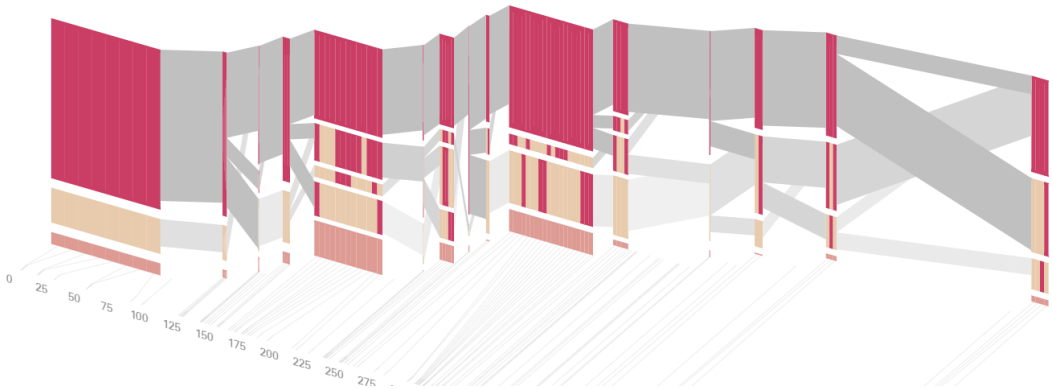
WHY TRANSITION?

Different representations support different tasks

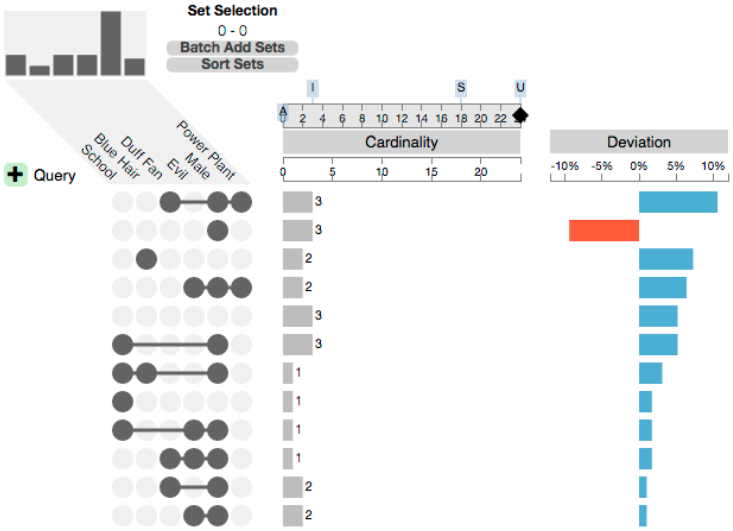
Bar chart vs stacked bar chart

Change ordering

Transition make it possible for users to track what is going on



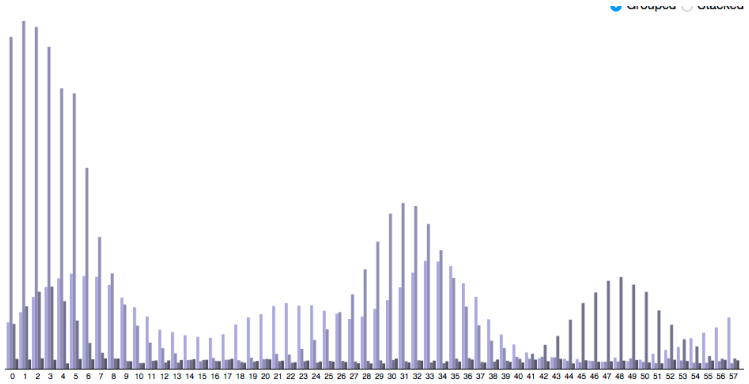
https://benfry.com/isometricblocks/



http://vcg.github.io/upset/

ANIMATED TRANSITIONS

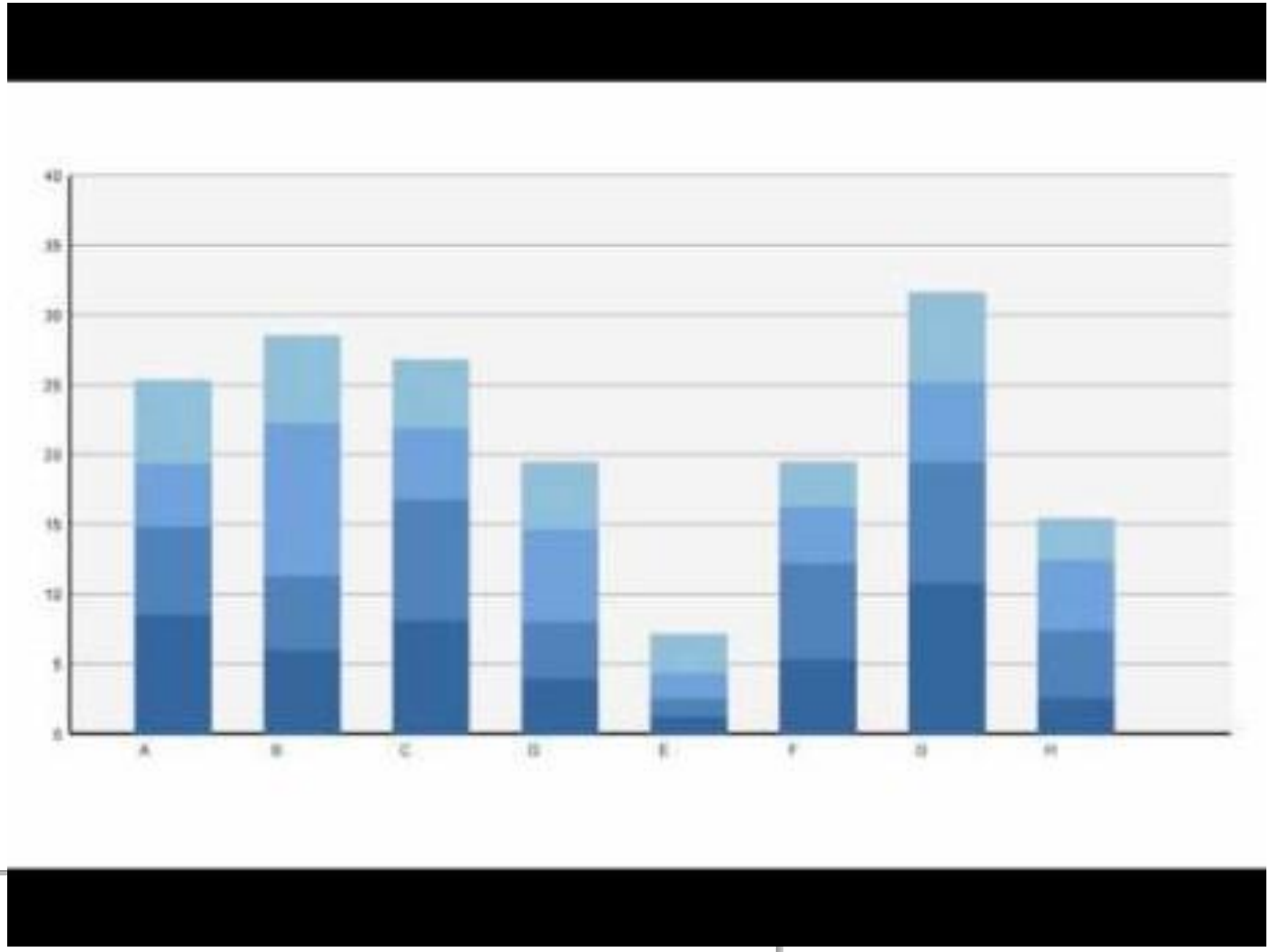
Smooth interpolation between states or visualization techniques



http://bl.ocks.org/mbostock/3943967

https://observablehq.com/@d3/zoomable-sunburst

WHY ANIMATED TRANSITION?



Animated Transitions in Statistical Data Graphics

Jeffrey Heer, George G. Robertson

Abstract—In this paper we investigate the effectiveness of animated transitions between common statistical data graphics such as bar charts, pie charts, and scatter plots. We extend theoretical models of data graphics to include such transitions, introducing a taxonomy of transition types. We then propose design principles for creating effective transitions and illustrate the application of

https://youtu.be/vLk7mlAtEXI

ANIMATION CAVEATS (WARNING)

Changes can be hard to track

Eyes over memory

Show all states in multiple views

NAVIGATION

NAVIGATION

Pan

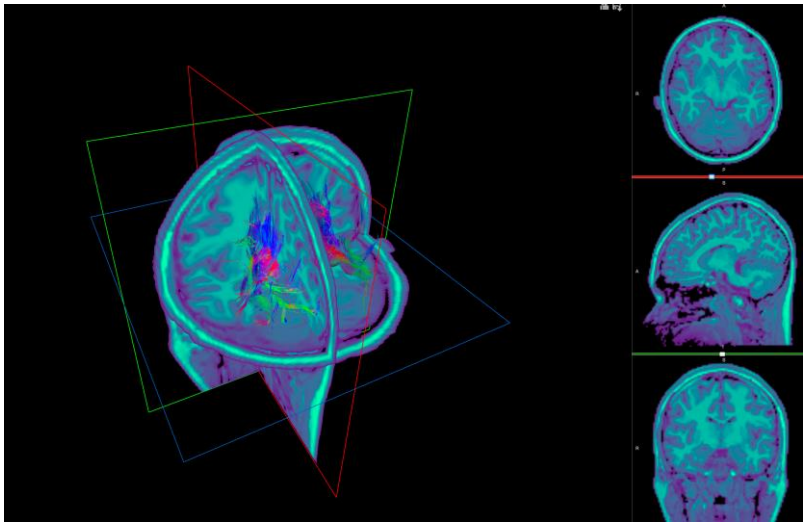
Move around

Zoom

Enlarge/make smaller

Rotate

https://www.google.com/maps/@40.5388417,19.7863469,14278660m/data=!3m1!1e3



http://x.babymri.org/example1/?scene=http://x.babymri.org/example1/scene.json

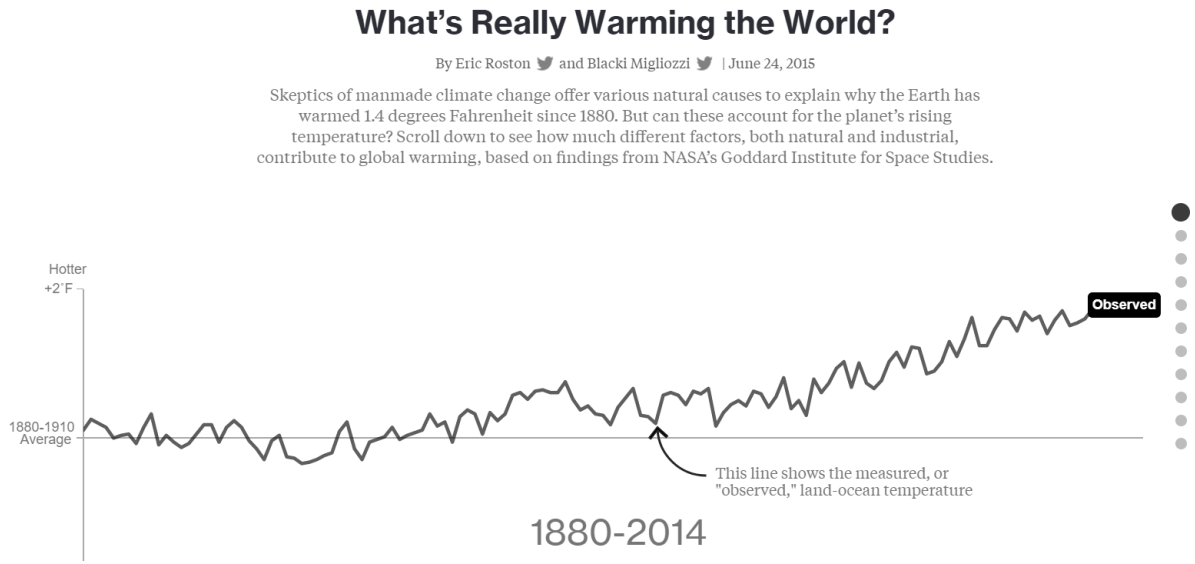
SCROLLTELLING

Telling an interactive story

Interaction by scrolling

Nice but

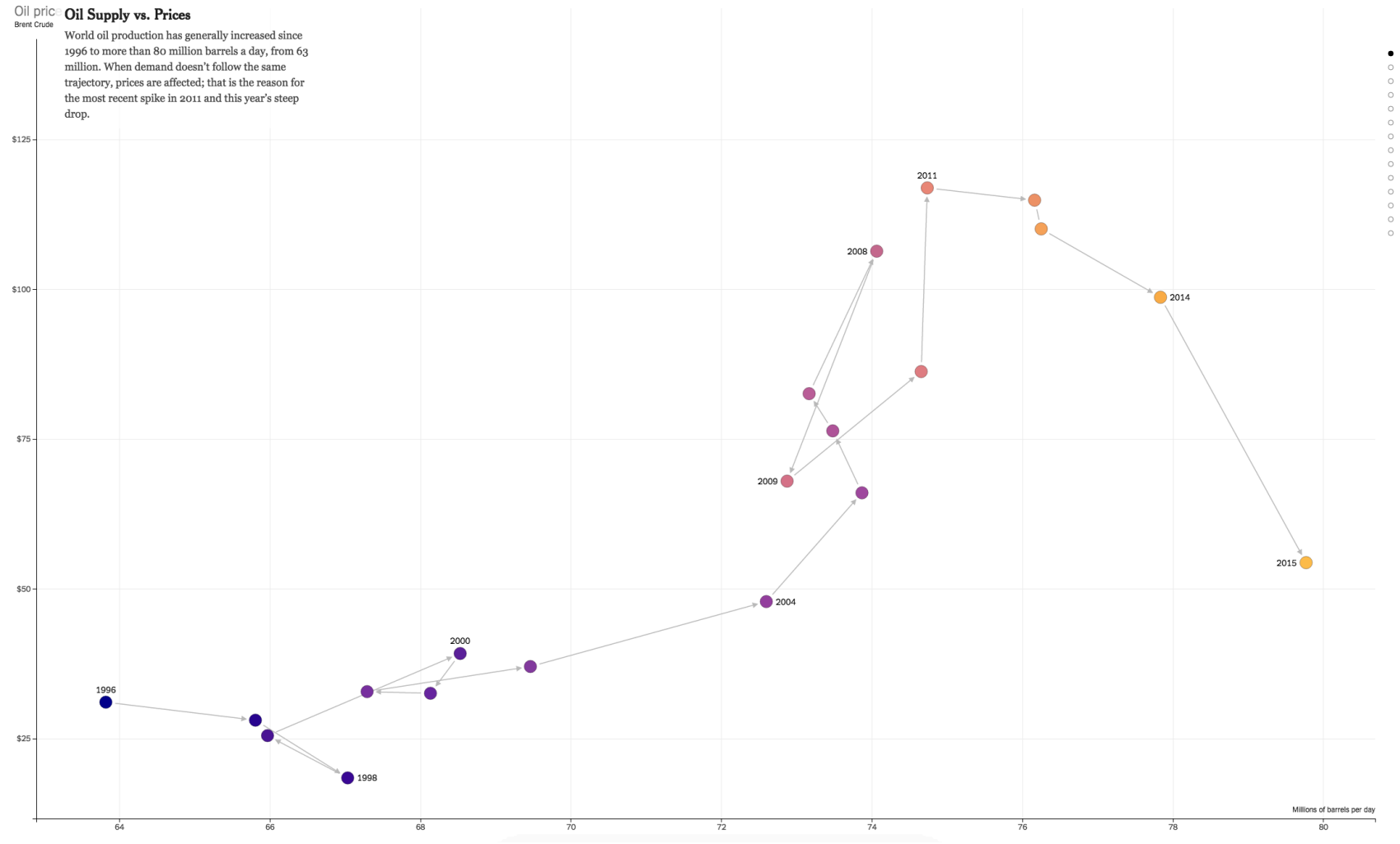
- Continuous scrolling vs discrete states
- Direct access
- Unexpected behavior



https://www.bloomberg.com/graphics/2015-whats-warming-the-world/

https://eagereyes.org/blog/2016/the-scrollytelling-scourge

SCROLLTELLING: OIL PRICES

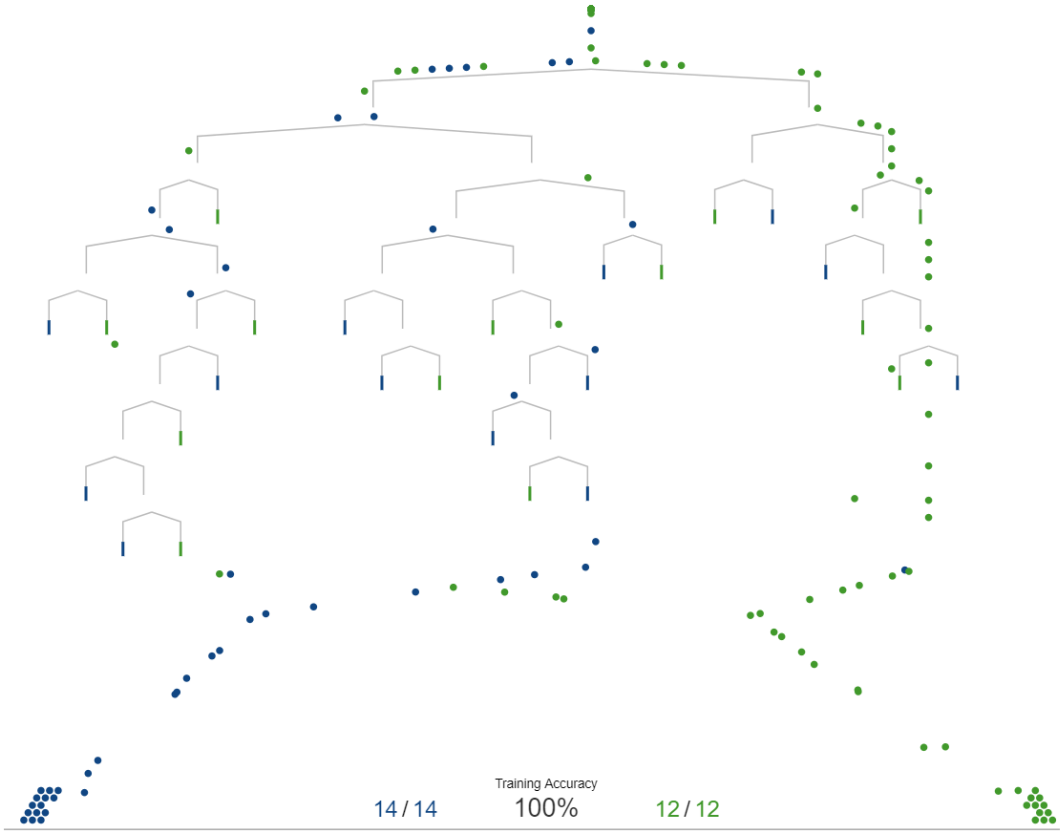


http://www.nytimes.com/interactive/2015/09/30/business/how-the-us-and-opec-drive-oil-prices.html?_r=0

SCROLLTELLING: INTRO TO ML

Making predictions

The newly-trained decision tree model determines whether a home is in San Francisco or New York by running each data point through the branches.



http://www.r2d3.us/visual-intro-to-machine-learning-part-1/

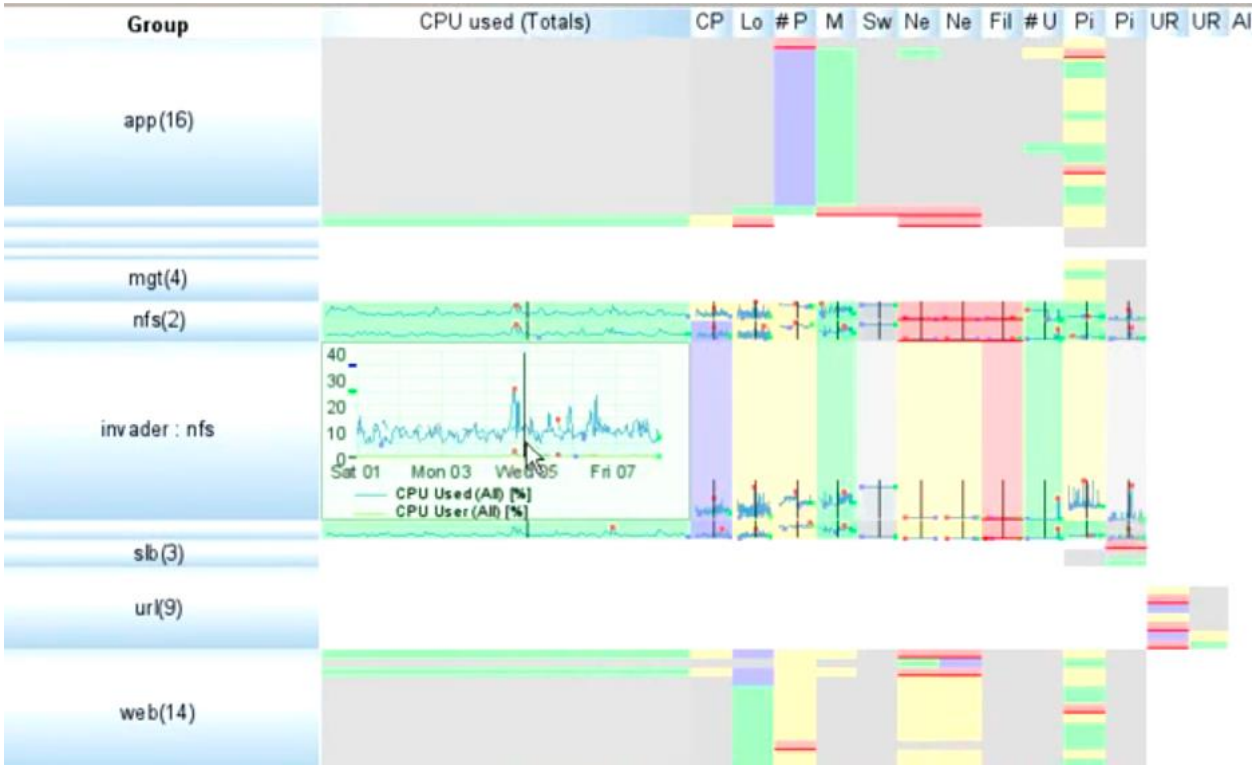
SEMANTIC ZOOMING

SEMANTIC ZOOMING

Update content on zooming

More detail as more space becomes available

Ideally readable at multiple resolutions



https://youtu.be/ld0c3H0VSkw

FOCUS + CONTEXT

FOCUS + CONTEXT

Carefully pick what to show

Hint at what you are not showing

FOCUS + CONTEXT

Synthesis of **visual encoding and interaction**

User selects regions of interest (focus) through navigation or selection

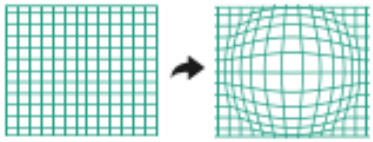
Provide context through

- aggregation
- reduction
- layering

- **Embed**
  - Elide Data

  - Superimpose Layer

  - Distort Geometry

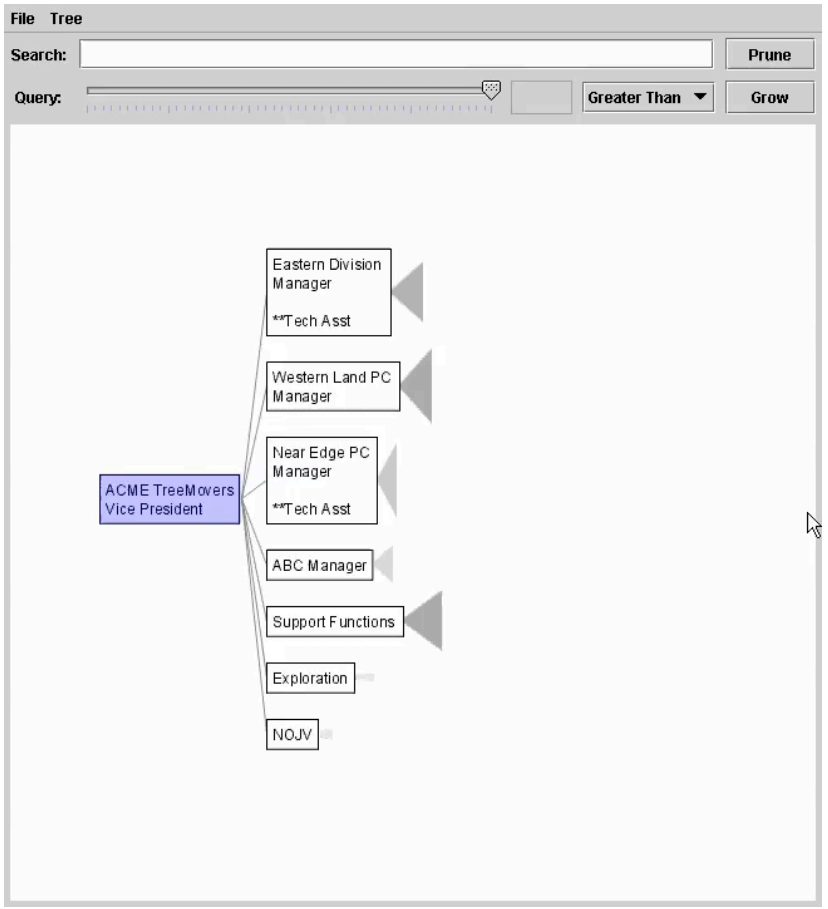


ELISION

Focus item shown in detail

Other items summarized for context

Example: SpaceTree



https://www.cs.umd.edu/hcil/spacetree/

https://youtu.be/F7-vJBsFBWw

DEGREE OF INTEREST (DOI)

Represent objects in the neighborhood in detail, and only major landmarks far away

Balance between local detail and global context

$$DOI(x) = I(x) – D(x,y)$$

I(x): interest in object x

D – a distance function to the current focus y of x

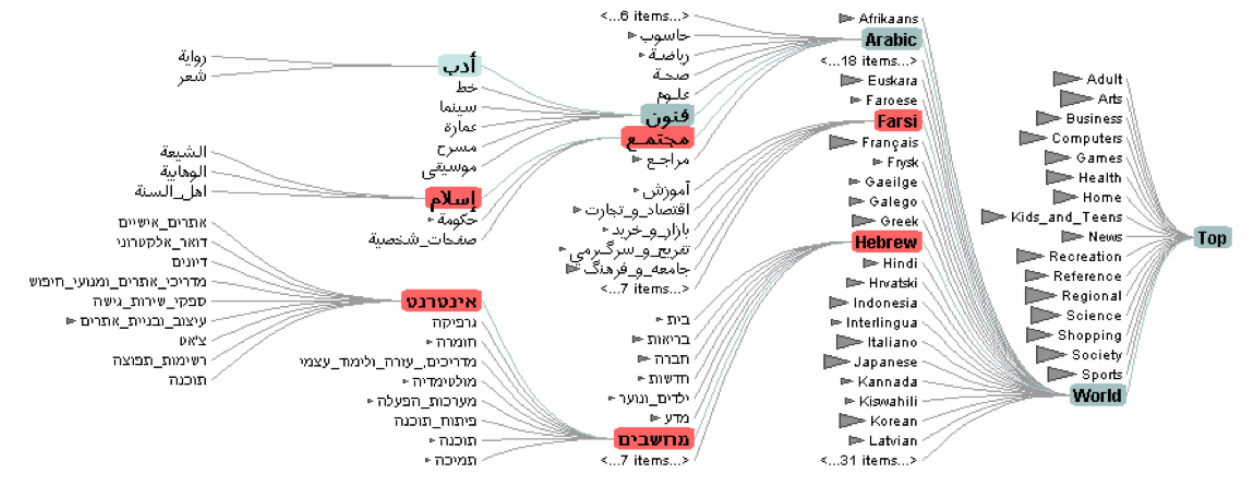
There may have many foci

DOI TREE

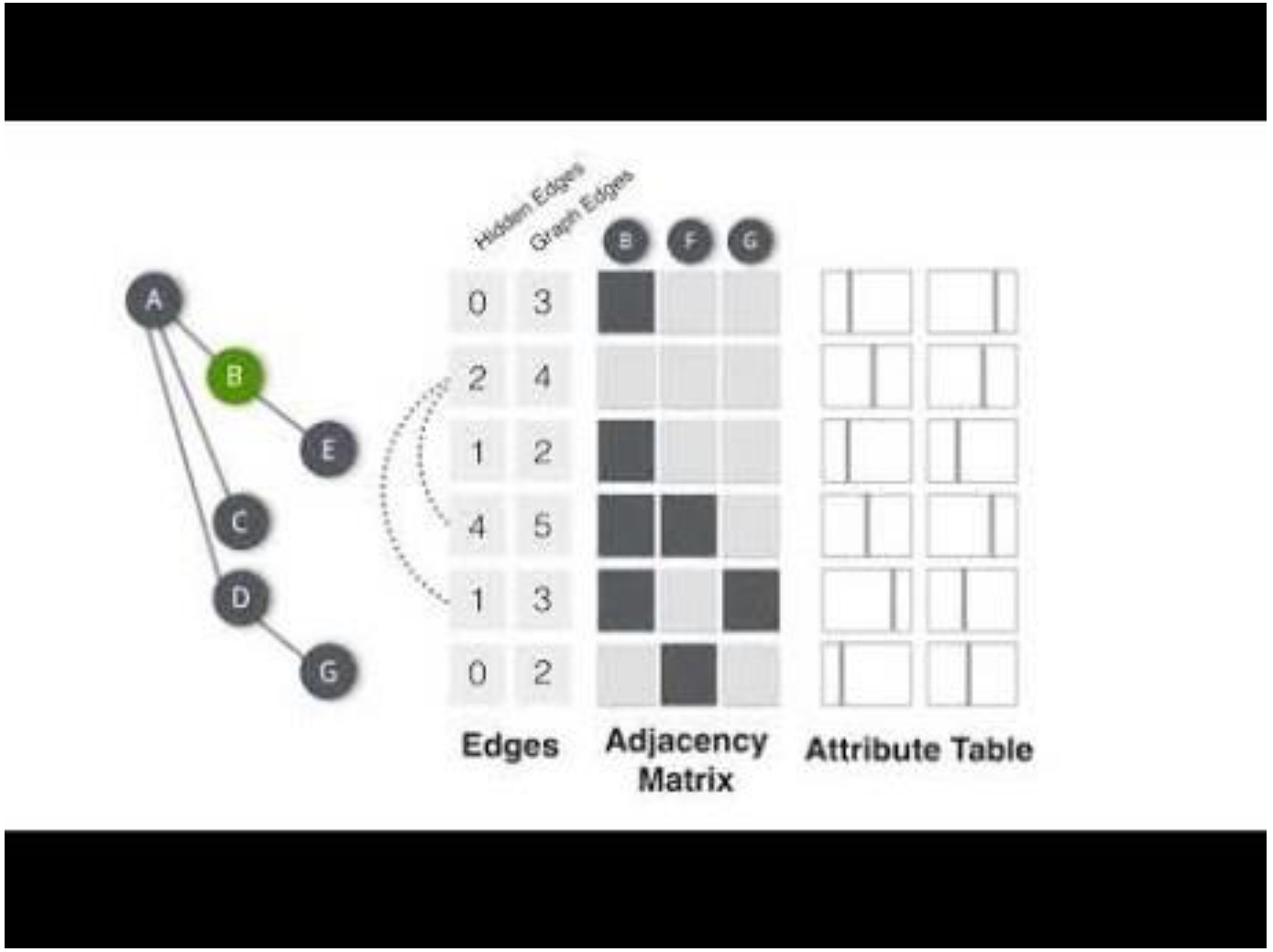
Interactive tree with animated transitions that fit within a bounded region of space layout depends on the user's estimated DOI

USE:

- logical filtering based on DOI
- Geometric distortion of node size based on DOI
- Semantic zooming on content based on node size
- Aggregate representations of elided subtrees



DOI: JUNIPER EXAMPLE



https://youtu.be/EAjNxFgsJ58

SUPERIMPOSE

Focus layer limited to a local region of view

Instead of stretching across the entire view

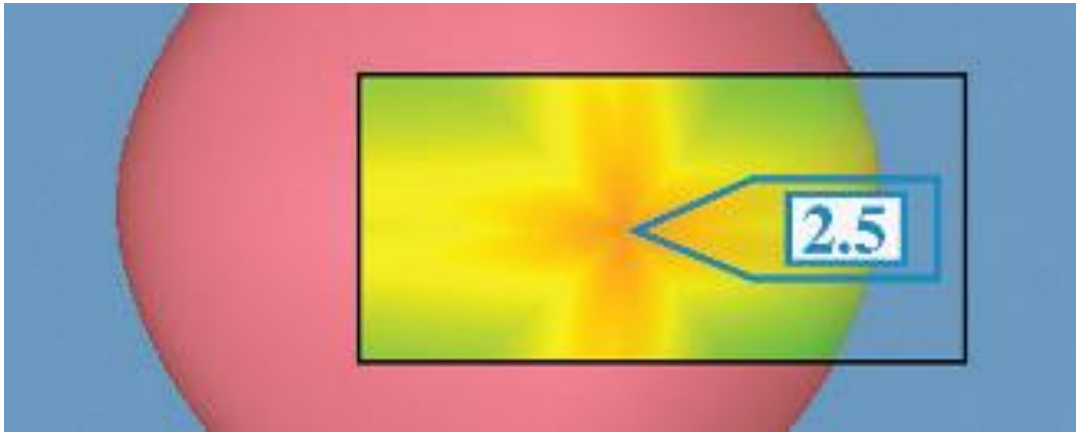
TOOLGLASS & MAGIC LENSES

Magic lense:

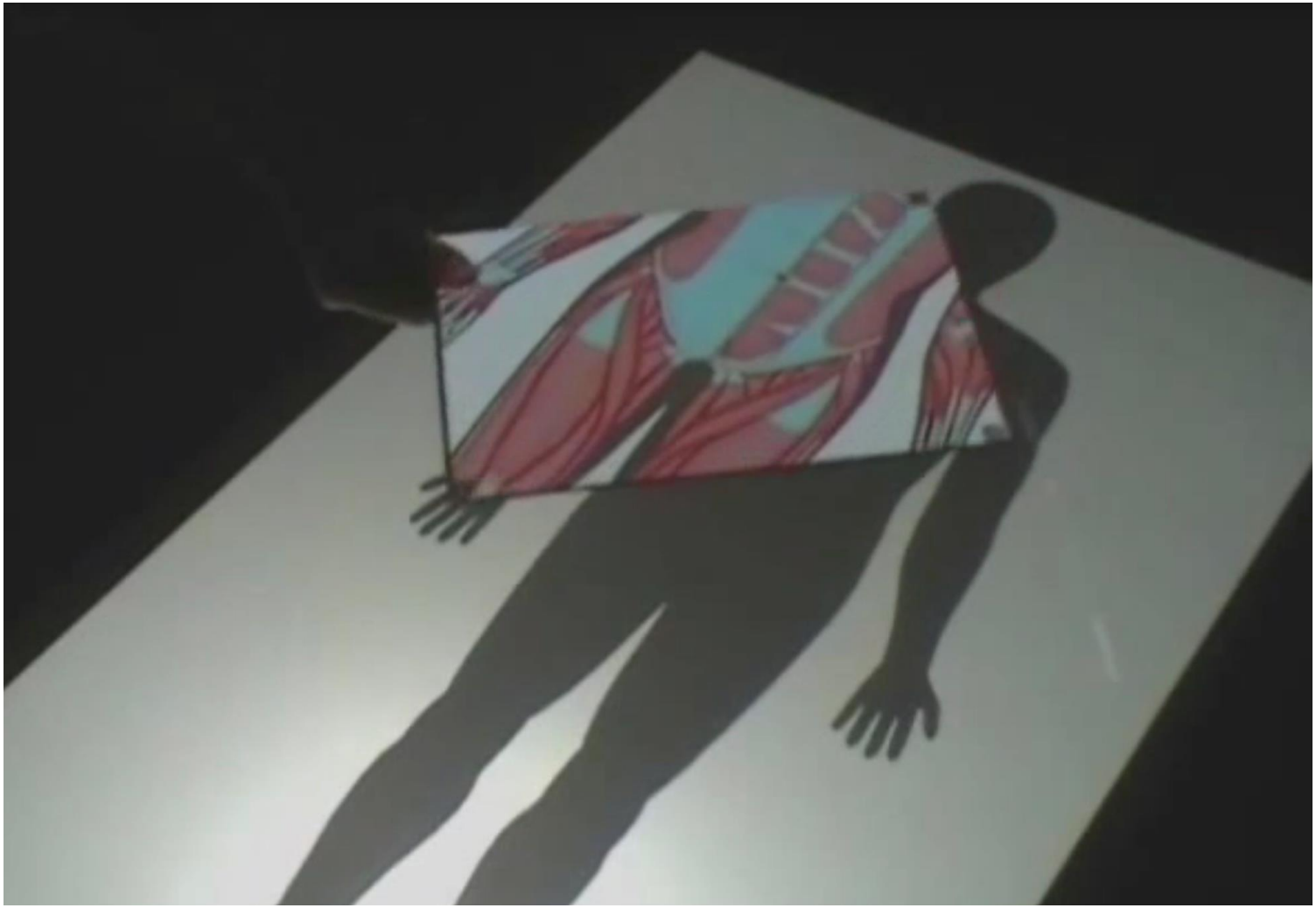
details/different data is shown

when moving a lens

over a scene

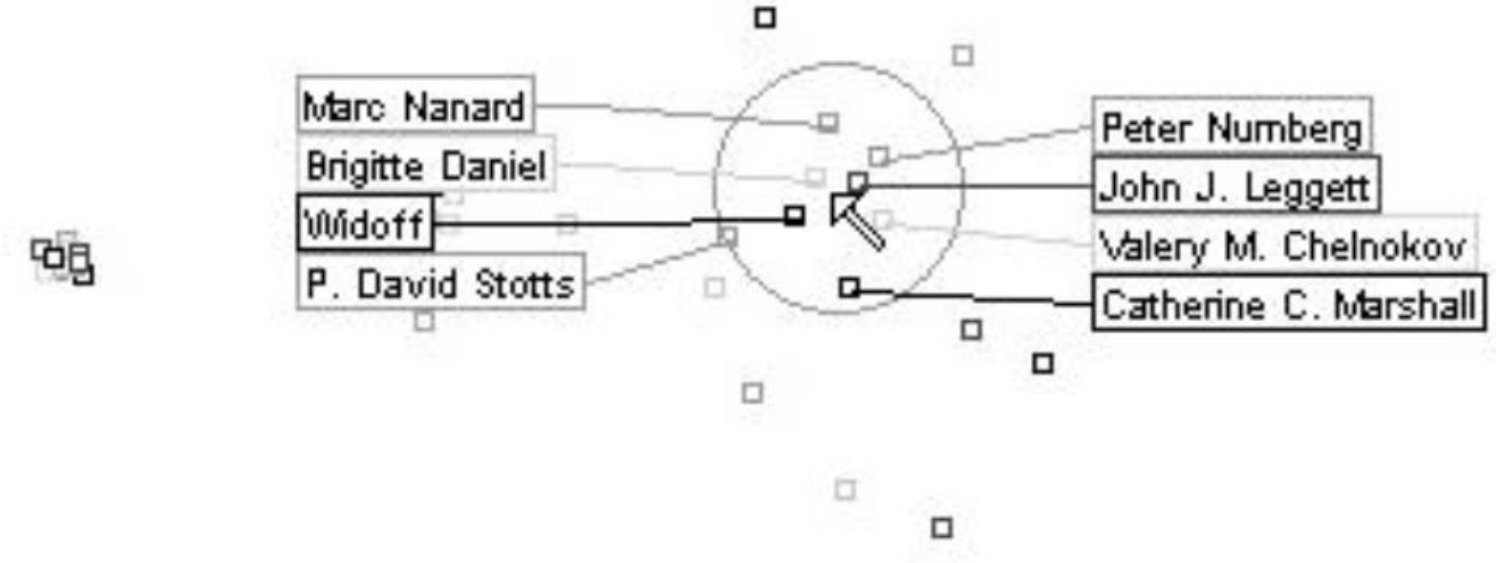


MAGIC LENSE



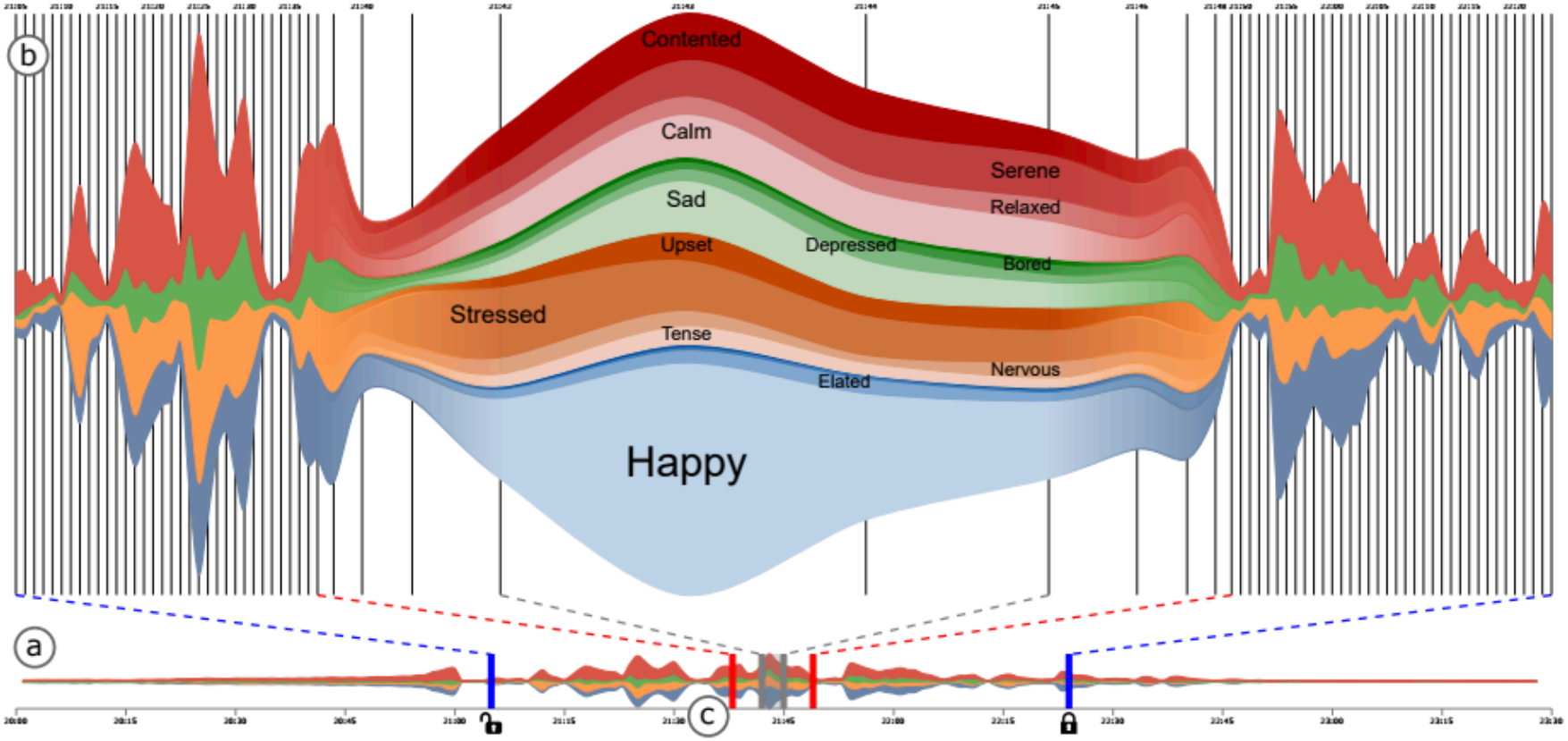
https://www.youtube.com/watch?v=h-mF4_OAhU0

MAGIC LENSE: LABELING



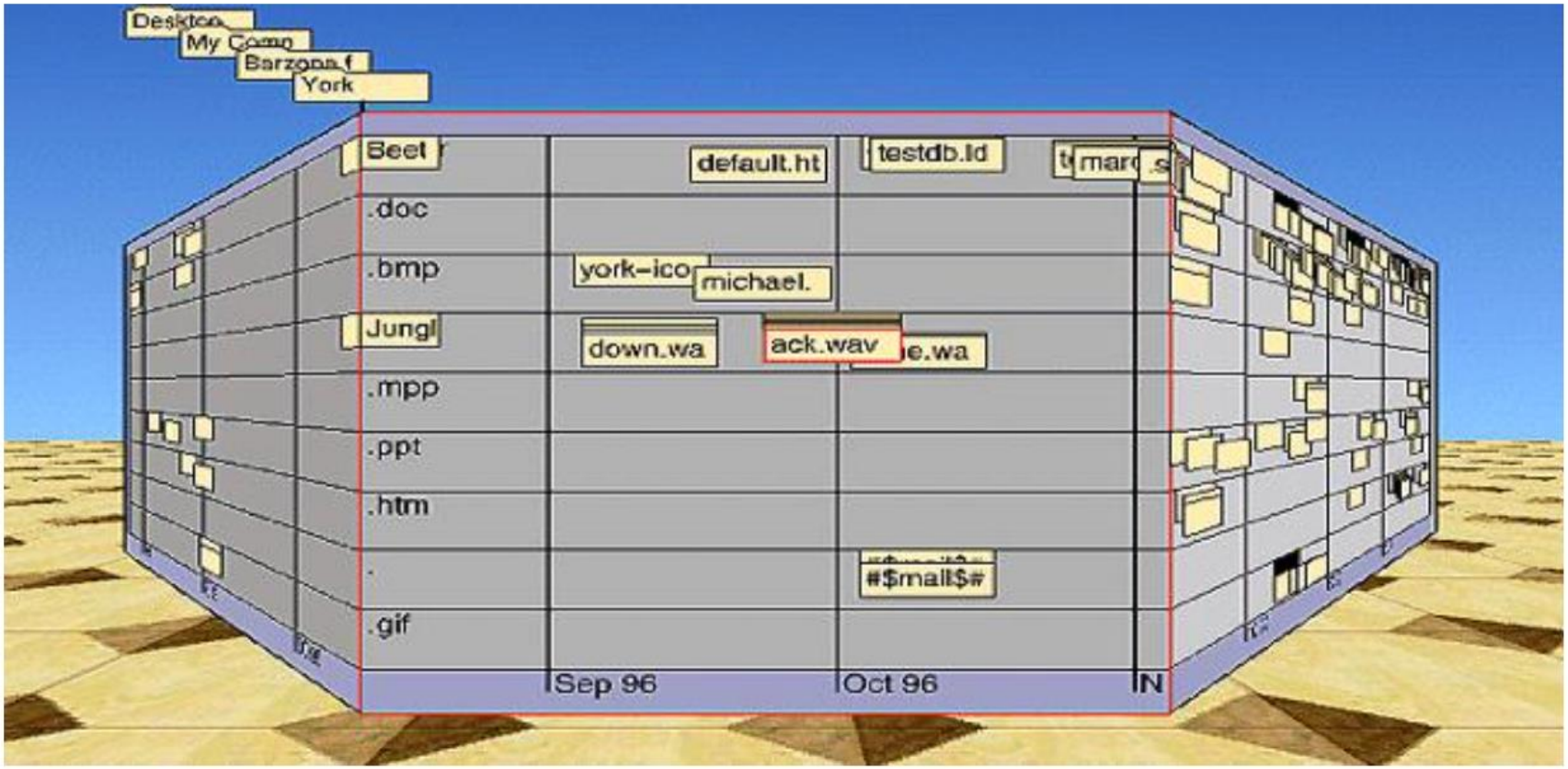
DISTORTION

Use geometric distortion of the contextual regions to make room for the details in the focus regions(s)



http://advanse.lirmm.fr/multistream/

PERSPECTIVE WALL



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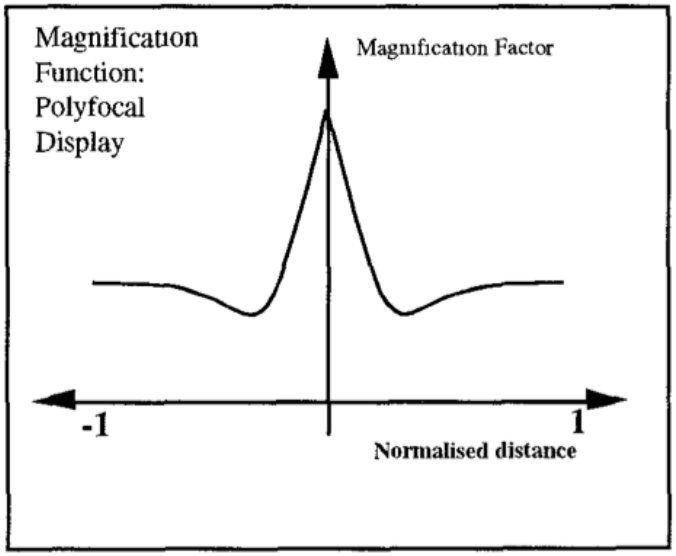
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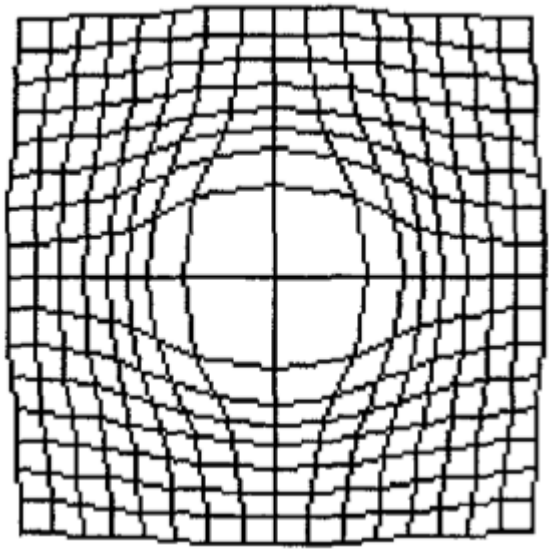

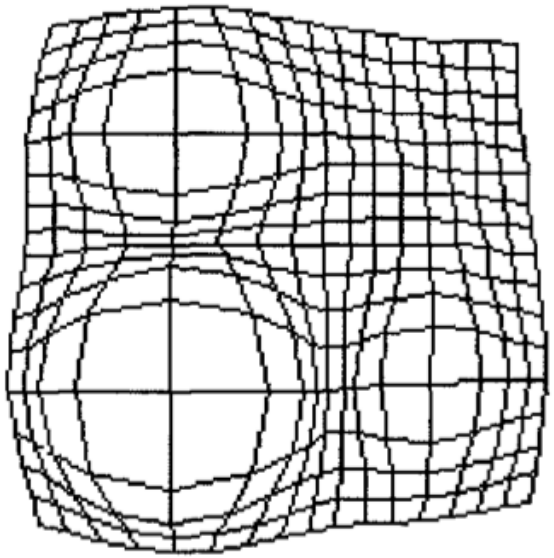

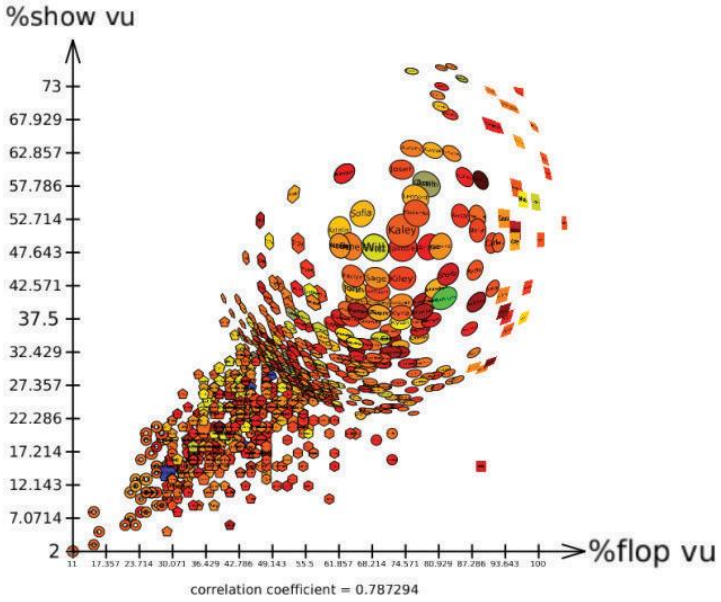
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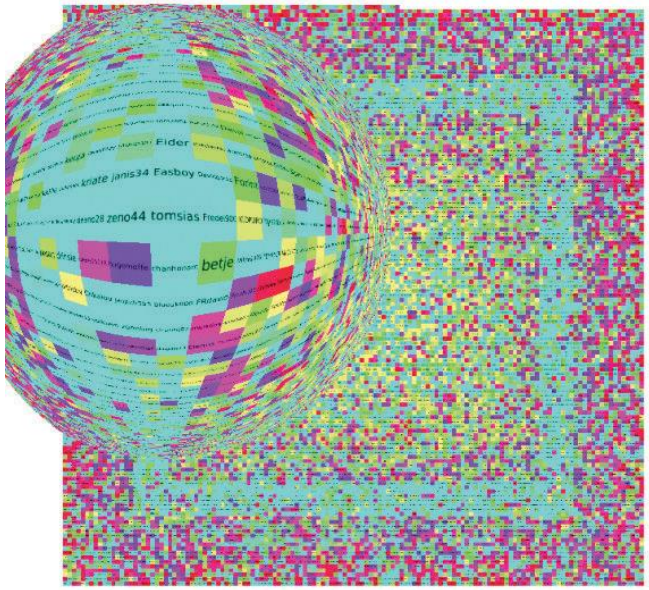
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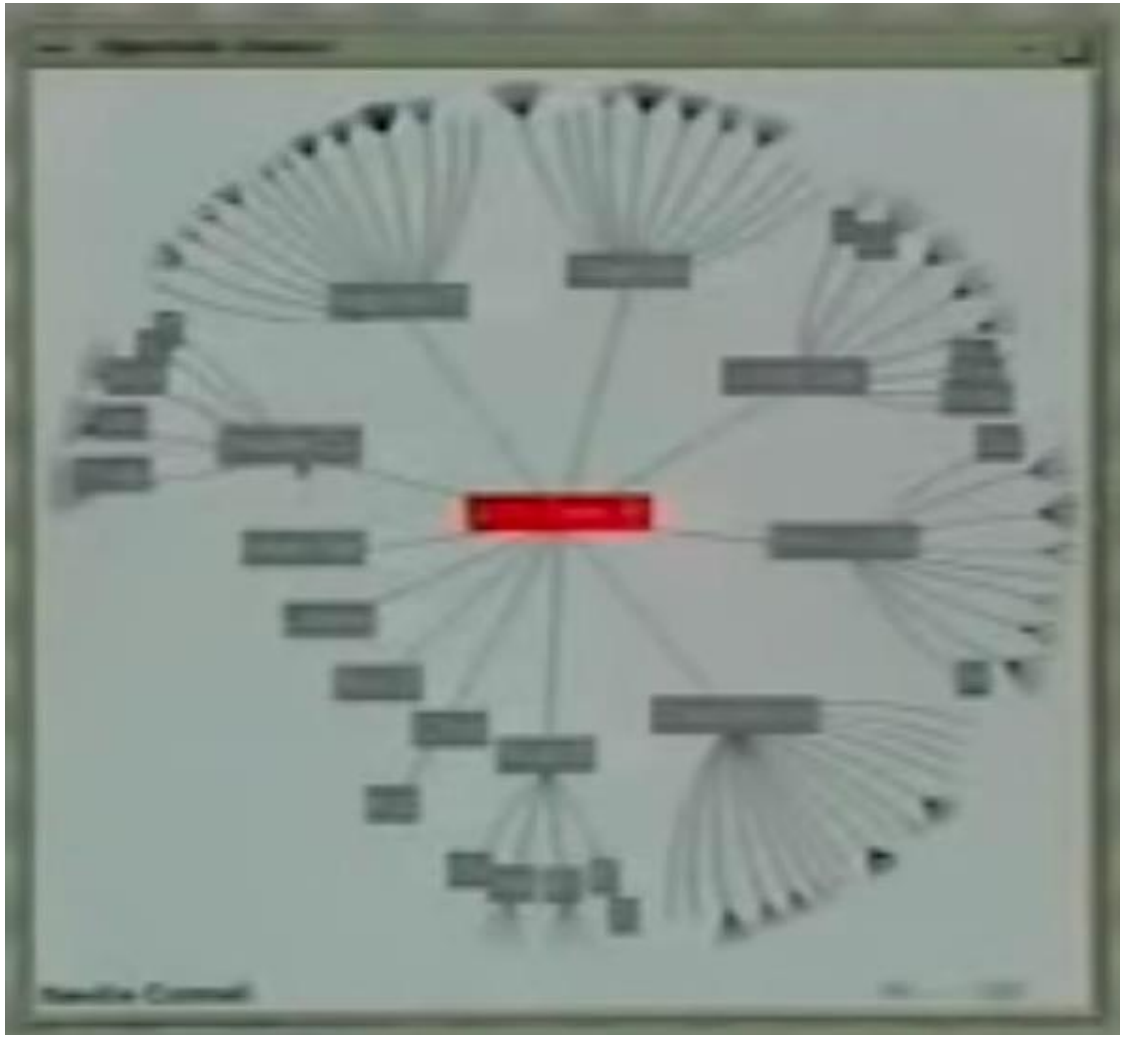
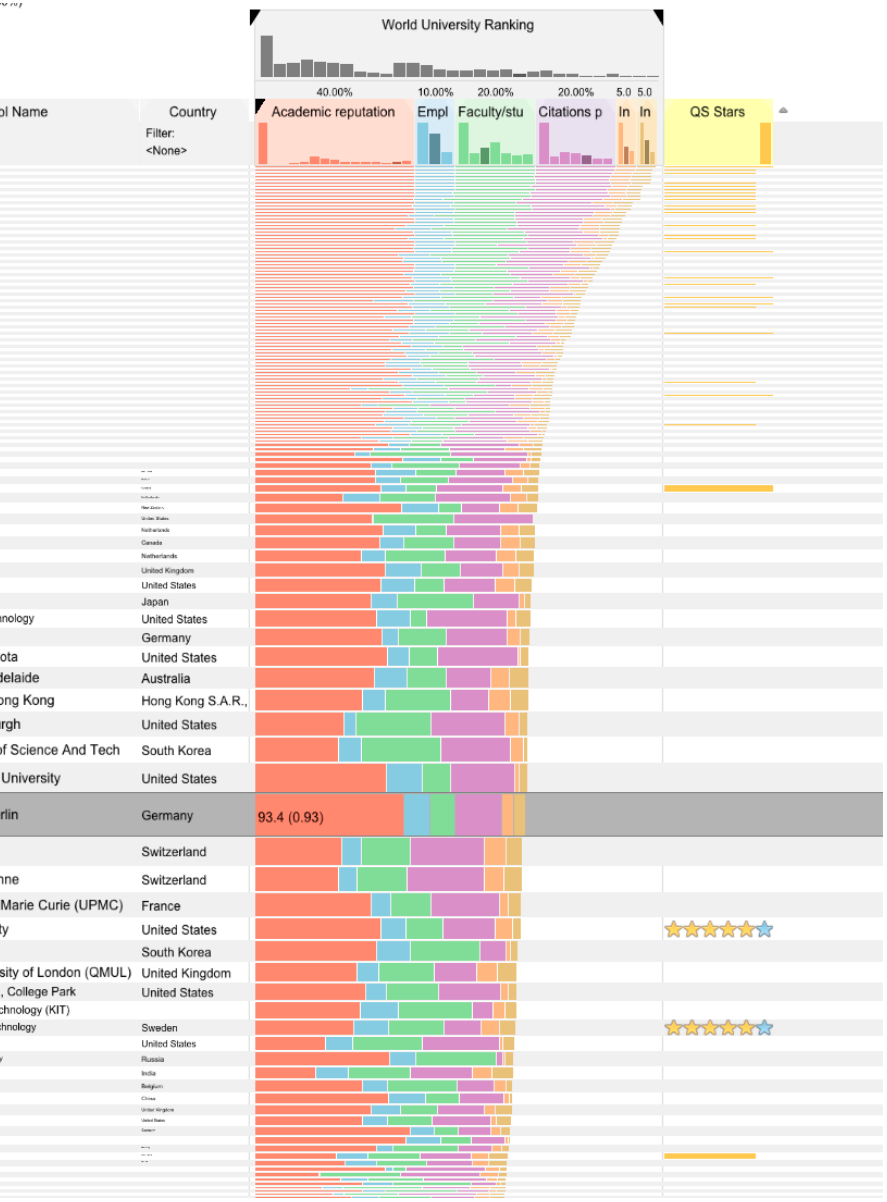
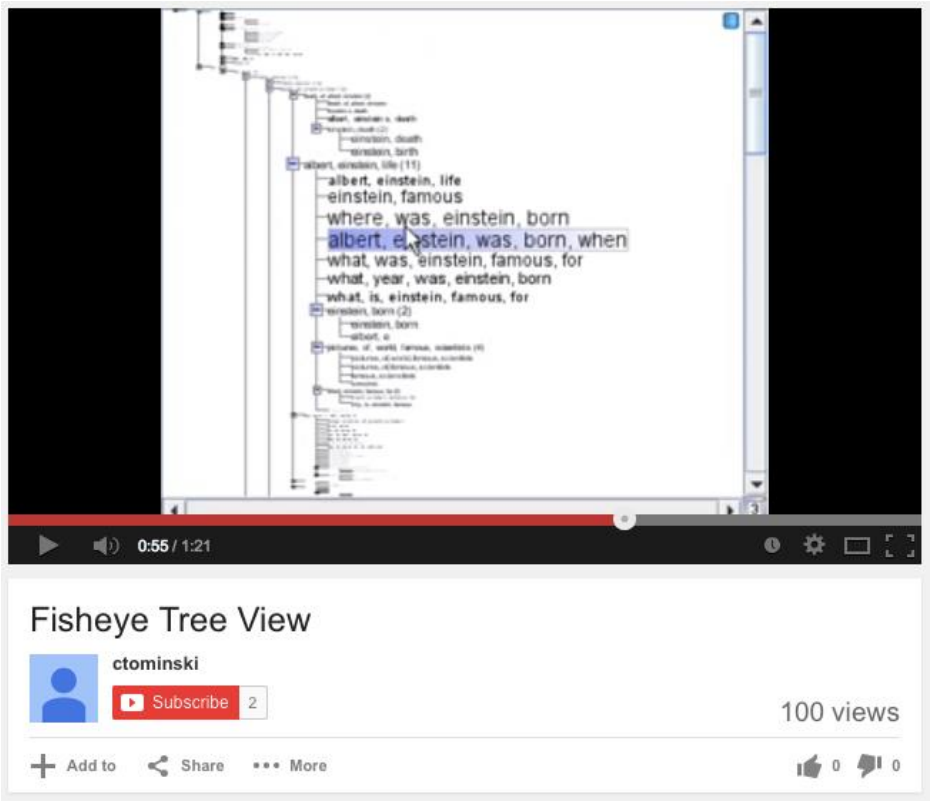


https://www.youtube.com/watch?v=pwpze3RF55o

http://pmcruz.com/information-visualization/data-lenses





https://www.youtube.com/watch?v=mTf7bPvdeJo

DISTORTION CONCERNS

Unsuitable for relative spatial judgements

Overhead of tracking distortion

Visual communication of distortion

  gridlines, shading

Target acquisition problem

  lens displacing items away from screen location

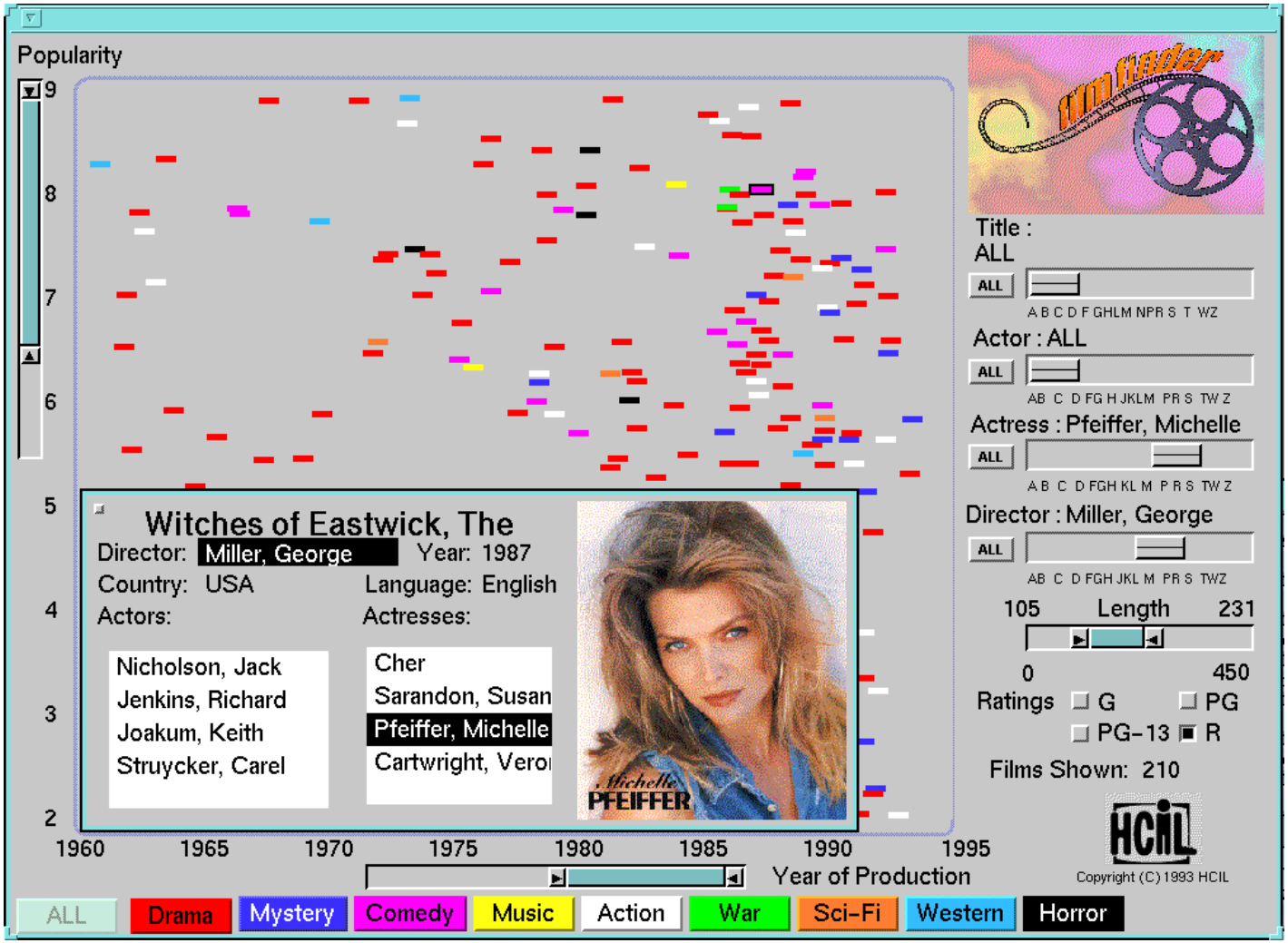
Mixed result compared to separate views and temporal naviation

OVERVIEW + DETAIL

OVERVIEW AND DETAIL

One view shows overview

Other show detail



[FilmFinder, Ahlberg & Shneiderman, 1994]

FILTERING & DYNAMIC QUERYING

MANTRA

Visual information seeking matra (Shneiderman, 1996)

Overview first,

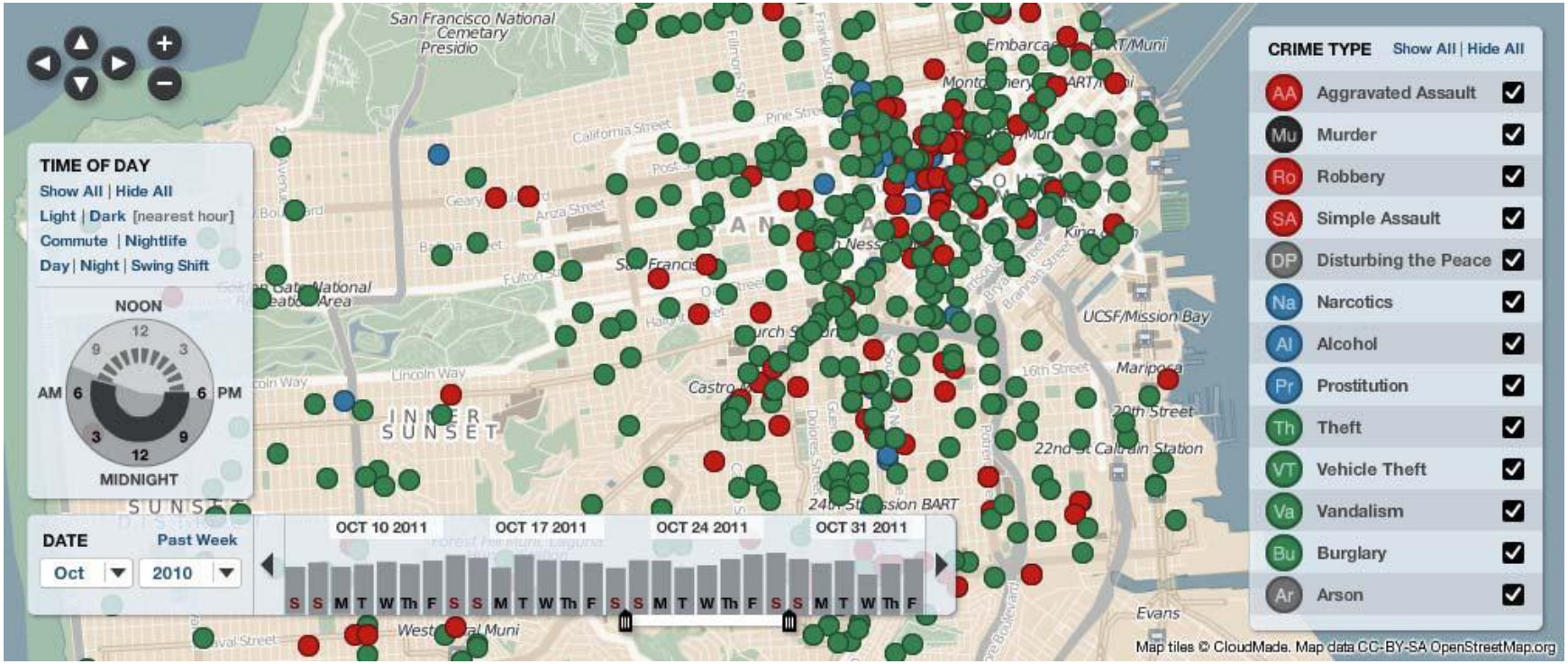
Zoom and filter,

Then details on demand

Related, history, extract

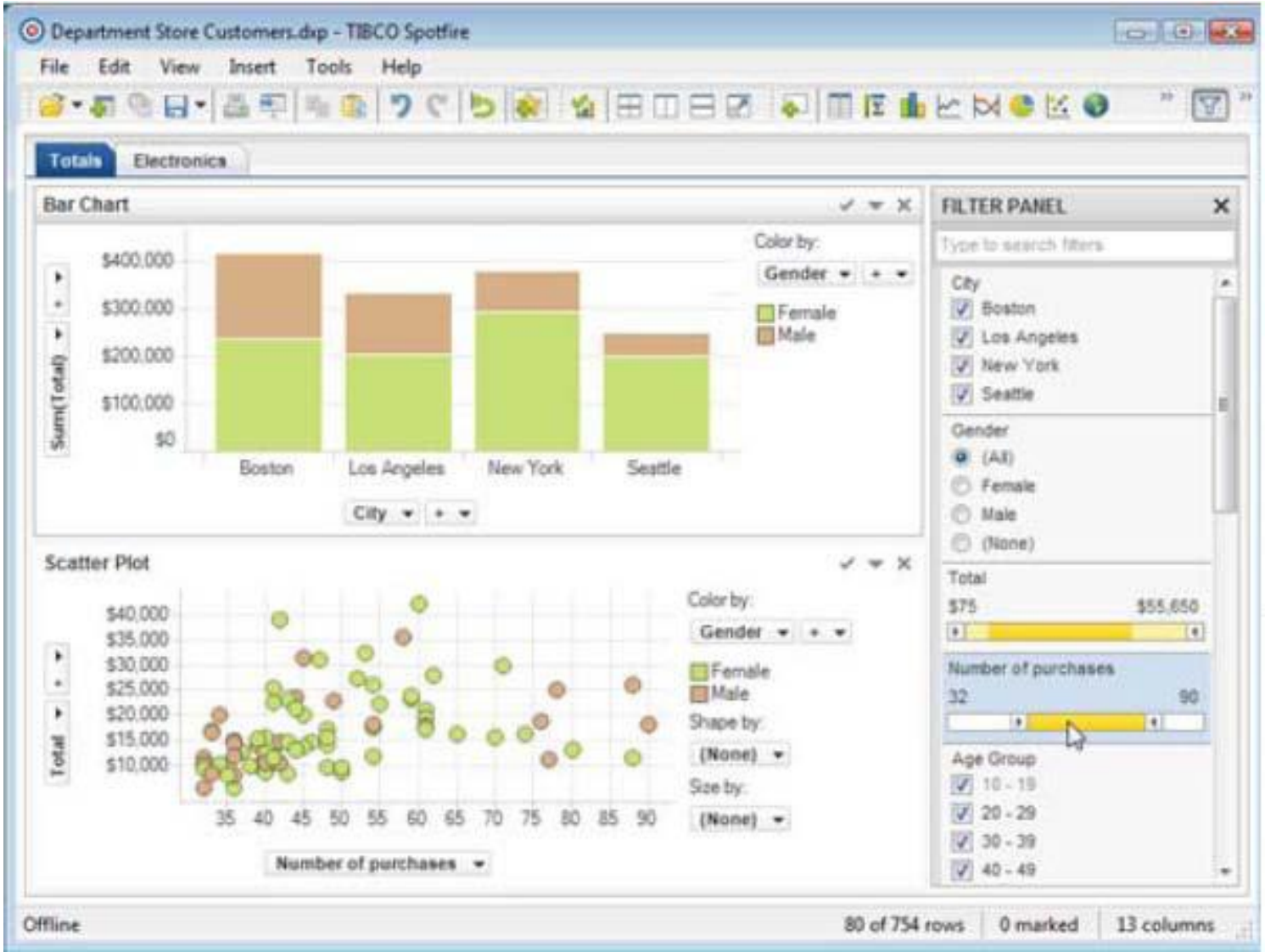
DYNAMIC QUERIES

Define criteria for inclusion/ exclusion

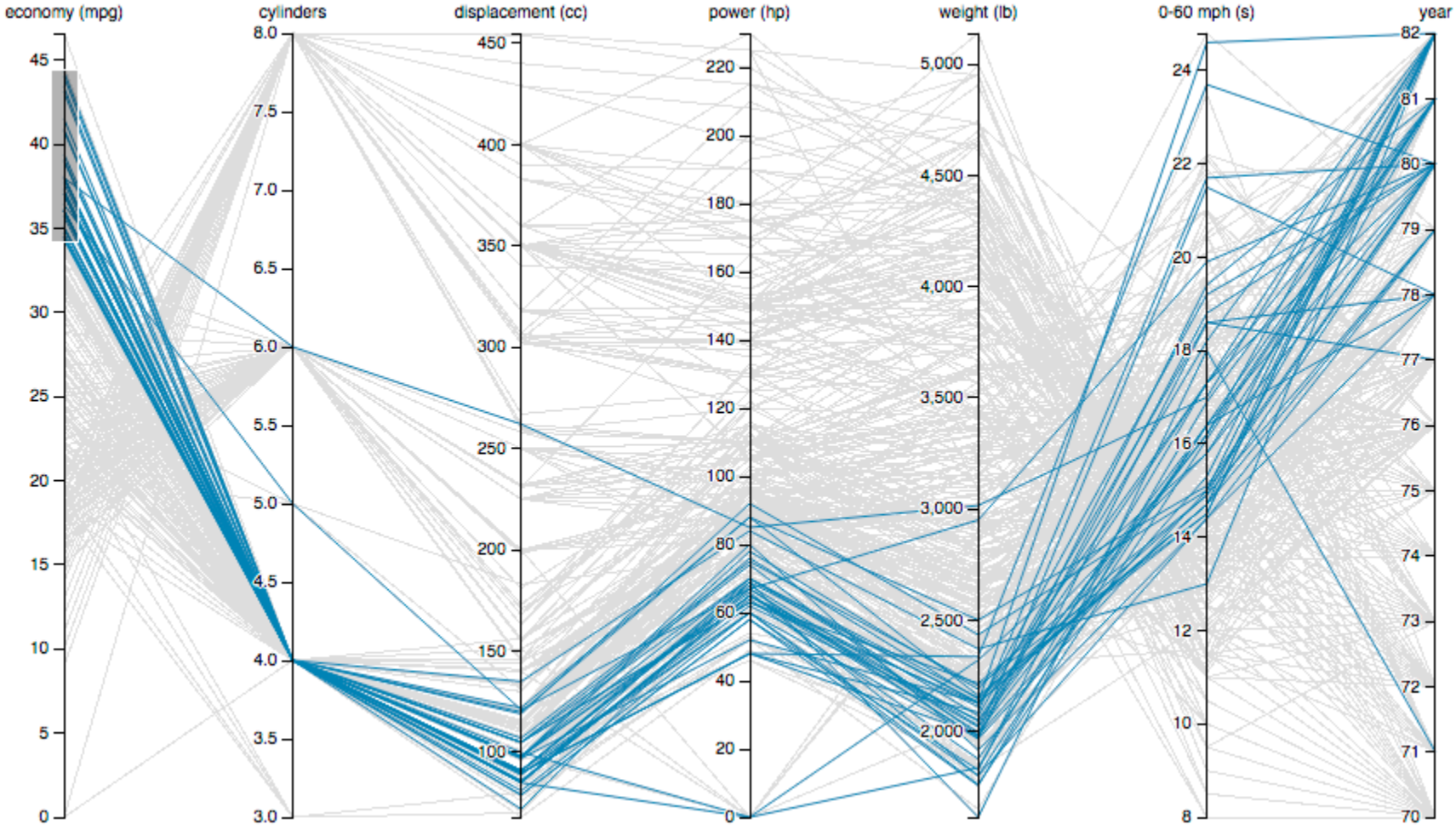


DYNAMIC QUERY WITH FILTER

Spotfire

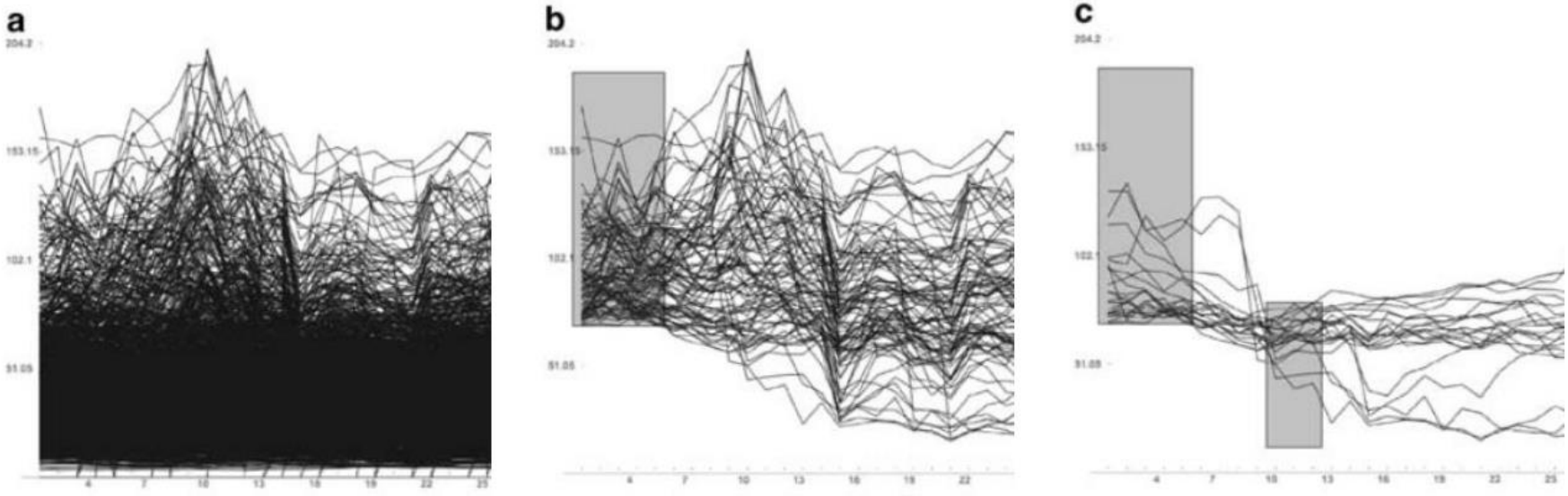


VISUAL QUERIES



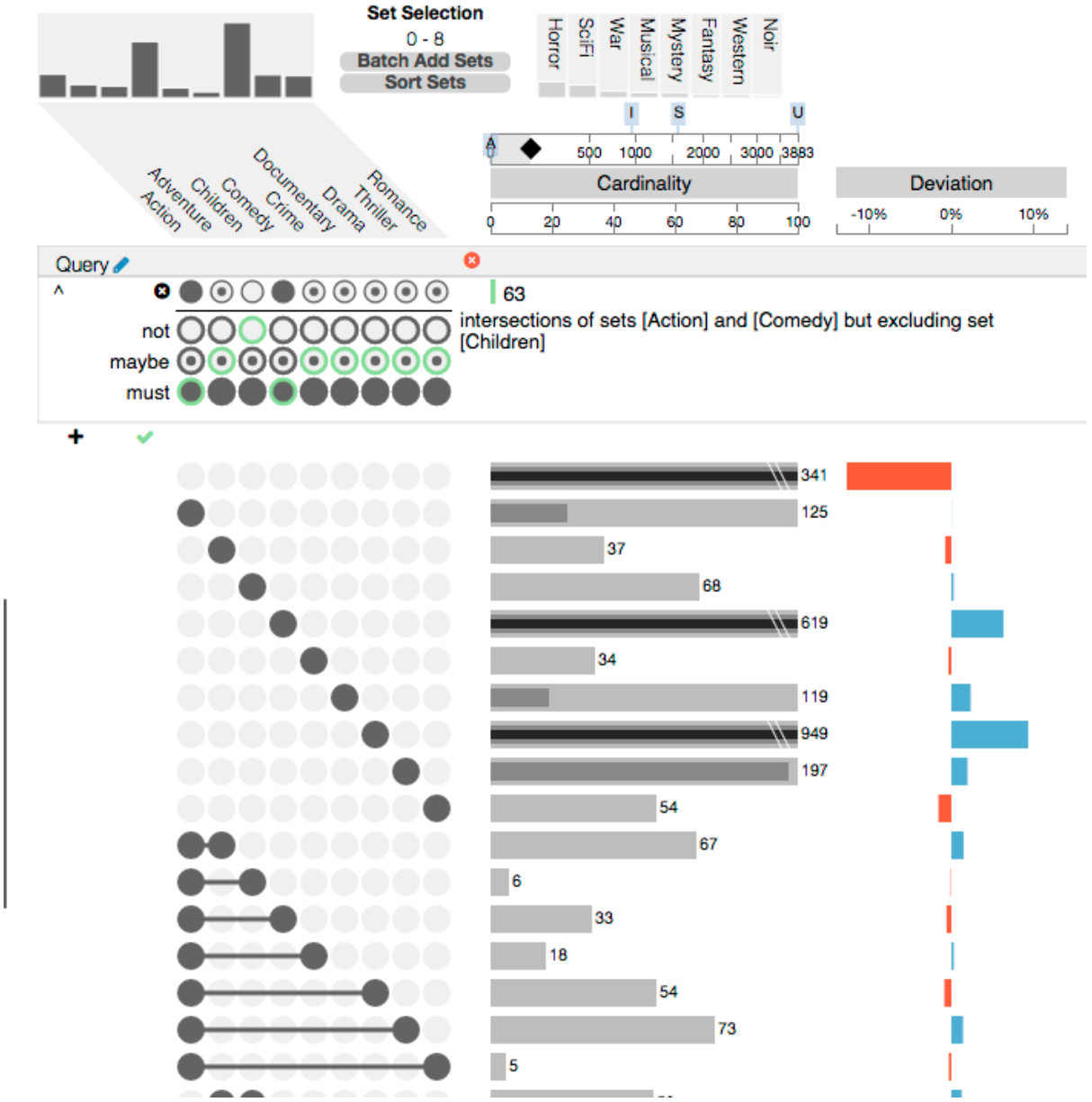
http://bl.ocks.org/jasondavies/1341281

VISUAL QUERIES



Time Searcher (Hocheiser, 2003)

QUERY INTERFACES



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Interactive Dynamics for Visual Analysis

A taxonomy of tools that support the fluent and flexible use of visualizations

Jeffrey Heer, Stanford University
Ben Shneiderman, University of Maryland, College Park

The increasing scale and availability of digital data provides an extraordinary resource for informing public policy, scientific discovery, business strategy, and even our personal lives. To get the most out of such data, however, users must be able to make sense of it: to pursue questions, uncover patterns of interest, and identify (and potentially correct) errors. In concert with data-management systems and statistical algorithms, analysis requires contextualized human judgments regarding the domain-specific significance of the clusters, trends, and outliers discovered in data.

Visualization provides a powerful means of making sense of data. By mapping data attributes to visual properties such as position, size, shape, and color, visualization designers leverage perceptual skills to help users discern and interpret patterns within data.[11] A single image, however, typically provides answers to, at best, a handful of questions. Instead, visual analysis typically progresses in an iterative process of view creation, exploration, and refinement. Meaningful analysis consists of repeated explorations as users develop insights about significant relationships, domain-specific contextual influences, and causal patterns. Confusing widgets, complex dialog boxes, hidden operations, incomprehensible displays, or slow response times can limit the range and depth of topics considered and may curtail thorough deliberation and introduce errors. To be most effective, visual analytics tools must support the fluent and flexible use of visualizations at rates resonant with the pace of human thought.

The goal of this article is to assist designers, researchers, professional analysts, procurement officers, educators, and students in evaluating and creating visual analysis tools. We present a taxonomy of interactive dynamics that contribute to successful analytic dialogues. The taxonomy

TABLE 1: **Taxonomy of interactive dynamics for visual analysis**

Data & View Specification	**Visualize** data by choosing visual encodings. **Filter** out data to focus on relevant items. **Sort** items to expose patterns. **Derive** values or models from source data.
View Manipulation	**Select** items to highlight, filter, or manipulate them. **Navigate** to examine high-level patterns and low-level detail. **Coordinate** views for linked, multi-dimensional exploration. **Organize** multiple windows and workspaces.
Process & Provenance	**Record** analysis histories for revisitation, review and sharing. **Annotate** patterns to document findings. **Share** views and annotations to enable collaboration. **Guide** users through analysis tasks or stories.

consists of 12 task types grouped into three high-level categories, as shown in table 1: (1) data and view specification (visualize, filter, sort, and derive); (2) view manipulation (select, navigate, coordinate, and organize); and (3) analysis process and provenance (record, annotate, share, and guide). These categories incorporate the critical tasks that enable iterative visual analysis, including visualization creation, interactive querying, multiview coordination, history, and collaboration. Validating and evolving this taxonomy is a community project that proceeds through feedback, critique, and refinement.

Our focus on interactive elements presumes a basic familiarity with visualization design. The merits and frailties of bar charts, scatter plots, timelines, and node-link diagrams, and of the visual-encoding decisions that underlie such graphics, are certainly a central concern, but we will largely pass over them here. A number of articles and books address these topics in great detail,[11,12,16,52] and we recommend them to interested readers.

Within each branch of the taxonomy presented here, we describe example systems that exhibit useful interaction techniques. To be clear, these examples do not constitute an exhaustive survey; rather, each is intended to convey the nature and diversity of interactive operations. Throughout the article the term *analyst* refers to someone who uses visual analysis tools and not to a specific person or role. Our notion of analyst encompasses anyone seeking to understand data: traditional analysts investigating financial markets or terrorist networks, scientists uncovering new insights, journalists piecing together a story, and people tracking various facets of their lives, including blood pressure, money spent, electricity used, or miles traveled.

DATA AND VIEW SPECIFICATION

To enable analysts to explore large data sets involving varied data types (e.g., multivariate, geospatial, textual, temporal, networked), flexible visual analysis tools must provide appropriate controls for specifying the data and views of interest. These controls enable analysts to selectively *visualize* the data, to *filter* out unrelated information to focus on relevant items, and to *sort* information to expose patterns. Analysts also need to *derive* new data from the input data, such as normalized values, statistical summaries, and aggregates.

VISUALIZE

Perhaps the most fundamental operation in visual analysis is to specify a visualization of data: analysts must indicate which data is to be shown and how it should be depicted. Historically, this process required custom programming of a specific visualization component. Within user interfaces, such visualization "widgets" are often presented in a *chart typology*, a palette of available visualization templates (bar charts, scatter plots, map views, etc.) into which analysts can slot their data. This method of interaction will be immediately familiar to users of spreadsheet programs: users select a chart type and assign data variables to visual aspects such as the X/Y axes and the size or color of visualized marks. A chart typology has the benefits of simplicity and familiarity, but it also limits the types of possible visualizations and makes it cumbersome to try out different visualizations of the same data.

Some visualization system designers have explored alternative approaches. Classic scientific visualization systems[1] and more recent platforms for artistic expression[9] use *data-flow graphs*, in which the visualization process is deconstructed into a set of finer-grained operators for data import, transformation, layout, coloring, etc. Analysts interactively chain these operators together

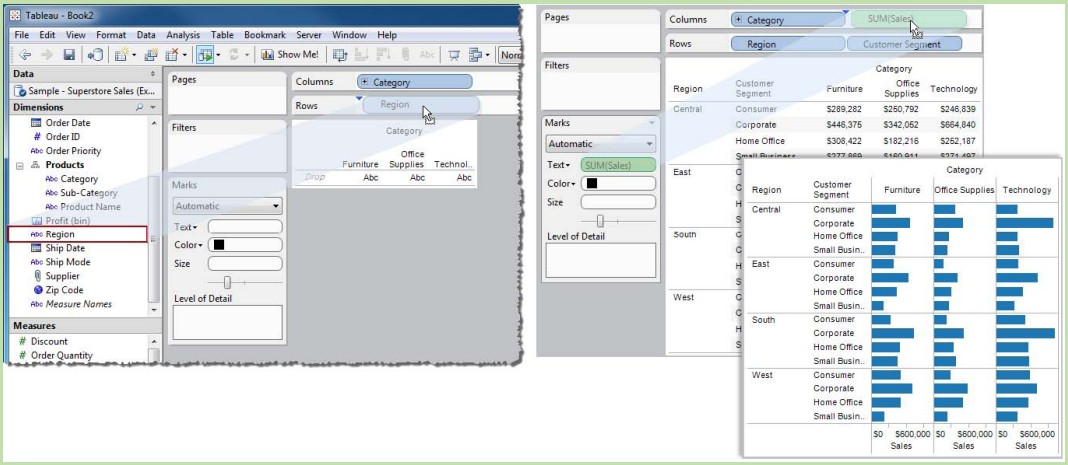
to construct novel displays. Through flexible combinations of operators, data-flow models can enable a larger space of visualization designs. Data-flow systems require more input effort than chart typologies, however, and may be limited by the set of available operators. In many cases, novel designs require analysts with programming expertise to develop new building blocks for the system.

Other systems are based on *formal grammars* for visualization construction. These grammars constitute high-level languages for succinctly describing how data should be mapped to visual features. By combining a handful of such statements, analysts can construct complex, customized visualizations with a high degree of design control. This approach is used by a number of popular data visualization frameworks such as Leland Wilkinson's *Grammar of Graphics*,[57] ggplot2 for the R statistical analysis platform,[56] and Protovis for HTML5.[10] Each of these requires at least minimal programming ability, however.

Tableau[51] (née Polaris[50]) provides an example of visualization specification by drag-and-drop operations: analysts place data variables on "shelves" corresponding to visual encodings such as spatial position, size, shape, and color (see figure 1). The visual specification is then translated into an underlying formal grammar that determines both the visualization design and corresponding queries to a database. This approach leverages the expressiveness of formal grammars while avoiding the need for programming. Another advantage is that formal grammars can be augmented with automated design facilities: a system can generate multiple visualization suggestions from a partial specification.[37,38,44] While systems based on formal grammars are both fluent and expressive, users need to understand the underlying generative model, which imposes a steeper learning curve than the more familiar chart typology.

FIGURE 1

Visual encoding via drag-and-drop actions in Tableau[51]



Fortunately, these methods are not mutually exclusive. Analysts can apply a data-flow system or formal grammar to define new components to include within a chart typology, leveraging the improved expressiveness of the former and the ease of use of the latter. Novel interfaces for visualization specification are still needed. A formal grammar that uses graphical marks (rectangles, lines, plotting symbols, etc.) as its basic primitives provides a conceptual model compatible with interactive design tools. New tools requiring little to no programming might place custom visualization design in the hands of a broader audience.

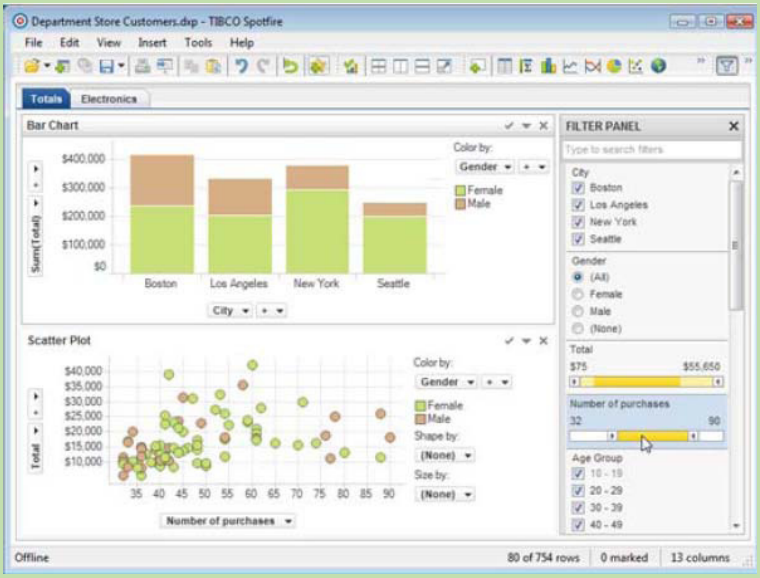
FILTER

Filtering of data values is intrinsic to the visualization process, as analysts rarely visualize the entirety of a data set at once. Instead, they construct a variety of visualizations for selected data dimensions. Given an overview of selected dimensions, analysts then often want to shift their focus among different data subsets— for example, to examine different time slices or isolate specific categories of values.

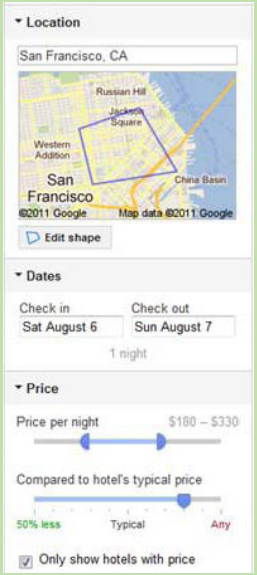
Designers have devised a variety of interaction techniques to limit the number of items in a display. Analysts might directly select (e.g., "lasso") items in a display and then highlight or exclude them; we discuss these forms of direct view manipulation later. Another option is to use a suite of auxiliary controls, or *dynamic query widgets*,[47] for controlling item visibility (see figures 2, 3, and 4). The choice of appropriate widget is largely determined by the underlying data type. Categorical or ordinal data can be filtered using simple radio buttons or checkboxes (when the number of distinct items is small), or scrollable lists, hierarchies, and search boxes with

FIGURE 2

Examples of dynamic query filter widgets



Spotfire



Google Hotel Search

autocomplete (when the number of distinct items is large or contains arbitrary text). Ordinal, quantitative, and temporal data can also be filtered using a standard slider (for a single threshold value) or a range slider (for specifying multiple endpoints). When coupled with realtime updates to the visualization, these widgets allow rapid and reversible exploration of data subsets. In figure 2, Spotfire (left) provides a variety of controls for filtering visualized data: checkboxes and radio buttons filter categorical variables, while range sliders filter numerical values; on the right, Google Hotel Search provides widgets for geographic, date, and price ranges. Query controls can be further augmented with visualizations of their own: figure 3 shows a range slider augmented with a histogram of underlying values.

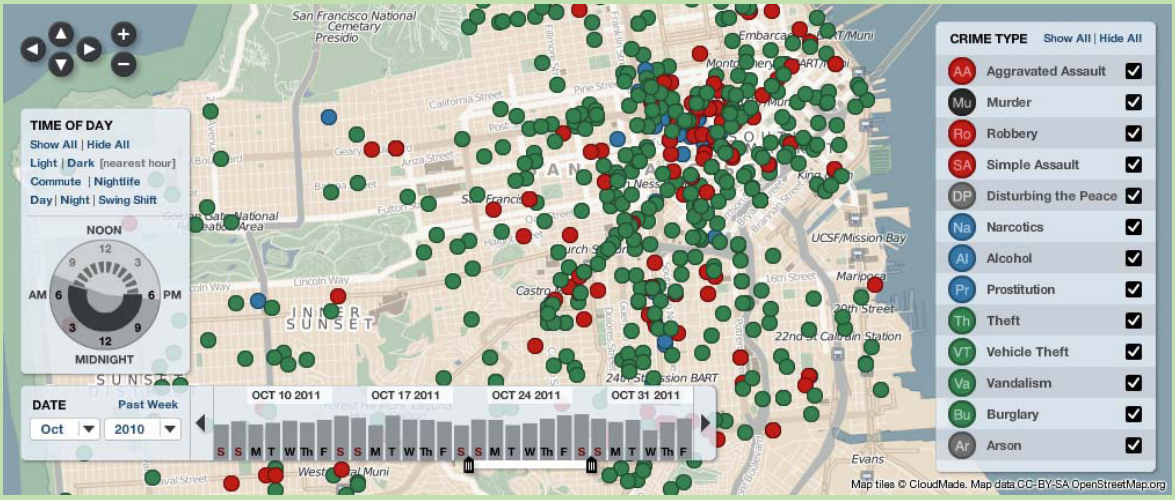
Expert analysts also benefit from more advanced functionality. For example, a search box might

FIGURE 3

Histogram sliders in NodeXL

FIGURE 4

Zoomable map from CrimeSpotting.org



support sophisticated query mechanisms, ranging in complexity from simple keyword search, to regular expression matching, to a full-fledged structured query language. While these additional mechanisms may not support rapid, incremental exploration as fluently as graphical widgets, they provide a means for expressing more nuanced criteria. Filtering also interacts with other operations: filtering widgets may operate over data sorted in a user-specified manner (see next section), or users might create derived values (see section after next) and filter based on the results.

SORT

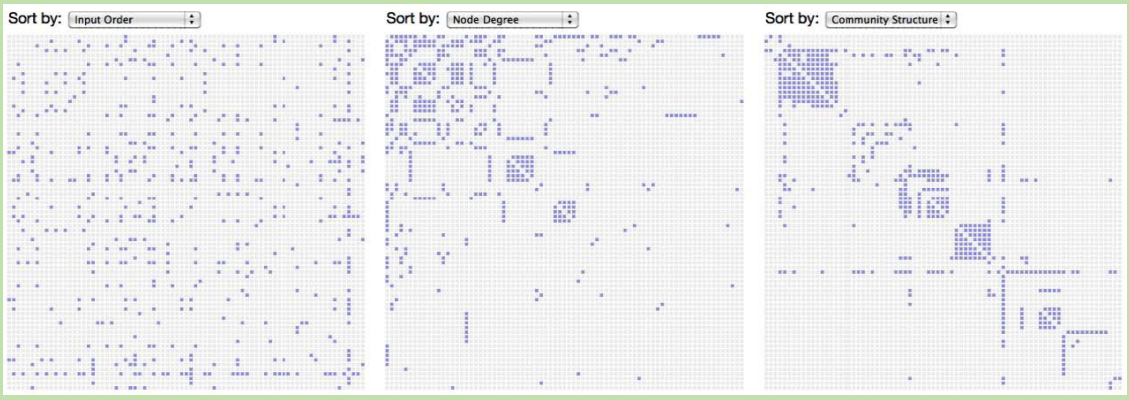
Ordering (or sorting) is another fundamental operation within a visualization. A proper ordering can effectively surface trends and clusters of values[5] or organize the data according to a familiar unit of analysis (days of the week, financial quarters, etc.). The most common method of ordering is to sort records according to the value of one or more variables. Sorting controls can be simple choices in a toolbar or clicks on the header of a table to produce ascending or descending sorts for numerical or textual values. Sometimes specialized sort orders such as weekday or month names are necessary to reveal important patterns.

Ordering becomes more complicated in the case of multiple view displays, in which both entire plots and the values they contain may be sorted to reveal patterns or anomalies. Sorting values consistently across plots (for example, by their marginal mean or median values) can reveal patterns while facilitating comparison among plots.

Some data types (e.g., multivariate tables, networks) do not always lend themselves to simple sorting by value. Such data may require more sophisticated *seriation* methods[20,57,58] that attempt to minimize a distance measure among items. The goal is to reveal underlying structure (e.g., clustering) within the data. An example is shown in figure 5, a matrix-based visualization of a social network. On the left, a matrix plot of a social network conveys little structure when the rows and columns (representing people) are sorted alphabetically. Interactively reordering the matrix by node degree reveals more structure (center). Seriating the matrix by network connectivity reveals underlying clusters of communities (right).

FIGURE 5

Reorderable matrices



DERIVE

As an analysis proceeds in iterative cycles, users may find that the input data is insufficient: variables may need to be transformed or new attributes derived from existing values. Common cases include normalization or log transforms to enable more effective value comparisons. Derived measures are often used to summarize the input data, ranging from descriptive statistics (mean, median, variance) to model fitting (regression curves) and data transformation (group-by aggregation such as counts or summations). While analysts can derive new values prior to importing data for visual analysis, the overhead of moving between tools stymies fluid, iterative exploration. As a result, visual analytics tools should include facilities for deriving new data from input data. Often this functionality is provided via a *calculation language*, similar to those found in spreadsheets or database query languages. Beyond these basic functions, hypothesis-testing methods (t-tests, ANOVA) can amplify the benefits of smooth integration of statistics and visualization.

Improved derivation methods present a promising frontier for visual analytics research. How can visual tools support flexible construction of more advanced models or derived values? Using programming-by-demonstration methods, analysts might annotate patterns (e.g., of network intrusion events[60]) from which the system can generalize pattern-recognition rules. Or, visual tools might automatically fit applicable statistical models to the data based on the current visualization state. For example, the nesting of variables within common "pivot" displays could be mapped to the structure of a linear model. More principled frameworks that wed visualization to modeling and forecasting are still emerging.

VIEW MANIPULATION

Once analysts have created a visualization through data and view specification actions, they should be able to manipulate the view to highlight patterns, investigate hypotheses, and drill down for more details. Analysts must be able to *select* items or data regions to highlight, filter, or operate on them. Large information spaces may require analysts to scroll, pan, zoom, and otherwise *navigate* the view to examine both high-level patterns and fine-grained details. Multiple, linked visualizations often provide clearer insights into multidimensional data than do isolated views. Analysis tools must be able to *coordinate* multiple views so that selection and filtering operations apply to all displays at once and *organize* the resulting dashboards and work spaces.

SELECT

Pointing to an item or region of interest is common in everyday communication because it indicates the subject of conversation and action. In the physical world, people coordinate their gestures, gaze, and speech to indicate salient items. For example, different hand gestures can communicate angle (oriented flat hand), height (horizontal flat hand), intervals (thumb and index finger in "C" shape), groupings (circling a region), and forces (accelerating fist).[27] In visual analysis, reference (or *selection*) remains of critical importance, but it is realized through a more limited set of actions, such as clicking or lassoing items of interest.

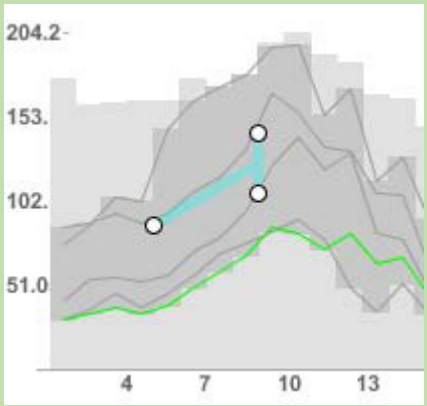
Common forms of selection within visualizations include mouse hover, mouse click, region selections (e.g., rectangular and elliptical regions, or free-form "lassos"), and area cursors (e.g., "brushes"[4] or dynamic selectors such as the bubble cursor,[18] which selects the item currently closest to the mouse pointer).

These selections often determine a set of objects to be manipulated, enabling highlighting, annotation, filtering, or details-on-demand. Note that interactive selection is closely related to filtering: selections can be used to identify items to remove from the display. The context of interaction must also be taken into account when choosing a selection method. For example, responding to hover events to provide details-on-demand is inappropriate when using touch-based input on a tablet or mobile phone.

Selections can also vary in terms of their expressive power. Most interfaces support selections of a collection of items. Though this approach is easy to implement, it does not allow analysts to specify higher-level criteria. A more powerful, albeit more complex, approach is to support selections as queries over the data.[22] Maintaining query structure increases the expressiveness of visualization applications. For example, rather than directly selecting the contained items, drawing a rectangle in a chart may specify a range query over the data variables represented by the X and Y axes. The resulting selection criteria can then be saved and applied to dynamic data (updating items may enter or exit a query region) or to a completely different visualization. Examples include querying stock-price changes in TimeSearcher[28] (see figure 6) and attribute ranges in parallel coordinates displays[30] (figure 7). In figure 6 an angular selection tool specifies a target slope (rate of change) and tolerance for a collection of stock prices. All time series with a similar slope over the queried time range are selected; shaded regions show envelopes of minimum and maximum values. The widget operates directly on the visualization: dragging the widget from left to right interactively queries other time windows. In figure 7 parallel coordinates plot multidimensional data as line segments among parallel axes. Here, an analyst has dragged along the axes to create interactive selections that highlight automobiles with low weight and high mileage.

FIGURE 6

Querying time-series by slope in TimeSearcher[28]



Designing more expressive selection methods remains an active area of research. For example, researchers have proposed methods to map mouse gestures over a time-series visualization to select perceptually salient data regions such as peaks, valleys, and slopes[35] (see figure 8) or to query complex patterns of temporal variation.[29] Initial selections can also be used as a starting point for more complex selections, as analysts might click a representative object and then formulate a broader selection based on the object's properties (e.g., "select all items blue like this one").[22] Of course, selection need not be limited to the mouse and keyboard: input modalities such as touch, gesture, and speech might enable new, effective forms of selection.

FIGURE 7

Selection queries in parallel coordinates

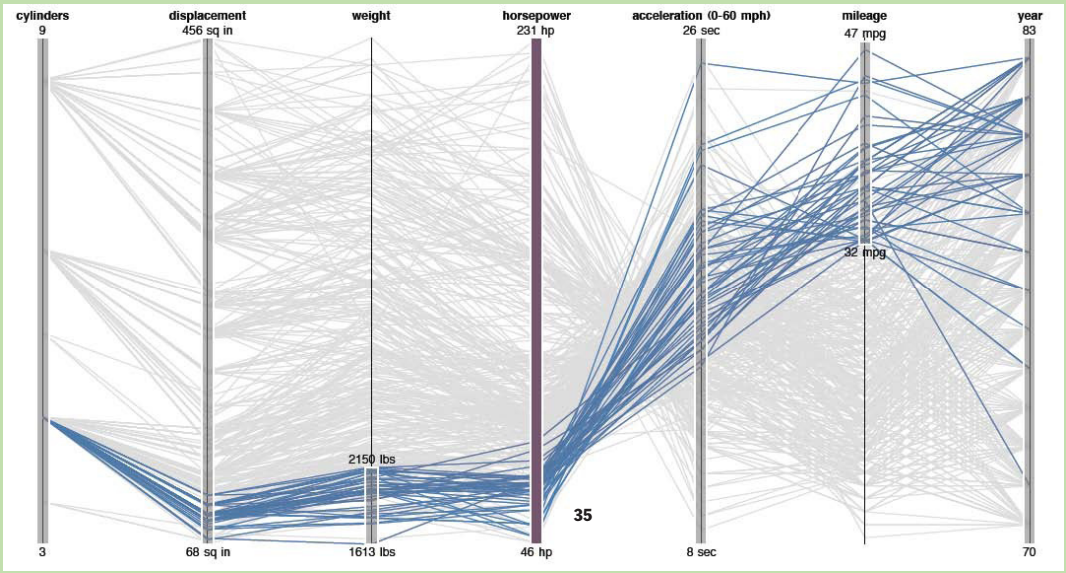
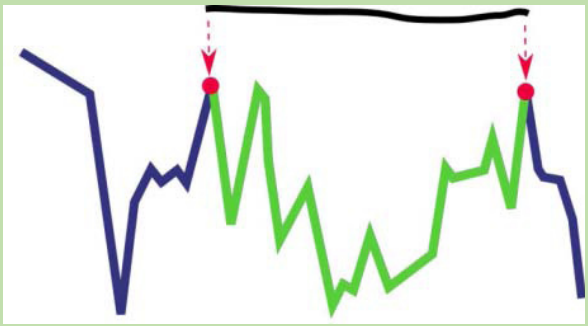


FIGURE 8

Perceptual interpretation of Ink annotations



NAVIGATE

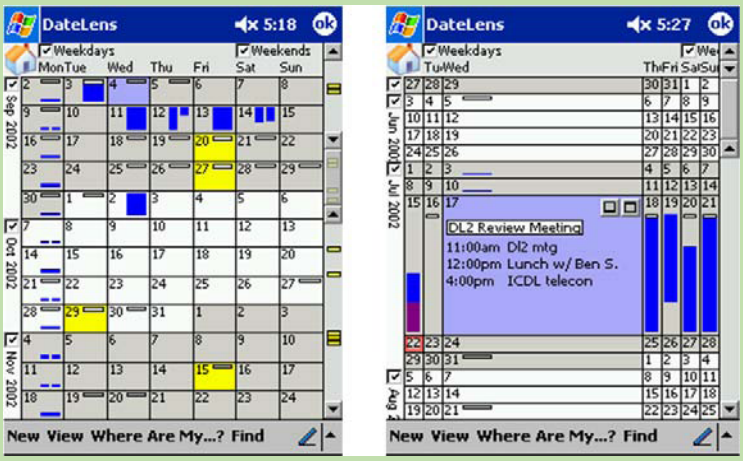
How analysts navigate a visualization is in part determined by where they start. One common pattern of navigation adheres to the widely cited visual information-seeking mantra: "Overview first, zoom and filter, then details-on-demand."[48] Analysts may begin by taking a broad view of the data, including assessment of prominent clusters, outliers, and potential data-quality issues. These orienting actions can then be followed by more specific, detailed investigations of data subsets. A common example is geographic maps: an overview might show an overall territory, followed by zooming into regions of interest. For example, the map in figure 4 depicts criminal activity by time and region. It shows all crimes committed after dark during the last week of October 2011. Dynamic query widgets enable filtering by time of day (left), date span (bottom), and type of crime (right). Pan (drag) and zoom (buttons and scroll wheel) controls enable view navigation. As an analyst zooms in on the map, the circular crime markers gain detailed labels— a form of *semantic zooming*.

Of course, starting with an expansive overview is not always advisable. A legal analyst researching for an upcoming trial may be wise to forego an overview of the entire history of U.S. court decisions. Instead, the analyst might start with the legal decisions most relevant to the current case, perhaps determined by keyword search, and expand the investigation to other, cited decisions. This form of navigation can be summarized as "Search, show context, expand on demand."[53]

In either case, visualizations often function as *viewports* onto an information space. Analysts need to manipulate these viewports to navigate the space. Common examples include scrolling or panning a display via scrollbars or mouse drag, and zooming among different levels using a zoom slider or scroll wheel (figure 4). Zooming need not follow a strict geometric metaphor: semantic zooming[7] methods can modify both the amount of information shown and how it is displayed as analysts move among levels of detail. In the calendar in figure 9, the display magnifies selected regions as analysts navigate from months to days to hours. Semantic zooming reveals more details within focal regions. Additionally, dynamic query widgets, such as range sliders for the X and Y axes

FIGURE 9

DateLens Focus-plus-Context Calendar[6]



of a scatter plot, can filter the visible data range and thus provide a form of zooming within a chart.

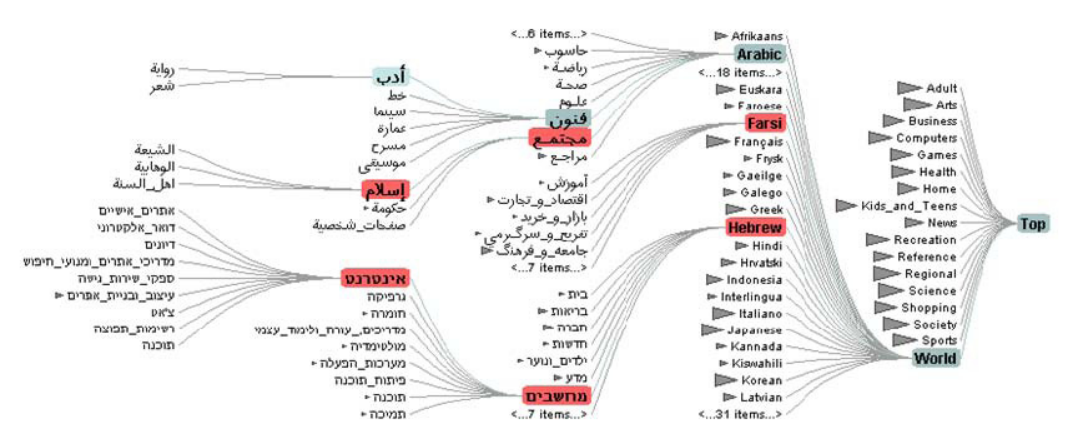
To aid navigation further, researchers have developed a variety of *focus plus context* methods. These "bifocal" views[49] provide a detailed view of a high-interest data region while retaining surrounding context to help keep analysts oriented. A second key idea is the use of *overview and detail* displays. For example, a geographic visualization might include a large zoomed-in map (the detail), while a smaller, zoomed-out map includes a rectangle showing the position of the zoomed-in view within the broader terrain (the overview). In this case, the detail view provides the focus, and the overview provides context. The benefits are highest when the zoom factor (ratio of overview to detail view) is 5-20.[42] When larger zoom factors are needed, intermediate overviews may also be helpful.

A different approach is to use *distortion* or magnification techniques that transform the entire display region such that contextual regions are demagnified. A simple example is the Mac OS X dock, which uses 1D fisheye distortion to show common applications; more sophisticated methods employ distortion in multiple dimensions. While often visually intriguing, complex distortion methods have yet to prove their worth in real-world applications: viewers can become disoriented by nonlinear distortions, which show no significant performance improvement over simpler methods such as zooming.[39]

In addition to manipulating display space, focus-plus-context methods can be applied directly to the data itself. The goal is to identify which data items are currently of high interest (focus), which are of high importance regardless of the current focus (context), and which can be safely removed from view. *DOI* (*degree-of-interest*) functions[17,24,53] calculate scores for information content based both on general importance (e.g., top-level categories within a hierarchy, or nodes with high centrality in a graph) and current interest (e.g., as indicated by mouse clicks, search queries, or proximity to other high-interest items). The distribution of DOI scores can then be used to selectively control the visibility of items based on the current view size and context of interaction, as in figure 10. As analysts click on or search for different items, the DOI scores

FIGURE 10

Degree-of-interest tree of a taxonomy with 600k items[24]



dynamically update to reveal relevant unseen data or hide irrelevant detail. A model of the analyst's current interest filters the display to the most relevant items. Low-interest items are elided but still accessible through aggregate representations. The interest estimates update as an analyst explores the taxonomy, initiating animated transitions between different views of the data.

Visualizations can provide cues to assist analysts' decisions of where and how to navigate. The controls for view manipulation have often been invisible, such as zooming/panning by mouse movement. Improved strategies facilitate discovery by analysts and provide visible indication of settings in legends or other ways, such as scrollbar positions, that provide informative feedback. An important challenge is to show selected items, even when they are not in view. For example, the results of a text search that are not currently in view might be shown by markers in the scrollbar[61] or the periphery of the display.[3,19]

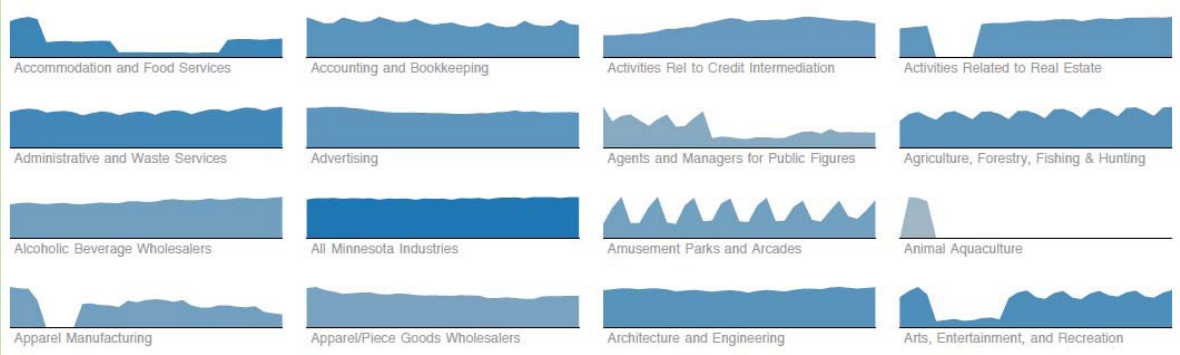
COORDINATE

Many analysis problems require *coordinated multiple views* that enable analysts to see their data from different perspectives. A public policy analyst studying educational attainment might produce a bar chart of people's ages, a map of locations, a textual list with education history, and a scatter plot showing income vs. education. By selecting a single item or a group in one view, analysts might see related details or highlighted items in the other views. This powerful approach to exploring multivariate data also enables drilling down into subgroups, marking sets, and exporting selections.

Multiview displays can facilitate comparison. For example, Edward Tufte[52] advocates the use of *small multiples*: a collection of visualizations placed in spatial proximity and typically using the same measures and scales. As in figure 11, these small multiples, also called *trellis plots*, enable rapid comparison of different data dimensions or time slices. The visualization shows employment figures by economic sector in Minnesota. The repetition of the chart form supports comparison among sectors. Plotting all the data in one chart would otherwise clutter and obscure individual trends. Selecting a point in time in one view highlights the corresponding point in all other views.

FIGURE 11

Small multiples display



Alternatively, multiple view displays can use a variety of visualization types— such as histograms, scatter plots, maps, or network diagrams— to show different projections of a multidimensional data set. An analyst constructs a complex patchwork of interlinked tables, plots, and maps in figure 12 to analyze the outcomes of elections in Michigan. Annotations indicate how selected data items correspond between visualization views. Accompanying items such as legends, histogram sliders, and scrollbars with highlighting markers can all provide multiple views onto the data. Automatically generated legends and axes are important for providing accurate annotations for analysts and meaningful explanations when visualizations are shared. Legends and axes can also become control panels for changing color palettes, marker attributes, variable ranges, or provenance information.[43]

Multiview displays can also enable interactive exploration across views. *Brushing and linking* is the process of selecting (brushing) items in one display to highlight (or hide) corresponding data in the other views.[4] In figure 13, a baseball analyst makes selections in one plot and corresponding items highlight in the others. On the left, selecting high-income players (top-right plot) shows little dependence on career length or fielding ability, but correlates with hitting performance. On the right, selecting the cluster of players who make more assists than put-outs (middle-left plot) reveals a strong dependence on position. Each visualization can thus serve as an input channel for revealing patterns across a data set. Linked selection enables rich, multidimensional reasoning by allowing analysts to assess how patterns in one view project onto the others. Analysts may wish to coordinate views in variety of ways:[40,55] selecting items in one view might highlight matching records in other views, or instead provide filtering criteria to remove information from the other displays. Linked navigation provides an additional form of coordination: scrolling or zooming one view can simultaneously manipulate other views.

Though comparing multiple visualizations requires viewers to orchestrate their attention and mentally integrate patterns among views, this process is often more effective than cluttering a single visualization with too many dimensions. Future studies of how analysts construct multiview

FIGURE 12

Multiple coordinated views in Improvise[55]

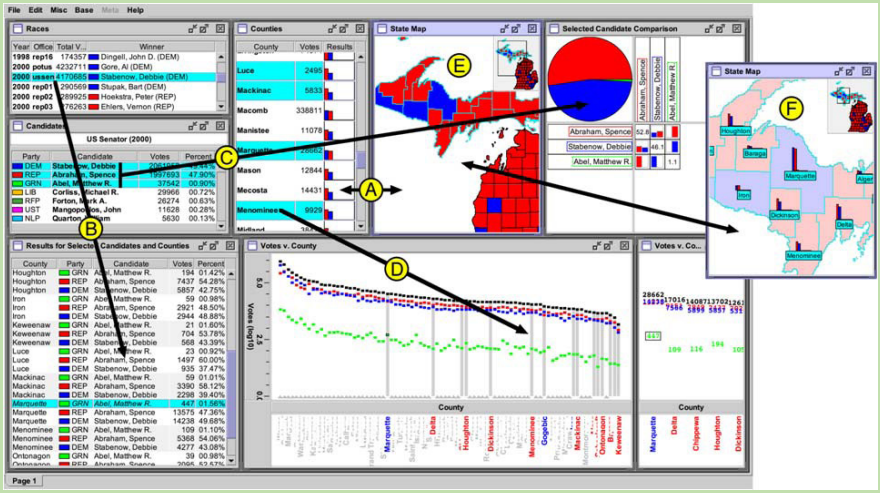
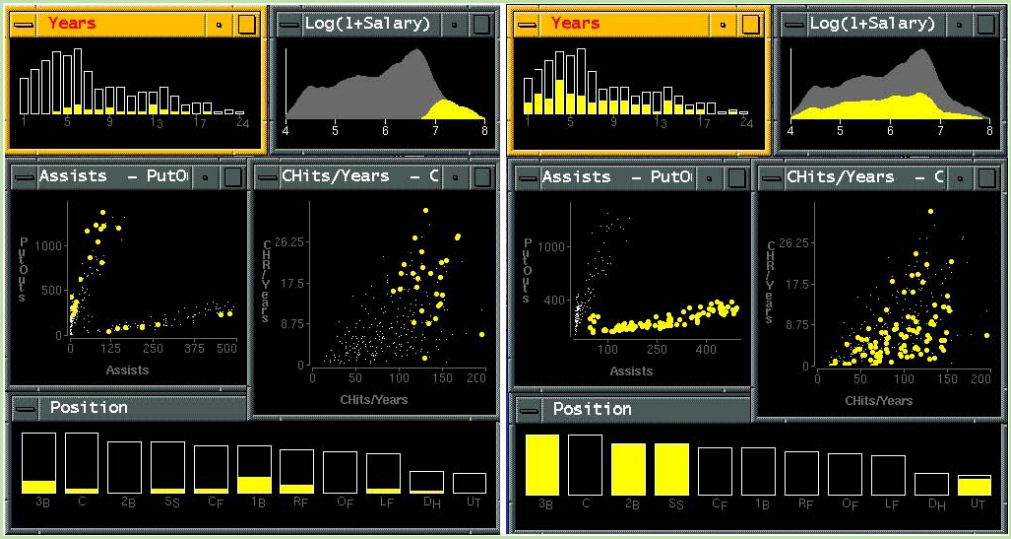


FIGURE 13

Brushing and linking of baseball statistics in GGobi



displays and specify coordination behaviors (e.g., highlighting, filtering) could provide designers with an understanding of how to build more effective tools. In addition, if designers ensure that rich multiview displays stay understandable, analysts are more likely to make compelling insights. Newcomers to an analysis, or even seasoned analysts simply returning from a coffee break, may become confused by the number of views and the potentially complicated set of coordinated queries between them. Visual analytics systems that provide access to coordination settings and replay the history of view construction can enhance understanding.

ORGANIZE

When analysts make use of multiple views they face the corresponding challenge of managing a collection of visualizations. As in traditional window-based interfaces, analysts may wish to open, close, maximize, and lay out different components. As purely manual window manipulation can be tedious, well-designed visual analytics tools simplify the organization of visualization views, legends, and interactive controls. For example, a human resources data set may show a scatter plot of salary by years of experience, plus a bar chart showing 10 age groups, and a treemap with seven corporate sites, each with 10-30 job titles. These three visualizations might give a large area for the scatter plot, with the bar chart and treemap to the right side stacked one above the other. A control panel with sliders, checkboxes, radio buttons, and a search box could be on the far right, with a details-on-demand window and annotation box across the bottom. This tiled approach allows analysts with sufficiently large displays to see all the information and selectors at once, minimizing distracting scrolling or window operations, while enabling them to concentrate on extracting and reporting insights. The coordination across windows means that slider movements or checkbox selections will

cause all views to update, allowing rapid exploration of just the employees at certain sites or specific job titles.

Typical systems allow analysts to add views, such as a second scatter plot, in ways that make modest changes to the existing window organization. An alternative approach is to add a new tab that contains the second scatter plot, so analysts can switch between the first and second set of windows. A common feature is to add trellised views, so multiple visualizations can be created at once— for example, separate bar charts showing age distributions for each of the seven corporate sites.

More advanced systems might aid this process through automated support[8] that enables multiple windows to be opened/closed as a group and lays them out in orderly ways. Useful methods include standard scatter plot matrices (showing all pairs of scatter plots) or custom generation of related views of interest (e.g., of data variables correlated to the visualized attributes). Desirable features are automatic (re)sizing as views are added or removed and layout routines to place related views in spatial proximity.

As larger and multiple displays become more common, layout organization tools will become decisive factors in creating effective user experiences. Similarly, the demand for tablet and smartphone visualizations will promote innovation in layout organizations that are compact and reconfigurable by simple gestures. Zooming, panning, flipping, and sequencing strategies will also improve analyst experiences and facilitate effective presentations to others.

PROCESS AND PROVENANCE

Visual analytics is not limited to the generation and manipulation of visualizations— it involves a process of iterative data exploration and interpretation. As a result, visual analytics tools that provide facilities for scaffolding the analysis process will be more widely adopted. Tools should preserve analytic provenance by keeping a *record* of analyst actions and insights so that the history of work can be reviewed and refined. Textual logs of activity have benefits, but visual overviews of activity can be more compact and comprehensible. If analysts can *annotate* patterns, outliers, and views of interest, they can document their observations, questions, and hypotheses. In a networked environment, analysts should be empowered to *share* results and discuss with colleagues, coordinate the work of multiple groups, or support processes that may take weeks and months. Moreover, analysis tools can explicitly *guide* novices through common analysis tasks, provide progress indicators for experts, or lead viewers through an analysis story.

RECORD

When analyzing data with visualizations, users regularly traverse the space of views in an iterative fashion. Exploratory analysis may result in a number of hypotheses, leading to multiple rounds of questions and answers. Analysts can generate unexpected questions that may be investigated immediately or revisited later. After conducting analysis, analysts may need to review, summarize, and communicate their findings, often in the form of reports or presentations.

To support iterative analysis, visual analysis tools can record and visualize analysts' *interaction histories*. At a minimum, applications should provide basic undo and redo support. While low-level input such as mouse and keyboard events are easy to capture, histories become much more valuable when they record high-level semantic actions. By modeling the space of user actions (view

specifications, sorting, filtering, zooming, etc.), richer logs can be constructed and visualized.[15,25,32,45] Common visual representations of analytic actions include both chronological ("timeline") and sequential ("comic strip") views. As shown in figure 14, a "comic strip" display retraces the steps taken in a visual analysis of business operations data.

Visual histories also reveal the hierarchical patterns of branching histories. Reading the graph shown in figure 15 in a snake-like fashion (first left-to-right, then right-to-left) reveals patterns of iterative exploration, branching, and backtracking in an analysis. Techniques for "chunking" related actions together can further reduce clutter.[25,36]

Visual histories can support a range of interactions. First, histories provide a convenient mechanism to revisit prior analysis states and resume incomplete explorations. Adding metadata such as comments, tags, or ratings to states can facilitate later review and sharing. Interactive histories can also capture a repeatable sequence of operations that can be named and saved as a

FIGURE 14

Visual analysis history[25]

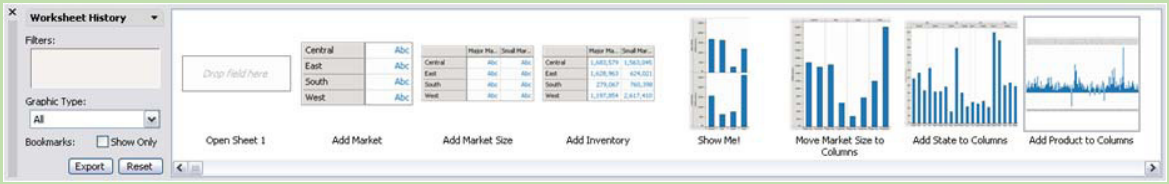
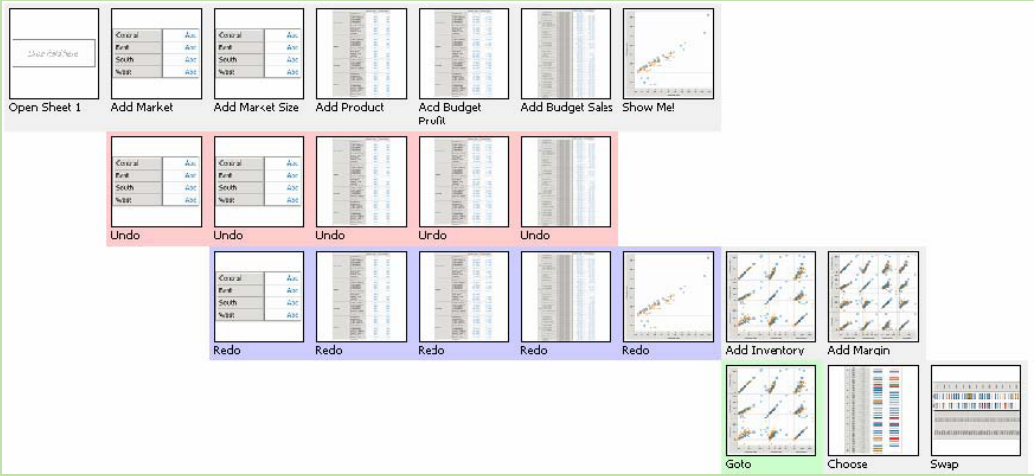


FIGURE 15

Behavior graph of branching analysis[25]



reusable macro. This powerful feature enables analysts who are dealing with many similar data sets to automate their efforts. Histories might spur sharing: analysts can export selected analysis trails, ranging from screen shots to interactive presentations, to external media. Finally, histories also provide a means to study analysts and model analytic processes.[32,45]

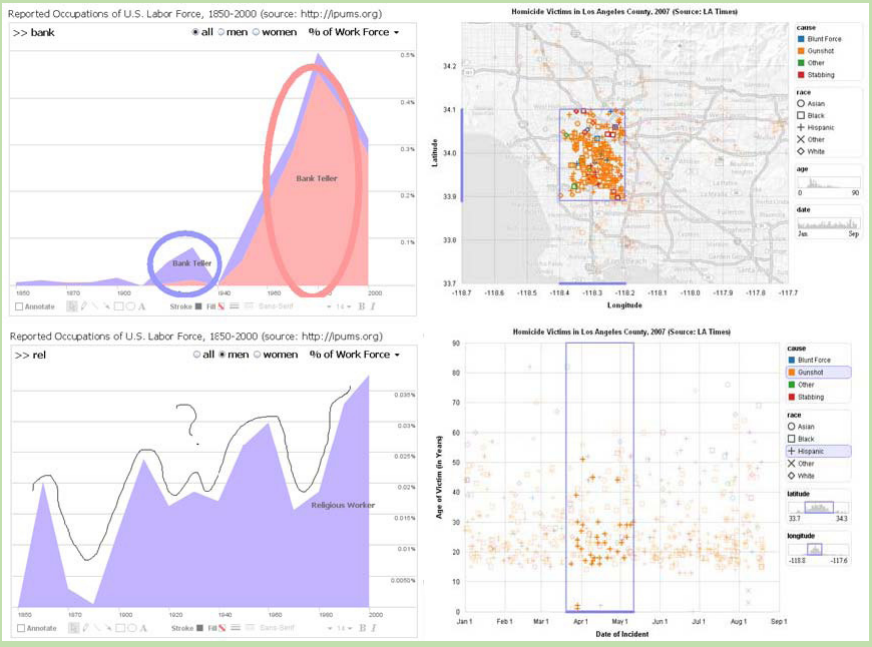
ANNOTATE

Interactive visualizations often serve not only as data exploration tools, but also as a means for recording, organizing, and communicating insights gained during exploration. One option is to allow textual annotation of states within a visual history. More expressive annotations are possible through direct interaction with the view, using the selection techniques discussed earlier. Analysts may wish to "point" to specific items or regions within a visualization and associate these annotations with explanatory text or links to other views.[26]

Freeform *graphical annotations* provide one expressive form of pointing.[26] Drawing a circle around a cluster of items or pointing an arrow at a peak in a graph can direct the attention of viewers. The angle or color of the arrow or shape of the hand-drawn circle may communicate emotional cues or add emphasis. The left side of figure 16 shows annotated occupational data: the top annotation highlights a gender reversal among bank tellers using color-coded ellipses; the bottom annotation expresses confusion regarding the erratic percentage of religious workers. Although such drawings allow a high degree of expression, they lack an explicit tie to the underlying data. Freeform annotations implemented as vector graphics can persist over geometric transformations such as panning and zooming, but if they are not "*data-aware,*" then they may become meaningless in the

FIGURE 16

Annotated visualizations



face of operations such as filtering or aggregation.

Annotations can be made data-aware when realized as selections, as seen in the right side of figure 16. In the top chart, selection queries anchor annotations of crime data. The bottom chart shows annotations transferred across a change in visual encodings: the selected geographic range is now conveyed using histogram sliders. These selections can be represented as a set of selected items, a declarative query, or both.[22] Data-aware annotations allow a pointing intention to be reapplied to different views of the same data, enabling reuse of references across different choices of visual encodings. Data-aware annotations may also enable analysts to search for all commentary or visualizations that reference a particular data item. As data-aware annotations are machine readable, they might also be used to export selected data or aggregated to identify data subsets of high interest.

SHARE

Researchers in visual analytics often focus on the perceptual and cognitive processes of a single analyst. In practice, real-world analysis is also a social process that may involve multiple interpretations, discussion, and dissemination of results.[26,54] The implication is clear: to support the analysis life cycle fully, visual analytics tools should support social interaction. At minimum, tools must be able to export views (png, jpg, ppt, etc.) or data subsets (csv, json, xls, etc.) for sharing and revisitation. An important capability is to export the settings for the control panels, so other analysts can see the same visualization. Figure 17 shows sense.us [26] one example of a collaborative visual analysis tool incorporating view sharing, annotation, and discussion. The system consists of (a) an interactive visualization, (b) a set of graphical annotation tools, (c) bookmark trails for saved views, (d) a text-entry field for adding comments (bookmarks can be dragged onto the text field to link views to a comment), (e) textual comments attached to the current view, and (f) a shareable URL that is updated automatically as the visualization state changes.

A simple but effective aid to collaboration is view sharing via *application bookmarking*: a visual analytics system should be able to model and export its internal state.[26,54] Unlike a static screen shot, bookmarking enables analysts to take up an exploration where their collaborators left off. View sharing often takes the form of an URL or similar identifier that allows a collaborator to navigate quickly to a view of interest. Seeing an identical view provides collaborators with a common ground for discussion. Annotation methods can be applied within such views to further collaboration. One challenge for effective view sharing concerns how to handle dynamic data: should a bookmarked view maintain a snapshot to historical data, provide access to the most current data, or both?

Another method of sharing and dissemination is to *publish* a visualization. Commercial tools such as Spotfire and Tableau can publish visualization dashboards as interactive Web pages. These Web-based components provide a subset of interactive functionality (e.g., selection, search, and drill-down) to enable some amount of follow-up analysis. Services such as IBM's Many Eyes[54] can be used to embed visualization applets in external Web sites. Publishing is particularly important for reaching larger audiences. While publishing is a necessary condition for broad sharing, it may not be sufficient by itself for engaging viewers.[21] Visualizations embedded within a blog or discussion forum can reach an established audience and may foster discussion more effectively than a centralized site.[13]

Other collaborative concerns depend on the context of use. Are collaborators working *synchronously* (same time) or *asynchronously* (different time)? Are they *co-located* (same place) or

distributed (different place)? Each of these configurations may require specialized strategies that consider the division of work, access control, presence indicators, and activity awareness.[21,26,31]

GUIDE

The exploration process is well understood for some traditional domains. For example, a very simple workflow might remove incomplete data items, sort, select high-value items, and report on these selections. Analysts, however, may need to develop new strategies that are formalized to guide newcomers and provide progress indicators to experts. Visual-analysis systems can incorporate *guided* analytics to lead analysts through workflows for common tasks.

Some processes are clearly linear, but many visual-analytics tasks require richer *systematic yet flexible* processes that allow analysts to take excursions while keeping track of what they have done. For example, SocialAction[41] organizes social-network analysis into a sequence of activities (e.g., rank nodes, plot nodes, find communities); the system allows analysts to skip steps selectively and keeps a record of which steps have been completed. In figure 18, the panel on the left suggests common steps to structure social network analysis and provides progress indicators.

In a related vein, experts often develop visualizations that are used by less knowledgeable team members, in much the same way that spreadsheet macros enable specialists to encode accounting or business practices for others. More research is needed to identify effective visual-analytics processes and enable expert analysts to create reusable workflows.

In recent years, journalists have been experimenting with different forms of *narrative* visualization[46] by structuring interactive graphics to tell stories with data. Visualizations from *The*

FIGURE 17

Collaborative visual analysis in Sense.us[26]

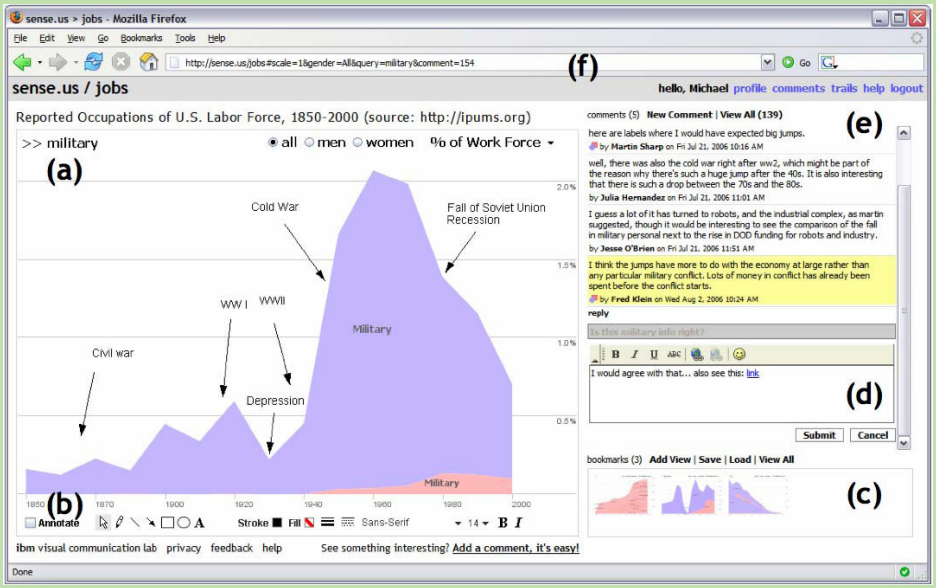
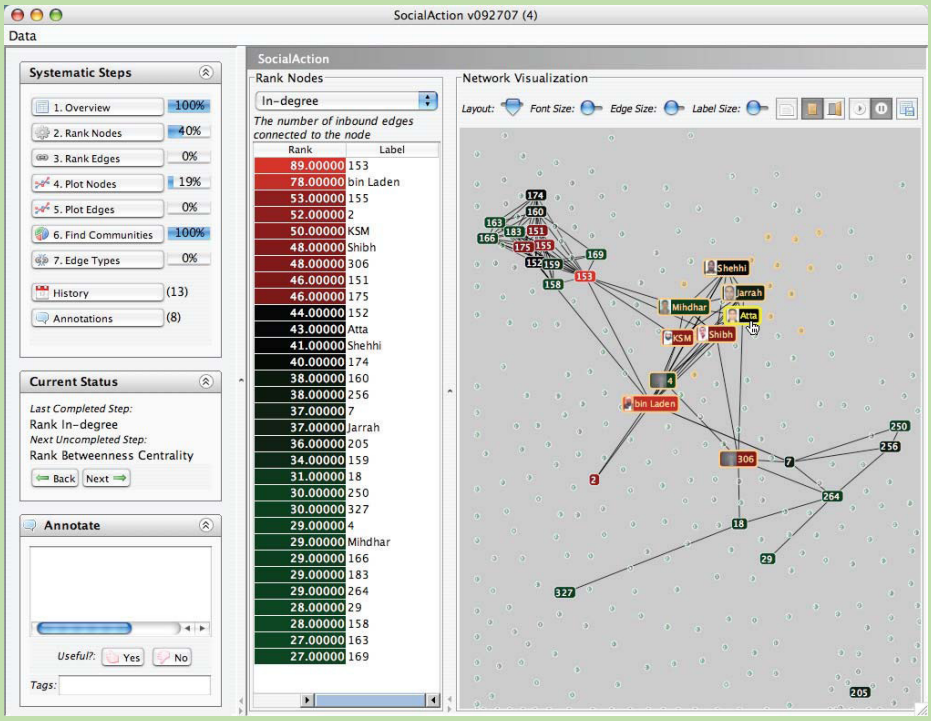


FIGURE 18

Systematic yet flexible analysis in SocialAction[41]



New York Times, The Washington Post, The Guardian, and other news sources often lead the viewer step by step through a linear narrative, guided by supporting text and annotations. In figure 19, for example, an interactive graphic uses staging and annotation to guide the reader through decades of budget predictions. At a story's conclusion, such visualizations provide interactive controls for further exploration. These narrative structures both communicate key observations from the data and cleverly provide a *tacit tutorial* of the available interactions by animating each component along with the story. By the time the presentation opens up for freeform exploration, the viewers have already seen demonstrations of the interactive controls. These and other forms of narrative visualization demonstrate how guided analytics can be used to disseminate data-driven stories to a general audience.

CONCLUSION

We hope this taxonomy and discussion will help advance visual analytics on multiple fronts. For students and newcomers to the field, the taxonomy provides an orienting, high-level introduction to the interactive concerns at the heart of successful visual analysis. We encourage interested readers to consult the systems, books, and research papers referenced in this article to develop a deeper understanding of these concerns. For developers, the taxonomy can function as a checklist of

items to consider when creating new analysis tools. For researchers, the taxonomy helps highlight critical areas that would benefit from further investigation, including new methods for interactive view specification, a closer integration of visualization and statistical algorithms, selection and annotation techniques that leverage data semantics, and effective approaches to guided analytics.

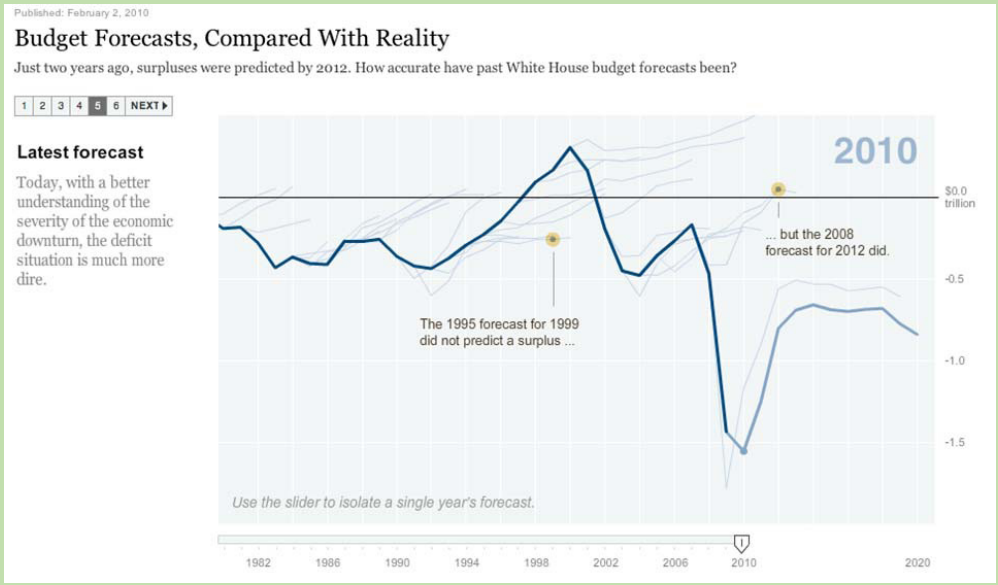
Of course, by attempting to provide an abstracted picture of a domain, taxonomies may be incomplete. In some cases, we separately categorize aspects that are closely related. Dynamic query widgets enabling data specification often serve as a means of view navigation. Selection techniques are also central to effective annotation schemes.

In other instances, we selectively omit material. First, we do not go into great depth regarding implementation details. Supporting realtime interactivity often requires careful attention to system design, especially for large data sets. While popular platforms for large data analysis such as MapReduce[14] achieve adequate throughput, their high latency and lack of online processing limit fluent interaction. The demands of truly interactive analysis pose important research challenges for the designers of analysis platforms, ranging from low-latency architectures to intelligent sampling and aggregation methods.[34]

Our taxonomy is also somewhat sparing in its discussion of the current frontier of visual analytics research. For example, how best to incorporate (semi-)automated statistical methods within a visualization environment is a central challenge. Our discussion of derived data only scratches the surface. A related concern is the task of *data wrangling*:[33] reformatting, cleaning, and integrating data sets so that they are amenable to visual analysis. Incorrect or improperly structured data diverts the attention and energy of trained analysts and presents a significant barrier to newcomers. As data cleaning requires nuanced human judgment based on domain knowledge ("is this outlier an error

FIGURE 19

Data storytelling by The New York Times



or a discovery?"), data wrangling is a necessarily interactive process combining statistical methods, visualization, and interaction techniques. This topic deserves a deeper treatment than we can provide within our compact taxonomy.

These concerns represent active areas of research, and we expect our characterization of the field to evolve in the years to come. Validating and evolving this framework is a community project that can proceed through feedback, critique, and refinement by visual analytics researchers and practitioners. We invite the insights and commentary of the visualization, statistics, database, and HCI (human-computer interaction) communities, and eagerly anticipate the continued flowering of improved tools for making sense of the wealth of data that surrounds us.

ACKNOWLEDGMENTS

We thank our colleagues and students for providing valuable comments on drafts: Maneesh Agrawala, Jason Chuang, Cody Dunne, John Guerra-Gomez, Pat Hanrahan, Sean Kandel, Diana MacLean, and Kostas Pantazos. This work was partially supported by National Science Foundation grants IIS-0968521, IIS-1017745, CCF-0964173 and SBE-0915645, NIH-National Cancer Institute grant RC1-CA147489, and ONC-SHARP grant on Cognitive Information Design and Visualization.

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LOVE IT, HATE IT? LET US KNOW

feedback@queue.acm.org

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University of Maryland, College Park. He was elected as a Fellow of the ACM in 1997 and a Fellow of the American Association for the Advancement of Science in 2001. He received the ACM SIGCHI Lifetime Achievement Award in 2001. He is a member of the National Academy of Engineering.

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VIEWS

DESIGN CHOICES FOR MULTIPLE VIEWS

Juxtapose and Coordinate Multiple Side-by-Side Views

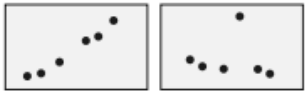
- Share Encoding: Same/Different
  - *Linked Highlighting*

- Share Data: All/Subset/None

- Share Navigation

		Data		
		All	Subset	None
Encoding	Same	Redundant	Overview/ Detail	Small Multiples
	Different	Multiform	Multiform, Overview/ Detail	No Linkage

Partition into Side-by-Side Views



Superimpose Layers

WHY MULTIPLE VIEWS?

Eyes beat memory

- Showing two views side by side are easier to compare than changing views over time

No single visual encoding is optimal for all possible tasks

- Use different encoding for one data

Too many to shown in one view

LINKED VIEWS

Multiple views are simultaneously visible and linked together

Actions in one view affect the others

LINKED VIEWS OPTIONS

Four options

- Highlighting: to link, or not
- Navigation: to share, or not
- Encoding: same or multiform
- Dataset: Shared all, subset, or none

→ Share Encoding: Same/Different

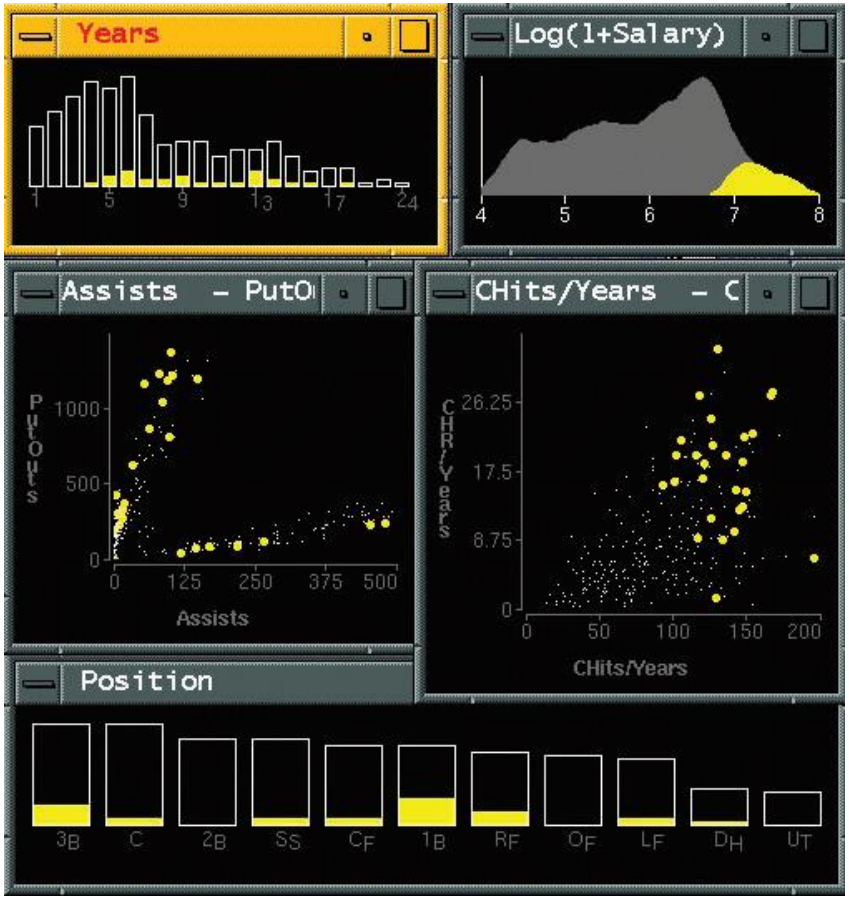
→ *Linked Highlighting*

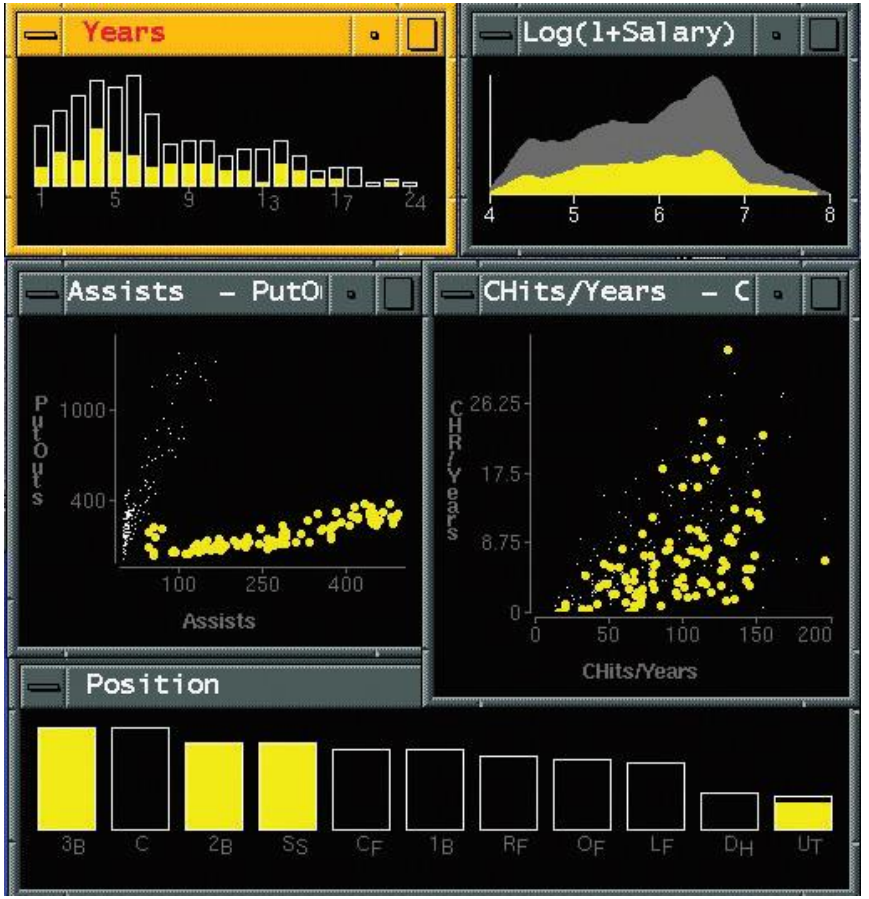
→ Share Data: All/Subset/None

→ Share Navigation

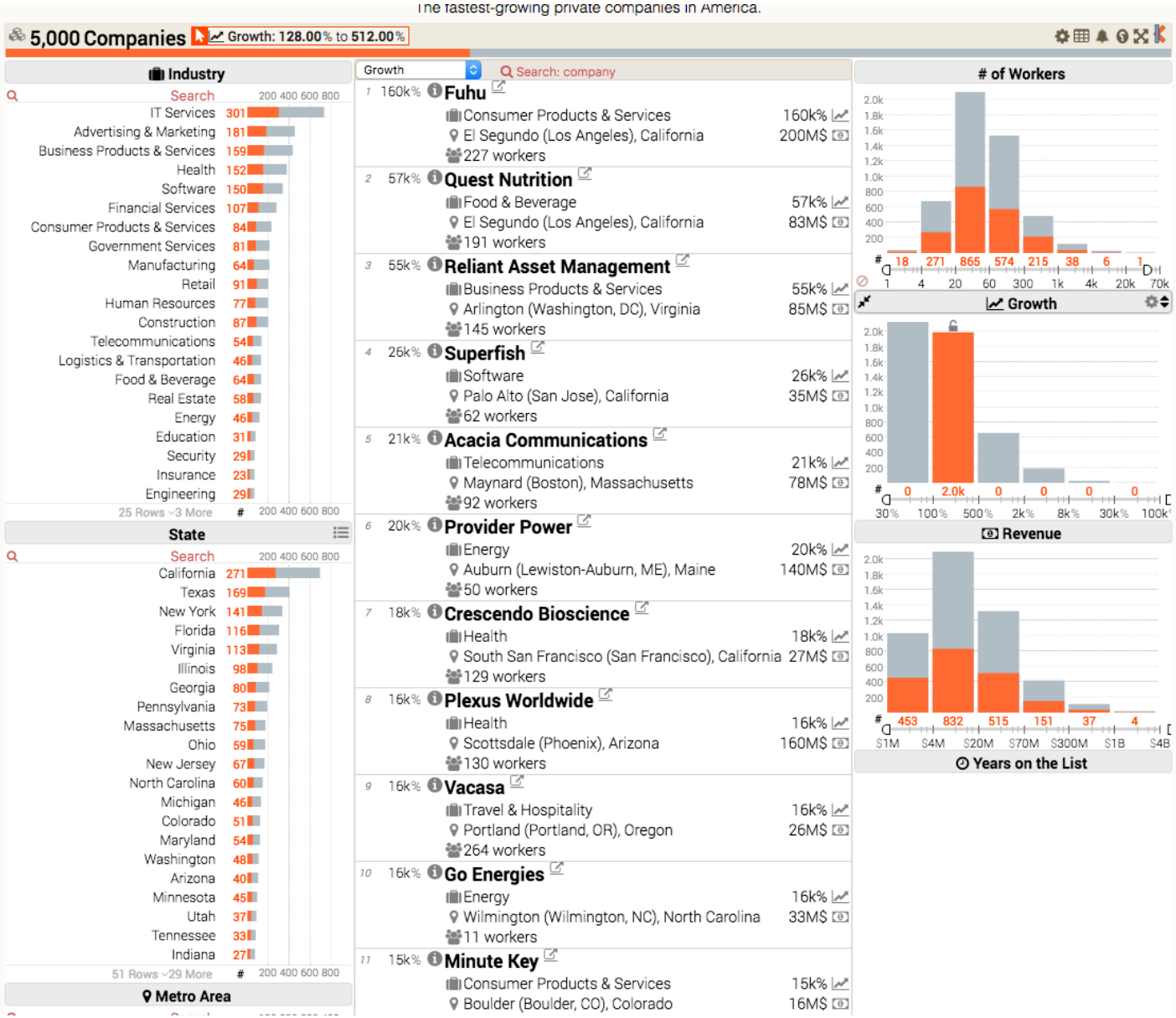
		Data		
		All	Subset	None
Encoding	Same	Redundant	Overview/ Detail	Small Multiples
	Different	Multiform	Multiform, Overview/ Detail	No Linkage

LINKED HIGHLIGHTING

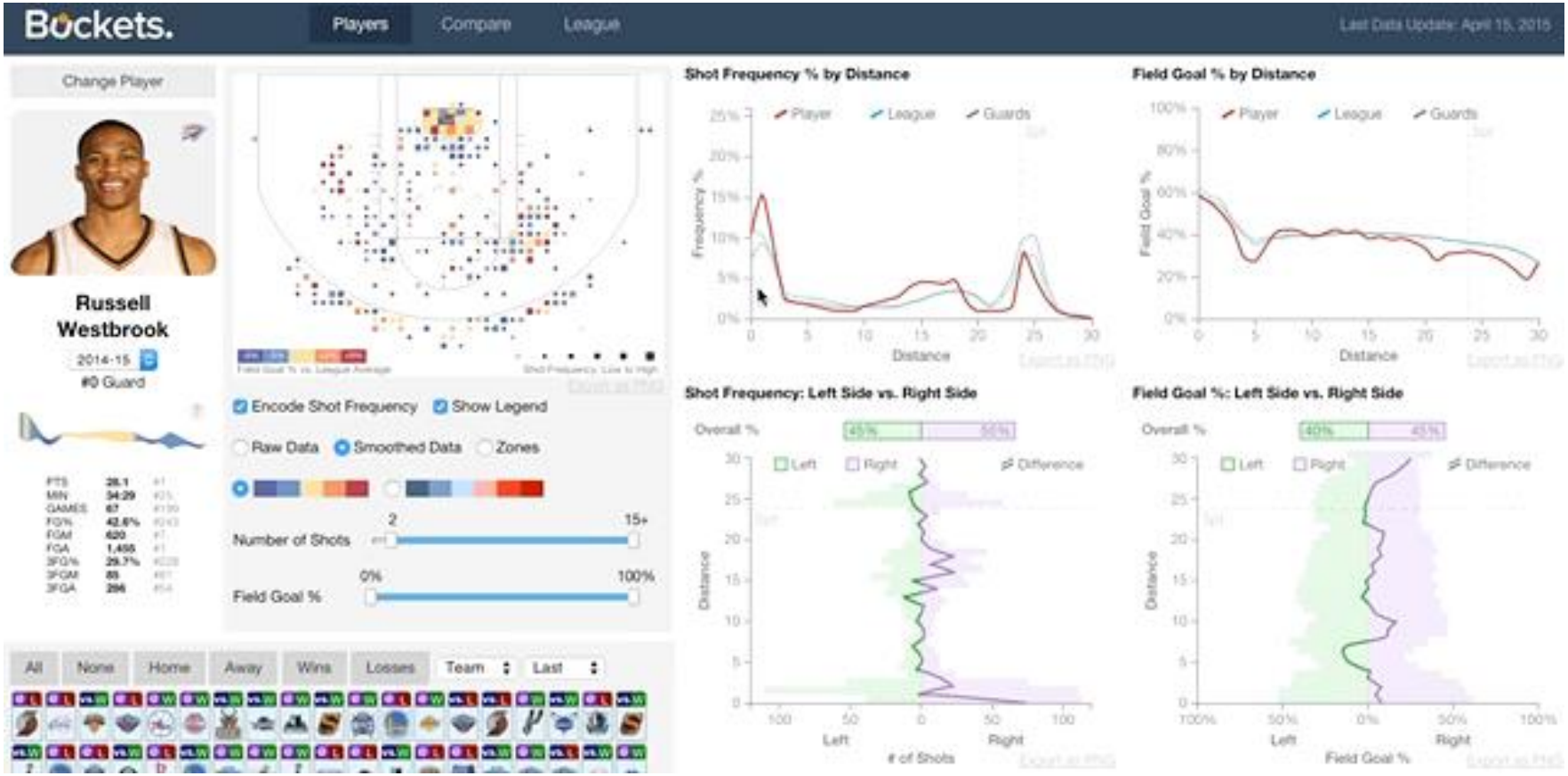




LINKED HIGHLIGHTING



MULTIDIRECTIONAL LINKING



https://buckets.peterbeshai.com/app/#/playerView/201935_2015

MULTIFORM

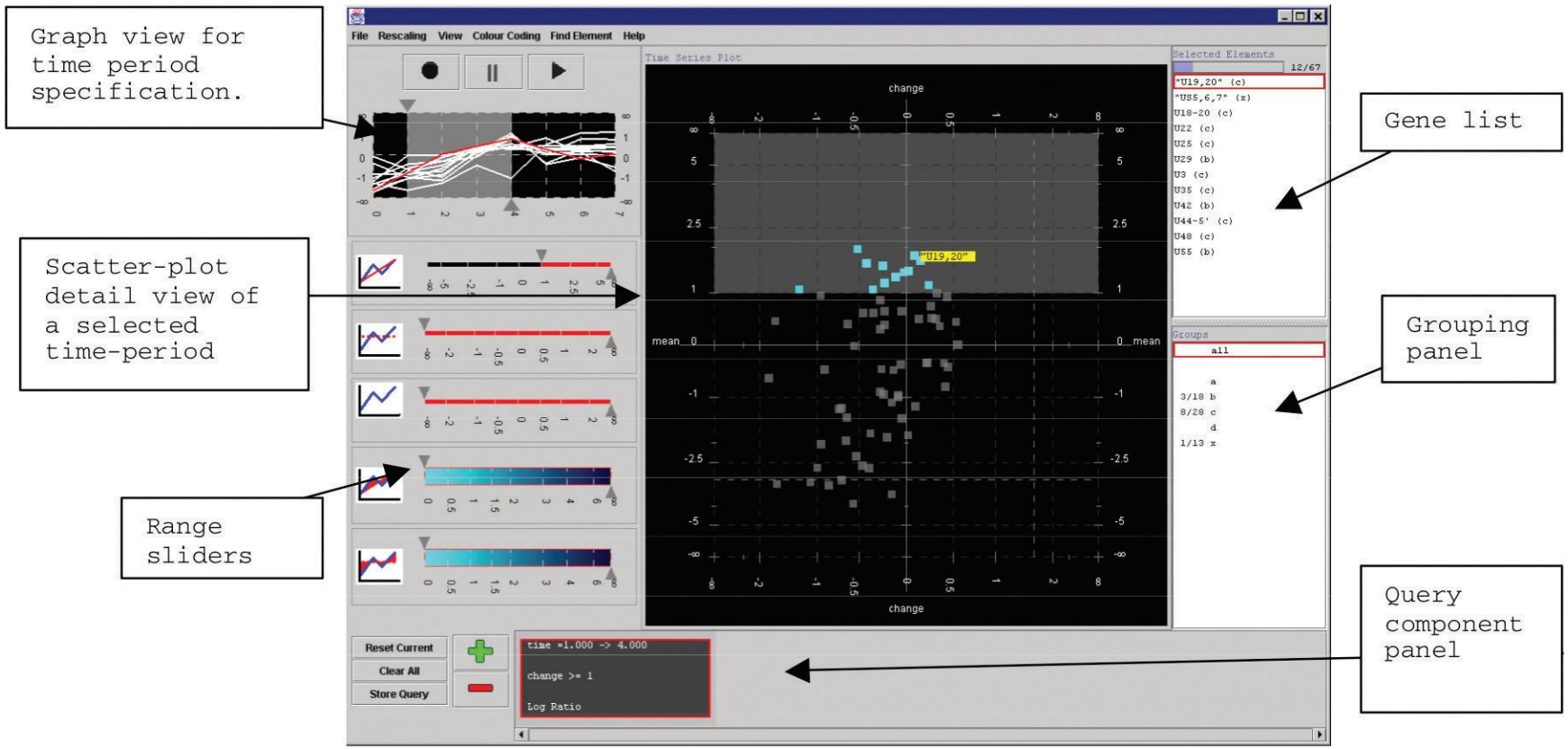
Different visual encodings are used between views

- shared data: either all data, or subset of data (overview + detail)

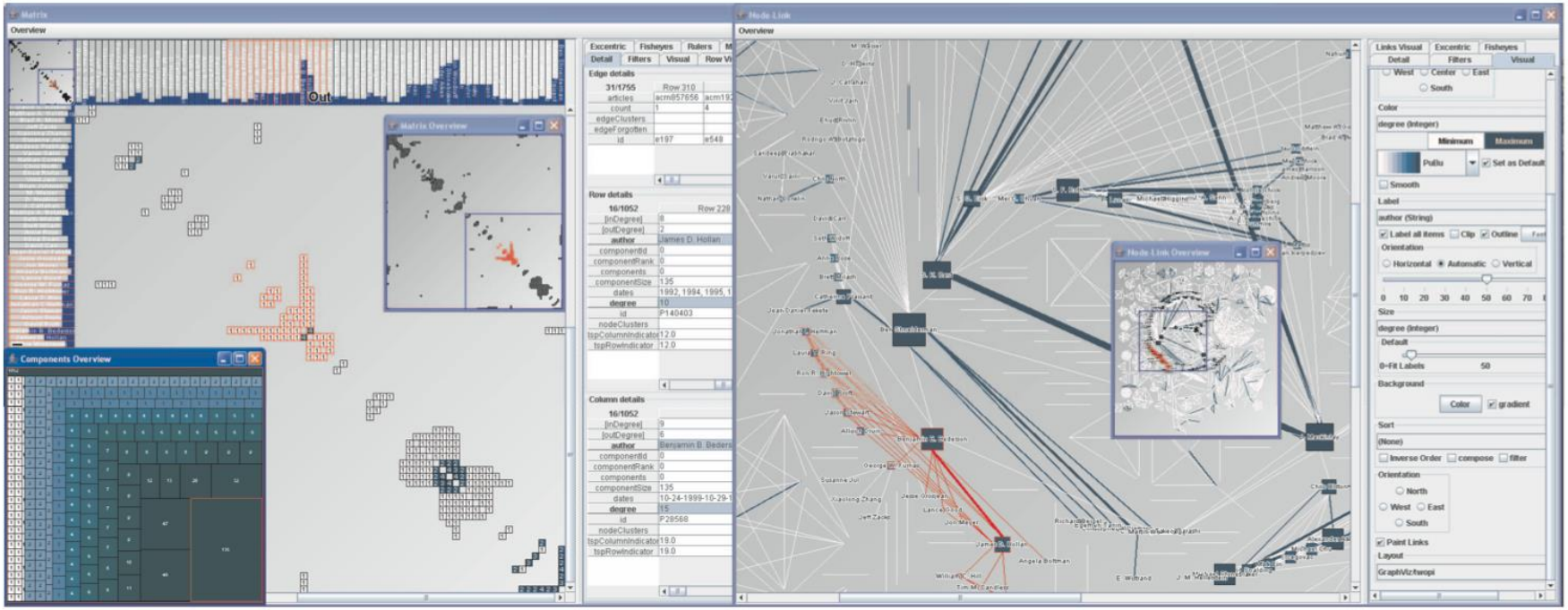
Rational:

- Single view has strong limits on the numbers of attributes that can be shown simultaneously
- Different views support different tasks

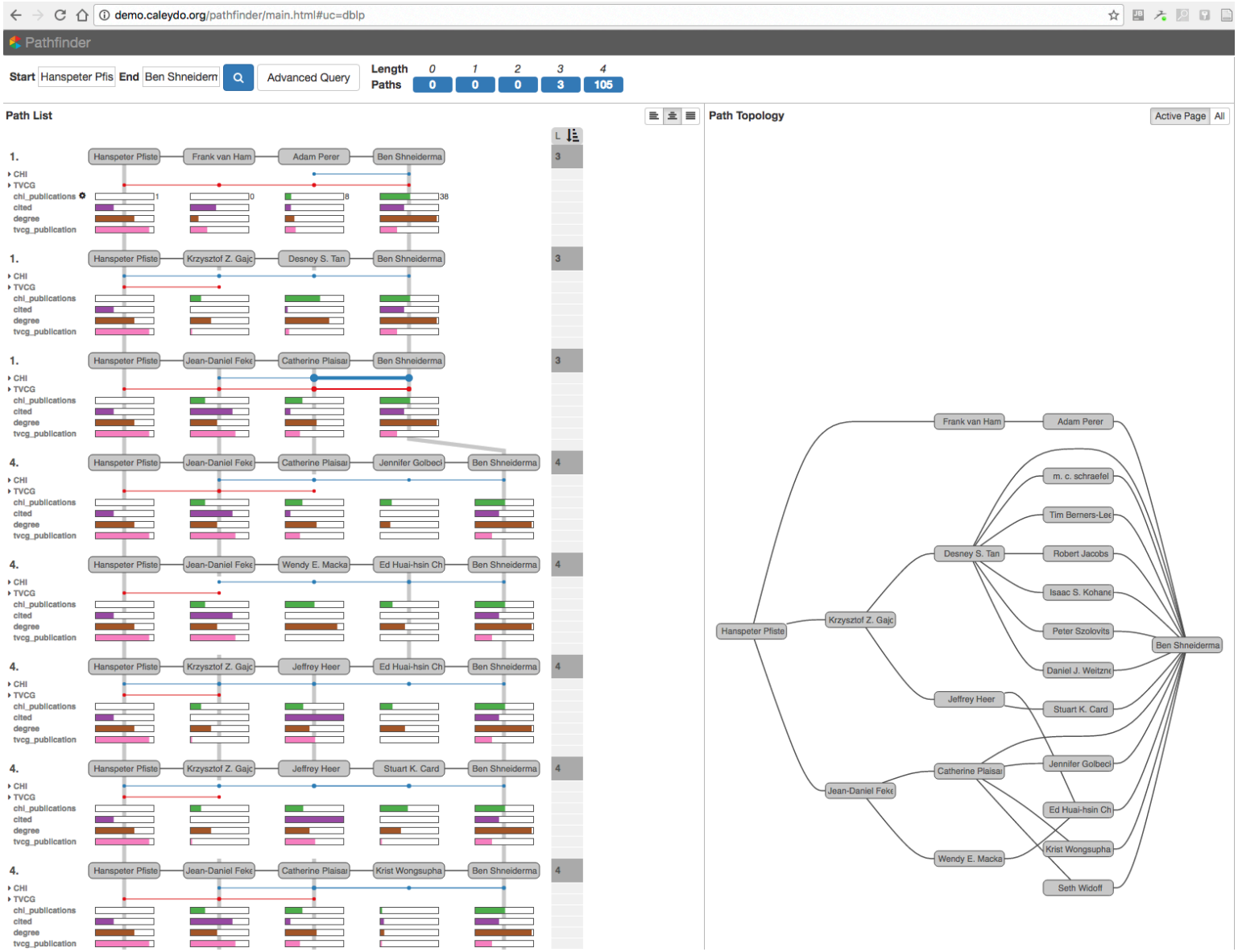
MULTIFORM OVERVIEW-DETAIL



MATRIX EXPLORER



MULTIFORM


demo.caleydo.org/pathfinder/main.html#uc=dblp
Pathfinder
Start Hanspeter Pfis End Ben Shneiderm
Advanced Query
Length 0 1 2 3 4
Paths 0 0 0 3 105
Path List
Path Topology
Active Page
All
Hanspeter Pfiste
Frank van Ham
Adam Perer
Ben Shneiderma
Krzysztof Z. Gajc
Desney S. Tan
Jean-Daniel Feke
Catherine Plaisa
Jennifer Golbeck
Wendy E. Macka
Ed Huai-hsin Ch
Jeffrey Heer
Stuart K. Card
Krist Wongsupha
m. c. schraefel
Tim Berners-Lee
Robert Jacobs
Isaac S. Kohane
Peter Szolovits
Daniel J. Weitzne
Seth Widoff
CHI
TVCG
chi_publications
cited
degree
tvcg_publication

OVERVIEW + DETAIL

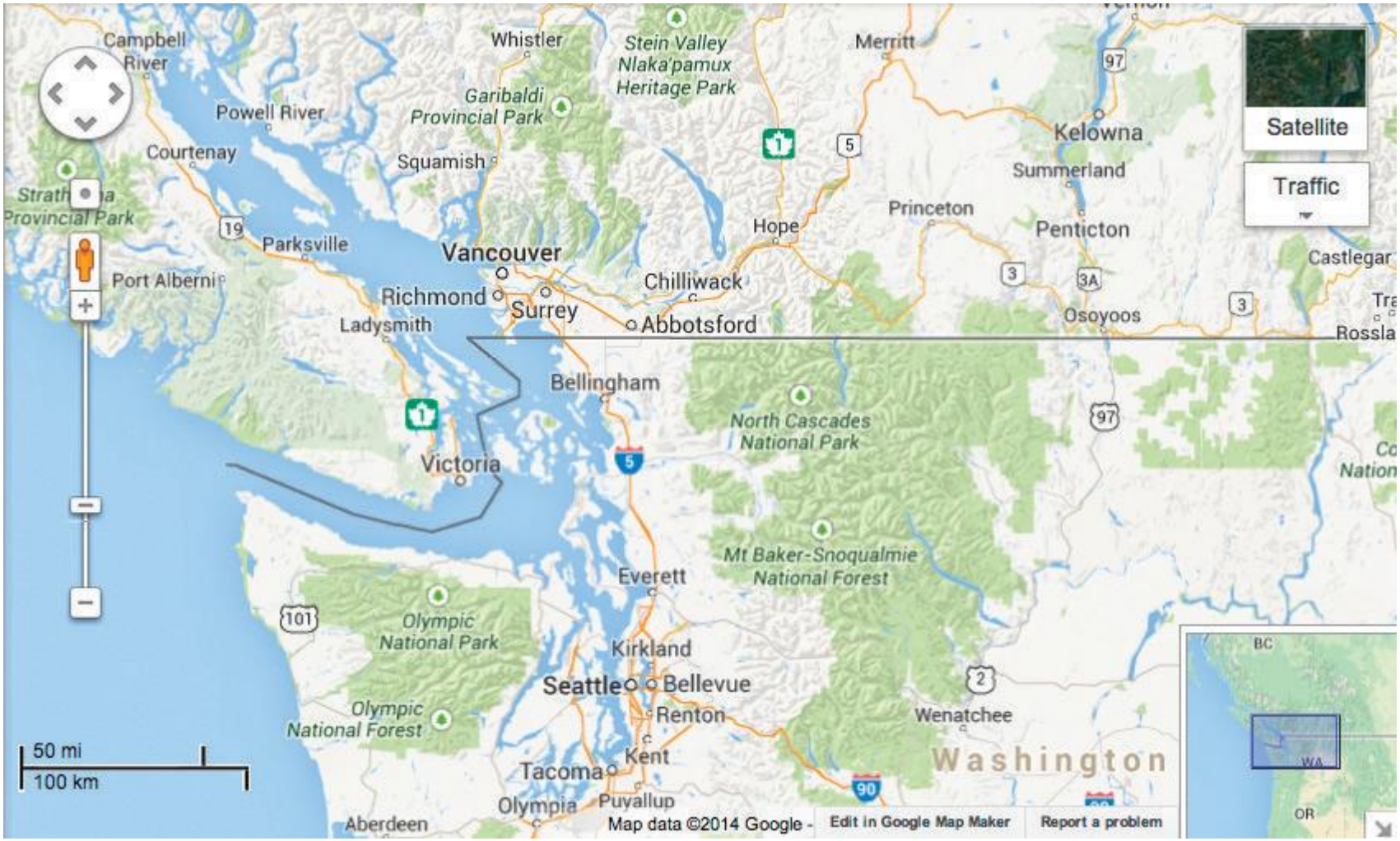
One view shows (often summarized) information about entire dataset

Additional view(s) show more detailed information about a subset of the data

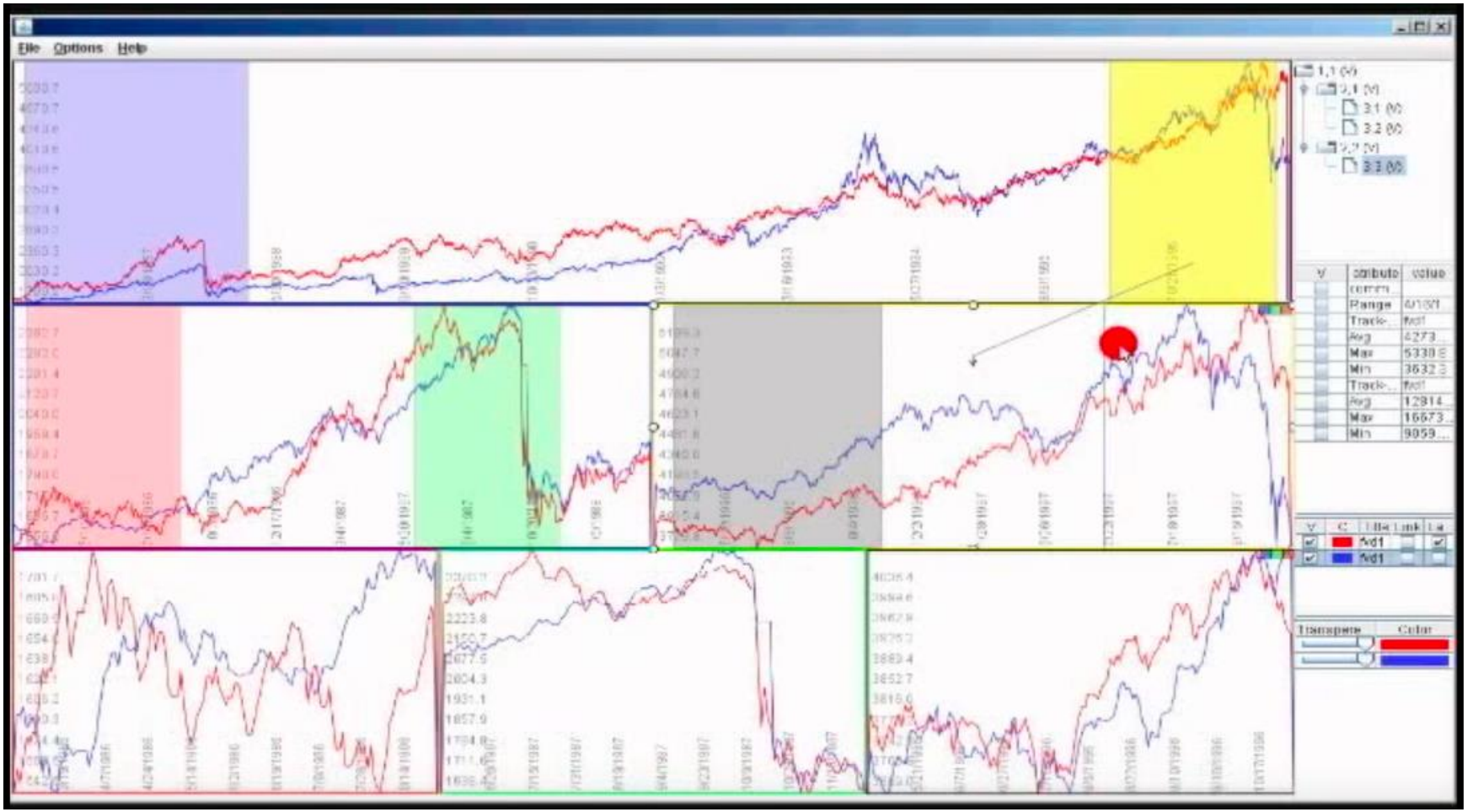
Rational:

For large or complex data, a single view of the entire dataset cannot capture fine details

MULTIFORM: OVERVIEW-DETAIL



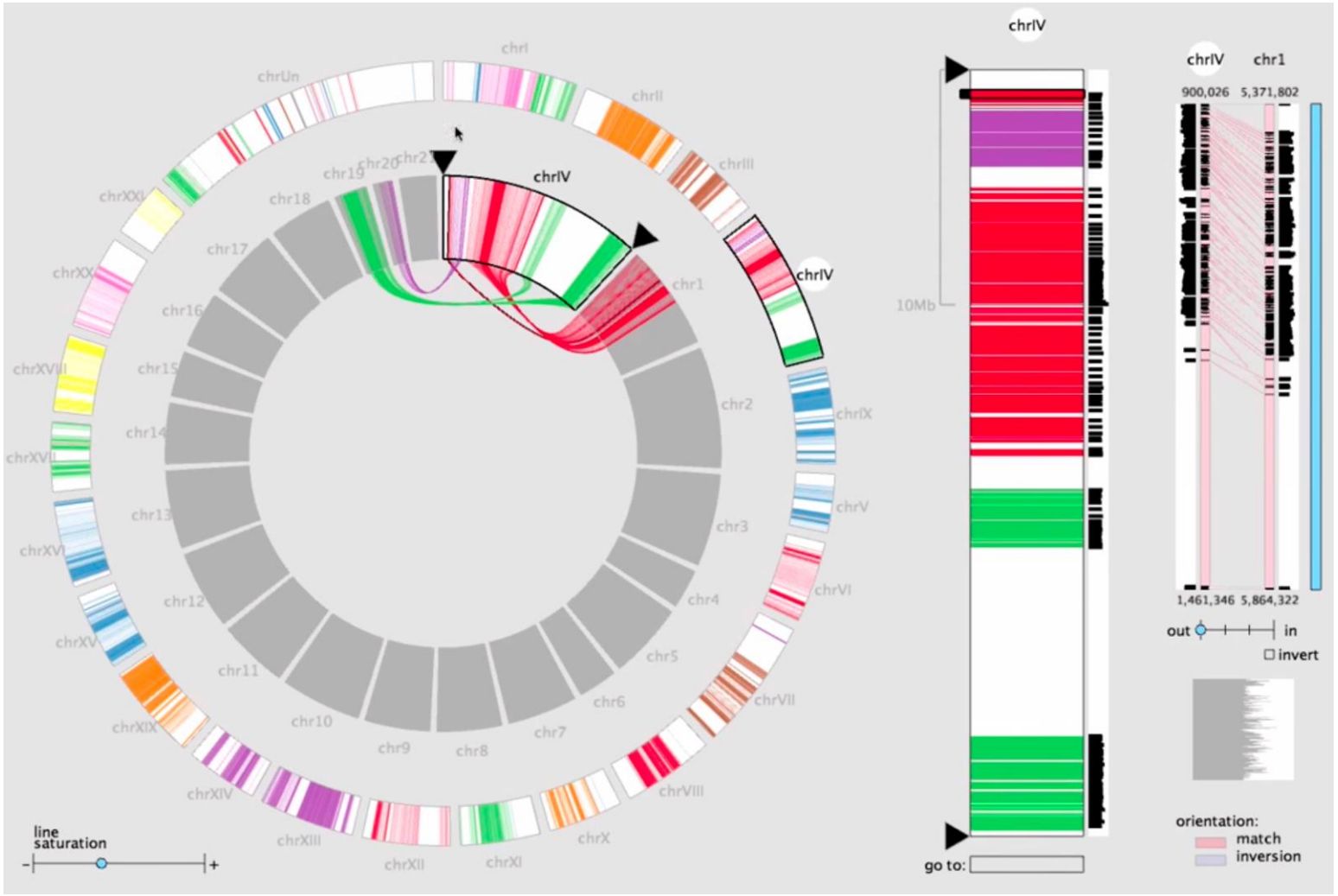
STACK ZOOMING



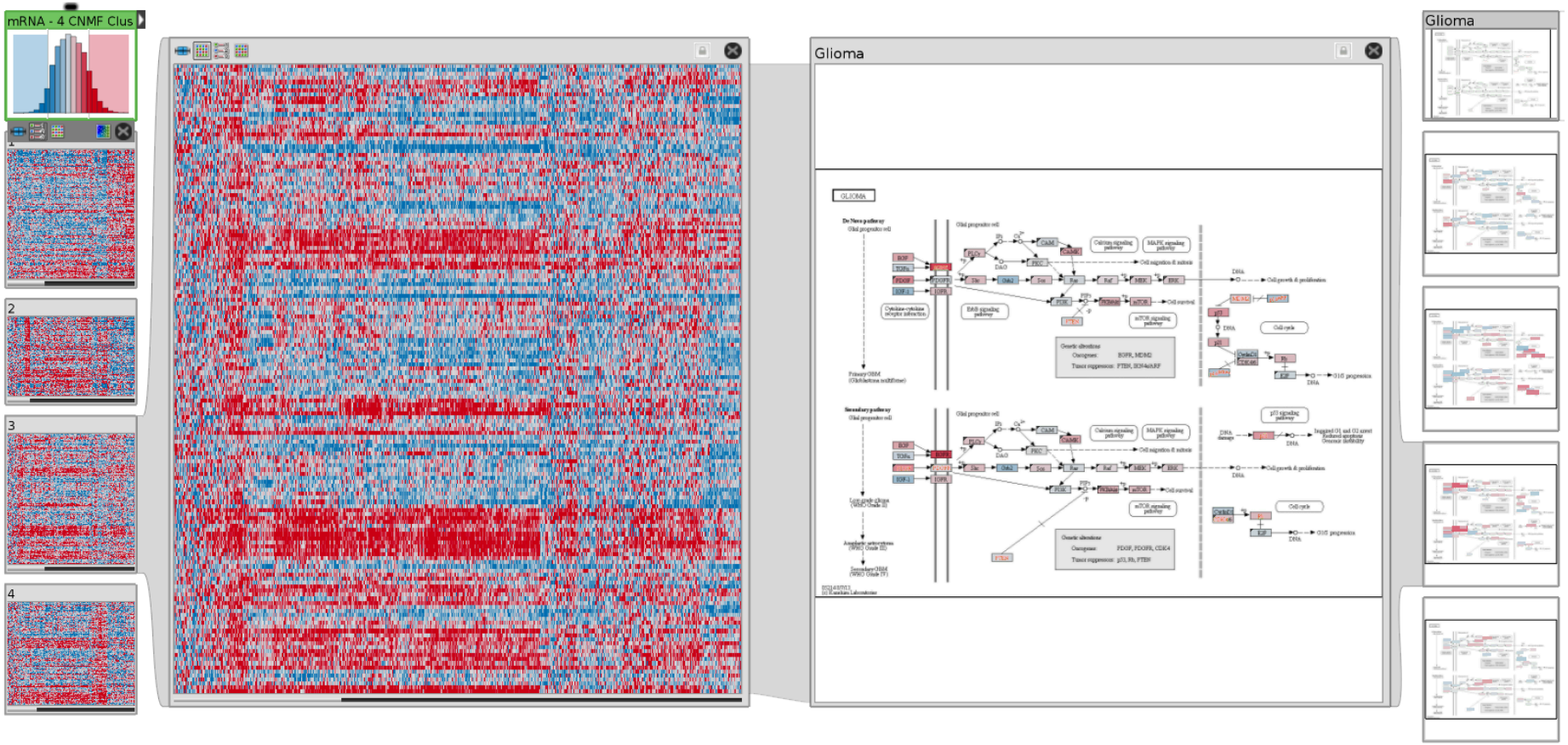
Same Data - Same Encoding, Different Resolution

[Javed & Emlqvist, PacificVis, 2010]

MIZBEE


chrI
chrII
chrIII
chrIV
chrV
chrVI
chrVII
chrVIII
chrIX
chrX
chrXI
chrXII
chrXIII
chrXIV
chrXV
chrXVI
chrXVII
chrXVIII
chrXIX
chrXX
chrXXI
chrUn
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
chr13
chr14
chr15
chr16
chr17
chr18
chr19
chr20
chr21
line saturation
-
+
chrIV
10Mb
go to:
chrIV
chr1
900,026
5,371,802
1,461,346
5,864,322
out
in
invert
orientation:
match
inversion

STRATOMEX



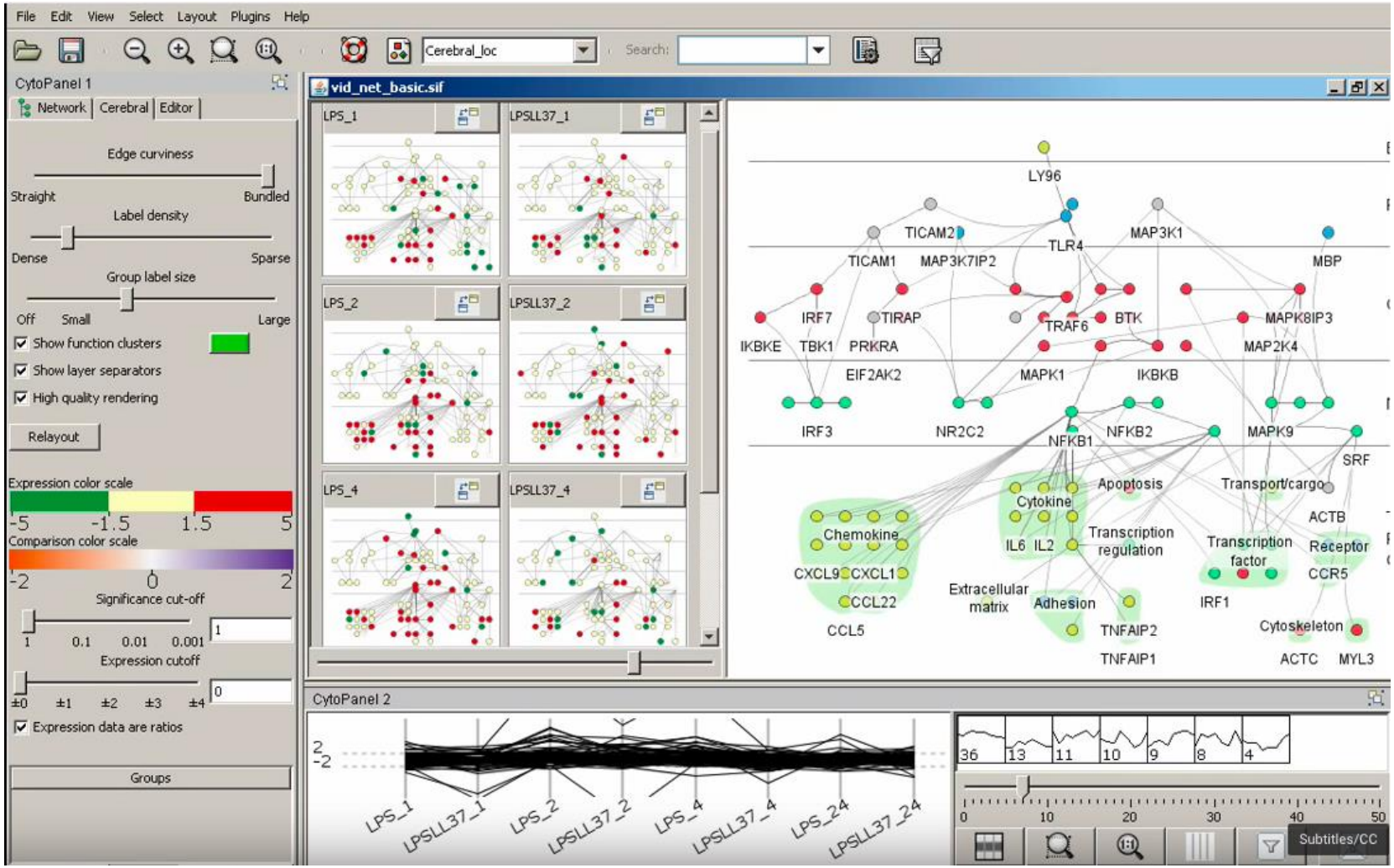
SMALL MULTIPLES

Each view uses the same visual encoding, but shows a different subset of the data

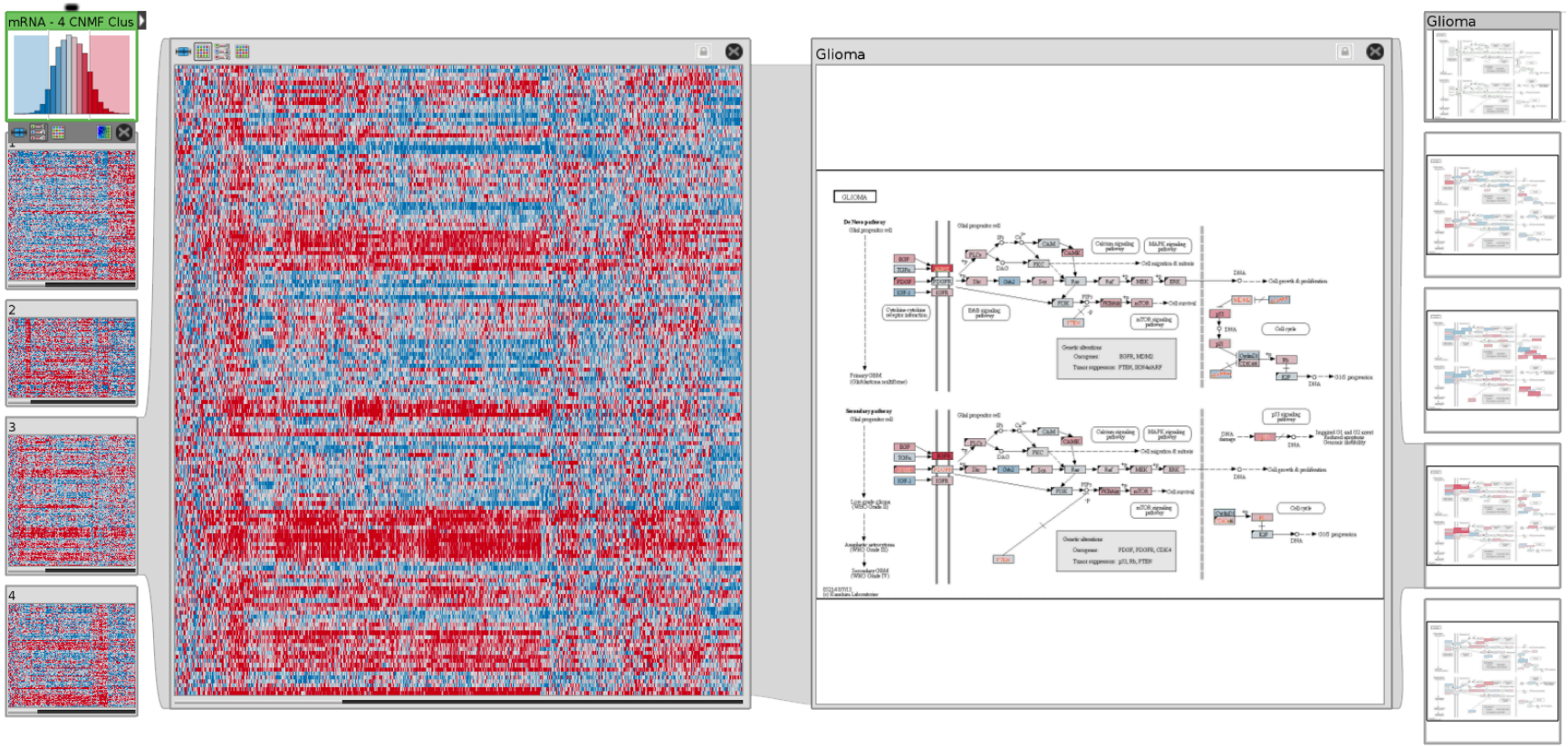
Rational:

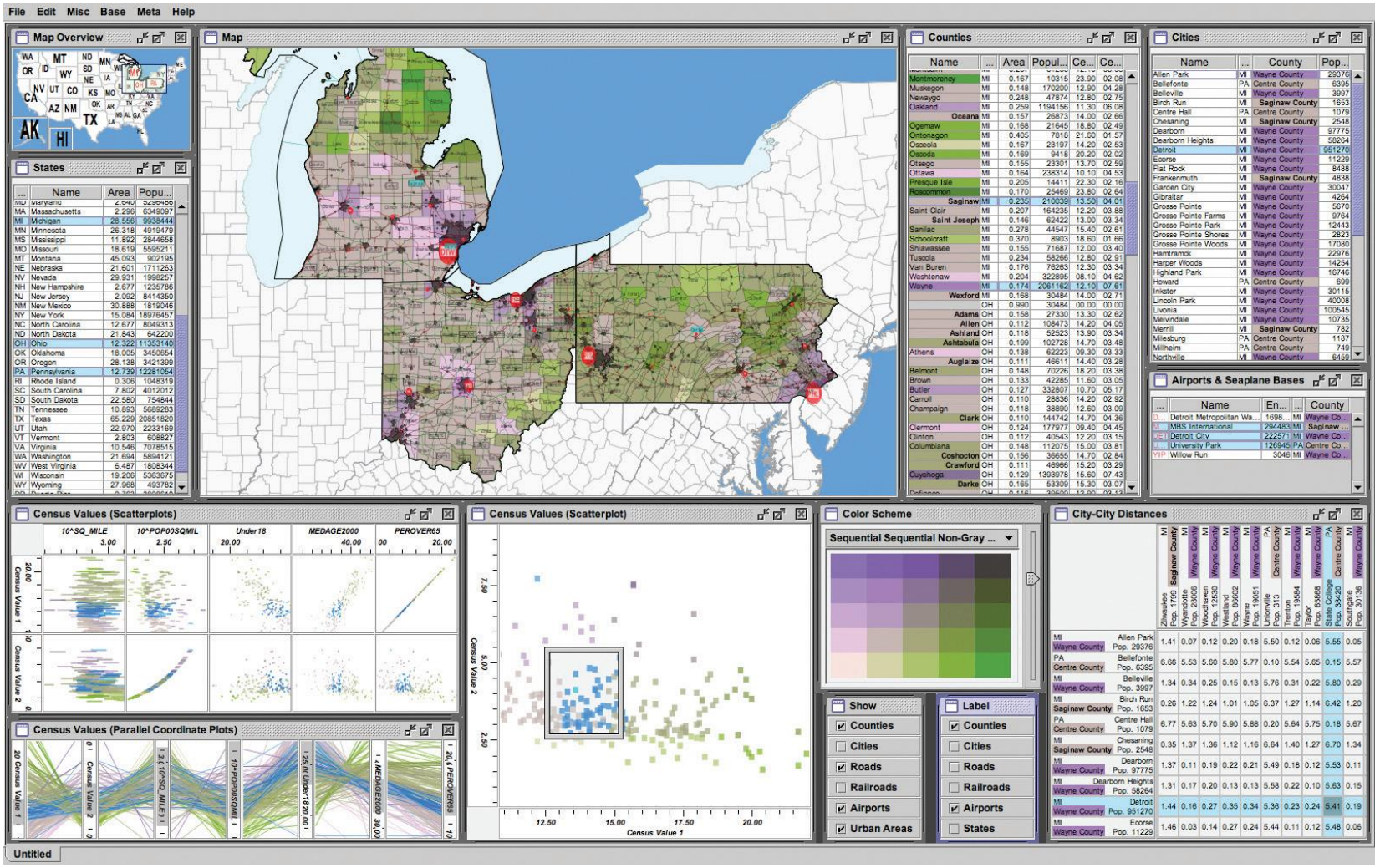
Quickly compare different parts of a dataset, relying on eyes instead of memory

SMALL MULTIPLES FOR GRAPH ATTRIBUTES

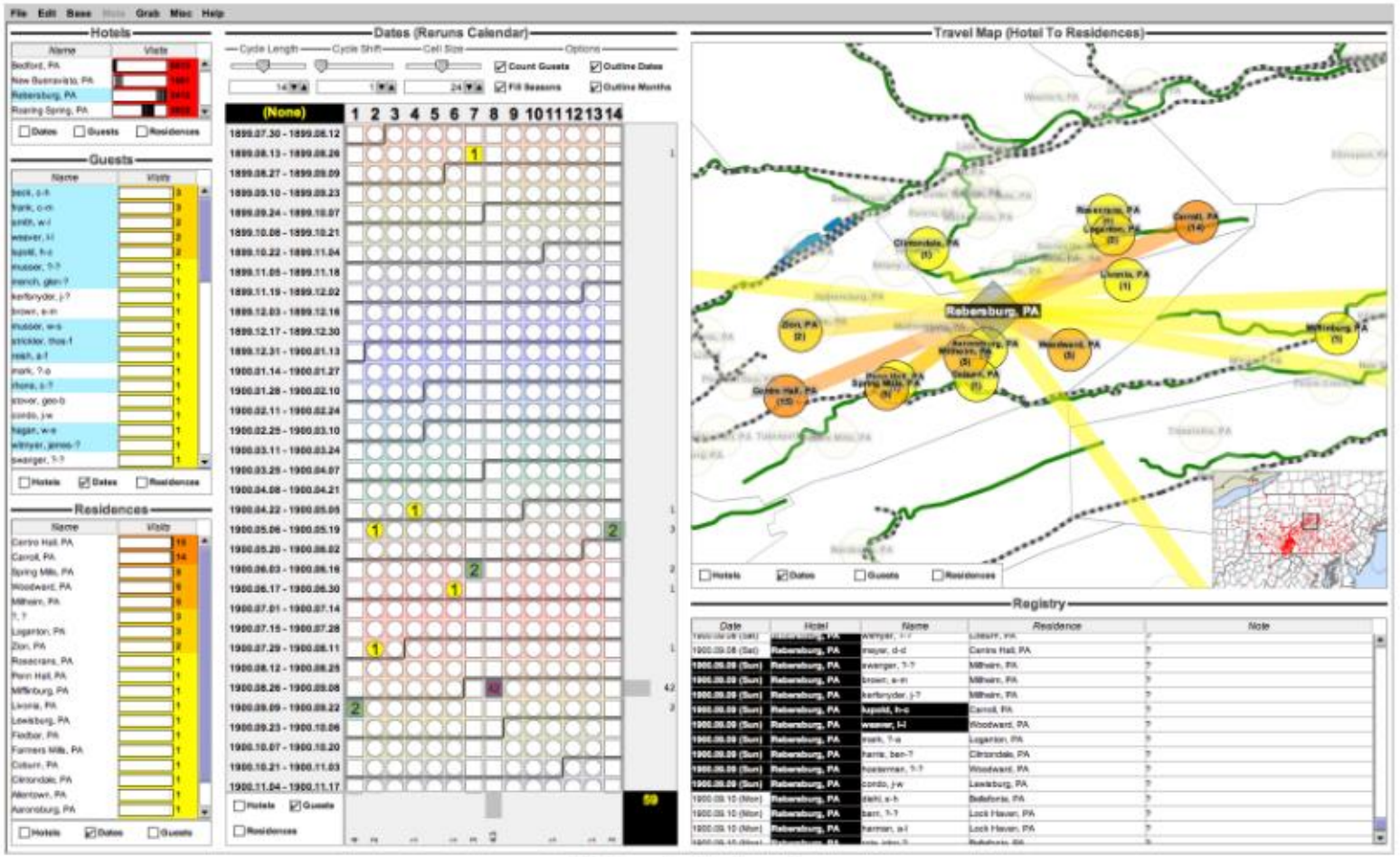


STRATOMEX




File Edit Misc Base Meta Help
Map Overview
Map
Counties
Cities
States
Airports & Seaplane Bases
Census Values (Scatterplots)
Census Values (Scatterplot)
Color Scheme
Sequential Sequential Non-Gray ...
City-City Distances
Census Values (Parallel Coordinate Plots)
Show
Counties
Cities
Roads
Railroads
Airports
Urban Areas
Label
Counties
Cities
Roads
Railroads
Airports
States
Census Value 1
Census Value 2
Untitled

VISUAL ANALYSIS OF HISTORICAL HOTEL VISITATION PATTERNS



Click image for full-size version.

https://www.youtube.com/watch?v=Tzsv6wkZoiQ

PARTITIONING

PARTITIONING

Action on the dataset that separates the data into groups

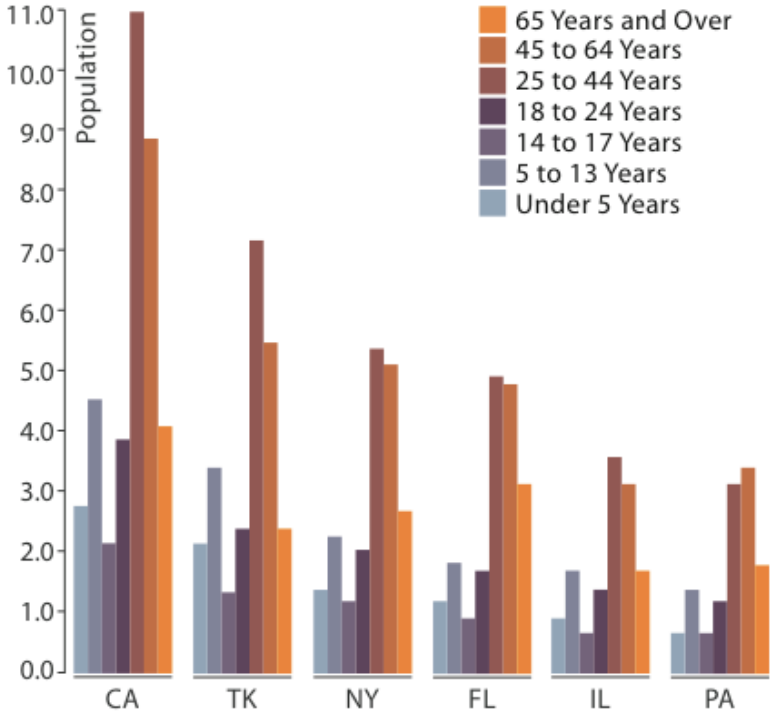
Design choices

- How to divide data up between views, given a hierarchy of attributes
- How many splits, and order of splits
- How many views (usually data driven)

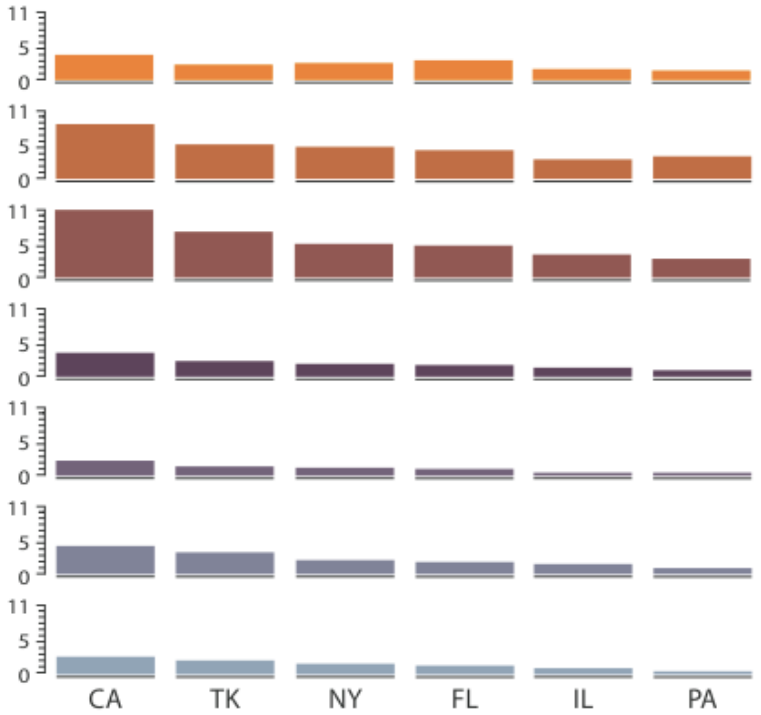
Partition attribute(s)

Typically categorical

AGE DISTRIBUTION BY STATE



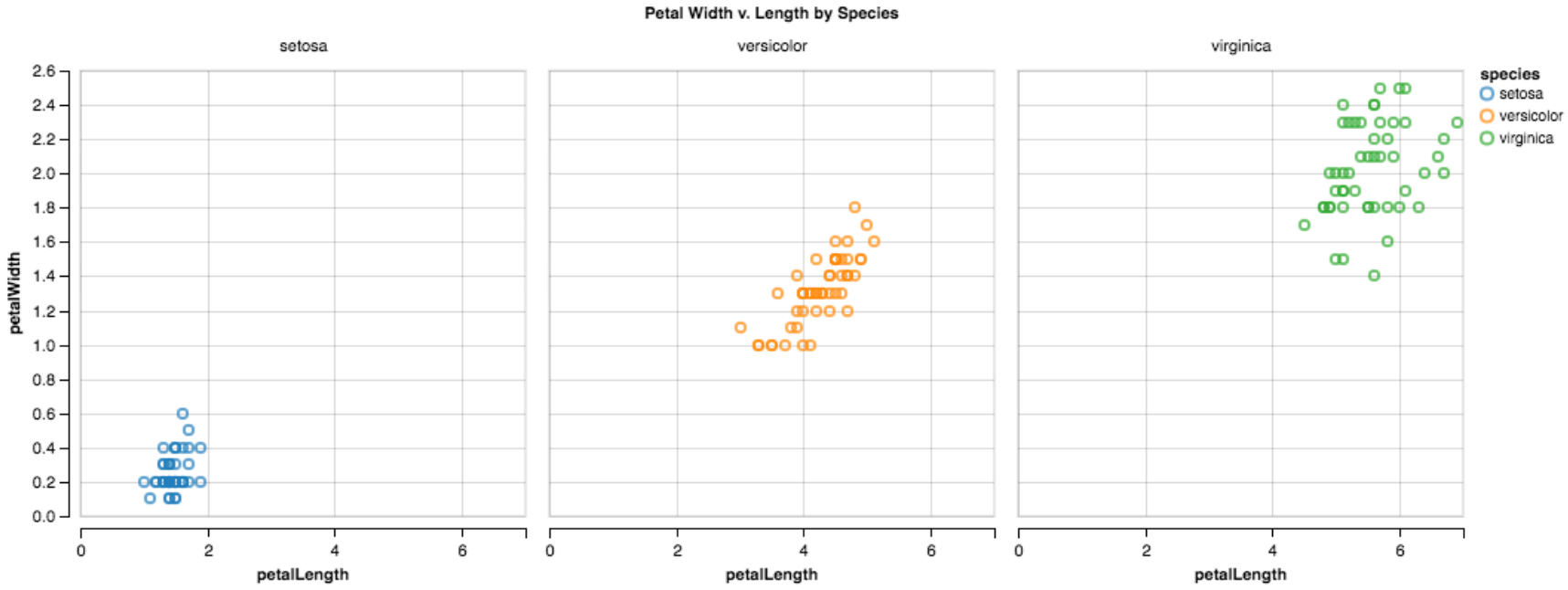
Partitioned by State



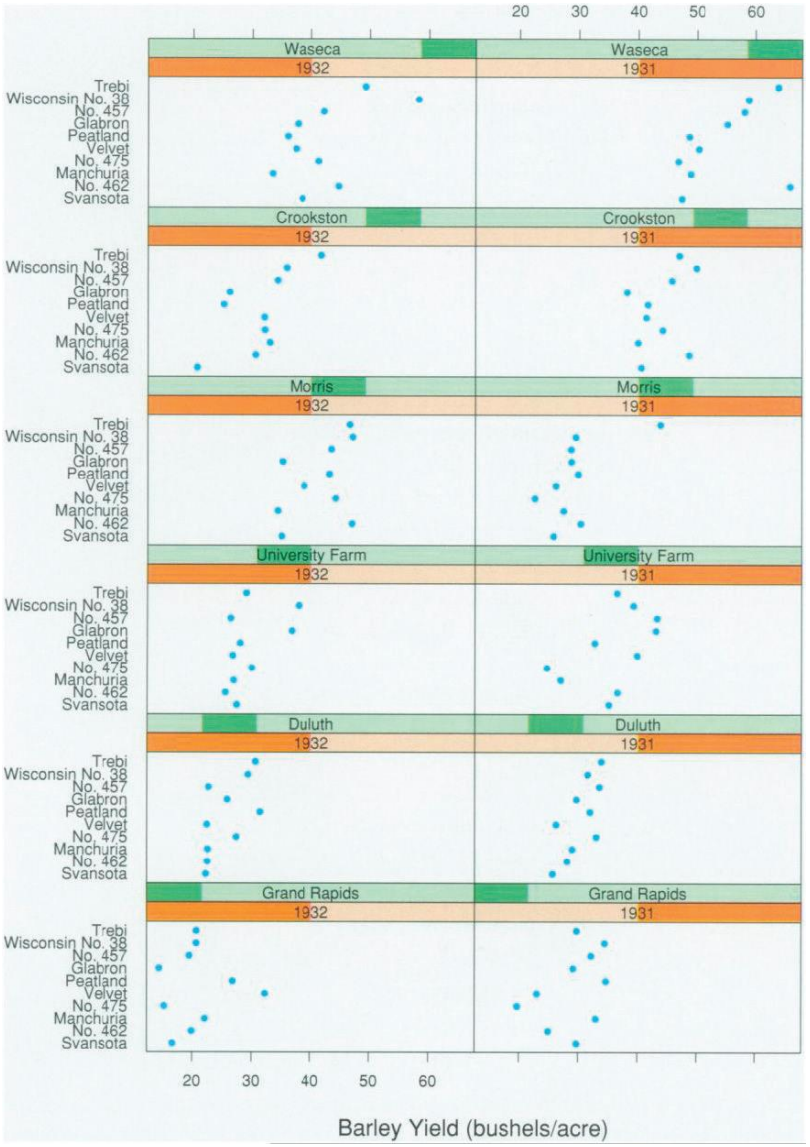
Partitioned by Age Group and State

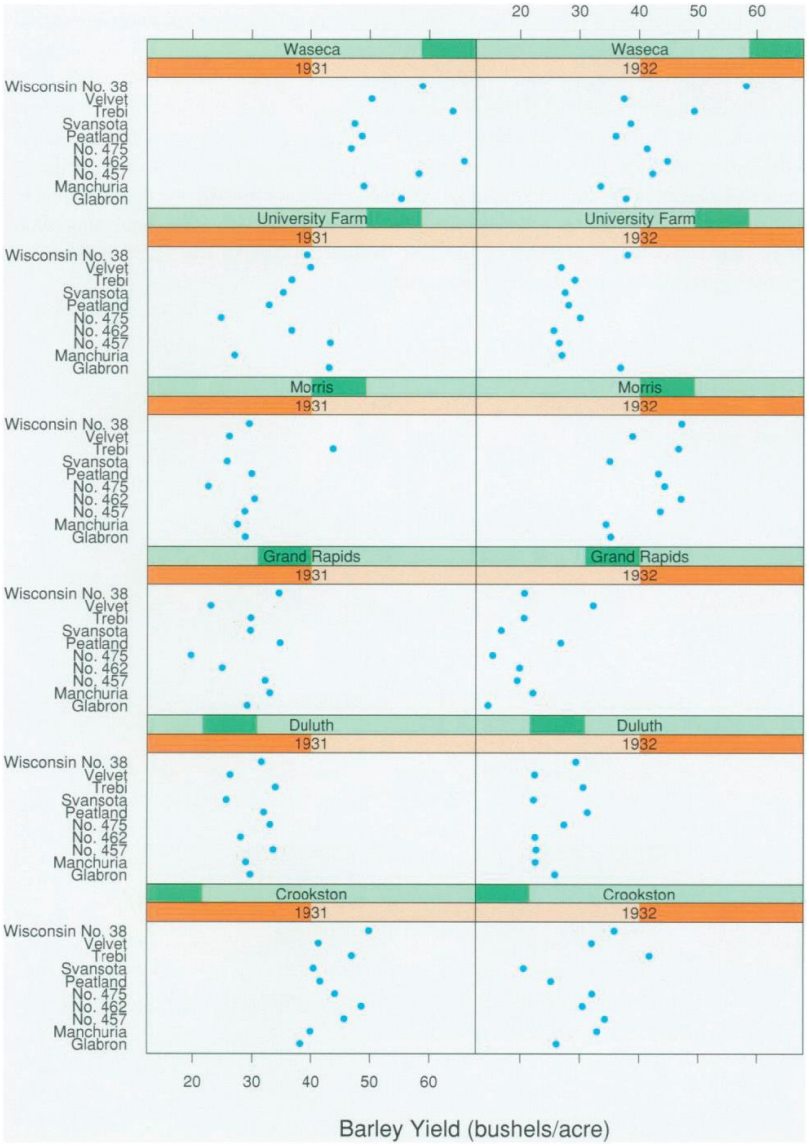
https://bl.ocks.org/mbostock/4679202

PARTITION BY CATEGORY



TRELLIS


Waseca 1932
Waseca 1931
Crookston 1932
Crookston 1931
Morris 1932
Morris 1931
University Farm 1932
University Farm 1931
Duluth 1932
Duluth 1931
Grand Rapids 1932
Grand Rapids 1931
Trebi
Wisconsin No. 38
No. 457
Glabron
Peatland
Velvet
No. 475
Manchuria
No. 462
Svansota
20 30 40 50 60
Barley Yield (bushels/acre)


Waseca 1931
Waseca 1932
University Farm 1931
University Farm 1932
Morris 1931
Morris 1932
Grand Rapids 1931
Grand Rapids 1932
Duluth 1931
Duluth 1932
Crookston 1931
Crookston 1932
Wisconsin No. 38
Velvet
Trebi
Svansota
Peatland
No. 475
No. 462
No. 457
Manchuria
Glabron
20 30 40 50 60
Barley Yield (bushels/acre)

TRELLIS PLOTS

panel variables

attributes encoded in individual views

partitioning variables

partitioning attributes assigned to columns and rows

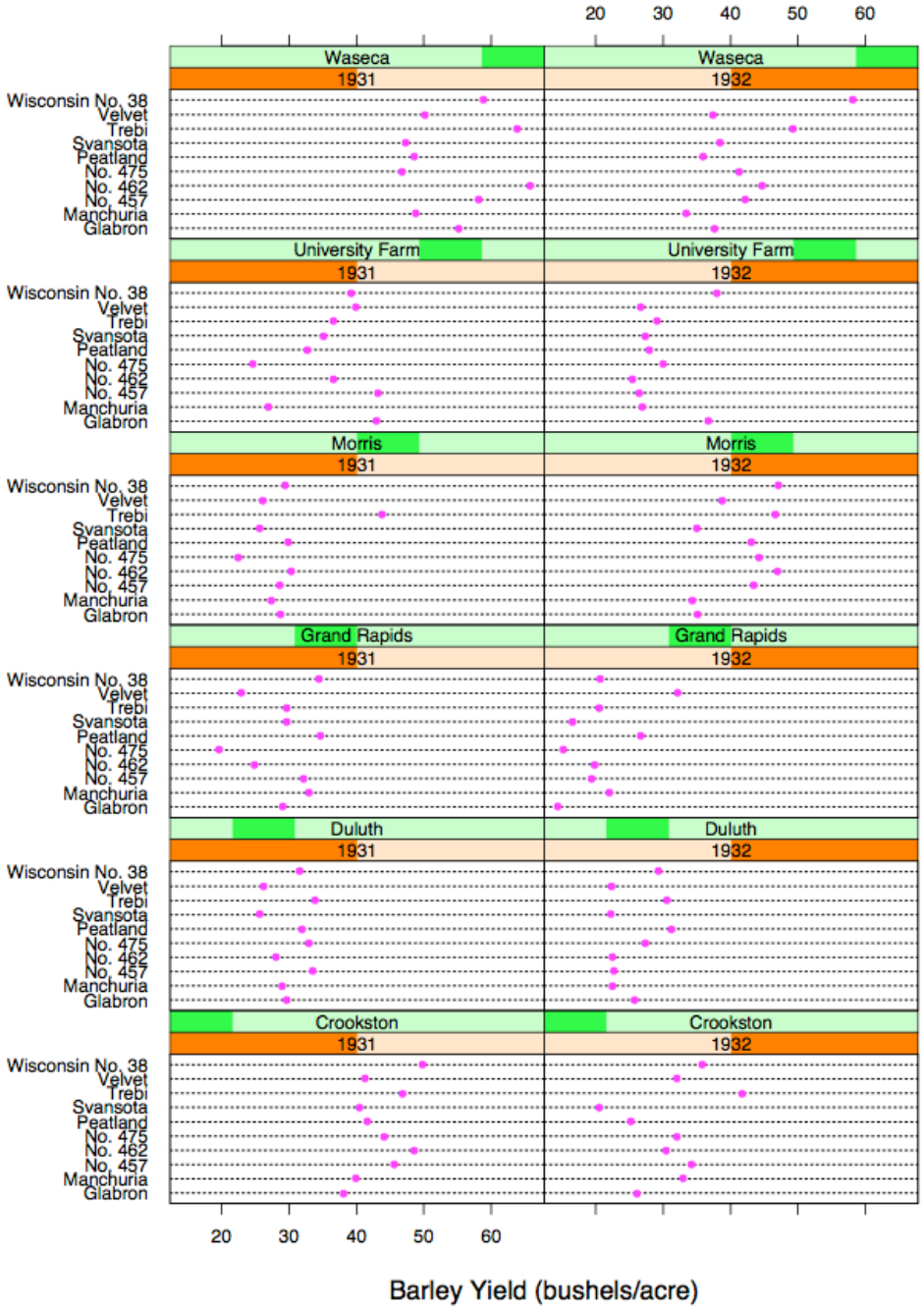
main-effects ordering

order partitioning variable based on derived data

support perception of trends and structure in data

Data
Barley Yields in two years across multiple farms for multiples barley strains

partitioning variables
Columns partitioned by year
Rows partitioned by farm

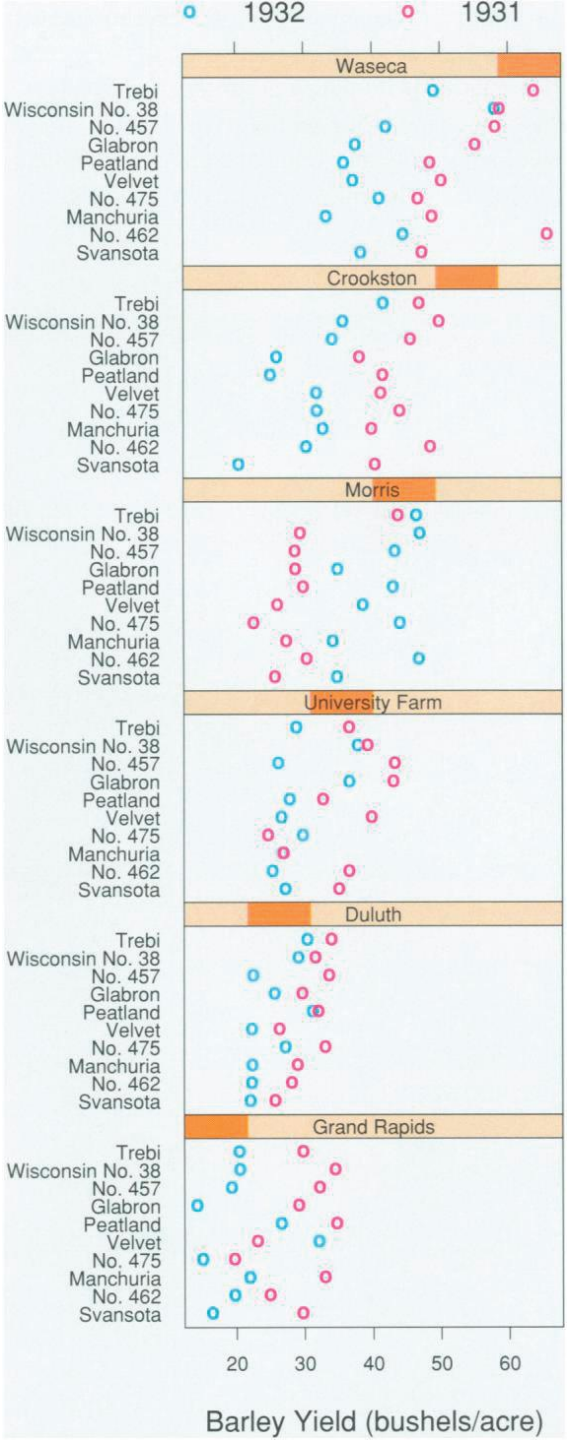


TRELLIS

Years in a single plot with year encoded by color

Showing strong evidence of an anomaly in the data

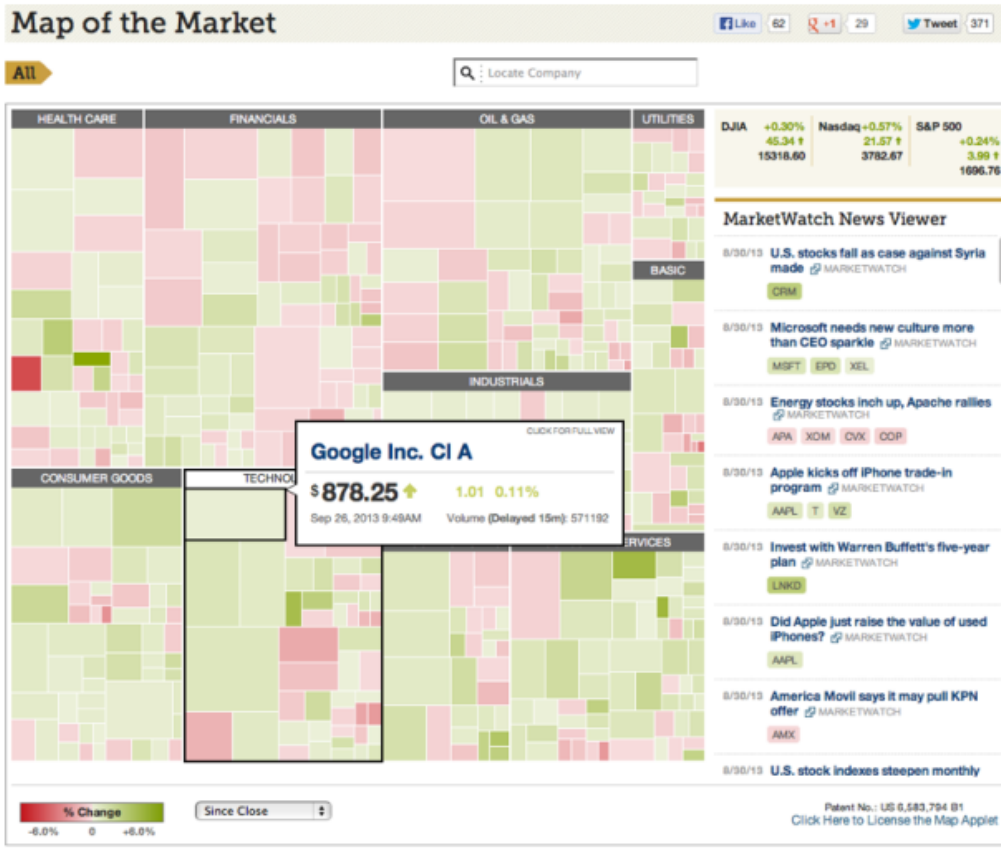
(the 3rd part)



RECURSIVE SUBDIVISION

partitioning:

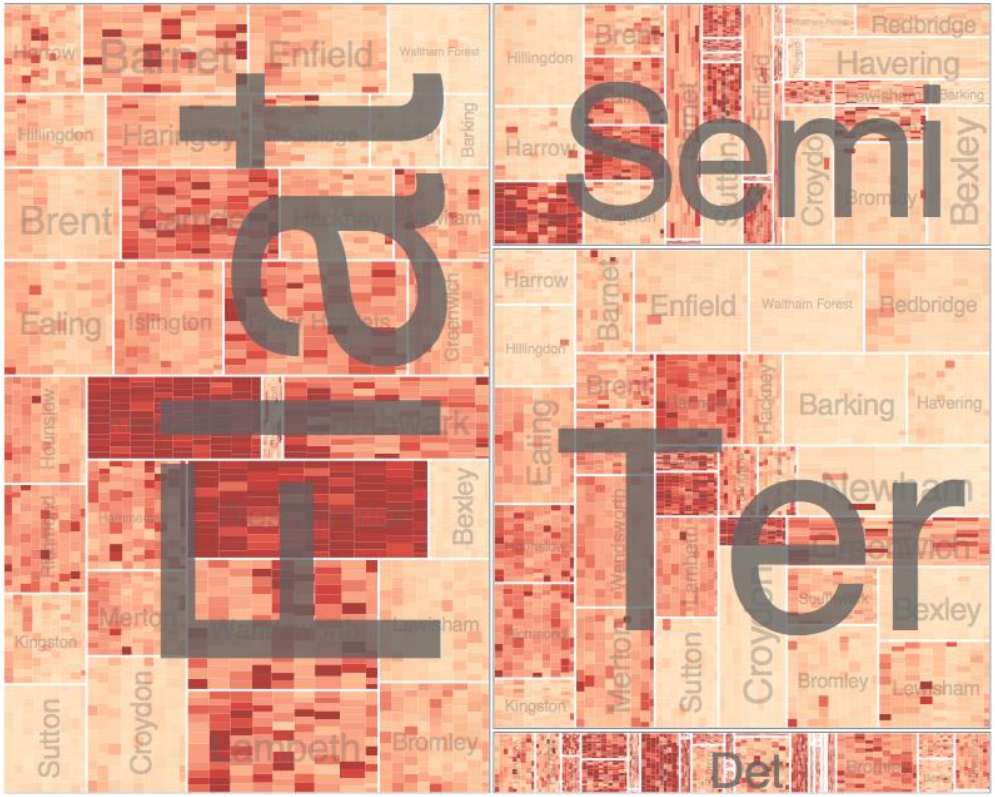
Flexibly transform data attributes into a hierarchy use **treemaps** as spacefilling rectangular layouts



Treemap

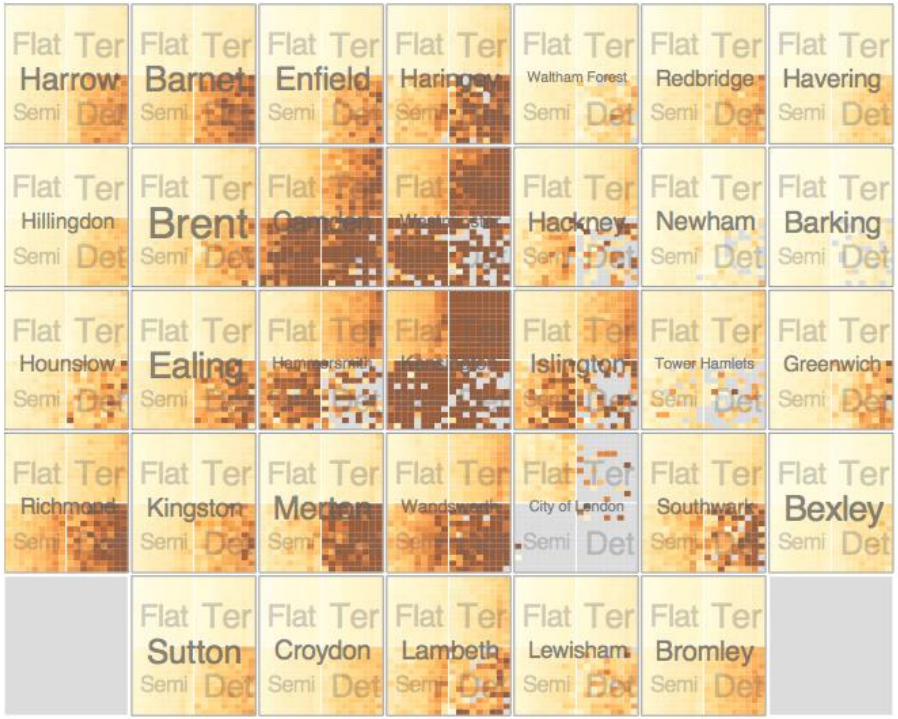
HIVE EXAMPLE: LONDON PROPERTY

partitioning attributes
house type
neighborhood
sale time
encoding attributes
average price (color)
number of sales (size)
results
between neighborhoods, different housing distributions
within neighborhoods, similar prices



HIVE EXAMPLE

partitioning attributes
neighborhood
house type
sale time (year)
sale time (month)
encoding attributes
neighborhood location (approximate)
average price (color)
n/a (size)
results
expensive neighborhoods near center of city



Configuring Hierarchical Layouts to Address Research Questions

Aidan Slingsby, Jason Dykes and Jo Wood
giCentre, Department of Information Science, City University London
http://www.gicentre.org/hierarchical_layouts/

https://vimeo.com/9870257

LAYERING

Combining multiple views on top of one another

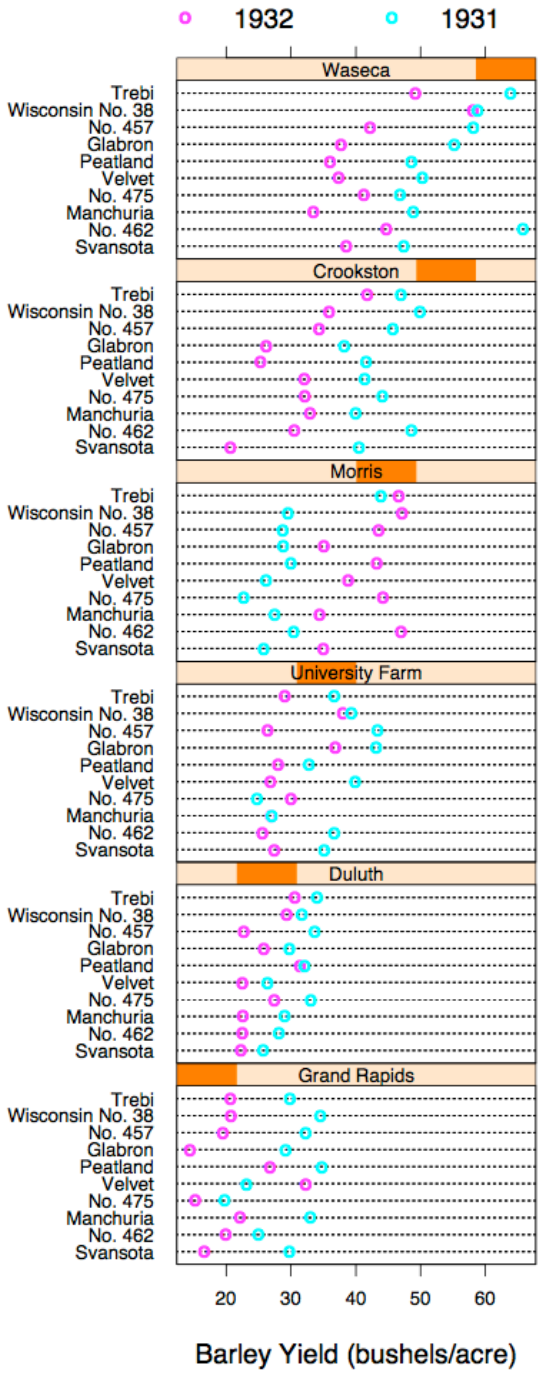
To form a composite view

Rational

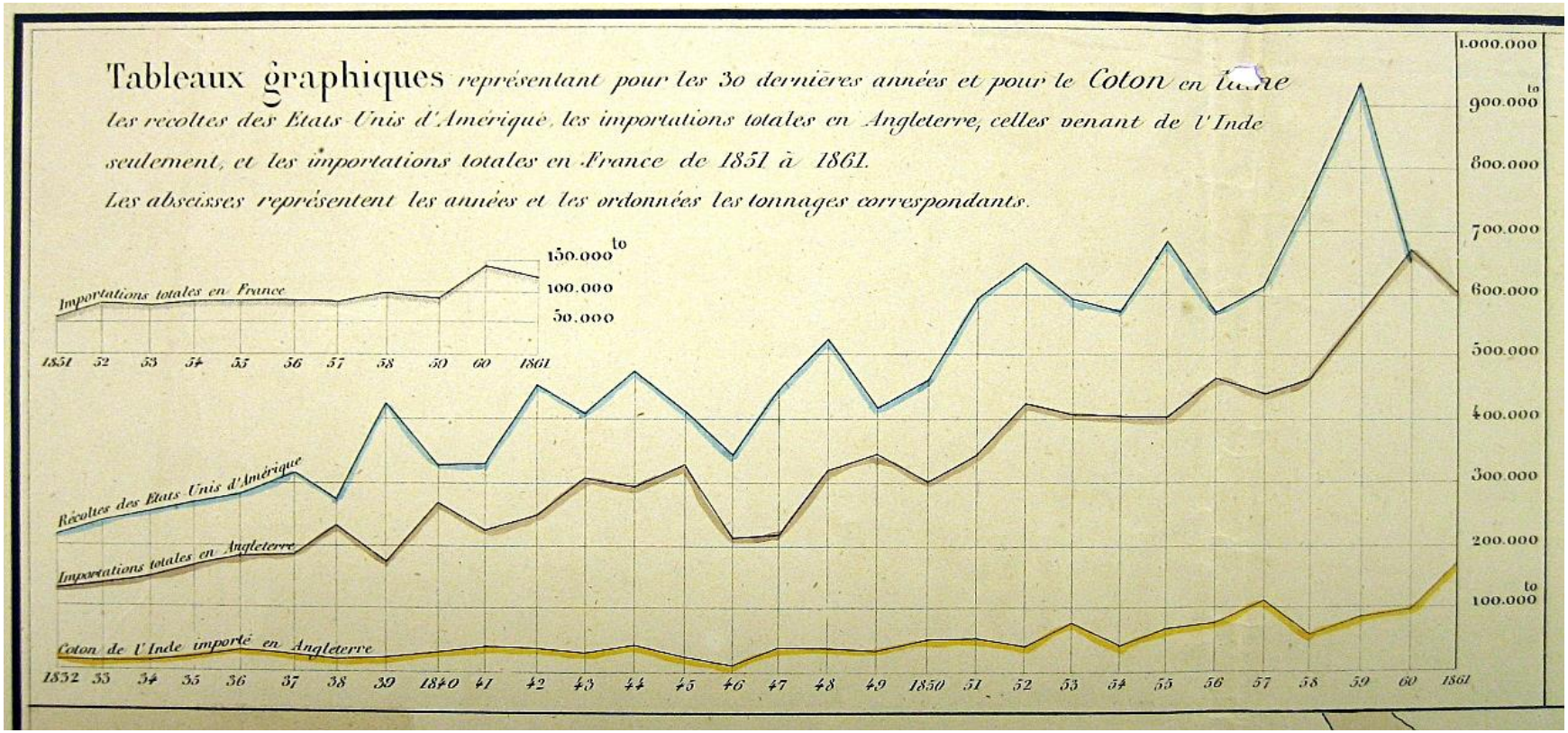
Supports a larger, more detailed view than using multiple views

Trade-off

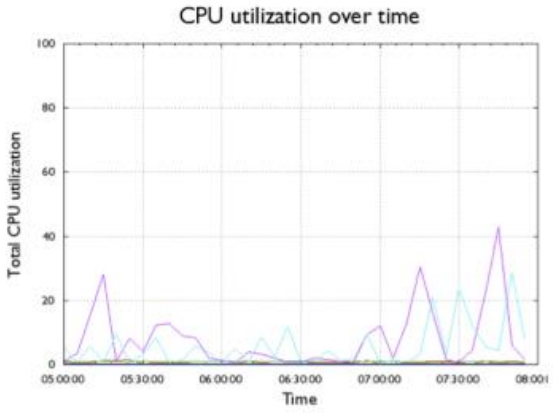
Layering imposes constraints on visual encoding choice as well as number of layers that can be shown

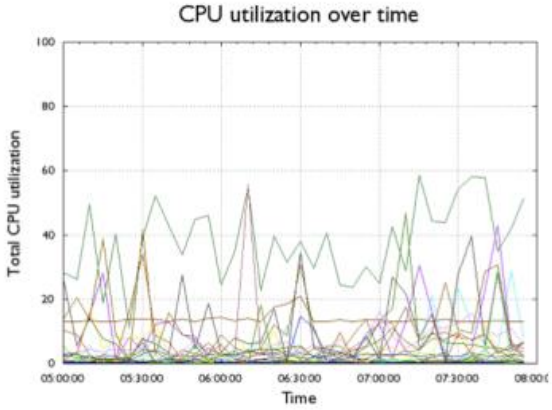


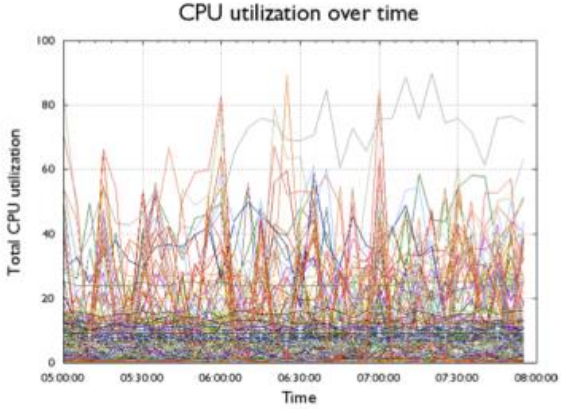
JOSEPH MINARD



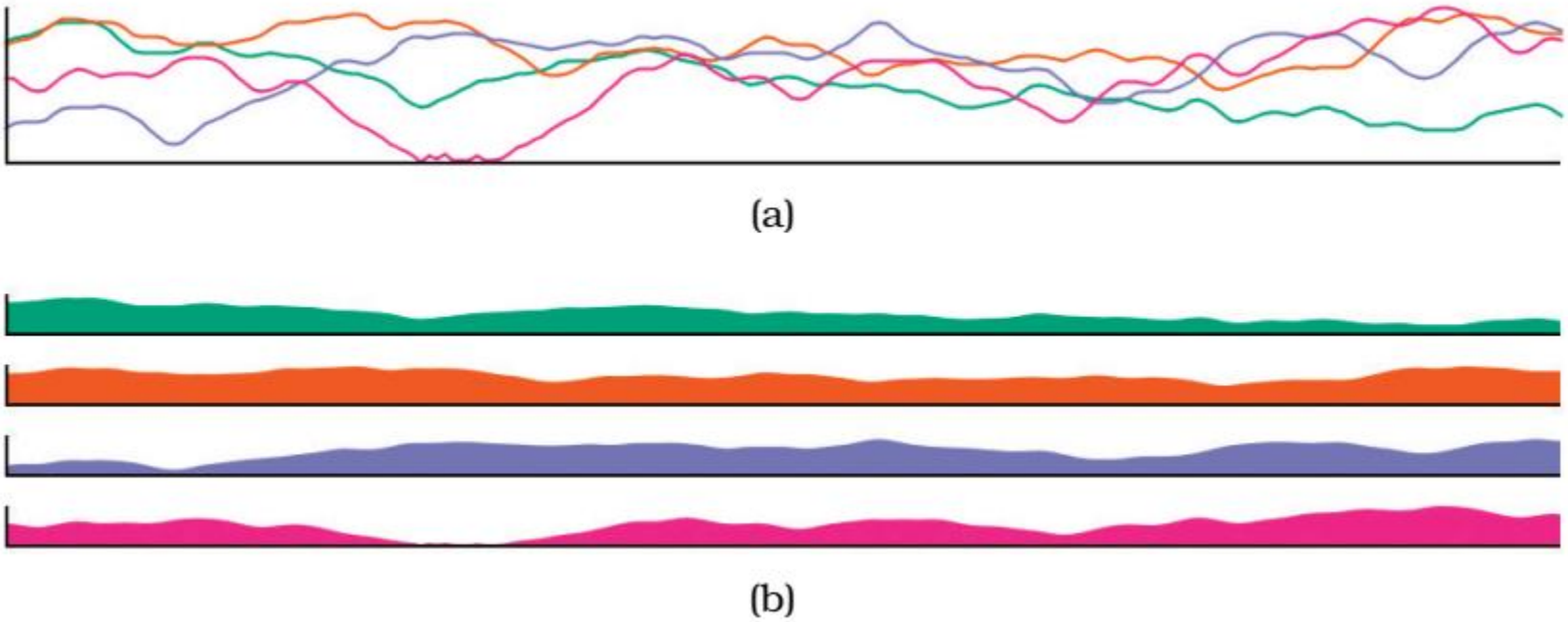
OVERLAYS


CPU utilization over time
Total CPU utilization
Time


CPU utilization over time
Total CPU utilization
Time


CPU utilization over time
Total CPU utilization
Time

SUPERIMPOSED VS JUXTAPOSED

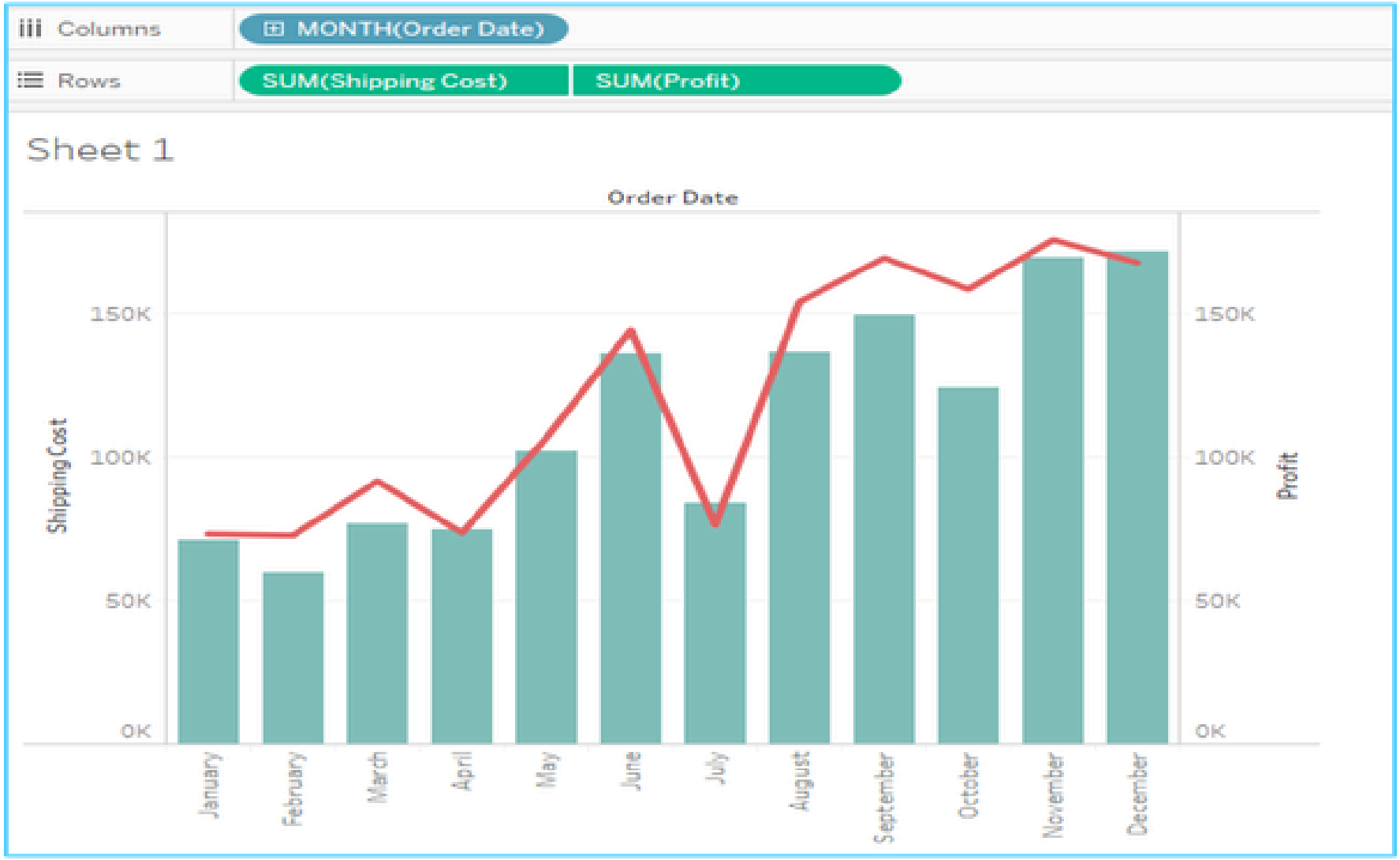


Empirical study comparing superimposed line charts to juxtaposed filled-area line charts. (a) Superimposed line charts performed best for tasks carried out within a local visual span. (b) Juxtaposed filled area charts were best for global tasks, especially as the number of time series increased.

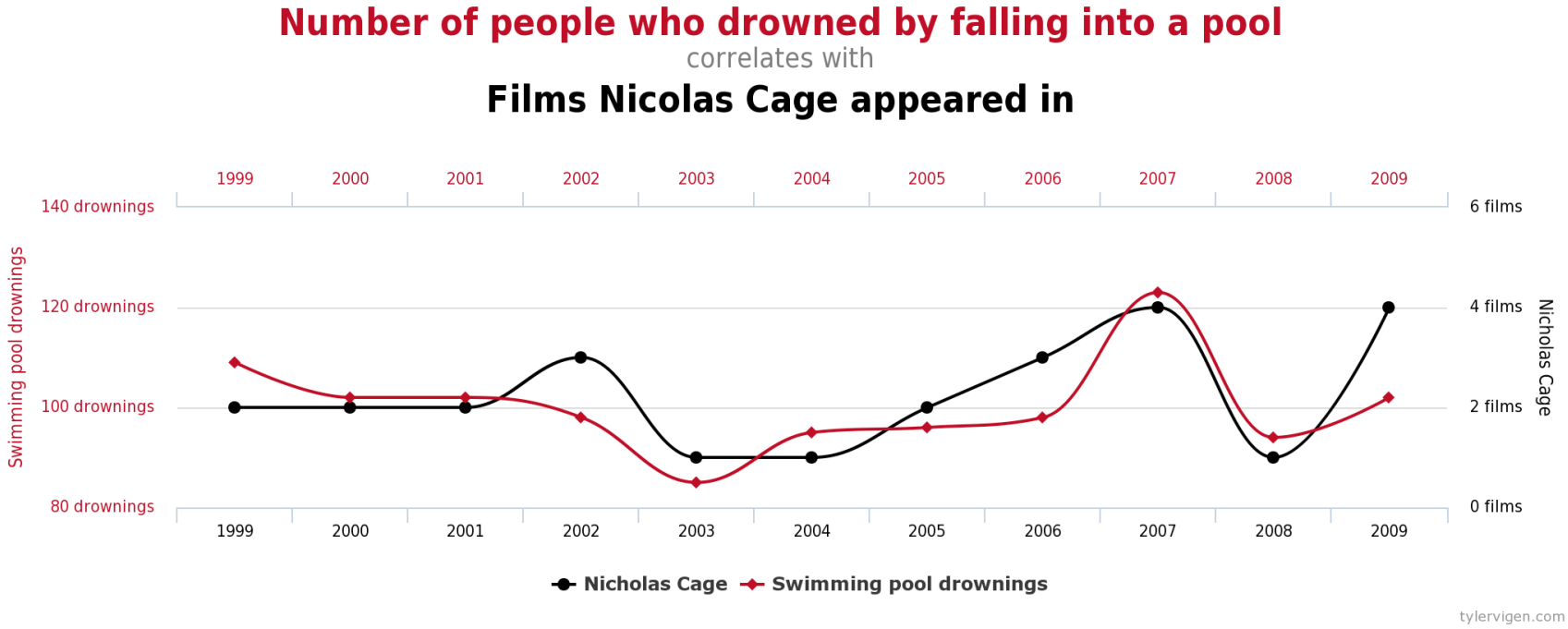
LAYERING

Petaluma
Tomales
1
Point Reyes Station
Point Reyes-Petaluma Rd
Novato Blvd
101
121
37
Novato
SAN PABLO BAY
Olema
Sir Francis Drake Boulevard
POINT REYES NATIONAL SEASHORE
San Rafael
80
580
1
PACIFIC OCEAN
Oakland
GOLDEN GATE
0
10 Kilometers
0
10 Miles
San Francisco

DUAL AXIS


Columns
MONTH(Order Date)
Rows
SUM(Shipping Cost)
SUM(Profit)
Sheet 1
Order Date
Shipping Cost
Profit
150K
100K
50K
0K
January
February
March
April
May
June
July
August
September
October
November
December

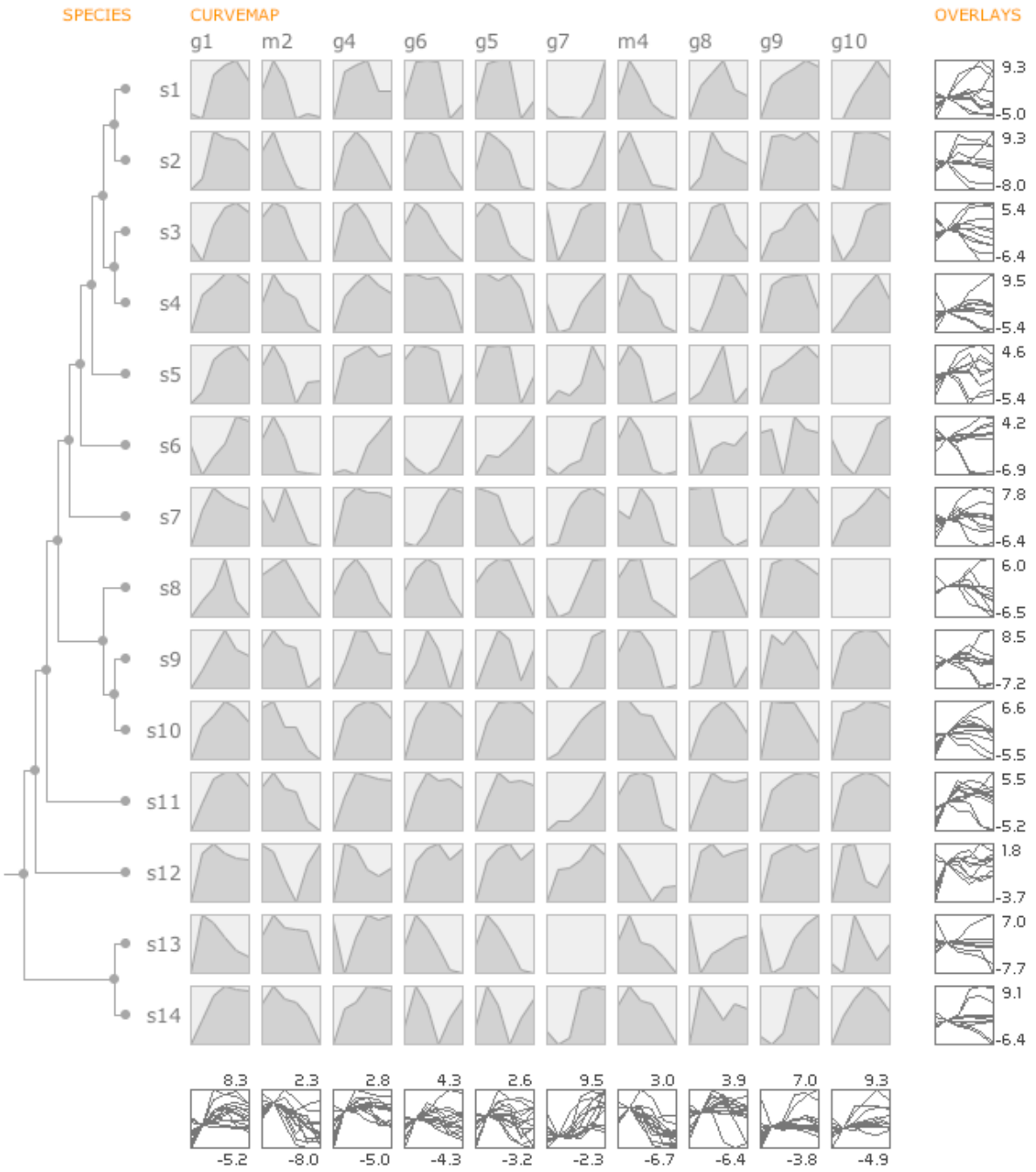
DUAL AXIS (DON'T)



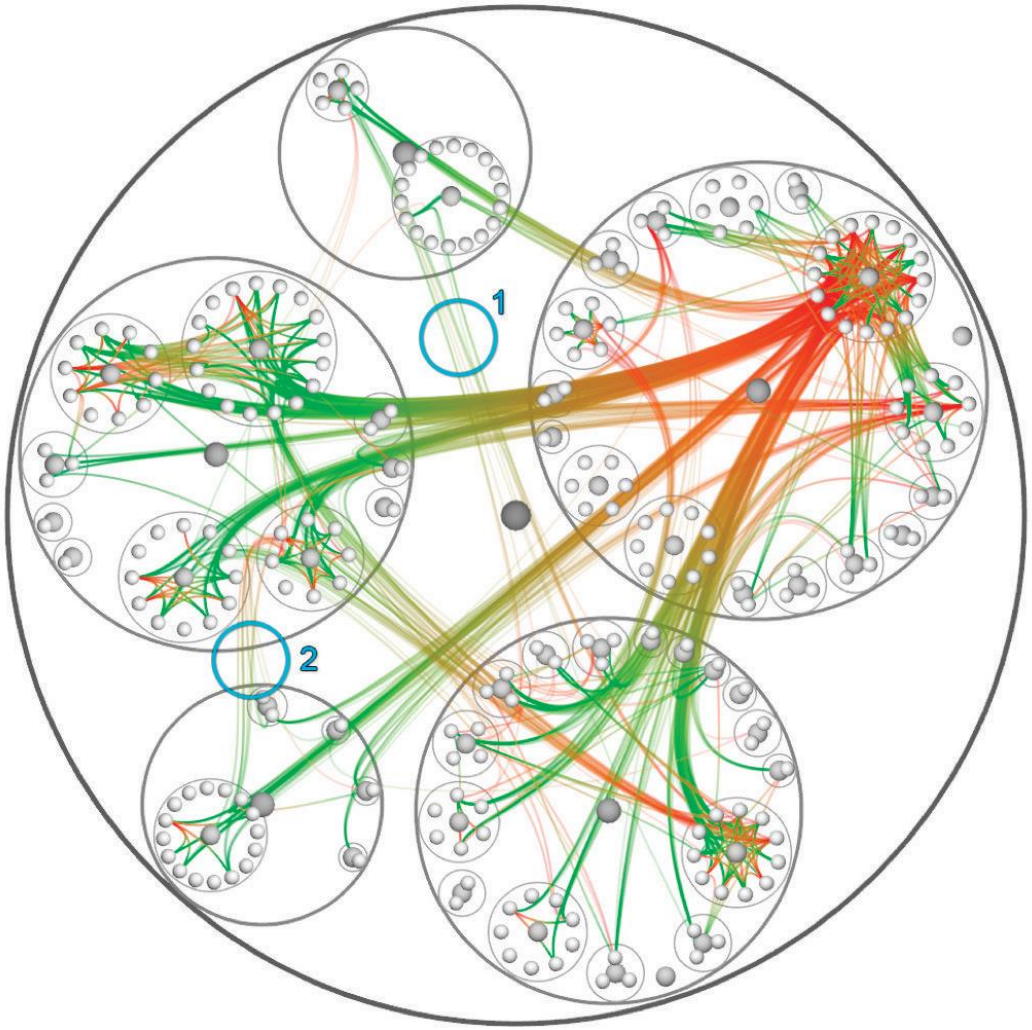
COMBINED

Partitioned + layered graph

Synchronized through highlighting

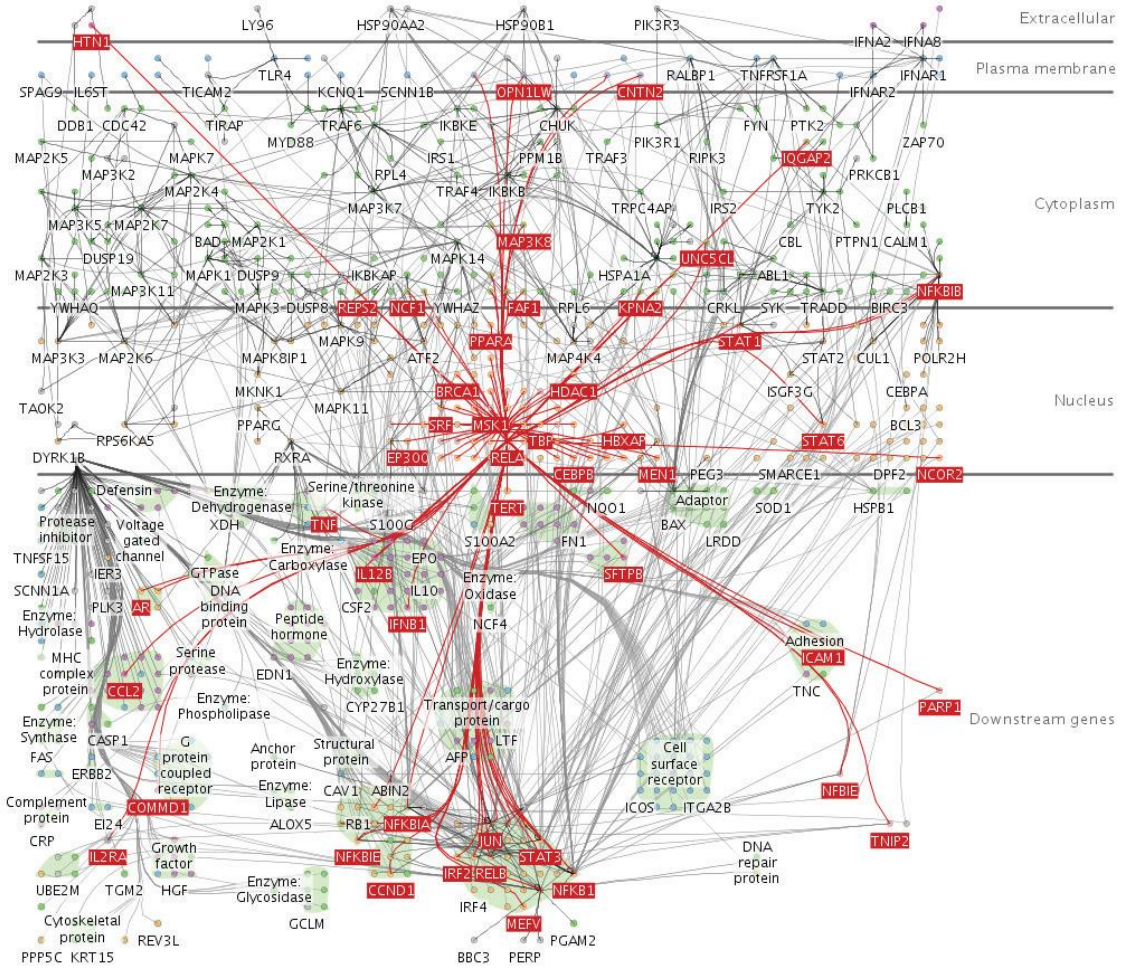


LAYERS

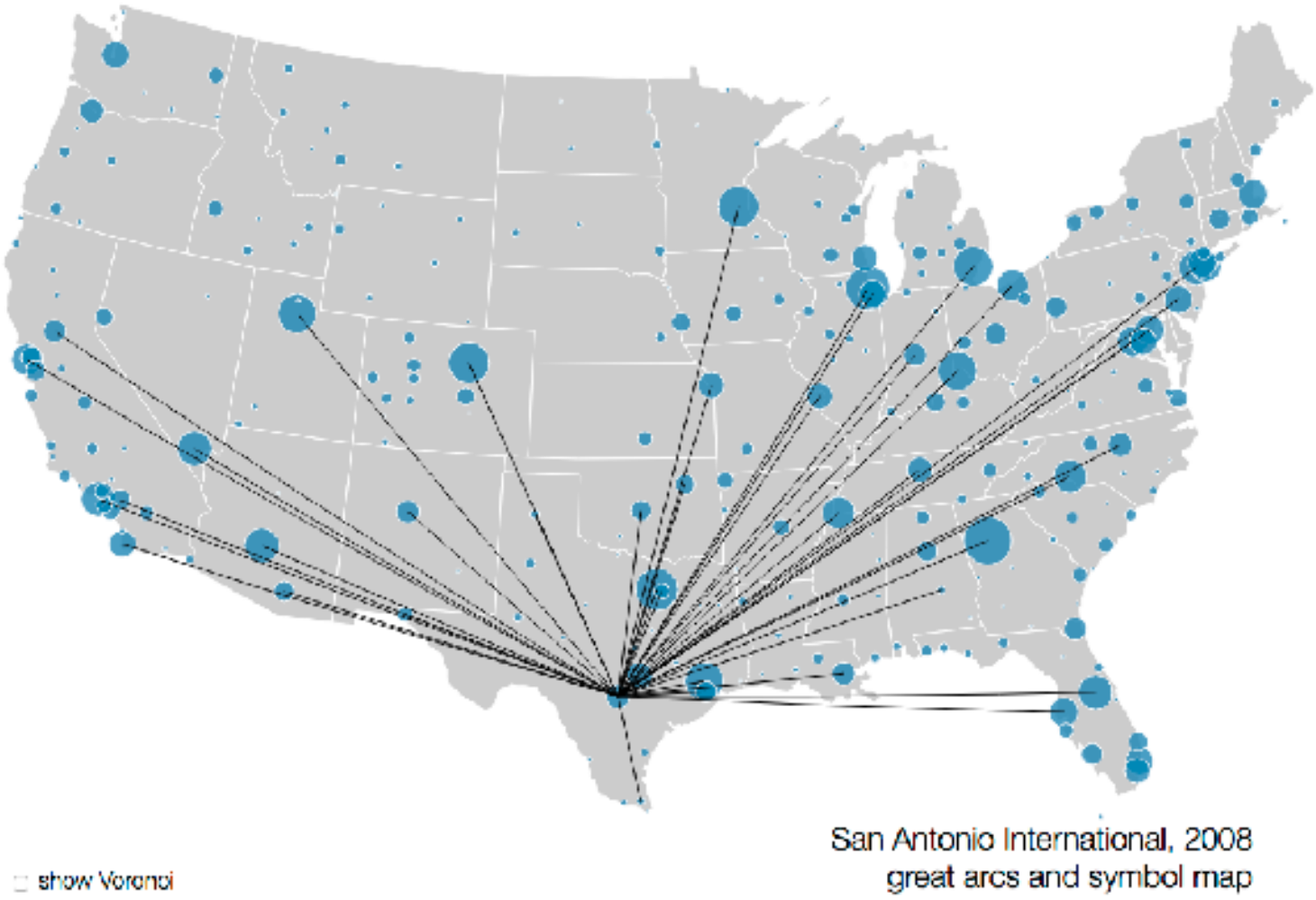


DYNAMIC LAYERS

in response to user selection.
The number of possible layers can be huge



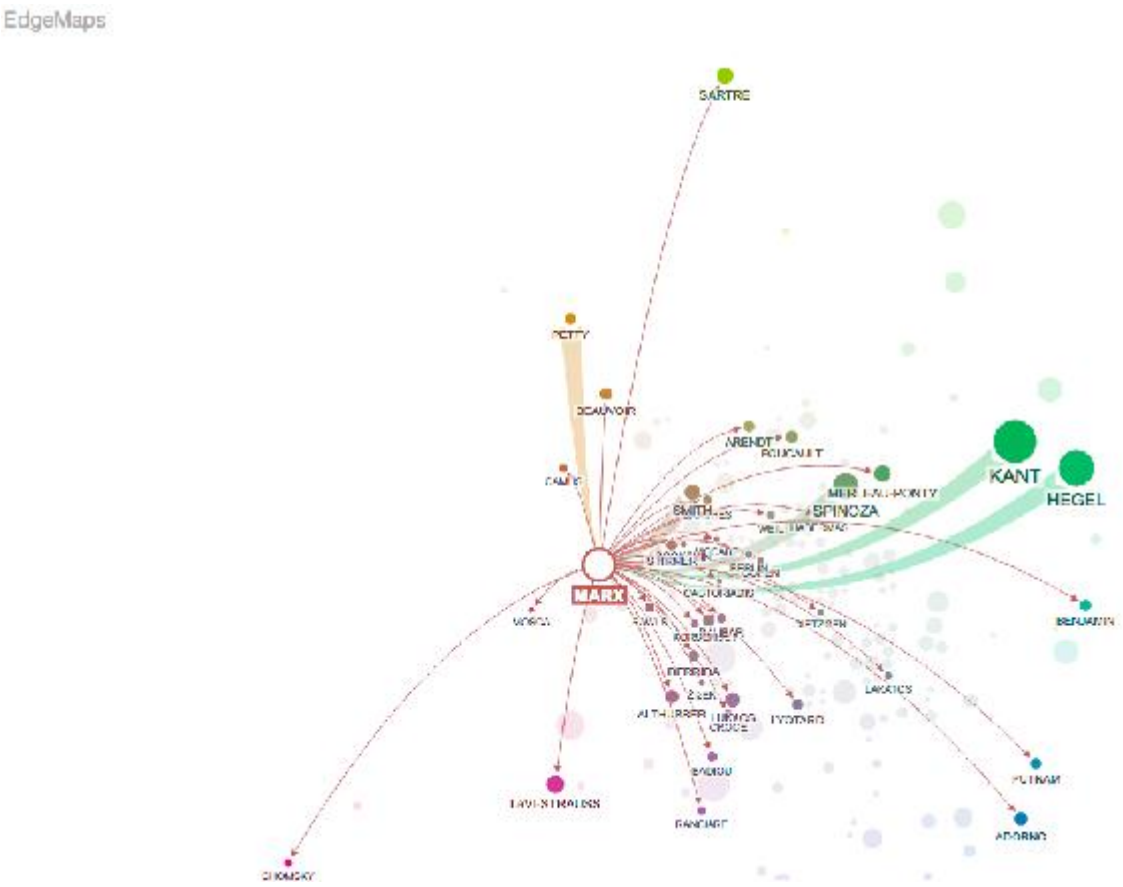
DYNAMIC VISUAL LAYERING



http://mbostock.github.io/d3/talk/20111116/airports.html

DYNAMIC VISUAL LAYERING

One-hop neighbor highlighting

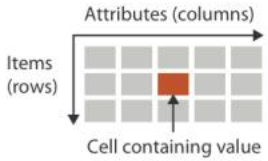


https://mariandoerk.de/edgemaps/demo/

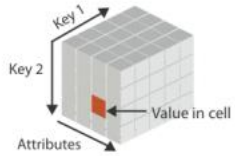
TABLES

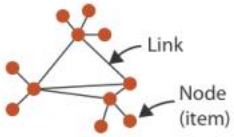
DATASET TYPES

Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

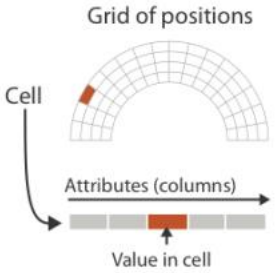


→ *Multidimensional Table*





→ *Trees*



EXERCISE: SKETCH 2 WAYS TO VIS EACH TABLE

	Age	Best 100 m	Furthest Jump	Sex
Amy	16	13.2	5.2	F
Basil	18	12.4	4.2	F
Clara	14	14.1	2.5	F
Desmond	22	10.01	6.3	M
Charles	19	11.3	5.3	M

	BPM T1	BPM T2	BPM T3
Amy	90	130	150
Basil	70	110	109
Clara	60	140	141
Desmond	84	100	108
Charles	81	110	130

SCALE OF TABLES

Need different approaches for “normal” and “high-dimensional” tables

How many dimensions?

~ 50 – tractable with “just” visualization

~1000 – need analytical methods

How many records?

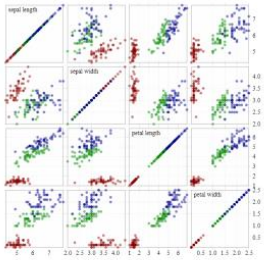
~ 1000 – “just” vis is fine

>> 10 000 – need analytical methods

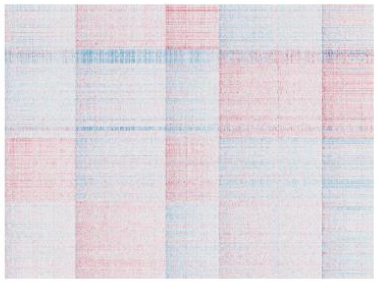
Homogeneity

- Same data type?
- Same scales?

ANALYTIC COMPONENT

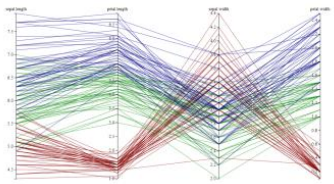


Scatterplot Matrices
[Bostock]

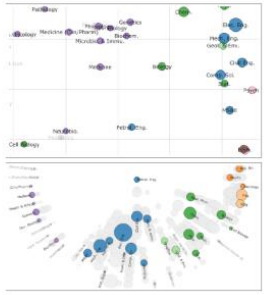


Pixel-based visualizations / heat maps

Multidimensional Scaling
[Doerk 2011]



Parallel Coordinates
[Bostock]



[Chuang 2012]

no / little analytics ⟷ **strong analytics component**

TECHNIQUES AND TASKS

Magnitude

Distribution

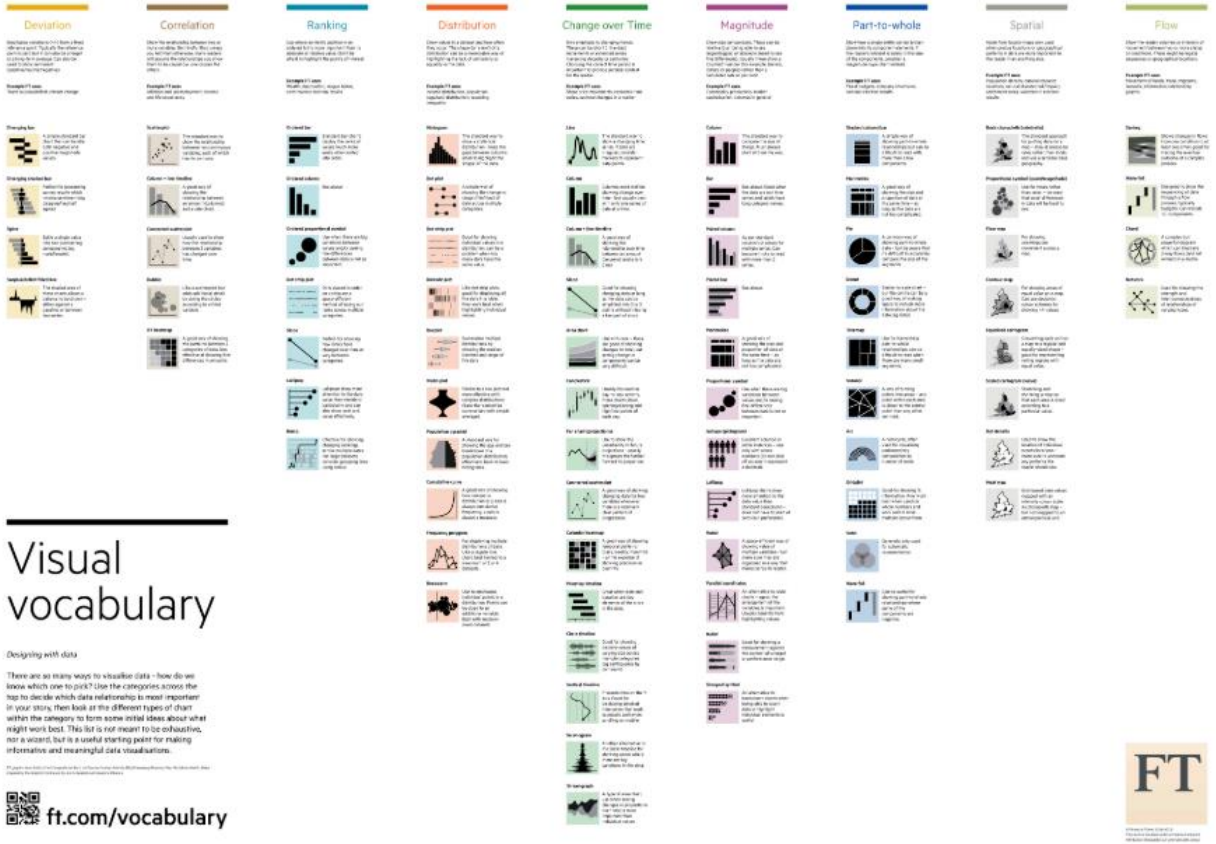
Ranking

Part to whole

Deviation

Correlation

Change over time



https://github.com/ft-interactive/chart-doctor/tree/master/visual-vocabulary
https://gramener.github.io/visual-vocabulary-vega/#/Magnitude/

MAGNITUDE

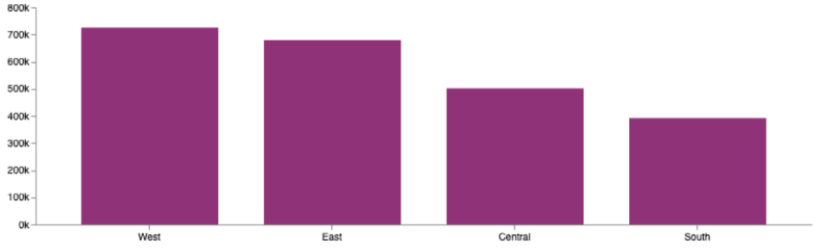
Show size comparison

- Relative: to see bigger/larger
- Absolute: to see differences

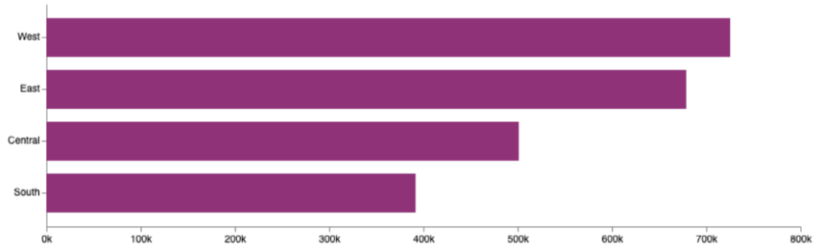
Usually for “counted” numbers

Rather than a calculated rate or per cent

BAR CHARTS

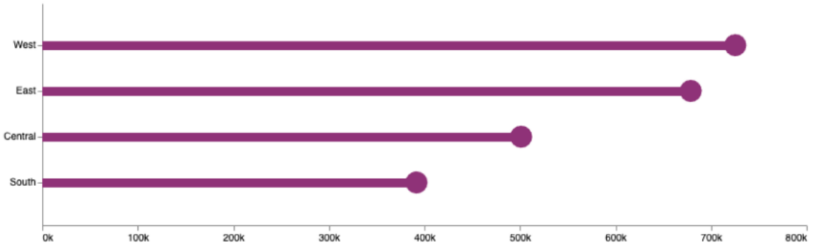


Vertical Bar Chart / Column Chart



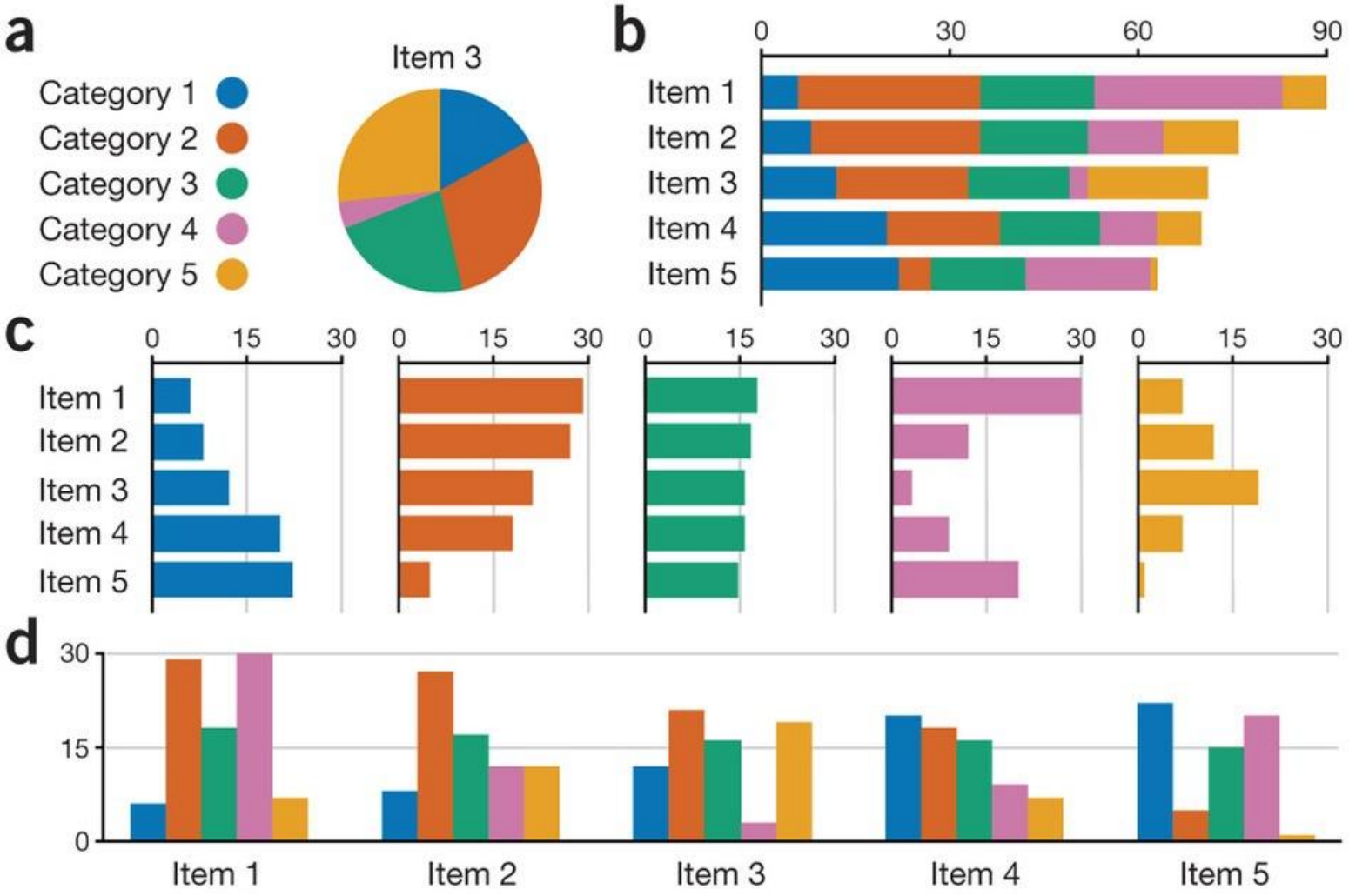
Horizontal Bar Chart

Grouped Bar Chart



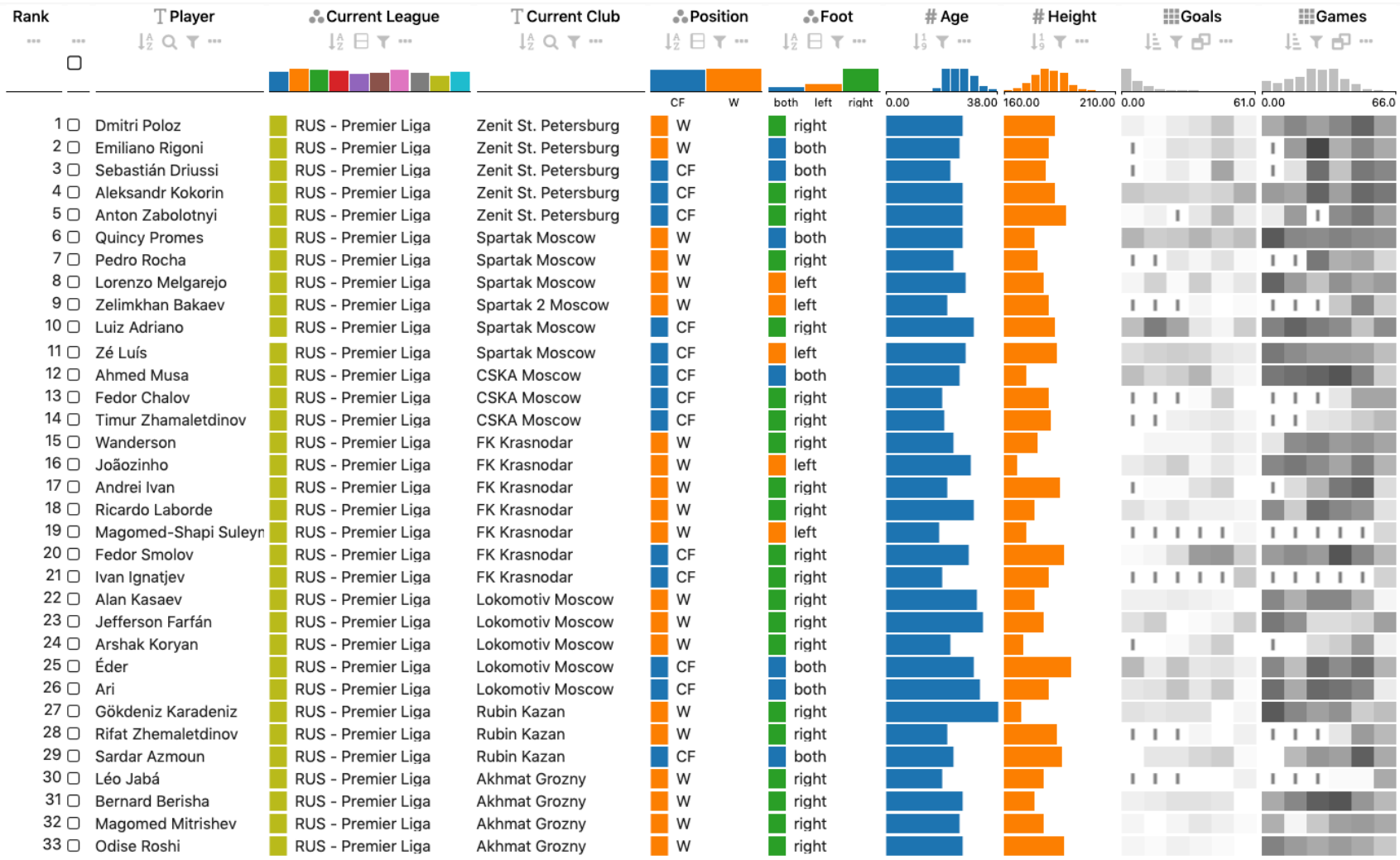
Lollipop Chart

BAR CHARTS TYPES



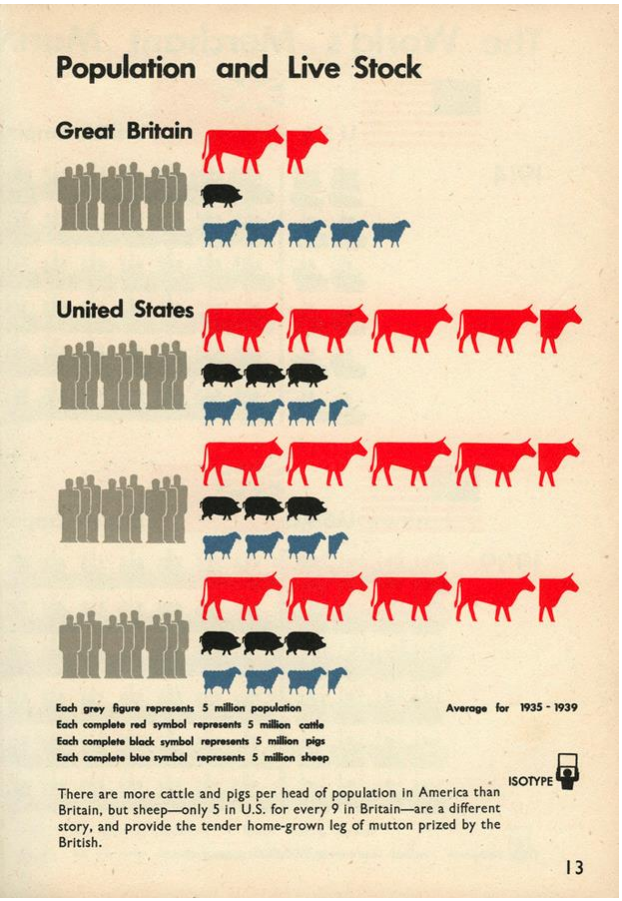
Streit & Gehlenborg, PoV, Nature Methods, 2014

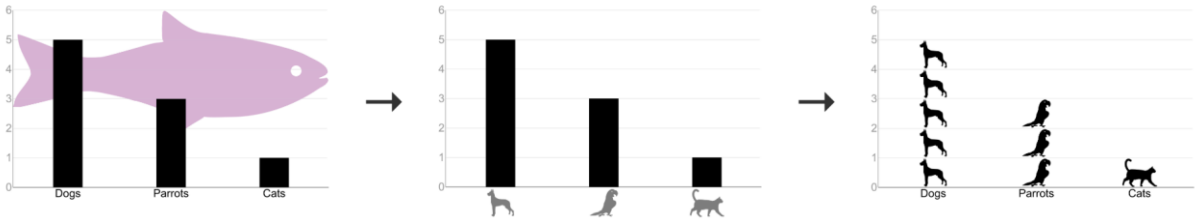
BAR CHARTS IN COMPLEX VIS



Rank	Player	Current League	Current Club	Position	Foot
1	Dmitri Poloz	RUS - Premier Liga	Zenit St. Petersburg	W	right
2	Emiliano Rigoni	RUS - Premier Liga	Zenit St. Petersburg	W	both
3	Sebastián Driussi	RUS - Premier Liga	Zenit St. Petersburg	CF	both
4	Aleksandr Kokorin	RUS - Premier Liga	Zenit St. Petersburg	CF	right
5	Anton Zabolotnyi	RUS - Premier Liga	Zenit St. Petersburg	CF	right
6	Quincy Promes	RUS - Premier Liga	Spartak Moscow	W	both
7	Pedro Rocha	RUS - Premier Liga	Spartak Moscow	W	right
8	Lorenzo Melgarejo	RUS - Premier Liga	Spartak Moscow	W	left
9	Zelimkhan Bakaev	RUS - Premier Liga	Spartak 2 Moscow	W	left
10	Luiz Adriano	RUS - Premier Liga	Spartak Moscow	CF	right
11	Zé Luís	RUS - Premier Liga	Spartak Moscow	CF	left
12	Ahmed Musa	RUS - Premier Liga	CSKA Moscow	CF	both
13	Fedor Chalov	RUS - Premier Liga	CSKA Moscow	CF	right
14	Timur Zhamaletdinov	RUS - Premier Liga	CSKA Moscow	CF	right
15	Wanderson	RUS - Premier Liga	FK Krasnodar	W	right
16	Joãozinho	RUS - Premier Liga	FK Krasnodar	W	left
17	Andrei Ivan	RUS - Premier Liga	FK Krasnodar	W	right
18	Ricardo Laborde	RUS - Premier Liga	FK Krasnodar	W	right
19	Magomed-Shapi Suleyn	RUS - Premier Liga	FK Krasnodar	W	left
20	Fedor Smolov	RUS - Premier Liga	FK Krasnodar	CF	right
21	Ivan Ignatjev	RUS - Premier Liga	FK Krasnodar	CF	right
22	Alan Kasaev	RUS - Premier Liga	Lokomotiv Moscow	W	right
23	Jefferson Farfán	RUS - Premier Liga	Lokomotiv Moscow	W	right
24	Arshak Koryan	RUS - Premier Liga	Lokomotiv Moscow	W	right
25	Éder	RUS - Premier Liga	Lokomotiv Moscow	CF	both
26	Ari	RUS - Premier Liga	Lokomotiv Moscow	CF	both
27	Gökdeniz Karadeniz	RUS - Premier Liga	Rubin Kazan	W	right
28	Rifat Zhemaletdinov	RUS - Premier Liga	Rubin Kazan	W	right
29	Sardar Azmoun	RUS - Premier Liga	Rubin Kazan	CF	both
30	Léo Jabá	RUS - Premier Liga	Akhmat Grozny	W	right
31	Bernard Berisha	RUS - Premier Liga	Akhmat Grozny	W	right
32	Magomed Mitrishev	RUS - Premier Liga	Akhmat Grozny	W	right
33	Odise Roshi	RUS - Premier Liga	Akhmat Grozny	W	right

ISO TYPE VISUALIZATION





Otto and Marie Neurath

http://steveharoz.com/research/isotype/

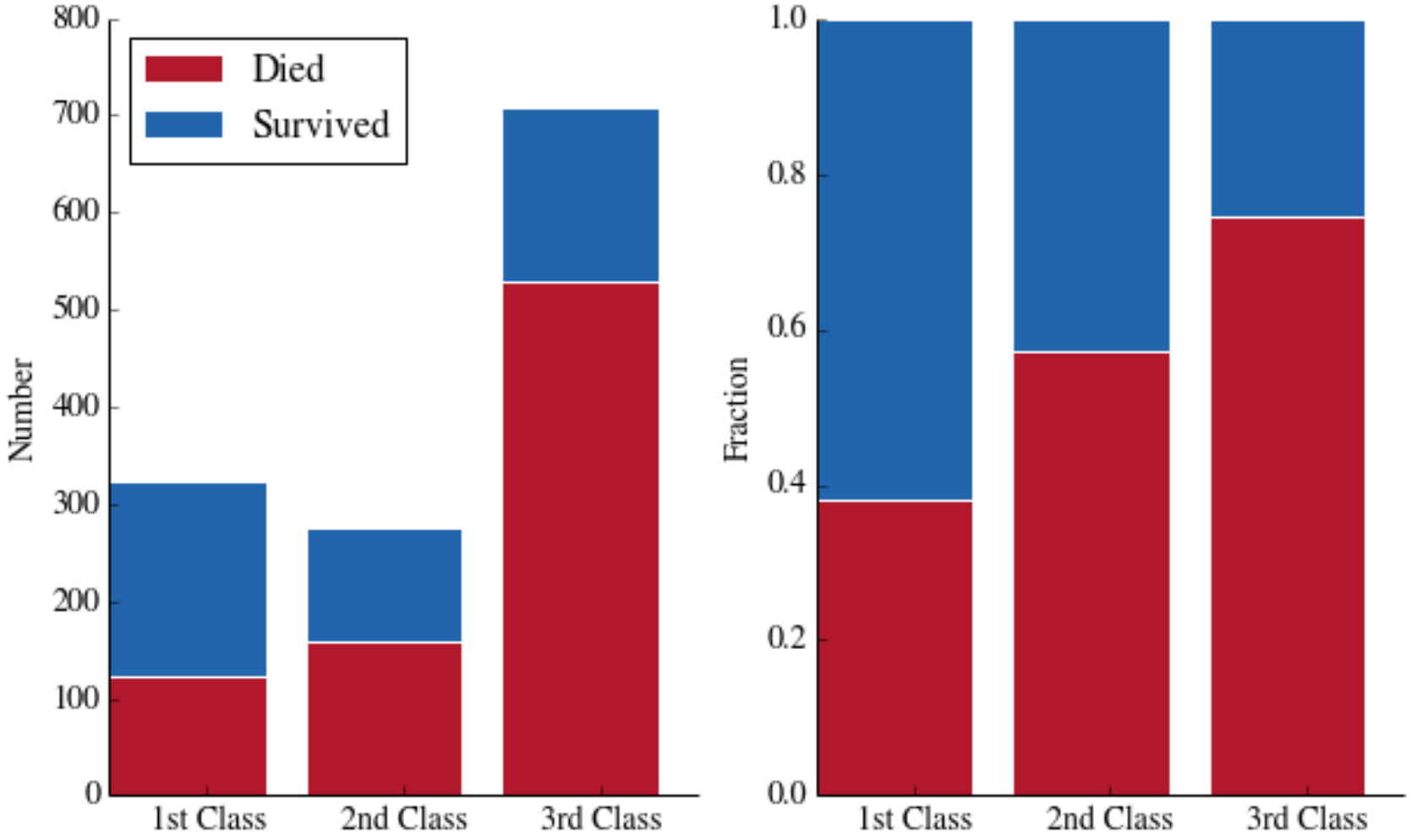
PART OF WHOLE

Show how a single entity can be broken down into its component elements

If the interest is solely in the size of the components

Consider magnitude-type chart

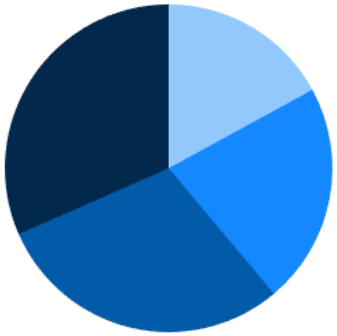
STACKED BAR CHART



PIE AND DONUT CHARTS

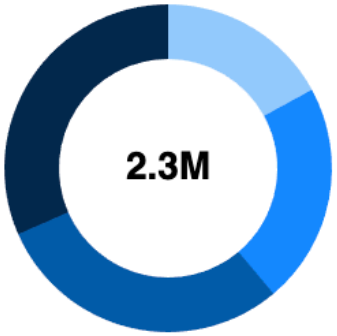
Pie Edit

A common way of showing part-to-whole data - but be aware that it's difficult to accurately compare the size of the segments.



Donut

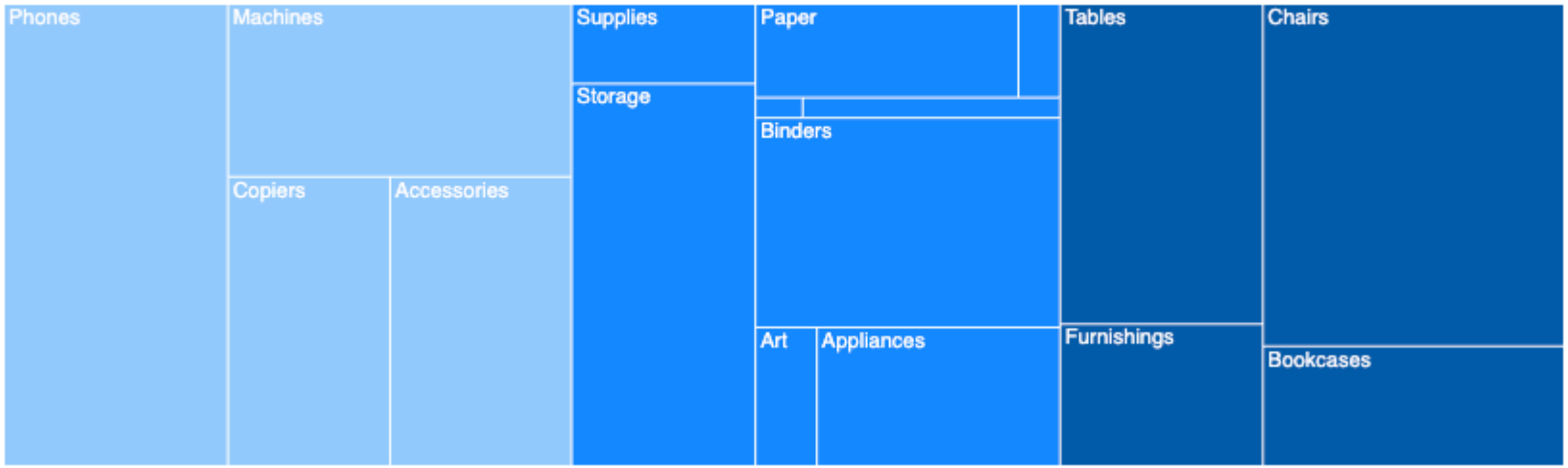
Similar to a pie chart - but the centre can be a good way of making space to include more information about the data (eg. total)



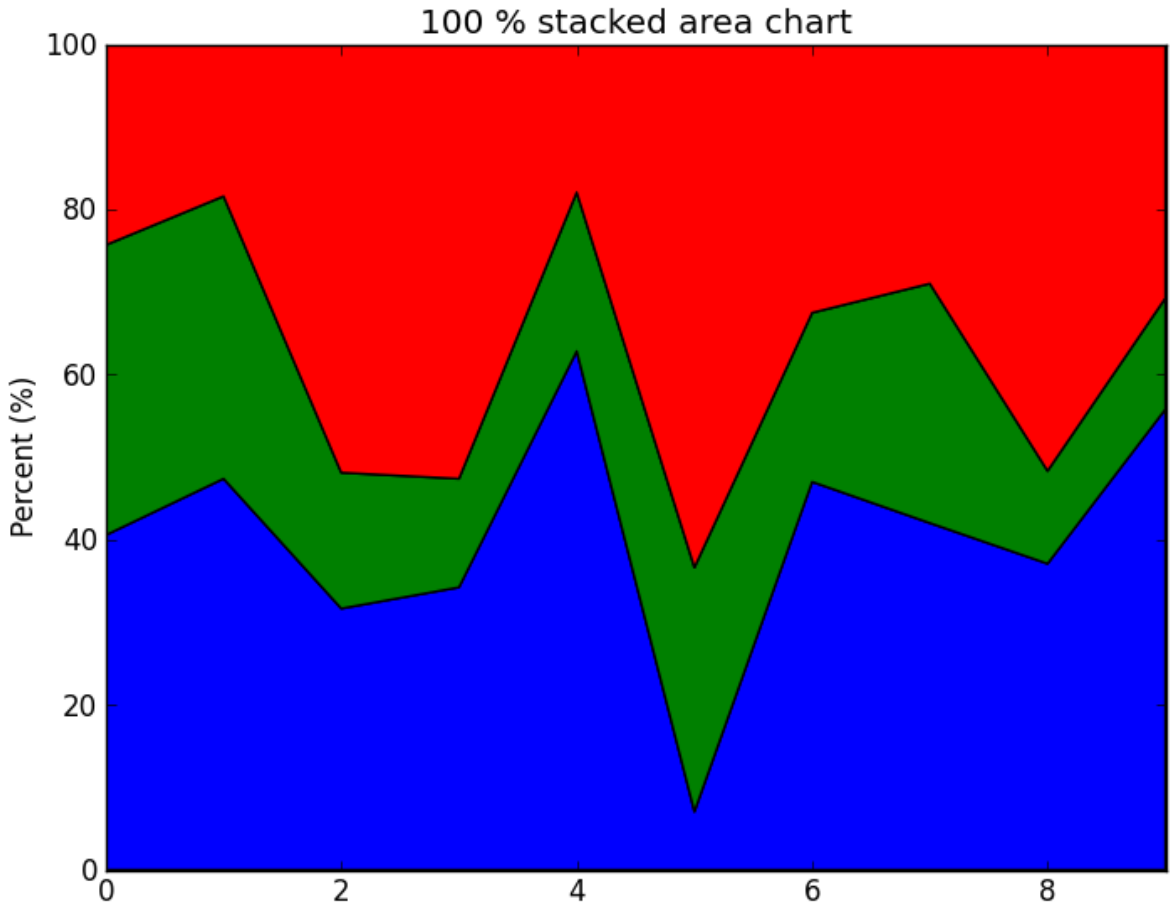
TREEMAP

Treemap

Use for hierarchical part-to-whole relationships; can be difficult to read when there are many small segments



STACKED AREA



DISTRIBUTION

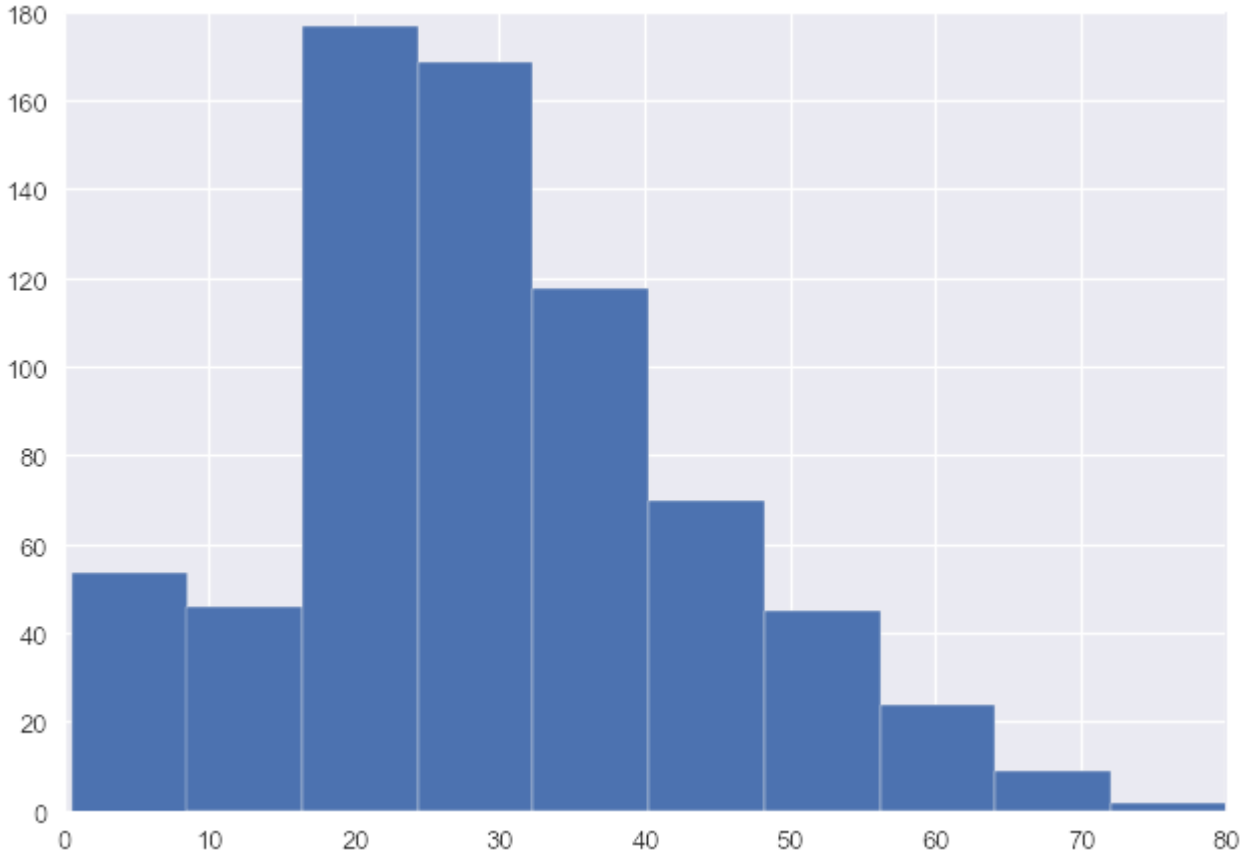
Aggregating large data vectors

Instead of showing all data points, show a data's distribution

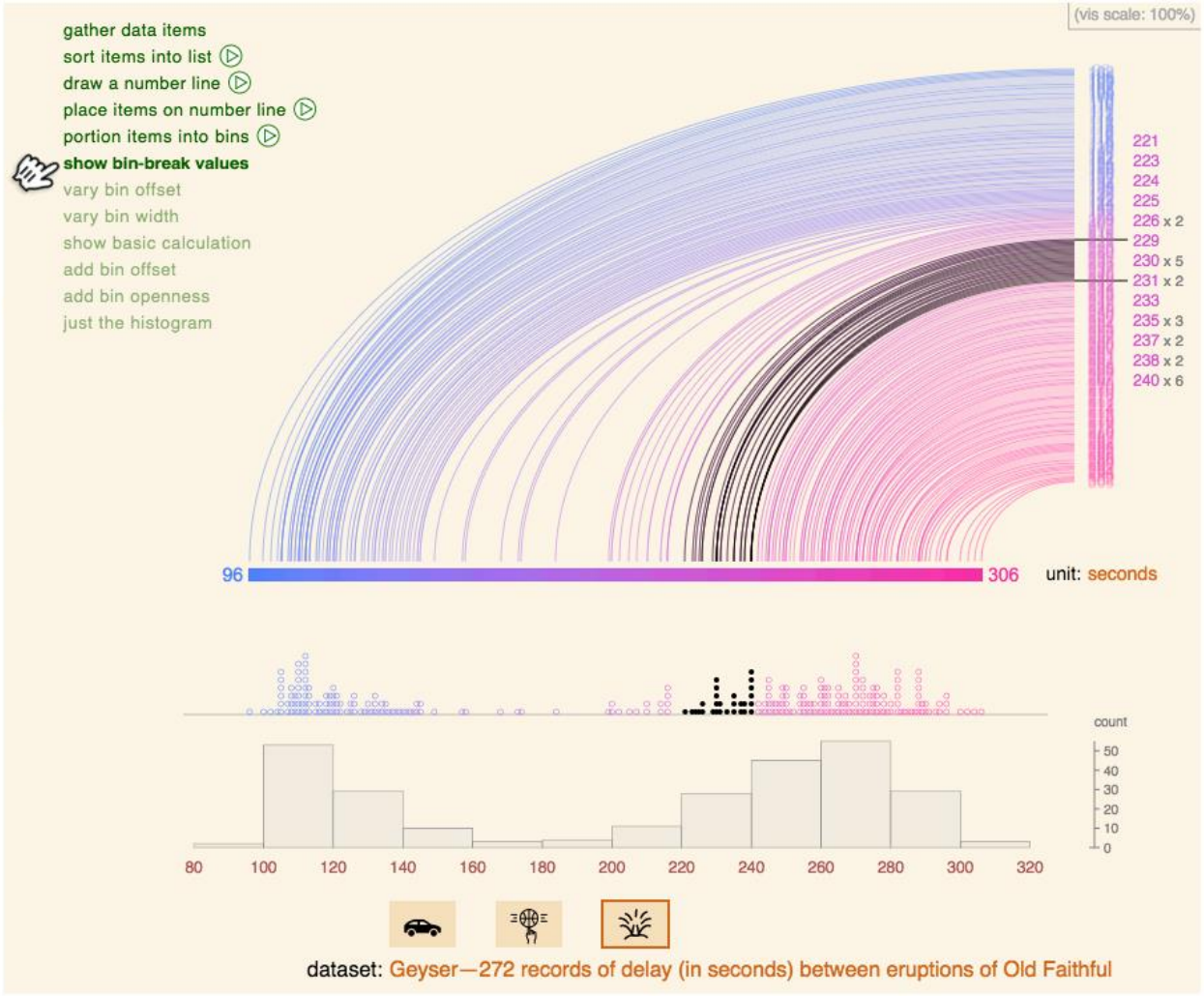
Pro: Compact representation

Con: Works only if data is "well behaved" for the type of distribution visualization

WHAT IS A HISTOGRAM



WHY HISTOGRAM IS HARD



http://tinlizzie.org/histograms/

HISTOGRAM

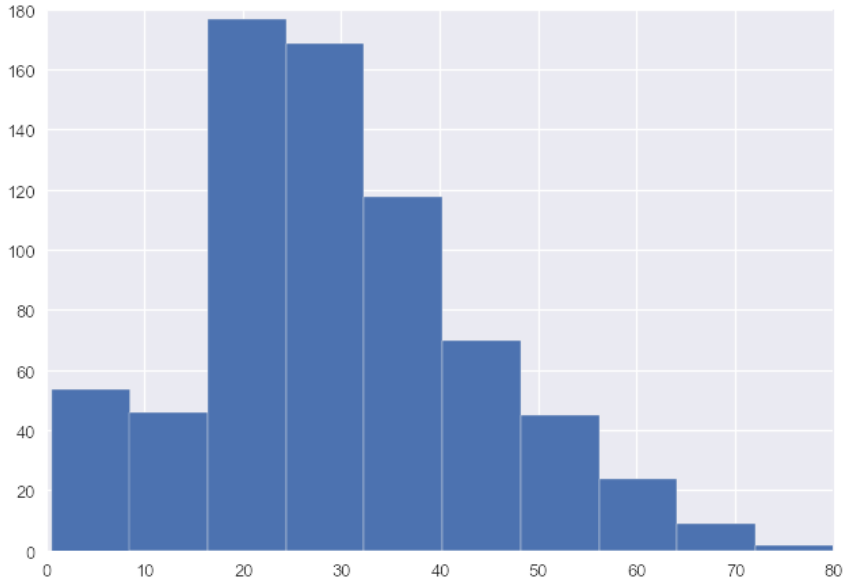
Good #bins hard to predict

Make interactive

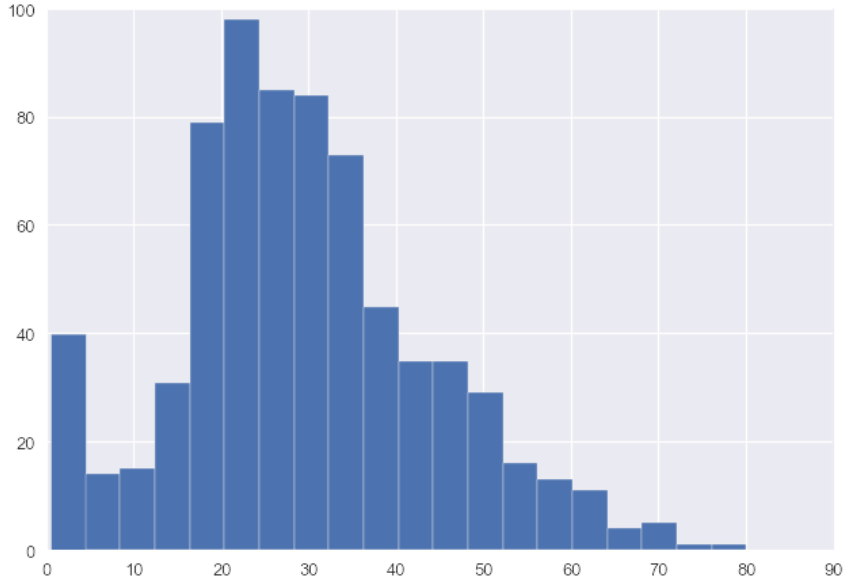
Rules of thumb:

#bins = sqrt(n)

#bins = $\log_2(2) + 1$

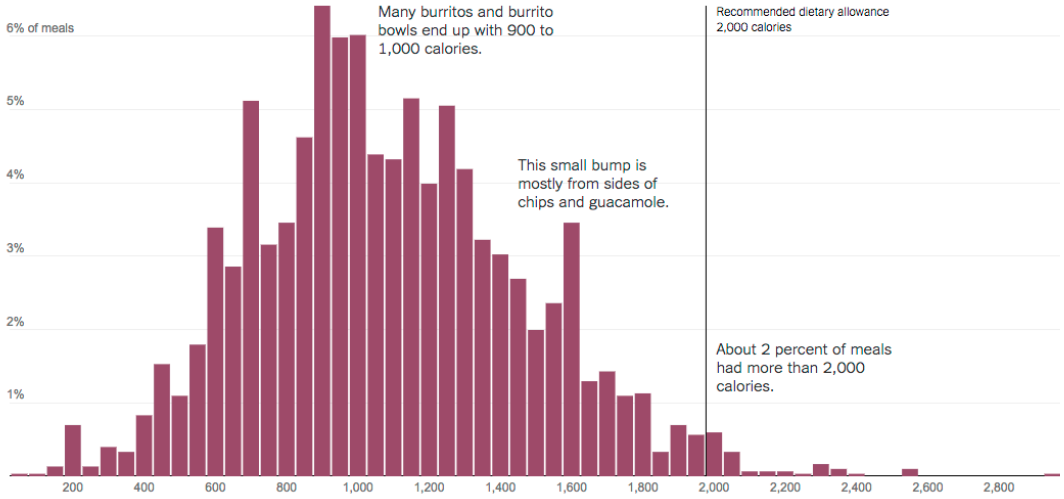


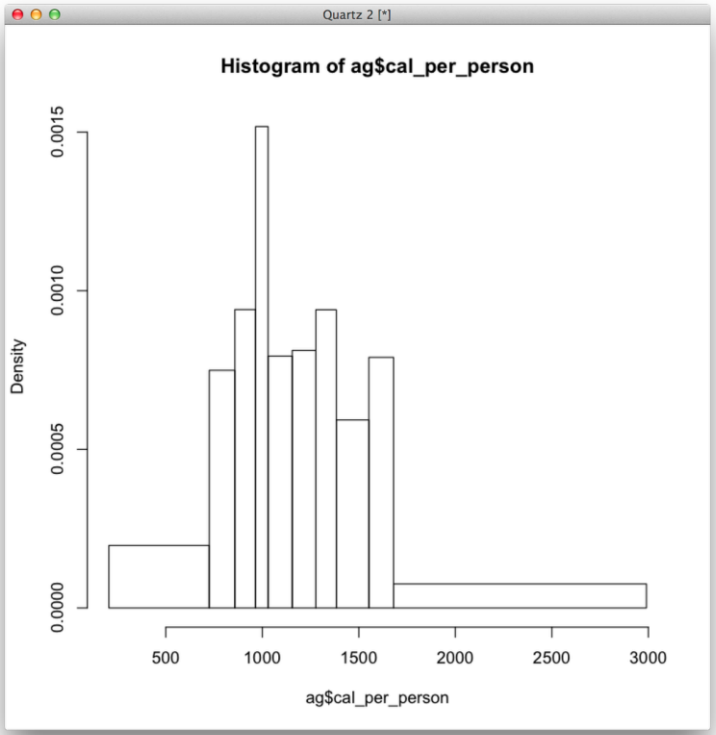
10 bins



20 bins

UNEQUAL BIN WIDTH

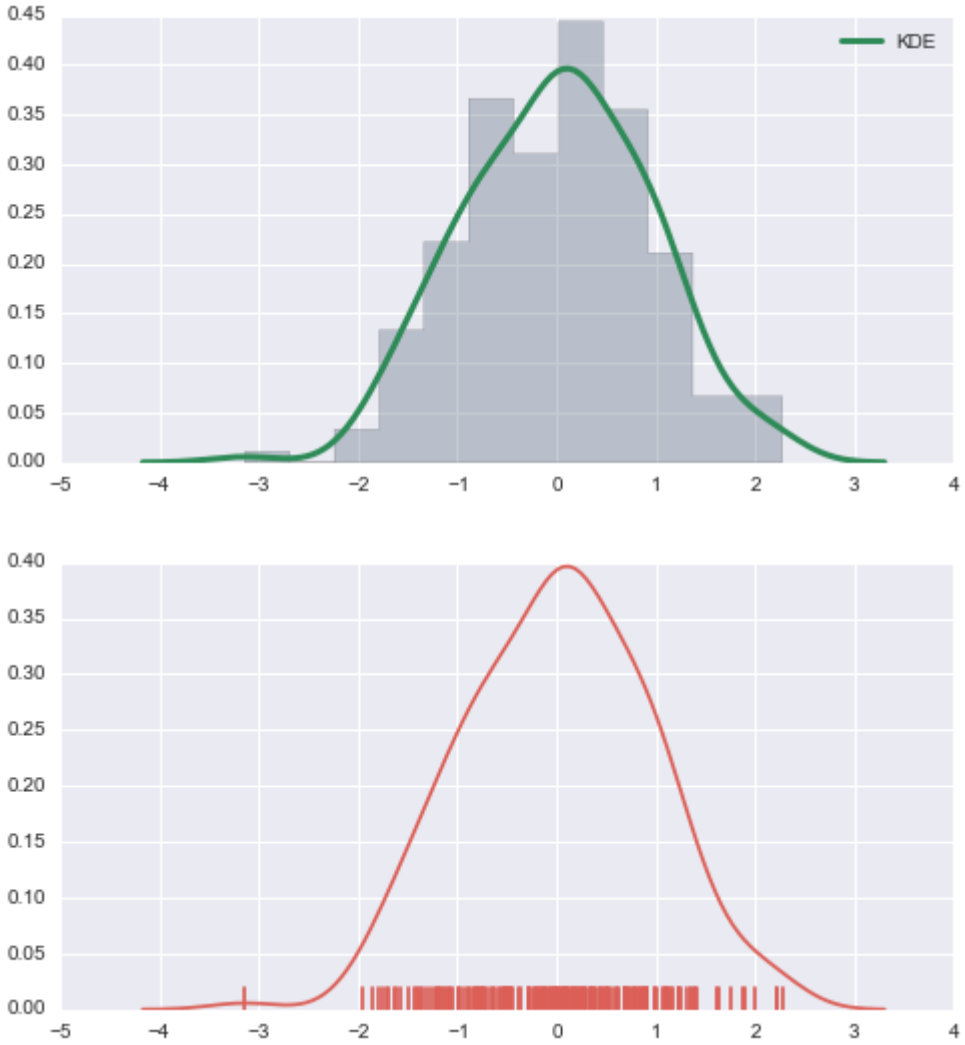


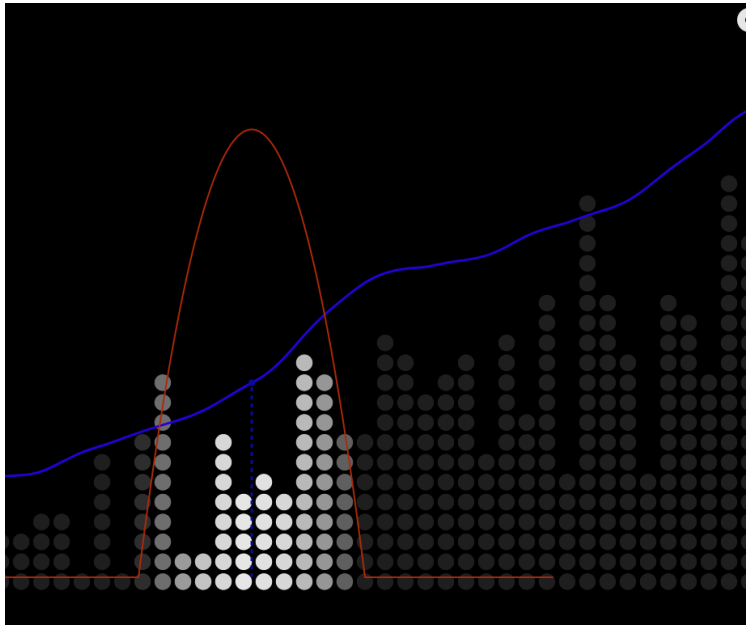


Can be useful if data is much sparser in some areas than others
Show density as area, not high

https://www.nytimes.com/interactive/2015/02/17/upshot/what-do-people-actually-order-at-chipotle.html?_r=1

DENSITY PLOTS (KERNEL DENSITY ESTIMATION)





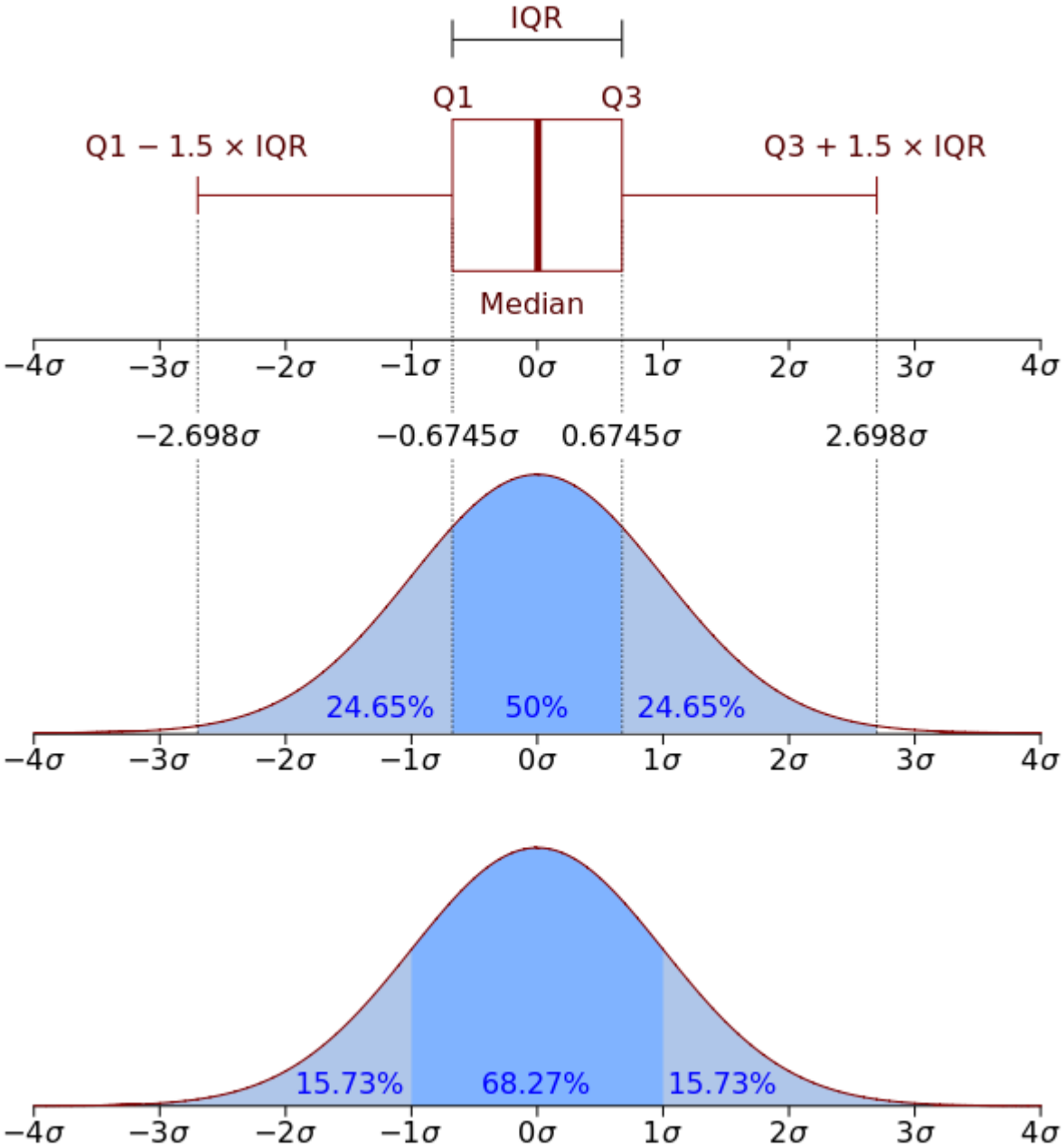
https://mathisonian.github.io/kde/

BOX PLOTS

aka Box-and-Whisker Plot

Show outliers as points!

Bad for non-normal distributed data
Especially bad for bi- or multimodal distributions



ONE BOXPLOT, FOUR DISTRIBUTIONS

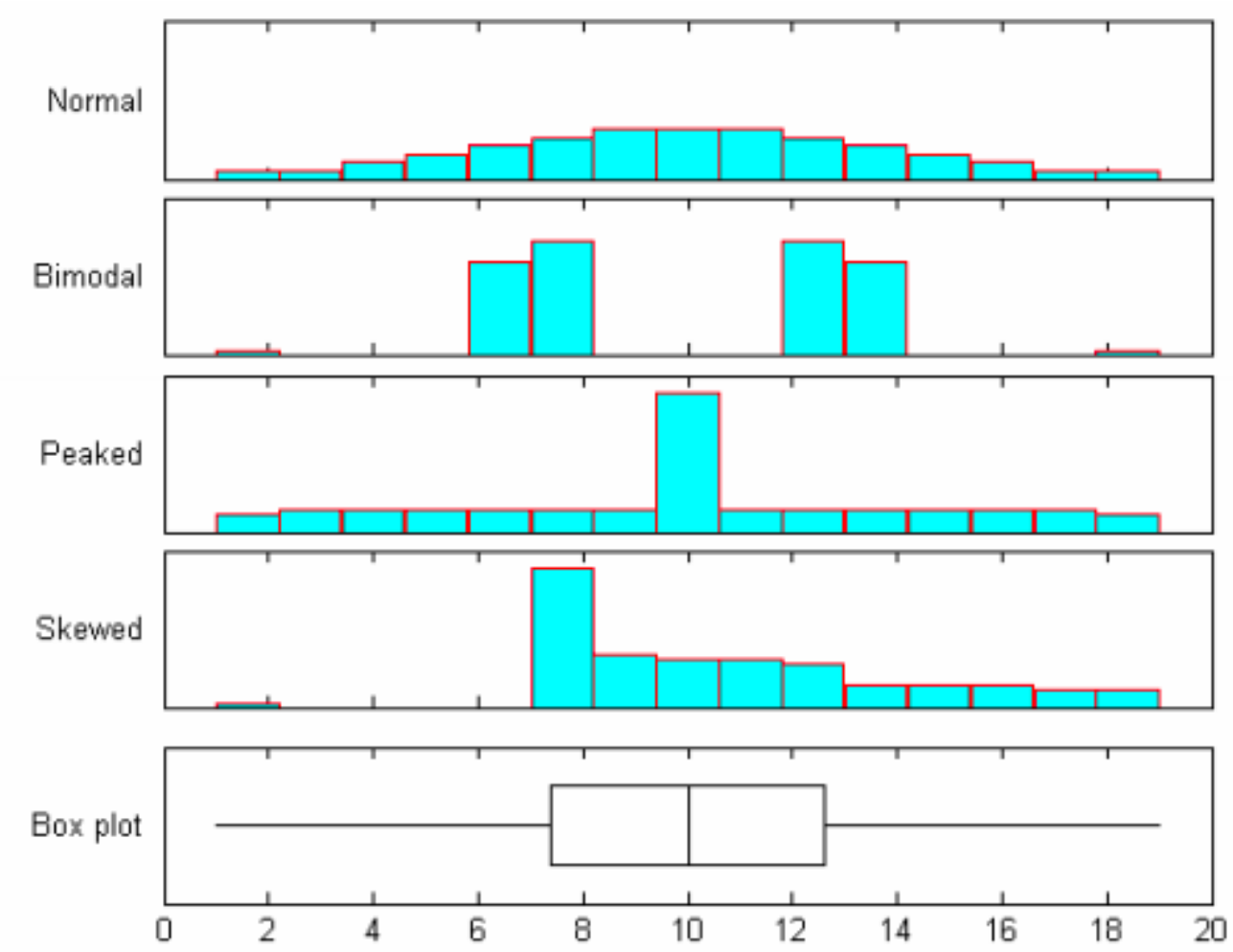


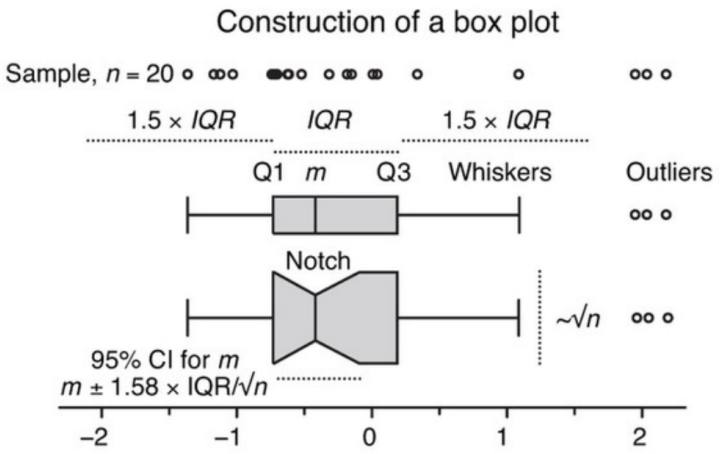
Figure 1: Histograms and box plot: four samples each of size 100

NOTCHED BOX PLOTS

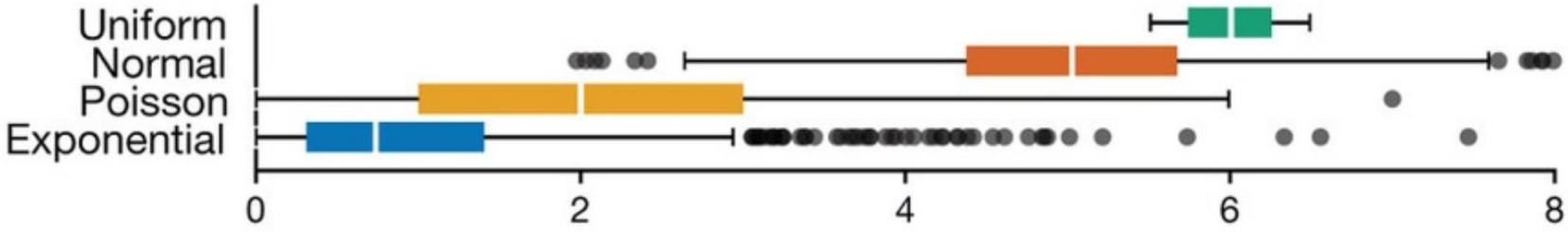
Notch shows
m +/- 1.5i x IQR/sqrt(n)

-> 95% Confidence Intervall

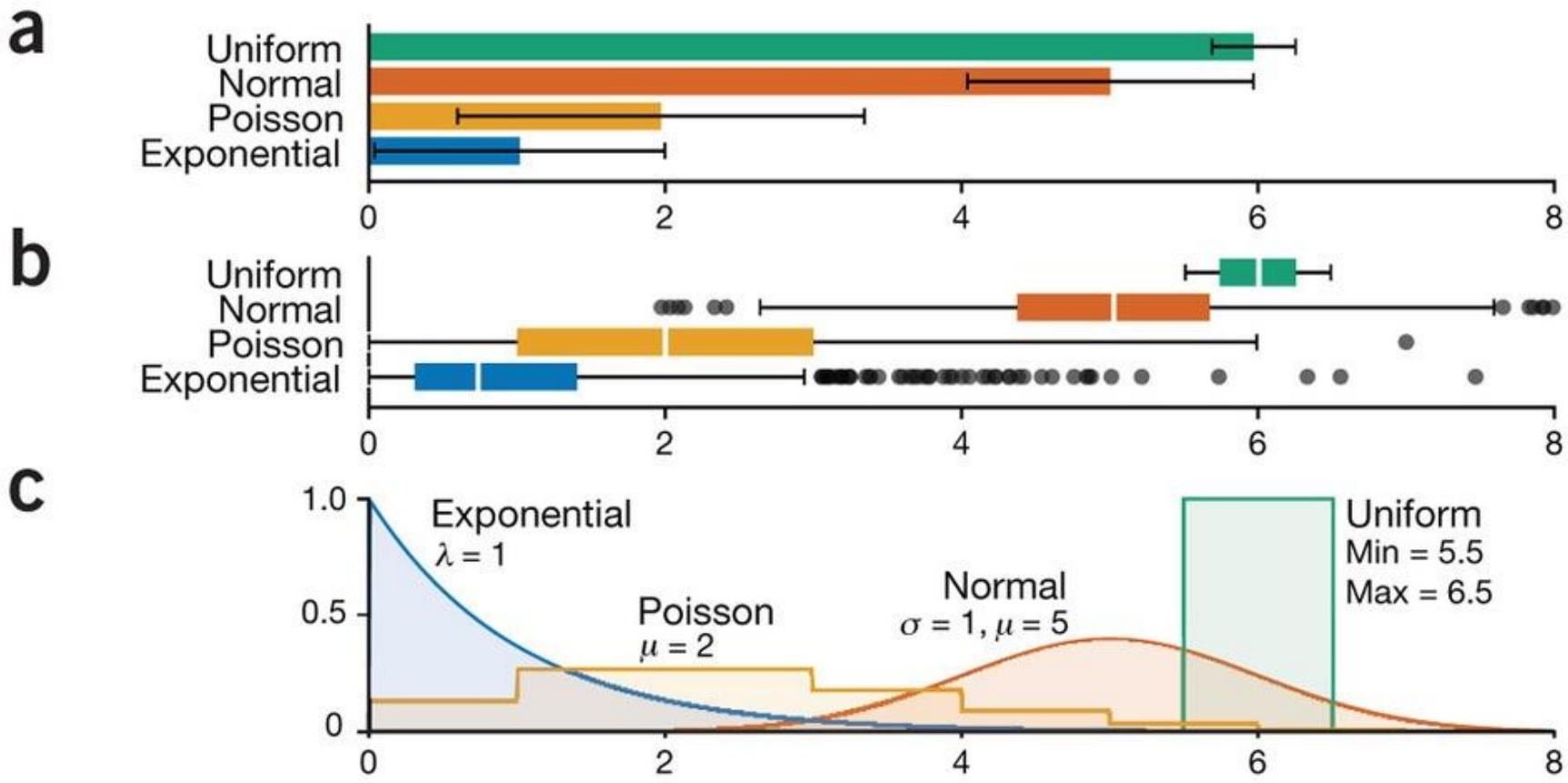
A guide to statistical significance.



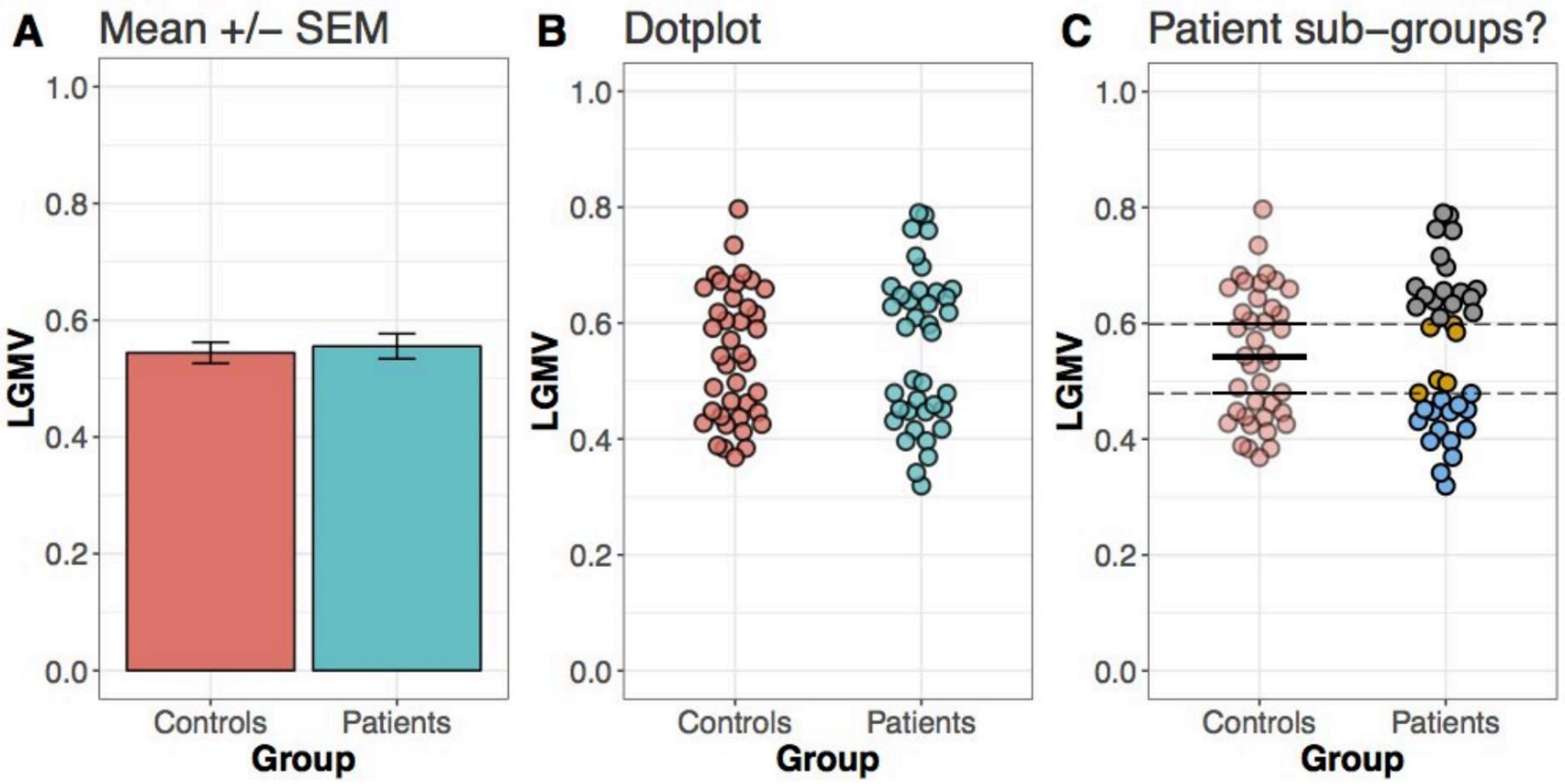
BOX PLOTS



COMPARISON

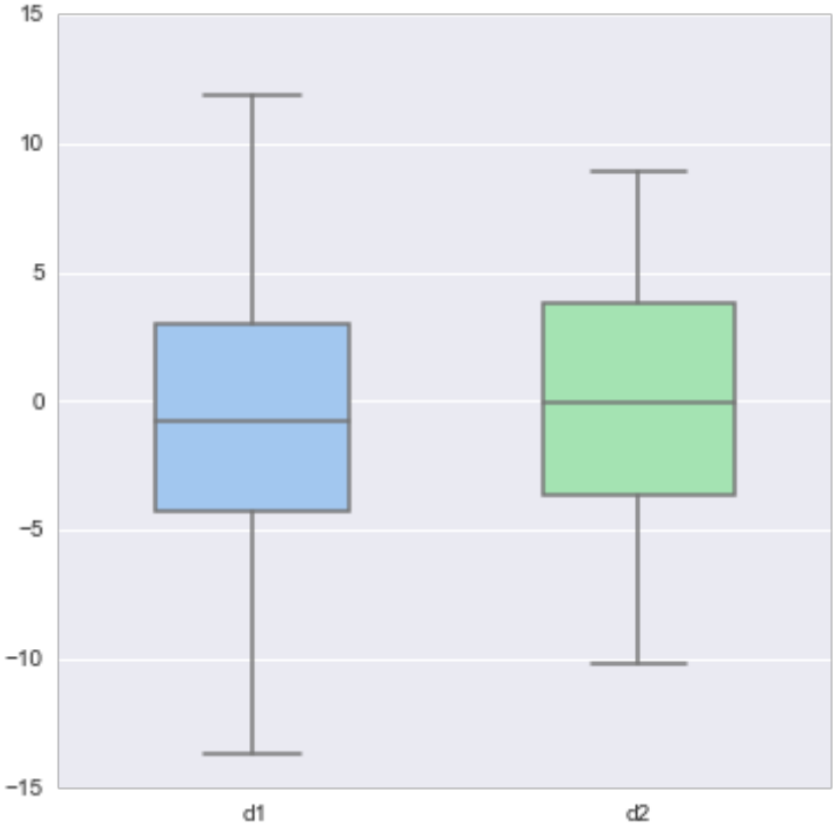

a
Uniform
Normal
Poisson
Exponential
0
2
4
6
8
b
Uniform
Normal
Poisson
Exponential
0
2
4
6
8
c
1.0
0.5
0
Exponential
λ = 1
Poisson
μ = 2
Normal
σ = 1, μ = 5
Uniform
Min = 5.5
Max = 6.5
0
2
4
6
8

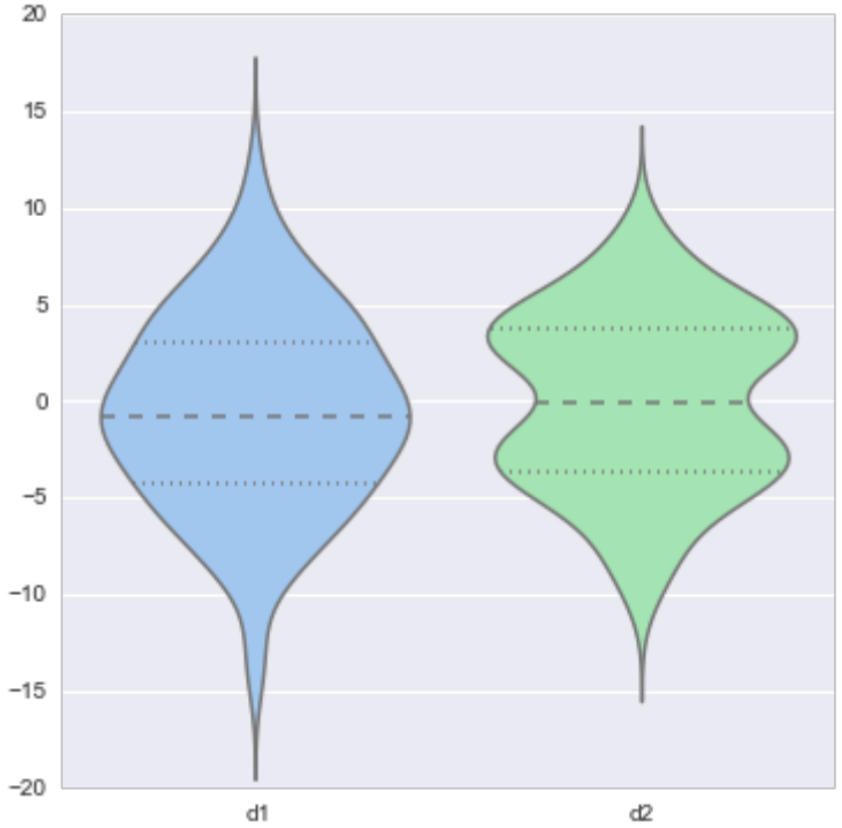
BAR CHARTS VS DOTPLOTS



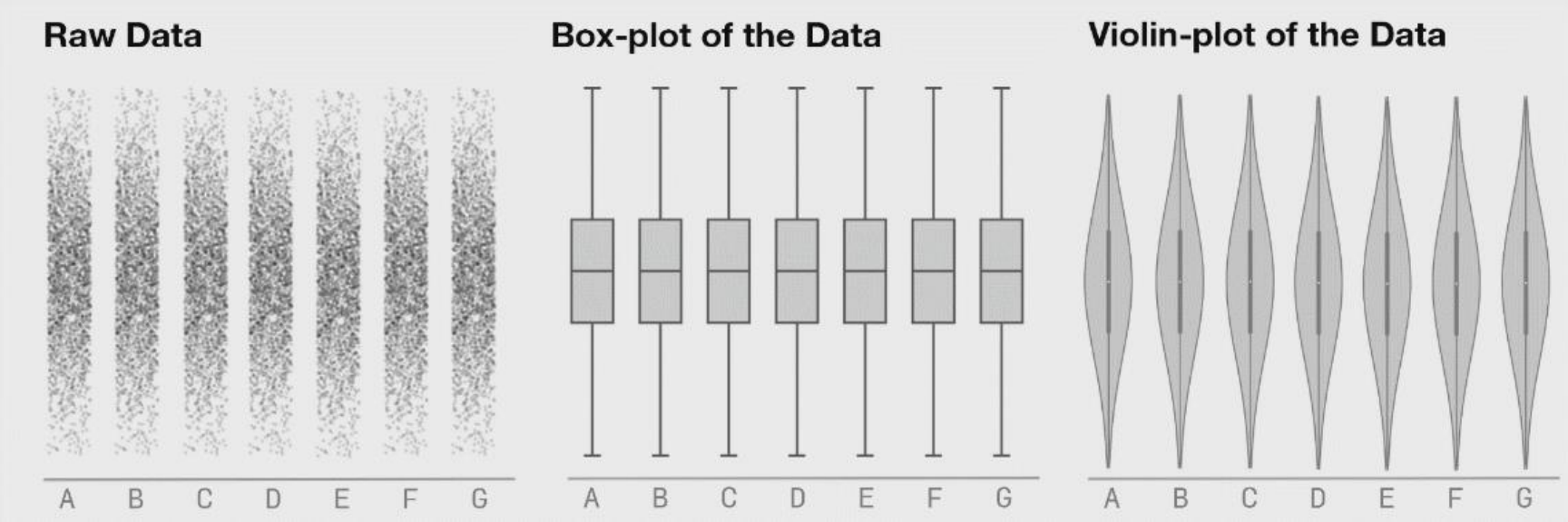
VIOLIN PLOT

Violin plot = box plot + probability density function





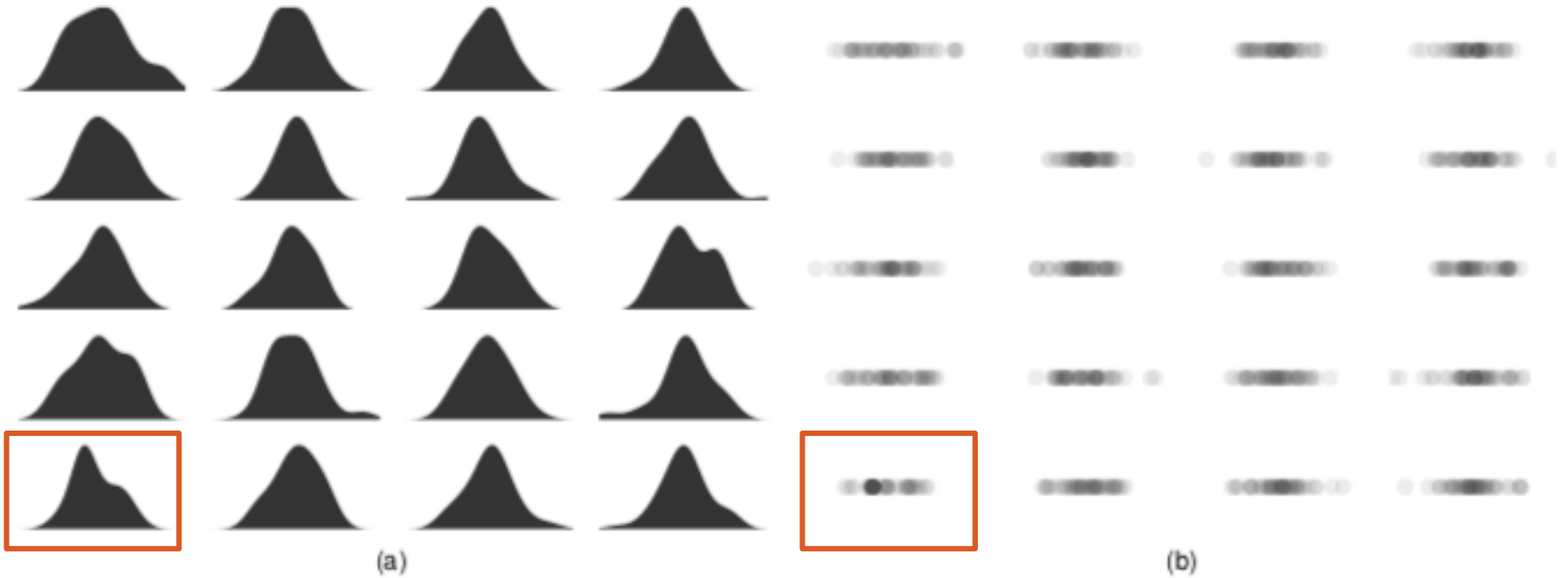
DIFFERENT DISTRIBUTIONS



https://blog.bioturing.com/2018/05/16/5-reasons-you-should-use-a-violin-graph/

DATA FLAWS

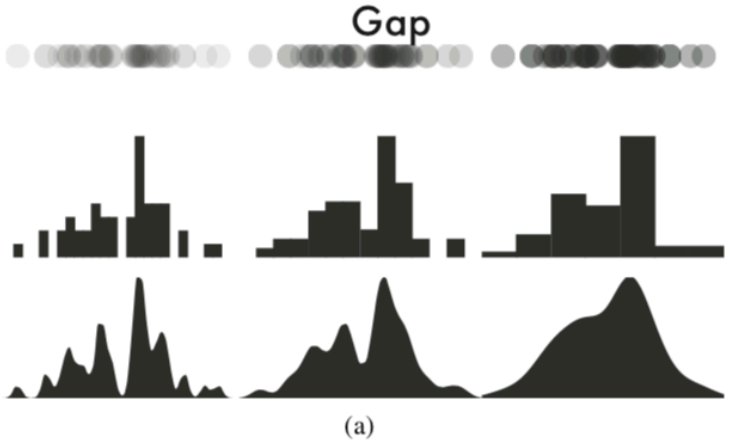
19 charts are random samples from a gaussian.
1 chart has 20% of samples with identical value



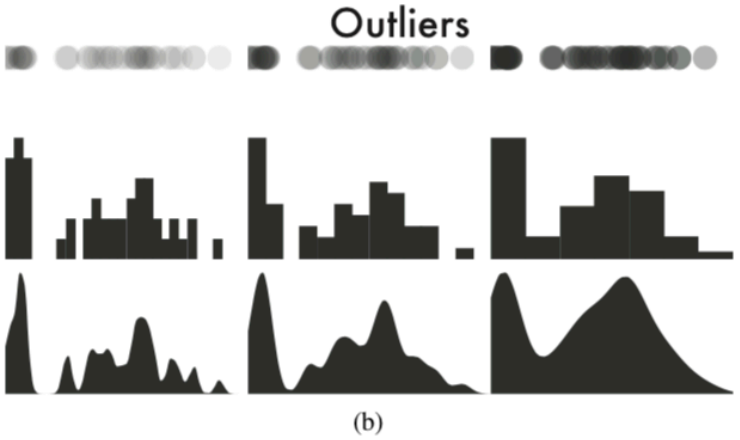
DETECTING DATA FLAWS

Tricky with aggregate visualization

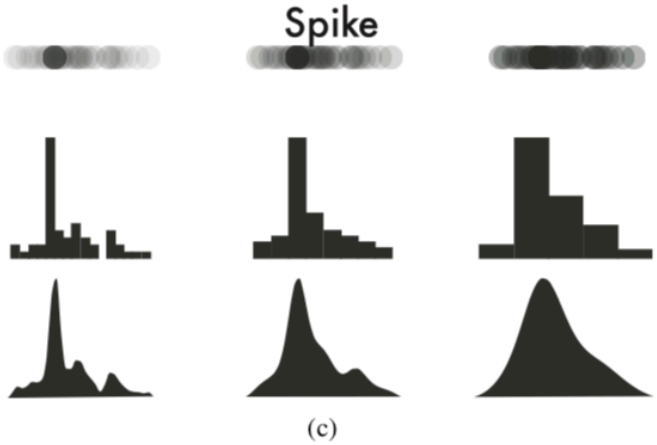
Bin size / kernel type / bandwidth / visualization choice all affect different situations



(a)



(b)

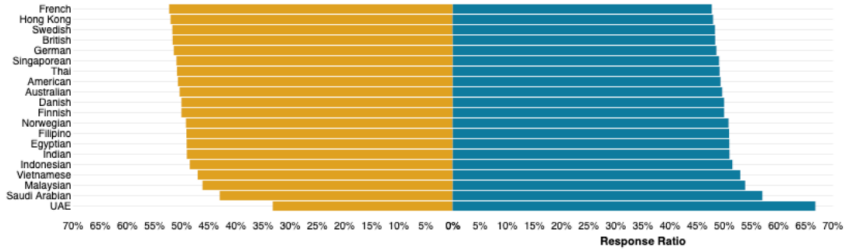


(c)

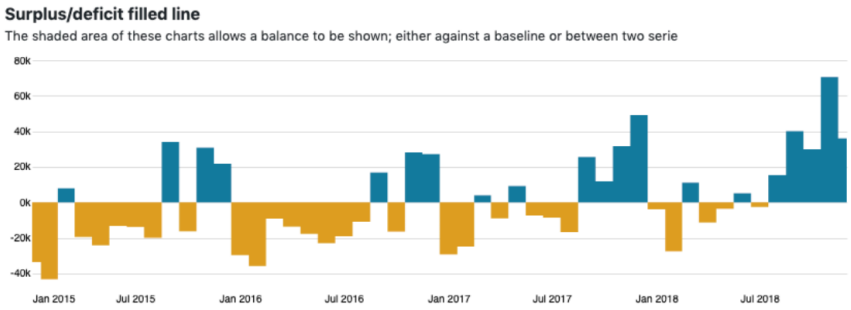
DEVIATION

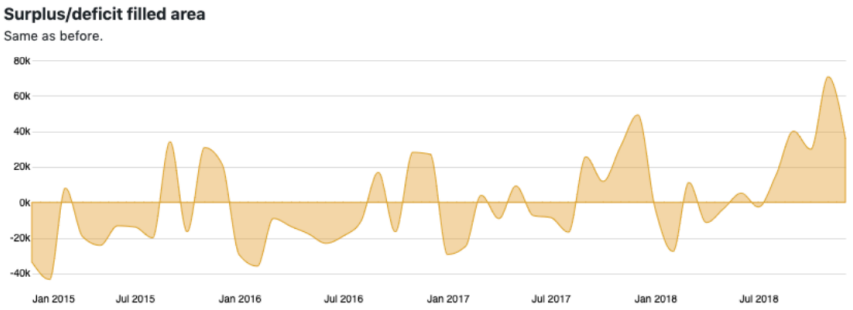
COMPARISON TO REFERENCE POINT

Diverging Bar Chart



Juxtaposing Two Variables (male/female)





CHANGE OVERTIME

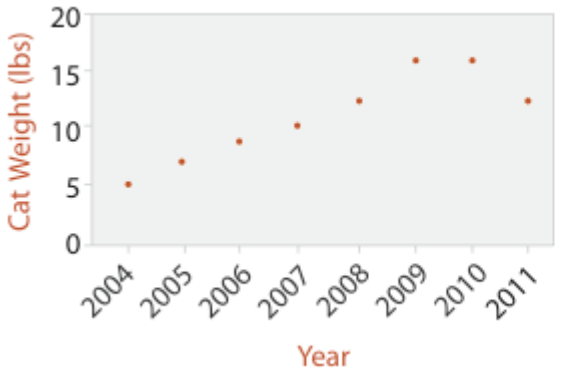
LINE CHART

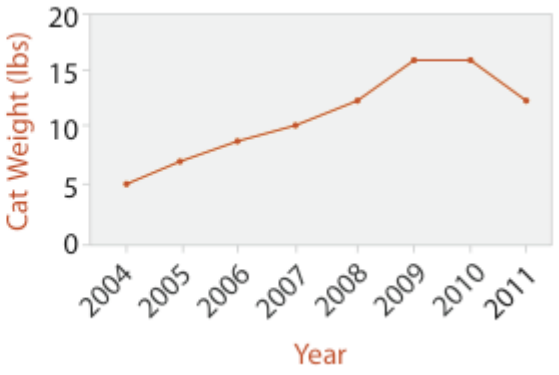
Simple

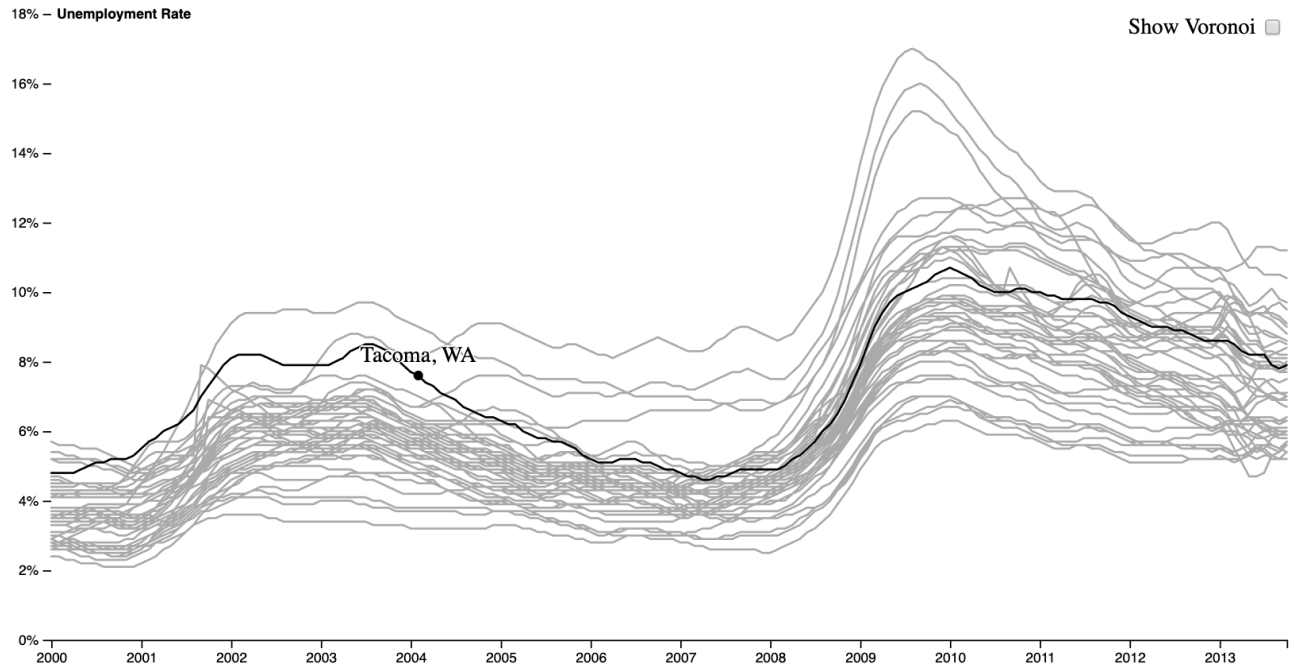
Familiar

Accurate

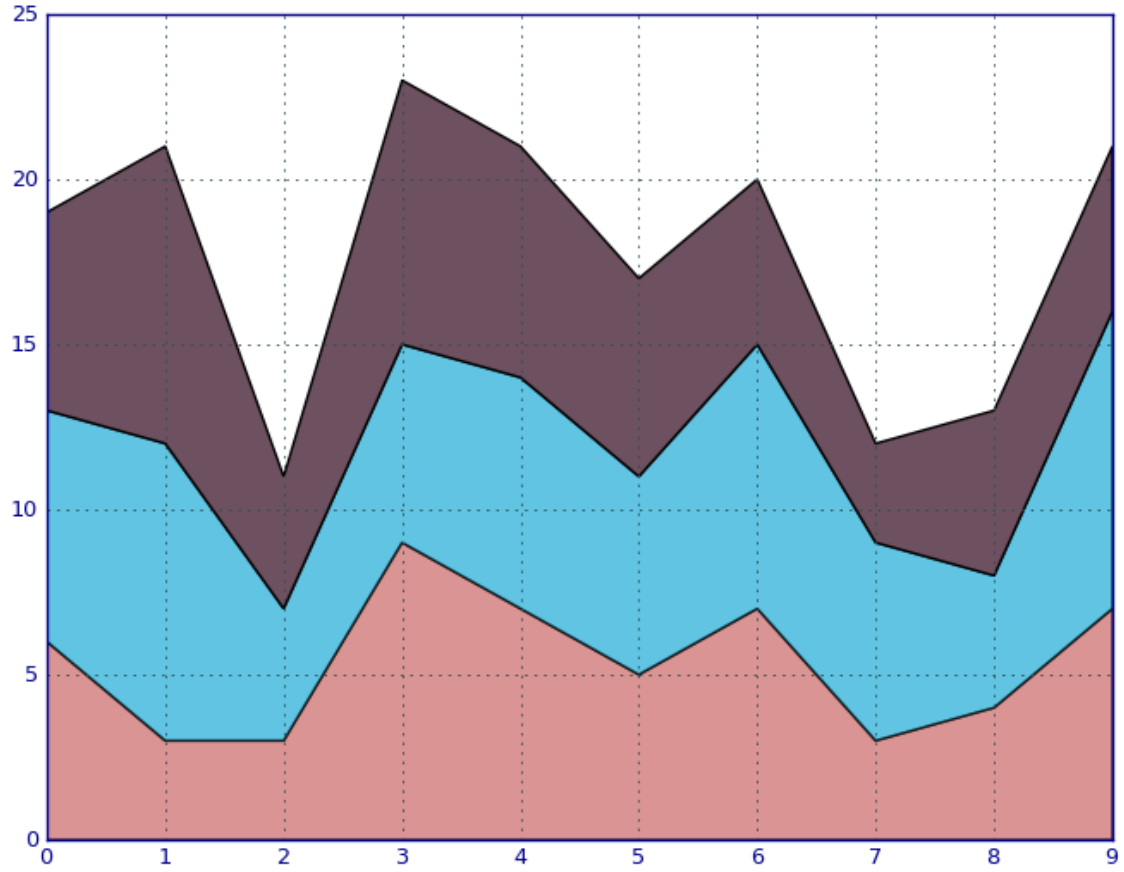
Fairly Scalable

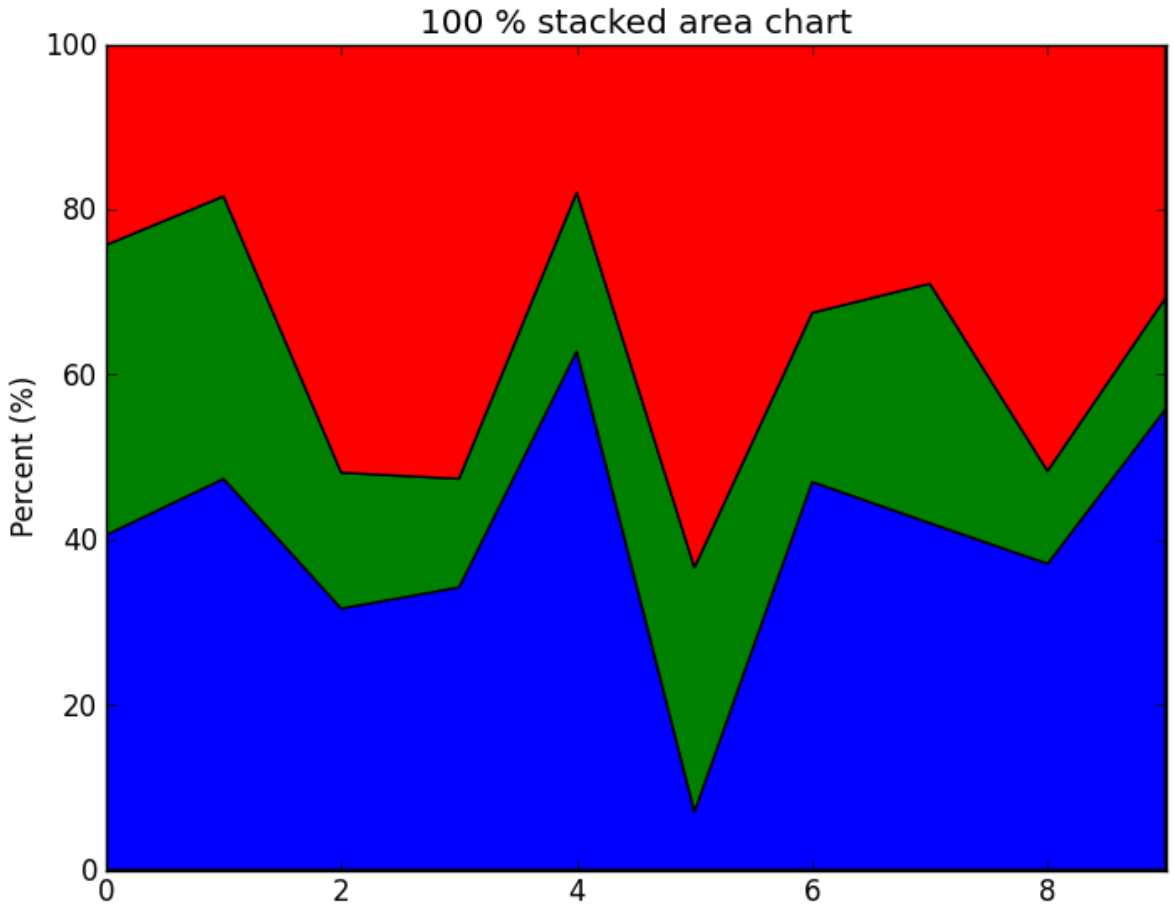




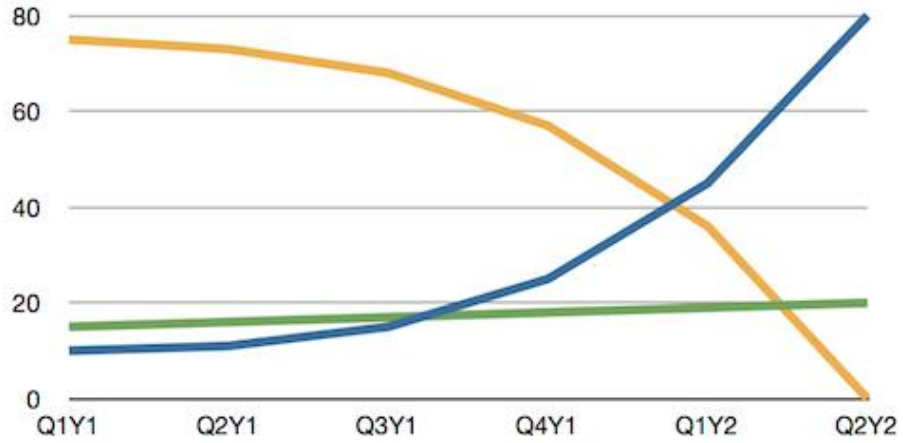


STACKED AREA

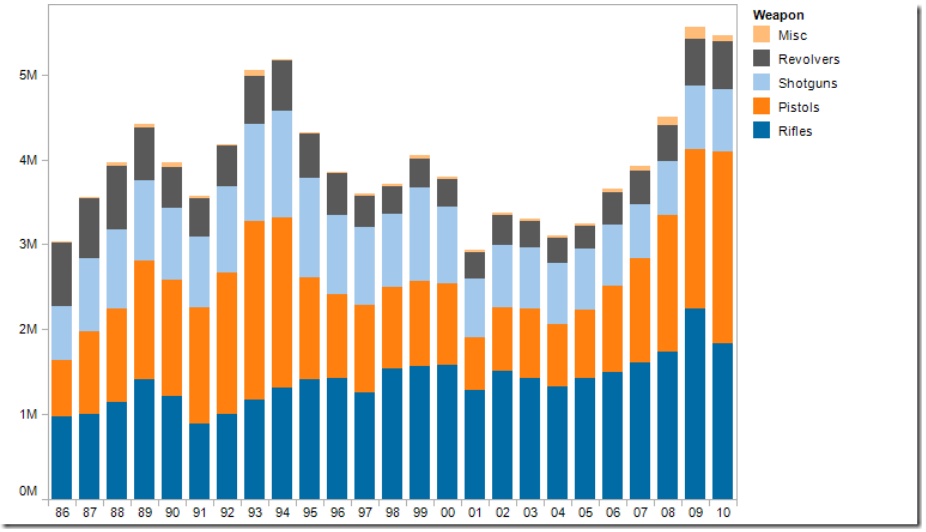


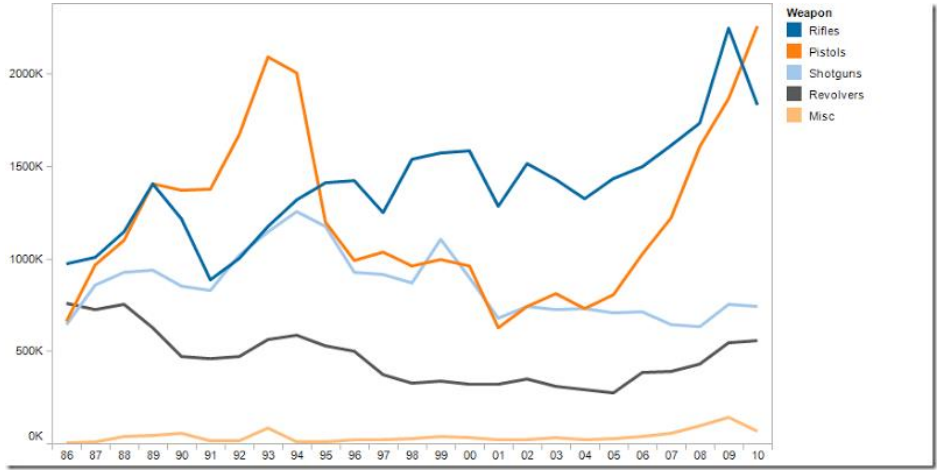

100 % stacked area chart
Percent (%)
100
80
60
40
20
0
0
2
4
6
8

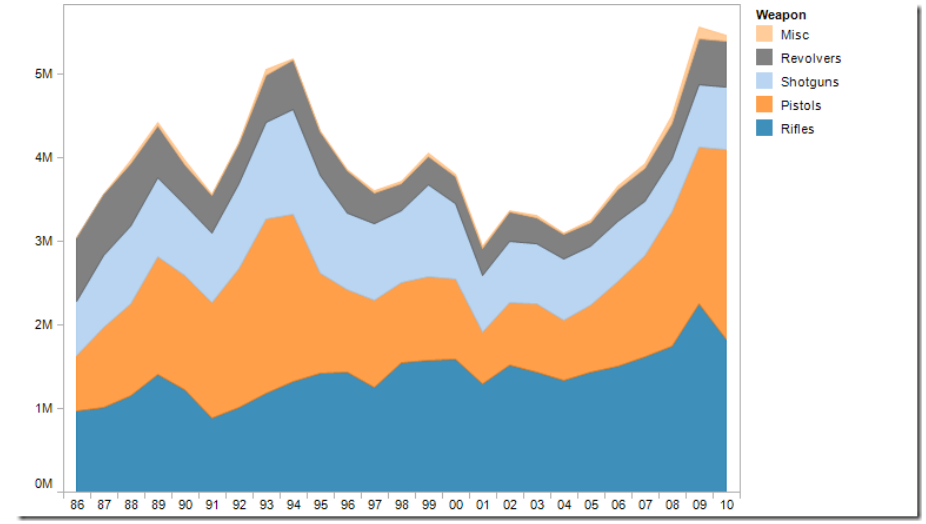
CAN YOU SPOT THE TREND?



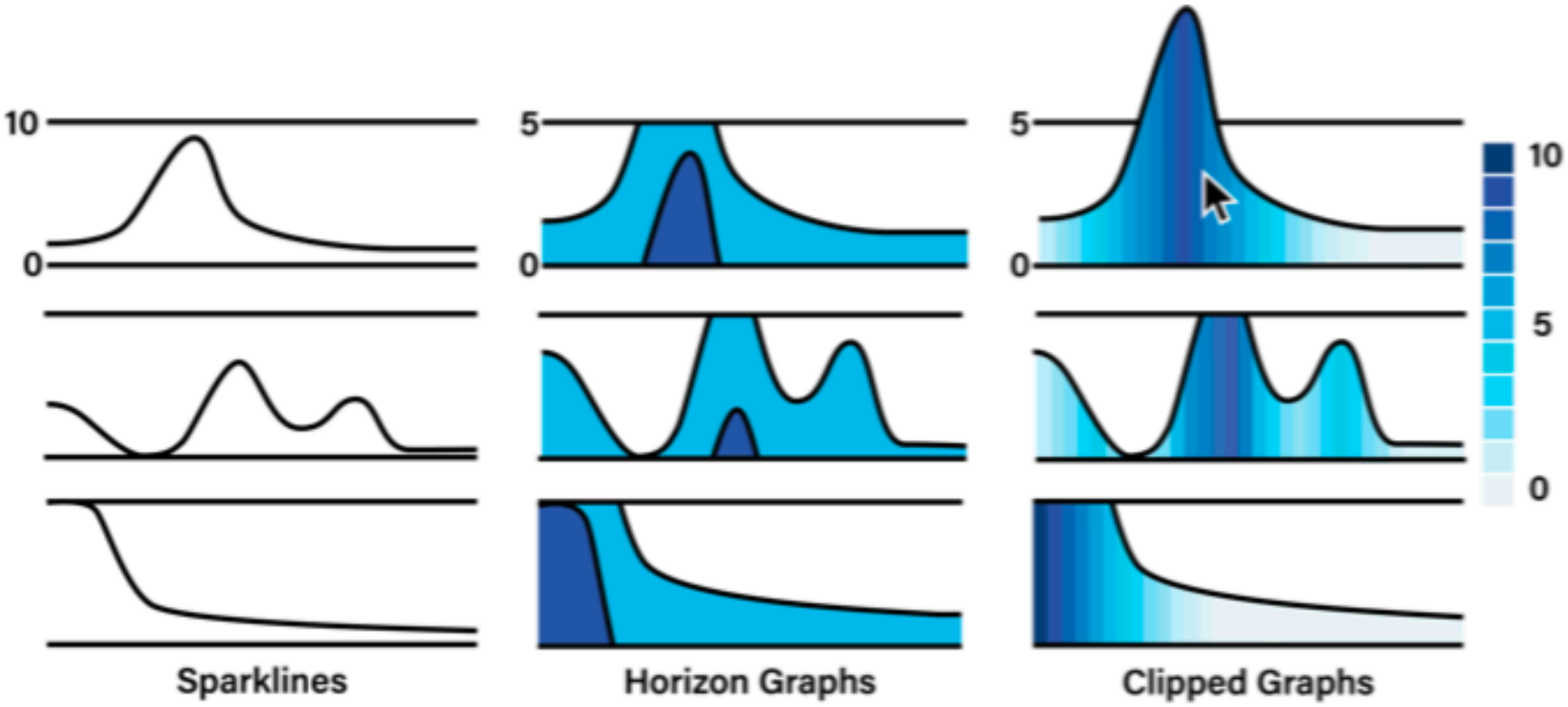
CAN YOU SPOT THE TREND?







MULTIPLE LINE CHARTS



SPARKLINES

Small line charts. can be embedded in text or part of a table

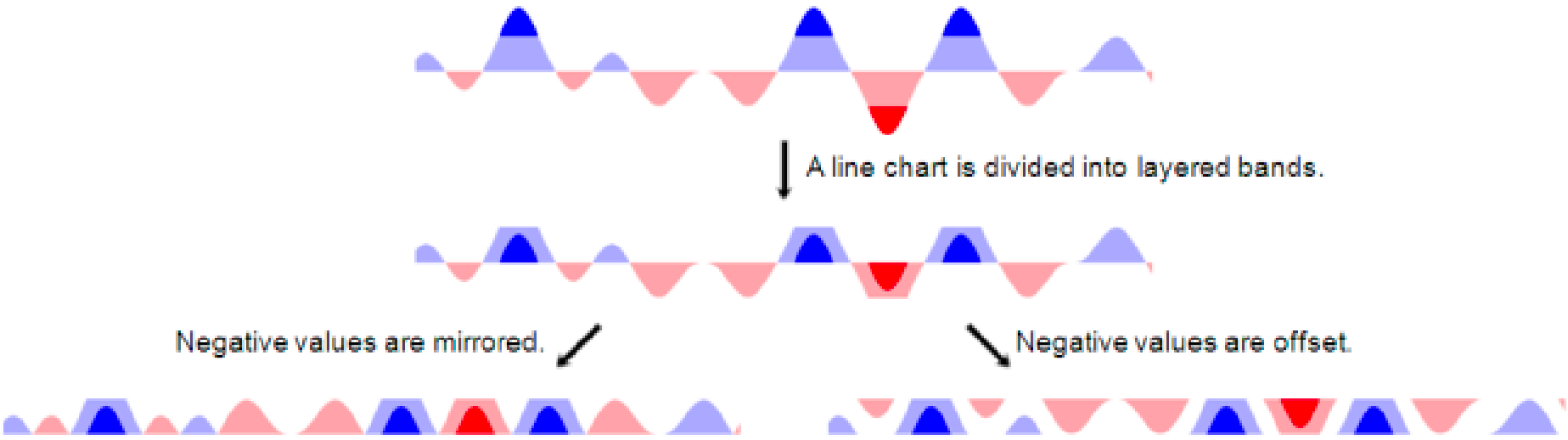
Symbol	Bid	Ask	Last	Change	T	Chart	Volume	High	Low	Value Change		Value	Gain	
DELL	89 3/4	89 13/16	89 3/4	+ 1 1/4			10,310,100	90 1/8	88 1/2	+1.41%	250	17,950	+273.72%	13,147
CPQ	48 7/16	48 9/16	48 7/16	- 13/16			25,628,700	51 1/4	1/4	-1.65%	-81	4,844	+60.79%	1,831
SDTI	26 1/4	26 3/8	26 3/8	+ 1/2			504,600	27 3/8	25 5/8	+1.93%	250	13,188	+133.15%	7,531
COMS	46 1/2	46 9/16	46 9/16	- 25/32			3,191,100	47 15/16	45 3/4	-1.65%	-102	6,053	+29.79%	1,389
LU	111 5/8	111 11/16	111 9/16	+ 1 9/16			5,104,600	112 5/8	110	+1.42%	78	5,578	+22.76%	1,034
YHOO	368 1/16	368 1/2	368 1/2	+ 17 1/4			3,787,800	381 3/16	280	+4.91%	431	9,213	-0.41%	-38
AOL	162 13/16	163	163	+ 8			10,008,500	164	158 1/2	+5.16%	280	5,705	+73.06%	2,408
CMGI	97 3/8	97 1/2	97 1/2	+ 5 7/8			1,323,800	98 1/2	93	+6.41%	705	11,700	+186.76%	7,620
SPLN	33 13/16	33 15/16	33 13/16	+ 7/16			300,200	34 3/4	33 5/8	+1.31%	88	6,763	+94.60%	3,288
BEAS	13 1/2	13 5/8	13 5/8	- 7/16			389,200	14 1/4	13 1/8	-3.11%	-44	1,363	-9.17%	-138
GNET	102	103 3/16	101 5/16	+ 6 1/8			307,600	108	97	+6.43%	613	10,131	+130.26%	5,731
RNWK	67	67 1/4	67	+ 2 3/4			1,233,900	69	64 15/16	+4.28%	275	6,700	+79.87%	2,975
MSFT	173 1/8	173 1/4	173 5/16	+ 1 3/4			13,284,500	174 7/16	170	+1.02%	175	17,331	+54.74%	6,131
INTC	133 3/4	133 13/16	133 13/16	- 3 1/8			8,094,300	137 1/2	133 3/8	-2.28%	-625	26,763	+65.20%	10,563
TOTAL								205,302	80,993	+1.63%	2,293	143,280	+79.41%	63,377

Mauricio Pochettino has lead Spurs on their best run 8TH 2ND in 24 years of the Premier League

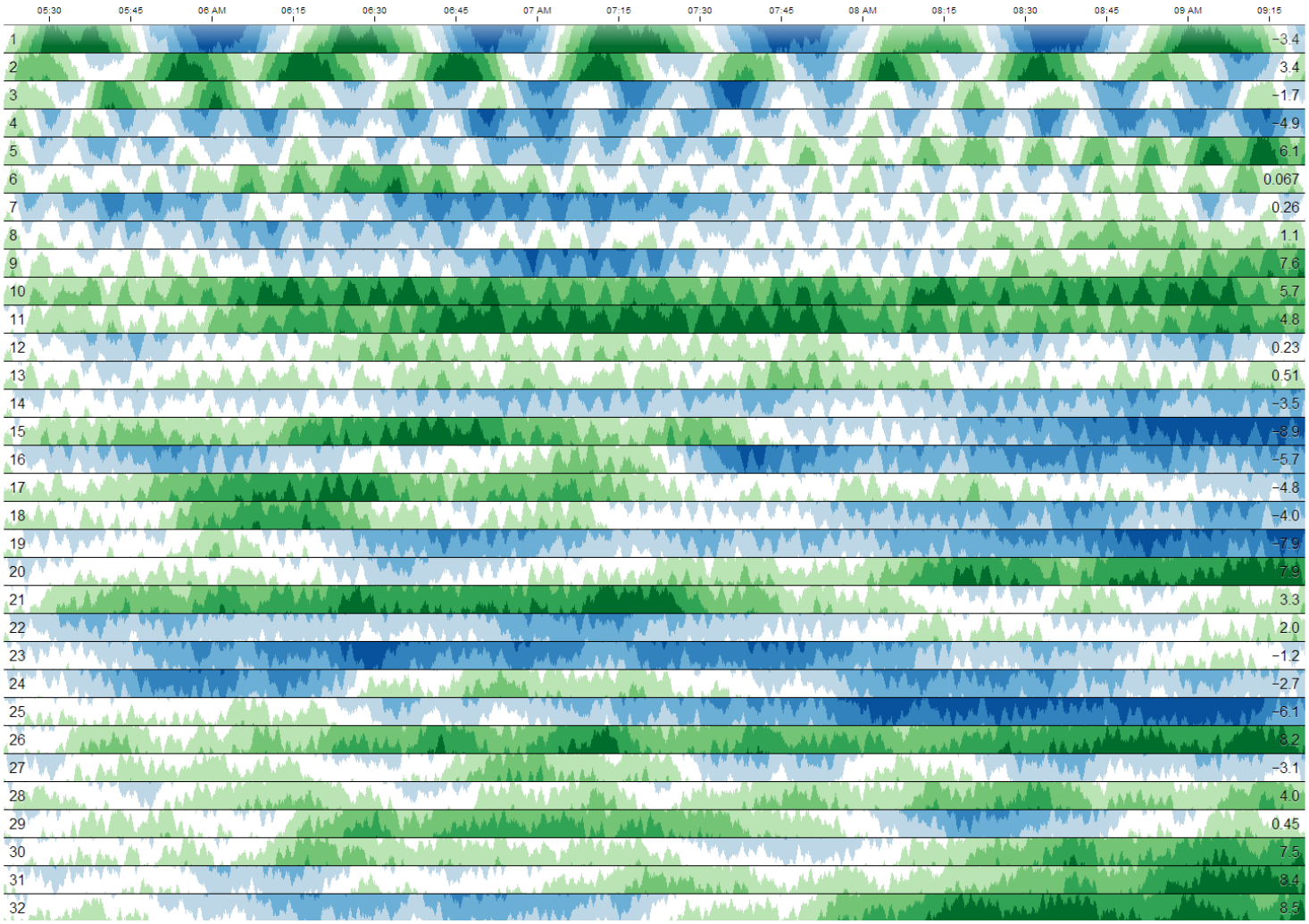
Alibaba stock is at 5 yr high 93.89 152.11 as of July 2017

The FTSE100 Brexit bounce 5562 7501 continues one year on from the vote last summer

HORIZONTAL GRAPH

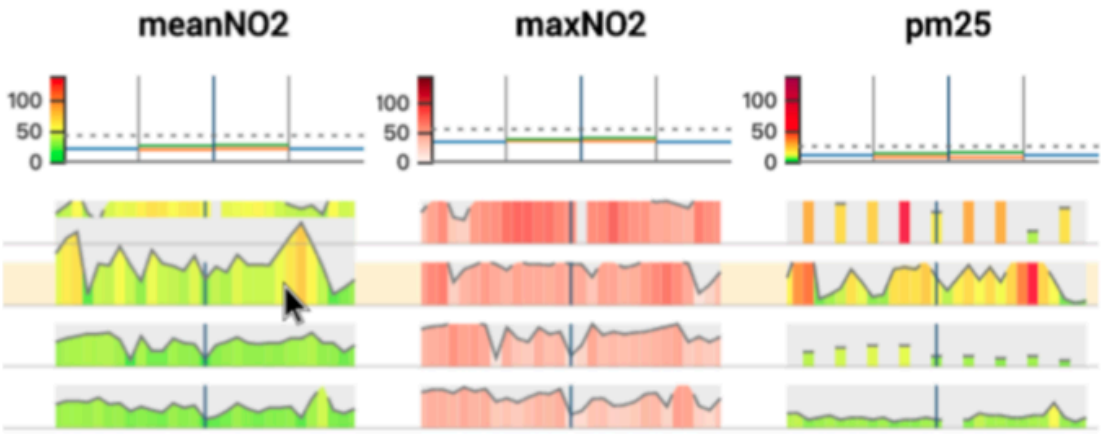


HORIZONTAL GRAPH

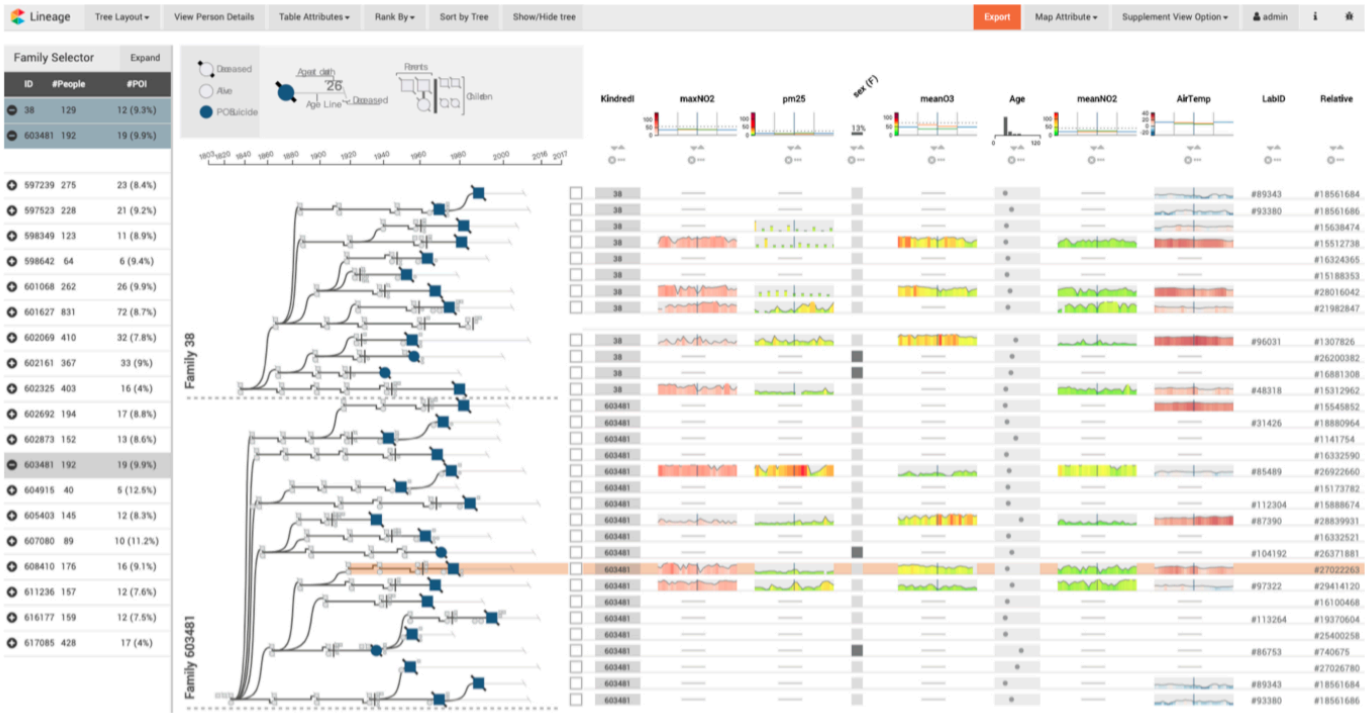


http://square.github.io/cubism/

CLIPPED GRAPHS



https://youtu.be/IxoaFe6yBMg

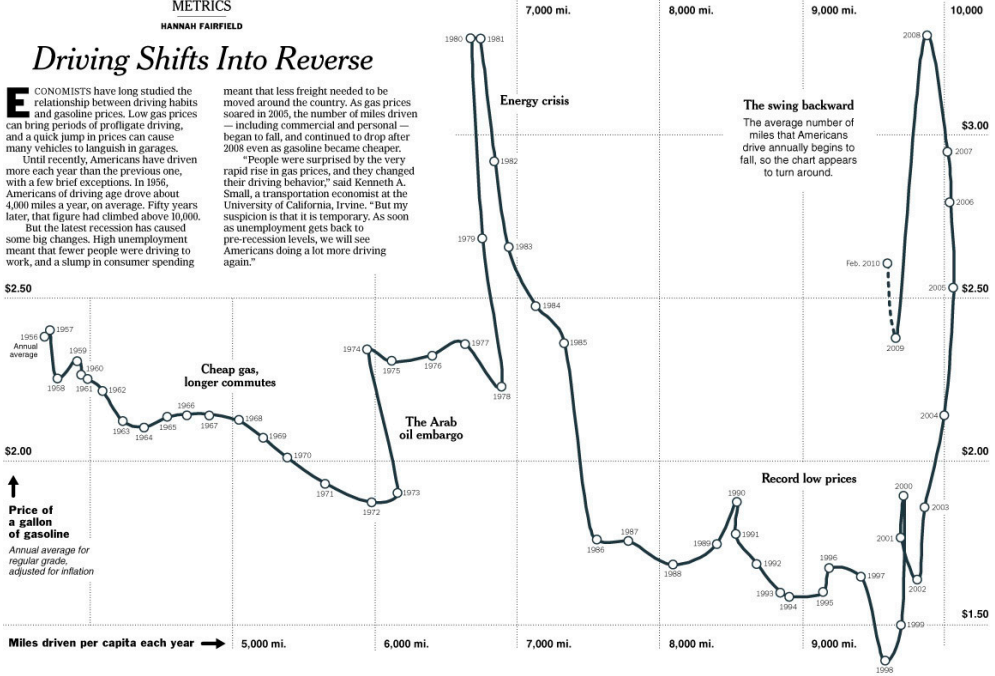


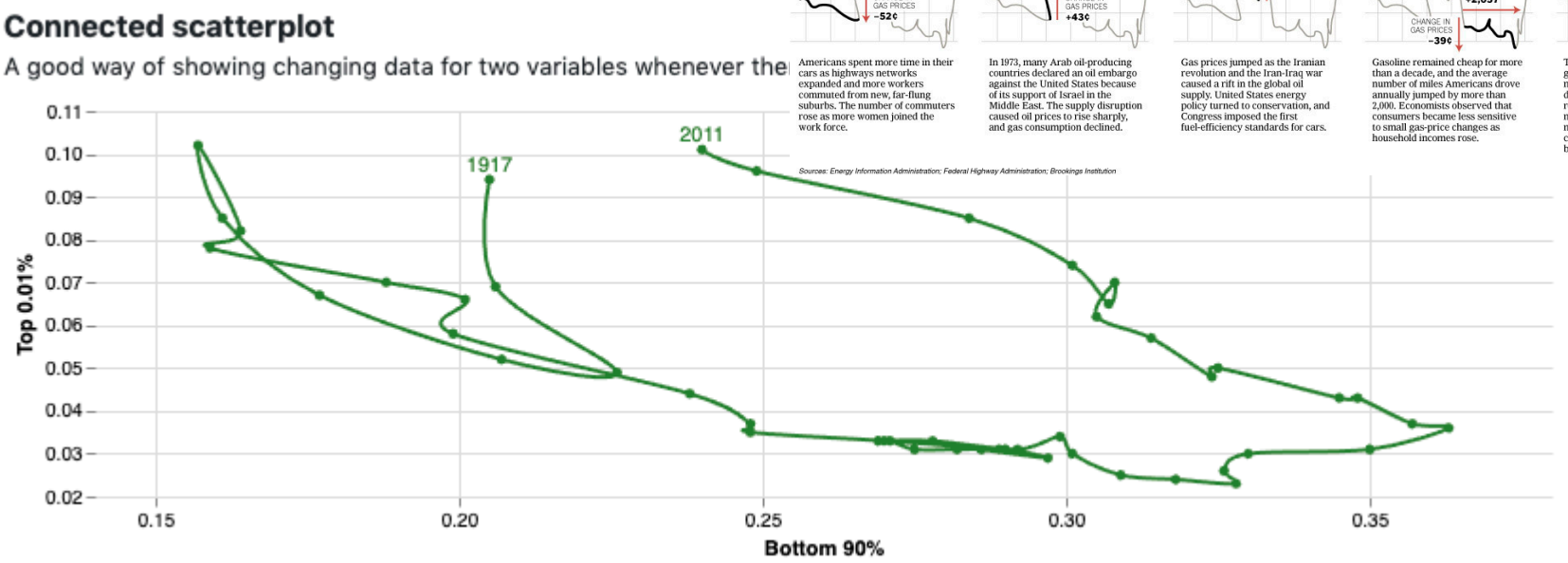
CONNECTED SCATTERPLOT

Two Variables + Time
Only one per Chart!
Labels important

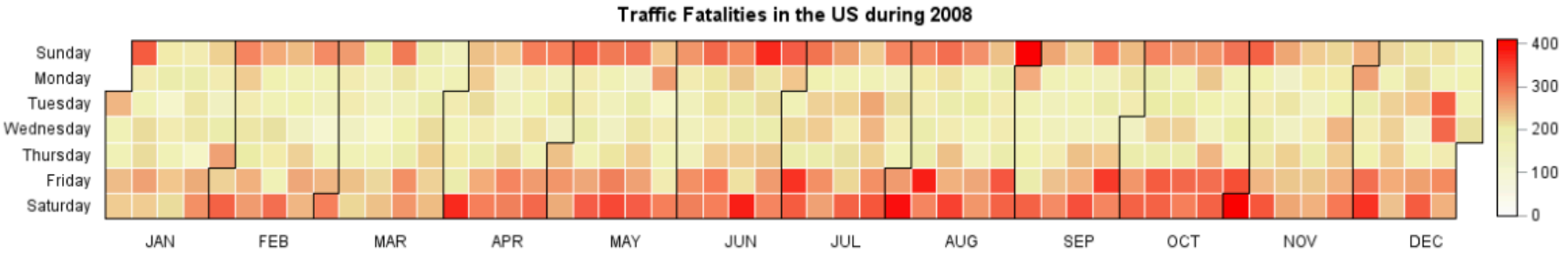
http://steveharoz.com/research/connected_scatterplot/

http://www.thefunctionalart.com/2012/09/in-praise-of-connected-scatter-plots.html



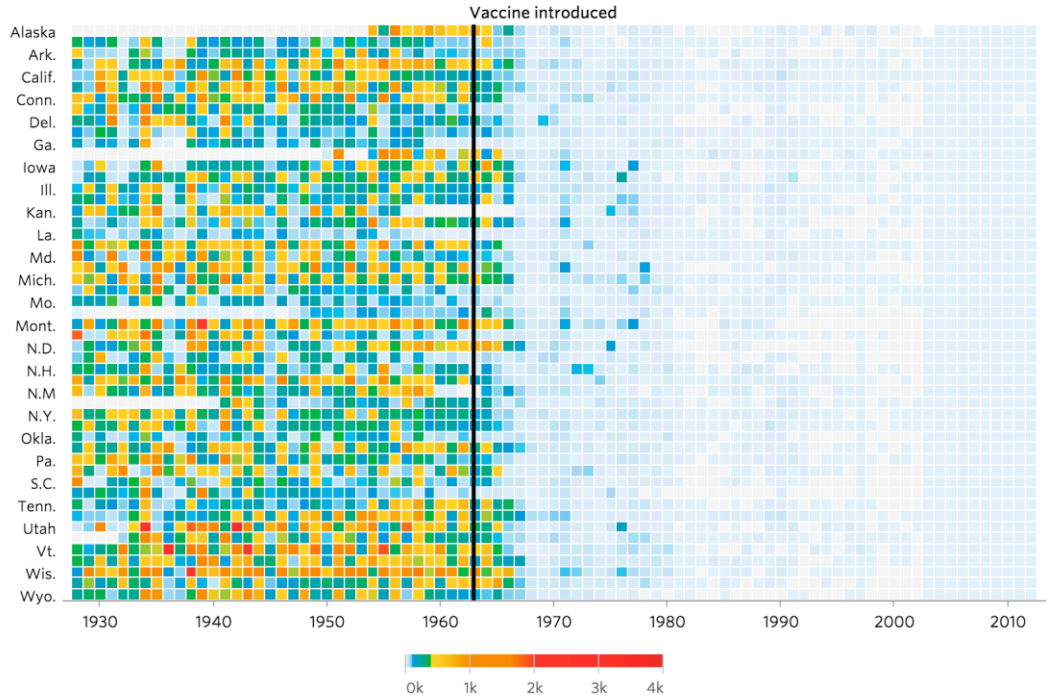


HEATMAP



The heat maps below show number of cases per 100,000 people.

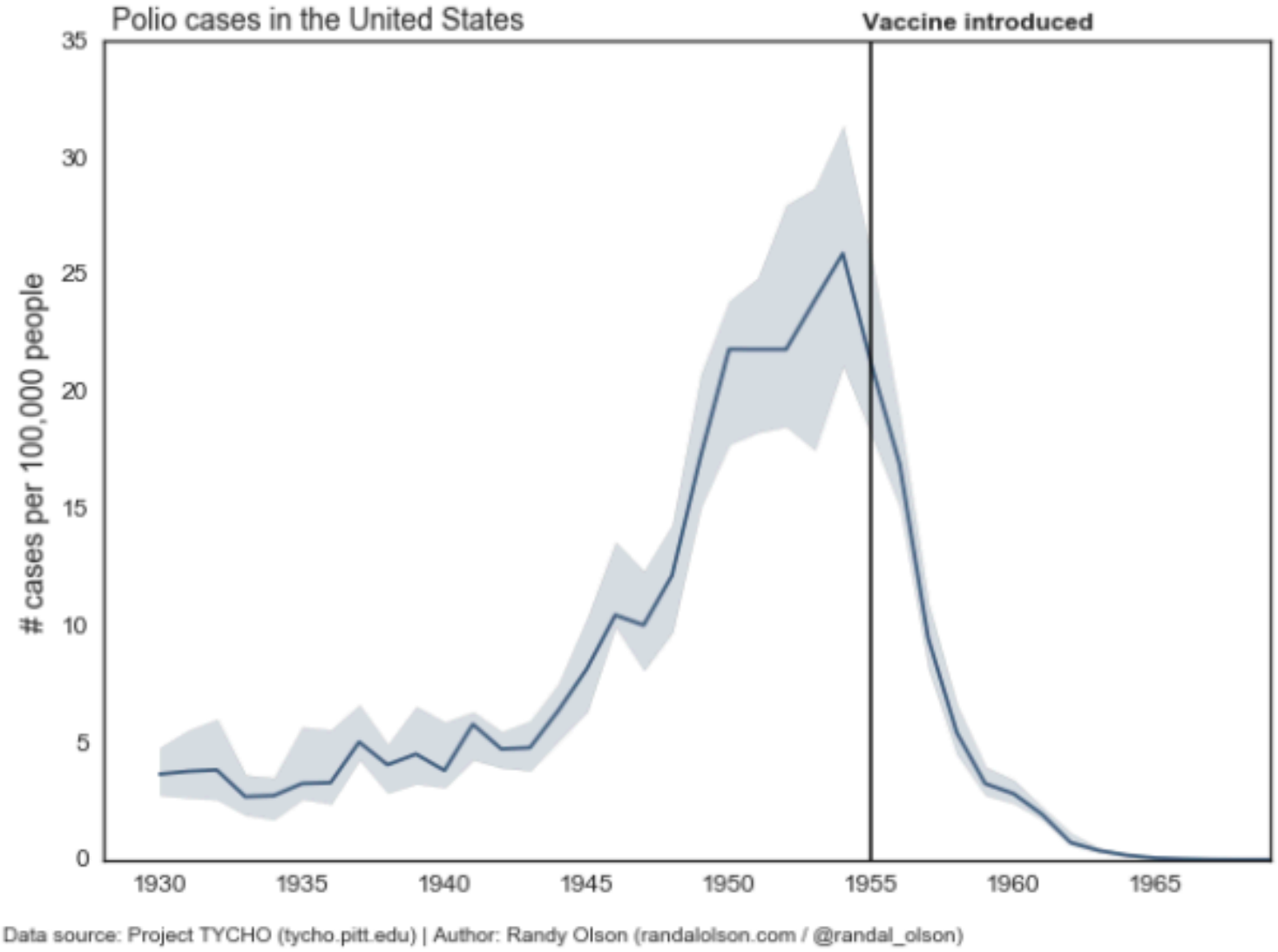
Measles



Note: CDC data from 2003-2012 comes from its Summary of Notifiable Diseases, which publishes yearly rather than weekly and counts confirmed cases as opposed to provisional ones.

https://www.informationisbeautifulawards.com/showcase/660-vaccines-and-infectious-diseases

SOMETIMES YOU CAN SHOW TOO MUCH DATA



http://www.randalolson.com/2016/03/04/revisiting-the-vaccine-visualizations/

RANKING

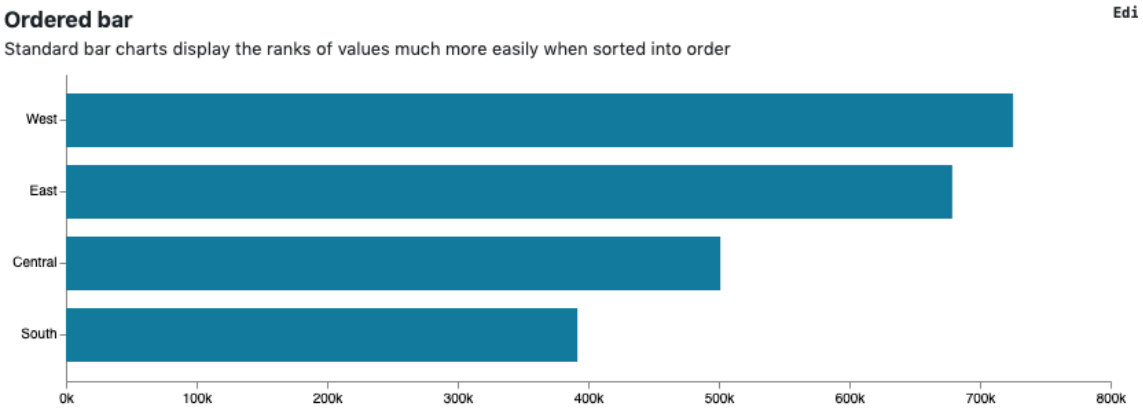
RANKING EXERCISE

	1	2	3	4	5	6	7
Bavaria	8	6	2	4	2	**1**	3
Dortmund	**1**	**1**	5	2	3	8	8
Leipzig	2	2	**1**	**1**	**1**	2	4
Leverkusen	5	5	4	8	7	6	7
Moenchengladbach	10	7	8	7	6	5	**1**
Wolfsburg	6	4	3	5	8	7	2

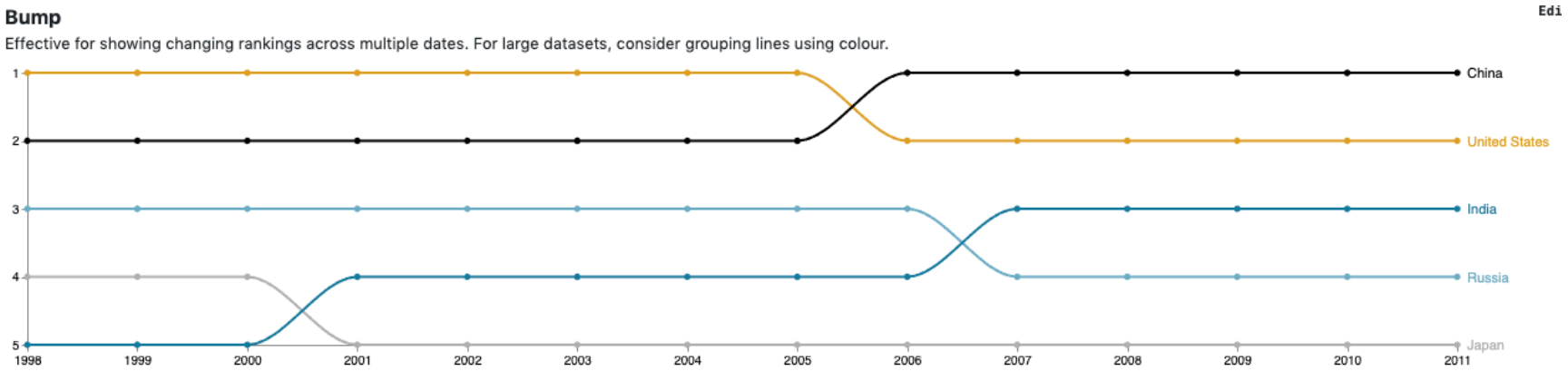
Design a visualization showing the ranking of these football clubs over time.

RANKING

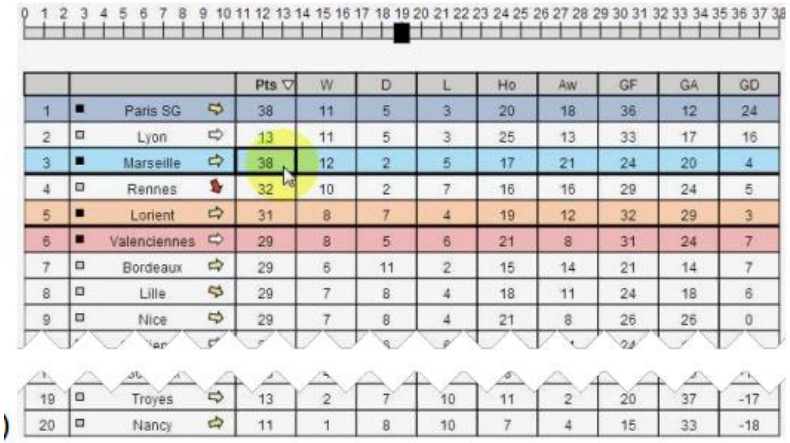
Magnitude Visualization + Sorting



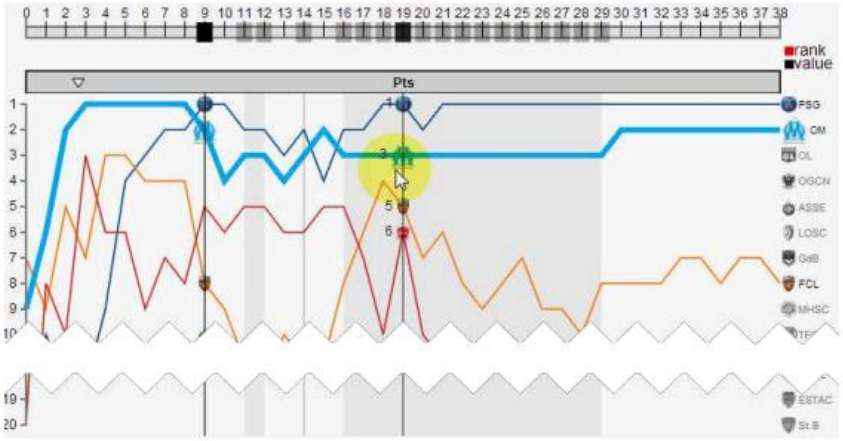
Bump Charts for Rankings over Time



TEMPORAL RANKINGS



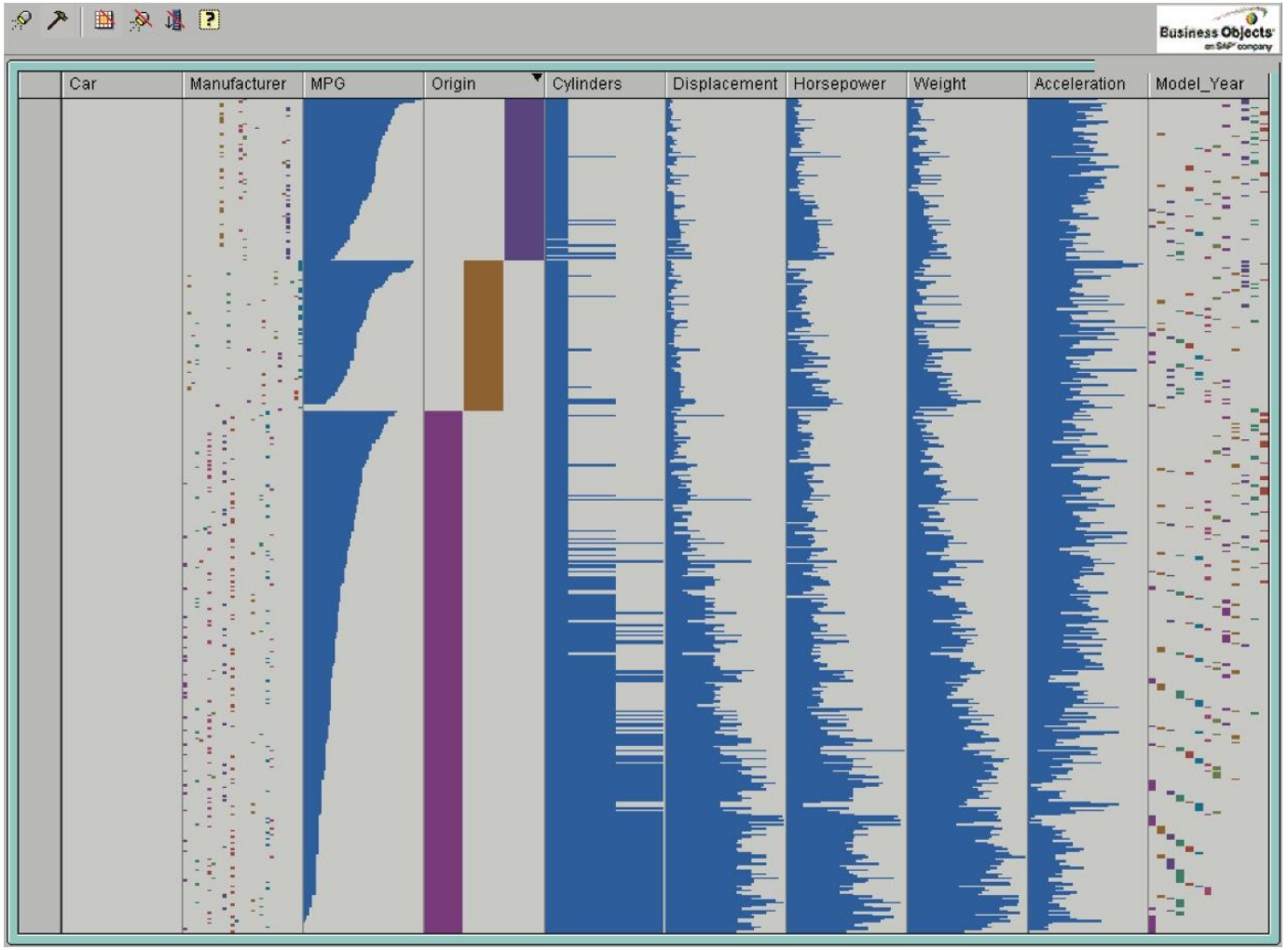
(b)



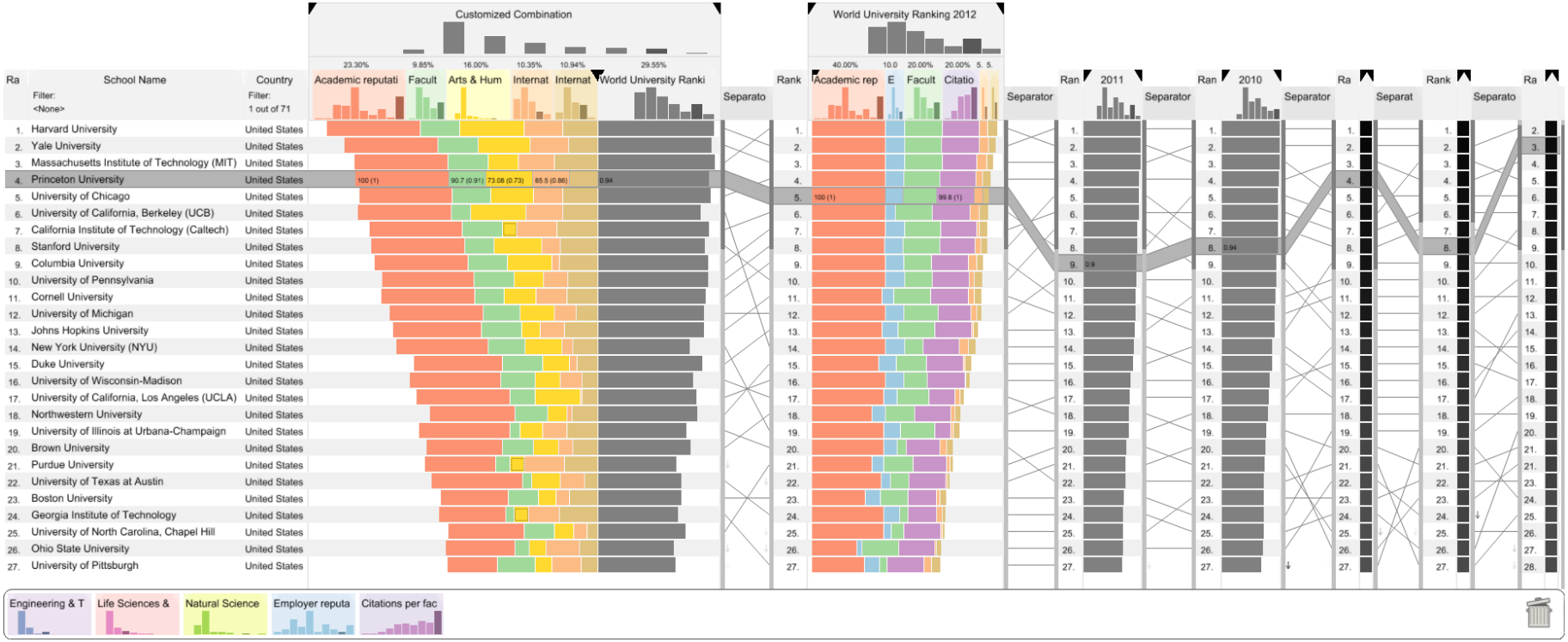
https://youtu.be/a0duFWu3Zp0

TABLE LENS

Interactive table based representation



LINEUP



http://lineup.caleydo.org

RANKING

Rank	University	Score
1.	MIT	
2.	Harvard	
3.	Princeton	
4.	Cambridge	
5.	Oxford	

MULTIPLE ATTRIBUTES

COMBINER FUNCTION

(Weighted) sum

Score = $w_a A + w_b B + w_c C$ → **Serial**

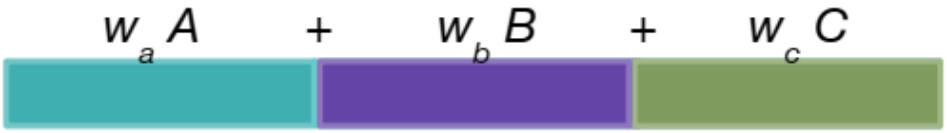
Maximum

Score = max (A, B, C) → **Parallel**

Product → **Complex**

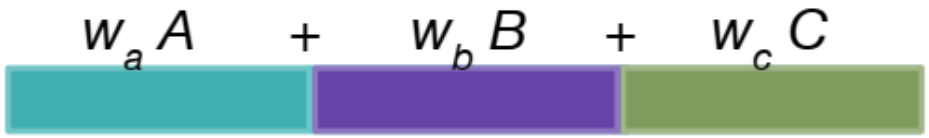
Nesting

Serial Combiner (as Stacked Bar)



Rank	University	A	B	C
1.	MIT			
2.	Harvard			
3.	Princeton			
4.	Cambridge			
5.	Oxford			

Serial Combiner (as Stacked Bar)

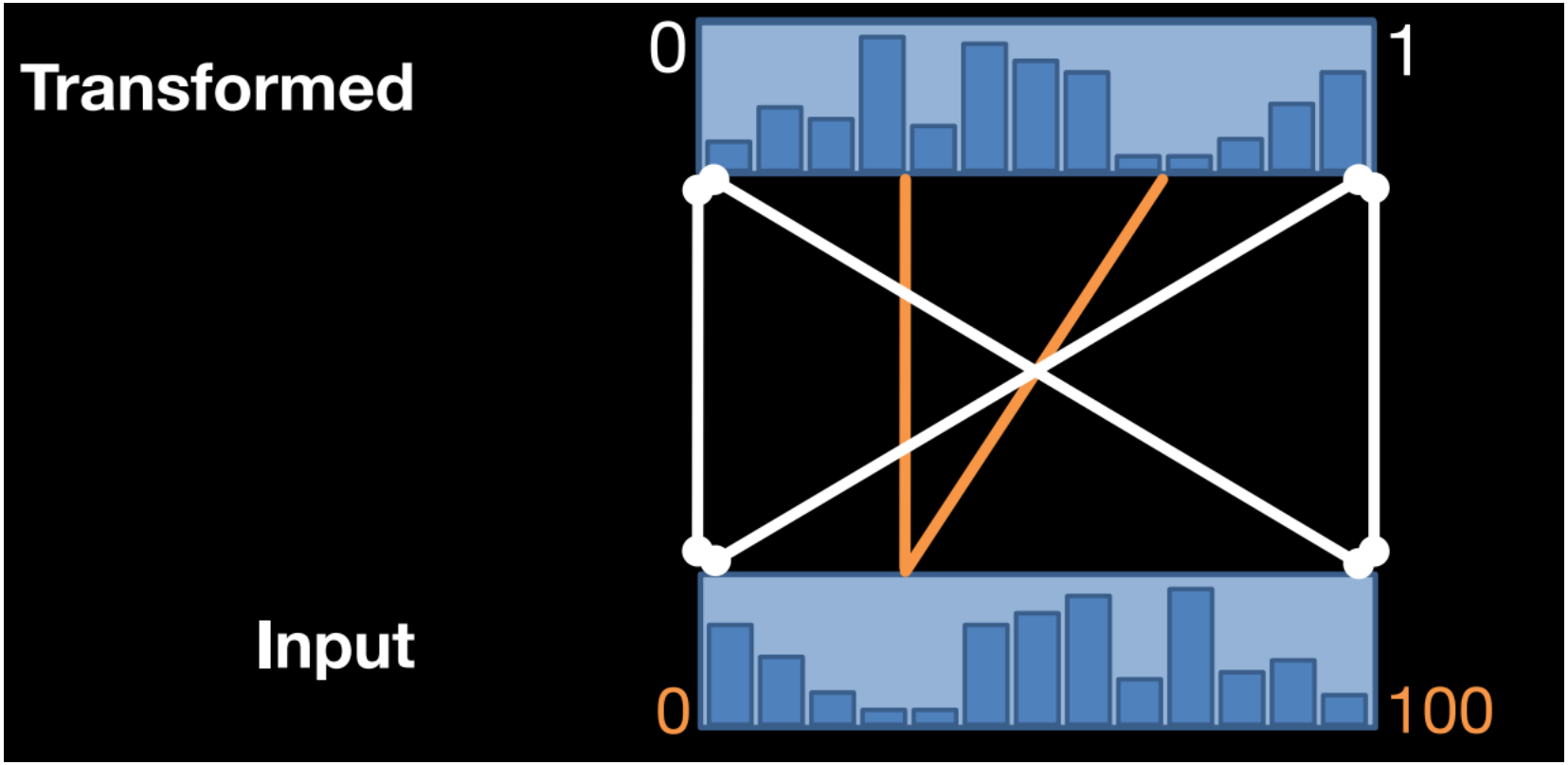


Rank	University	A	B	C
1.	MIT			
2.	Harvard			
3.	Princeton			
4.	Cambridge			
5.	Oxford			

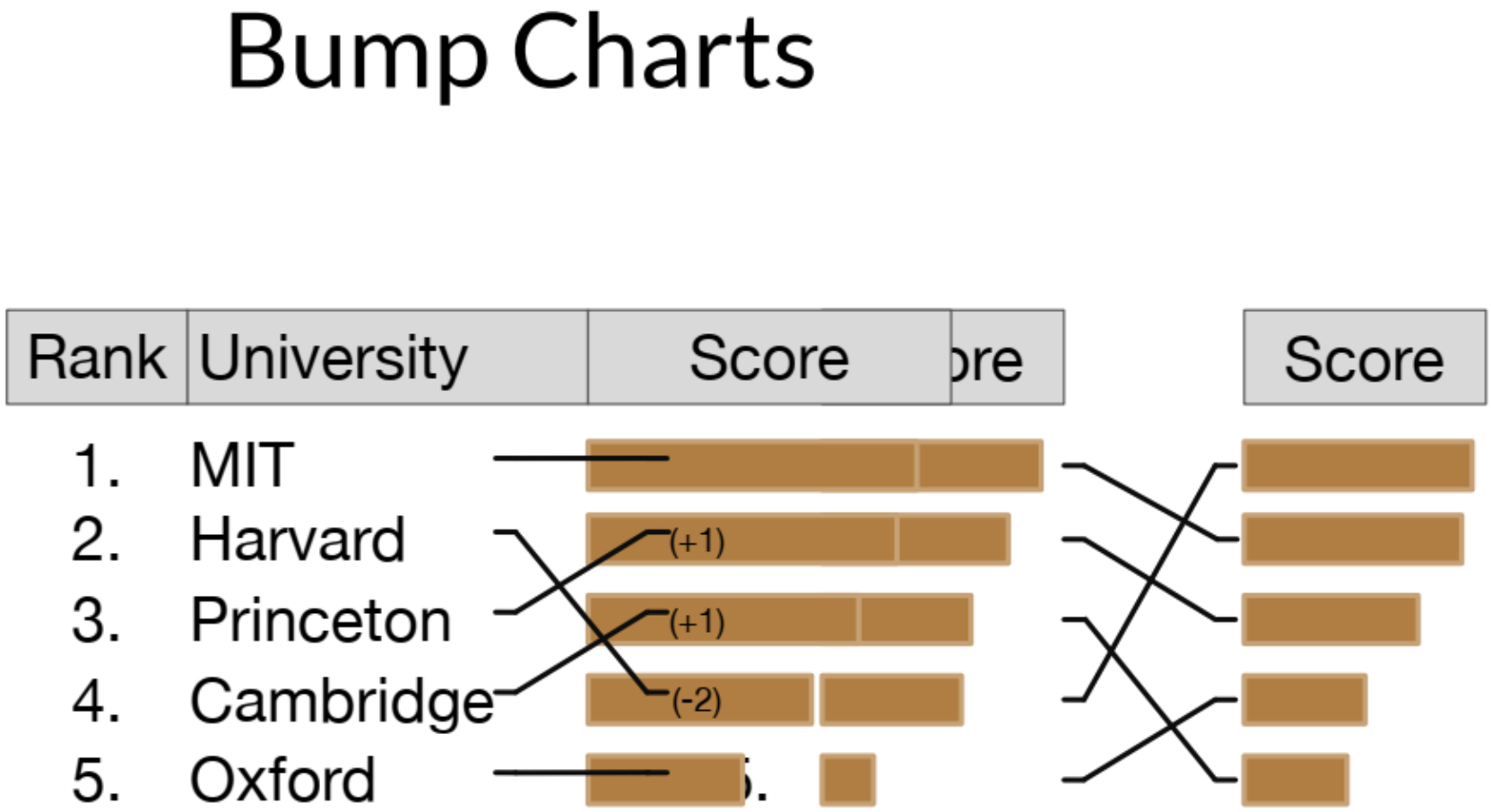
Serial Combiner (as Stacked Bar)

Rank	University	A	B	C
1.	MIT			
2.	Harvard			
3.	Princeton			
4.	Cambridge			
5.	Oxford			

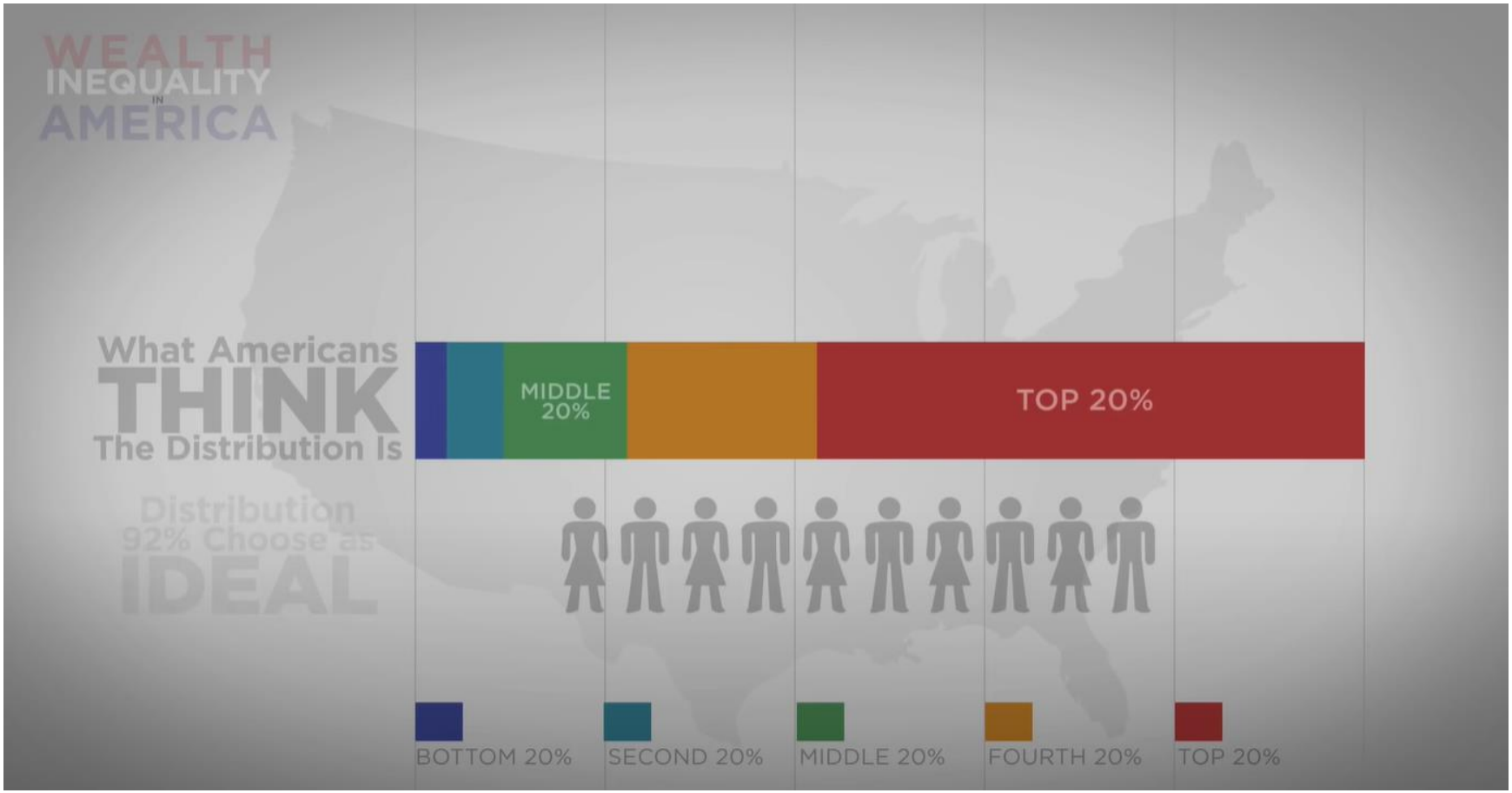
MAPPING ATTRIBUTES TO SCORES



BUMP CHARTS


Bump Charts
Rank
University
Score
ore
Score
1. MIT
2. Harvard
3. Princeton
4. Cambridge
5. Oxford
(+1)
(+1)
(-2)

STORYTELLING



https://www.youtube.com/watch?v=QPKKQnijnsM

STORYTELLING

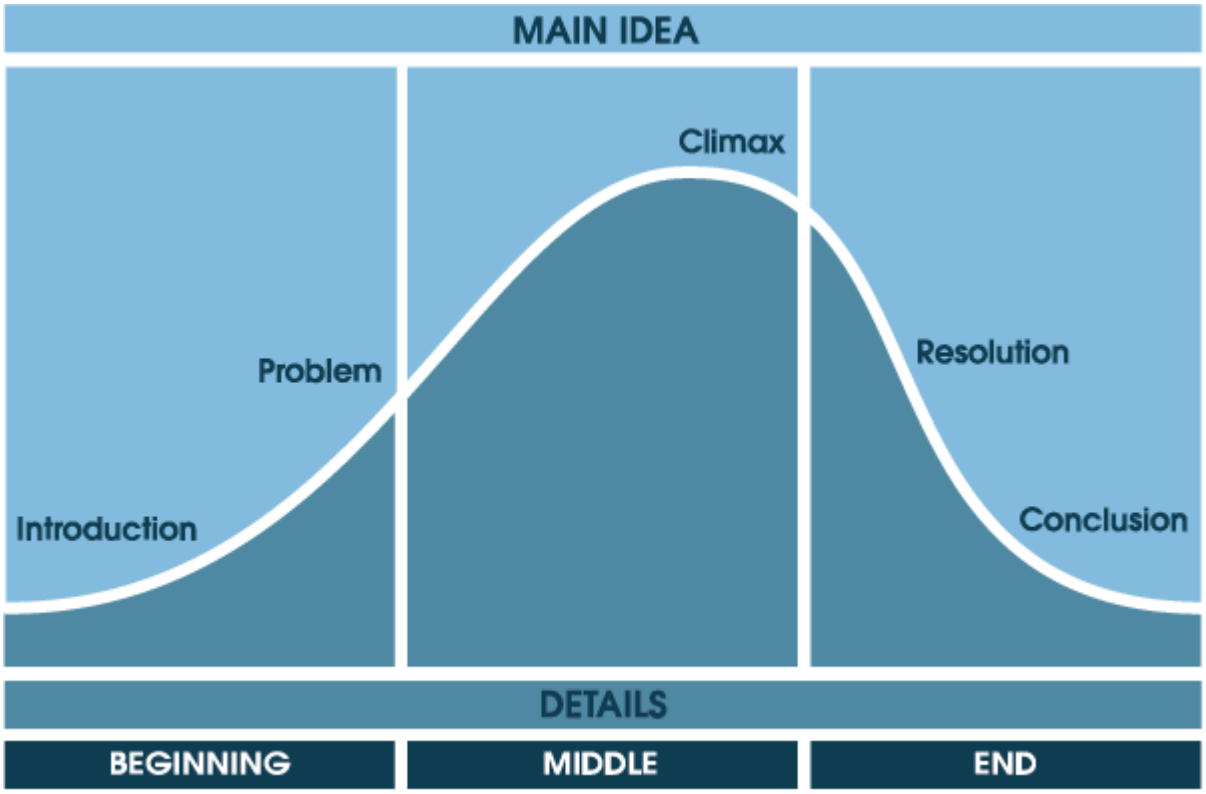
“The world’s second-oldest profession”

Good stories do more than provide facts and data

- They situate and give context
- They engage
- They educate

Who / What / Where

Why / How

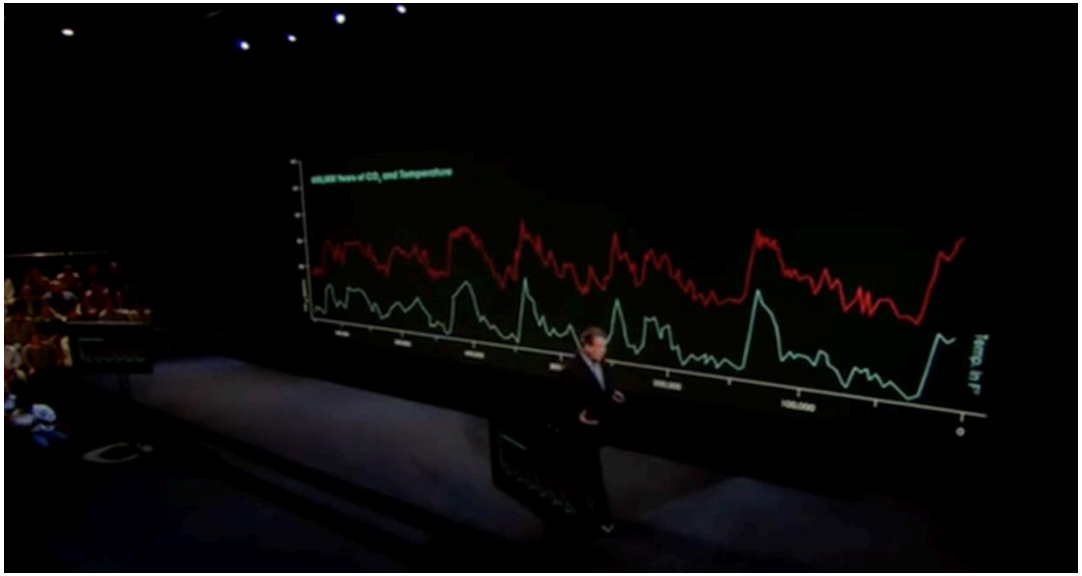


STORYTELLING WITH DATA/VIS

Underscore your arguments with Data/Facts

Leverage the power of Visualization

- Show trends
- Show correlations
- Show outliers
- Convey magnitudes



https://youtu.be/-JIuKjaY3r4

COMPONENTS OF A DATA STORY

Introduction

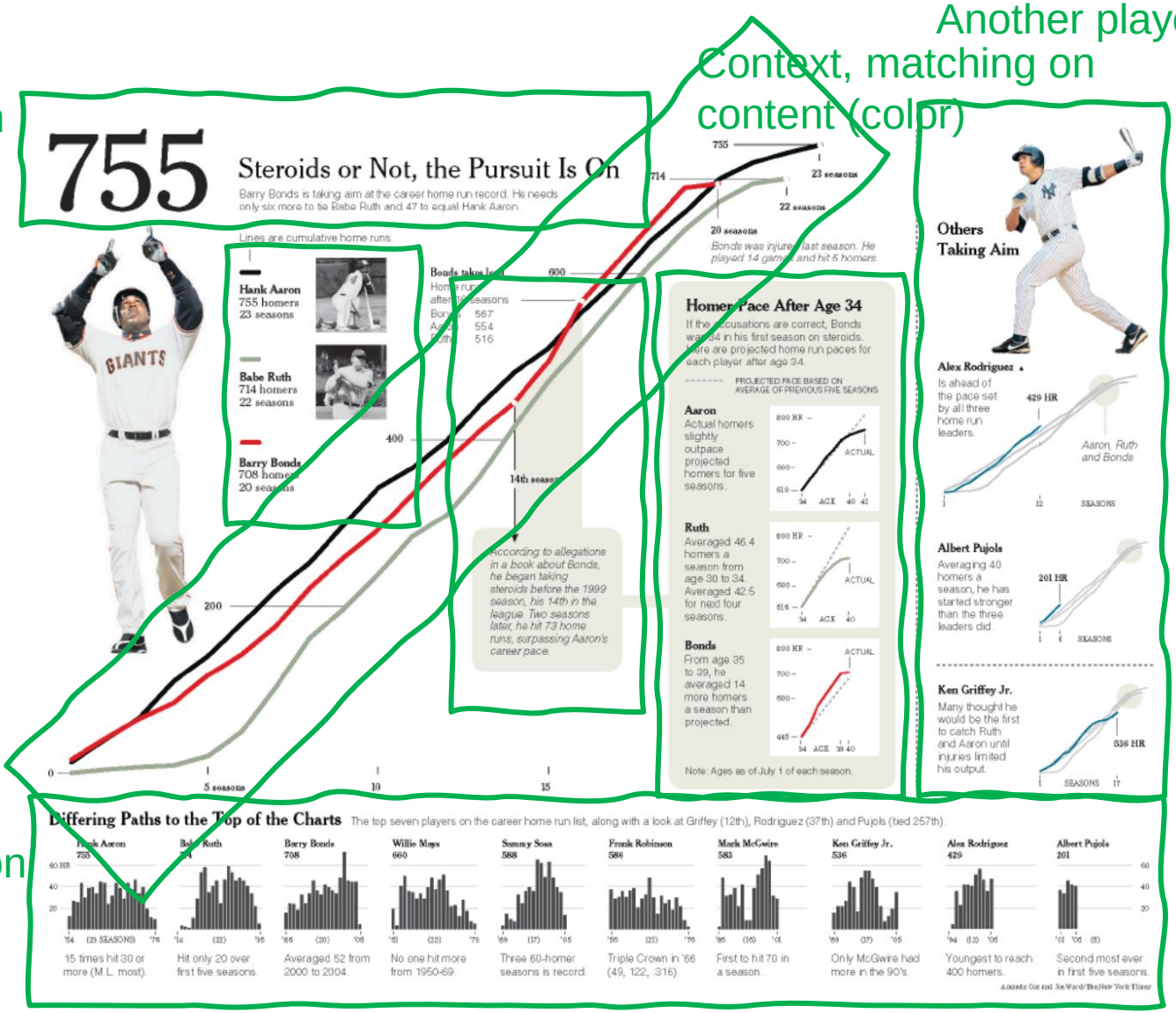
Context

Main story

Annotation of key points

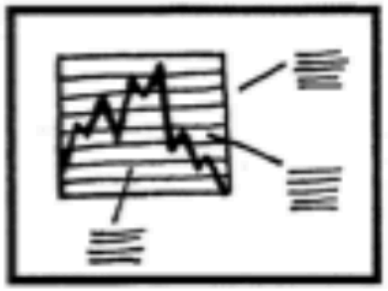
Context, matching on content (color)

Another player

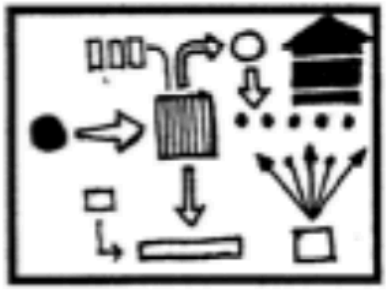


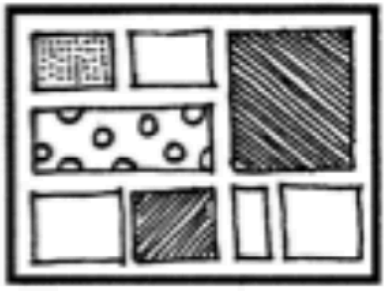
GENRE

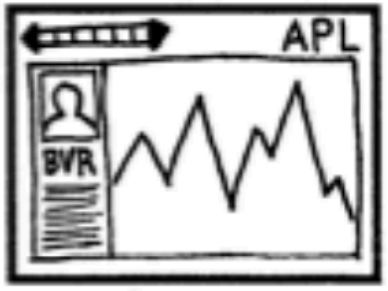
Magazine Style


Annotated Chart

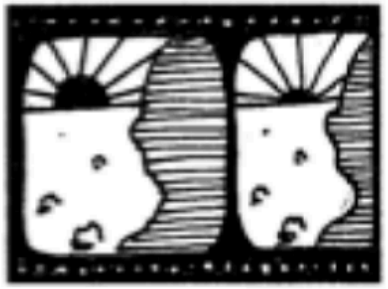
Partitioned Poster


Flow Chart


Comic Strip



Slide Show


Film/Video/Animation

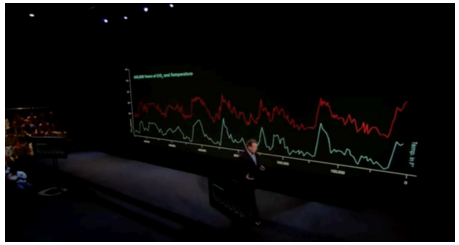
AUTHOR OR READER DRIVEN

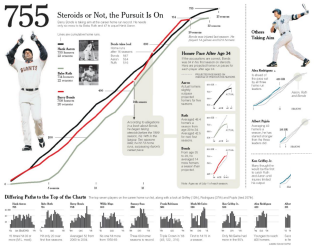
Author driven — **Reader driven**

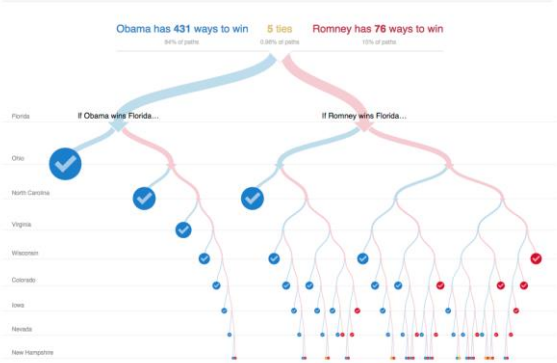
Linear ordering

Heavy messaging

No interactivity







No ordering

No messaging

Free interactivity

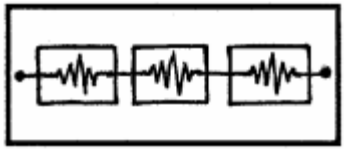
AUTHOR OR READER DRIVEN

Martini Glass structure

start with author driven, open up for exploration

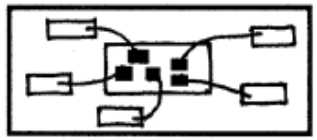
Interactive slideshow

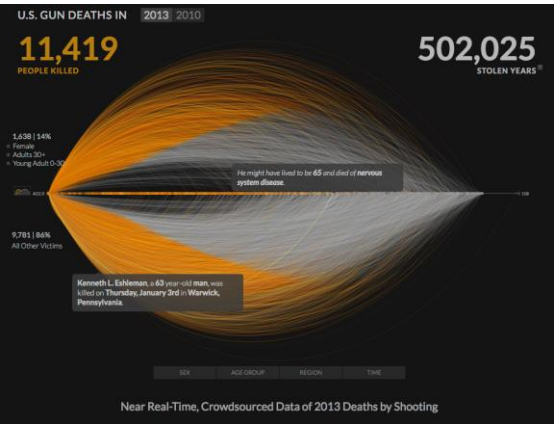
Split into multiple scenes, allow interaction mid-way

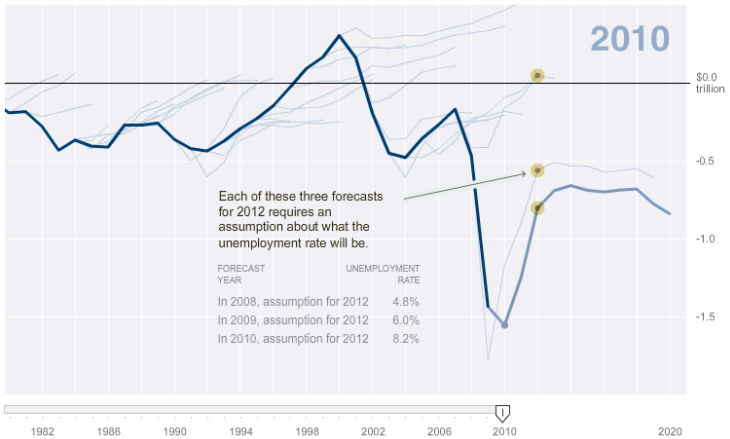


Drill-down story

Let reader decide which path to follow, all paths are annotated







STRATEGY FOR STORYTELLING

LAYOUT PRINCIPLE

Descriptive titles

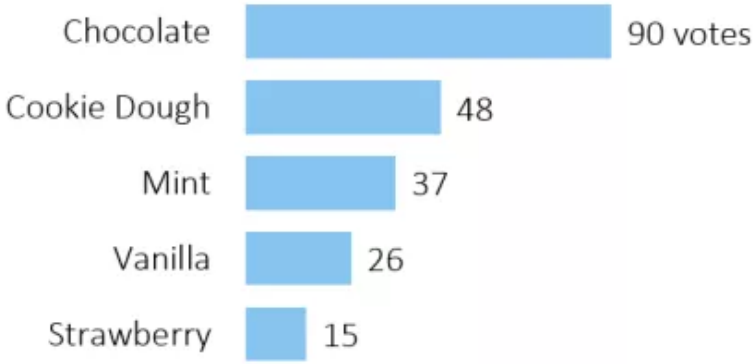
Descriptive subtitles

Annotations

Saturation

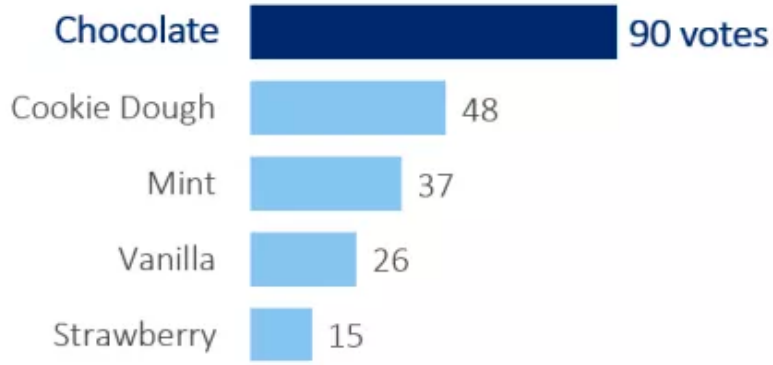
DESCRIPTIVE TITLE AND SATURATION

Ice cream flavor preferences based on 2014 survey of elementary school students (n=216)



or

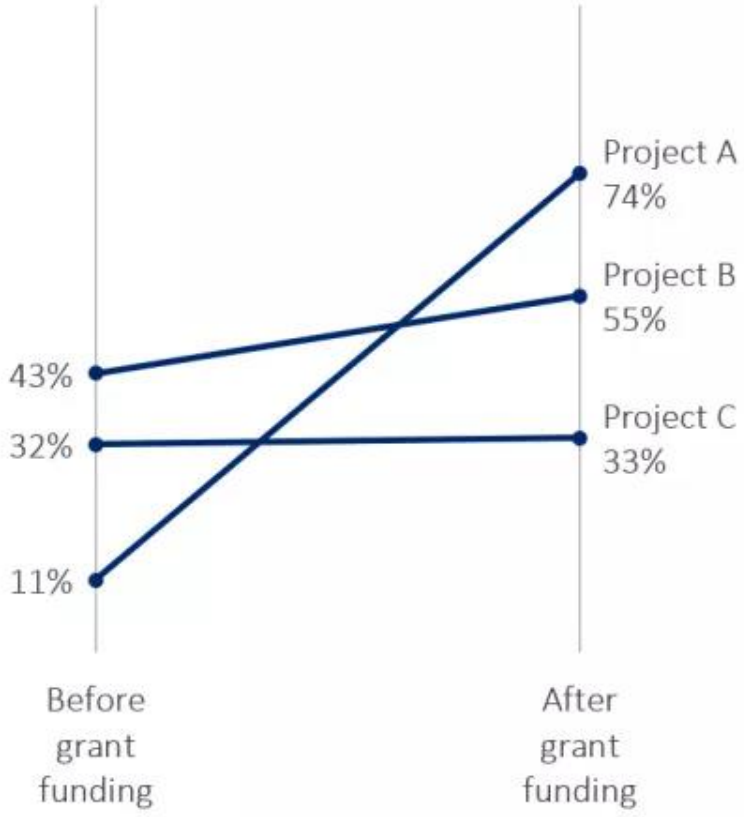
Chocolate was most popular flavor among elementary students surveyed



Source: 2014 survey of elementary school students (n=216)

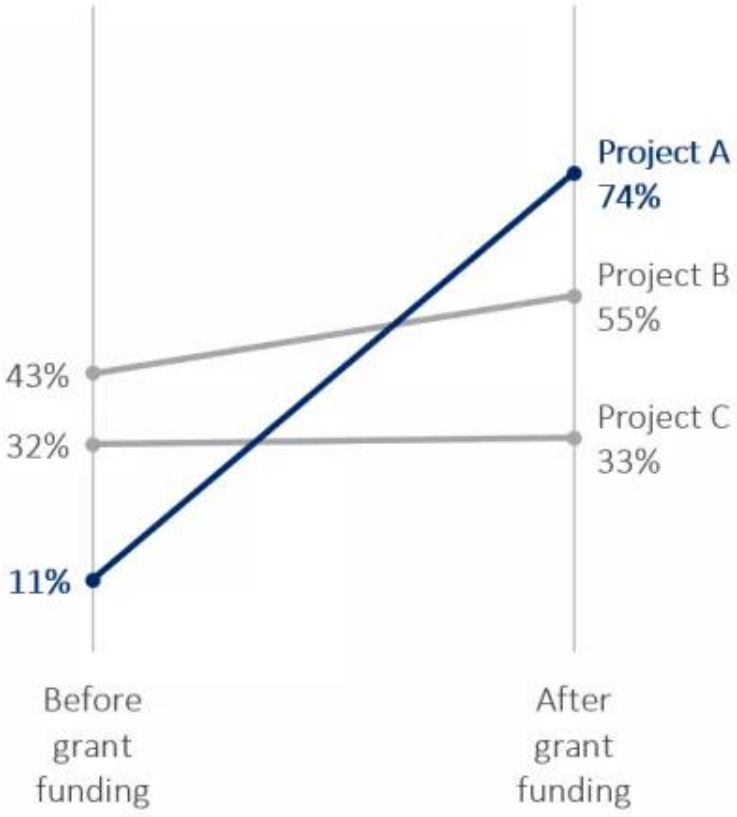
DESCRIPTIVE TITLE AND **SATURATION** EMPHASIZE HOW PROJECT A IS PERFORMING PARTICULARLY WELL

Project results before and after implementation of grant



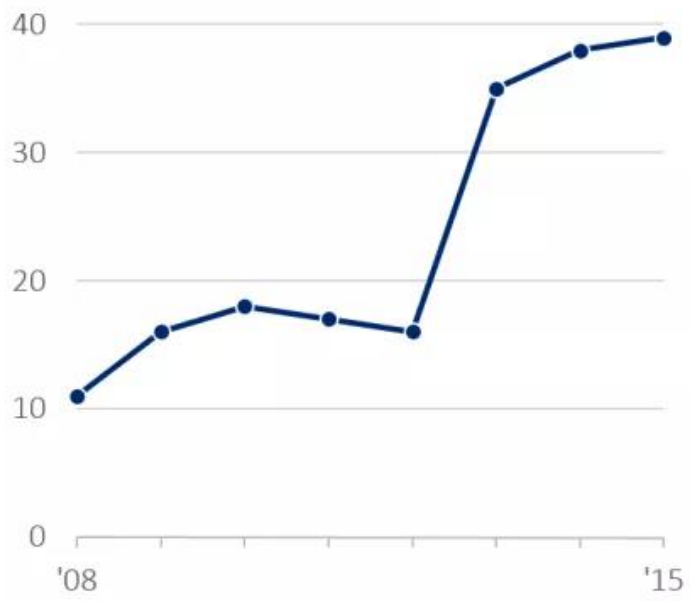
or

Project A had the greatest gains after the four-year grant funding



DESCRIPTIVE TITLE, **DESCRIPTIVE SUBTITLE,** AND **ANNOTATION** FOR INCREASED UNDERSTANDING

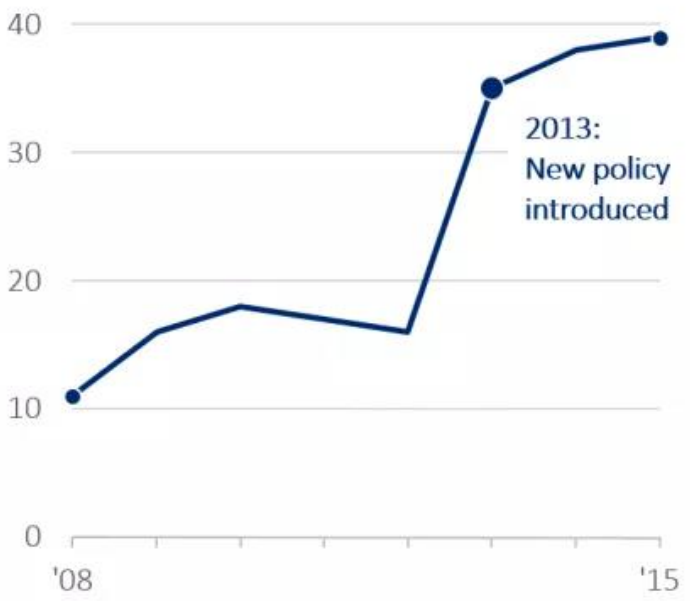
Number of studies funded each year



or

We're funding more studies each year

Beginning in 2013, we set aside new funding to measure the effectiveness of our initiatives – and we evaluated 39 of our programs in 2015 alone.



INTERACTIVITY

Navigation

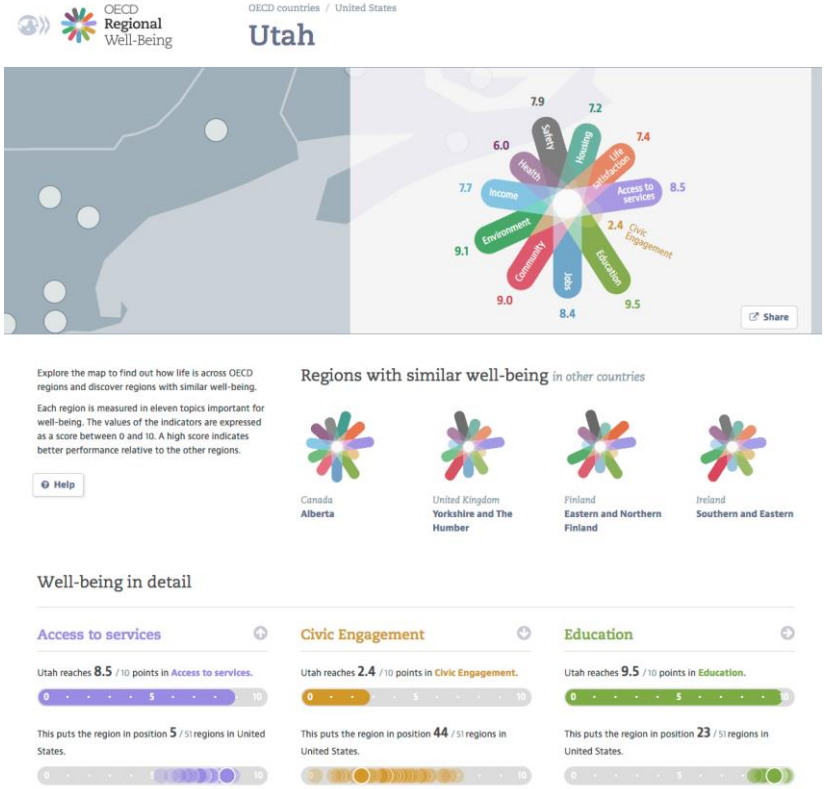
Steppers, Scrolling,
Play/Fwd/Rwd, etc.

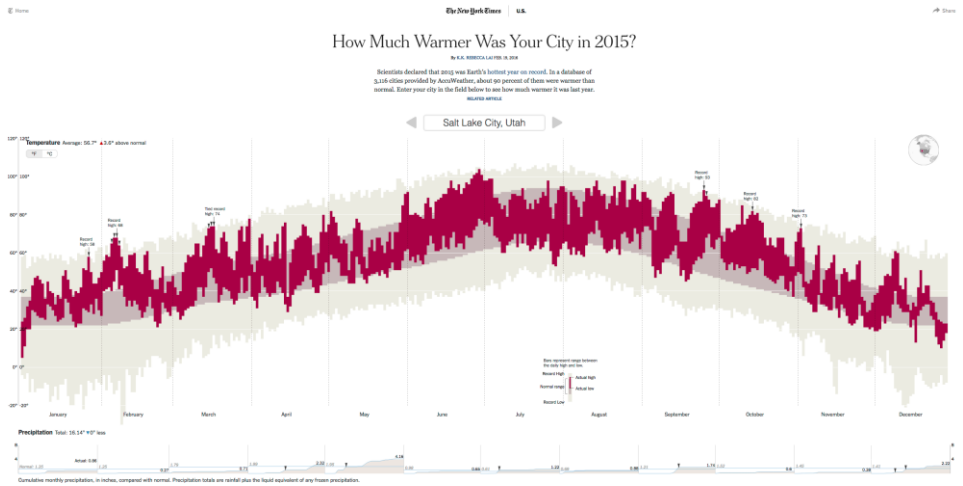
Details on Demand

Highlights when desired

Making it relevant to the reader

What do you think?
Who are you?
Where do you live?



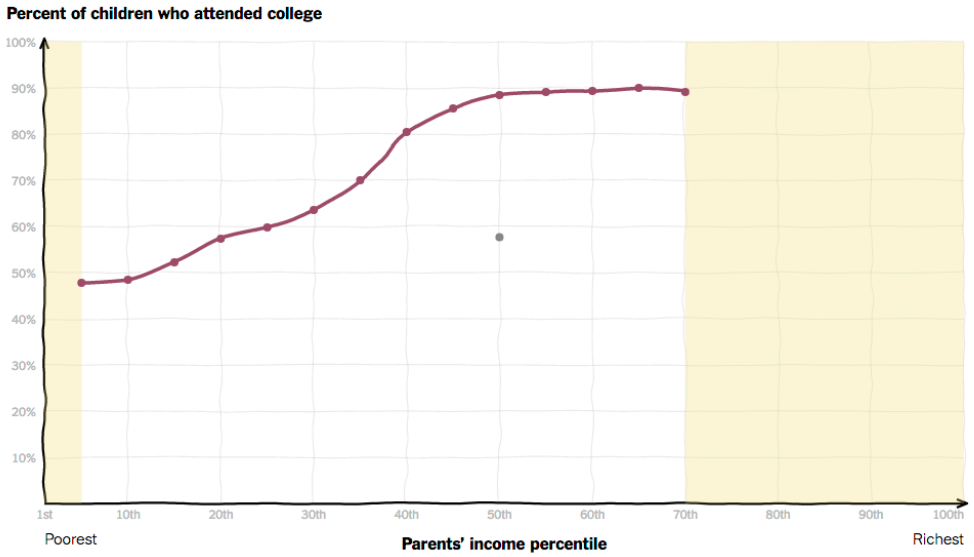


ASK FOR OPINIONS/ PRIOR KNOWLEDGE

Ask reader to draw a trend
Juxtapose with

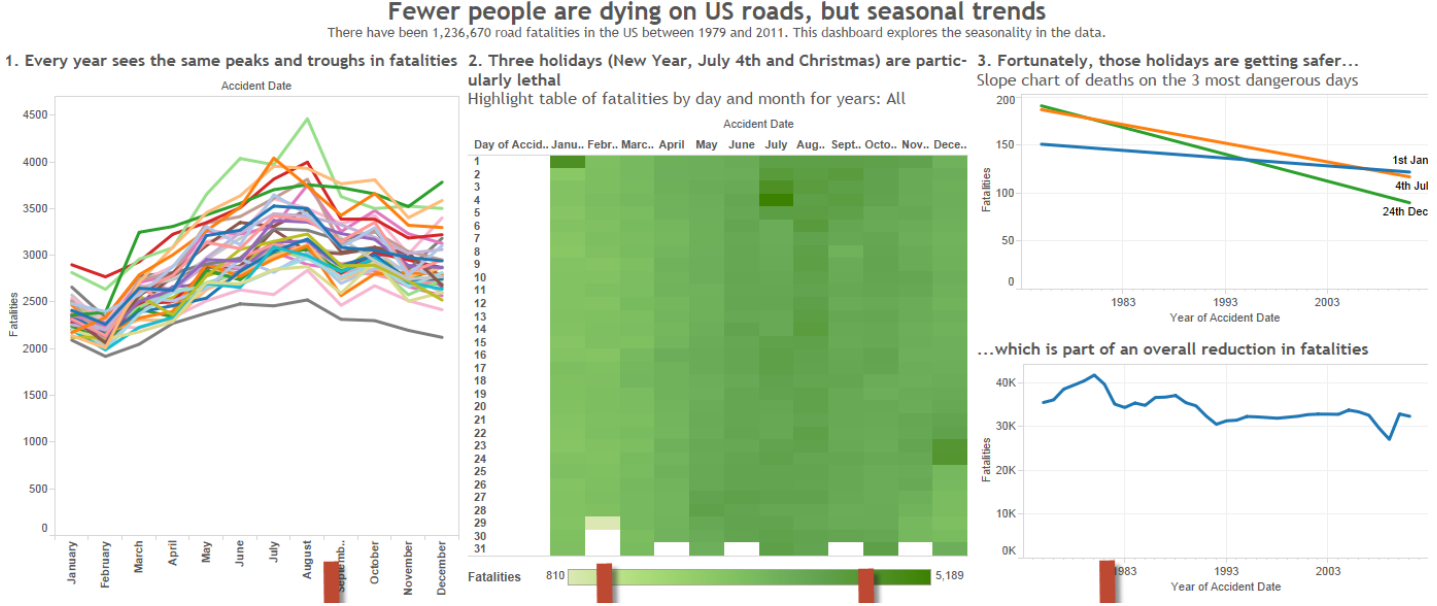
reality

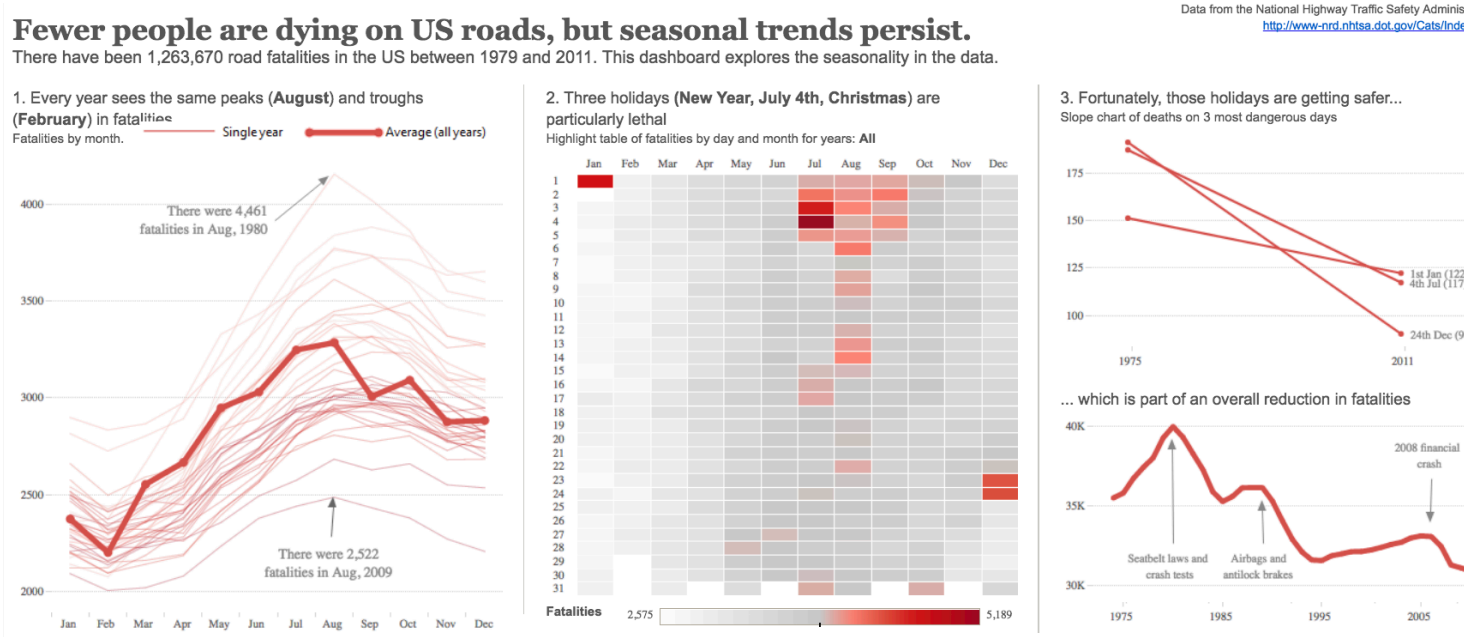
what other people think



DESIGN CONSIDERATION

Fewer Colors
Averages for Context
Better Scales
Richer Annotations





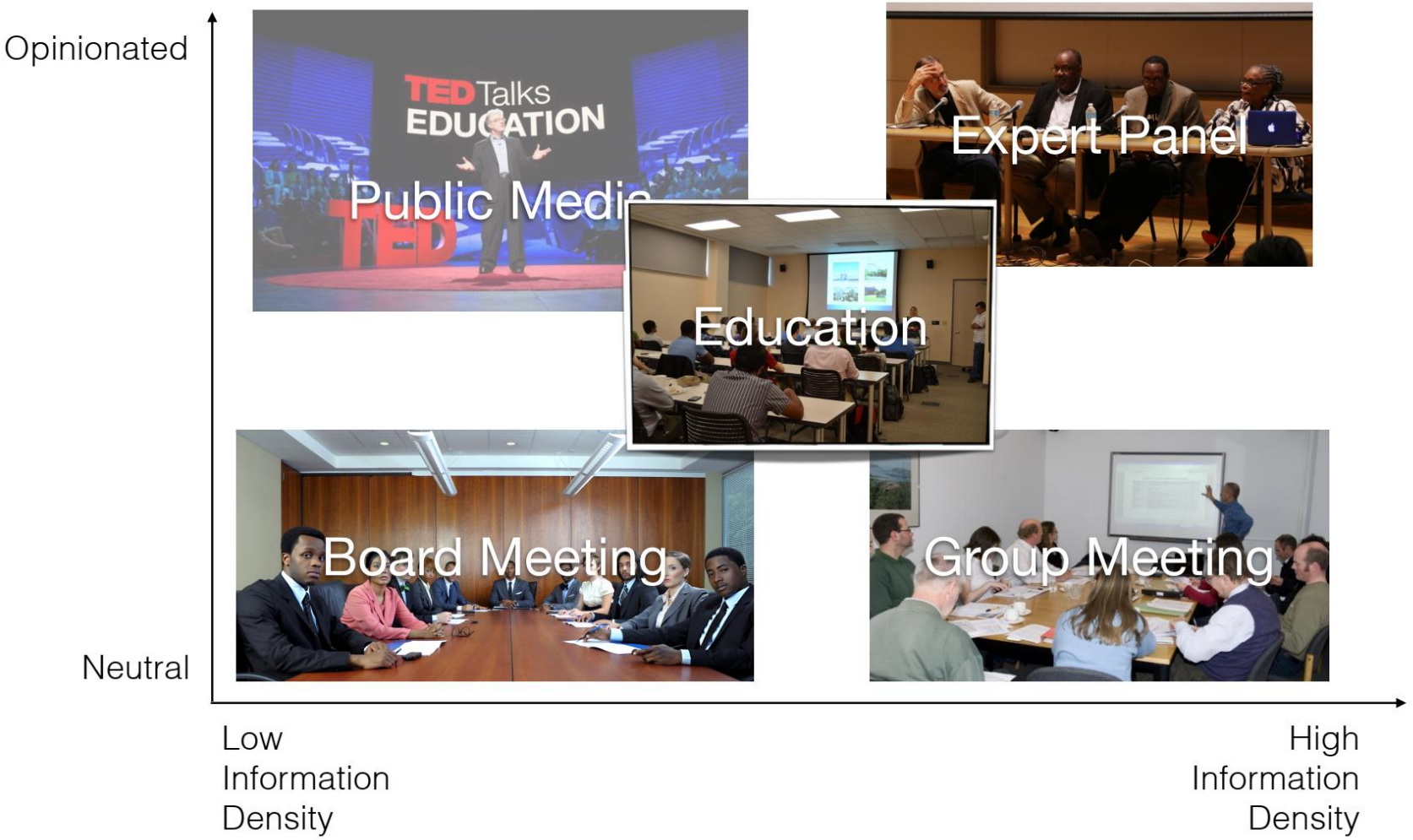
ENGAGEMENT

KNOW YOUR AUDIENCE

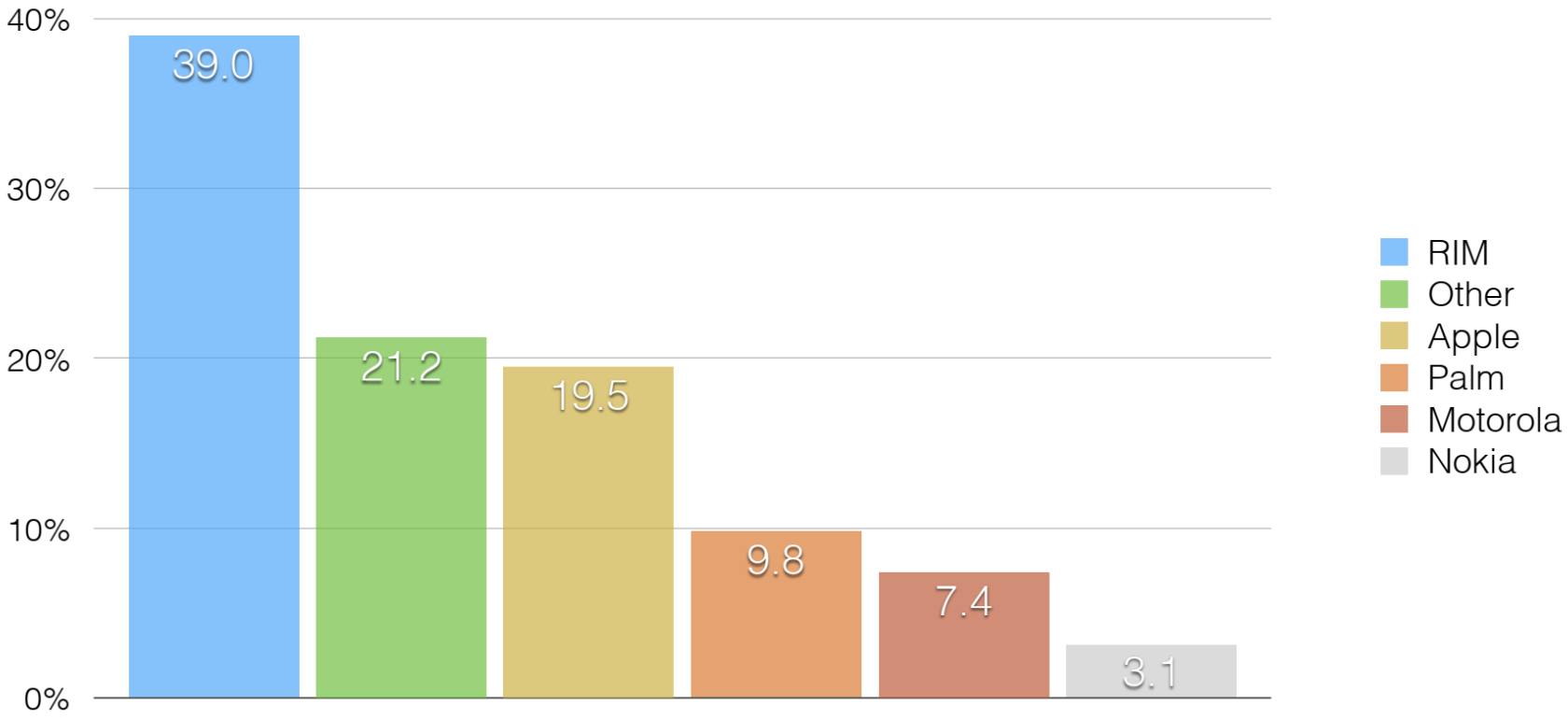
People you don’t know are difficult to influence

What do they know?
What motivates them?
What experiences do you share?
What are common goals?
What insights can you give them?

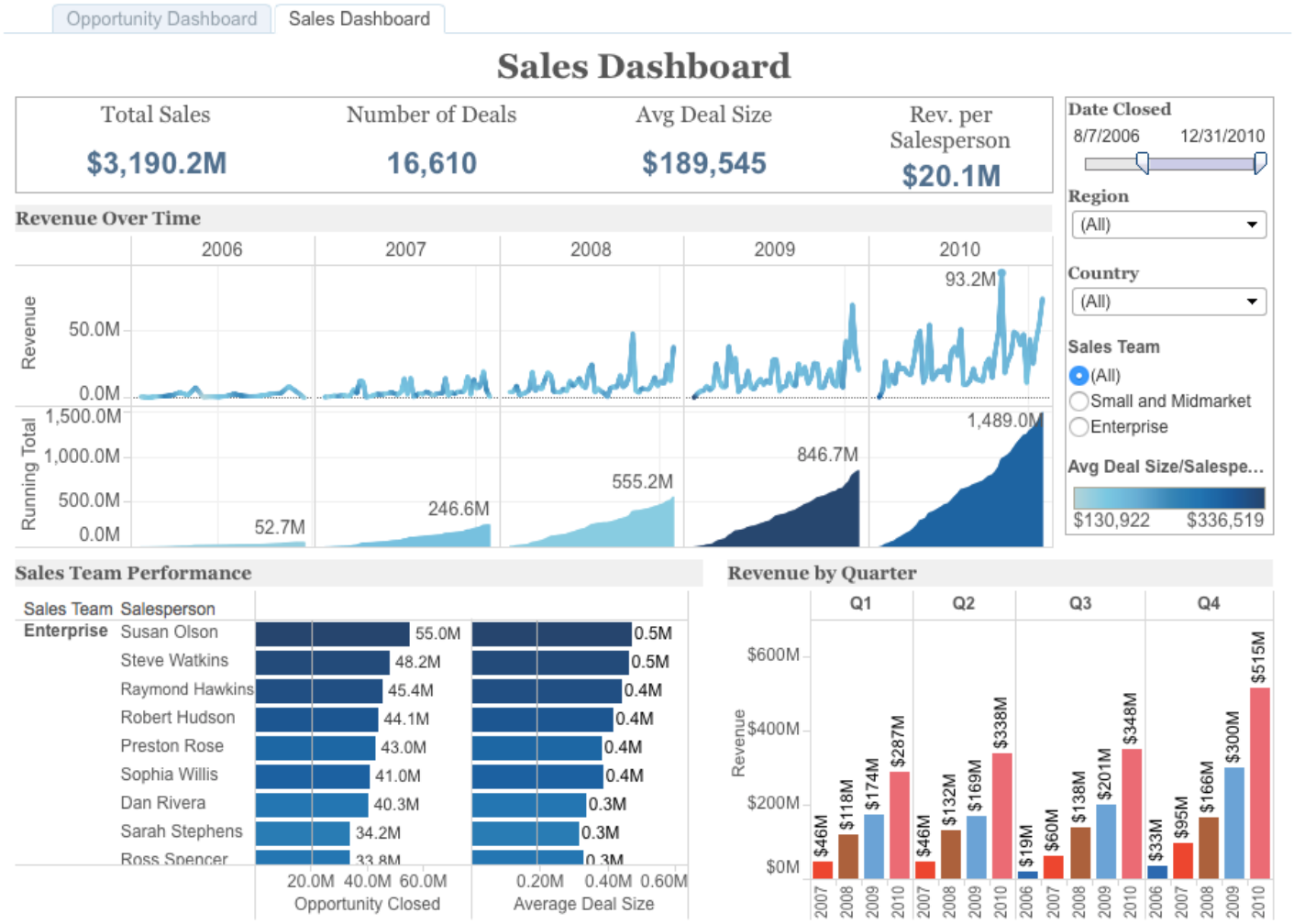
TARGET AUDIENCE



TARGET AUDIENCE?

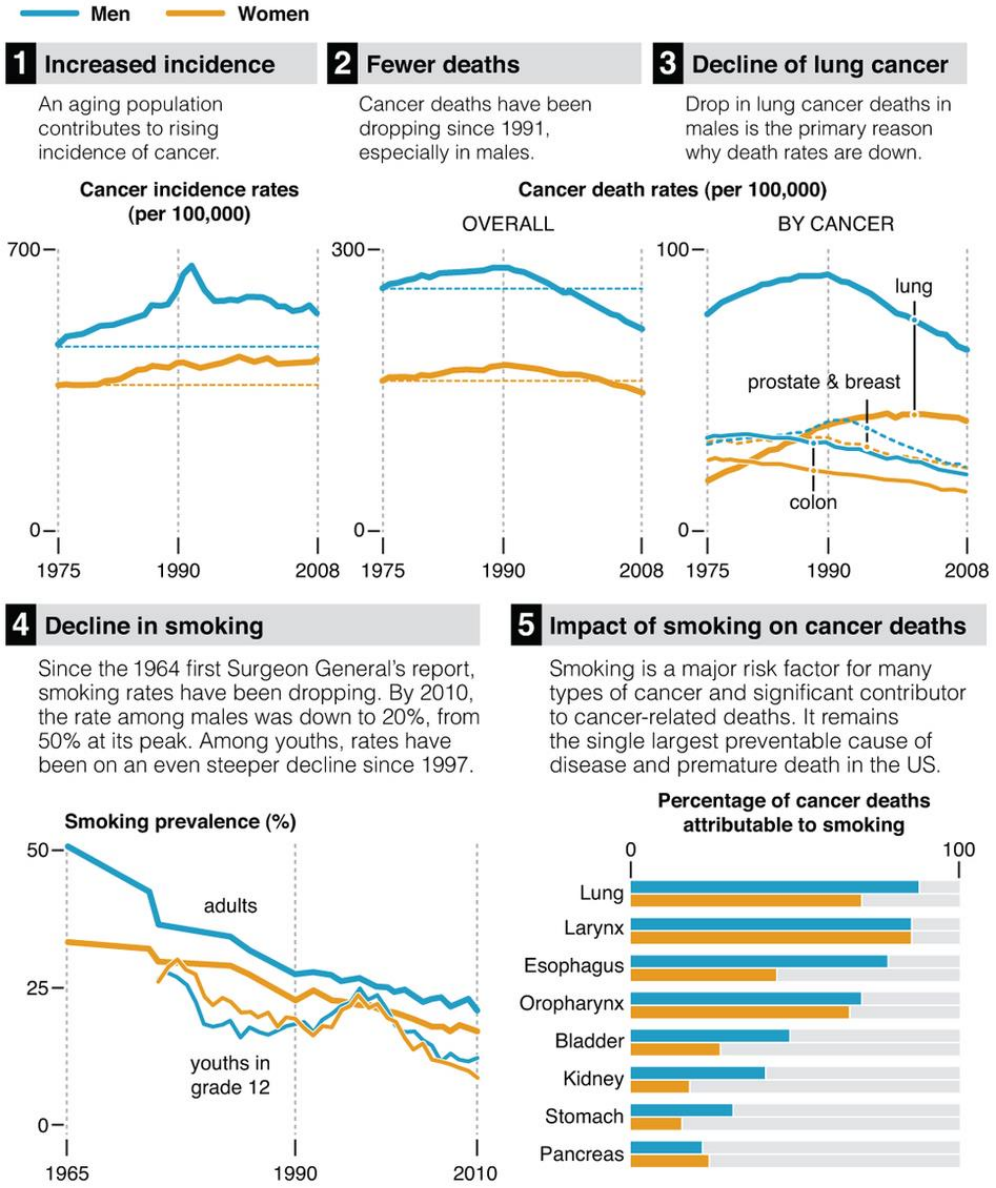


TARGET AUDIENCE?



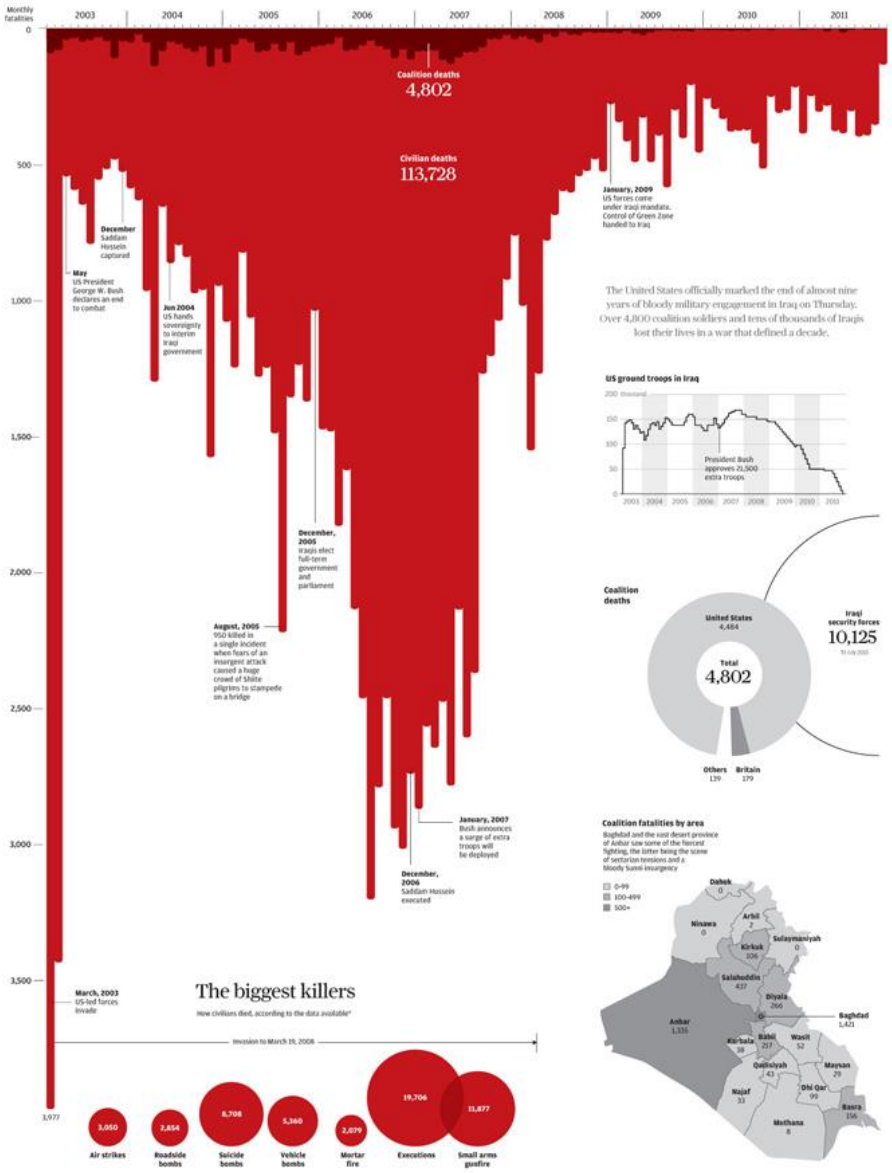
WHERE THERE'S SMOKE—THERE'S CANCER

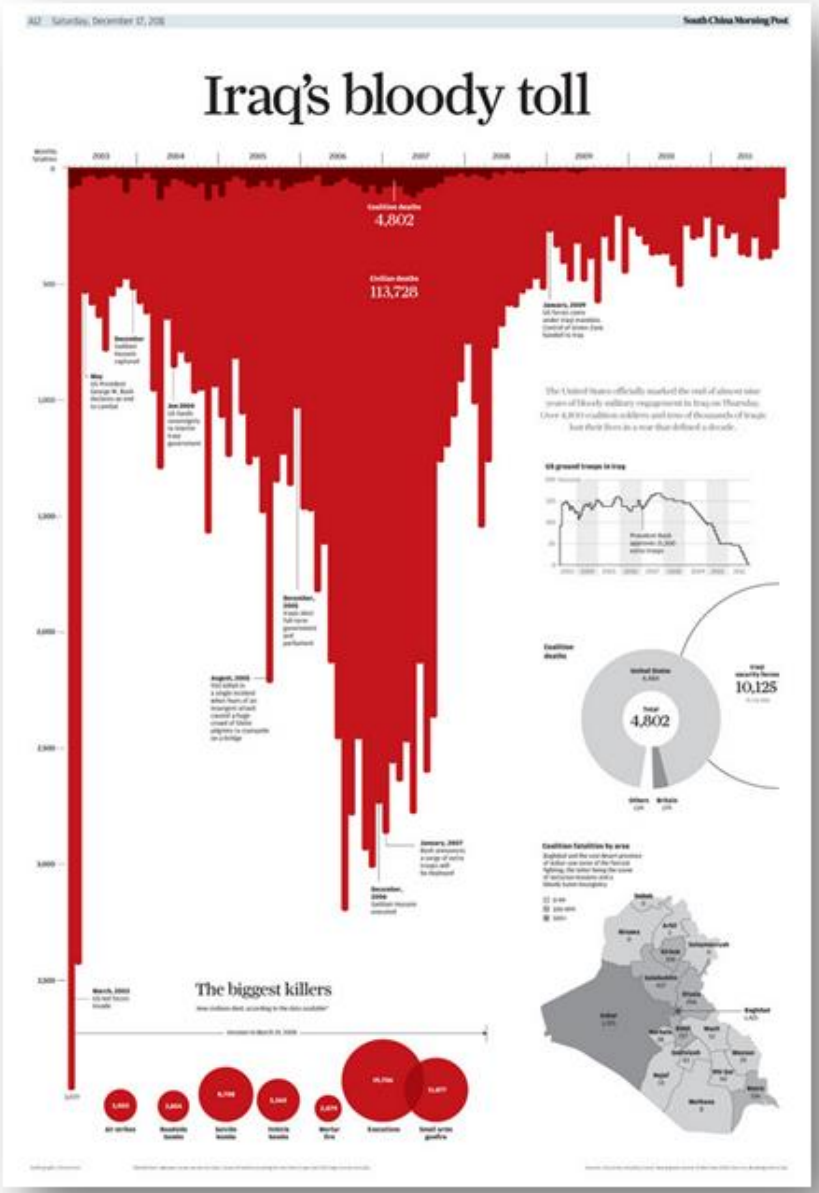
Cancer rates are up, but mortality is down. New diagnostics and treatments are responsible for part of this trend. But the greatest single contributing factor is the decline in smoking—rates are at their lowest level in 50 years.

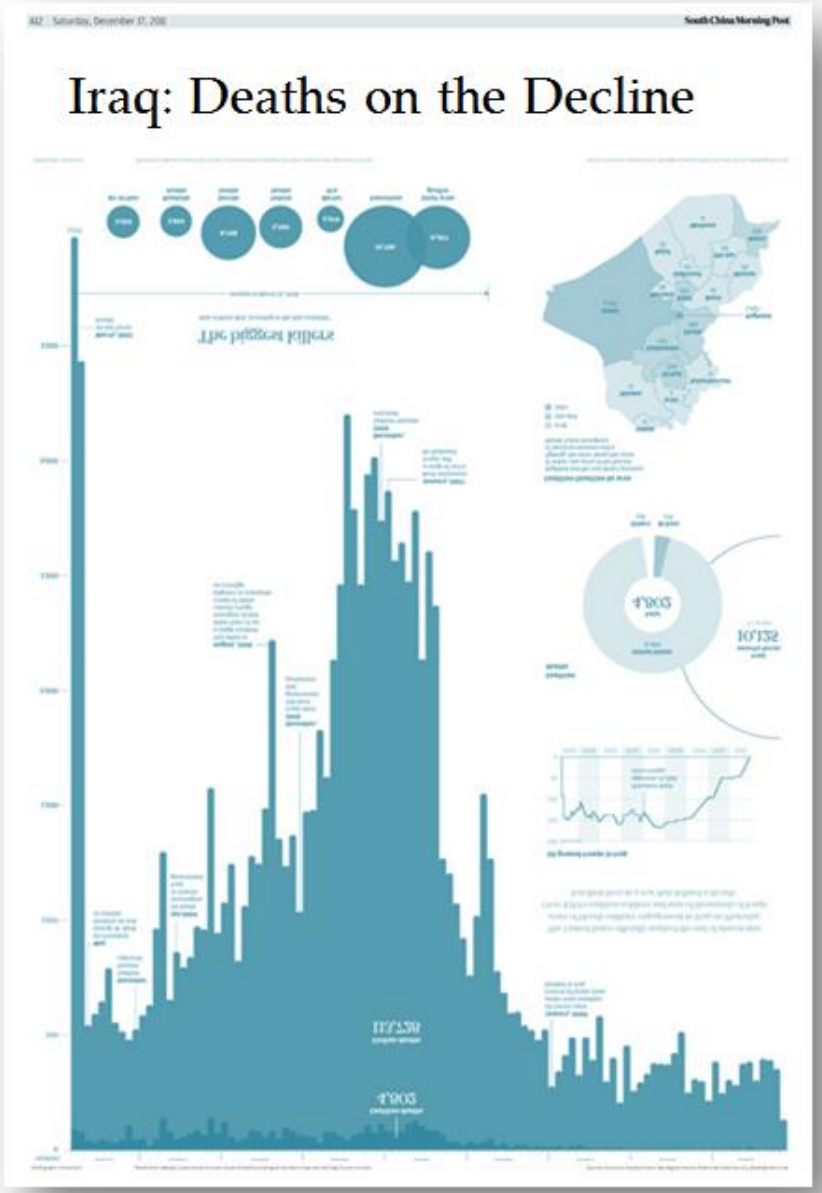


source: American Cancer Society Cancer Statistics 2012; Monitoring the Future (University of Michigan).

Iraq's bloody toll

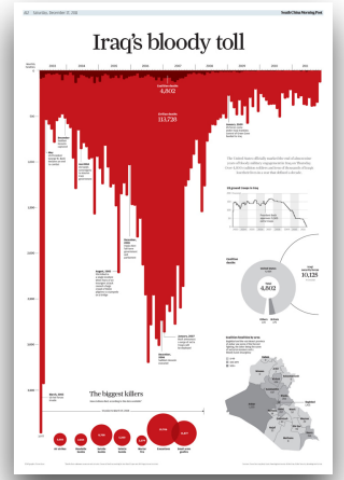


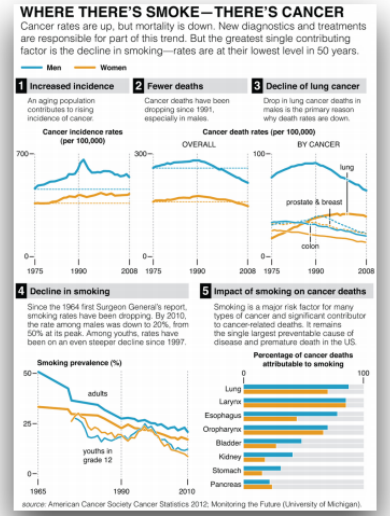

Iraq's bloody toll
South China Morning Post
Coalition deaths
4,802
Civilian deaths
113,728
10,125
The biggest killers

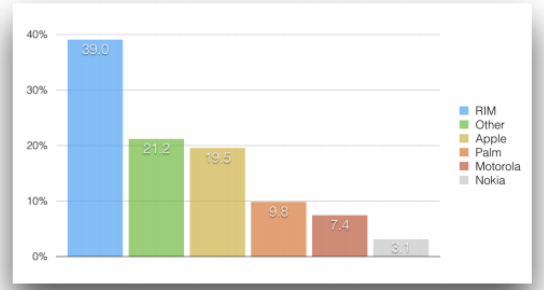

Iraq: Deaths on the Decline
South China Morning Post

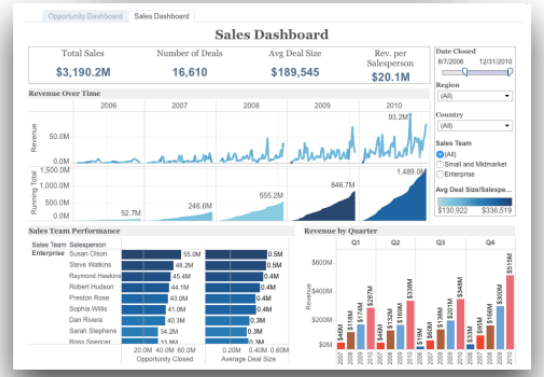
Opinionated

Neutral





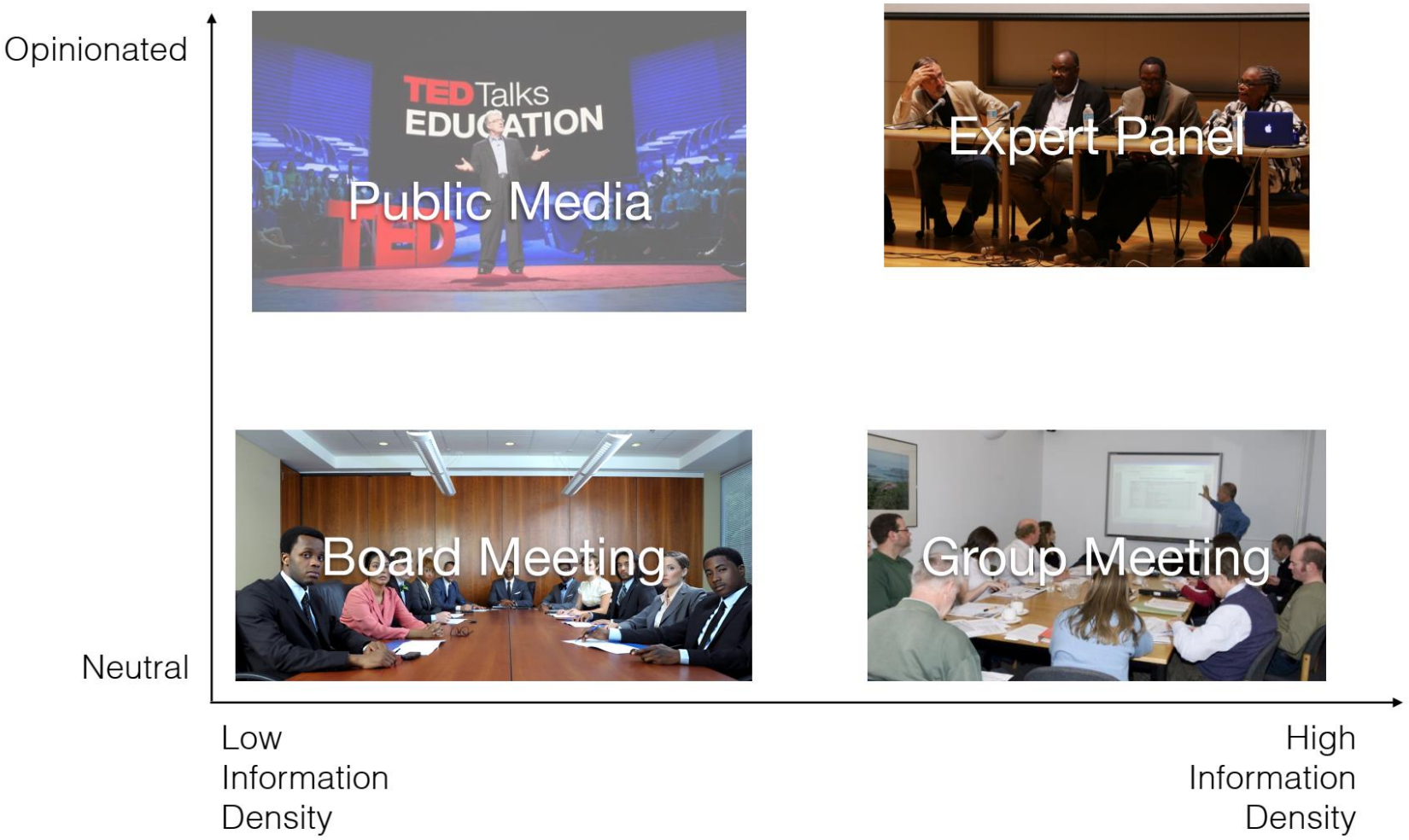




Low Information Density

High Information Density

TARGET AUDIENCE



EXERCISES

What's the Genre?
(Magazine, Annotated Chart, Partitioned Poster, Flow Chart, Comic Strip, Slide Show, Video/Animation)

Is it Author or Reader Driven?

Can you make out a particular structure?
(Martini Glass, Interactive Slideshow, Drill-Down Story)? Why?

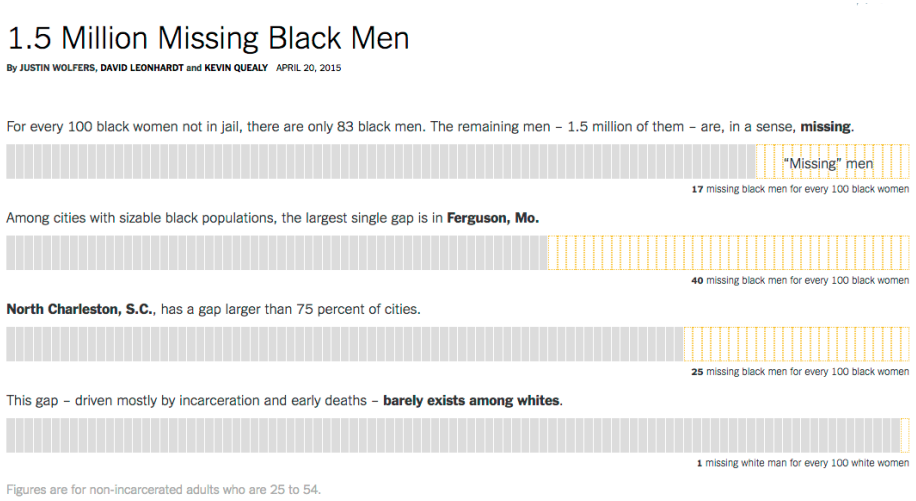
Does it have a good Introduction?
What medium is used for the introduction?

Is the story opinionated or neutral?

Does it use titles, legends, and highlights well?

Does it use interactivity well?

Is the visualization well designed?



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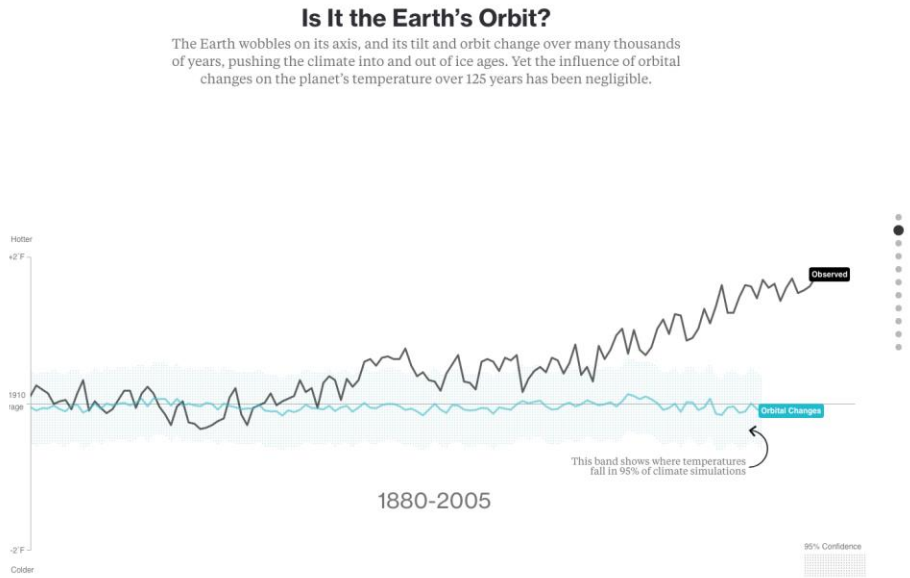

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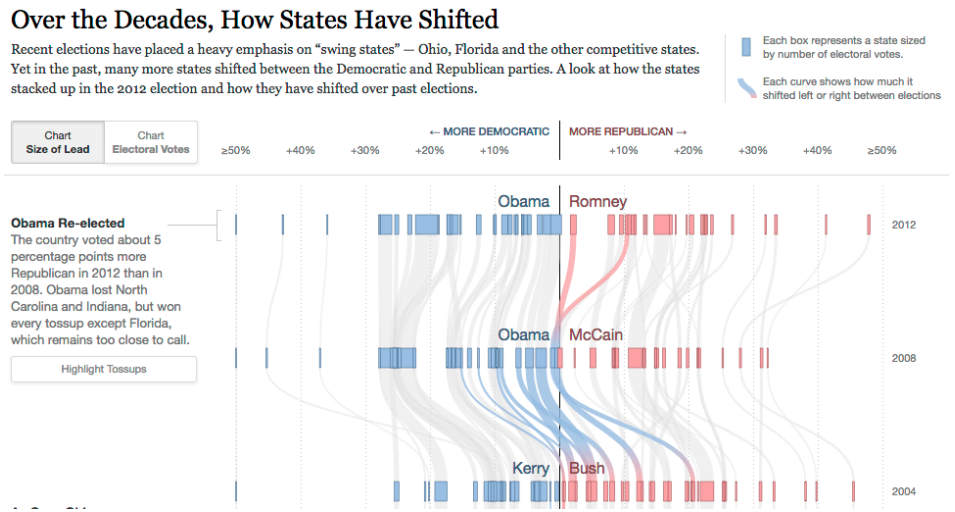


EVALUATION

PROBLEM-DRIVEN VS TECHNIQUE-DRIVEN

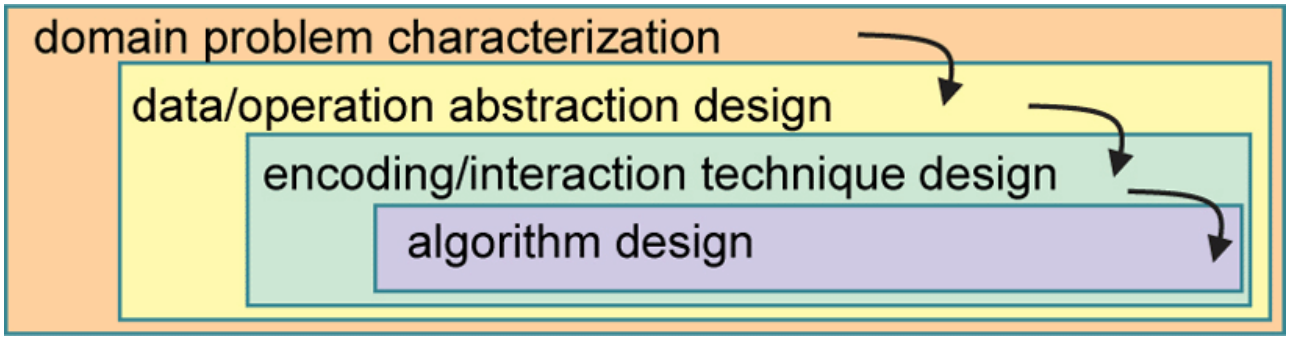
Problem-driven

- top-down approach
- identify a problem encountered by users
- design a solution to help users work more effectively
- sometimes called a design study

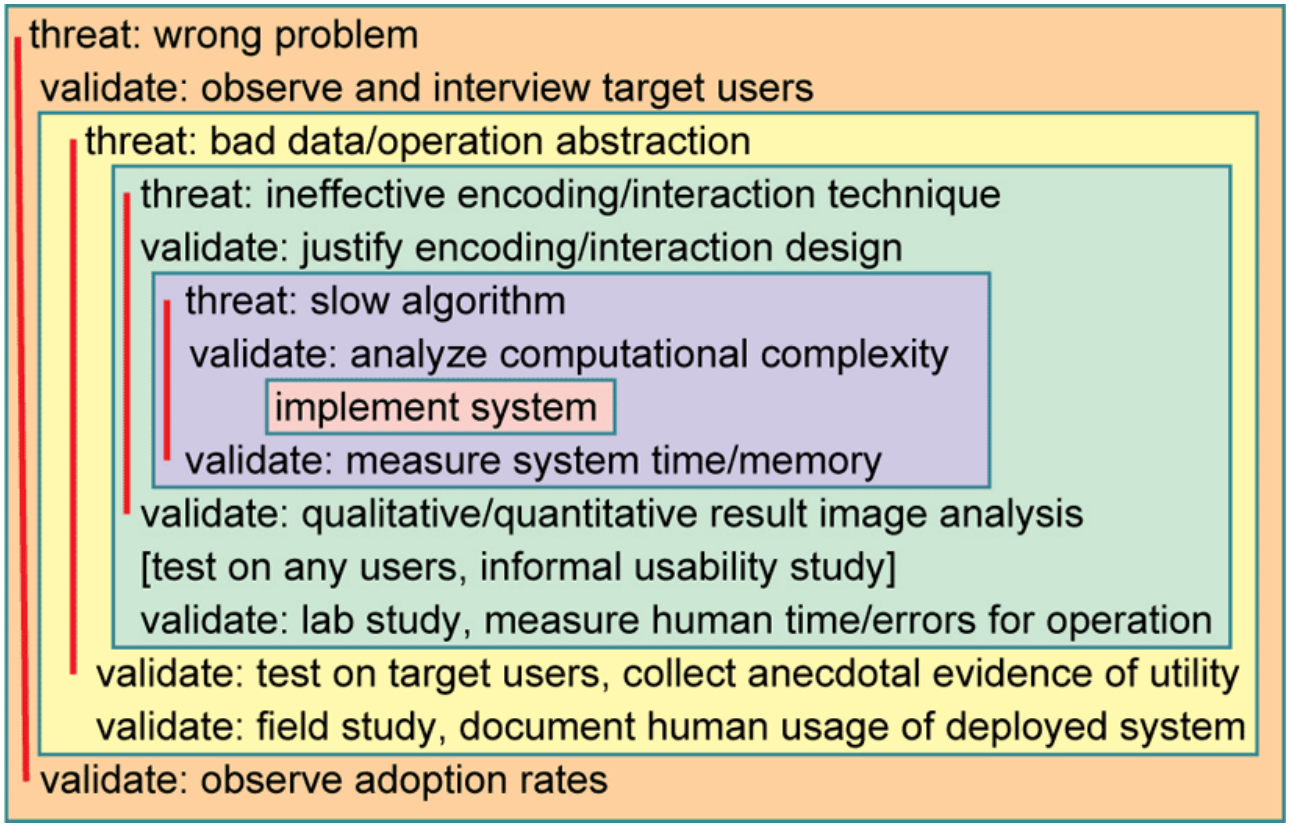
Technique-driven

- bottom-up approach
- invent new visualization techniques or algorithms
- classify or compare against other idioms and algorithms

THE NESTED MODEL

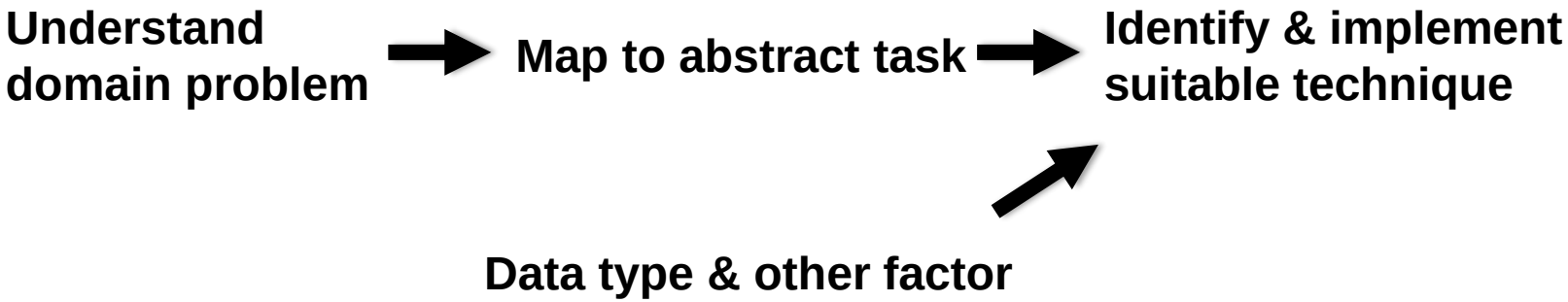


Design



Threats & evaluation

DESIGN PROCESS



DOMAIN CHARACTERIZATION

Details of an application domain

Group of users, target domain, their questions, their data

- varies wildly by domain
- must be specific enough to continue with

Cannot just ask people what they do

- introspection is hard

DOMAIN PROBLEM CHARACTERIZATION

Infinite numbers of domain tasks

Can be broken down into simpler abstract tasks

We know how to address the abstract tasks

Identify task – data combination:
solutions probably exist

EXAMPLE: FIND GOOD MOVIES

I want to identify good movies in genres I like

Domain: general population, movie enthusiasts

DATA & TASK ABSTRACTION

The what-why, map into generalized terms

Identify tasks that users wish to perform or already do

Find data types and good model of the data

Sometimes must transform the data for a better solution

this can be varied and guided by the specific task

EXAMPLE: FIND GOOD MOVIES

What is a good movie for me?

- Highly rated by critics?
- Highly rated by audiences?
- Successful at the box office?
- Similar to movies I liked?
- Specific genres?

Data sources: IMDB, Rotten Tomatoes, …

ENCODINGS & INTERACTIONS

The design of visualization techniques

- Visual encodings
- Interactions

Ways to create and manipulate the visual representation of data

Decisions on these may be separate or intertwined

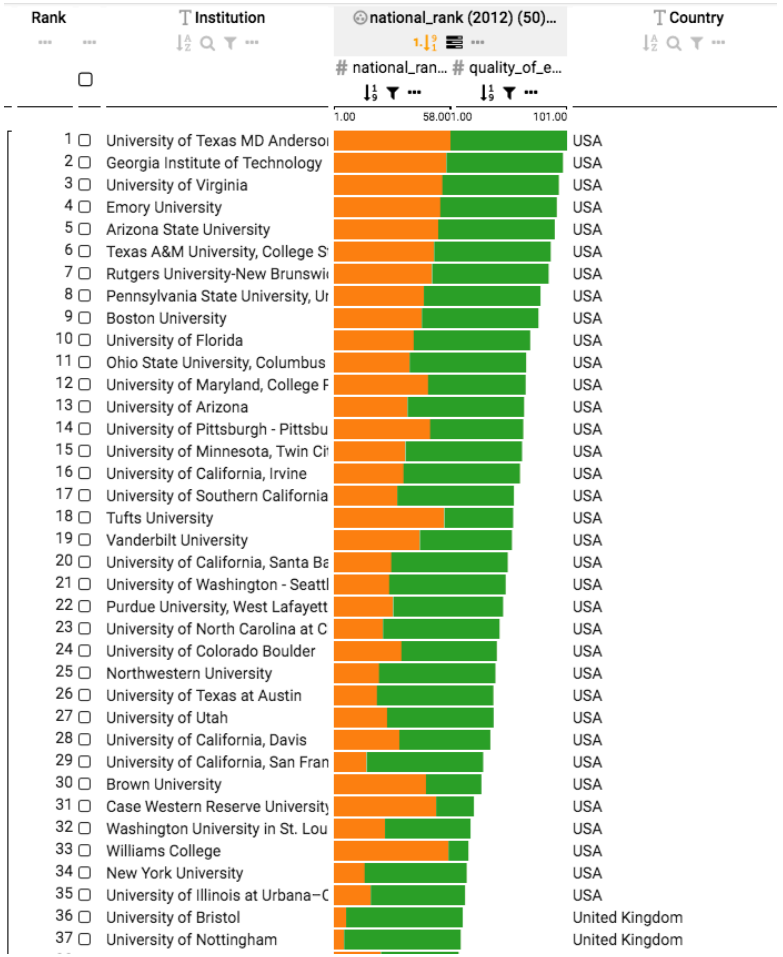
Visualization design principles drive decisions

EXAMPLE: FIND GOOD MOVIES

Combination of audience ratings and critics ratings, filtered by genre

Idiom:

- **stacked bar chart for ratings**
- **filter interface for genre**



TASKS

Analyze

- High-level choices
- Consume vs produce

Search

- Find a known/unknown item

Query

- Find out about characteristics of item
- By itself or relative to others

HIGH-LEVEL ACTIONS: ANALYZE

Consume

Discover vs present

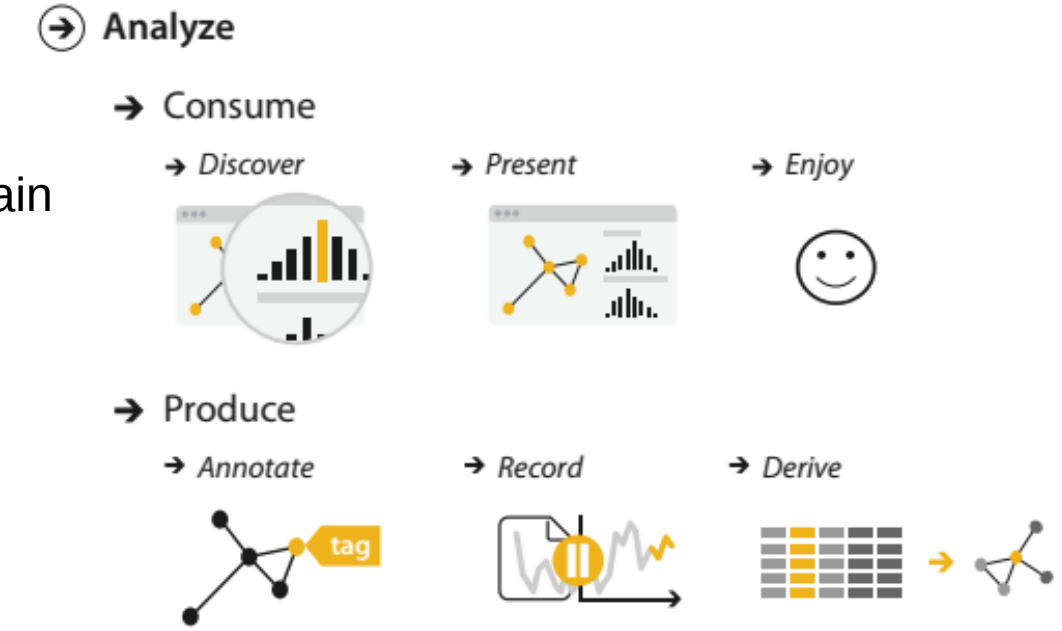
Classic split: explore vs explain

Enjoy: casual, social

Produce

Annotate, record

Derive: crucial design choice



MID-LEVEL ACTIONS: SEARCH, QUERY

Search: what does user know?

Target, location

Search

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

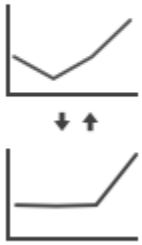
How much of the data matters?

One, some, all

Query

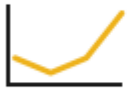
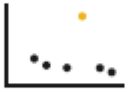
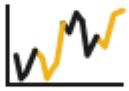
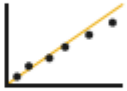
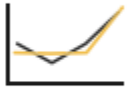
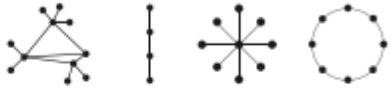
→ Identify

→ Compare



→ Summarize

LOW LEVEL: TARGETS

- **All Data**
  - → Trends 
  - → Outliers 
  - → Features 
- **Attributes**
  - → One
    - → *Distribution*
    - → *Extremes*
  - → Many
    - → *Dependency*
    - → *Correlation* 
    - → *Similarity* 
- **Network Data**
  - → Topology 
  - → *Paths*
- **Spatial Data**
  - → Shape

DESIGNING VISUALIZATIONS

WHAT IS DESIGN?

creating something new to solve a problem

can be used to make buildings, chairs, user interfaces, etc.

design is used in many fields

many possible users or tasks

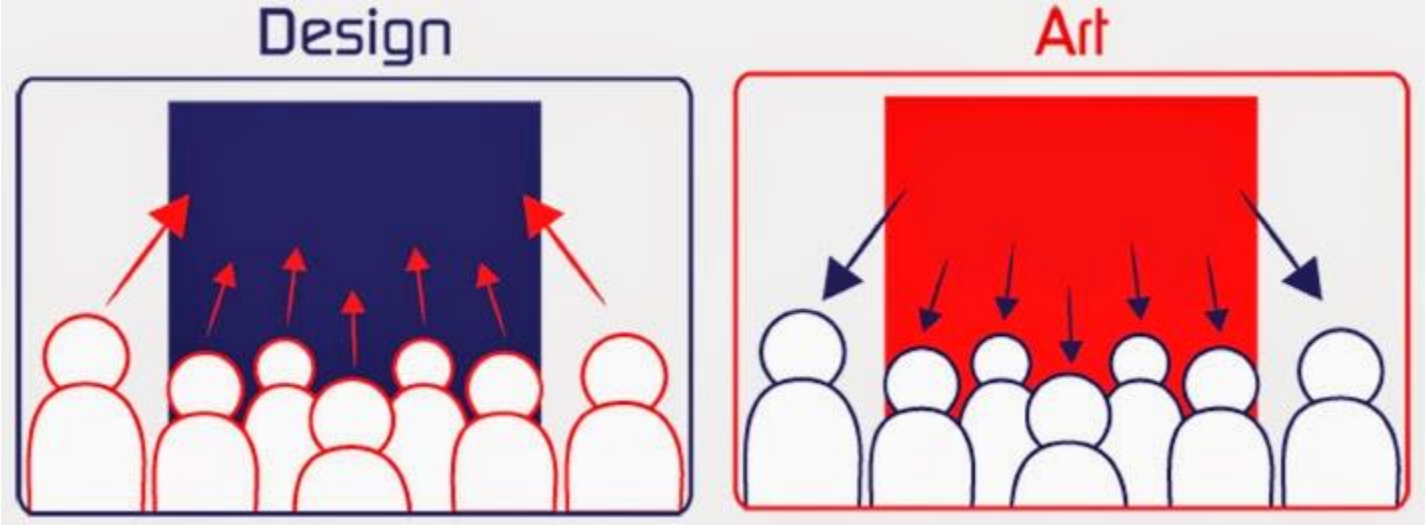
WHAT IS DESIGN NOT?

just making things pretty

art – appreciation of beauty or emotions invoked

something without a clear purpose

building without justification or evidence



FORM & FUNCTION

Commonly: “Form follows function”

Function can constrain possible forms

Form depends on tasks that must be achieved

“The better defined the goals of an artifact, the narrower the variety of forms it can adopt”

- Alberto Cairo

WHEN DO WE DESIGN?

Wicked problems

- No clear problem definition
- Solutions are either good enough or not good enough
- Multiple solutions exist, not true/false
- No clear point to stop with a solution

Examples of non-wicked (“tame”) problems

- Mathematics, chess, puzzles

WHY DOES DESIGN MATTER FOR VIS?

Many ineffective visualization combinations

Users with unique problems & data

Variations of tasks

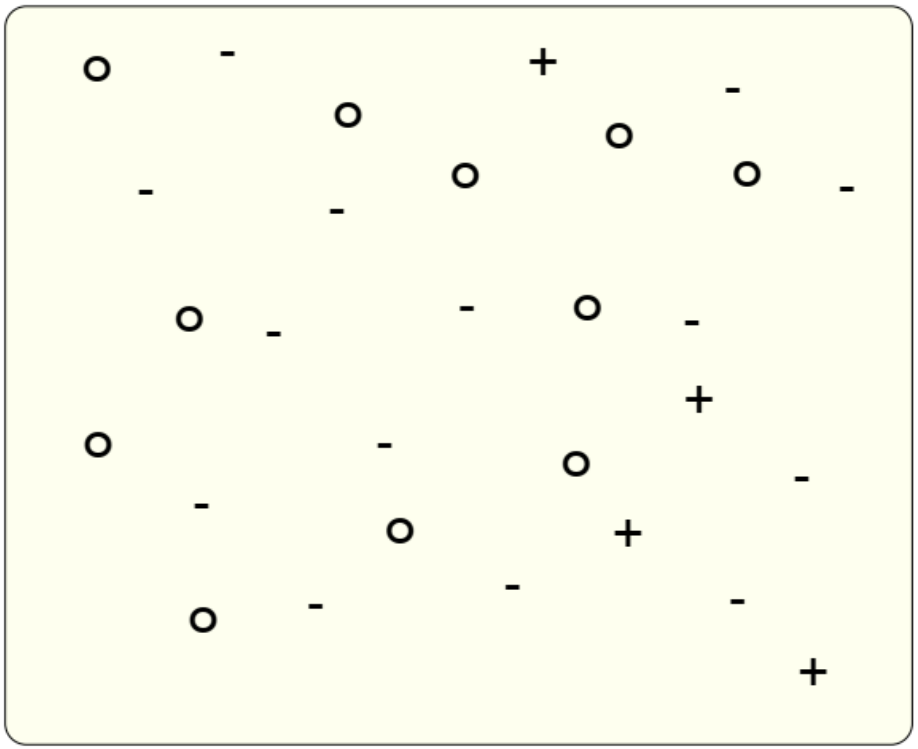
Large design space

Pitfall

Premature Design Commitment

by T. Munzner

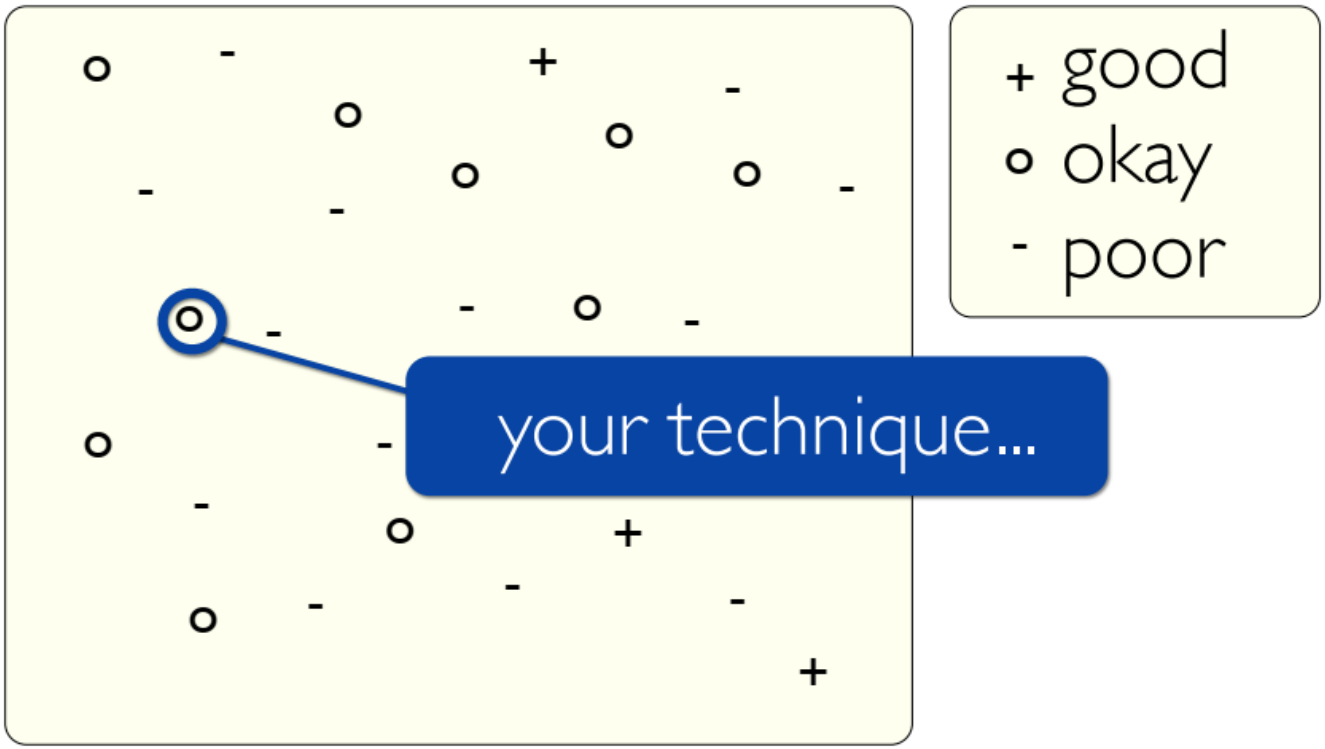
METAPHOR
Design Space



+ good
o okay
- poor

by T. Munzner

METAPHOR
Design Space



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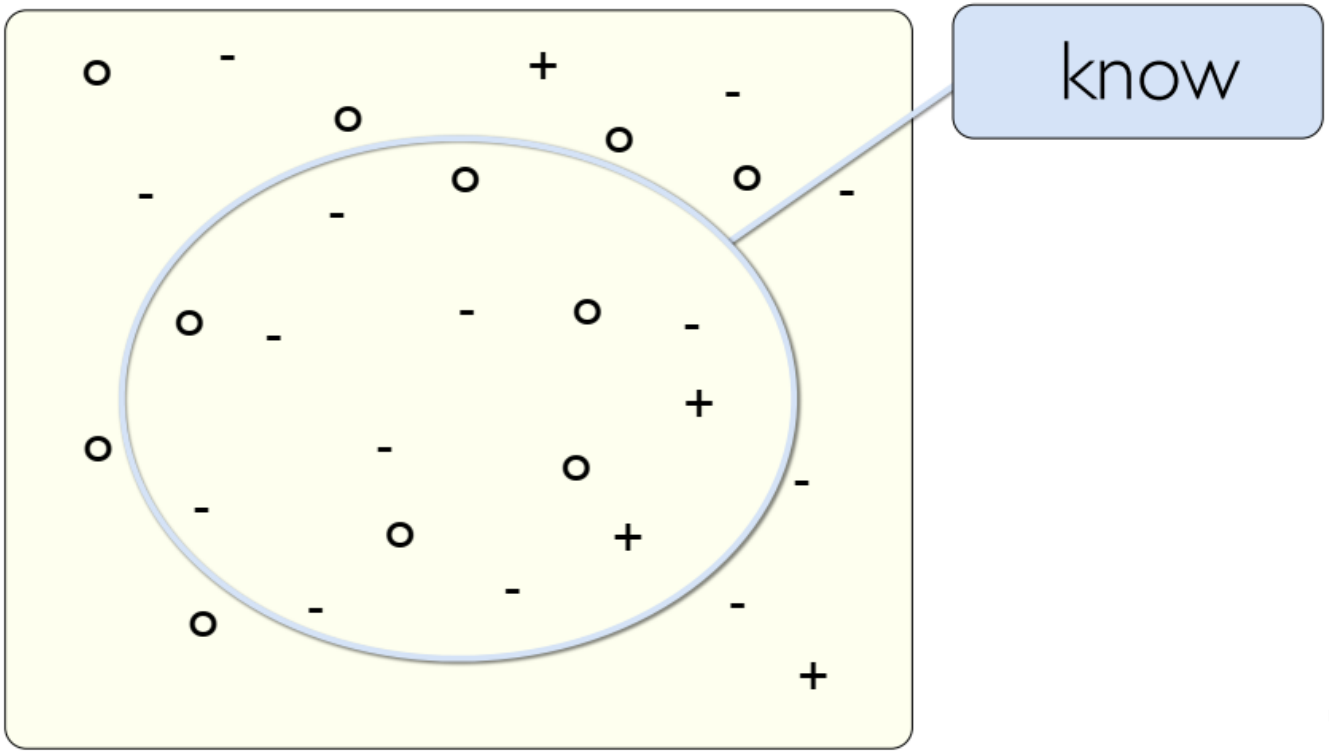
METAPHOR
Design Space



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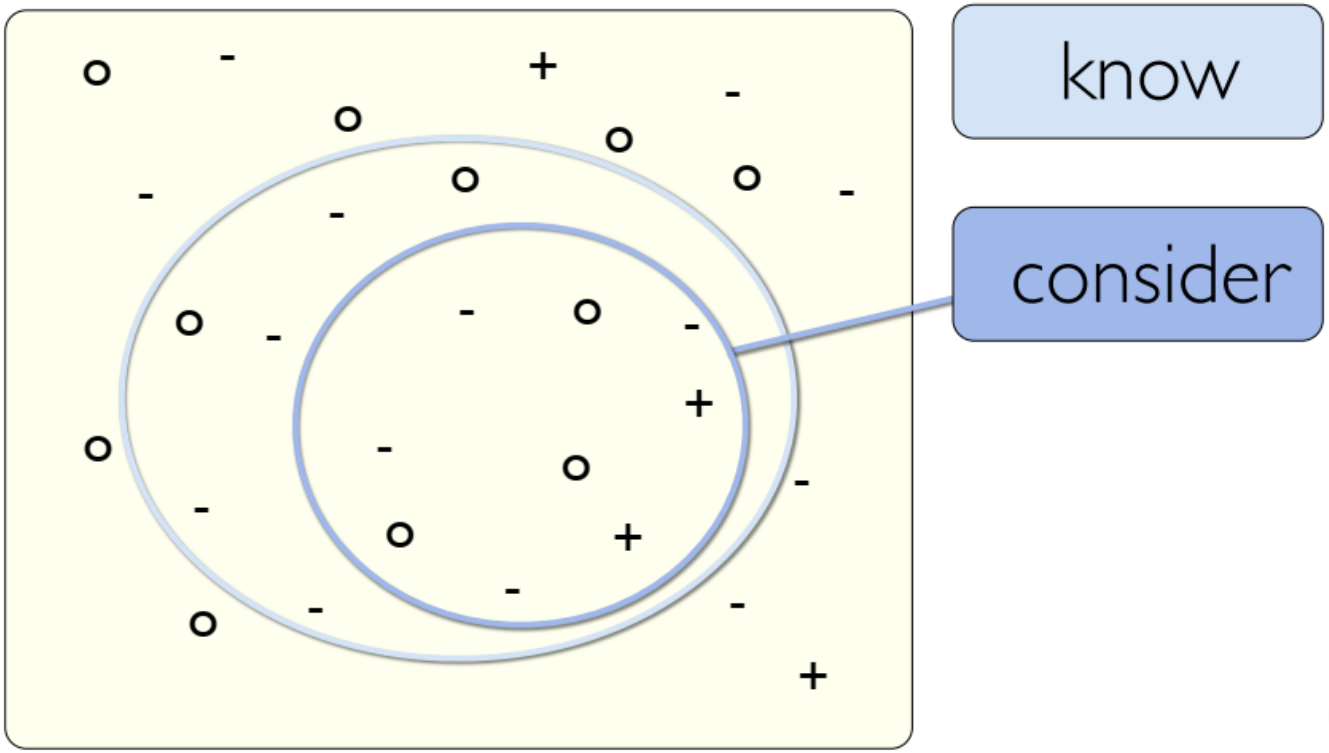
METAPHOR

Design Space



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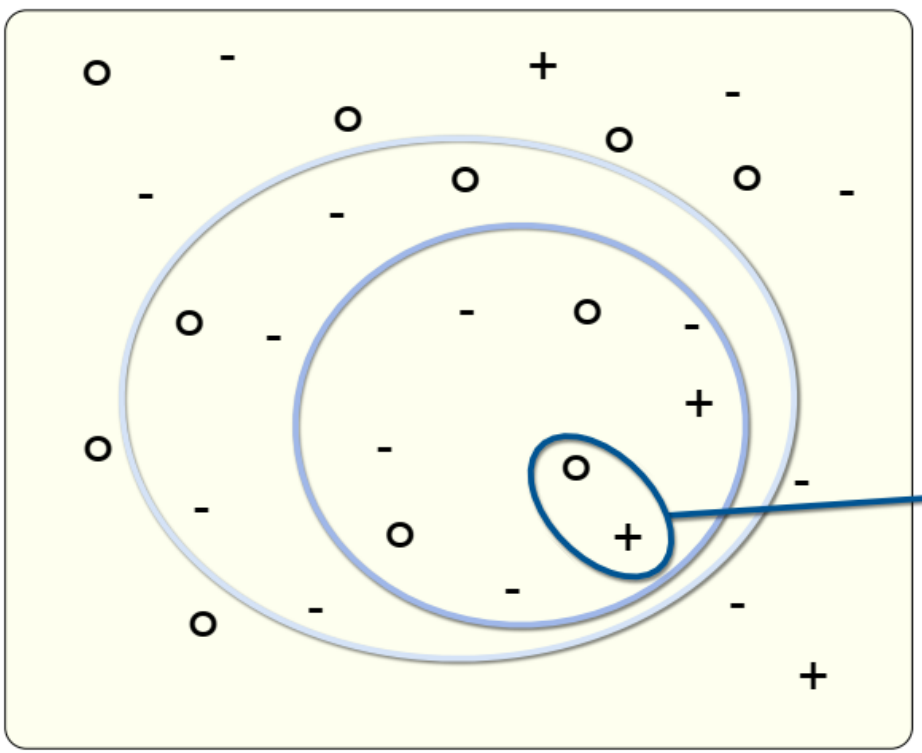
METAPHOR
Design Space



know

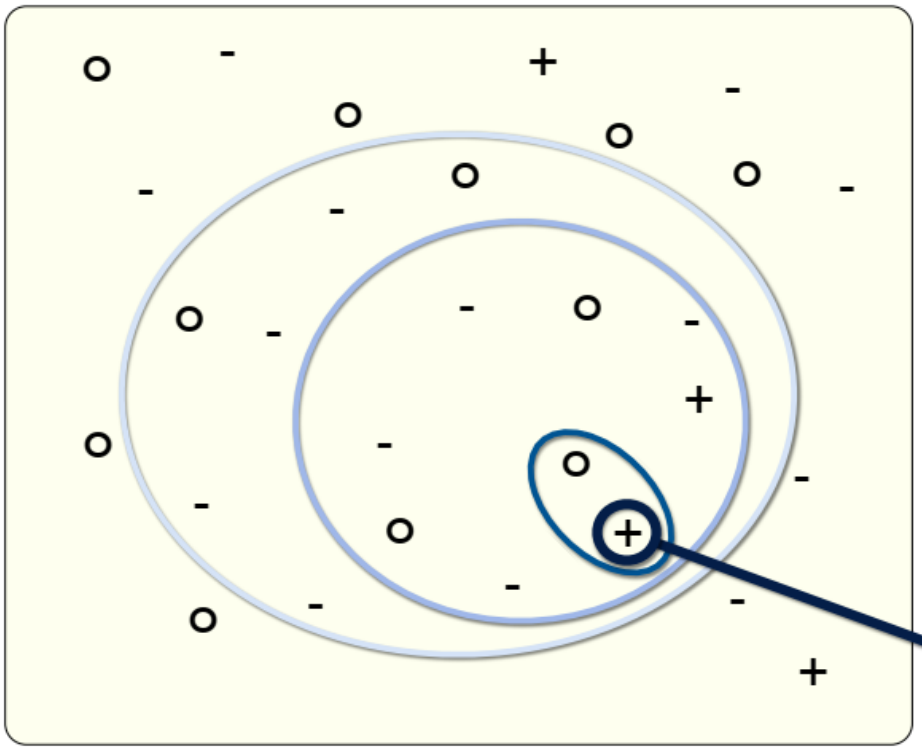
consider

propose

by T. Munzner

METAPHOR

Design Space



know

consider

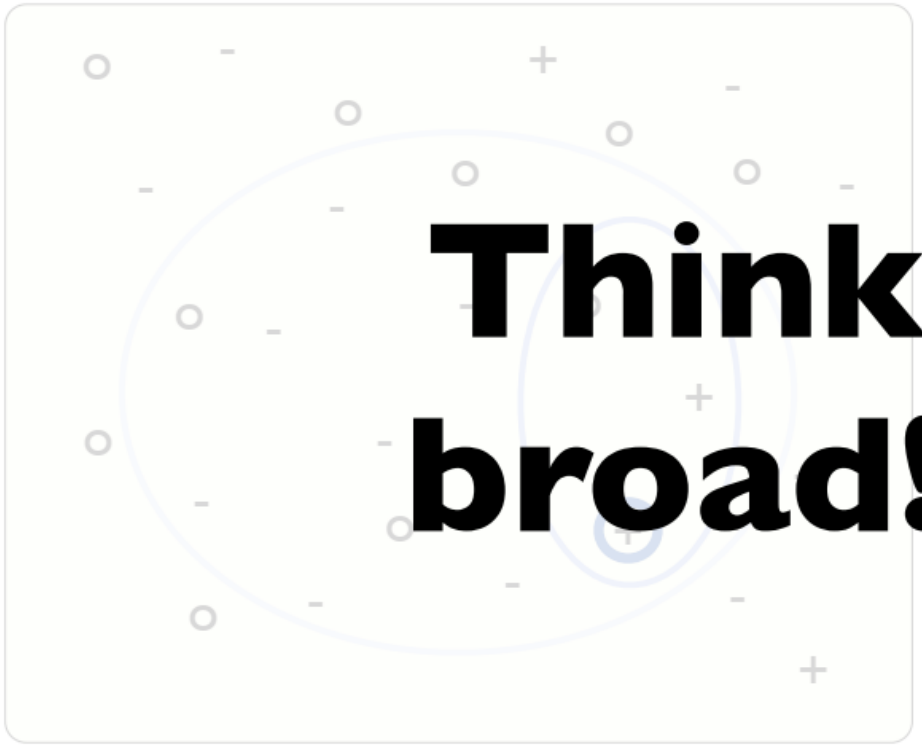
propose

select

by T. Munzner

Metaphor

Design Space



Think broad!

+ good
o okay
- poor

consider

propose

select

by T. Munzner

40

EVALUATION METHODS

Controlled experiment

Laboratory, crowd-sourced

Interviews/questionnaires

Unstructured, structured, semi-structured

Field observation, lab observation

Video/audio analysis

Coding/classification of user behavior (speech, gestures)

Log analysis

Algorithmic performance measurement

EVALUATION METHODS

Heuristic evaluation

Judge compliance with recognized metrics/usability methods (heuristics)

Usability testing, e.g., thinking aloud tests

Wizard of Oz

Human simulates response of system

Test functionality before it's implemented

Eye tracker evaluation

Expert evaluation

Insight-based evaluation

Case studies

QUANTITATIVE VS QUALITATIVE EVALUATION

Quantitative methods

Objective metrics, measurements

Use numbers/statistics for interpreting data

Qualitative methods

Subjective metrics

Description of situations, events, people, interactions, and observed behaviors, the use of direct quotations from people about their experiences, attitudes, beliefs, and thoughts

Focused on understanding how people make meaning of and experience their environment or world

INTERNAL VS EXTERNAL VALIDITY

Internal validity – can you trust your experiment

High when tested under controlled lab conditions

Observed effects are due to the test conditions (and not random variables)

External validity – is your experiment representative of real world usage

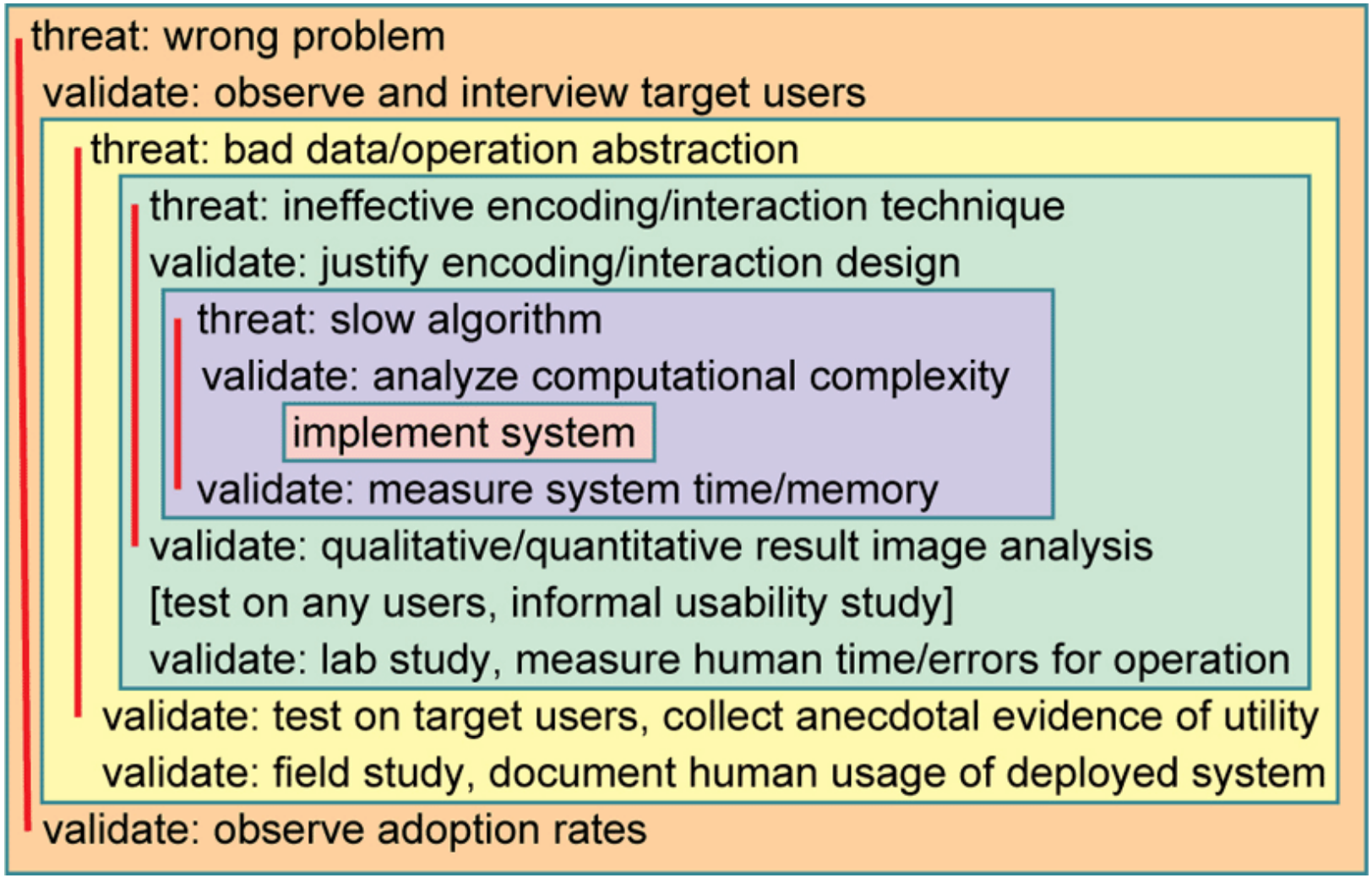
High when interface is tested in the field, e.g. handheld device tested in museum

Results are valid in real world

The trade-off

The more akin to real-world situations, the more experiment is susceptible to uncontrolled sources of variation

SCOPE OF EVALUATION



SCOPE OF EVALUATION

Pre-design

To understand potential users' work env. And workflow

Design

To scope a visual encoding and interaction design space based on human perception and cognition

Prototype

to see if a visualization has achieved its design goals, to see how a prototype compares with the current state-of-the-art systems or techniques

Deployment

to see how a visualization influences workflow and work processes, to assess the visualization's effectiveness and use in the field

Re-design

to improve a current design by identifying usability problems